



METODOLOGÍA PARA LA MEDICIÓN DE LA HIDRATACIÓN DEL CEMENTO ADICIONADO CON CENIZA VOLANTE A PARTIR DE IMPEDANCIA ELÉCTRICA

Simón Darío Peñaranda Sanjuán

Universidad Nacional de Colombia
Facultad de Ingeniería, Departamento de Ingeniería Civil y Agrícola
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Simón Darío Peñaranda Sanjuán

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“Siempre acabamos llegando a donde nos esperan”

Jose Saramago

A mis padres Victor y Maria, mi hermano Victor A. y a Naysla, quienes durante este tiempo han sido mi apoyo constante y motivación del día a día para nunca darme por vencido sin importar las dificultades del camino.

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Resumen

Metodología para la medición de la hidratación del cemento adicionado con ceniza volante a partir de impedancia eléctrica

El presente trabajo de investigación se centra en el estudio del uso y comportamiento de los electrodos impresos en 3D SPEs para su implementación en la evaluación de las propiedades de materiales cementantes, siendo el principal objetivo del trabajo identificar las diferentes fases y procesos producto de la hidratación de las pastas de cemento cuando estas contienen adiciones de ceniza volante dentro de su composición. Esto a causa de que durante las últimas décadas se ha venido evidenciando un alto potencial con el uso de las técnicas de ensayos de espectroscopia de impedancia electroquímica (EIS), los cuales surgen de la necesidad del estudio y predicción del comportamiento en estado fresco y la durabilidad del concreto, lo cual ha generado que de igual forma durante estos años se hayan desarrollado diferentes metodologías que permitan su medición principalmente mediante pruebas de laboratorio algo complejas.

Por esta razón, en el presente documento se realiza el estudio de los electrodos SPEs base polipropileno, los cuales como se podrá observar en los capítulos 4 y 5 presentaron un adecuado comportamiento que permitió realizar las mediciones de resistividad eléctrica a lo largo de su fase inicial de hidratación, logrando identificar a partir de los datos obtenidos durante las primeras 24 horas de medición los puntos críticos de la curva que definen las 4 fases principales producto de los procesos de hidratación de la mezcla; logrando así encontrar esos momentos claves donde se producen los cambios físicos y químicos dentro de una muestra, al igual que la afectación y cambios que se generaron al proceso cuando la muestra fue adicionada con ceniza volante. Adicionalmente, se realizan mediciones de la resistividad real de las muestras hasta el día 42, donde se identifica que las referencias de electrodos utilizados durante el ejercicio (los cuales están diseñados

para realizar mediciones en medios acuosos) pueden llegar a presentar novedades de funcionamiento pasadas 72 horas, ya que se logran identificar variaciones en comparación con los ensayos realizados mediante el método tradicional de las barras de carbono.

Adicionalmente, en el presente trabajo se desarrollaron mediciones de la liberación del calor acumulado durante la fase de hidratación de las pastas de cemento, para lo cual fue necesaria la adecuación de una cámara semi adiabática construida según las indicaciones de norma BS EN 196-9:210. *“Methods of testing cement. Heat of hydration. Semi-adiabatic method”* (HidrocemUN), la cual fue calibrada y verificada de acuerdo con un calorímetro adiabático de la referencia comercial I-CAL 8000, llegando a obtener buenos resultados en las muestras desarrolladas.

Finalmente, a partir de los resultados alcanzados se logra definir una metodología la cual permite a partir de las mediciones de resistividad eléctrica mediante el uso de electrodos SPEs base polipropileno, la definición de modelos lineales con los que se logran identificar los tiempos de fraguado inicial y final, al igual que los tiempos donde se inicia la pérdida de manejabilidad de la muestra. Logrando adicionalmente identificar las diferentes fases de la hidratación de las pastas de cemento en función de los procesos fisicoquímicos desarrollados en cada una de estas, al igual que la correlación que existe entre dichas mediciones con la evolución del calor de hidratación acumulado de las muestras a partir de modelos bilineales.

Palabras clave: (Electrodos impresos en 3D, ceniza volante, espectroscopia de impedancia electroquímica, hidratación y calorimetría).

Abstract

Methodology for measuring the hydration of cement added with fly ash from electrical impedance

This research work focuses on the study of the use and behavior of 3D SPEs printed electrodes for their implementation in the evaluation of the properties of cementitious materials, the main objective of the work being to identify the different phases and processes resulting from hydration. Of cement pastes when they contain additions of fly ash within their composition. This is due to the fact that during the last decades a high potential has been shown with the use of electrochemical impedance spectroscopy (EIS) testing techniques, which arise from the need to study and predict the behavior in the fresh state and the durability of the concrete, which has generated that in the same way during these years different methodologies have been developed that allow its measurement mainly through somewhat complex laboratory tests.

For this reason, in this document the study of polypropylene-based SPE electrodes is carried out, which, as can be seen in chapters 4 and 5, presented an adequate behavior that allowed electrical resistivity measurements to be carried out throughout its initial phase of hydration, managing to identify from the data obtained during the first 24 hours of measurement the critical points of the curve was define the 4 main phases resulting from the hydration processes of the mixture; this managing to find those key moments where physical and chemical changes occur within a sample, as well as the affectation and changes that were generated to the process when the sample was added with fly ash. Additionally, measurements of the real resistivity of the samples are carried out until day 42, where it is identified that the electrode references used during the exercise (which are designed to carry out measurements in aqueous media) may present past operating

novelties 72 hours since variations can be identified in comparison with the tests carried out using the traditional method of carbon bars.

Additionally, in the present work, measurements of the release of heat accumulated during the hydration phase of the cement pastes were developed, for which it was necessary to adapt a semi-adiabatic chamber built according to the indications of the BS EN 196-9 standard: 210. "Methods of testing cement. Heat of hydration. Semi-adiabatic method" (*HidroCemUN*), which was calibrated and verified according to an adiabatic calorimeter of the commercial reference I-CAL 8000, obtaining good results in the developed samples.

Finally, based on the results achieved, it is possible to define a methodology which allows, from electrical resistivity measurements through the use of polypropylene-based SPE electrodes, the definition of linear models with which it is possible to identify the initial setting times and end, as well as the times where the loss of manageability of the sample begins. Achieving additionally to identify the different phases of the hydration of the cement pastes based on the physicochemical processes developed in each of these, as well as the correlation that exists between said measurements with the evolution of the accumulated heat of hydration of the samples from of bi linear models.

Keywords: (3D printed electrodes, fly ash, electrochemical impedance spectroscopy, hydration and calorimetry).

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Lista de Símbolos y abreviaturas

A continuación, se presenta la lista de símbolos y abreviaturas que nos ayudaran a comprender mejor algunos párrafos de textos.

Símbolos con letras latinas

Símbolo	Término	Unidad SI	Definición
R	Resistencia eléctrica	$\Omega \cdot m$	$\rho \frac{L}{A}$
I	Corriente	Amp	Ec. 2.7
E	Potencial	W	Ec. 2.6
A	Área transversal de la carga	m^2	Ec. 2.8
L	Longitud	m	Ec. 2.8
Z	Impedancia	Ω	Ec. 2.10
V	Voltaje	v	Ec. 1 – Anexo 1
K	Constante de calibración		Ec. 5 – Anexo 1

Símbolos con letras griegas

Símbolo	Término	Unidad SI	Definición
ρ	Resistividad	$\frac{A \cdot R}{L}$	Ec. 2.8 y 2.9
φ	Ángulo de fase	rad	
ω	Frecuencia	Hz	
Φ	Ángulo de fase	$^{\circ}$	
σ	Conductividad real	Ω	Ec. 2.11
ϵ_r	Permitividad relativa	Ω	Ec. 2.12

Subíndice

Subíndice	Término
i	Fraguado inicial
f	Fraguado final
m	Perdida de manejabilidad
s	Punto de saturación
real	Impedancia real
img	Impedancia imaginaria
e	Potencial eléctrico

Abreviaturas

Abreviatura	Término
Ca	Calcio
Si	Sílice
H ₃ O	Hidroxilos
Na	Sodio
K	Potacio
EIS	Espectroscopia de impedancia electroquímica
C ₃ A	Aluminato tricálcico
C ₄ AF	Ferroaluminato tetracálcico
C ₃ S	Silicato tricálcico
C ₂ S	Silicato bicálcico
CO ₂	Dioxido de carbono
OH-	Hidróxilos
ITZ	Zona de transición intersticial
Ca(OH) ₂	Hidróxido de calcio
TSA	Trisulfoaluminato de calcio hidratado
MSA	Monosulfoaluminato de calcio hidratado
CP	Cemento puro
CV	Ceniza volante
ART	Alta resistencia temprana
CCC	Camino de conducción continua
ICP	Caminos aislados
DCP	Caminos de conducción discontinuos
CA	Corriente alterna
CC	Corriente continua
Rb	Resistividad eléctrica real
FA	Ceniza Volante
A/C	Relación agua/cemento
Δt	Delta de temperatura
SPEs	Electrodo serigrafiado
PET	Sustrato de tereftalato de polietileno
d	Días
K	Constante de calibración cámara semi-adiabática

Ag	Planta
AgCl	Cloruro de plata
F _i	Fraguado inicial
F _f	Fraguado final
P _m	Perdida de manejabilidad
P _s	Punto de saturación
P1	Punto crítico 1
P2	Punto crítico 2
P3	Punto crítico 3
Lol	Perdida por ignición
W _e	Trabajo eléctrico
P _e	Potencial eléctrico

Introducción

Desde el inicio del uso de los materiales base de cemento, ha sido una incógnita y motivo de investigación la forma en que estos materiales anhidros al entrar en contacto con el agua desarrollan las importantes cualidades ingenieriles que hoy en día son de gran importancia para el avance y desarrollo de la humanidad. Sin embargo, ha sido este acelerado proceso de desarrollo y la necesidad de disminuir la huella de carbono de la producción cementera por lo que ha surgido la necesidad de avanzar en la obtención de un material que brinde propiedades físicas similares a las del cemento Pórtland con menores costos ecológicos, ya que según el centro de estudios británico Chatham House, la industria cementera es la fuente de aproximadamente el 8% de las emisiones de dióxido de carbono (CO₂) del mundo (Rodgers, 2018). Razón por la cual durante los últimos años se han venido desarrollando diferentes investigaciones que permitan mitigar esta condición; donde gran parte de este objetivo se ha logrado gracias al uso en su mayoría de subproductos industriales como lo son la ceniza volante, la cual proviene principalmente de la quema de carbón mineral en plantas termoeléctricas.

Si este subproducto es tratado, procesado y adicionado de forma correcta en el cemento, tienen la capacidad de mejorar propiedades como la resistencia mecánica, la durabilidad, entre otras propiedades. Sin embargo, una vez este material es adicionados en el cemento, no solo las propiedades físicas en estado endurecido de este cambian; se ha logrado identificar que también se presentan modificaciones en las características del material en estado fresco, alterándose así los procesos reactivos y de liberación de energía que se dan dentro de las mezclas, lo cual se debe principalmente a la entrada de las partículas de material anhidro en contacto con el agua, generando un ambiente altamente alcalino, donde abundan los iones de Calcio (Ca), Sílice (Si), Hidroxilos (OH), los cuales son un medio propicio para que los granos de ceniza inicien sus procesos de reacción liberando iones de sodio (Na), potasio (K), aluminio (Al) y sílice (Si) a la fase líquida de la hidratación.

Aunque hoy en día se tiene pleno conocimiento sobre cómo se desarrolla el proceso de reacción de la ceniza volante en un ambiente alcalino y la forma en la que dicho material afecta los tiempos de hidratación del cemento, aun se tiene limitada información sobre la forma en que ciertos minerales como el sodio, potasio y los sulfatos influyen en los procesos de hidratación del cemento (Lizarazo, 2018).

Por esto, gracias a los avances que se han venido presentando en ramas de la tecnología del concreto a partir del estudio de las propiedades de las mezclas en estado fresco, representadas por metodologías como la resistividad eléctrica bulk a partir de la espectroscopia de impedancia electroquímica (EIS), se ha logrado estudiar de manera detallada el comportamiento fisicoquímico de las mezclas una vez se inicia la fase de hidratación de los minerales. Siendo este un método eléctrico que surge como una alternativa rápida y de bajo costo para muchas aplicaciones del estudio del concreto, como lo fue el análisis de la penetración de cloruros en el concreto, patología causante de los problemas más comunes de corrosión del refuerzo. Sin embargo, durante los últimos años se logra demostrar que a partir del uso de los ensayos electroquímicos la relación de estas propiedades con la micro-estructura del material, correlacionando así características como la porosidad, la difusividad y el transporte iónico a dicha propiedad (Lizarazo, Ramírez, et al., 2016).

Más allá de los importantes avances logrados con la implementación de esta técnica en el estudio de las propiedades de las mezclas de concreto, desde su aparición en las primeras investigaciones realizadas por McCarter en 1988, ha surgido la inquietud sobre cuál es la mejor forma para poder llevar a cabo estas mediciones, llevando a que a lo largo de los años se hayan realizado diversas propuestas de medición (mediciones de corriente continua, corriente alterna, métodos de electrodos, métodos sin contacto, entre otras) que permitan resolver las diferentes dificultades que pueden llegar a presentar estos ensayos, teniendo como común denominador el alto costo de los equipos a utilizar.

Por esta razón, en el presente trabajo se desarrolló la implementación del desarrollo de una metodología basada en la implementación de electrodos serigrafiados impresos en 3D, los cuales nos permitieron mediante su uso el estudio y correlación de las principales propiedades de hidratación de las pastas de cemento en edades tempranas y tardías con las mediciones tradicionales de tiempos de fraguado y evolución del calor liberado de las

mezclas, lográndose de esta forma desarrollar una metodología para el análisis de la hidratación de las pastas de cemento a partir de los diferentes cambios en el comportamiento eléctrico de las pastas de cemento con el paso del tiempo.

1. Preliminares

En el presente capítulo se da un breve panorama del problema a abarcar con el desarrollo del presente trabajo de investigación, de igual forma, se sustenta la estructura con que se da desarrollo al documento. Para esto se presentarán los ítems de identificación del problema, justificación, objetivos y estructura de la tesis.

1.1 Identificación del problema

Debido a que el cemento sigue siendo hoy en día uno de los materiales de mayor demanda y uso por la humanidad, existe una presión en el sector de la producción cementera por lograr mejorar las propiedades físicas y mecánicas que dan a dicho material la importancia que tiene para el desarrollo de la civilización.

Esto ha llevado a que en las áreas de investigación en tecnología del cemento se hayan aplicado nuevas tecnologías y herramientas de medición y control que permiten profundizar y comprender mejor las diferentes etapas de la producción del cemento y concreto, donde el estado fresco y la hidratación de las mezclas han sido uno de los principales objetos de investigación. Lo anterior debido a los innumerables vacíos que aún existen sobre la forma con que se llevan a cabo las diferentes reacciones fisicoquímicas durante la hidratación.

Por esta razón, comprender como se da la hidratación de las mezclas y como lograr mejorar los procesos de medición en estas etapas tempranas es un paso fundamental, ya que en la actualidad los procesos comúnmente utilizados para realizar dichas mediciones son ensayos con una alta complejidad y costo. Pero recientemente se han implementado nuevas técnicas como el calor de hidratación en cámaras adiabáticas y semi-adiabáticas; sin embargo, se sigue presentando la necesidad de desarrollar nuevas técnicas que

permitan complementar estas mediciones; por lo que en el presente trabajo se busca el desarrollo de una metodología que permita el estudio de los procesos de hidratación en las mezclas de cemento a partir de la aplicación de las diferentes metodologías de ensayos de impedancia electroquímica (EIS) como medida complementaria a estos métodos tradicionales mencionados anteriormente.

1.2 Justificación

Los procesos de desarrollo de las propiedades de hidratación en el cemento hidráulico inician cuando este entra en contacto con el agua, generando una disolución de los componentes anhidros del cemento y el desarrollo de un grupo de reacciones fisicoquímicas exotérmicas (Lura et al., 2010), durante las cuales se dan procesos de cambios mineralógicos dentro de la mezcla. Estos procesos están acompañados de la liberación de energía en forma de calor, la cual es asociada principalmente con la reacción de los aluminatos tricálcicos (C_3A) y los silicatos tricálcicos (C_3S), los cuales una vez comienzan sus procesos de hidratación aumentan su área específica, disminuyendo el espacio de los poros existente en la mezcla, dando como resultado una disminución del volumen inicial de la mezcla (Lura et al., 2010). Igualmente, la disminución del volumen inicial de la mezcla se ve afectado por la pérdida de agua que se utiliza en la hidratación del cemento anhidro y la que se pierde en los procesos de evaporación. Este cambio de volumen dará como resultado una contracción masiva de la pasta de cemento, efecto que de no ser controlado adecuadamente podría ocasionar patologías de fisuración en el concreto endurecido y por ende la pérdida significativa de una de sus principales propiedades físicas como lo es la resistencia a la compresión.

Considerando lo descrito anteriormente, es importante aclarar, que gran parte de los problemas asociados con la hidratación del cemento se pueden mitigar si se comprende el desarrollo de estos una vez el material anhidro entra en contacto con el agua. Aunque dichos efectos han sido estudiados décadas atrás, aún se tienen grandes vacíos sobre la forma en que se dan las reacciones y procesos de liberación de energía al interior de la mezcla. Es por esta razón que mediante la presente propuesta de grado se busca el desarrollo de una metodología en la que a partir de mediciones de la resistividad eléctrica

durante las diferentes etapas de hidratación (edades tempranas y tardías) de una mezcla de cemento adicionado con ceniza volante, se pueda estudiar desde otro punto de vista los procesos de hidratación y desarrollo de minerales.

1.3 Objetivos

A continuación, se presentan los objetivos principales planteados en el presente trabajo de investigación.

1.3.1 Objetivo general

- Desarrollar una metodología para la medición de la hidratación del cemento hidráulico adicionado con ceniza volante a partir de la impedancia eléctrica.

1.3.2 Objetivos específicos

- Evaluar las propiedades eléctricas durante el proceso de hidratación del cemento hidráulico adicionado con ceniza volante mediante espectroscopia de impedancia electroquímica.
- Realizar la medición de la hidratación de las pastas de cemento hidráulico adicionadas con ceniza volante a partir de métodos tradicionales como los tiempos de fraguado con agujas de Vicat y mediciones de calorimetría en cámara semi-adiabática.
- Establecer un modelo que permita determinar la hidratación del cemento Pórtland adicionado con ceniza volante (FA) a partir de técnicas eléctricas.

1.4 Estructura de la tesis

El presente documento cuenta con 5 capítulos que abarcan desde el marco teórico y la descripción de diferentes contextos que permiten comprender mejor los procesos, ensayos y análisis que se desarrollaron durante la tesis, hasta los resultados y conclusiones obtenidas de los ensayos e hipótesis planteadas. A continuación, se describen cada uno de estos 5 capítulos.

Capítulo 1. Marco Teórico: incluye el estado del arte con referencia a los procesos de hidratación de las mezclas de cemento anhidro y cómo influyen en estos procesos las

adiciones minerales de algunos materiales como lo son las cenizas volantes. Finalmente se hace un recuento sobre que son los ensayos de espectroscopia de impedancia electroquímica y como se pueden implementar en el estudio de las mezclas cementantes.

Capítulo 2. Metodología: en el desarrollo de este capítulo se explicará el procedimiento de medición y ensayo desarrollado con la finalidad de obtener la información necesaria para poder definir una metodología de medición de la resistividad que permita estudiar los procesos de hidratación. Durante esta sección se hará énfasis en los procesos de adecuación y calibración del calorímetro existente en el Laboratorio de Estructuras de la Universidad Nacional de Colombia.

Capítulo 3. Resultados Obtenidos: durante este capítulo, se muestran los resultados obtenidos de los ensayos realizados (Agujas de Vicat, calorimetrías y EIS) para las muestras de cemento adicionadas con diferentes porcentajes de ceniza volante, esto con el fin de poder medir los procesos de hidratación y las variaciones que se pueden presentar en cada una de las mezclas.

Capítulo 4. Desarrollo y análisis de la metodología para la medición de la hidratación a partir de los ensayos de resistividad: una vez terminados los ensayos propuestos y analizados los datos obtenidos, se presenta una metodología que permitió estudiar los procesos de hidratación en las pastas de cemento a partir de las mediciones de resistividad eléctrica en las muestras de cemento con y sin adiciones minerales. Finalmente, se realiza un análisis específico sobre los alcances y limitaciones de la metodología desarrollada.

Capítulo 5. Conclusiones y recomendaciones: finalmente, en este capítulo se presentan las conclusiones generales obtenidas luego de realizado el trabajo de investigación. Adicionalmente, se dan algunas recomendaciones para tener en cuenta para los próximos trabajos que se realicen en dicha área.

2. Marco Teórico

En el desarrollo del presente capítulo se busca presentar algunas definiciones básicas sobre el cemento y sus principales propiedades, lo cual facilitará la comprensión de términos y etapas de los procesos en que el cemento adquiere sus principales propiedades; de igual forma, se presentará el estado del arte sobre los temas relacionados a las mediciones de resistividad eléctrica mediante espectroscopia de impedancia electroquímica y los diferentes usos que se le viene dando hoy en día.

2.1 Generalidades del cemento

El cemento hidráulico es un polvo que al ser mezclado con agua forma una masa plástica fácil de moldear que fragua con el tiempo y se endurece gradualmente generando un aumento de la resistencia. Este polvo puede ser clasificado en varias categorías o grupos, según la composición mineralógica obtenida durante la producción, lo cual puede variar en función del proceso tecnológico utilizado durante la clinquerización; para esta clasificación se pueden seguir los parámetros de evaluación definidos por normas como la ASTM C150 (*Standard Specification for Portland Cement*).

El cemento Pórtland o cemento tipo I (según las clasificaciones recomendadas en la norma ASTM C150), es un cementante inorgánico e hidráulico formado por silicatos y aluminatos de calcio (Lizarazo-marriaga, 2010) producto de fusión a altas temperaturas de los óxidos de sílice, calcio, alúmina y hierro (ver capítulo 2.2); los cuales al entrar en contacto con el agua sufren un proceso de transformación físicoquímico, que le da las propiedades físicas que han hecho de este producto uno de los materiales más importantes para la humanidad y su desarrollo.

Aunque describir la composición del clinker o cemento no es nada fácil debido a las innumerables variables que pueden afectar la producción, es fácil notar que la suma de

cuatro componentes producto de la fusión de los óxidos (CaO , SiO_2 , Al_2O_3 y Fe_2O_3) es por regla general superior al 95%. De esta forma se pueden presentar los procesos de cristalización de las fases del cemento Pórtland en función de cuatro componentes (Kurdowski, 2014). Estos cuatro componentes fundamentales del cemento Pórtland son definidos como:

- Silicato tricálcico (C_3S) o Alita
- Silicato bicálcico (C_2S) o Belita
- Aluminato tricálcico (C_3A) o Celita
- Ferroaluminato tetracálcico (C_4AF) o Felita

La alita y la belita son las fases de mayor presencia (componentes encargados de dar las propiedades de resistencia a corto y largo plazo en las mezclas), y pueden llegar a ocupar hasta el 80% del total del cemento, lo cual varía en función de los procesos de clinkerización de minerales usados y el tiempo en que el material permanece en el horno (Kurdowski, 2014), entre otros factores de importancia.

Finalmente, es importante mencionar que el 5% de la composición restante se considera como subproductos relacionados con las impurezas de la materia prima a la hora de la clinkerización. A pesar de que el volumen de estos subproductos sea muy bajo, pueden llegar a tener efectos significativos en las propiedades de las mezclas en estado fresco y endurecido. Tal como lo describe Newman (2007) en sus escritos, algunos de los componentes secundarios más representativos son:

Tabla 2-1: Subproductos de la clinkerización estimados por Newman

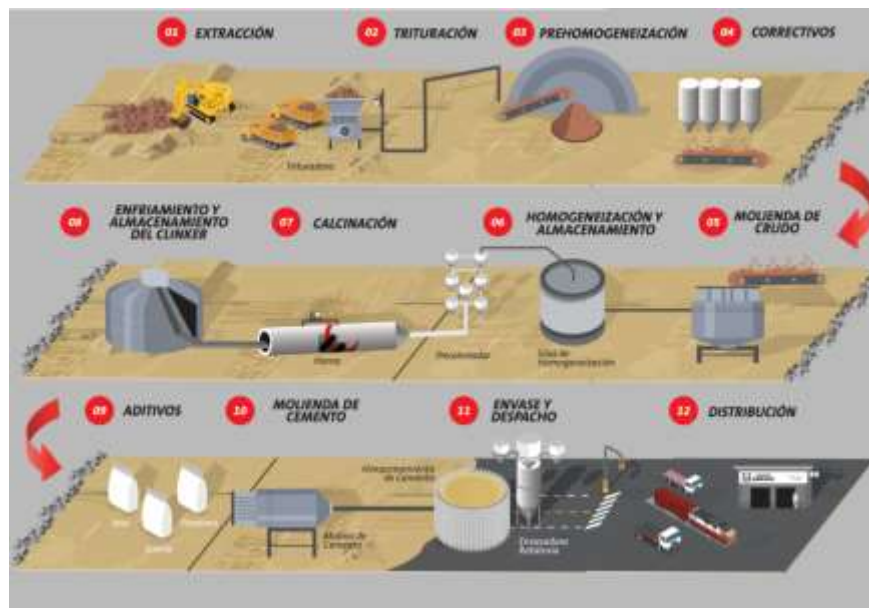
Componente	Porcentaje
MgO	1.50%
SO_3 (S)	1.20%
K_2O	0.72%
Pérdida al fuego	0.50%
TiO_2	0.30%
Na_2O	0.18%
P_2O_5	0.15%
Mn_2O_3	0.05%

Nombre de la fuente: (Kurdowski, 2014)

2.2 Proceso de fabricación del cemento

Los procesos de fabricación del cemento, al igual que la tecnología con que estos son producidos, tienen una incidencia directa sobre la composición mineral final y por ende en el comportamiento principalmente de los procesos de hidratación. Tal como lo describe la Asociación de Productores de Cemento (ASOCEM) (2018), el proceso de fabricación se puede dividir en tres grandes etapas (figura 2-1), las cuales se pueden explicar cómo:

Figura. 2-1: Etapas de producción del cemento.

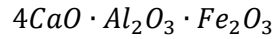
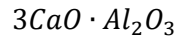
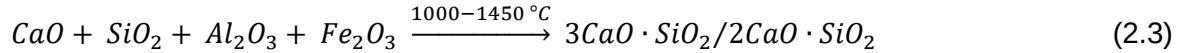
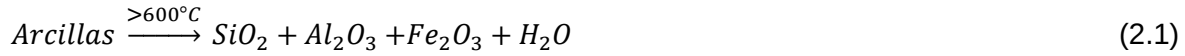


Nombre de la fuente: Holcim Mexico, 2020.

- Etapa 1, explotación y homogenización del material: etapa donde el material es extraído desde las canteras y transportado hasta la planta de producción, aquí se realiza un primer proceso de trituración y molienda del material, con lo que se busca que sea más fácil la dosificación y el control de calidad de las muestras.
- Etapa 2, clinkerización de la muestra: esta se puede definir como la parte principal para la producción del cemento, pues es aquí, donde a partir de elevar la temperatura por encima de 1400°C en la mezcla previamente homogenizada, se da inicio a los procesos de clinkerización de minerales y obtención de los minerales básicos del cemento, los cuales son productos de la descomposición de las arcillas

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(ecuación 2.1) y calizas (ecuación 2.2), lo cual nos da como resultado los siguientes productos (Lizarazo, 2018):



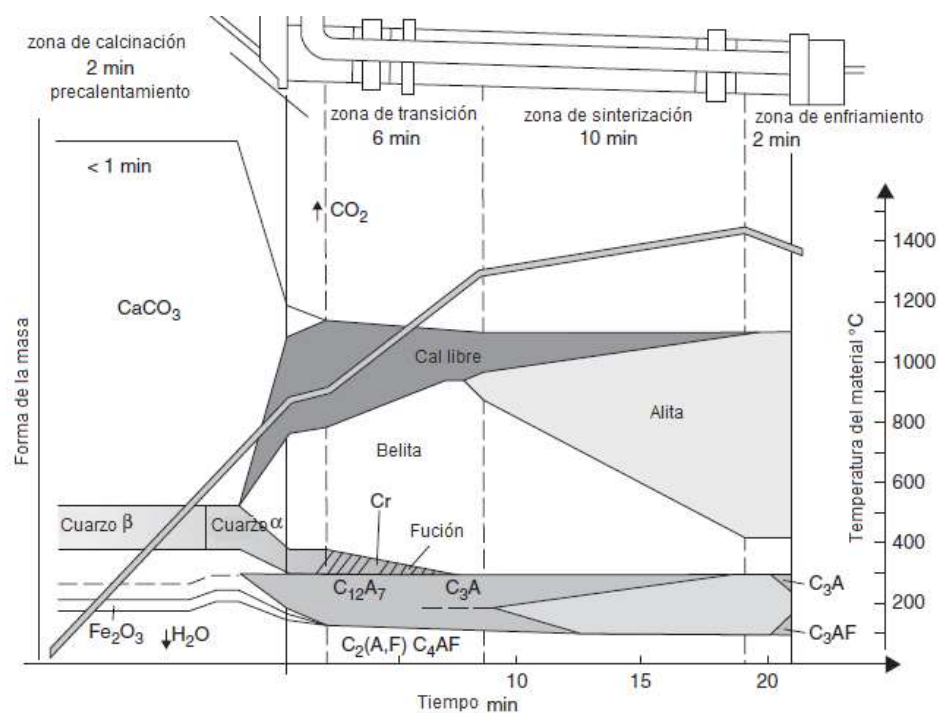
Finalmente, este nuevo material es enfriado rápidamente hasta alrededor de los 200°C para obtener como resultado el clinker. En esta etapa, es importante resaltar que como se puede ver en la ecuación 2.2, uno de los subproductos de la transformación de las calizas es el dióxido de carbonó (CO_2), el cual, sumado a los residuos del calentamiento de los hornos y transporte de material, entre otros, hacen que la industria cementera sea una de las mayores generadoras de CO_2 en el mundo, produciendo cerca del 8% del total mundial (Rodgers, 2018).

- Etapa 3, molienda del material: finalmente, las partículas de clinker son sometidas a nuevos procesos de molienda, donde son adicionados materiales como yeso y en algunas ocasiones adiciones minerales. Estos materiales son molidos en conjunto con el clinker para así tener una mejor homogenización del material previo a su empaclado, dando como resultado la obtención de cementos hidráulicos. Este proceso de molienda y como tal la finura del material es una característica muy importante, ya que define comportamientos específicos del material durante la etapa de hidratación, así como una influencia sobre las propiedades reológicas de la muestra.

Es importante aclarar, que durante el proceso de clinkerización de los minerales (ver figura 2.2), el cemento almacena una gran cantidad de energía debido a los incrementos de temperatura a los que es llevado; esta energía será liberada en forma de calor una vez el

cemento anhidro entra en contacto con el agua y se de inicio a las diferentes reacciones fisicoquímicas producto de los procesos de hidratación. Estos procesos pueden ser monitoreados a través del calor específico, el cual es un parámetro importante de las mezclas que requiere de un control continuo durante los procesos de fundición (Lura et al., 2010), ya que su control es un indicador para garantizar una correcta postura y fraguado de las mezclas; pero también puede ser utilizado como un parámetro de medición que permite conocer las características de los cementos (Li et al., 2003a).

Figura. 2-2: Proceso de clinkerización de minerales.



Nombre de la fuente: Adaptado de kurdowski, 2014.

2.3 Adiciones minerales

Las adiciones minerales hacen referencia a materiales silico-aluminosos que pueden ser de origen natural o artificial, las cuales surgen a raíz de la necesidad de producir concretos con menor huella de carbono. Lo anterior considerando que las adiciones minerales artificiales son en su mayoría subproductos industriales, los cuales se presentan como una

muy buena alternativa de material cementante suplementario (SCM) (Whiting & Nagi, 2003).

Una de las adiciones minerales artificiales más utilizadas actualmente es la ceniza volante (*fly ash*), la cual surge de la quema de carbones minerales utilizados principalmente para la generación eléctrica. Las cenizas volantes están clasificadas como adiciones minerales activas – puzolánicas, lo que significa que estas tienen altos contenidos de materiales silico-aluminosos y adicionalmente han sufrido un proceso de tratamiento termico. Este material requiere ser finamente molido y estar en la presencia de un ambiente altamente alcalino como el que se da con la presencia del hidróxido de calcio, para así poder desarrollar productos de hidratación similares al cemento Pórtland (Lizarazo-marriaga, 2010).

2.3.1 Características de las cenizas volantes

De acuerdo con la Norma ASTM C-618, “la Ceniza Volante se define como un residuo finamente dividido que resulta de la combustión de carbón pulverizado que es recolectado por medio de precipitadores electrostáticos”, estas se clasifican en tipo N, F o C de acuerdo con los requerimientos químicos (tabla 2.2) y físicos (tabla 2.3) dados en la norma, los cuales son influenciados por las propiedades químicas del carbón empleado. Actualmente, en Colombia es más común encontrar cenizas de clase F; las cuales a diferencia de las de las demás clases, rara vez presentan propiedades cementantes cuando se mezclan solo con agua (Comité ACI 232.2R-96, 1996).

Tabla 2-2: Requerimientos químicos para la clasificación de las cenizas según norma ASTM C-618

Criterio	Clase		
	N	F	C
$\text{SiO}_2 + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3$ (min %)	70	70	50
SiO_3 (max %)	4	5	5
Contenido de humedad (max %)	3	3	3
Perdida al fuego, LOI, (máx %)	10	6	6

Tabla 2-3: Requerimientos físicos para la clasificación de las cenizas según norma ASTM C-618

Criterio	Clase		
	N	F	C
Finura: Cantidad retenida en malla húmeda 45Um (No.325) (máx %)	34	34	34
Strength activity index: Con cemento portland a 7 días (min % control)	75	75	75
Con cemento portland a 28 días (min % control)	75	75	75
Requerimiento de agua (max % control)	115	105	105
Estabilidad: Expansión o contracción en autoclave (max %)	0.8	0.8	0.8
Requerimientos de uniformidad, Densidad: máxima variación con respecto al promedio (max %)	5	5	5
Porcentaje retenido en malla 45um (No. 325): máxima variación con respecto al promedio (máx %)	5	5	5

Por lo general, las cenizas volantes son más finas que el cemento y consisten principalmente en partículas esféricas vítreas, así como residuos de hematita y magnetita, carbón y algunas fases cristalinas formadas durante el enfriamiento, proceso que marca de manera considerable la calidad de estos materiales, ya que en función de la velocidad con que se realice, se tendrán mayor cantidad de fases vítreas (Whiting & Nagi, 2003). Este contenido vítreo, es uno de los factores que más influyen en la calidad del material, junto con otros cómo la finura, el bajo contenido de carbón y álcalis (Kurdowski, 2014).

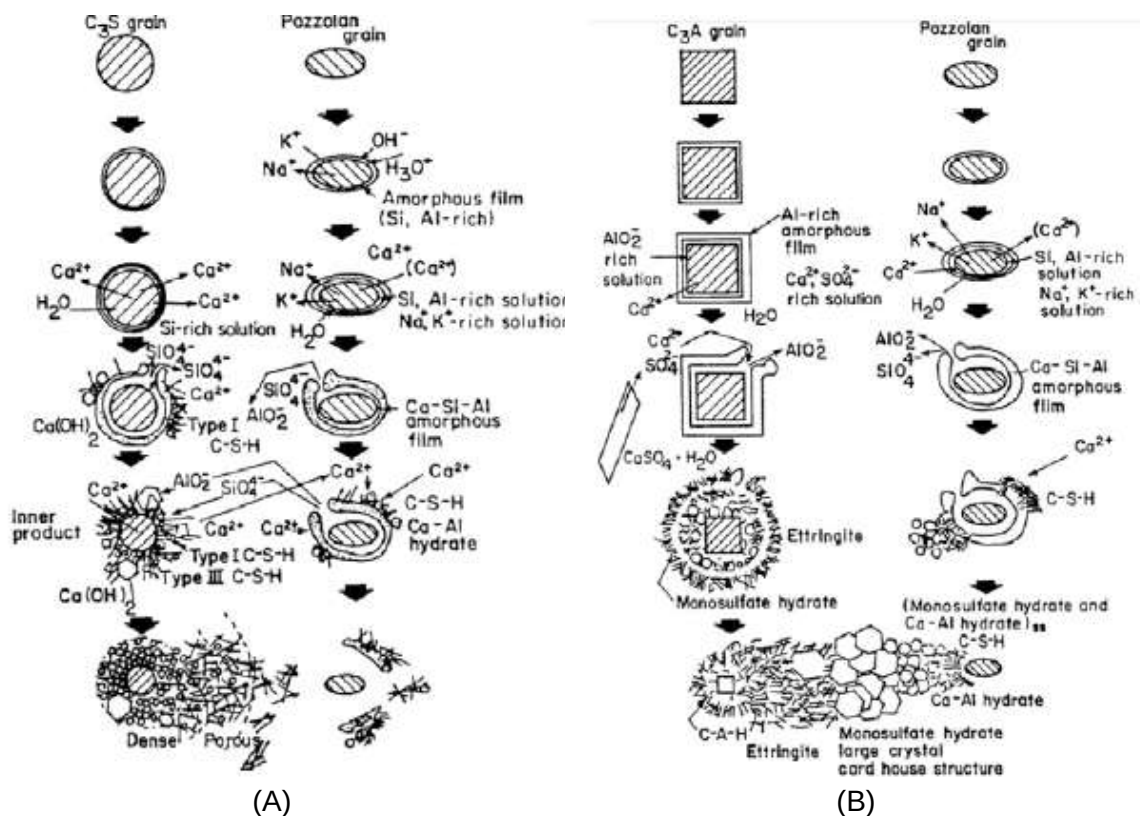
2.3.2 Hidratación de las cenizas volantes

El proceso de hidratación de la ceniza volante implica reacciones con los componentes del cemento (Fonseca Barrera, 2016). Dicho proceso se inicia con la disolución de las capas más superficiales de los granos de ceniza, donde se da la liberación de iones de sodio y potasio, lo que genera el inicio de la de hidrolisis de la ceniza al interactuar con el H_3O^+ presente en la solución acuosa, generando así un ambiente alcalino. Esto induce a la formación de una costra rica en iones de sílice, aluminato, sodio y potasio sobre los granos

de ceniza, la cual a medida que interactúa con los iones de Ca^{2+} va generando un aumento del espesor y la presión osmótica de dicha capa (Lizarazo, 2018).

Finalmente, superada la presión osmótica en los granos de ceniza debido a las diferencias de concentraciones iónicas con el medio acuoso, se da la liberación de los iones de silicatos y aluminatos a la solución, los cuales comienzan a reaccionar con los iones de calcio presente, induciendo principalmente a la formación de gel C-S-H (ver figura 2-3) y a la formación de etringita en el caso de presencia de yeso.

Figura. 2-3: Proceso de hidratación de la ceniza volante con C_3S (A) y C_3A (B).



Nombre de la fuente: (Ramezaniapour, 2013).

2.3.3 Efecto de las cenizas volantes en las mezclas de cemento.

Una vez las cenizas volantes son adicionadas a las mezclas de concreto, son afectadas tanto las propiedades en estado fresco como en estado endurecido de la mezcla. En

función de los porcentajes de adición o reemplazo de ceniza volante, propiedades del concreto como los tiempos de fraguado, el calor de hidratación liberado, la resistividad, entre otras (estado fresco) o la resistencia a la compresión, la permeabilidad, entre otras (estado endurecido) pueden llegar a sufrir modificaciones que favorecen la mezcla en el caso en que las cenizas sean dosificadas de la forma correcta. Lo anterior se podrán profundizar en investigaciones como las realizadas por Garzón (2014) y Fonseca (2016) en la Universidad Nacional de Colombia Sede Bogotá, donde se estudió y evaluó el comportamiento de algunas de estas propiedades para diferentes cenizas y diferentes dosificaciones.

Con el fin de comprender mejor el alcance del presente proyecto, se quiere resaltar que debido a su efecto puzolánico y a las propiedades físicas de las cenizas, estos materiales influyen en la microestructura de la matriz del cemento, en su porosidad y en la composición química de la solución de poro. Por tanto, se ejercen una influencia sobre la resistividad eléctrica del cemento y concreto. Esta influencia sobre la resistividad eléctrica está relacionada principalmente con los cambios en la microestructura del material; los cuales en 1994 fueron estudiados por Ehtesham, quien trabajó sobre la química de la solución de poros y el refinamiento de la estructura de los poros, llegando a la conclusión que al reemplazar el 30% del cemento por cenizas volantes, la concentración de iones hidroxilo (OH^-) se reduce y la distribución de poros es más fina que en el concreto simple, encontrando que el radio de poro promedio de la matriz de cemento con cenizas volantes fue 30% menor que el de la matriz de cemento simple. Finalmente, se concluyó que la resistividad eléctrica del concreto adicionado con cenizas volantes es más alta que la del concreto normal por un factor de aproximadamente 2.2 veces (Whiting & Nagi, 2003).

2.4 Hidratación del cemento

La reacción del cemento con el agua son un proceso muy complejo, el cual durante mucho tiempo ha sido objeto de numerosas discusiones y controversias, las cuales hasta el día de hoy no han logrado centrarse en una explicación totalmente sólida de la forma en que se desarrolla. Muchos de los estudios relacionados a este proceso han revelado que el mecanismo a edades tempranas es más complejo, puesto que no se trata solo de la ruptura de los enlaces en la estructura cristalina por las moléculas de agua (proceso endotérmico con efecto térmico correspondiente a la energía del sistema) y al transporte de iones a la

solución con su hidratación simultánea (un proceso exotérmico); ya que una vez el cemento anhidro entra en contacto con el agua, se dan una serie de complejas reacciones fisicoquímicas y termodinámicas que transforman los componentes dándoles nuevas características y propiedades (Kurdowski, 2014).

Aunque hoy en día no se tengan completamente claros los diferentes procesos que se dan durante la hidratación, varios autores concuerdan en que durante las primeras etapas de este proceso se pueden identificar condiciones específicas como lo son: la transformación de fases anhidras a fases hidratadas mediante reacciones químicas, la ganancia de resistencia y pérdida de consistencia debido a los productos de la hidratación, aumento de la temperatura y liberación de calor, cambios volumétricos y el aumento en la resistividad eléctrica por cambios en la microestructura de la muestras (Lizarazo-marriaga, 2010); los cuales han sido factores de estudio que han ayudado a responder muchas de las incógnitas que han surgido con respecto a los procesos de hidratación.

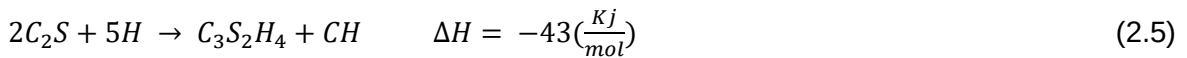
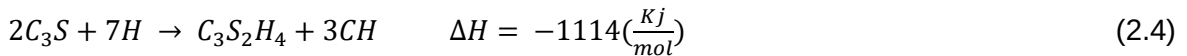
Lo anterior da entender la gran complejidad que tienen los procesos de hidratación, complejidad que puede aumentar a medida que se tienen más componentes en la mezcla (adiciones minerales, aditivos, etc); pero que de igual forma se han convertido en factores que han impactado los requerimientos tecnológicos que implican su estudio, puesto que con el paso del tiempo y la aparición de nuevas herramientas tecnológicas, se han permitido nuevos avances en la comprensión y explicación del proceso.

Aun así, durante los procesos de hidratación del cemento hidráulico (en especial el cemento Pórtland), existen principalmente cuatro compuestos minerales que lo conforman (C_3S , C_2S , C_3A y C_4AF), y cuyas reacciones de hidratación difieren significativamente entre sí. Una cantidad considerable de estudios ha demostrado que los productos de hidratación del cemento Pórtland son química y físicamente cercanos a los productos de hidratación del compuesto individual en condiciones de reacción similares. Aunque los productos de hidratación pueden influirse entre sí o interactuar con otros compuestos minerales en el sistema agua-cemento, finalmente los errores inducidos por la suposición de hidratación independiente son bajos.

Por último, ya que la hidratación de los cuatro compuestos minerales principales del cemento Pórtland proporciona una base para describir las características de hidratación del cemento (Swaddiwudhipong et al., 2002), a continuación, se explicará de forma individual el comportamiento de cada uno de estos compuestos una vez entran en contacto con el agua.

2.4.1 Hidratación de los silicatos de calcio

La hidratación de los silicatos de calcio, también conocidos como Alita (C_3S) y Belita (C_2S), puede ser considerada como las de mayor relevancia, no solo por representar cerca del 80% de los minerales del cemento, si no, que son estos los encargados de producir el gel C-S-H o los silicatos de calcio hidratados, los cuales son encargados de brindar entre otras propiedades, la resistencia a la compresión de la mezcla. Es importante mencionar que durante los procesos de hidratación los silicatos se liberan considerables cantidades de energía (Ec. 2.4 y 2.5), la cual se ve reflejada en un aumento de la temperatura del material.



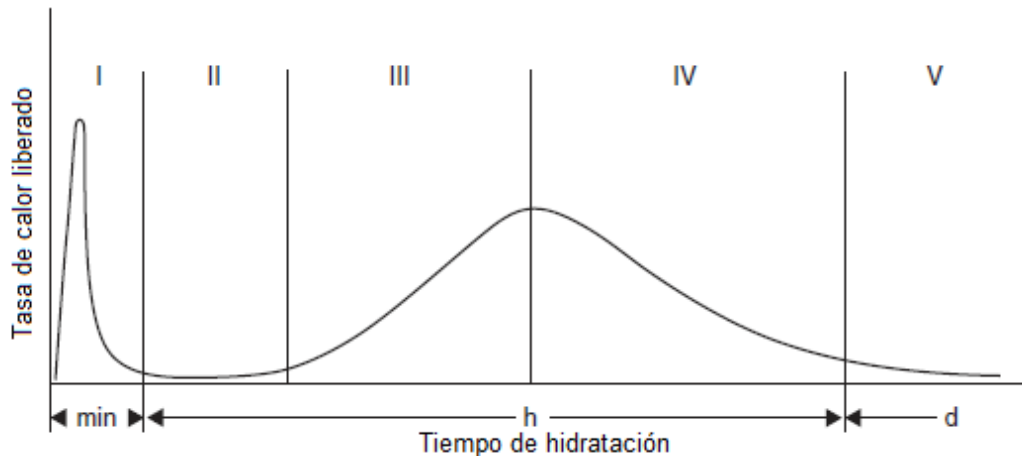
Donde ΔH , representa el delta en el calor de hidratación alcanzado.

Aunque los procesos de hidratación de estos minerales se dan en periodos de tiempo diferente, ya que como lo mencionan autores como Lizarazo (2018), una vez pasados 28 días se puede considerar que hasta 80% de la Alita ya ha completado su hidratación, mientras que en el caso de la Belita es solo hasta el día 14 cuando se da inicio de dicho proceso. Actualmente se ha llegado a la conclusión de que el gel C-S-H producto de ambos minerales es físicamente igual.

Según kurdowski (2014), la cinética de la reacción es bastante compleja e incluye varios procesos, pero en el caso de la secuencia de reacción del C_3S se puede seguir más fácilmente tomando como referencia la curva calorimétrica de la figura 2.4, donde se muestra la velocidad de evolución de calor en función del tiempo. El flujo de calor de la

muestra es proporcional a la velocidad de reacción y es medido fácilmente en el caso de dicho producto, que es el compuesto más representativo del cemento, ya que comprende el 40 - 70 % del peso de este. Identificando que la hidratación de C_3S comprende las cinco etapas que pueden verse en la figura 2.4.

Figura. 2-4: Curva calorimetría de hidratación del silicato tricálcico.



Nombre de la fuente: Adaptado de kurdowski, 2014.

Tabla 2-4: Descripción procesos de hidratación.

Periodo	Estado de reacción	Proceso químico	Comportamiento cinético general
Temprano	I. Periodo de pre-inducción	Hidrolisis inicial: Liberación de iones	Muy rápido Control químico
	II. Periodo de inducción	Disolución continua, formación temprana de C-S-H	Nucleación o control de difución
Medio	III. Periodo de aceleración (periodo de pre-inducción)	Crecimiento inicial de productos de hidratación permanente	Rápido Control Químico
	IV. Periodo de desaceleración	Crecimiento continuo de productos de hidratación, desarrollo de la microestructura	Química y control de difución
Tardío	V. Periodo de difución (periodo de estado estacionario)	Densificación gradual de la microestructura	Muy lenta control de difución

También se debe mencionar que adicional del gel C-S-H, en el proceso de hidratación del cemento, se produce Hidróxido de calcio $Ca(OH)_2$, el cual es conocido en la literatura cómo Pórtlandita, y que debido a sus características cristalinas puede llegar a afectar las

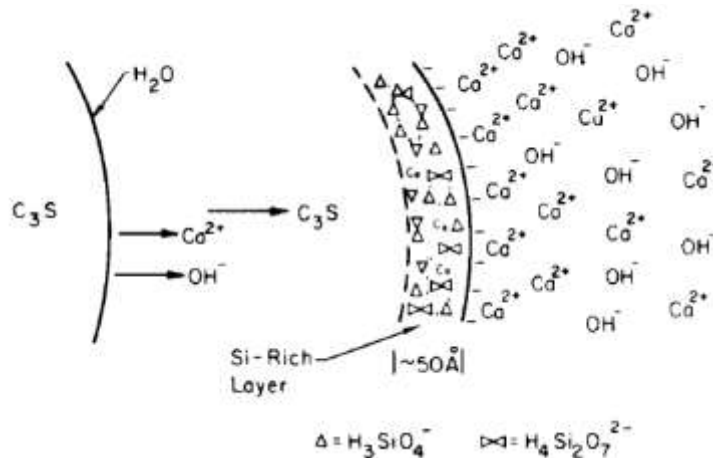
propiedades físicas de la mezcla en caso de que su proporción sea mayor a los límites ya establecidos (Lizarazo-marriaga, 2010).

Por último, Con respecto a la curva de hidratación del silicato tricálcico (figura 2-4), se puede observar que las 5 etapas pueden ser fácilmente identificables como: hidrólisis inicial, período de inducción, aceleración, desaceleración y estado estable.

Primera etapa (hidrólisis inicial):

Donde el primer contacto con agua da comienzo a la hidrólisis, proceso en que los iones de calcio y los iones de hidróxido se liberan rápidamente sobre la superficie de cada partícula (la reacción es rápida por la gran superficie en contacto con agua). Llegando inicialmente a un punto en que los valores de pH se elevan a más de 12, marcando el momento en que comienza a disminuir la velocidad de reacción (después de aproximadamente 15 minutos). Sin embargo, debe comprender la forma en que inicialmente el silicato tricálcico se disuelve superficialmente y suministra los iones silicato $\text{H}_2\text{SiO}_4^{2-}$, $(\text{OH})^-$ y Ca^{2+} a la mezcla para iniciar el proceso (Claisse, 2016), lo cual representando esquemáticamente en la figura 2-5.

Figura. 2-5: Disolución del C_3S para formar capa rica en silicato.

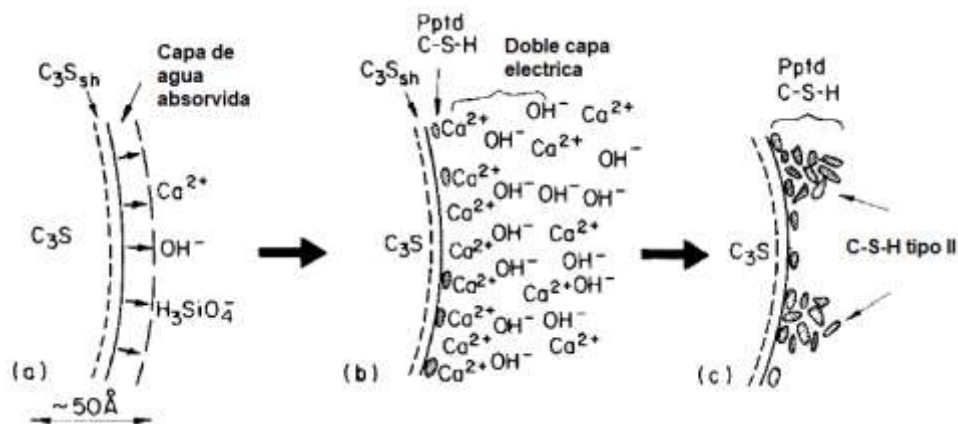


Nombre de la fuente: Adaptado de kurdowski, 2014.

Como se puede ver, al cabo de pocos segundos la solución es sobresaturada con respecto al silicato de calcio hidratado, el cual precipita rápidamente. En la terminología cementera el hidrato obtenido se representa como C–S–H, lo cual puede presentar una

estequiometría que se aproxima a $C_{1.5}\text{-S-H}_2$ (Kurdowski, 2014). En la figura 2-6 se muestra una representación esquemática del mecanismo de disolución y precipitación en la hidratación del C_3S que conduce a la formación de C-S-H. También, se puede ver en la curva de flujo térmico que la velocidad de disolución disminuye rápidamente en los primeros instantes, lo que como ya se mencionó anteriormente, es debido al rápido aumento del pH producido por la liberación de iones $(OH)^-$.

Figura. 2-6: Mecanismos de disolución-precipitación en la hidratación del C_3S que conduce a la formación de C-S-H.



Nombre de la fuente: Adaptado de kurdowski, 2014.

En este proceso la relación Ca/Si del silicato anhidro es igual a 3, mientras que la del silicato hidratado que precipita es 1.5, por lo cual el balance global de las reacciones corresponde a la liberación en la solución de los iones $(OH)^-$ y Ca^{2+} excedentes. Estos iones se van a acumular en la solución, que va a alcanzar la sobresaturación con relación a la Pórtlandita, $Ca(OH)_2$. Cuando las concentraciones de Ca^{2+} y OH^- alcancen unos valores críticos, entonces el $Ca(OH)_2 = CH$ empezará a cristalizar la solución con la consiguiente formación de C-S-H y la reacción de C_3S continúa de nuevo con rapidez (Kurdowski, 2014).

Segunda etapa (periodo de inducción):

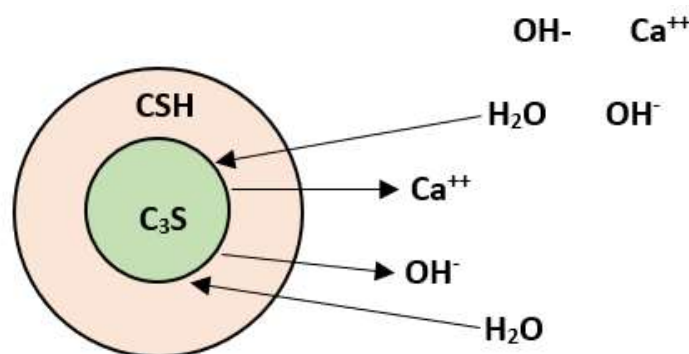
Finalmente, la hidrólisis se ralentiza rápidamente, pero continúa durante todo el período de inducción. Este período (latente) es causado por la necesidad de lograr una cierta concentración de iones en la solución antes de los núcleos cristalinos a partir de los cuales crecen los productos de hidratación. Este fenómeno es una característica de muchas

reacciones químicas y es dominado por el control de nucleación. Periodo durante el cual el consumo de agua y la cantidad de hidratos formados son muy pequeños, lo que explica porque el concreto presenta propiedades de manejabilidad y trabajabilidad, es decir, se mantiene en un estado plástico durante un determinado tiempo (Kurdowski, 2014).

Aunque en la literatura se encuentran algunas teorías que describen el principio y la terminación del periodo de inducción, este se basan en 4 hipótesis para la fase inicial del proceso (barrera de hidratos, defectos de red, reacción de las capas límites y doble capa eléctrica) y 2 hipótesis para la fase de terminación (nucleación de C-S-H y nucleación de $\text{Ca}(\text{OH})_2$). Las cuales podrán ser profundizadas en la literatura, ya que para el presente documento nos centraremos en la teoría de la barrera física de la difusión.

Teoría que se basa en la formación de una capa de C-S-H alrededor de la partícula de C_3S por donde el $\text{Ca}(\text{OH})_2$ tiene que salir (figura 2-7). Dicha capa retarda la velocidad de reacción ya el agua tiene que pasar a través de la capa de C-S-H en contracorriente con el $\text{Ca}(\text{OH})_2$. Como hay concentraciones elevadas de $\text{Ca}(\text{OH})_2$ solubilizadas en el agua, el potencial químico se disminuye continuamente hasta que la solución está sobresaturada con respecto a la Pórtlandita, $\text{Ca}(\text{OH})_2$. Por lo anterior, la velocidad de la hidratación de C_3S no puede volver a crecer antes de que el $\text{Ca}(\text{OH})_2$ empiece a cristalizarse y así se mueve el equilibrio de la reacción hacia la derecha, que es lo que marca el inicio de la tercera etapa (Kurdowski, 2014).

Figura. 2-7: Esquema de difusión de las barreras físicas.



Nombre de la fuente: propia.

Tercera etapa (periodo de aceleración):

Después de 1 a 3 horas, es decir, al final del período de inducción, se inicia la solidificación o fraguado inicial. Aquí, el C_3S empieza rápidamente a hidratarse de nuevo, ya que el $Ca(OH)_2$ inicia su cristalización (durante esta etapa el CH comienza a cristalizar sobre la superficie del C_3S , formando un recubrimiento sobre el grano y procede nuevamente con la reacción del C_3S) y así se mueve el equilibrio de la reacción hacia la derecha (Ec. 2.6). Adicionalmente, durante esta etapa la velocidad de hidratación alcanza un máximo al final del período de aceleración, este máximo corresponde con el pico de la evolución de calor, dando en este tiempo (2 a 8 horas) que el fraguado final ya ha ocurrido y comienza el endurecimiento inicial. Debido a que la cristalización de $Ca(OH)_2$ lleva a este compuesto fuera de la capa de C-S-H (figura 2-8), la velocidad de difusión de $Ca(OH)_2$ crece y, por consiguiente, crece la difusión de H_2O hacia el núcleo de C_3S , aumentando la velocidad de hidratación (Kurdowski, 2014).

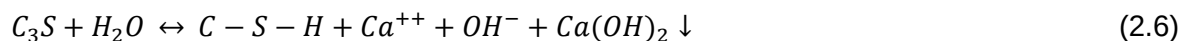
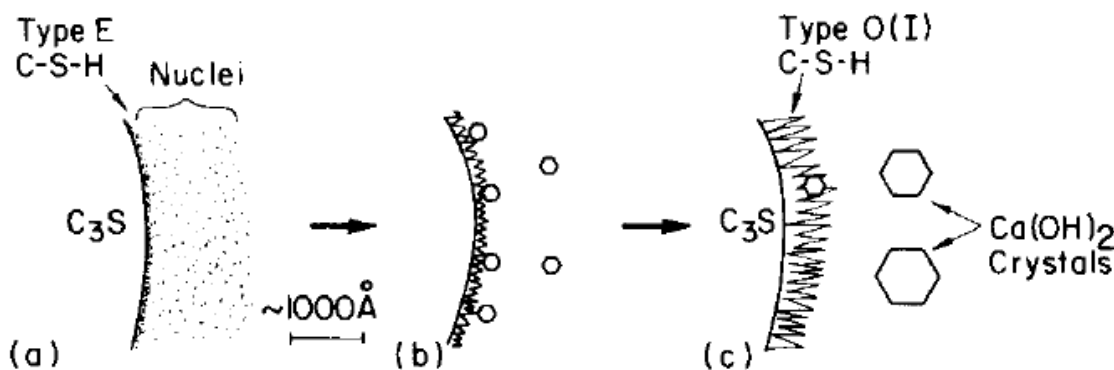


Figura. 2-8: Cristalización del $Ca(OH)_2$ fuera de la capa de C-S-H.



Nombre de la fuente: Adaptado de kurdowski, 2014.

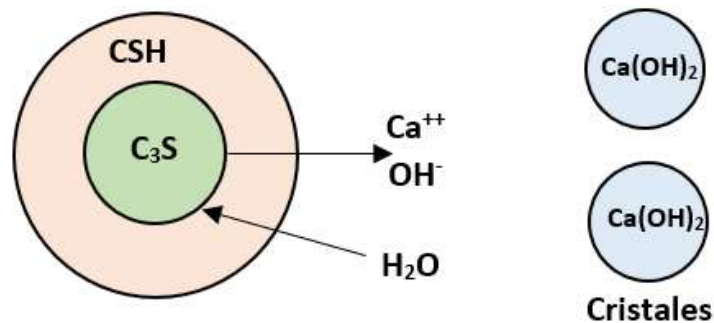
Cuarta etapa (periodo de desaceleración):

En continuación del proceso de hidratación del C_3S , el espesor de la capa de hidrato C-S-H aumenta y forma una barrera a través de la cual el agua debe fluir para alcanzar el C_3S no hidratado y a través de la cual los iones deben difundirse para alcanzar los cristales en crecimiento (momento en que la velocidad de reacción es igual a la velocidad de difusión). A partir de este momento, mientras el espesor de la capa sigue creciendo y el movimiento

a través de la capa C-S-H determina la velocidad de la reacción, la hidratación queda controlada por la velocidad de difusión dentro de la capa. Luego la velocidad de reacción empieza a disminuir hasta que llegue a un estado estable (etapa 5) después de 12 a 24 horas (Li, Wei, & Li, 2003a).

Las reacciones controladas por difusión son generalmente bastante lentas y la velocidad disminuye cuando aumenta el espesor de la barrera de difusión, motivo por el cual la hidratación tiene la tendencia a acercarse al 100 % de forma asintótica.

Figura. 2-9: Momento en que la velocidad de reacción es igual a la velocidad de difusión.



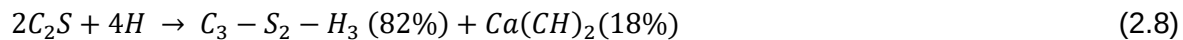
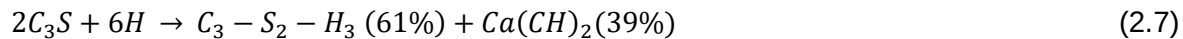
Nombre de la fuente: propia.

Quinta etapa (periodo de difusión estacionario):

Durante esta etapa, la difusión es tan lenta que la velocidad de hidratación está controlada únicamente por la velocidad de difusión. Como el espesor de la capa sigue creciendo, la velocidad de difusión sigue disminuyendo hasta que no haya más C_3S a hidratar (Li et al., 2003b).

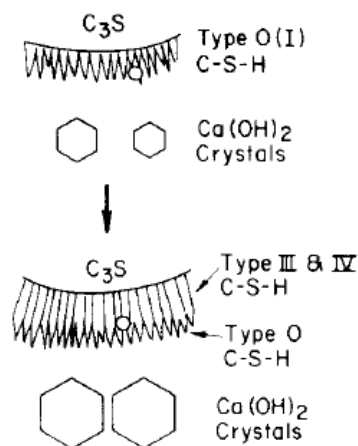
En cuando a la hidratación de C_2S aunque es similar a la de C_3S , su proceso es más lento, esto debido a que es un compuesto menos reactivo que el C_3S . Además, el calor de hidratación de C_2S es menor que el de C_3S y por eso la curva calorimétrica es difícil de medir experimentalmente. Sin embargo, la curva presenta un comportamiento similar a la de C_3S (figura 2-4), con la particularidad que el segundo pico tiene una menor intensidad. Sin embargo, en las ecuaciones 2.7 y 2.8 se puede ver la tasa de hidratos que se convierten en C-S-H y $Ca(OH)_2$, lo cual, aunque en el caso del $2C_2S$ pueda representar

una mayor cantidad, es importante tener en cuenta que su volumen en el cemento es menor.



Estas ecuaciones (2.7 y 2.8) pueden ser considerarse sólo como un punto de partida en la discusión que trata sobre el mecanismo de reacción porque, las relaciones calcio/sílice (C/S) en el gel C-S-H, así como la relación hidrogeno/sílice (H/S) son variable; y es el hidróxido de calcio la única fase hidratada bien definida (Kurdowski, 2014).

Figura. 2-10: Representación esquemática del crecimiento de los productos de hidratación durante la etapa 5.



Nombre de la fuente: Adaptado de kurdowski, 2014.

Finalmente, se debe decir que la hidratación, como otras reacciones químicas, es sensible con respecto a la temperatura, ya que la velocidad de reacción aumenta cuando esta variable aumenta. Sin embargo, la dependencia entre la velocidad y la temperatura está relacionada con el grado de reacción. Donde la etapa en la que la temperatura tiene más importancia es en el período de aceleración (etapa 3) ya la velocidad de reacción es controlada químicamente. En la etapa 5, en que la hidratación es controlada completamente por difusión, la reacción no es tan sensible con respecto a la temperatura, aunque el coeficiente de difusión puede variar con el aumento de esta (Kurdowski, 2014).

2.4.1.1. Conceptos a tener en cuenta

Silicatos de calcio hidratados o Gel C-S-H: Como se mencionó anteriormente, el gel C-S-H es el encargado de dar las principales propiedades mecánicas que caracterizan al concreto, ya que se encarga de la adherencia de las pastas con los agregados de las mezclas.

Este material tiende a representar hasta el 60% del volumen hidratado de la mezcla. Destacando que se caracteriza por ser una estructura amorfa con una alta rigidez y superficie específica de alta resistencia a la deformabilidad; la cual se conforma por silicatos laminares que constan de planos poliédricos de CaO, tetraedracadenas de SiO y capas intermedias con agua o cationes como Na⁺, K⁺ o Ca²⁺, similar a una estructura de tobermorita. Las propiedades C-S-H están determinadas por la relación C/S que puede llegar a variar de 0.6 a 1.8 en el gel C-S-H sintético, por encima y por debajo de estos límites de C/S se forman otras fases como la Pórtlandita y SiO_{2(am)} (Wolter et al., 2019).

Hidróxido de calcio Ca(OH)₂ ó Pórtlandita: este es un material muy cristalino con una forma prismática hexagonal, el cual puede llegar a ocupar hasta el 25% del volumen hidratado de la mezcla. Aunque este material no aporta beneficios a las propiedades mecánicas de la matriz cementante, se debe resaltar que este es el encargado de mantener el pH de la pasta en valores altos (12-13), lo que funciona como una reserva alcalina, ayudando a mantener a los concretos reforzados protegidos contra la corrosión electroquímica (Calleja, 2001).

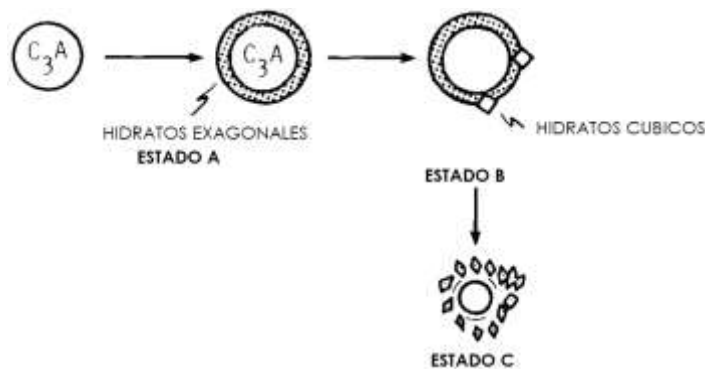
Sulfoaluminato de calcio 3CaO·Al₂O₃·3CaSO₄·32H₂O o Etringita: surge como un producto de la hidratación del sulfoaluminato tricálcico. Una vez se inician los procesos de hidratación este material es el primero en cristalizarse, caracterizándose por tener una forma de partículas alargadas y laminares. La etringita tiene la función de dar plasticidad a la pasta de cemento para poder manipularse; a su vez es importante para controlar el tiempo de fraguado y el desarrollo de la resistencia. Sin embargo, una vez se agota el sulfato en la mezcla y la concentración de aluminatos aumenta, la etringita gradualmente se transforma en monosulfato (León et al., 2012).

2.4.2 Hidratación de los aluminatos

Entre los minerales presentes en el cemento de Pórtland, es el aluminato tricálcico (C_3A) es el más reactivo con el agua, llegando a tener una influencia significativa sobre la hidratación temprana y la reología del cemento. Este, tiene la particularidad que mientras se va dando la transformación de los aluminatos de calcio, se liberan grandes cantidades de calor, lo cual en caso de no controlarse puede llegar a afectar los tiempos de fraguado de las mezclas, pudiendo ocasionar un proceso de fraguado relámpago. Es por esta razón que una vez se ha logrado enfriar el clinker a temperaturas entre 100°C y 180°C y antes del proceso de molienda, se adicionan pequeñas cantidades de sulfato de calcio (yeso o hemihidrato), los cuales tienen la función de retardar los procesos de hidratación de la celita, que con el paso del tiempo se transformara en sulfoaluminato tricálcico o también conocido como etringita (Kurdowski, 2014).

Cuando se habla de la hidratación del C_3A en ausencia de yeso, se debe hablar de 4 proceso de hidratación, en donde existe una reactividad inicial alta seguida de un período de baja reactividad. Ello puede ser debido a que los productos iniciales de la hidratación (hidratos hexagonales) forman una barrera protectora sobre la superficie de las partículas de C_3A . Seguidamente, la conversión de los hidratos hexagonales a cúbicos rompe la barrera protectora y la hidratación prosigue de nuevo rápidamente. El momento en el que los hidratos hexagonales se convierten en cúbicos depende de las condiciones de la reacción (temperatura, presencia de núcleos de C_3AH_6 , etc).

Figura. 2-11: Estado de hidratación del aluminato tricálcico en ausencia de yeso.

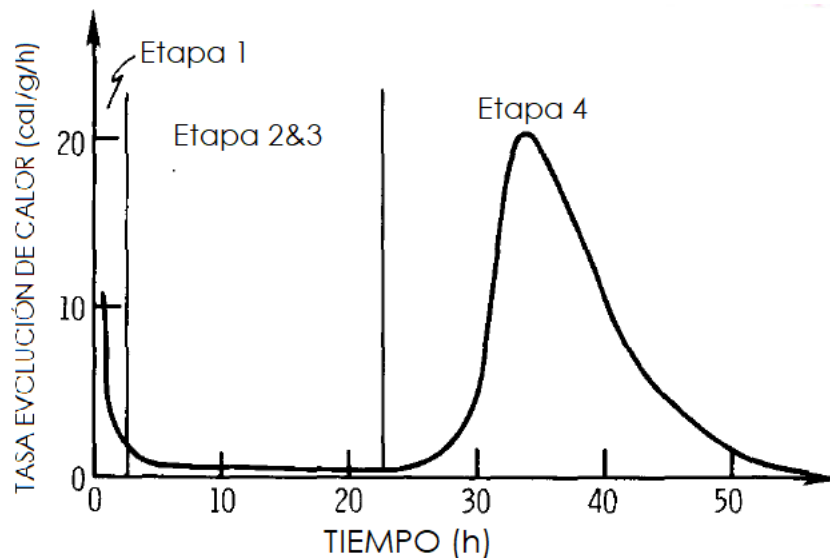


Nombre de la fuente: Adaptado de kurdowski, 2014.

La conversión de los hidratos hexagonales a cúbicos conduce a un aumento de la porosidad y a una rotura de la microestructura, que da lugar a una pérdida de resistencia de la pasta. Sin embargo, cuando hablamos de la hidratación del C_3A en presencia de yeso y cal, se tiene que las reacciones de hidratación son reemplazadas por unas más lentas, las cuales se desarrollan según una secuencia en la que existen variaciones de velocidad.

Cuando se estudia la curva calorimétrica de la hidratación de C_3A que se muestra en la figura 2-12, la cual es cualitativamente parecida a la curva de C_3S , aunque las reacciones comprendidas son distintas y la cantidad de calor desarrollado en el caso del C_3S es mucho más elevada. El primer pico de calor del C_3A se debe a la formación de etringita y el segundo pico coincide con la conversión de etringita a monosulfato, donde el tiempo que pasa antes de que la conversión empiece depende de la cantidad de sulfato disponible. Es importante destacar que cuando hay una gran cantidad de yeso disponible en el sistema, la etringita puede quedarse estable mucho más tiempo y la conversión a monosulfato ocurrirá entre 12 a 36 horas en casi todos los cementos.

Figura. 2-12: Curva de evolución de calor del C_3A en presencia de yeso y cal.



Nombre de la fuente: Adaptado de kurdowski, 2014.

Por último, como se puede observar en la figura 2-12, se tiene que en el proceso de hidratación del C_3A existen 4 etapas fácilmente identificables, las cuales se puede definir

como periodo inicial, periodo de retardo, periodo de espesamiento del yeso y reacciones a largo plazo, las cuales son etapas que se caracterizan por:

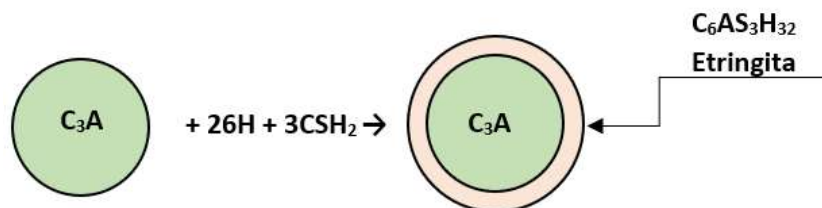
Periodo inicial

Durante esta etapa el C_3A y el yeso se disuelven rápidamente en el agua, reacciones que vienen acompañadas de un desprendimiento de calor importante y de la formación de una solución muy sobresaturada, donde los iones aquí formados se combinan instantáneamente para formar cristales de etringita o sal de Candlot (trisulfoaluminato de calcio hidratado “TSA”) (Kurdowski, 2014)



Estos compuestos aparecerán bien cristalizados, donde su forma será de agujas muy finas o “erizos”, formando así una capa de etringita que cubre las partículas de C_3A , e impidiendo la difusión de los iones SO_4^{-2} , $(OH)^-$ y Ca^{2+} .

Figura. 2-13: Periodo inicial de la hidratación del C_3A .



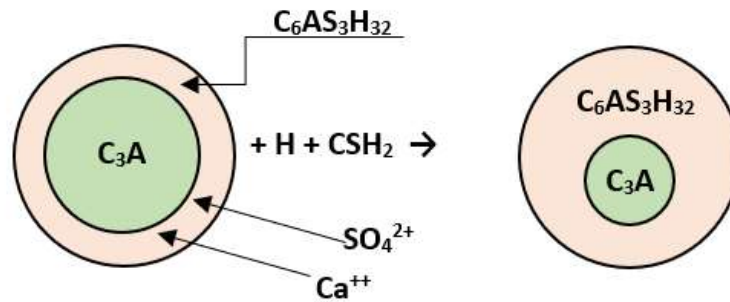
Nombre de la fuente: propia

Periodo de retardo

Este periodo hace referencia a un tiempo de reacción lenta, durante el cual la etringita continua con su proceso de formación. El C_3A posee un "periodo durmiente" de similares características al del C_3S . Sin embargo, en este caso la velocidad de reacción aumenta de forma significativa en el transcurso del periodo de fraguado del silicato, lo que es debido no solo a los procesos de activación térmica, sino también a la formación de compuestos mixtos (silico-aluminatos de calcio) (Kurdowski, 2014).

La duración de esta segunda etapa depende de la cantidad de yeso disponible en la mezcla, esto debido a que la etringita solamente es estable mientras hay suficiente exceso de SO_4^{2-} . Es importante tener presente que en este proceso la velocidad de reacción se retarda debido a que los iones SO_4^{2-} y Ca^{++} tienen que atravesar por difusión la capa de etringita antes de que puedan reaccionar con el C_3A formando nuevamente etringita (figura 2-14).

Figura. 2-14: Periodo de retardo durante la hidratación del C_3A .



Nombre de la fuente: propia

Periodo de espesamiento del yeso.

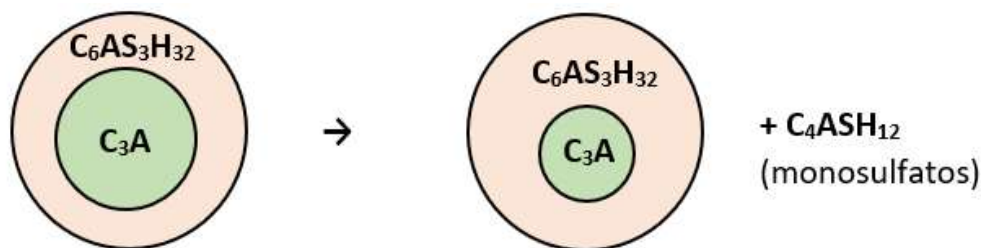
La etringita es un producto de hidratación estable cuando dentro de la solución hay suficiente cantidad de sulfato disponible. Si la concentración de SO_4^{2-} disminuye dentro de la solución, la etringita se hace inestable y puede llegar a convertirse en monosulfato. La formación de la etringita consume tres moles de sulfato de calcio por cada mol de aluminato tricálcico. Pero, considerando que el contenido en yeso de los cementos está limitado por las normas y, que en general es inferior a la proporción estequiométrica mencionada anteriormente, se puede afirmar que existe un exceso de aluminato tricálcico frente al yeso (Wang & Lee, 2012).

Por esta razón, al cabo de 10 a 24 horas, es decir, después de fraguado del cemento, el yeso se ha consumido totalmente. Esto hace decrecer la concentración de iones sulfato y calcio; por lo que la solución se vuelve subsaturada con respecto a la etringita, la cual va a disolverse creando una nueva fuente de iones sulfato, que con el aluminato que aun queda presente, formará un nuevo compuesto definido como monosulfoaluminato de calcio hidratado (MSA) (Wang & Lee, 2012).



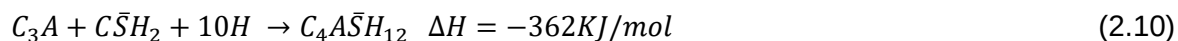
Esta reacción provoca una rápida disolución del aluminato anhidro, el cual alimenta de iones aluminato la solución; trayendo consigo una caída en el contenido de iones de sulfato y de la conductividad eléctrica, acompañando de un pico pronunciado de flujo térmico. En los cementos se observa igualmente durante este periodo que el ferroaluminato también reacciona rápidamente (Lizarazo, Ramírez, et al., 2016).

Figura. 2-15: Formación monosulfatos a partir de la reacción de la etringita con el C3A



Nombre de la fuente: propia

Una vez la etringita empieza a convertirse en monosulfato, la velocidad de hidratación del C_3A comienza a crecer otra vez, esto asociado a que se produce la rotura de la barrera protectora durante la conversión de etringita a monosulfato, permitiendo que el C_3A pueda reaccionar rápidamente de nuevo (Wang & Lee, 2012). Este proceso se puede describir a través de la siguiente ecuación:



Reacciones a largo plazo:

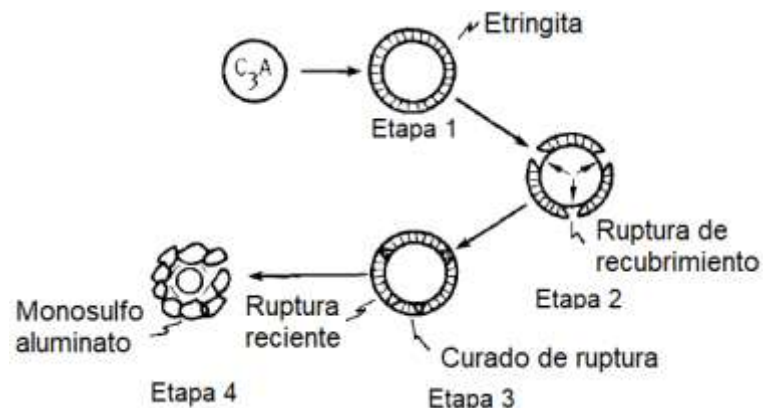
En el concreto, la etringita se transforma totalmente en monosulfoaluminato al cabo de algunas semanas, y este último al cabo de un mes reacciona lentamente con el aluminato y el ferroaluminato en exceso, para así generar un producto hidratado complejo el cual contiene los iones aluminato, ferrato, sulfato, calcio e hidróxilo (Wang & Lee, 2012).

Tabla 2-5: Formación de productos de la hidratación del C_3A .

Reacción Molar CSH_2/C_3A	Productos estables formados durante la hidratación
3	Etringita
3 - 1	Etringita + monosulfato aluminato
1	Monosulfato aluminato
<1	C_4ASH_{12} - C_4AH_{13} (Solución sólida)
0	C_3AH_6 (Hidrogranate)

Como producto de la hidratación del C_3A se tiene la formación de diferentes productos estables (tabla 2-5), los cuales vienen dados en función de la relación molar (Yeso / Aluminado). Allí, se observa la formación principalmente de dos compuestos, la etringita y el monosulfato, aunque al finalizar el fraguado (final de la etapa de inducción) la tendencia será cada vez mayor a estabilizar el monosulfato. Es importante tener en cuenta que las condiciones que provocan la formación de uno u otro componente durante el fraguado vienen marcadas por la cantidad de sulfato en la solución y más concretamente por los valores de la relación (SO_4^{2-} / AlO_2^-) (Wang & Lee, 2012).

Finalmente, en esta etapa es importante tener en cuenta que cuando la dosificación de yeso es inadecuada, es posible que exista todavía C_3A sin reaccionar, cuando la etringita se ha convertido en monosulfato. En estos casos, las fases AFm (monosulfato) que se forman tienen composiciones entre C_4ASH_{12} y C_4AH_{13} , los cuales pueden ser fases mezcladas o soluciones sólidas.

Figura. 2-16: Cinética y secuencia de la hidratación del C_3A .

Nombre de la fuente: Adaptado de kurdowski, 2014.

2.4.3 Hidratación de los ferroaluminatos

La secuencia en la cual el C_4AF forma sus productos de hidratación en presencia de yeso, es igual a la de C_3A ; Sin embargo, en este caso las reacciones son más lentas y la evolución de calor no es tan grande. Para esto es poco común que el C_4AF reaccione con la suficiente rapidez para poder causar fraguado rápido, donde adicionalmente, la presencia del yeso retardara la hidratación del C_4AF . Se debe tener presente que las modificaciones en la composición de la fase de ferrito afectan únicamente la velocidad de hidratación; por lo que cuando el contenido de F aumenta, la velocidad de la hidratación disminuye (Swaddiwudhipong et al., 2002).

En función de esto, se puede pensar que los óxidos de hierro pueden sustituir a los óxidos de aluminio en los productos de la hidratación. Pero se debe tener presente que durante esta fase no hay la suficiente cal en el C_4AF para poder formar las fases de AFm sin que también se formen óxidos de hierro o aluminio amorfos. En la práctica, estas reacciones no tienen importancia debido a que en el cemento la cantidad de Ca^{2+} y OH^- de otras hidrataciones es tan grande que hay poca posibilidad de que llegue a darse (Swaddiwudhipong et al., 2002).

2.5 Estructura, microestructura y mecanismos de transporte del concreto

En cuanto a las pastas de cemento, la microestructura está formada por diferentes productos de hidratación, principalmente hidratos de silicato de calcio e hidróxido de calcio; y por una red de poros llena de diferentes iones alcalinos. Donde se tiene que los productos de la hidratación, los cuales al igual que los agregados se consideran como aislantes. Por lo tanto, la conductividad general del hormigón está relacionada con el flujo de iones a través de la red de poros, los cuales proporcionan caminos de transporte de cargas eléctricas. Sin embargo, para comprender mejor la composición de la estructura y microestructura de las pastas de cemento y las mezclas de concreto, se debe partir de los componentes que conforman una muestra, los cuales según (Sanchez de Guzmán, 2001) se pueden definir como:

- **Cemento:** es el material principal de toda mezcla ya que es el encargado de dar las propiedades aglutinantes a esta; el porcentaje de este puede variar entre 7% al 15% en mezclas de concreto y entre 60% a 70% en pastas de cemento.
- **Agua:** puede variar entre 7% al 15% en mezclas de concreto y entre 25% a 35% en pastas de cemento.
- **Agregados:** en el caso de las mezclas de concreto puede ser del 59% al 76%.
- **Aire:** en cuando a este componente el cual ocupa entre el 1% y 15% del volumen de la mezcla y el cual puede ser aire atrapado naturalmente en la mezcla, o adicionado intencionalmente con el uso de aditivos.

La definición de las proporciones dependerá de las características de cada uno de estos materiales y de la tipología de concreto a la que se quiera llegar, lo cual definirá los rasgos distintivos de la mezcla en estado endurecido. Adicionalmente, la dosificación tiene incidencia directa sobre la microestructura, la cual tiene una gran influencia sobre algunas propiedades del material que definen en gran medida la forma y magnitud en la que se da el transporte y flujo de iones mientras se desarrollan los procesos fisicoquímicos de hidratación. Las propiedades más relevantes para las pastas de cemento son:

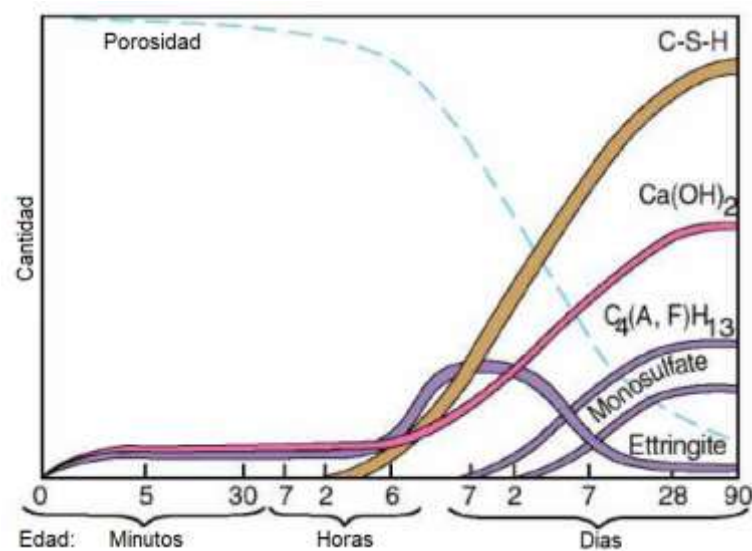
- **Permeabilidad:** esta propiedad está fuertemente relacionada a la porosidad de la muestra y consiste en que la muestra pueda ser atravesado por un fluido debido a una diferencia de presiones entre las caras expuestas; el movimiento de estos fluidos puede ser comprendido por medio de la ley de Darcy (Sanchez de Guzmán, 2001).

Aunque en el caso de la permeabilidad se debe profundizar en el aporte de los agregados, la pasta cementante y la interfaz entre estos, para el presente documento se estudiará únicamente la permeabilidad de las pastas cementantes; propiedad que está principalmente definida por el agua presente en la mezcla, la cual determinará aspectos como los espacios totales libres dentro de la matriz y los espacios sin cubrir dentro de la muestra. Lo anterior, se debe al agua consumida durante la hidratación y a la que se pierde por los procesos de evaporación (Mehta, P., & Monteiro, 2006).

- **Porosidad:** hace referencia a pequeños vacíos que se presentan en las mezclas, estos pueden estar aislados o entrelazados unos con otros formando complejas

redes de conducción dentro de la muestra (permeabilidad). Los poros pueden ser calificados en función a su origen como: poros de gel en el cemento, poros capilares, vacíos grandes, poros asociados a la zona de transición interfacial (ITZ), microvacíos y discontinuidades. En cuanto a sus efectos en las propiedades del concreto, la porosidad se puede clasificar como: porosidad en la matriz de pasta o en la ITZ (Higuera Flórez, 2016).

Figura. 2-17: Evolución de la cantidad de poros y productos resultantes de la hidratación en función del tiempo.



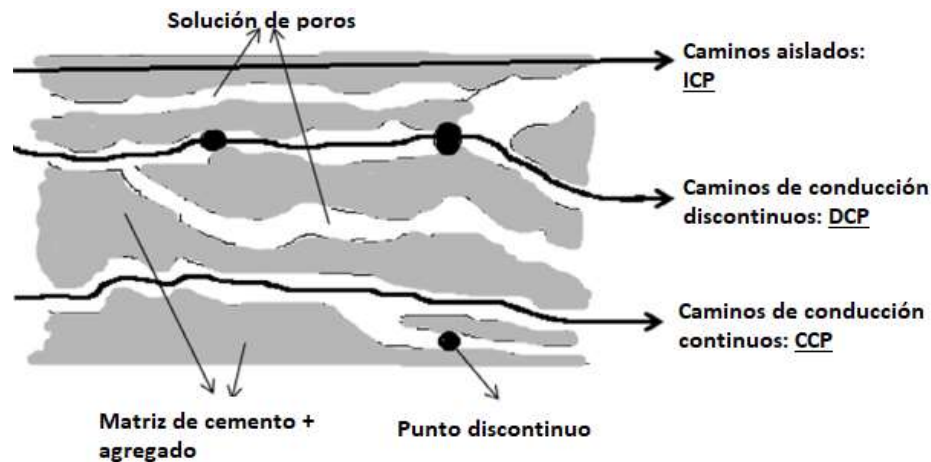
Nombre de la fuente: Adaptado de(Whiting & Nagi, 2003)

- **Adsorción:** El termino hace referencia a la cantidad de agua o sustancias que puede llegar a penetrar dentro de los poros saturables de la matriz cementante o de pasta (Sanchez de Guzmán, 2001). Este valor está fuertemente relacionado con la porosidad y la permeabilidad que pueda llegar a tener el material, ya que su variable se define como el porcentaje de agua que penetra el material con respecto al peso de muestra seca.
- **Hermeticidad:** esta propiedad hace énfasis en la capacidad del material de retener el agua sin escapes visibles, en donde al igual que la permeabilidad, esta propiedad se ve favorecida por bajas relaciones a/c, buenas gradaciones del material, la

inclusión de aire, no presentar fisuras ni vacíos, entre otras variables que también ayuda a la hermeticidad (Sánchez de Guzmán, 2002).

En lo que corresponde a la presente investigación, existe una combinación de estas propiedades que será de mayor relevancia; y que está relacionada con la porosidad y la permeabilidad de la muestra. Como se mencionó anteriormente, la red de comunicación dentro de la matriz cementante hace referencia a pequeños espacios que existe entre las partículas de cemento, los que una vez comienzan a darse los procesos de hidratación, pueden extenderse a lo largo de toda mezcla formando extensas interconexiones que sirven como caminos para el transporte de agua y materiales iónicos producto de los procesos de hidratación de las partículas de cemento (Figura. 2-6).

Figura. 2-18: Modelo idealizado de la conducción en el cemento.



Nombre de la fuente: Adaptado de Lizarazo-Marriaga et.al, 2013.

Esta red de poros es de vital importancia para los procesos de hidratación del cemento, y de verse afectada o interrumpida durante las primeras horas de la mezcla podría conllevar a un falso fraguado del material; trayendo consigo pérdidas en las propiedades del cemento. De igual forma, una vez la mezcla ha alcanzado su fraguado final y aun cuenta con un alto contenido de caminos de conducción continuos, se podrían ocasionar serios problemas a futuro en las estructuras con patologías relacionadas a la permeabilidad, penetración de cloruros, exudación, retracción plástica, entre otras patologías de suma importancia e impacto.

2.5.1 Mecanismos y propiedades de transporte en el concreto.

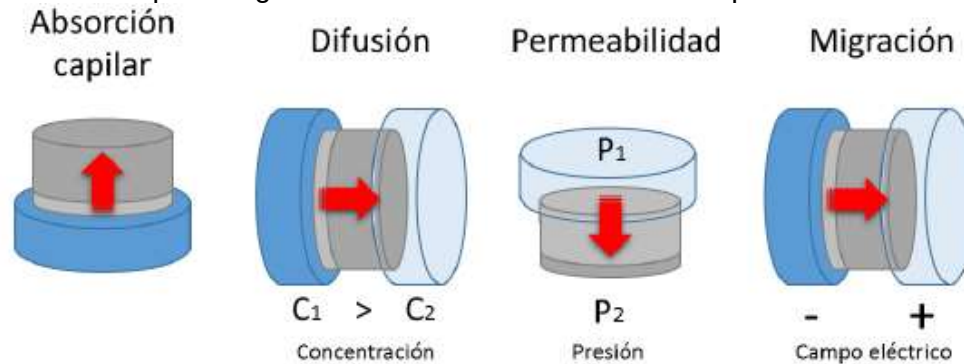
El transporte de líquidos y de iones dentro del concreto se da por medio de las redes de poros existentes en la matriz cementante, tal como se ha venido mencionando atrás. Estos movimientos pueden ser inducidos por diversos principios físicos, dentro de los cuales se destacan:

Absorción capilar: Desplazamiento ascendente de líquido por succión a través de un capilar como consecuencia de la tensión superficial del líquido. El fenómeno se da en materiales no saturados (Fonseca Barrera, 2016).

Difusión: Está asociada al movimiento de moléculas o iones bajo la acción de un gradiente de concentración. De forma más estricta, se puede mencionar que está relacionada al potencial químico de una zona de alta concentración iónica a una zona de baja concentración. Esto implicará que no es necesario que exista un flujo a través de la muestra para que se de este movimiento iónico de un lugar a otro en las muestras (Claisse, 2016).

Permeabilidad: Movimiento de un líquido en presencia de un gradiente de presión. Tal como se mencionó anteriormente, esta es una de las propiedades más importantes de las mezclas de cemento y está fuertemente relacionada con la porosidad que pueda llegar a presentar el material y la forma en la que estos poros llegan a interconectarse (Fonseca Barrera, 2016).

Migración o electromigración: Este fenómeno hace referencia al movimiento de iones bajo la acción de un campo eléctrico. Dicho campo eléctrico no solo puede estar asociado a fuentes externas, ya que también puede ser causado por el diferencial de potencial eléctrico a causa de la corrosión, como es el caso de picaduras que ocurre en el acero de refuerzo (Higuera Flórez, 2016).

Figura. 2-19: Esquemas gráficos de los mecanismos de transporte

Nombre de la fuente: Fonseca, 2016.

Es muy importante comprender la forma en que se realiza el transporte de sustancias dentro de las matrices del cemento, sobre todo lo relacionado a los agentes agresivos que afectan la durabilidad de los concretos. Actualmente se ha establecido que la difusión se puede considerar como uno de los fenómenos principales para entender la forma en que se dan estos procesos. Este es un fenómeno que obedece a la ley de Fick, aunque en el presente documento no se busca profundizar en estos aspectos, podrán ser consultados los trabajos realizados por (Higuera Flórez, 2016) o (Claisse, 2016), los cuales se encuentran relacionados en la bibliografía del presente documento.

2.6 Espectroscopia de Impedancia electroquímica (EIS)

La técnica de espectroscopia de impedancia electroquímica EIS por sus siglas en inglés “*Electrochemical impedance spectroscopy*”, se fundamenta en la respuesta eléctrica de los materiales. Por lo que para entender la magnitud de dicha técnica se deben comprender los siguientes conceptos.

2.6.1 Resistividad eléctrica

La resistividad eléctrica (o su inversa, la conductividad) es una medida de oposición que presenta un circuito a una corriente cuando se aplica una tensión. En el caso de las pastas de cemento o concretos, es una propiedad que está directamente relacionada con la permeabilidad de los fluidos y la difusión de iones a través de superficies porosas (Whiting & Nagi, 2003). Esta propiedad, extiende el concepto de resistividad a los circuitos

de corriente alterna (CA) que poseen tanto magnitud, como fase, a diferencia de la resistencia, que solo tiene magnitud. Cuando un circuito es alimentado con corriente continua (CC), su impedancia será igual a la resistencia, lo que puede ser interpretado como la impedancia con ángulo de fase cero (Mendoza et al., 2010).

Así mismo, se debe tener en cuenta que, aunque el uso de las propiedades eléctricas de los materiales base de cemento surgen como una opción rápida y económica para obtener resultados ante problemáticas como el transporte de iones cloruros en estructuras de concretos, los cuales de no ser identificados y/o corregidos en edades tempranas representan enormes costos de mantenimiento y reparación, razón por la que surge la necesidad de encontrar una forma rápida y económica de evaluar la migración del ion cloruro.

Es aquí donde gracias a la facilidad de correlacionar estas propiedades eléctricas con la microestructura del concreto, que se da inicio al uso de ensayos fundamentados en EIS, como las mediciones de resistividad y conductividad eléctrica de la muestra. Metodología que desde hace varias décadas ha venido tomando gran relevancia en la tecnología del concreto, ya que a partir del desarrollo de investigaciones se ha buscado obtener correlaciones de propiedades relacionadas con: hidratación del cemento, evaluación de la integridad estructural, caracterización de materiales, optimización de materiales, evaluación de la corrosión de las barras de refuerzo, evaluación de la durabilidad y propiedades relacionadas con el transporte del concreto, entre otros (Lizarazo, Ramírez, et al., 2016).

Finalmente, cuando hablamos de las propiedades resistivas de los materiales cementantes, es importante tener en cuenta que la resistividad eléctrica de las mezclas con cemento (pastas o concretos) están fuertemente relacionadas con la microestructura de su matriz, su estructura de poros, porosidad y distribución del tamaño de los poros. Esto, también es función de la concentración de iones y su movilidad en la solución de los poros (Hunkeler, 1996; Bürchler, et al., 1996). Tomando relevancia aspectos como la química del cemento, el contenido de cemento, la relación agua-cemento, el uso de aditivos y materiales cementantes suplementarios, los cuales se convierten en factores que también influyen en la microestructura de la matriz de cemento, así como en la química

de la solución de poros y, que por lo tanto tendrán una incidencia directa sobre la resistividad eléctrica del material (Mendoza et al., 2010).

2.6.2 Medición de la resistividad eléctrica de una muestra (Espectroscopia de impedancia electroquímica)

El estudio de los ensayos de impedancia eléctrica parte de investigaciones previas sobre las propiedades eléctricas de los materiales, lo cual da espacio para la división de estas metodologías en dos áreas. En la primera, se utilizan electrodos los cuales se conectan a una muestra base cemento para realizar mediciones directas de resistividad eléctrica, pero debido a que cualquier aplicación de corriente directa a una muestra de cemento y/o concreto produce un efecto de polarización del electrolito e induce a la formación de oxígeno e hidrógeno en los electrodos (Lizarazo, 2018), se cree que estos métodos limitan los resultados precisos para la resistividad eléctrica del cemento y/o concreto. En la segunda área, para superar las desventajas de los métodos de corriente continua (CC), se aplica una corriente alterna (CA) de alta frecuencia (> 1000 Hz) para medir la resistividad eléctrica (Li et al., 2003a) y evitar los fenómenos de polarización en la muestra.

Actualmente existen muchas metodologías y procedimientos de ensayos; sin embargo, generalmente todos parten de describir la resistencia eléctrica (R), utilizando circuitos de corriente alterna. En donde para un circuito de DC la relación entre la corriente (I) y el potencial (E) están dados por la ley de Ohm (Ec. 2.6).

$$E (V) = I(Amp) * R(ohm) \quad (2.6)$$

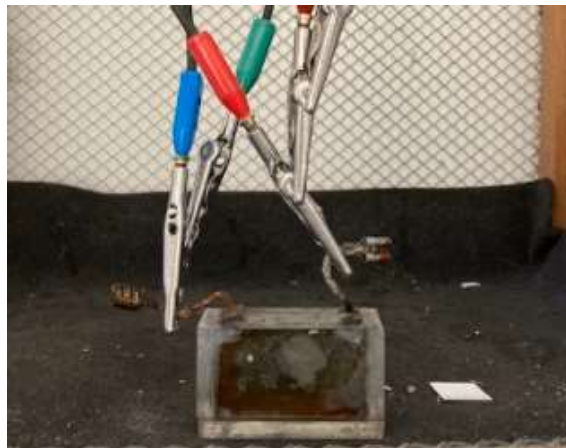
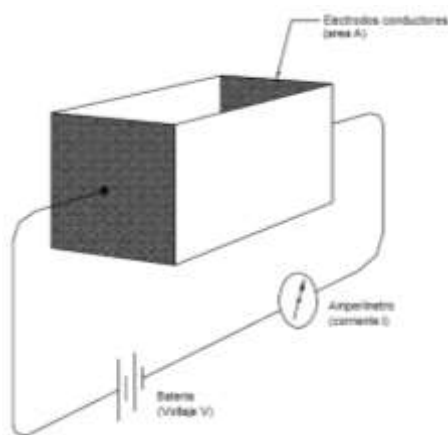
En el caso de la aplicación de CA, la expresión equivalente hará referencia a la impedancia (Z) del material (Ec. 2.7).

$$E (V) = I(Amp) * Z(ohm) \quad (2.7)$$

Donde es necesario destacar que la impedancia en un circuito de CA es función de la frecuencia de la señal aplicada, la cual puede ser expresada en Herz (Hz) o (s^{-1}); y a su vez, dicho valor constará de la amplitud de la señal, el potencial y el ángulo de fase que esta pueda llegar a presentar.

En el caso del ensayo más común y sencillo, la resistividad eléctrica puede definirse según (Esbach, 1975) como la resistencia eléctrica de un conductor de volumen unitario y sección transversal constante en el que la corriente se distribuye de manera continua y uniforme (figura 2-20). Esta medición de la resistividad en laboratorio para las muestras cementantes es relativamente sencilla y consta de la aplicación y medición de un voltaje y corriente aplicados en una muestra de dimensiones definidas (alto, largo y ancho) para así calcular la resistividad. Alternativamente, se pueden usar varios dispositivos de electrodo para realizar el ensayo, siendo los más comunes los electrodos de carbono. Finalmente, la resistividad se calcula a partir de la ecuación 2.8.

Figura. 2-20: (a) Esquema de medición de la resistividad eléctrica. (b) Mediciones de resistividad eléctrica realizadas en la UNAL.



Nombre de la fuente: (a) Pórtland Cement Association (b) Propia.

$$\rho = \frac{A \cdot R}{L} \quad (2.8)$$

Donde;

ρ = Resistividad (ohm*cm)

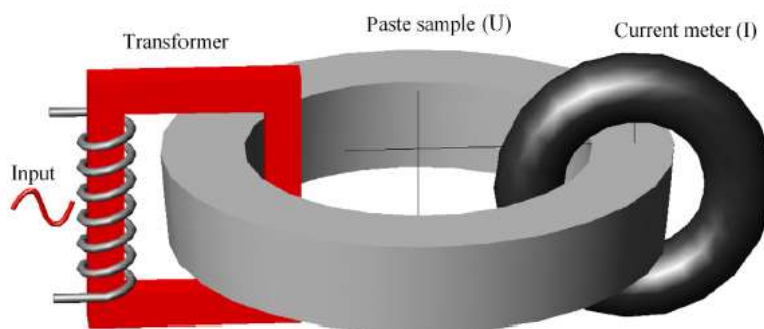
A = Área de la sección (cm²)

R = Resistencia (ohm)

L = Longitud de la probeta (cm)

Aunque en estos casos se elimina el efecto de polarización, el uso de CA no soluciona por completo los problemas de contacto y liberación de gas asociados al método. Por esta razón, y para resolver este problema de contacto, los investigadores suelen usar fuerza externa para sujetar los electrodos a la muestra de concreto desde los dos extremos (Lizarazo, Ramírez, et al., 2016). Sin embargo, como alternativa, a mediados del año 2000, surge un nuevo dispositivo de medición de resistividad eléctrica el cual no requiere estar en contacto con la muestra (figura 2-21), metodología que ha sido puesta a prueba en investigaciones como la realizadas por (Yousuf et al., 2017), (Li et al., 2003a) entre otros, donde se evalúa la metodología de los ensayos sin contacto.

Figura. 2-21: Sistema para la medición de resistividad eléctrica sin contacto.



Nombre de la fuente: (Yousuf et al., 2017).

La figura 2-21 representa un esquema del dispositivo de medición de resistencia sin contacto, el cual adopta el principio de los transformadores, eliminando los electrodos utilizados en las pruebas convencionales de resistividad. Dicha metodología parte inicialmente del concepto de que un transformador consta de dos bobinas de alambre enrolladas en el mismo núcleo (donde la bobina primaria es la entrada del transformador y la bobina secundaria la de salida), y su funcionamiento está fundamentado en el principio de la inductancia mutua, lo cual hace que se induzca un voltaje en la bobina secundaria. Por esta razón, en el sistema la muestra a base de cemento es un anillo con una sección transversal rectangular, el cual actúa como el anillo secundario del transformador.

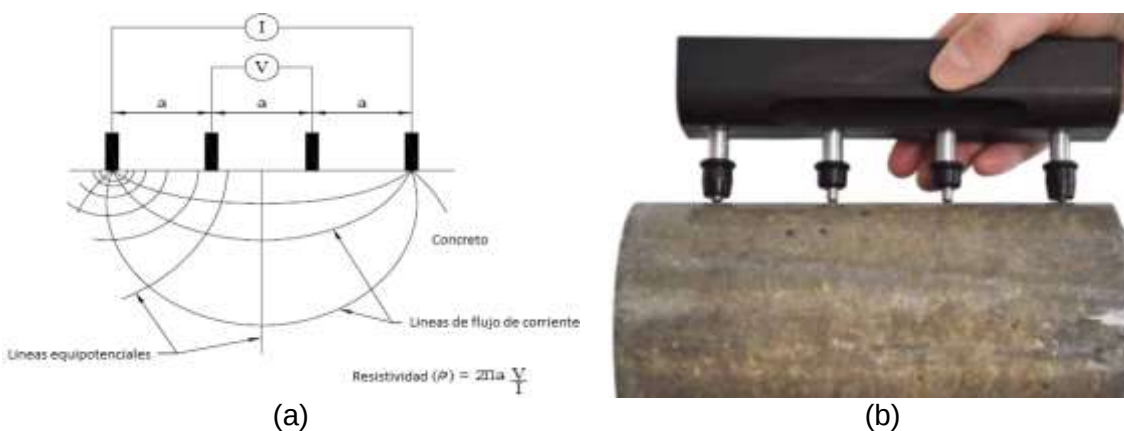
Para esto, el ensayo consta de la aplicación de una onda sinusoidal de CA en la bobina primaria mediante un generador de frecuencia, donde esta onda es amplificada para luego pasar a través del núcleo principal del transformador, creando así un campo magnético. Esta inducción electromagnética ocurre para inducir un voltaje toroidal en la muestra del

anillo de concreto (segunda bobina). Mientras tanto, la corriente que pasa a través de la muestra es detectada por el medidor de corriente, tras lo cual la resistividad de la pasta de cemento se puede calcular de acuerdo con la ley de Ohm (Yousuf et al., 2017). Debido a que no hay electrodos, los problemas asociados como la polarización, la liberación de gas y contacto no serán una variable que afecte la medición.

Finalmente, se tienen algunas técnicas de medición de campo para la resistividad del concreto, las cuales se adaptaron inicialmente de la tecnología utilizada para la medición in situ de la resistividad del suelo relacionadas con la fórmula de Wenner (Wenner, 1916); el cual, parte de los instrumentos que generalmente utilizan una técnica de sonda de superficie de 4 clavijas, aunque también se han empleado otras configuraciones. La aplicación de este método al concreto ha tenido que superar varias dificultades, que incluyen:

- La carbonatación de la superficie del hormigón que puede introducir una resistividad adicional.
- La distancia entre electrodos y longitudes debe ser suficiente para referir la medición a una cierta profundidad de cobertura.
- La distancia del punto de medición a las varillas, ya que si están en el camino de la corriente se introduce un error.
- El modelo está fundamentado en una teoría de medio infinito como lo es el suelo.

Figura. 2-22: a. Esquema de metodología de las 4 clavijas. b. Ensayo de resistividad en concretos para el método de las 4 clavijas.



Nombre de la fuente: (Feliu et al., 1996).

Aunque los equipos modernos suelen tener en cuenta los dos primeros de estos inconvenientes, mientras que el tercero depende de cada estructura en particular (Dunn et al., 2010), el desarrollo de las mediciones de campo ha ido evolucionando constantemente en la búsqueda de nuevas metodologías que permitan de forma más ágil y sencilla realizar estas medias. Destacándose propuestas como la realizada por Feliu, quien describe un método de dos electrodos que a partir del uso de las barras de refuerzo como contra electrodo y un pequeño disco metálico colocado sobre la superficie del hormigón (Feliu et al., 1996), puede calcular la resistividad mediante una ecuación simple (Ec. 2.9).

$$\rho = 2 * R * D \quad (2.9)$$

Donde;

ρ = Resistividad (ohm*cm)

A = Diametro del electrodo (cm)

R = Resistencia (ohm)

Este método se basa en estimar la caída óhmica de la resistencia (R), entre un disco de radio pequeño que es colocado en la superficie de un electrolito y un contraelectrodo mucho más grande colocado en el infinito, donde Newman demostró teóricamente que la resistencia eléctrica medida entre el electrodo y contraelectrodo es una función de la resistividad del electrolito (Ec. 2.9).

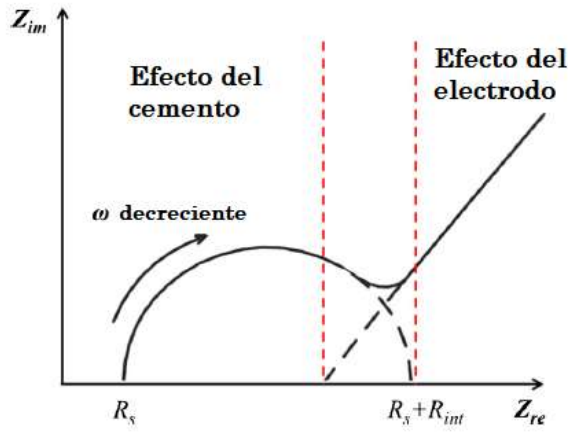
2.6.3 Consideraciones de las mediciones de impedancia electroquímica.

Cuando hablamos de los ensayos de impedancia electroquímica en el concreto, lo primero que debemos entender es que el concreto es esencialmente un material aislante en estado seco al horno y un semiconductor cuando está saturado, por lo que el efecto del contenido de humedad sobre la resistividad eléctrica es quizás el más fuerte de los muchos parámetros que influyen en esta propiedad y donde según (Whiting & Nagi, 2003), el cambio de resistividad en el rango de posibles contenidos de humedad es drástico.

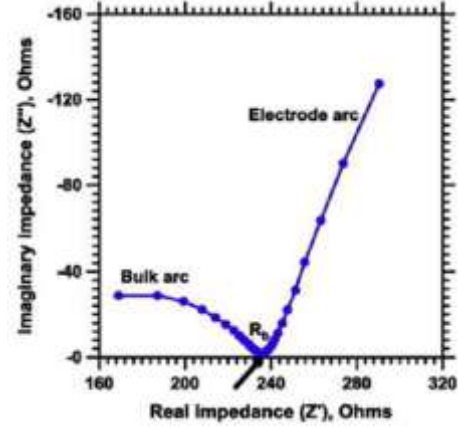
Considerando las precisiones realizadas anteriormente, algunos investigadores han relacionados que la resistividad eléctrica es una indicación de la cantidad de humedad en los poros y el tamaño y tortuosidad del sistema de poros; indicador que se ve fuertemente afectada por la cantidad de cemento en la mezcla, la relación A/C, el curado y los aditivos utilizados. Aquí, es importante resaltar que, aunque el nivel de cloruro no afecta en magnitud la resistividad ya que en estado natural existen muchos iones disueltos en el agua de los poros, algunos iones de cloruro más no hacen una gran diferencia. Sin embargo, se debe tener en cuenta que los cloruros en el concreto pueden ser higroscópicos, es decir, favorecerán que en la matriz se retenga agua. Razón por la que a pesar de todo se suele señalar a los cloruros de reducir la resistividad de una muestra cementante (Dunn et al., 2010).

Otra de las consideraciones importantes que se deben tener en el momento del ensayo, son las frecuencias de CA con las cuales se trabaja; y es que en condiciones experimentales normales los resultados del espectro de impedancia en muestras saturadas de concreto o mortero dan dos semicírculos claramente identificados como se ve en la figura. 2-23. En este gráfico complejo (gráfico de Nyquist), el tamaño y las características de los bucles están íntimamente relacionados con el comportamiento capacitivo de los materiales. Donde a frecuencias intermedias (aquellas superiores a 10^3 o 10^4 Hz, dependiendo de qué tan cerrados estén los poros), domina la impedancia del electrolito a través de los poros continuos y discontinuos (DCP y CCP) de la matriz. Las propiedades dieléctricas de alta frecuencia del material (ICP) dominan a frecuencias superiores a 10^5 Hz. Por último, las frecuencias entre 10^0 y 10^4 Hz se pueden ver afectadas por el comportamiento de la interfaz electrodo-electrolito (Higuera Flórez, 2016).

Por último, es importante hacer énfasis en que la presentación de la impedancia en el formato de Nyquist que se mencionó anteriormente, y que se ha utilizado en mediciones de propiedades eléctricas de CA, es solo uno de los cuatro parámetros conectados dentro del campo de la espectroscopia de inmitancia. Estos niveles son la impedancia $Z(\omega)$ y su admitancia recíproca $Y(\omega)$; permitividad relativa $\epsilon_r(\omega)$ y su módulo eléctrico recíproco $M(\omega)$ (Suryanto et al., 2016).

Figura. 2-23: Curva de Nyquist para las muestras de cemento.

Nombre de la fuente: (Zhu et al., 2017)



Nombre de la fuente: (Hu et al., 2019)

Resaltando que la impedancia ($Z(\omega)$), de un sistema cementante sujeto a un campo eléctrico sinusoidal a una frecuencia angular (ω), se puede escribir como:

$$Z(\omega) = Z'(\omega) - iZ''(\omega) \quad (2.10)$$

donde la componente real $Z'(\omega)$ es la resistencia y la componente imaginaria $Z''(\omega)$ es la reactancia. Indicando que, a cualquier frecuencia la respuesta eléctrica de tal sistema resultará de los fenómenos superpuestos de conducción y polarización. Estos se cuantifican, respectivamente, por la conductividad global, $\sigma(\omega)$ (Ec. 2.11), y la parte real de la permitividad relativa, $\epsilon'_r(\omega)$ (Ec. 2.12), que se desvinculan de la resistencia y la reactancia por:

$$\sigma(\omega) = \frac{Z'(\omega)}{Z'(\omega)^2 - Z''(\omega)^2} * \frac{L}{A} \quad (2.11)$$

$$\epsilon'_r(\omega) = \frac{1}{\epsilon_0 \omega} * \frac{Z''(\omega)}{Z'(\omega)^2 - Z''(\omega)^2} * \frac{L}{A} \quad (2.12)$$

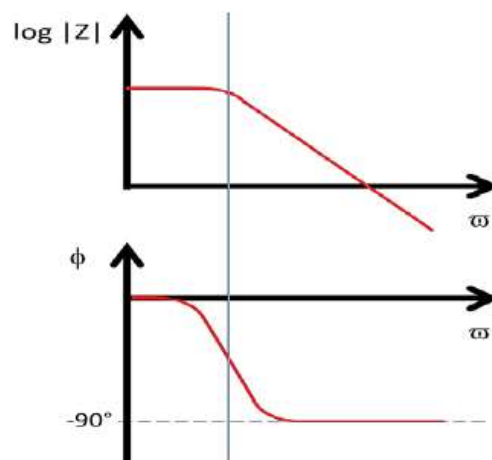
Considerando que la impedancia se presenta como un número complejo que se compone de una parte real (resistencia) y una parte imaginaria (reactancia capacitiva), la forma más adecuada de interpretación de la información es mediante una gráfica en dos ejes, en donde la parte real corresponde al eje X y la parte imaginaria corresponde al eje Y, dando como resultados un diagrama de Nyquist. En dicho gráfico cada punto se puede representar

como un vector, en donde la longitud corresponde al valor de $|Z|$, y el ángulo que forma el vector y el eje real se conoce como el ángulo de fase (ϕ). Finalmente, como se puede ver en la (figura 2-23), este diagrama en el que la frecuencia aumenta de derecha izquierda se puede estudiar de forma desglosada a partir de las siguientes ramas (Suryanto et al., 2016):

- Una rama en el lado de baja frecuencia (lado derecho) de la curva;
- Una meseta intermedia débilmente desarrollada que da una región de valle en forma de U; y,
- Un arco semicircular en el lado de alta frecuencia (lado izquierdo) cuyo centro está hundido debajo del eje $Z'(\omega)$.

Pero, considerando que el diagrama de Nyquist no permite conocer el valor de la frecuencia que fue usada para obtener la impedancia en un punto dado, es necesario generar diagramas de Bode (figura 2-24), en donde se grafica en el eje X el logaritmo en base 10 de la frecuencia (ω) y en el eje Y se grafica el valor del logaritmo en base 10 de la impedancia $|Z|$ o del ángulo de fase (ϕ) (Vázquez, 2007). El empleo de este tipo de diagramas permitirá determinar el comportamiento de los parámetros evaluados en función de la frecuencia.

Figura. 2-24: Diagrama de Bode para muestras de concreto.



Nombre de la fuente: (Ramirez Arrienta, 2017)

2.6.4 Aplicación de las mediciones de resistividad eléctrica en la evaluación de la hidratación del cemento

Después de ser utilizada por primera vez para estudiar las propiedades electroquímicas de las pastas de cemento por McCarter en 1988 (Hu et al., 2019), la técnica de impedancia de CA se ha aplicado ampliamente en la investigación de la hidratación temprana, la evolución microestructural y el comportamiento frente a la corrosión de los materiales cementantes. Todo esto gracias a la ventaja de ser una técnica no destructiva de fácil desarrollo; pero, con el paso de los años se ha venido demostrado que la medición de la resistividad eléctrica es una técnica prometedora en campos que permiten estudiar la variación de la estructura de los poros y la conductividad iónica durante la hidratación de los materiales, entre otros campos de acción que vienen siendo estudiados (Suryanto et al., 2016).

Por ende, para el desarrollo de la presente investigación se tratará el estudio de la aplicación de técnicas EIS como la resistividad, conductividad y la capacitancia desde la evaluación de la hidratación de las pastas de cemento adicionadas con ceniza volante en edades tempranas (≤ 24 horas) y en edades tardías (> 24 horas).

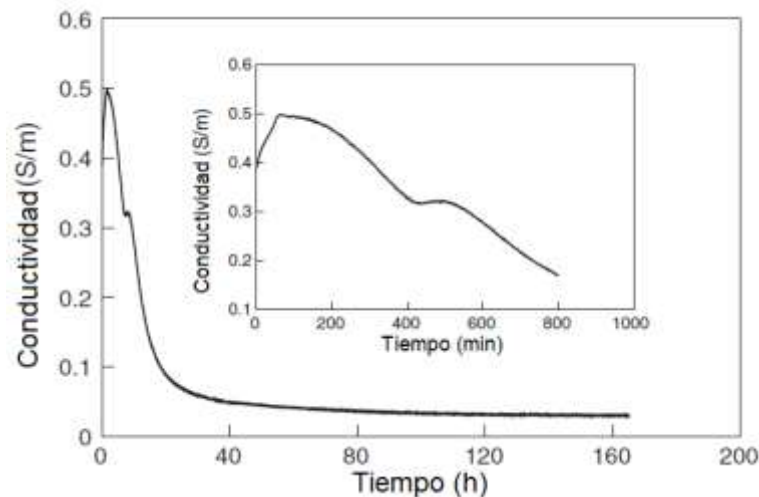
2.6.4.1 Mediciones en edades tempranas

Las mediciones de resistividad bulk, conductividad y capacitancia eléctrica en edades tempranas para las pastas de cemento, son estudios que se han basado principalmente en la comparación de los procesos de hidratación de las mezclas; lo cual en el caso de las pastas de cemento hacen referencia de un proceso fisicoquímico que conduce a la formación de hidratos, presentando a medida que se desarrollan características termodinámicas, cinéticas y estructurales.

Uno de los procesos más representativos que se lleva a cabo durante la hidratación de los cementantes es el fraguado, el cual es un evento de tiempo definido que corresponde a la transición de la pasta de blanda a dura, generando una serie de cambios en el estado físico de la muestra y las concentraciones iónicas dentro del agua de medición. Lo anterior, llega a verse reflejado en el cambio de la conductividad eléctrica de una muestra, razón por la cual en algunos trabajos de investigación se han utilizado parámetros como la conductividad y/o la resistividad eléctrica de los materiales a base de cemento para

analizar el control de la hidratación en función de las propiedades dieléctricas (Hu et al., 2019).

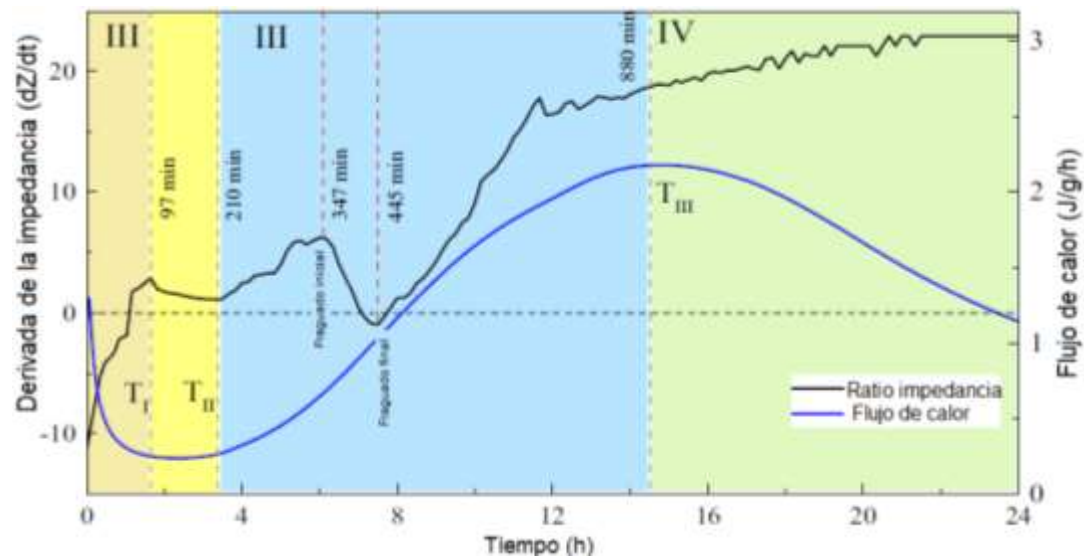
Figura. 2-25: Cambio de la conductividad eléctrica de los materiales base cemento con el tiempo de curado.



Nombre de la fuente: (Shen et al., 2016).

Como se puede ver en la figura. 2-25, la conductividad eléctrica (la cual representa físicamente el inverso de la resistividad eléctrica) inicialmente presenta bruscamente un aumento en sus valores, llegando a alcanzar su máximo valor una vez se llega a aproximadamente la primera hora de medición. Este rápido aumento de la conductividad eléctrica durante el inicio de la medición se atribuye a la disolución de iones (Ca^{+} , OH^{-} , K^{+} , Na^{+} , etc) presentes en el cemento. Sin embargo, a medida que la hidratación del cementante va avanzando, los productos de esta crecen rápidamente y absorben los iones alcalinos y SO_4^{2-} en la solución de los poros, por lo que la concentración de iones y la conductividad eléctrica comienzan a disminuir significativamente. Adicionalmente, el esqueleto sólido de los productos de hidratación comienza a llenar los poros separados y por ende a reducir su conectividad, lo que nuevamente resulta en una disminución de la conductividad eléctrica (Shen et al., 2016).

Figura. 2-26: Correlación entre flujo de calor y curva derivada de impedancia para pasta de cemento



Nombre de la fuente:(Chi et al., 2019).

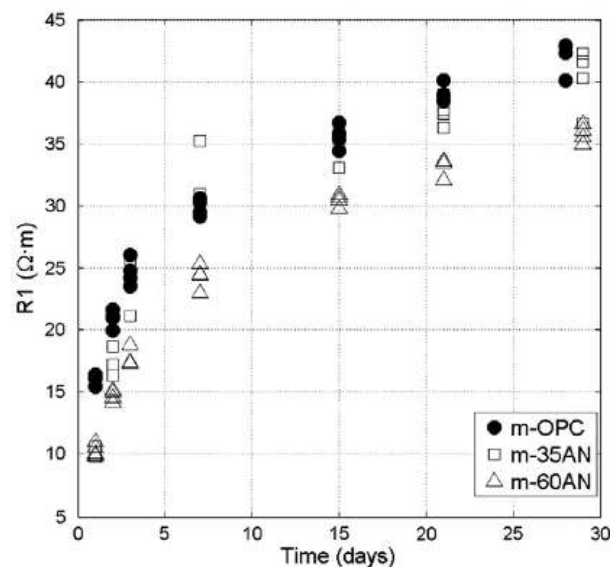
Esta disminución general de la conductividad con el tiempo se debe entre otras cosas al refinamiento continuo de la estructura de los poros como resultado de la hidratación en curso y la reacción puzolánica. Es por esto que la conductividad se puede considerar como una medida de todos los procesos de pérdida que operan dentro del material, pudiéndose llegar a cuantificar la energía disipada por el movimiento de las cargas en un campo eléctrico aplicado (Suryanto et al., 2016).

Con base en estos principios, investigaciones como la realizada por (Chi et al., 2019) han logrado encontrar importantes correlaciones entre el flujo de calor liberado durante la hidratación de las pastas de cemento y la derivada de la impedancia eléctrica de la muestra (figura 2-26). Esto al igual que (Chi et al., 2021; Liu et al., 2021; Yousuf et al., 2020), quienes a partir de las mediciones de resistividad eléctrica han desarrollado importantes modelos que permiten la relación de los tiempos de fraguado y fases de hidratación de las pastas de cemento con los cambios en las curvas de resistividad y conductividad de la mezcla. Investigaciones que han sido tomadas como base para el desarrollo del presente trabajo de investigación.

2.6.4.2 Mediciones en edades tardías

En cuando a las mediciones en edades tardías de las pasta de cemento, las principales investigaciones se han centrado en el estudio de la evolución de la resistividad real (R_b) de los materiales, y es que el principio de la hidratación de los materiales cementantes tiene correlación directa con los procesos de ganancia de resistencias en función de la edad, ya que se espera que con el paso de los días y el avance de los procesos de hidratación, se vaya generando una matriz interna de mayor complejidad y calidad en la cual se disminuya la existencia de caminos de conducción continua (Husain et al., 2017), lo cual deberá ver reflejado un incremento de la resistividad de la muestra en el tiempo y de igual forma en una disminución de la resistividad eléctrica de la muestra (figura 2-27).

Figura. 2-27: Evolución de la resistividad eléctrica (R_b) en el tiempo según investigación realizada por de Cruz et al.



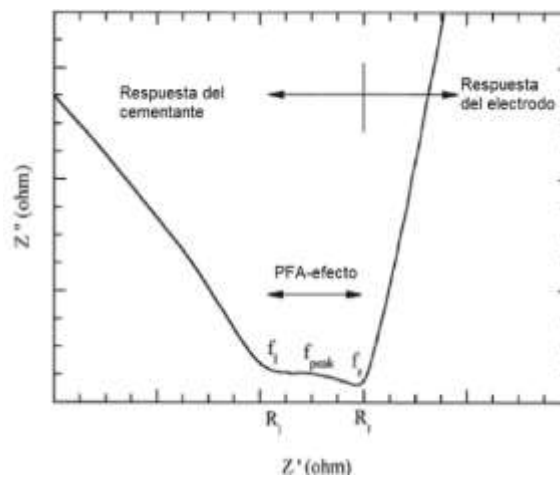
Nombre de la fuente: (Cruz et al., 2013)

Por esta razón, una de las teorías más estudiadas y relacionadas con la adición de FA a las mezclas de concreto, son los beneficios obtenidos en aspectos como la prolongación de la manejabilidad de la mezcla y la reducción de los picos de temperatura de está. La aparición de nuevas tecnologías como el EIS y el desarrollo de diferentes investigaciones han logrado identificar beneficios adicionales en el uso de esta adición, los cuales han estado enfocados al desarrollo de una mejor microestructura de la matriz cementante en edades tardías que se ve reflejado en la ganancia de características como la resistencia,

disminución de permeabilidad al agua y ion cloruro, entre otros beneficios vinculados con las propiedades de durabilidad de la mezcla (Suryanto et al., 2017).

Una de estas características relacionados con las edades tardías de las mezclas y que se revisará durante el presente trabajo de investigación están fundamentada en el análisis de los diagramas de Nyquist y el uso de la ceniza volante como adición. En investigaciones como las desarrolladas por McCarter (1988), se ha documentado que con la adición de FA baja en cal en las pastas de cemento, puede llegar a inducir cambios importantes en la respuesta eléctrica del sistema. Como se muestra en la figura 2-28, existe una región de meseta entre el electrodo y los arcos de la matriz de cemento en el diagrama de Nyquist, el cual se va haciendo más notorio a medida que aumenta la edad de la muestra y el contenido de ceniza en la mezcla. Fenómeno que con el tiempo se ha venido atribuyendo a la mejora en los efectos de polarización de doble capa en la superficie de las partículas de FA y sobre el cual a finales de 2004 se logró tener importantes avances para su demostración gracias a las investigaciones de (Hu et al., 2019).

Figura. 2-28: Gráficos de Nyquist para sistemas de cenizas volantes con bajo contenido de cal.



Nombre de la fuente: (Hu et al., 2019)

Adicional a las propiedades mencionado anteriormente, otros investigadores han centrado sus estudios en la comprensión de la evolución de la impedancia real (R_b) de una muestra con el tiempo, partiendo de hipótesis de que este valor debería presentar un aumento en

el tiempo de forma proporcional al desarrollo de los procesos de hidratación y previo refinamiento de la microestructura. Sin embargo, llaman la atención investigaciones como las realizadas por (Chi et al., 2019) y (Husain et al., 2017), quienes identificaron en el uso de adiciones minerales como la ceniza volante en las mezclas de concreto, que aunque los valores de R_b medidos en las muestras suelen ser inferiores a los valores de R_b medidos en las muestras sin adición; identificando adicionalmente que las muestras adicionadas con ceniza volante continúan desarrollando un aumento de los valores de R_b una vez superadas edades de 28 días, lo cual no se ve reflejando con igual claridad en las muestras sin adición, las cuales tienden presentar valores constantes de R_b una vez superados los 14 días de edad.

Condición que ve reflejada en estudios como los realizados por (Fonseca Barrera, 2016), quien a partir de mediciones de permeabilidad al ion cloruro a edades superiores a los 300 días, logró encontrar que al adicionar ceniza volante a las muestras de concreto, estas presentaban una continua mejora en su resistencia a la penetración del ion cloruro; propiedad que está directamente relacionada con el desarrollo y refinamiento de la red de poros y microestructura interna de la muestra.

3. Metodología

La metodología que se presente a continuación fue definida en función de los objetivos específicos planteados en el numeral 1.3.1 para el desarrollo de este trabajo de investigación. Con esto, se busca hacer una comparación entre las mediciones de resistividad eléctrica en muestras de cemento tipo ART adicionado con ceniza volante, con las que a partir de los métodos tradicionales utilizados para evaluar la hidratación como lo son los tiempos de fraguado con agujas de Vicat y las curvas de calor específico acumulado mediante ensayos de calorimetría. Se pretende encontrar una relación que permita definir una metodología de medición y estudio de la hidratación de los cementos a partir de ensayos EIS como la resistividad bulk y la capacitancia de una muestra.

Inicialmente, se realizó la calibración de la cámara semi-adiabática (HidroCemUn) del Laboratorio de Estructuras. Para esto se utilizó Cemento Pórtland (ver anexo B), con el cual se realizaron ensayos de forma paralela en el equipo diseñado en Universidad Nacional de Colombia y los equipo I-Cal 8000 de la empresa Toxemet. En cuanto a la realización de los ensayos propuestos para el desarrollo de la investigación, se utilizó Cemento ART, el cual fue suministrado por Cementos ALION (ver anexo B) y cenizas volantes termopaipa suministradas por CEMEX (ver anexo B). Finalmente, se utilizó agua potable con una temperatura entre 18°C a 20°C.

Con estos materiales realizaron mediciones en 3 muestras definidas de la siguiente forma:

Muestra 1: 100% cemento Pórtland.

Muestra 2: 85% de cemento Pórtland con adición del 15% de cenizas volantes.

Muestra 3: 70% de cemento Pórtland más adición del 30% de cenizas volantes.

Una vez definidas las muestras a utilizar, las etapas en que se desarrolló el presente trabajo de investigación fueron las siguientes:

3.1 Medición de la hidratación en pastas de cemento hidráulico adicionadas con ceniza volante a partir de métodos tradicionales como los tiempos de fraguado con agujas de Vicat y calorimetría semi-adiabática.

3.1.1 Ensayos para determinar el tiempo de fraguado

El desarrollo de la primera fase de la presente investigación consistió en realizar las mediciones tradicionales de consistencia normal y tiempos de fraguado con el aparato de Vicat. Para los ensayos mencionados anteriormente se tomaron como referencia los lineamientos de las Normas Técnicas Colombianas NTC 110 “Método para determinar la consistencia normal de cemento (aparato de Vicat)”, la cual permitió conocer las relaciones A/C para obtener la consistencia normal indicada para cada una de las muestras en estudio. También se usó la norma NTC 118 “Método para determinar el tiempo de fraguado del cemento hidráulico mediante el aparato de Vicat” para los tiempos de fraguado inicial y final para cada una de las muestras definidas.

Aunque los ensayos de consistencia normal fueron realizados bajo las indicaciones de la NTC 110, fue necesario modificar los tiempos de mezclado que en esta norma se especifican, esto debido a que al usar una mezcla de cemento ART con una adición de ceniza volante fue necesario mezclar más la muestra con el fin de garantizar una mejor homogenización del material. Con esto, el periodo de tiempo de mezclado fue de 3 minutos y 15 segundos (60 segundos más que lo indicado en la normal), el cual se distribuyó de la siguiente manera:

Tabla 3-1: Tiempos de mezclado usados para las muestras

Tiempo (s)	Actividad
30	Hidratación cemento + agua
30	Mezclado a velocidad lenta
15	Recolección de material con espátula
60	Mezclado a velocidad rápida
15	Recolección de material con espátula
45	Mezclado a velocidad rápida

Finalmente, se resalta que las relaciones A/C definidas para cada una de las muestras, se establecieron a parte de la consistencia normal de cada una de ellas, decidiendo así que se esté sea el parámetro constante y la cantidad de agua el variable. Esta A/C se utilizara para realizar los ensayos de calorimetría y resistividad eléctrica.

3.1.2 Ensayos de calorimetría en cámara semi-adiabatica (HidroCemUn)

Para realizar los ensayos de calorimetría, se usó como referencia la norma BS EN 196-9:210. *“Methods of testing cement. Heat of hydration. Semi-adiabatic method”*. En este ensayo, se tomó como referencia la metodología y el equipo de medición (cámara semi-adiabatica) que fue desarrollado en el año 2011 por la estudiante Luisa Natalia Peña Leal (figura 3-1), el cual por el paso de tiempo fue necesario actualizar tecnológicamente, lo cual nos implicó realizar un nuevo proceso de calibración del equipo.

Por lo anterior, surgió la necesidad de realizar algunas mejoras en el equipo como lo fueron el sistema de aislamiento, donde se adicionaron dos capas de protección térmica las cuales constan de una capa de poliuretano expandido y una caja de madera de 5mm de espesor que sirve como protección exterior de la cámara. Adicionalmente, se cambió el equipo de registro de temperatura, adquiriendo un PicLog 6 TC-08 de compañía PicoTechnology que cuenta con 8 canales de lectura para termocuplas tipo K (figura 3-2), las cuales son de fácil acceso en el mercado y pueden ser desechadas una vez terminado el ensayo.

Figura. 3-1: Cámara semi-adiabática construida en el 2011. (a) foto exterior y (b) foto al interior de la cámara.



(a)



(b)

Nombre de la fuente: Peña, Luisa. 2011

Figura. 3-2: Adecuaciones realizadas a la Cámara semi-adiabática. (a) foto exterior y (b) foto al interior de la cámara.



(a)



(b)

Nombre de la fuente: Propia.

Una vez realizados los ajustes constructivos del equipo, éste debió ser calibrado nuevamente (ver anexo A) con el fin de afinar los valores de las constantes (K), los cuales están en función de la temperatura pico alcanzada por cada muestra. A continuación, se describe brevemente la metodología de ensayo de la cámara.

- Previamente al comienzo del ensayo, se deberán iniciar las mediciones de temperatura (aproximadamente 10 a 15 min), esto con el fin de que en el equipo se estabilicen las lecturas de temperatura.

- Preparar los vasos de medición como se indica en el anexo A, antes de depositar las respectivas muestras (se utilizaron 3 mediciones por cada muestra definida, utilizando así la totalidad de las cámaras para evitar la variación en los datos).
- Introducir cada muestra dentro de los vasos Dewar e introducir la punta de la termocupla dentro de la muestra (corroborar que esta quede bien inmersa en el interior de la muestra).
- Puesto cada vaso en su respectiva sub-cámara, cerrar correctamente las dos tapas del equipo e iniciar la toma de las lecturas de temperatura (se recomienda dejar dos termocuplas para la lectura de la temperatura ambiente, una dentro de la cámara interior y la otra en la cámara exterior).
- Una vez terminado el ensayo, los datos deberán ser guardados y exportados en formato .CSV, con el fin de facilitar su procesamiento. Aquí se deberán tener en cuenta las ecuaciones de calibración de cada cámara (ecuaciones (3.1), (3.2) y (3.3)), las cuales permiten calcular el calor específico de cada muestra.

$$K1 = -0.00014t + 0.0681 \quad (3.1)$$

$$K2 = -0.000094t + 0.0694 \quad (3.2)$$

$$K3 = -0.00054t + 0.0841 \quad (3.3)$$

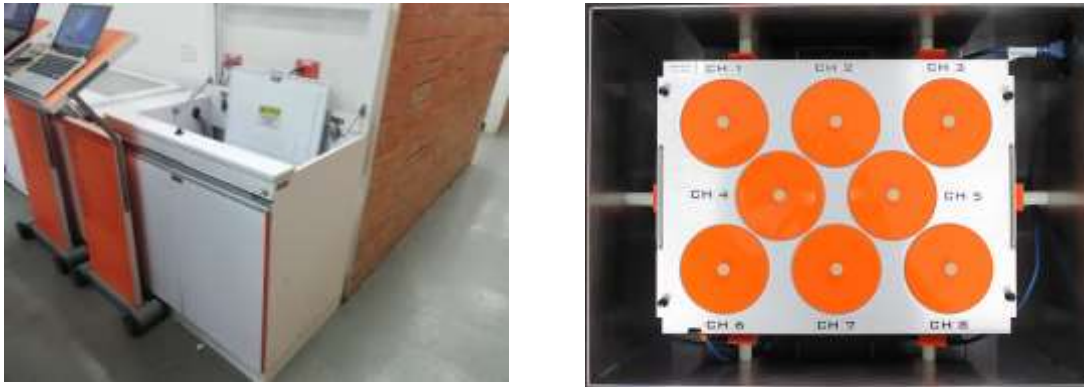
Donde Δt hace referencia al delta de temperatura de la muestra.

Adicional a las mediciones realizadas con el cámara semi-adiabática HidroCemUn, se hicieron mediciones con un equipo I-CAL 8000 fabricado por la compañía CALMETRIX (figura. 3-3), de propiedad de Toxement S.A., lo que permitió realizar una comparación entre los valores obtenidos por cada uno de los equipos y verificar el funcionamiento de la cámara.

Finalmente, se debe hacer énfasis en que el desarrollo de estos ensayos es de gran importancia en el momento de definir la metodología para la medición de la hidratación del cemento a partir de métodos eléctricos, ya que a partir de ellos se pueden identificar las etapas asociadas con los procesos de hidratación de lo cemento como son los tiempos de fraguado inicial y final, la temperatura máxima de hidratación, puntos de inflexión en la

liberación de calor, entre otras. Estos serán de importancia en el momento de analizar las curvas de resistividad de cada una de las mezclas propuestas.

Figura. 3-3: Calorímetro adiabático CALMETRIX, I-CAL 8000.



Nombre de la fuente: Propia.

3.2 Evaluación mediante el uso de metodologías de espectroscopia de impedancia electroquímica (EIS) las propiedades eléctricas de los procesos de hidratación del cemento hidráulico adicionado con ceniza volante en edades tempranas y tardías.

Para realizar estas mediciones se utilizó un potenciostato-galvanostato marca Gamry-750, con licencia de EIS hasta 5000 KHz, con la particularidad que en los ensayos no se utilizaron electrodos de barras de carbono, ya que en busca de mejorar el procedimiento de medición se implementaron electrodos serigraficos (RRPE1001C), los cuales son fabricados mediante impresión 3D (figura 3-3) por la compañía PINE RESEARCH. Es importante resaltar que en lo concerniente a los objetivos de la presente investigación, solo se utilizaran los electrodos con base en tereftalato de polietileno.

Figura. 3-4: Electrodo semigrafico impreso en 3D.



(a). Electrodo de referencia, (b). Electrodo de trabajo y (c). Contraelectrodo.

Nombre de la fuente: PINE RESEARCH.

Los electrodos serigrafados (SPEs) son dispositivos de medida electroquímica que se desarrollan a partir de la impresión de distintos tipos de tintas sobre sustratos de plástico o cerámica, los cuales permiten a partir de su uso un análisis in-situ con una elevada repetitividad, sensibilidad y precisión. La composición de las diversas tintas (carbono, plata, oro, platino) empleadas en la fabricación del electrodo determinan la selectividad y sensibilidad del mismo, lo cual debe estudiarse a partir del uso y finalidad que se quiera dar a estos sensores.

En el caso de los electrodos utilizados para la presente investigación (figura 3-4), estos cuentan con un sustrato de tereftalato de polietileno (PET), electrodos de trabajo y contraelectrodos de carbón, y un electrodo de referencia incorporado de plata / cloruro de plata (Ag / AgCl). Adicionalmente, un material dieléctrico aislante azul que recubre el sustrato de PET alrededor de los electrodos. Es importante resaltar que estos equipos son de bajo costo, por lo que se dejaron introducidos en la mezcla con el fin de estudiar las propiedades de la misma durante largos periodos de tiempo y de igual forma evaluar durabilidad a largo plazo.

Es importante tener en cuenta que en la actualidad el principal uso que se viene dando a estos elementos está relacionado con las áreas químicas y biomédicas, esto gracias a la facilidad de tamaño, fabricación y versatilidad de uso a la hora de realizar tareas como:

- Mediciones de glucosa en la sangre.
- Mediciones de compuestos fenólicos en plantas, animales y humanos.
- Detección de nitritos y fosfatos.
- Detección de metales pesados como el plomo y el mercurio.

- Detección de herbicidas en el agua.

Aplicaciones que se han logrado gracias a ventajas que brindan estos electrodos como:

- Cero mantenimiento: la ideología de estos electrodos está relacionada a un solo uso. Por lo tanto, no es necesario realizar mantenimiento ni pretratarlos antes del ensayo.
- Alta reproducibilidad: los electrodos serigrafiados son fácilmente reproducibles, esto los convierte en excelentes candidatos para realizar un alto número de ensayos en serie sin tener mayores afectaciones en la medición.
- Bajo costo: Comercialmente el valor de estos equipos en Colombia ronda entre 3 a 4 dólares por electrodo.
- Tamaño reducido: gracias a sus pequeñas dimensiones, éstos pueden ser utilizados en zonas de difícil acceso o para aplicaciones portátiles.
- Volumen de muestra reducido: gracias a sus dimensiones, permiten obtener resultados con bajos volúmenes de muestra. Esto los hace candidatos ideales para aplicaciones donde obtener muestras es costoso como, por ejemplo, los biosensores.
- Diseño flexible: gracias al método de fabricación, el diseño de los electrodos es altamente flexible.

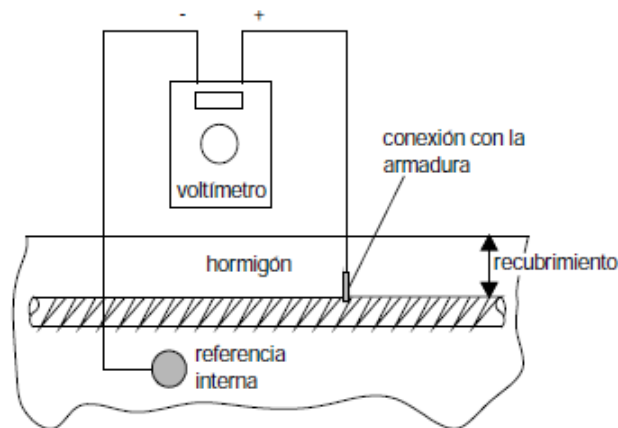
3.2.1 Calibración de la metodología de mediciones con el electrodo serigrafiado de referencia RRPE1001C

Es importante tener en cuenta que en la literatura aún no se encuentran aplicaciones de uso de los electrodos serigrafiados en el área de la tecnología del concreto, esto debido a que el principal uso que se viene dando a estos equipos está ligado a otras disciplinas. Por esta razón es importante comprender el uso de estos sensores y sobre todo desarrollar una metodología que permita la interpretación de las lecturas realizadas a partir de su uso.

Para el desarrollo del presente estudio se partirá de las investigaciones en metodologías de campo realizadas por Feliu desde 1966, las cuales parten del principio de uso de un electrodo circular embebido en la matriz del concreto y el uso del acero de refuerzo como

electrodo de trabajo (figura 3-5), tal y como se describe en el numeral 2.6.2 del presente documento.

Figura. 3-5: Esquema de metodología en campo propuesta por Feliu



Nombre de la fuente: (Broomfield, 2007)

Adicionalmente, se realizaron mediciones comparativas y de forma simultánea con el método tradicional de las barras de carbono, lo cual permitió tener un comparativo del comportamiento de las dos metodologías de resistividad eléctrica.

3.2.2 Evaluación de las propiedades eléctricas en edades tempranas.

Para el desarrollo del presente objetivo, se buscó realizar una serie de mediciones en las muestras de cemento descritas anteriormente. Dichas mediciones se hicieron durante las primeras 24 horas de iniciado el proceso de hidratación del cemento anhidro y consistieron en el registro de la impedancia eléctrica de las pastas de cemento, realizando 4 mediciones por cada una de estas, con frecuencias de corriente alterna constantes de 1.000Hz, 10.000Hz, 50.000Hz y 100.000Hz. Lo anterior, permitió el estudio del comportamiento del material para así definir la frecuencia con que se obtienen los mejores datos para sentar las bases teóricas del comportamiento eléctrico de las muestras.

Aquí se evaluarán propiedades como la resistividad, conductividad, ángulo de la fase y capacitancia de las pastas de cemento a ensayar, en las cuales se evaluarán esos cambios

en el comportamiento eléctrico del material con los tiempos relevantes identificados durante sus periodos de hidratación.

3.2.3 Evaluación de las propiedades eléctricas en edades tardías.

Con el fin de evaluar las propiedades eléctricas de las mezclas de concreto, se realizaron también ensayos potencioestáticos de resistividad eléctrica en edades de 0 h, 3 h, 24 h, 3 d, 7 d, 14 d, 21 d, 28 d, 36 d y 56 d, los cuales fueron evaluados en el rango de frecuencias de CA de 0.1 kHz a 5000 kHz. Donde a partir de los diagramas de Nyquist y Bode se buscó conocer el cambio en comportamiento resistivo de las muestras a medida que avanzan los procesos de hidratación. Durante este tiempo, las muestras fueron curadas en agua, según las recomendaciones de la norma NTC 1377 (*Elaboración y curado de especímenes de concreto para ensayos de laboratorio*).

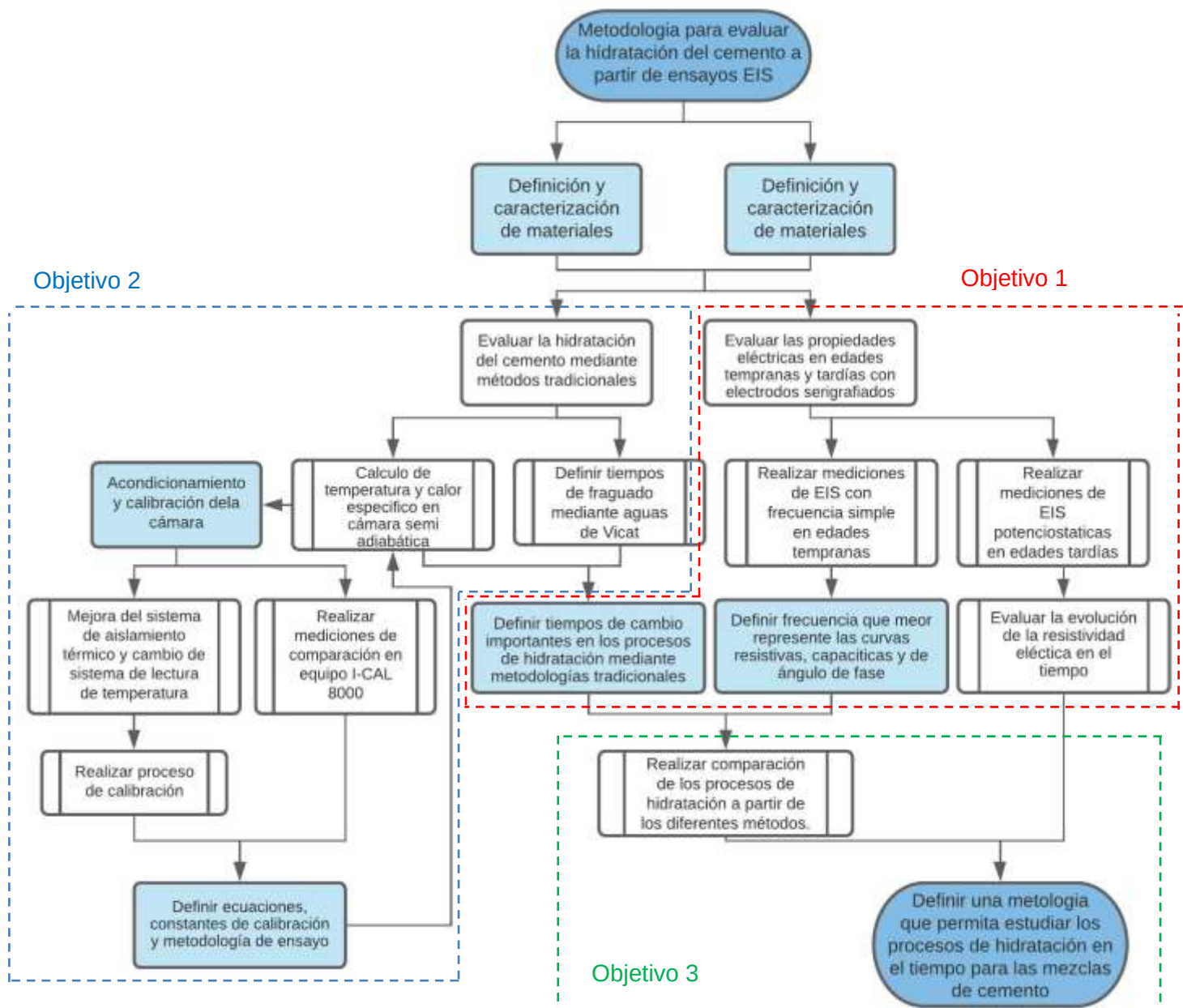
3.3 Metodología para el estudio de la hidratación del cemento hidráulico a partir de ensayos de resistividad.

Finalmente, se realizó una comparación entre los resultados obtenidos con las metodologías convencionales, de donde se obtuvieron datos relevantes como los tiempos de fraguado inicial y final, picos de temperatura, calor específico, entre otros valores relevantes, los cuales fueron traslapados con las curvas de resistividad eléctrica obtenidas de ensayos realizados durante las primeras 24 horas de preparada la mezcla. Para esta condición se usó una frecuencia constante, la cual fue definida en función de los resultados obtenidos durante el numeral 3.2.

Lo anterior, permitió identificar la forma en que se representan los tiempos y etapas más importantes de la hidratación de los cementos en una curva de resistividad eléctrica, pudiendo así definir una metodología que permite realizar estas mediciones de forma rápida y precisa, implementado el uso de nuevas herramientas de bajo costo como lo son los electrodos serigráficos tipo SPE.

En la figura 3-6, se presenta un resumen gráfico de la metodología propuesta para alcanzar los objetivos planteados.

Figura. 3-6. Resumen de procedimiento propuesto para definir una metodología para la medición de la hidratación de las pastas de cemento a partir de mediciones de impedancia eléctrica.



Nombre de la fuente: Propia.

4. Resultados obtenidos

A continuación, se presentan los resultados de los ensayos realizados y se evalúa su comportamiento con respecto a los objetivos planteados en esta investigación.

4.1 Ensayos de consistencia normal y tiempos de fraguado.

En esta sección, se presentan los resultados correspondientes a los ensayos de consistencia normal y tiempos de fraguado realizados a las muestras definidas en el capítulo 3 del presente documento, con esto se busca definir las relaciones A/C que nos permiten tener las consistencias normales de las mezclas previamente definidas, con las cuales se realizaron los ensayos de calorimetría e impedancia electroquímica.

Por su parte, los tiempos de fraguado se usaron con uno de los principales parámetros de comparación para la evaluación del comportamiento de las mezclas en cada uno de los ensayos. Dichos valores se representan a continuación.

Figura. 4-1: Ensayos de aguas de Vicat en muestras: a. 100% cemento, b. 85% cemento + 15% ceniza y c. 70% cemento + 30% ceniza.



(a)



(b)



(c)

Tabla 4-1: Ensayos de consistencia normal y tiempo de fraguado para la mezcla N°1.

Muestra N°1 (100%CP)		Muestra N°1 (100%CP)	
Relación A/C	0.27	Tiempo (min)	Penetración (mm)
Penetración (mm)	10	60	40.0
		70	40.0
		80	40.0
		90	32.0
		100	30.0
		<u>110</u>	<u>26.0</u>
		<u>120</u>	<u>23.0</u>
		130	21.5
		140	19.0
		150	18.0
		160	14.5
		170	6.5
		180	1.5
		190	1.0
		200	0.5
		210	0.5
		220	0.0
		230	0.0
		240	0.0

Figura. 4-2: Representación gráfica de los tiempos de fraguado muestra 1.

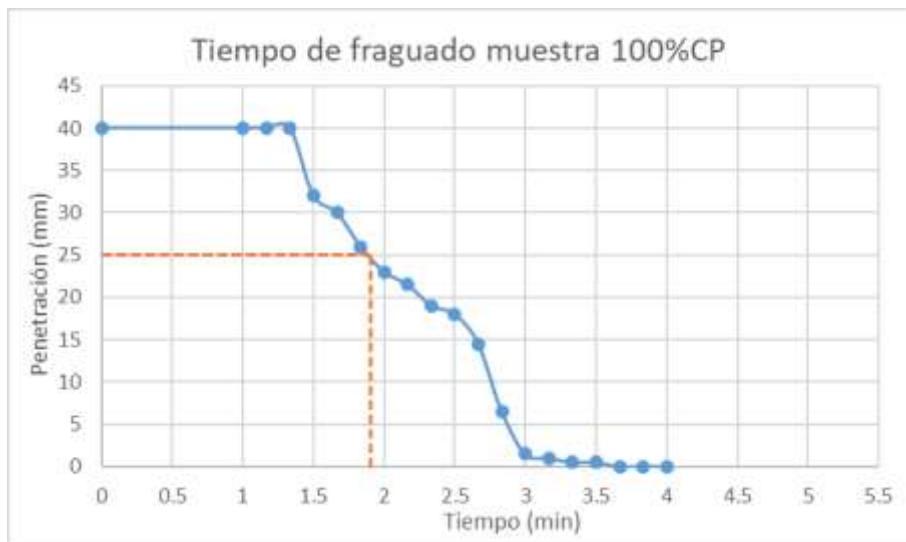


Tabla 4-2: Ensayos de consistencia normal y tiempo de fraguado para la mezcla N°2.

Muestra N°2 (85%CP - 15%CV)		Muestra N°2 (85%CP - 15%CV)	
Relación A/C	0.285	Tiempo (s)	Penetración (mm)
Penetración (mm)	10	60	40.0
		70	40.0
		80	40.0
		90	40.0
		100	40.0
		110	40.0
		120	36.0
		130	34.0
		140	28.0
		<u>150</u>	<u>25.5</u>
		<u>160</u>	<u>23.0</u>
		170	20.0
		180	18.5
		190	15.0
		200	12.0
		210	1.0
		220	1.0
		230	0.5
		240	0.5
		250	0.5
		260	0
		270	0
		280	0
		290	0

F_i (min)	154.50
F_i (h)	2.58
F_f (min)	260.00
F_f (h)	4.33

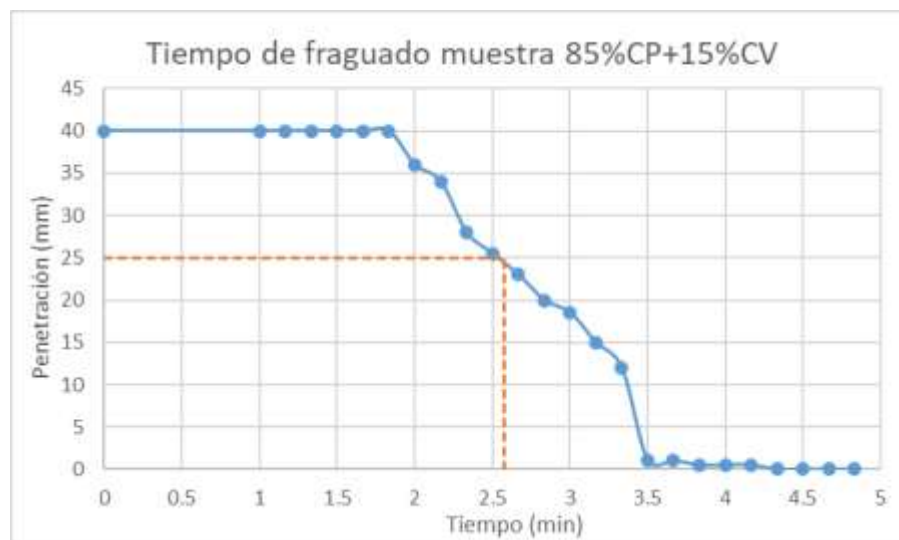
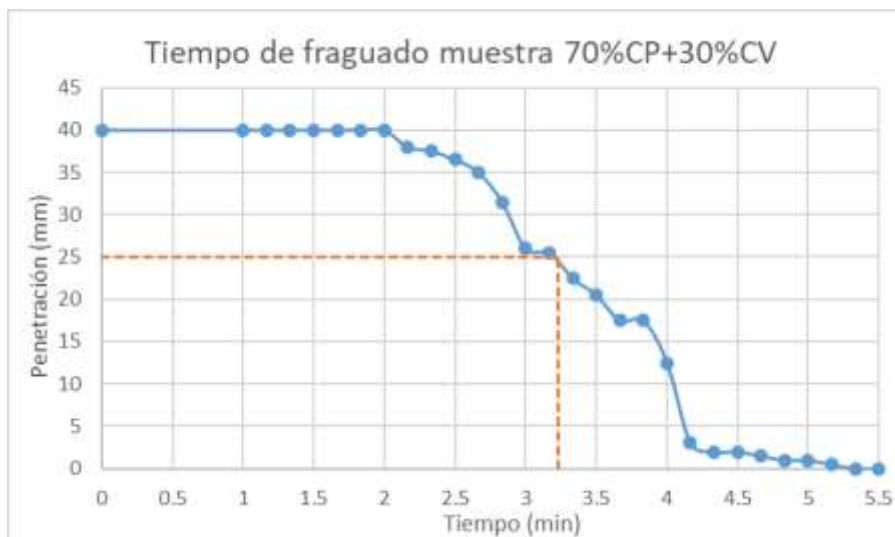
Figura. 4-3: Representación gráfica de los tiempos de fraguado muestra 2.

Tabla 4-3: Ensayos de consistencia normal y tiempo de fraguado para la mezcla N°3.

Muestra N°3 (70%CP - 30%CV)		Muestra N°3 (70%CP - 30%CV)	
Relación A/C	0.31	Tiempo (s)	Penetración (mm)
Penetración (mm)	10	60	40.0
		70	40.0
		80	40.0
		90	40.0
		100	40.0
		110	40.0
		120	40.0
		130	38.0
		140	37.5
		150	36.5
		160	35.0
		170	31.5
		180	26.0
		<u>190</u>	<u>25.5</u>
		<u>200</u>	<u>22.5</u>
		210	20.5
		220	17.5
		230	17.5
		240	12.5
		250	3.0
		260	2.0
		270	2
		280	1.5
		290	1
		300	1
		310	0.5
		320	0
		330	0

F_i (min)	193.83
F_i (h)	3.23
F_f (min)	290.00
F_f (h)	4.83

Figura. 4-4: Representación gráfica de los tiempos de fraguado muestra 3.



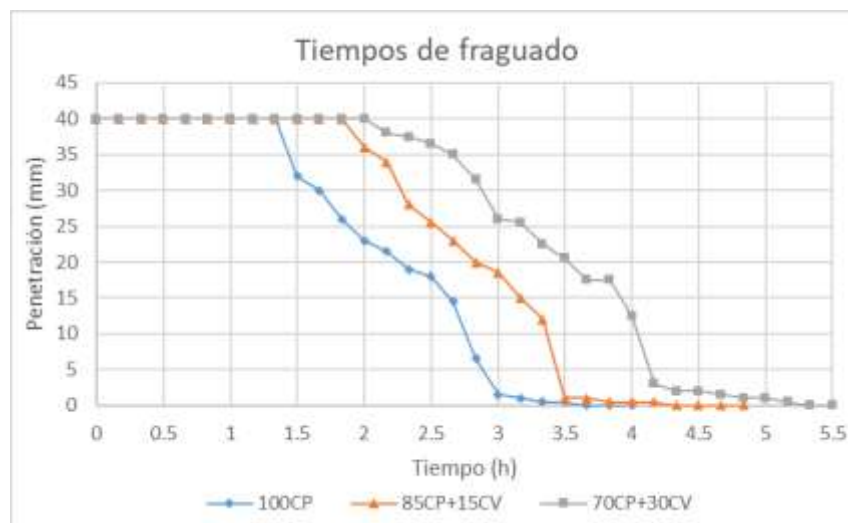
Finalmente, en la tabla 4-4 se presenta un resumen de los valores obtenidos para los tiempos de fraguado de cada una de las muestras en estudio.

Tabla 4-4: Resumen de tiempos de fraguado para las muestras en estudio.

	M1: 100% Cemento	M2: 85% Cemento + 15% Ceniza	M3: 70% Cemento + 30% Ceniza
Fraguado Inicial (min)	1.91	2.58	3.23
Fraguado Final (min)	3.67	4.33	4.83
Inicio de perdida de maneabilidad (min)	1.50	2.00	2.17
Relación A/C	0.270	0.285	0.310

Para el desarrollo del presente trabajo se incluirá el punto definido como inicio de perdida de manejabilidad (Pm), como el punto en que la muestra de pasta comienza a presentar resistencia a la penetración de la aguja de Vicat.

Figura. 4-5: Variación de los tiempos de fraguado en muestras en estudio.



Como se puede ver en la tabla 4-4 y la figura 4-5 y al igual que lo han explicado autores como Long et al (2005), Kurdowsku (2014), Sadique et al (2016) , entre otros, quienes coinciden en que el uso de la ceniza volante en las mezclas de concreto puede generar un efecto retardante en el inicio de los procesos de fraguado de la mezcla. Para este trabajo se tuvo retardos de aproximadamente 40 minutos para una adición de 15% de ceniza y aproximadamente 80 minutos cuando la adición de la ceniza se llevó hasta el 30%. Este

efecto se encuentra relacionado con la complejidad en los procesos de hidratación de la ceniza, lo cual como se puede ver en la figura 2-3, requiere de procesos de reacción de la puzolana con los iones para su previa hidrólisis e inicio en la liberación de iones K^+ y Na^+ a la solución acuosa (Lizarazo, 2018).

4.2 Calorimetrías en cámara semi-adiabática HidroCemUN

4.2.1 Validación de mediciones en la cámara semi-adiabática HidroCemUN

De acuerdo con los ensayos planteados y con el fin de evaluar el correcto funcionamiento de la cámara HidroCemUn, se realizaron mediciones de control con cemento tipo Pórtland, ensayos que fueron realizados según lo relacionado en la tabla 4-5. Las muestras fueron ensayadas de forma simultánea en los dos equipos I-CAL 8000 propiedad de TOXEMENT S.A y en la cámara desarrollada por el Laboratorio de Estructuras de la Universidad Nacional de Colombia “HidroCemUN”.

Tabla 4-5: Muestras y relaciones A/C utilizadas para los ensayos de calibración

Muestra	Relación A/C
N1	0.29
N2	0.30
N3	0.31
N4	0.32
N5	0.33
N6	0.34
N7	0.35
N8	0.36

Es importante mencionar, que en cuanto a las pruebas realizadas en la cámara semi-adiabática HidroCemUN, solo fueron ensayadas las muestras N4 a N8. Decisión que se toma por similitud encontrada en los resultados de las muestras N1 a N4 y los bajos inventarios de material cementante disponible para realizar los ensayos.

Realizados los ensayos en el equipo de TOXEMENT S.A, se tuvieron los siguientes resultados:

Figura. 4-6: Curvas de calor especifico vs tiempo obtenidas en el equipo N°1 de TOXEMENT S.A.

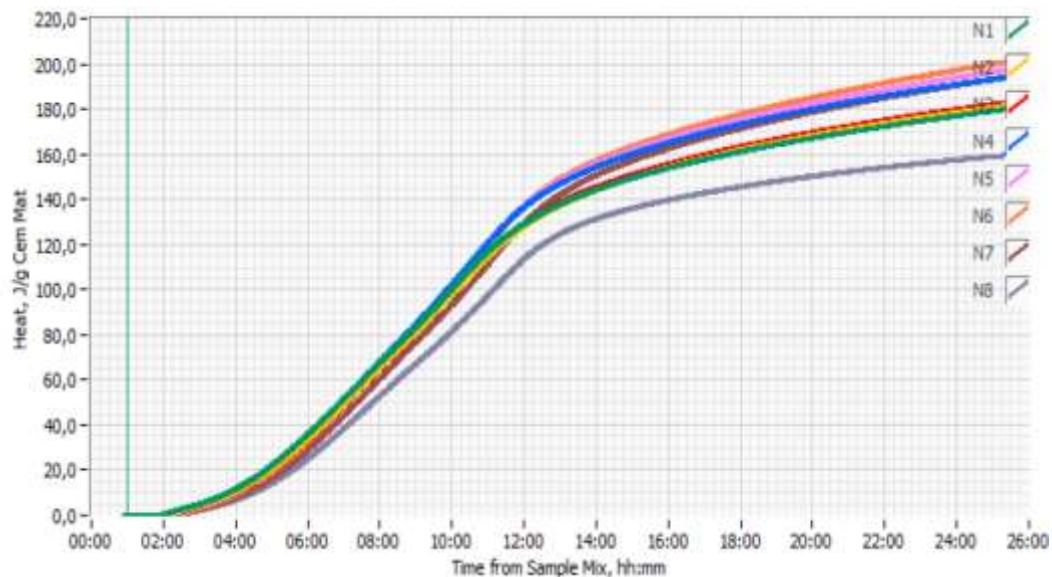
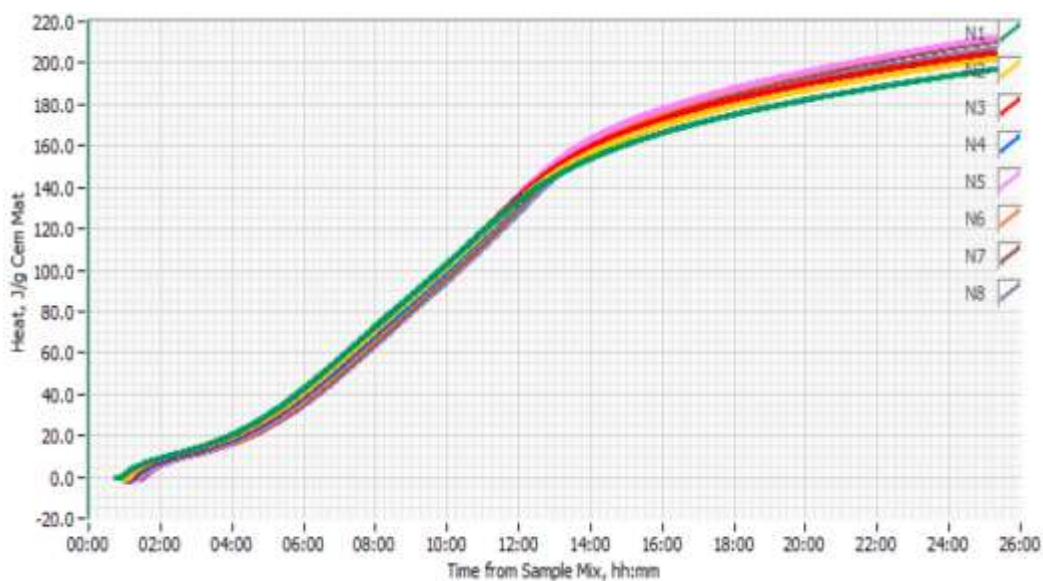


Figura. 4-7: Curvas de calor especifico vs tiempo obtenidas en el equipo N°2 de TOXEMENT S.A.



Inicialmente, se puede observar que aunque los ensayos fueron realizados al mismo tiempo y utilizando los materiales bajo la misma condiciones (temperatura inicial, humedad, etc), existe una variación en los resultados de las muestras N1, N2, N3 y N8; donde para el equipo N°1 el calor especifico liberado osciló entre 160 a 180 J/g pero en el equipo N°2

este valor supero los 200 J/g. Posteriormente, si se analizó el inicio de las gráficas, donde se puede ver que existe una ligera variación durante las primeras 2 horas de realizada la mezcla; esto se puede asociar a factores como el proceso de mezclado de cada una de las muestras, ya que estas fueron realizadas por dos personas de forma simultánea con el fin de iniciar las mediciones al mismo tiempo, lo cual es indicio de la alta sensibilidad del ensayo a agentes externos como el proceso de mezclado.

Tabla 4-6: Resumen ensayos de calorimetría para muestra N4.

RESUMEN DE RESULTADOS MUESTRA: Muestra N4 - R.A/C=0.32			
	Muestra N°1	Muestra N°2	Muestra N°3
Peso de muestra (g)		111.80	105.31
Tem. máx (°C)		48.44	46.02
Delta Tem. (°C)		26.5	24.2
Tiempo para T _{máx} (s)		28410.0	29310.0
Constante K		0.067	0.072
Área bajo la curva		703875.41	619814.28
Calor verdadero (Q)		47170.08	44621.67
Calor específico (J/g)		182.05	184.91
Fecha de ensayo	3/11/2021		

Figura. 4-8: Curvas de calor específico vs tiempo para la muestra N4 obtenidas en el equipo HidroCemUn.



En el desarrollo del ensayo de la muestra N4 se presentaron novedades con respecto a la termocupla instalada en la cámara número 1, esto se puede ver reflejado en el valor de calor específico acumulado el cual no es acorde con los valores obtenidos en las cámaras N° 2 y 3. Por esta razón, en la figura 4-4 no se presenta la curva de calor acumulado para dicha cámara.

Tabla 4-7: Resumen ensayos de calorimetría para muestra N5.

RESUMEN DE RESULTADOS MUESTRA: Muestra N5 - R.A/C=0.33			
	Muestra N°1	Muestra N°2	Muestra N°3
Peso de muestra (g)	99.07	100.97	109.86
Tem. máx (°C)	49.07	50.93	50.90
Delta Tem. (°C)	22.7	25.9	25.4
Tiempo para T _{máx} (s)	24480.0	24660.0	24660.0
Constante K	0.066	0.067	0.071
Área bajo la curva	574530.62	657115.49	639051.47
Calor verdadero (Q)	37819.28	44074.38	45615.17
Calor específico (J/g)	188.37	188.35	181.18
Fecha de ensayo	9/03/2021		

Figura. 4-9: Curvas de calor específico vs tiempo para la muestra N5 obtenidas en el equipo HidroCemUn.



Tabla 4-8: Resumen ensayos de calorimetría para muestra N6.

RESUMEN DE RESULTADOS MUESTRA: Muestra N6 - R.A/C = 0.34			
	Muestra N°1	Muestra N°2	Muestra N°3
Peso de muestra (g)	110.10	87.49	102.41
Tem. máx (°C)	53.56	50.46	52.61
Delta Tem. (°C)	28.1	25.6	27.8
Tiempo para T _{máx} (s)	21840.0	22260.0	22020.0
Constante K	0.065	0.067	0.070
Área bajo la curva	720478.95	655853.55	709534.65
Calor verdadero (Q)	47038.63	44006.92	49795.14
Calor específico (J/g)	210.80	217.04	212.18
Fecha de ensayo	11/02/2021		

Figura. 4-10: Curvas de calor específico vs tiempo para la muestra N6 obtenidas en el equipo HidroCemUn.



Tabla 4-9: Resumen ensayos de calorimetría para muestra N7.

RESUMEN DE RESULTADOS MUESTRA: Muestra N7 - R.AC=0.35			
	Muestra N°1	Muestra N°2	Muestra N°3
Peso de muestra (g)	91.30	102.10	106.60
Tem. máx (°C)	47.82	51.66	50.65
Delta Tem. (°C)	21.9	26.6	26.6
Tiempo para T _{máx} (s)	24900.0	24030.0	24480.0
Constante K	0.066	0.067	0.071
Área bajo la curva	509659.50	673187.70	688265.85
Calor verdadero (Q)	33591.15	45105.80	48742.99
Calor específico (J/g)	181.53	190.63	199.53
Fecha de ensayo	10/02/2021		

Figura. 4-11: Curvas de calor específico vs tiempo para la muestra N7 obtenidas en el equipo HidroCemUn.

Tabla 4-10: Resumen ensayos de calorimetría para muestra N8.

RESUMEN DE RESULTADOS MUESTRA: Muestra N8 - R.AC=0.36			
	Muestra N°1	Muestra N°2	Muestra N°3
Peso de muestra (g)	102.44	108.60	88.96
Tem. máx (°C)	46.74	48.95	44.74
Delta Tem. (°C)	21.6	24.5	20.0
Tiempo para T _{máx} (s)	26280.0	25140.0	26640.0
Constante K	0.066	0.067	0.074
Área bajo la curva	504283.65	602740.80	456038.85
Calor verdadero (Q)	33252.46	40502.25	33785.64
Calor específico (J/g)	160.17	160.92	165.73
Fecha de ensayo	9/02/2021		

Figura. 4-12: Curvas de calor específico vs tiempo para la muestra N8 obtenidas en el equipo HidroCemUn.



Una vez analizadas las gráficas presentadas en las figuras 4-9 a 4-12, que corresponden a los ensayos de calorimetría hechos con el equipo HidroCemUn, se realizó la comparación de los resultados a partir de las cinco zonas de transición que presenta el concreto en sus procesos de hidratación y liberación de energía (hidrolisis inicial, inducción, aceleración, desaceleración y estado estable, las cuales fueron descritas en el numeral 2.4 del presente documento).

Aquí es importante aclarar que debido a la naturalidad de los ensayos realizados y el tiempo de medición que fue de 24 horas aproximadamente, la zona de hidrolisis inicial no se logró identificar, por su parte, la zona de estado estable es ligeramente identificable. Esto, considerando que la primera zona se da durante un periodo de tiempo muy corto y se requiere de mediciones desde el primer contacto del agua con el cementante. En el caso de estado estable se requiere de la prolongación de las mediciones por varios días, razón por la cual dichas graficas se analizaron a partir del periodo de hidrolisis inicial, aceleración y desaceleración principalmente. Zonas que fueron explicadas en el numeral 2.4 y que se pueden clasificar para el presente ejercicio en función de los siguientes elementos:

- Zona de hidrolisis inicial: hace referencia al periodo en el que las partículas de cemento inician con los procesos de hidrolisis, dando como resultado la primera liberación de iones a la mezcla. Durante el proceso no se tienen cambios significativos en la temperatura de la muestra o liberación de calor de esta. Para el desarrollo de las presentes graficas se puede identificar esta zona en el rango de 0 a 4 horas.
- Zona de aceleración: es definida como el sector donde se da el inicio a los procesos significativos de liberación de calor, esta etapa está marcada con la formación del primer gel C-S-H y el inicio de los procesos de fraguado de concreto y perdida de manejabilidad de la mezcla. En las gráficas, esta zona se da entre las 4 a 12 horas.
- Zona de desaceleración: finalmente, esta zona hace referencia a la finalización de los procesos de hidrolisis del cemento. Donde para este punto, aunque ya se ha logrado el fraguado final de la mezcla, aún se encuentra un remanente de calor acumulado dentro de la misma. Durante esta etapa se puede mencionar que el gel C-S-H comienza sus procesos de ganancia de resistencia. En las gráficas, esta zona se puede identificar a partir de las 12 horas y puede llegar hasta las 36 o 48 horas, en donde según la literatura se da inicio de la zona estable.

Por último, aunque se pueden llegar a presentar variaciones de hasta el 10% en la estimación del calor especifico acumulado entre los dos equipos (HidroCemUN y I-CAL 8000), se puede identificar que las zonas de transición descritas anteriormente o los puntos de inflexión en las curvas se presentan en similares rangos de tiempo, resaltando que, de igual forma las curvas del calor especifico acumulado presentan tendencia de gran

similitud. Esto, nos muestra la confiabilidad y el buen funcionamiento logrado por el equipo desarrollado en la Universidad Nacional de Colombia.

4.2.2 Ensayos de calorimetría de las muestras en estudio.

Una vez se lograron tener valores de calor específico acumulado coherentes entre los dos equipos; se realizaron los ensayos de las muestras de interés en estudio para el presente trabajo de investigación. Es importante resaltar que de igual forma se realizó un ensayo de comparación en uno de los equipos de I-CAL 8000, donde los resultados obtenidos son los siguientes:

Figura. 4-13: Curva de calor específico acumulado en equipo I-CAL 8000.

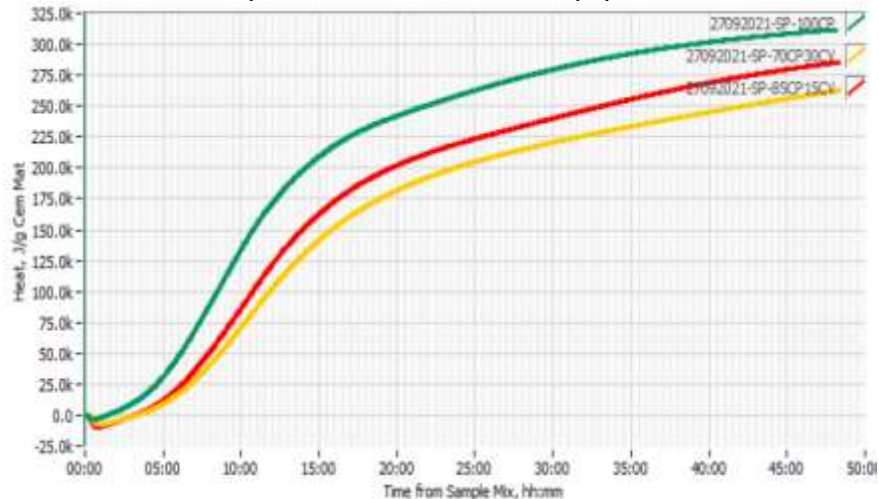


Figura. 4-14: 8 Curva de flujo térmico en equipo I-CAL 8000.

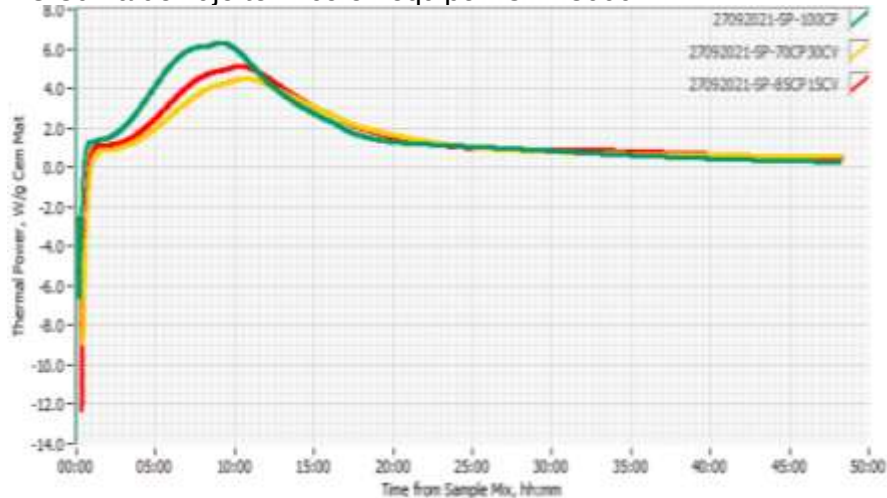


Tabla 4-11: Resumen ensayo de calorimetría para la muestra de 100% cemento.

RESUMEN DE RESULTADOS MUESTRA: Ensayo 100% CP			
	Muestra N°1	Muestra N°2	Muestra N°3
Peso de muestra (g)	105.19	102.69	92.12
Tem. máx (°C)	50.12	50.84	46.48
Delta Tem. (°C)	30.33	31.22	26.98
Tiempo para Tmáx (s)	29760	29760	29760
Constante K	0.065	0.067	0.071
Área bajo la curva	920791.2	959441.67	843381.72
Calor verdadero (Q)	59912.84	63889.41	59551.18
Calor específico (J/g)	281.02	268.47	282.09
Fecha de ensayo	7/06/2021		

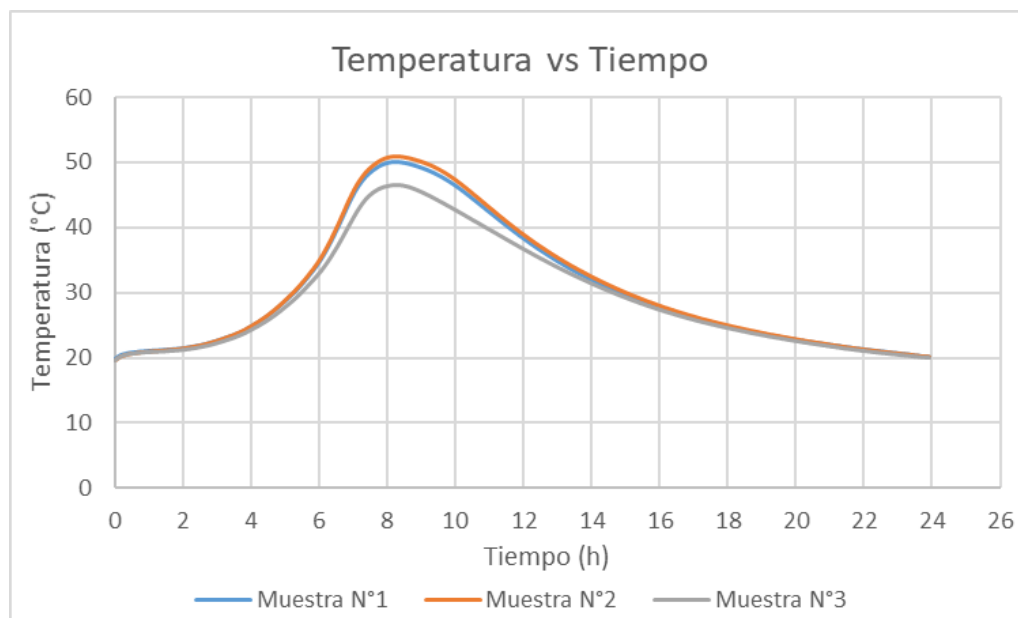
Figura. 4-15: Evolución de la temperatura para muestra de 100% cemento.

Figura. 4-16: Evolución del calor acumulado para muestras de 100% cemento.

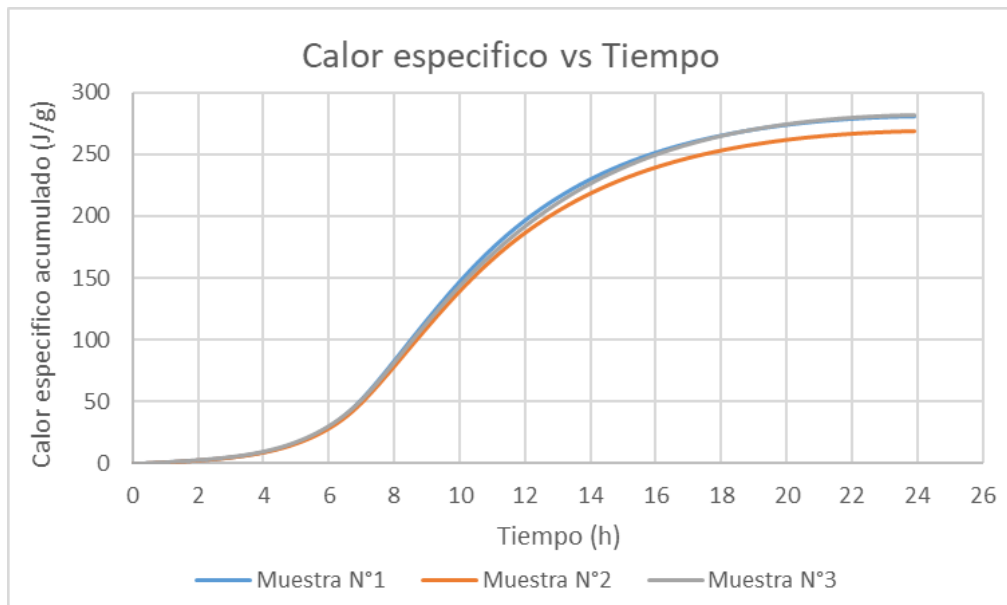


Tabla 4-12: Resumen ensayo de calorimetría para la muestra de 85% cemento y 15% ceniza volante.

RESUMEN DE RESULTADOS MUESTRA: Ensayo 85CP-15CV			
	Muestra N°1	Muestra N°2	Muestra N°3
Peso de muestra (g)	101.24	103.68	95.08
Tem. máx (°C)	46.49	48.80	45.41
Delta Tem. (°C)	25.05	27.26	23.98
Tiempo para Tmáx (s)	28620	28140	28380
Constante K	0.066	0.067	0.072
Área bajo la curva	741490.23	804841.23	694601.94
Calor verdadero (Q)	48638.13	53881.17	50088.09
Calor específico (J/g)	237.05	224.23	229.89
Fecha de ensayo	7/08/2021		

Figura. 4-17: Evolución de la temperatura para muestra de 85% cemento y 15% ceniza volante.

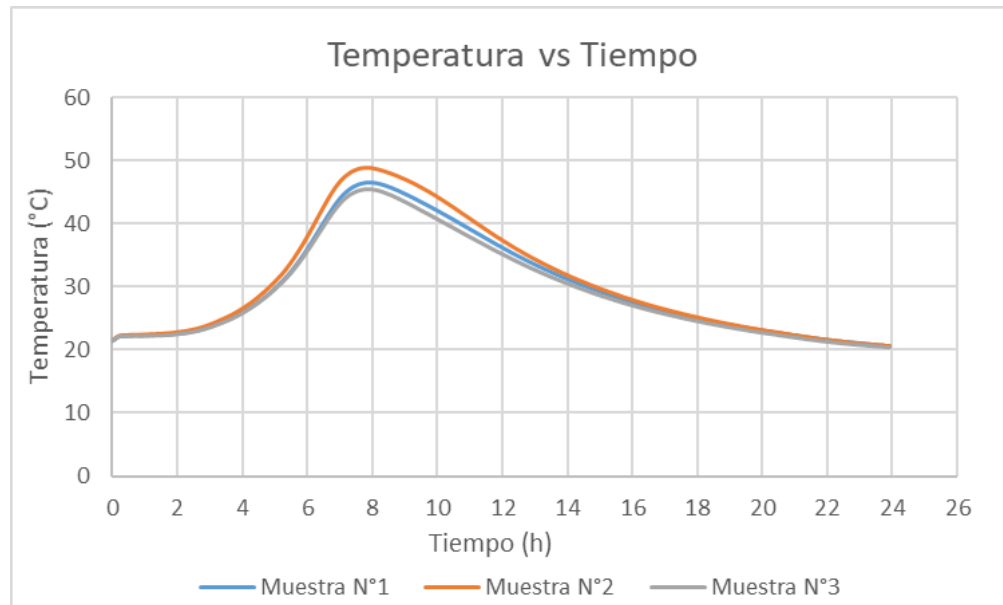


Figura. 4-18: Evolución del calor acumulado para muestras de 85% cemento y 15% ceniza volante.

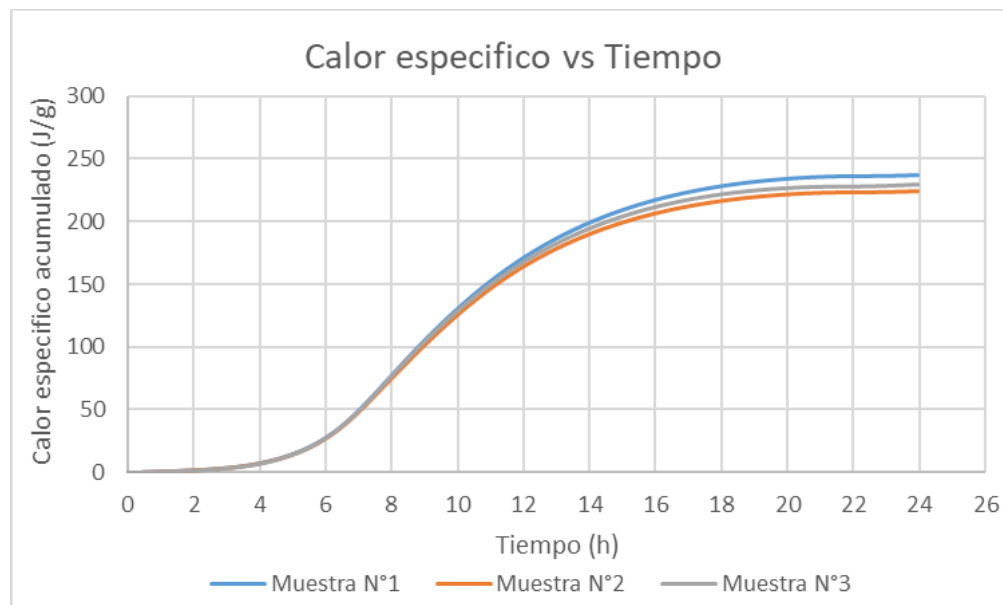


Tabla 4-13: Resumen ensayo de calorimetría para la muestra de 70% cemento y 30% ceniza volante.

RESUMEN DE RESULTADOS MUESTRA: Muestra 70%CP+30%CV			
	Muestra N°1	Muestra N°2	Muestra N°3
Peso de muestra (g)	98.33	104.36	97.31
Tem. máx (°C)	36.98	39.96	38.32
Delta Tem. (°C)	16.89	19.97	18.39
Tiempo para T _{máx} (s)	36360	35460	35580
Constante K	0.066	0.068	0.075
Área bajo la curva	597202.20	703605.45	635480.37
Calor verdadero (Q)	39660.68	47565.88	47600.34
Calor específico (J/g)	199.01	196.67	213.47
Fecha de ensayo	7/07/2021		

Figura. 4-19: Evolución de la temperatura para muestra de 70% cemento y 30% ceniza volante.

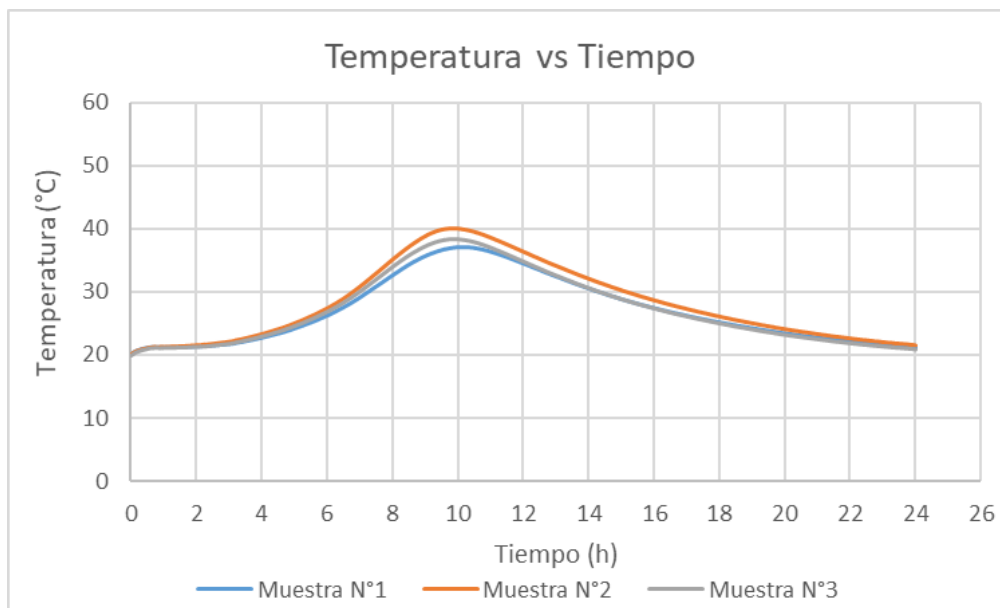
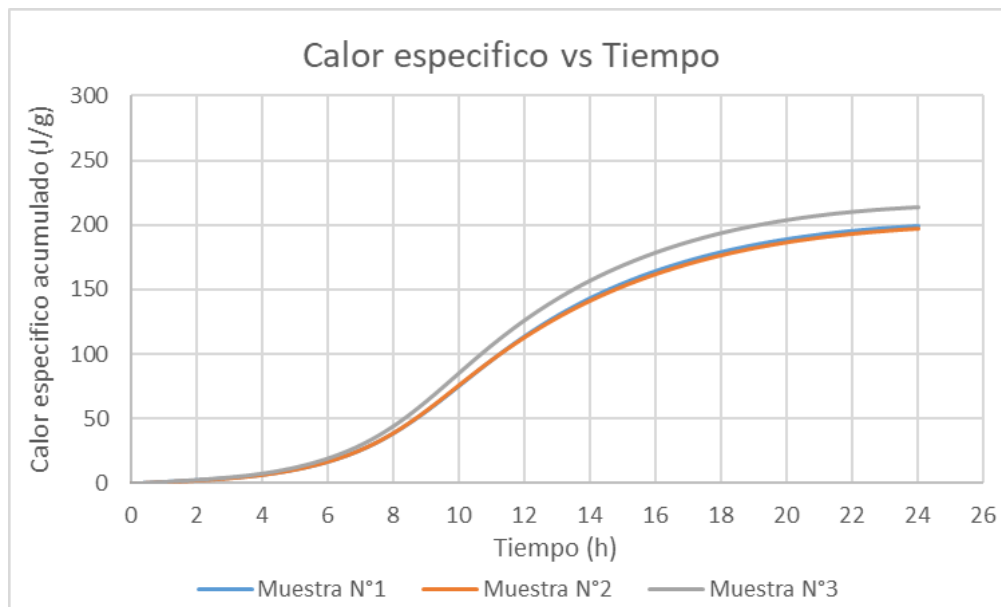


Figura. 4-20: Evolución del calor acumulado para muestras de 70% cemento y 30% ceniza volante.



Finalmente, se presenta el resumen de los resultados obtenidos.

Figura. 4-21: Evolución del Δ temperatura para las muestras en estudio en equipo HidroCemUN.

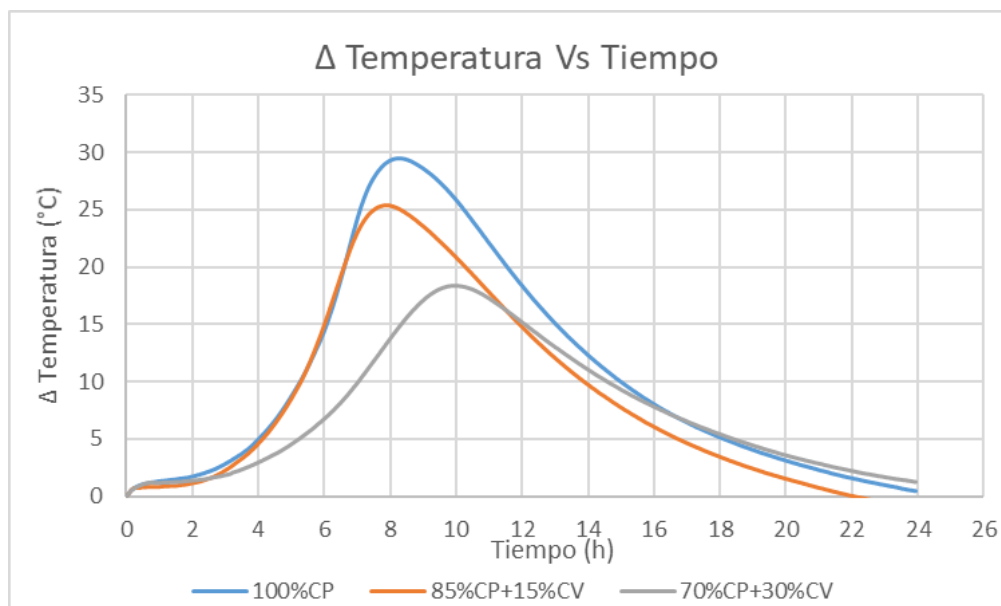
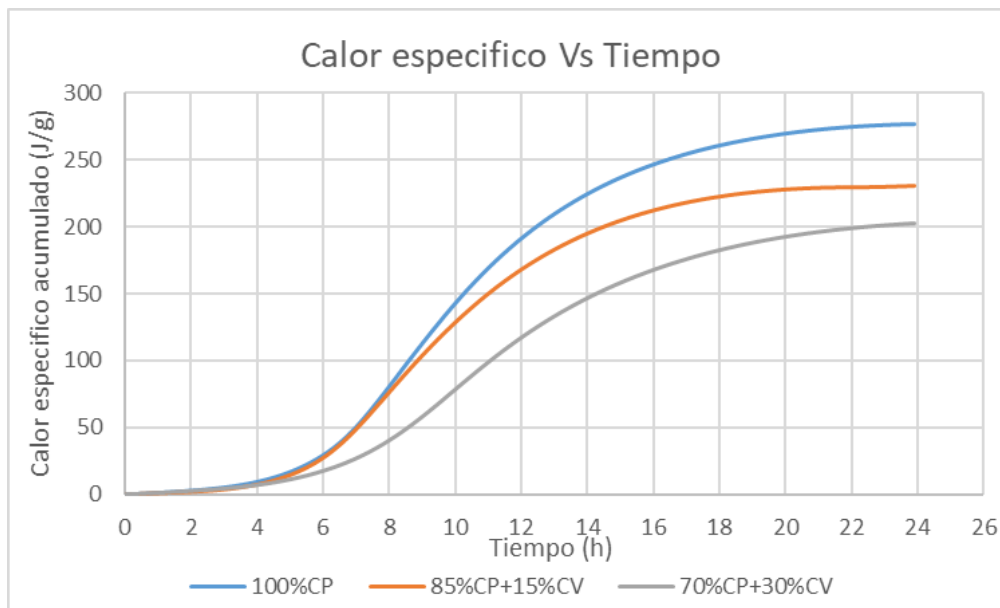


Figura. 4-22: Evolución del calor específico acumulado para las muestras en estudio en equipo HidroCemUN.



Como era de esperarse y como muchos autores lo han logrado identificar en sus estudios; la adición de ceniza volante al concreto influye considerablemente en la disminución de la temperatura máxima alcanzada por la mezcla y de igual forma en la cantidad de calor liberado. Como se puede ver en la figura 4-22, un reemplazo del 15% de ceniza volante, influyó en una disminución de 2.5 °C y 46.8 J/g con respecto a la temperatura máxima alcanzada por la muestra de 100% cemento. Por su parte, al aumentar el reemplazo de ceniza hasta el 30%, la disminución en temperatura fue de 10.7 °C y en cuanto calor acumulado total a las 24 h la disminución fue de 74.15 J/g.

Valores considerables si se trae a colación la importancia del control de la temperatura en el momento de fundir elementos masivos, en donde deficiencias en el control de esta propiedad puede llegar a causar problemas de fisuración en el elemento y por ende afectar la resistencia y estabilidad del mismo. Por otra parte, llama la atención que, aunque con el reemplazo de 15% de ceniza se logra una disminución en el pico de temperatura y un aumento en el tiempo del fraguado inicial, no se evidencia variación en la delta de tiempo necesario para alcanzar la temperatura máxima de la mezcla; situación diferente cuando

el reemplazo de ceniza es del 30%, en donde se logra que este pico de temperatura se corra en el tiempo por cerca de 1.7 horas respecto a la muestra con cemento al 100%.

Adicionalmente, en investigaciones realizadas por autores como Christensen (2001), Liu et al (2021) y Yousuf et al (2020), los cuales buscaron encontrar correlaciones entre los tiempos de fraguado y propiedades como la temperatura liberada, el calor específico, pulso ultrasónico, entre otras, que permitan de alguna forma más práctica conocer las condiciones de hidratación en la mezcla de concreto a partir de dichos ensayos. Se ha encontrado que algunos autores han hecho énfasis en la importancia de la interpretación de la pendiente de la curva del Δ temperatura durante sus primeras horas de hidratación. Identificándose marcadores que permiten resaltar los tiempos de “fraguado inicial” (F_i) y “fraguado final” (F_f) según la definición de la NTC 118 a partir de la gráfica $D\Delta\text{temp}/dt$.

En base a lo descrito anteriormente, en las figuras 4-23, 4-24 y 4-25 se realizan dichos análisis, con los cuales se evaluó la correlación con respecto a los tiempos de fraguado según la norma NTC 118, incluyendo también un límite definido como tiempo de inicio de pérdida de manejabilidad (P_m).

Figura. 4-23: Análisis grafico derivada del calor específico y derivada de Δ temperatura vs tiempo para la muestra 100% CP.

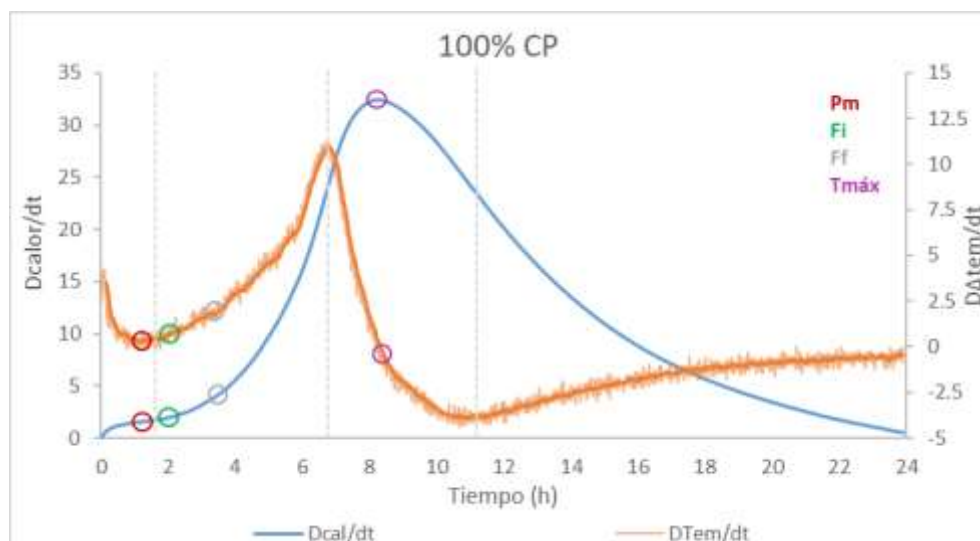


Figura. 4-24: Análisis grafico derivada del calor especifico y derivada de Δ temperatura vs tiempo para la muestra 85% CP + 15% CV.

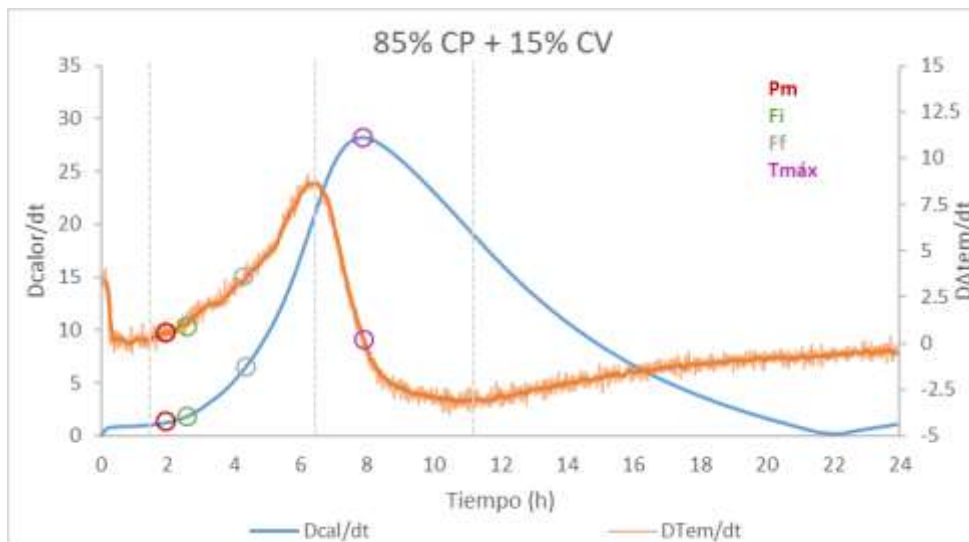
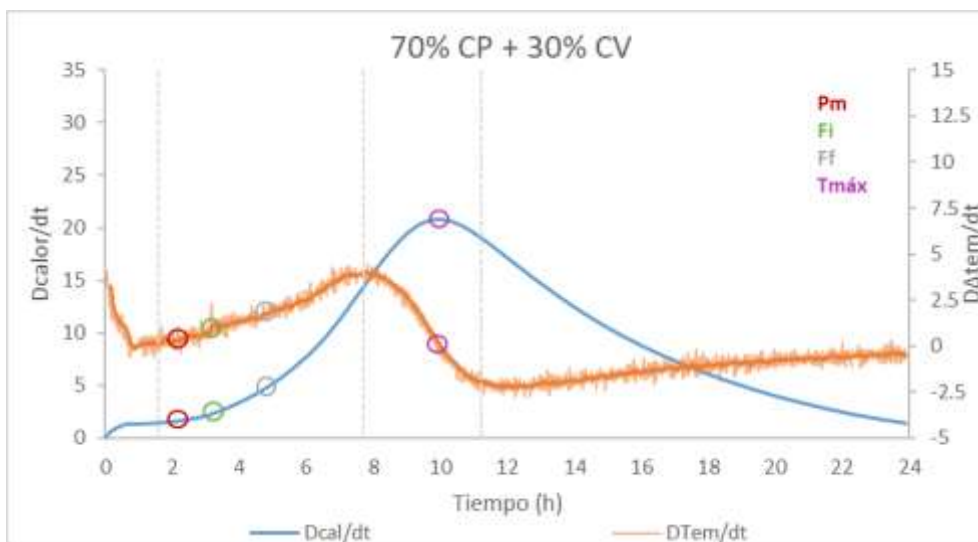


Figura. 4-25: Análisis grafico derivada del calor especifico y derivada de Δ temperatura vs tiempo para la muestra 70% CP y 30% CV.



En las gráficas presentadas anteriormente, se tienen los valores de tiempo en horas en el eje X; para el eje Y_1 se tiene $Dcalor/dt$ en ($Dcalor/dt$) y en el eje Y_2 $D\Delta temperatura/dt$ ($D\Delta tem/dt$). Gráficas en las que llama la atención principalmente la relación existente entre el tiempo Pm y el primer cambio de pendiente en las gráficas analizadas, y es que como lo menciona Christensen, estos cambios en los tiempos de fraguado de la mezcla más allá

de representar una alteración en la manejabilidad del material, representan un proceso que se da debido a diferentes afectaciones físico – químicas en la mezcla, las cuales pueden ser identificadas a travez de las diferentes propiedad del material, como lo es el caso de temperatura y el calor especifico liberado por la muestra.

4.3 Medición de la resistividad eléctrica mediante EIS

4.3.1 Calibración de los electrodos serigrafiados.

Cómo se describe en el numeral 3.2.1, para la calibración de la metodología de medición con los electrodos SPEs, se realizaron mediciones en paralelo de la resistencia eléctrica de una muestra de pasta durante 24 horas con una frecuencia de 10.000 Hz, utilizando las dos metodologías de medición (electrodos serigrafiados y barras de carbono). Esto con el fin de poder comparar los resultados de forma paralela y disminuir la variabilidad de los datos.

Para realizar el ensayo de calibración se dejó embebida una termocupla en cada una de las probetas (figura 4-27), con el fin de evaluar la evolución de la temperatura en cada una de las muestras y así poder descartar variables del ensayo. Adicionalmente, como se puede ver en la figura 4-26, se diseñaron 2 cajas de icopor como barreras aislantes de polipropileno en su interior, para así disminuir la afectación de la temperatura ambiente a las muestras en estudio (especialmente durante la noche).

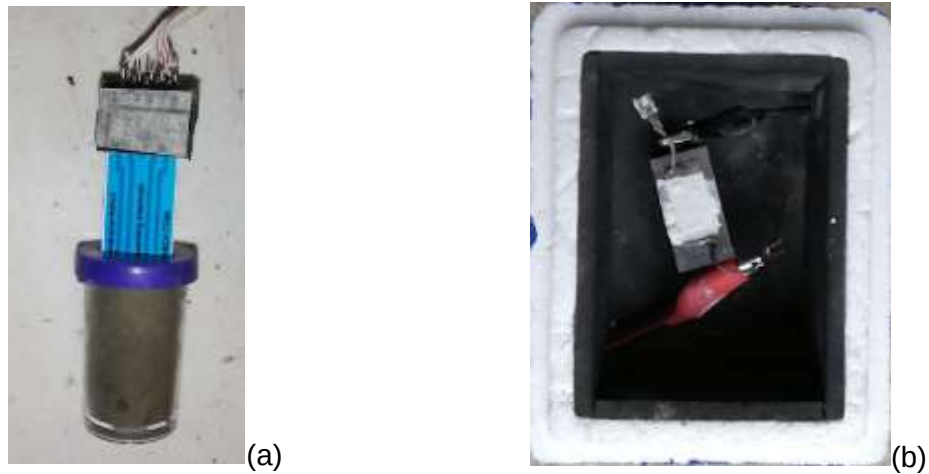
Figura. 4-26: Montaje de ensayo de resistividad en caja aislante a temperatura ambiente.



Nombre de la fuente: Propia

Es importante tener claro, que para realizar los ensayos se deben utilizar celdas de Faraday debido a que en estos se manejan pulsos eléctricos muy bajos, los cuales pueden ser fácilmente afectados por la interferencia eléctrica del exterior. Finalmente, el montaje de las nuestras muestras se ve reflejado en las figuras 4-27 a y b.

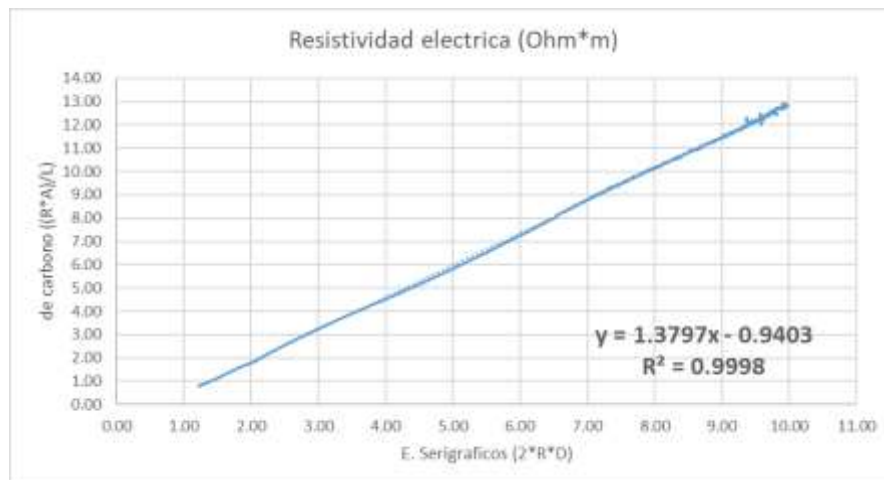
Figura. 4-27: Montaje de ensayos de resistividad eléctrica para electrodos serigrafiados (a) y para barras de carbono (b)



Nombre de la fuente: Propia

Igualmente, se debe tener en cuenta que en los ensayos de resistividad eléctrica con los electrodos serigrafiados, se utilizó el montaje de un modelo de 3 puntas, a diferencia de los ensayos de barras de carbono para los cuales se utilizó el modelo de 2 puntas.

Figura. 4-28: Grafica de correlación entre ensayo de resistividad a partir de ensayos con electrodos serigrafiados y barras de carbono.



Adicionalmente, con el fin de monitorear el correcto proceso de hidratación de las muestras, se realizó el seguimiento de la temperatura de dichos ensayos. Aquí se pudo observar que, aunque las muestras tuvieron un delta de temperatura entre ellas de 2 °C, el cual estuvo asociado a condiciones de sitio en el laboratorio y al volumen de la muestra ensayada, se puede ver que ambas probetas presentaron una tendencia de temperatura con un comportamiento similar durante todo el ensayo.

Figura. 4-29: Grafica de comparación de temperatura entre ensayo de resistividad a partir de ensayos con electrodos serigrafiados y barras de carbono.



Sin embargo, para el ejercicio presentado en la figura 4-28, la resistividad eléctrica del método de las barras de carbono fue calculado a partir de la ecuación 2.8 y en el caso de la resistividad para los electrodos serigrafiados se calculó a partir de la ecuación 2.9. Resultados que cómo se puede ver en la figura 4-28, presentan una buena correlación entre las lecturas tomadas por cada uno de los métodos. Sin embargo, aunque la correlación entre valores es del 0.9998, existe un desfase en las lecturas, por lo que se hace necesario que los valores de resistividad para el método de los electros SPes sean ajustado.

Para definir dicho ajuste, se partirá de la ecuación de correlación en las dos metodologías (ecu. 4.1)

$$\rho_{Ajustado} = (1.3797 * \rho) - 0.9403 \quad (4.1)$$

Reemplazando la ecuación la ecuación 2.8 en la ecuación 4.1 tendremos:

$$\rho_{Austado} = (1.3797 * (2 * R * D)) - 0.9403 \quad (4.2)$$

En donde el valor de D para todos los electrodos SPE será igual a 0.002 m, la cual, reemplazada en la ecuación 4.2 finalmente nos dará:

$$\rho_{Austado} = (5.5188 * 10^{-3} R) - 0.9403 \quad (4.3)$$

Una vez se logra obtener la ecuación 4.3, se puede ver en la figura 4-28 que esta ecuación presenta una mejor tendencia, por lo que la ecuación 4.3 fue definida como la de mejores resultados de resistividad eléctrica obtenidos a partir del uso de electrodos SPE.

Figura. 4-30: Resistividad eléctrica vs tiempo para los diferentes métodos de medición y sus respectivas ecuaciones.



Finalmente, una vez definida la ecuación para la evaluación de la resistividad eléctrica a partir de los electrodos SPE, se realizó la evaluación del % de error de las lecturas, comparado con el ensayo realizado mediante el método de las barras de carbono, en donde se puede ver en la figura 4-31 que este valor no supera en promedio el 1.1% de error, lo cual da garantía y seguridad de los resultados obtenidos mediante el uso de los electrodos SPE.

Figura. 4-31: Variación del % de error en el tiempo para la resistividad eléctrica medida con electrodos SPE.



4.3.2 Mediciones en edades tempranas con frecuencia constante

Para el inicio de las mediciones de la resistividad eléctrica de las muestras de concreto en estado fresco, inicialmente y como se menciona en el numeral 3.2.2, se realizaron mediciones con diferentes frecuencias de CA con el fin de evaluar bajo cuál de estas mediciones se lograba obtener lecturas en las que la impedancia dependiese en menor medida de la parte imaginaria del ensayo. Basados en la literatura se estudiaron frecuencias de 1000, 10000, 50000 y 100000Hz para cada una de las muestras, obteniendo los siguientes resultados.

4.3.2.1 Análisis de curvas de resistividad bulk.

Figura. 4-32: Resistividad eléctrica para diferentes frecuencias de CA: (a) muestra M1 (100%CP), (b) muestra M2 (85%CP+15%CV) y (c) muestra M3 (70%CP+30%CV).



(a)



(b)



(c)

Inicialmente, se logra comprobar que según indican autores como Lizarazo et al (2014), Higuera Flórez (2016) y Chi et al (2019) cuando se desea evaluar la resistividad eléctrica de una pasta de cemento, es la frecuencia de 10000Hz la que permite obtener mejores y más ajustados resultados. Esto se puede observar en la figura 4-32 a, b y c, donde para esta frecuencia los resultados de resistividad de cada una de las muestras fueron más estables. Por esta razón, en el análisis de las primeras 24 horas la resistividad bulk, conductividad y capacitancia se medirá bajo esta frecuencia de CA.

Con base en lo mencionado anteriormente, se realizará la comparación del desarrollo de la resistividad eléctrica global de las muestras evaluadas (figura 4-33), donde se observa que aunque las tres muestras presentan una tendencia similar, la muestra conformada solamente por cemento (M1), inicialmente y hasta un tiempo cercano a las 3.5 horas, presenta una menor resistividad que las muestras adicionadas con ceniza volante (M2 y M3) (figura 4-34). Lo anterior puede estar sustentado desde un punto de vista tanto físico (estructura interna de la matriz de la mezcla) como químico (comportamiento químico de la mezcla).

En este orden de ideas, para las primeras 3.5 horas de hidratación de las muestras se tiene desde el punto de vista físico la hipótesis de que las muestras adicionadas con ceniza volante llegan a presentar una matriz más densa, asociado esto con la diferencia del diámetro con respecto al cemento, lo que genera que, aunque la matriz este saturada en su totalidad, se presenten menos caminos de conducción. Sumado esto, al inicio del proceso de hidratación de las partículas de ceniza volante, donde estas absorben parte de los iones de Na^+ , K^+ , OH^- y H_3O^+ presentes en la solución de poros para su proceso de hidratación (procesos químicos), afectando la concentración iónica de la solución y por ende la capacidad de conducción de carga eléctrica a través de la matriz.

De esta forma, se estima que a través de la matriz de la muestra realizada sin el uso adición, se da la conducción eléctrica con mayor facilidad gracias a que está presenta una concentración de iones más alta en la solución de poros, al igual que mayor número de caminos de conducción continuos para el transporte de estos iones (Bentz, D; Clifton, J; Ferraris, C; Garboczi, 1999).

Figura. 4-33: Variación de la resistividad eléctrica en 24h para las muestras estudiadas.

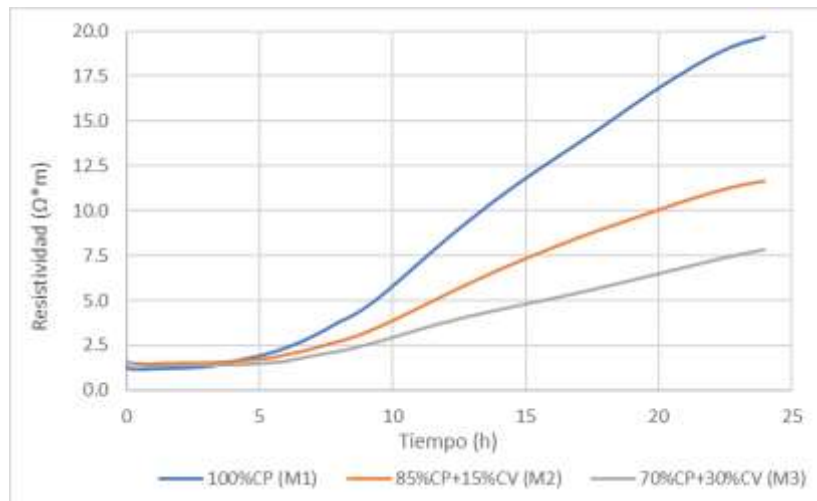
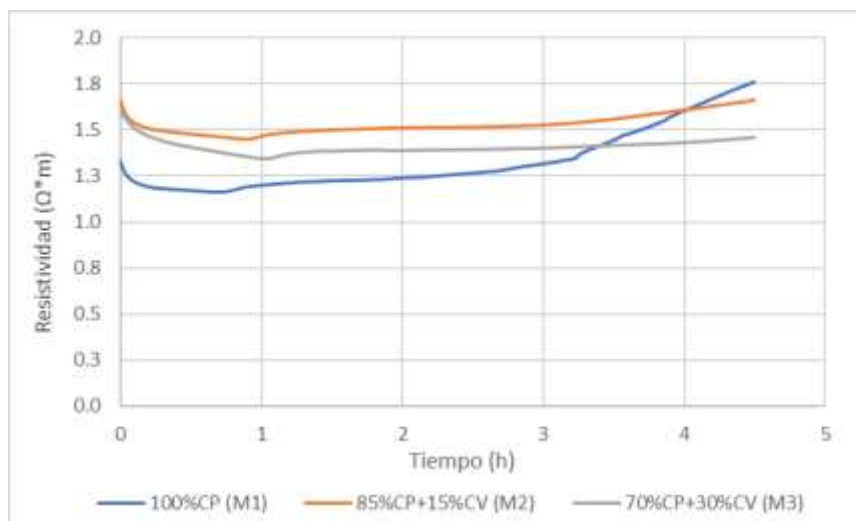


Figura. 4-34: Variación de la resistividad eléctrica en fase inicial para las muestras estudiadas.



Sin embargo, algunos autores como Chi et al (2019), Hu et al. (2019) y Suryanto et al. (2016), observaron los mismos patrones no solo durante las primeras horas de la hidratación, ya que para ellos también fue evidente que una vez superado este periodo de las 3.5 horas, las muestras adicionadas con ceniza volante no presentan variaciones significativas en las tasas de aumento de la resistividad, cómo si pasa en las muestras conformadas con 100% cemento, las cuales a las 24 horas pueden presentar valores de resistividad de hasta dos veces el valor logrado por una muestra adicionada con ceniza

volante. Razón que en la literatura ha sido relacionada con un retraso en diferentes niveles ya que la ceniza volante no hidráulica solo actúa como un "relleno", en lugar de participar en el proceso de hidratación durante las edades tempranas de la muestra (Chi et al., 2019).

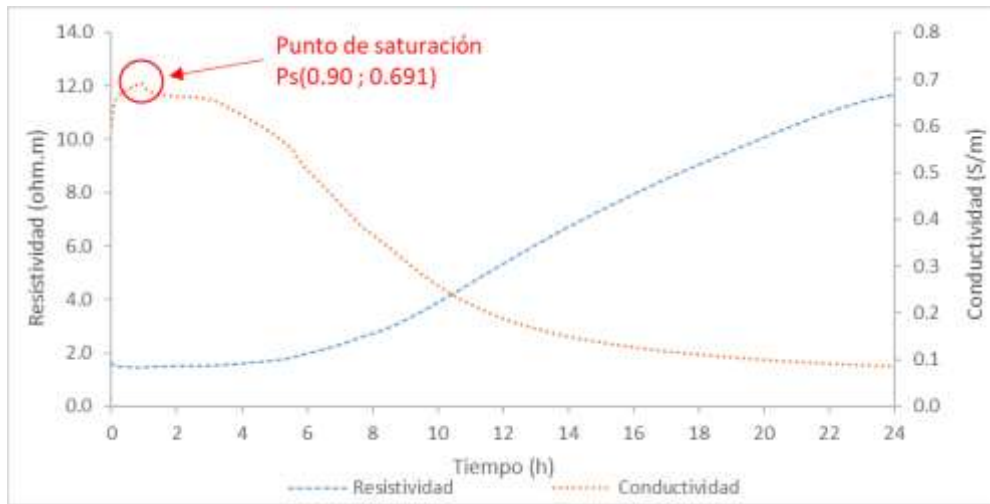
Sin embargo, esta hipótesis también puede ser analizada desde la incidencia del punto de saturación (Ps) de cada una de las mezclas, ya que según definiciones realizadas por Yousuf & Xiaosheng, (2020) y Yousuf et al. (2020), en este momento la matriz de cemento está conformada por poros completamente conectados con un grado de saturación del 100%, donde dicho punto se logra identificar de forma más sencilla si analizamos las gráficas de conductividad de las muestras (figura 4-35 a, b y c). Aquí, para definir el principio del punto Ps se parte de que en dicho momento el contenido iónico es el que gobierna el desarrollo de la resistividad y/o conductividad debido al aumento gradual de iones en la solución; efecto que es causado por el proceso persistente de disolución de los minerales de clínker, lo cual genera que cuando la solución esta sobresaturada principalmente por iones CH, los hidratos comiencen a precipitar sobre la superficie de las partículas de cemento (inicio de la pérdida de manejabilidad) (Tang et al., 2014).

Concluyendo que en estos procesos a mayor cantidad de cemento liberarán más iones, lo cual hará que el punto Ps se presente en un periodo de tiempo más corto.

Figura. 4-35: Variación de la resistividad y la conductividad eléctrica de las muestras estudiadas durante las primeras 24 horas de hidratación. (a) muestra M1, (b) muestras M2 y (c) muestras M2.



(a)



(b)



(c)

Por esta razón, basados en lo descrito anteriormente y lo representando en las figuras 4-35 a, b y c, se concluye que el punto Ps es equivalente al momento en que la conductividad eléctrica de la muestra presenta su máximo valor. Lo cual, también representa el inicio de la precipitación de los productos de hidratación “Pórtlandita”, pudiendo considerarse que es en este punto donde se da el inicio de los procesos de hidratación de la muestra si se consideran la pendiente descendente que representa en la gráfica (Chi et al., 2019).

Finalmente, en relación con lo observado en la evaluación de los tiempos de fraguado (tabla 4-4) y las hipótesis planteadas para la variación de la resistividad eléctrica (figura 4-33 y 4-34), se puede ver cómo es la muestra M1 (100% cemento) a pesar de presentar los valores de conductividad más altos en el punto de saturación, la muestra que desarrolla el Ps en un menor tiempo (0.23 horas antes que la muestra M2 y 0.34 horas que la muestra M3); por lo que considerando lo anteriormente mencionado, hará que los procesos de precipitación de los productos de la hidratación se inicien en un menor tiempo. Esto se puede identificar fácilmente en la figura 4-34, en donde para la conductividad de la muestra M1, cuando $t = 3.125$ horas, se presenta un cambio considerable en la pendiente de la línea, lo que nos indica un rápido inicio de ganancia de resistividad en la muestra (cabe resaltar que para la muestra M1 se obtuvo que el F_r se dio cuando $t=3.67$ horas).

4.3.2.2 Modelo conceptual para el estudio de la hidratación de las pastas de cemento a partir de la resistividad bulk de la muestra.

Una vez hecho el análisis y revisión de las curvas de resistividad y conductividad bulk de las muestras en estudio, y para el seguimiento de uno de los principales objetivos del presente trabajo (desarrollo del modelo conceptual de hidratación), se realizó a partir de estos datos una correlación que permite identificar las diferentes etapas de fraguado de una pasta de cemento en función de la variación de la capacidad para la conducción eléctrica.

Para este análisis se partió de las investigaciones realizadas por autores como (Li et al., 2003b; Shen et al., 2016; Tang et al., 2014; Yim et al., 2017; Yousuf et al., 2020; Yousuf & Xiaosheng, 2020), quienes en sus estudios logran definir diferentes modelos que permiten a partir de la variación de las pendientes de la resistividad bulk en el tiempo, encontrar una correlación para identificar los cambios en las etapas de hidratación de las muestras analizadas (tiempos de fraguado). Por lo que para nuestro análisis nos centraremos en la identificación de los puntos críticos propuestos por Yousuf et al. (2020) los cuales son definidos como:

Punto crítico P1: este punto hace referencia al valor más bajo de resistividad alcanzado durante el periodo de hidratación y ocurre cuando la solución de poros de la pasta de

cemento alcanza la sobresaturación con respecto al CH. Es por esta razón que en la literatura se afirma que este punto es equivalente a la resistividad eléctrica en el punto de saturación (ver figura 4-35 a, b y c), lo cual corresponde al tiempo de inicio de la precipitación de los productos de hidratación. Es importante resaltar que para este momento las pastas están aún en estado fluido y a punto de entrar en estado plástico. Finalmente, se estima que hasta el transcurso de dicho punto, la resistividad desarrollada por la muestra está controlada principalmente por las propiedades químicas (Tang et al., 2014).

Punto crítico P2: en cuanto al punto crítico P2, se tiene que una vez se inicia el refinamiento gradual de la porosidad capilar en el sistema y por ende la formación de productos de hidratación, la pasta de cemento inicia su proceso de pérdida de la consistencia, dando fin a su fase en estado plástico. Este cambio en el estado de solidificación de la pasta se debe principalmente al aumento en la proporción de sólidos conectados en la matriz, estimándose por lo tanto que el inicio del fraguado ocurrirá cuando se forme la primera vía conectada de hidratos (Li et al., 2003b).

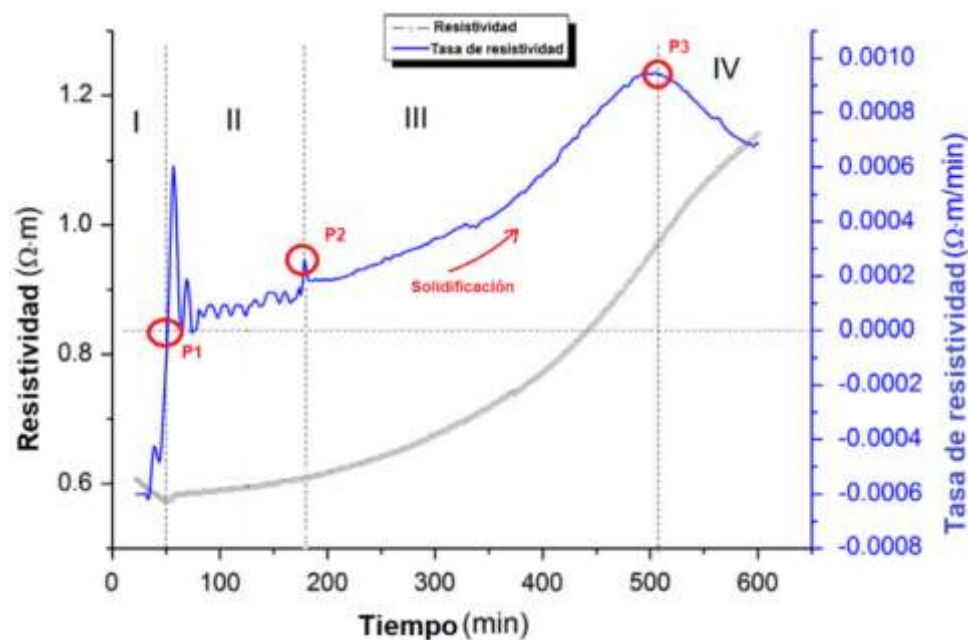
En este punto, la rápida formación de gel C-S-H durante el período acelera la tasa de desarrollo de la resistividad, llevando a que se pueda relacionar el punto crítico P2 con la incidencia del tiempo de fraguado final de las pastas (F_f determinado mediante el método de las agujas de Vicat); considerándose como el inicio de la aceleración de la resistividad debido a la percolación de los hidratos (se estima que en este punto comienza a tomar relevancia la parte física o el esqueleto de la matriz en la conductividad eléctrica). Finalmente, marcando el tiempo del punto crítico P2 en la curva de velocidad, se puede detectar el rango de fraguado de la pasta de cemento, ya que, durante el desarrollo de este periodo, la pasta pasa de un estado plástico a convertirse en un material rígido (Yousuf et al., 2020).

Punto crítico P3: finalmente, para el punto P3 se tiene que con el crecimiento continuo de los productos de hidratación en la matriz, los poros inician un proceso para des percolarse y así la pasta evoluciona en un proceso de ganancia de resistencia (paso final al estado sólido); por lo que este punto nos indica el momento en que la pasta comienza su proceso de ganancia de resistencia. Durante este periodo la percolación de los sólidos y la

expansión de los cúmulos expansivos gobiernan el proceso de fraguado; lo cual se ve reflejado en un nuevo cambio en la tendencia de la curva de resistividad, desarrollando un proceso de continuo aumento de las propiedades resistivas (Yousuf et al., 2020).

Por lo tanto, en consideración de lo anterior y según los argumentado por Yousuf et al. (2020), el período de fraguado de una pasta de cemento puede definirse como el tiempo para alcanzar el punto P2 en la curva de tasa de desarrollo de resistividad, el cual ocurre aproximadamente en el medio de los puntos P1 y P3.

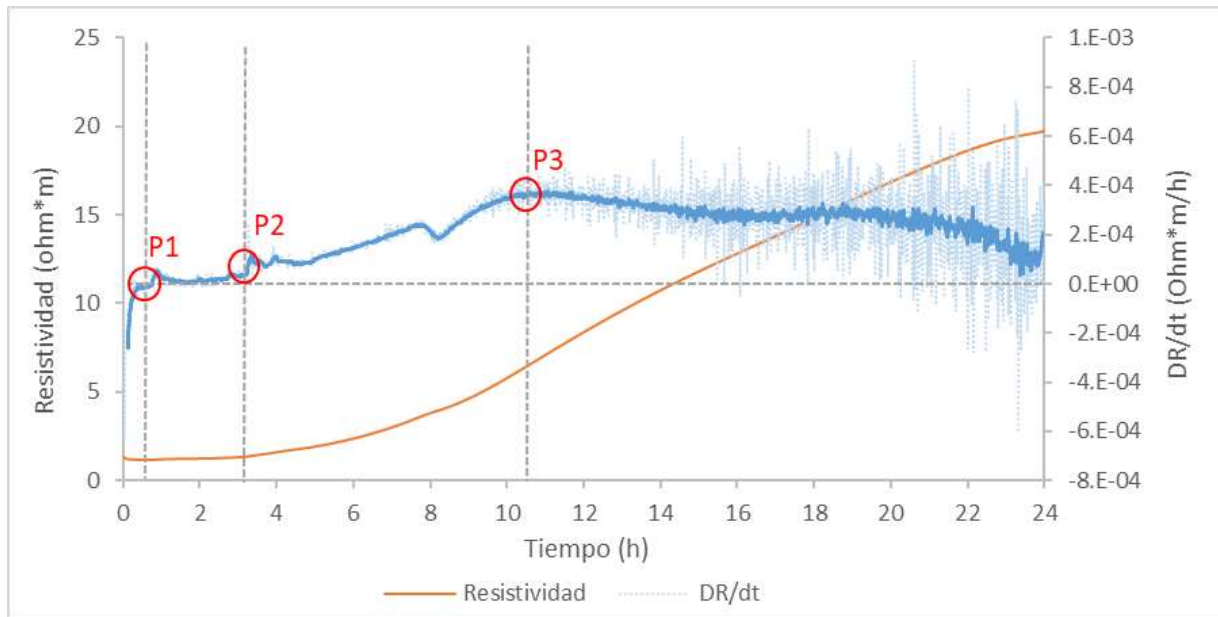
Figura. 4-36. Identificación de los puntos críticos en curva de derivada de la resistividad.



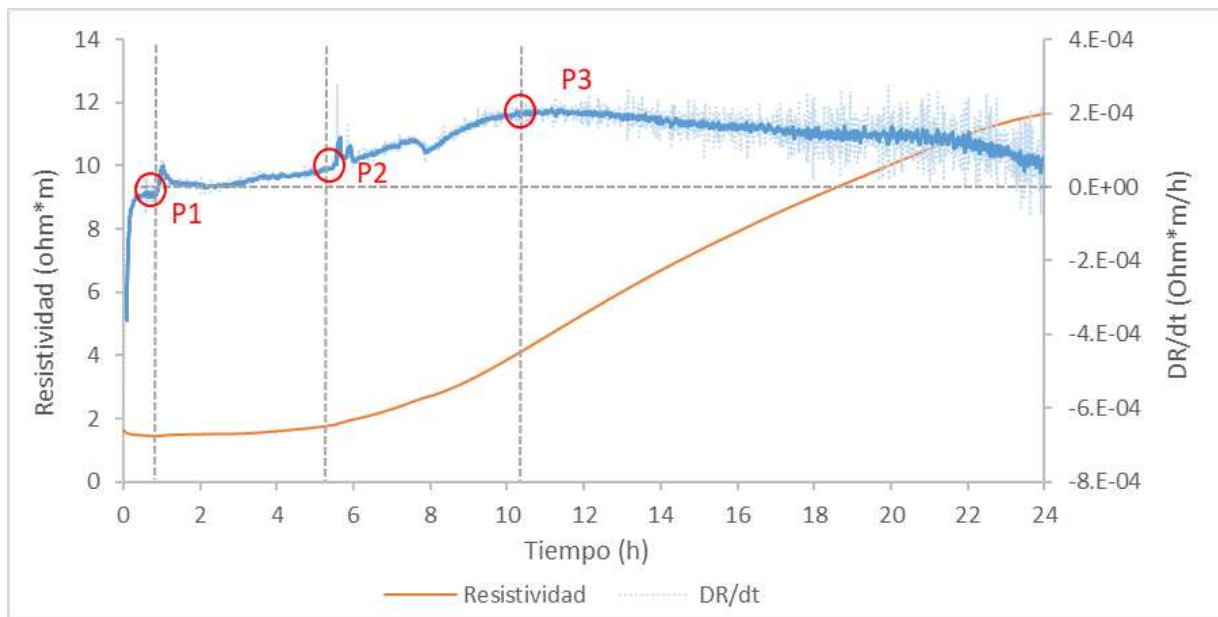
Nombre de la fuente: adaptado de (Tang et al., 2014)

Finalmente, en base a lo descrito anteriormente y lo representado en la figura 4-36 a, b y c, se procede con el análisis de los puntos críticos en las gráficas de derivada de la resistividad de las 3 muestras objeto del presente trabajo (M1, M2 y M3).

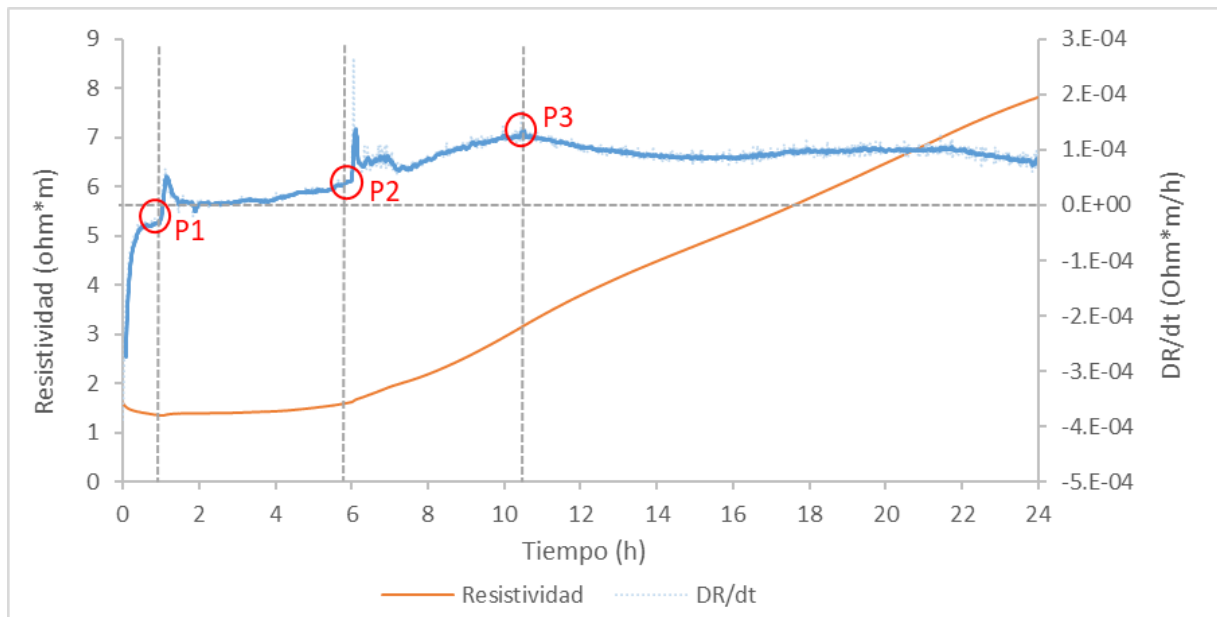
Figura. 4-37: Análisis de los puntos críticos sobre curva de derivada de la resistividad con respecto al tiempo. (a) muestra M1, (b) muestras M2 y (c) muestras M3.



(a)



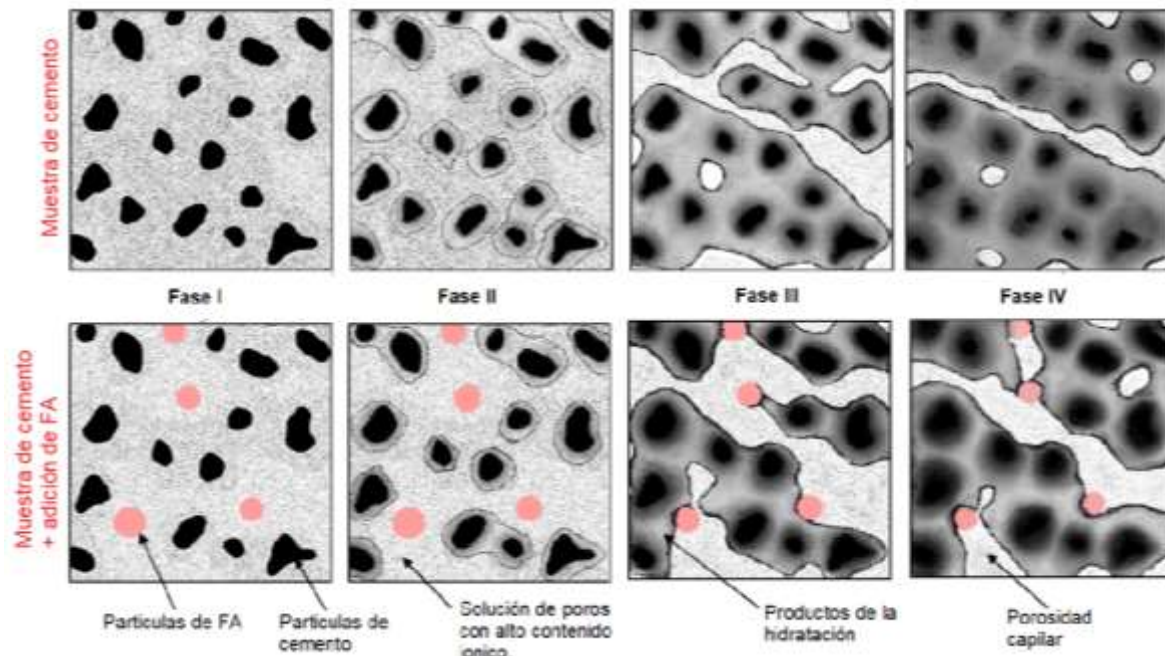
(b)



(c)

Como se puede ver en las figuras 4-37 a, b y c, para las muestras estudiadas se pueden identificar los tres puntos críticos de acuerdo con lo indicado en la literatura, puntos que representan tres cambios significativos en la evolución de la resistividad de las muestras. Sin embargo, para una mejor interpretación de los tiempos definidos por los puntos críticos, la curva se puede dividir en cuatro fases (figura 4-38), durante las cuales se desarrollan diferentes procesos de hidratación que pueden llevar a comprender de forma más sencilla la manera en la que se da la hidratación de las pastas y de igual forma el efecto de la adición de la FA en sus diferentes dosificaciones. Se debe mencionar que autores como Chi et al. (2019) llegaron a relacionar estas 4 fases con las etapas relacionadas a las curvas de flujo de calor producto de la hidratación del cemento y que fueron descritas en el capítulo 2.4 del presente documento.

Figura. 4-38: Desarrollo de la hidratación en las diferentes fases definidas para muestras de cemento y adicionadas con FA.



Nombre de la fuente: adaptado de (Yousuf & Xiaosheng, 2020)

Fase 1: esta fase viene definida hasta la obtención del punto P1 (punto de saturación de la mezcla) y está marcada por el inicio del contacto del cemento con el agua, en donde los granos de cemento comienzan a disolverse y se inicia la liberación de iones al medio acuoso (por ejemplo, OH^- , K^+ , Na^{2+} , SO_4^{2-} y Ca^{2+}). Esto genera cambios en el agua, la cual se convierte en una solución acuosa con una concentración específica de iones que es controlada mediante la relación a/c utilizada en el diseño y la composición química del cemento.

Durante este periodo la concentración de iones aumenta constantemente por el proceso continuo de disolución de los granos de cemento hasta sobresaturar la solución con respecto a CH. Es importante resaltar que, durante el desarrollo de esta fase, los poros capilares permanecen completamente conectados, donde el sistema contiene una cantidad escasa de productos de hidratación. Por lo tanto, se cree que durante la fase líquida la resistividad global está gobernada por la concentración iónica presente en la solución de poros hasta llegar al punto P1 (Yousuf et al., 2020).

Finalmente, el punto P1 ocurre como resultado de un cambio en la tendencia de la resistividad de creciente a decreciente, lo que indica el inicio de la precipitación de los primeros productos de la hidratación, marcando así el comienzo de la precipitación de hidratos en la superficie de las partículas de cemento después de la sobresaturación con respecto a la fase de disolución. Más adelante estos nuevos hidratos precipitados actúan como aislante de la electricidad y provocaran que la resistividad deje de disminuir y por ende comience a aumentar (Yousuf et al., 2020).

Fase 2: una vez alcanzado el punto P1, la curva de resistividad tiende a nivelarse, esto a causa de que se entra en una etapa donde existe un equilibrio dinámico entre la disolución de iones y la precipitación de hidratos; destacando que la formación de hidratos es un proceso que consume iones y afecta la conductividad del sistema. Aquí el gel C–S–H y CH con otros hidratos ocupan continuamente los espacios entre las partículas de cemento a través de la reacción, disminuyendo los caminos conductores de la corriente eléctrica y provocando un aumento de la resistividad. Cabe señalar que tanto el fraguado inicial como el final del cemento ocurrirán durante la tendencia de aceleración de la resistividad.

Adicionalmente, durante esta etapa de crecimiento continuo de productos, se atrapan los poros capilares y comienzan a formarse pequeños poros, los cuales son afectados por el aumento en la fracción de volumen de sólidos, disminuyendo significativamente la conectividad de estos poros que rigen el desarrollo de la resistividad eléctrica del cemento; como se mencionó en la sección anterior, la tendencia de la resistividad cambió nuevamente en un tiempo de hidratación particular de un estado ligeramente estático a una fase de aceleración, marcando otro punto que se produce en la curva de tasa de resistividad eléctrica (punto P2) (Yousuf et al., 2020). Es importante resaltar que una vez se entra en el proceso de precipitación de productos de la hidratación, es el C–S–H el principal producto de hidratación que contribuye al endurecimiento de la pasta y al desarrollo de las propiedades mecánicas del cemento.

Por último, llama la atención cómo en esta fase comienzan a ser más notorio los efectos de la inclusión de la FA en la mezcla, lo cual ya presentaba sus primeros indicios en el tiempo requerido para lograr el punto de saturación de cada mezcla. Sin embargo, es en esta fase donde se presenta una de las mayores implicaciones, la cual está relacionada con el aumento en el periodo de tiempo que dura el desarrollo de la fase II; y es que para

la mezcla M1, la duración de la fase 2 es de aproximadamente 3.19 horas, tiempo que aumenta hasta 4.55 horas en el caso de M2 y hasta 4.84 horas en el caso de M3.

Fase 3: al llegar al punto P2 la pasta de cemento entra en estado sólido, esto gracias a que continua la expansión de cúmulos sólidos llegando a formar una columna vertebral sólida en la matriz de la pasta, lo que permite que el sistema pueda llegar a soportar cargas mecánicas. Lo anterior viene relacionado con la percolación de geles C–S–H para compensar la liberación de iones por transformación AFt–AFm, lo que genera que la resistividad continúe aumentando después del punto P2 y la pasta como material endurecido continúe ganando resistencia a través de la reacción continua agua-cemento. La tasa de resistividad creciente que se muestra en la figura 4.37 a, b y c continúa acelerando hasta llegar al punto P3 donde se desaceleró y comienza un ligero proceso de estabilización de esta (Yousuf et al., 2020).

Fase 4: como se mencionó anteriormente, una vez se llega al punto P3, la tasa de resistividad presenta una disminución en la aceleración, lo cual viene acompañado de un decrecimiento. Para especular sobre la tendencia decreciente de la tasa de resistividad después del punto P3, se deben considerar factores tanto físicos como químicos. Donde según algunos autores, durante la etapa de endurecimiento de la pasta de cemento, la etringita (AFt) se transforma en monosulfato (AFm) debido al consumo de yeso, esta transformación tiene como resultado la liberación de iones Ca^{+2} y SO_4^{2-} , lo que aumenta la conductividad del sistema y genera que se presente una leve disminución en la resistividad. Mientras tanto, la hidratación de C_3S continúa, filtrando el camino del gel C–S–H a través de los espacios entre los granos de cemento.

Finalmente, la estrangulación de los poros en un tiempo de hidratación y que viene asociado al impacto de los hidratos en la superficie de las partículas de cemento, al control de la difusión de iones en los hidratos y a las partículas de cemento sin reaccionar, desaceleran la velocidad de reacción y por ende la resistividad. En esta fase, la fracción de volumen de los sólidos aumentara hasta que el sistema carece de espacio adecuado y fuentes de reactivos para el crecimiento adicional de hidratos, lo que dificulta la evolución de la resistividad del sistema (Yousuf et al., 2020).

Estas 4 fases mencionadas anteriormente, nos pueden llevar a concluir que la vía conductora de la corriente eléctrica a través de la pasta de cemento está dominada por la resistividad de las fases líquida, así como por la fracción de volumen y la conectividad de cada una de las fases mencionadas. Durante el tiempo de hidratación, la resistividad de la fase líquida disminuye hasta el punto de saturación, mientras que la fracción de volumen y la conectividad fueron minimizadas por la percolación de los productos de hidratación. Una vez se comienza a disminuir la conectividad de los poros, reduciéndose de igual forma los poros capilares en pequeños poros a través de la hidratación del cemento, se comienza a generar mayores dificultades para que la corriente pase por completo en el sistema, representándose esto un aumento de la resistividad ya que se forma el primer camino de un sólido conectado, evidenciándose un aumento de la resistividad debido al aumento en el número de aisladores.

4.3.3 Mediciones en edades tardías con múltiples frecuencias

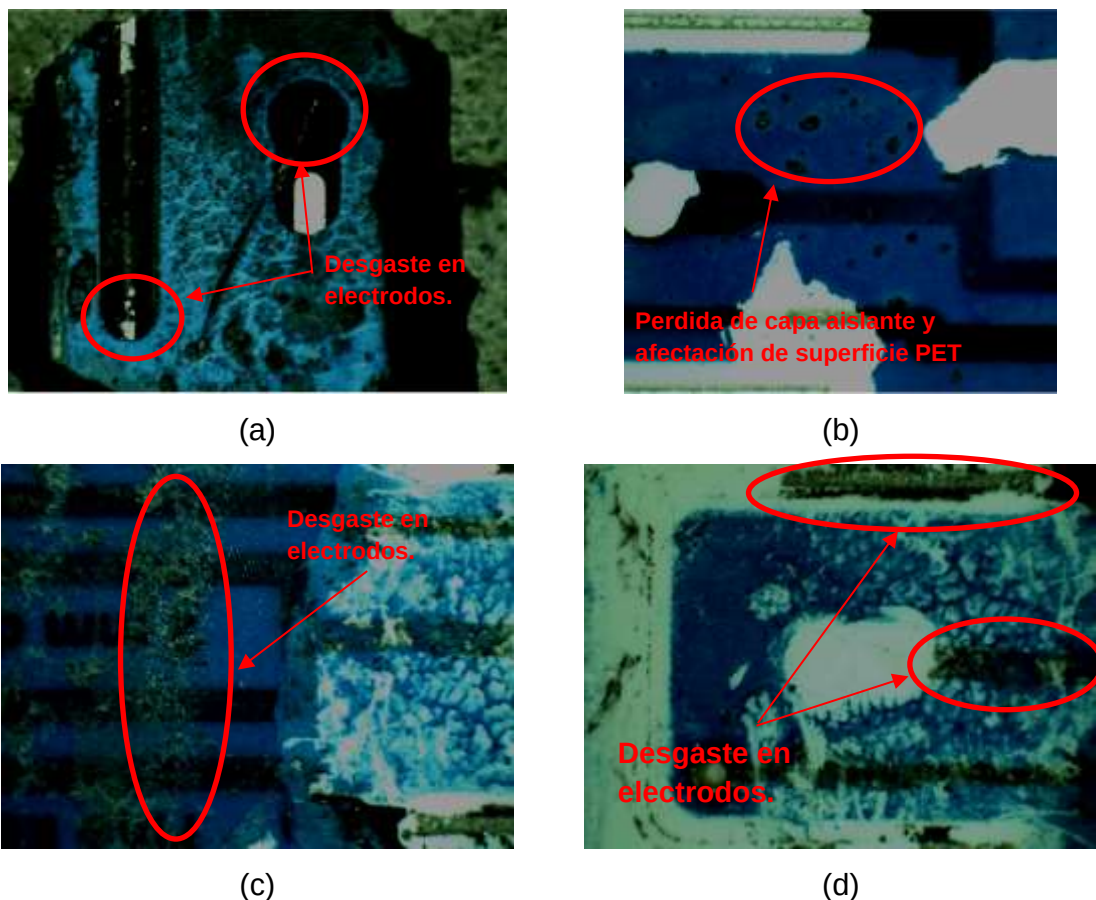
En cuanto al estudio de la resistividad eléctrica en edades tardías, es importante mencionar que estas mediciones se ven afectada principalmente por dos parámetros: la resistividad eléctrica de la fase líquida y el desarrollo de los poros a través del sistema, como lo aclara la relación empírica de la ley de Archie (Archie, 2003). Por lo tanto, las mediciones de la resistividad eléctrica de las pastas de cemento pueden considerarse un método apropiado para observar el desarrollo de la microestructura de un sistema de pasta de cemento (Yousuf et al., 2017). Por esta razón para la presente investigación, se desarrollaron mediciones de resistividad de las tres muestras definidas con anterioridad en edades de 0 días, 0.125 días, 1 día, 3 días, 7 días, 14 días, 21 días, 28 días, 42 días y 56 días, las cuales serán analizadas principalmente a partir de los diagramas de Nyquist.

Sin embargo, para el presente ejercicio se realizaron mediciones utilizando electrodos serigrafiados impresos en 3D (referencia RRPE1001C) y el método tradicional de electrodos de barras de carbono. Presentándose como novedad que una vez pasados 3 días, se comienzan a evidenciar algunas inconsistencias en las mediciones tomadas con los electrodos serigrafiados (figura 4-40); novedad principalmente atribuida a que para el presente trabajo de investigación fue utilizada una referencia de electrodos fabricados en un sustrato de tereftalato de polietileno con un recubrimiento dieléctrico, materiales que

están principalmente diseñados para el trabajo en un medio acuoso, por lo que una vez sometidos al ambiente alcalino de las mezclas de cemento, se comienzan a presentar desgaste en sus componentes (figura 4-39).

Con base en lo anterior, para este tipo de mediciones se recomienda utilizar electrodos fabricados sobre sustratos cerámicos como los ofertados por PINERESERCH bajo las referencias RRPE2001PT o RRPE2001AU, los cuales son equipos que presentan una mayor resistencia a ambientes agresivos o de temperaturas altas, como lo pueden ser las mezclas de cemento.

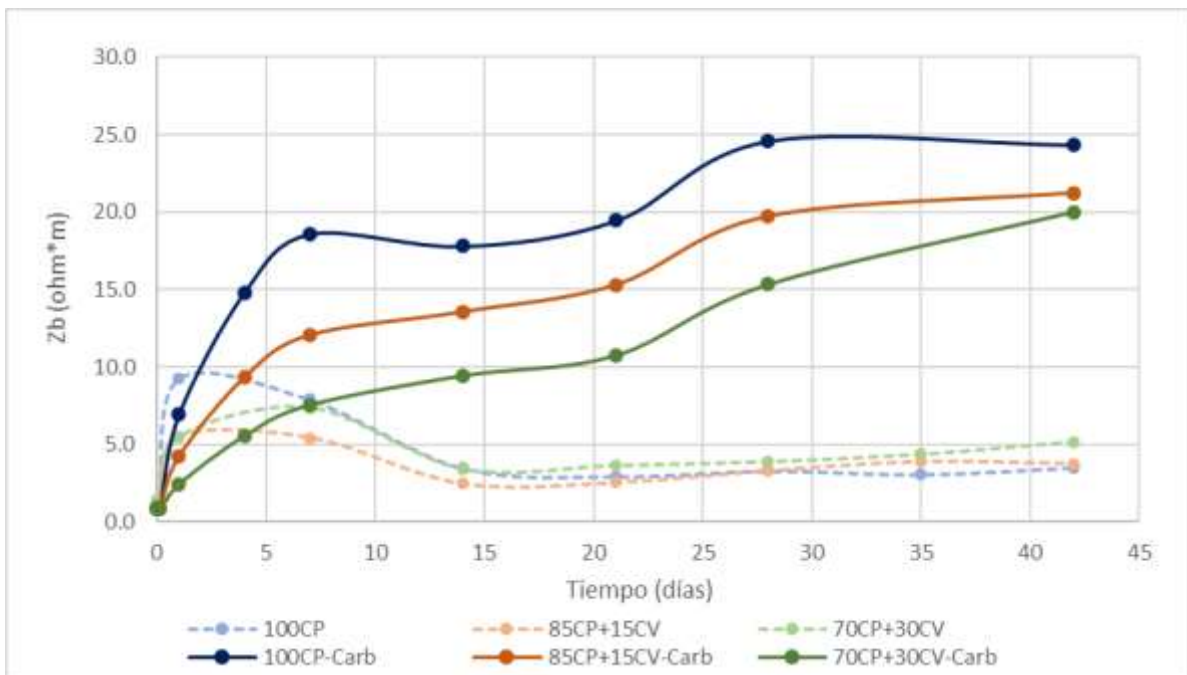
Figura. 4-39: Deterioro en electrodos serigrafiados una vez pasados 45 días de realizadas las mezclas de concreto.



Como se puede evidenciar en la figura 4-40, donde se realiza la comparación de las mediciones de R_b para las lecturas tomadas con electrodos serigrafiados y el método

tradicional de los electrodos de carbón. Se evidencia que el caso de los electrodos SPE una vez tomadas las lecturas a 3 días, se presentan una disminución atípica en los valores de Rb de las muestras, lo cual va en dirección opuesta de las hipótesis y teorías estudiadas por los autores citados anteriormente, quienes en sus investigaciones han observado comportamientos opuestos. Esta hipótesis se logra comprobar una vez realizadas las mediciones con los electrodos de carbón donde se observa un constante aumento de Rb a medida que aumenta la edad de las muestras.

Figura. 4-40: Comparación de mediciones de Rb para lecturas tomadas con electrodos serigráficos y electrodos de carbono.



Una vez observados los datos presentados en la figura 4-40, es importante resaltar que más allá de que la muestra realizada con 100% cemento (M1) presentan valores de Rb más altos que las demás muestras estudiadas (M2 y M3), se tiene los valores de Rb para M1 una vez superadas las edades de 28, no continúan desarrollando evolución en su resistividad real. Lo cual puede ser un indicio de que la muestras ha alcanzado un punto en donde la microestructura interna ya ha terminado su proceso de evolución, comportamiento que se puede asociar al uso de cementos tipo ART, los cuales están conformados en su mayoría por clinker.

Sin embargo, cuando se analizan las mezclas que fueron adicionadas con ceniza volante, se observa en las lecturas tomadas a 42 días una tendencia de incremento en los valores de R_b , lo cual según las hipótesis planteadas nos indica que aún se continúan dando procesos de cambio dentro de la matriz cementante. Comportamiento que viene relacionado con lo ya se ha mencionado a lo largo del presente trabajo y que basados en investigaciones realizadas por otros autores, nos confirma que los procesos de hidratación de las mezclas adicionadas con FA suelen desarrollarse de forma más lenta debido a la naturalidad de la hidratación de las cenizas; teniendo casos de referencia como las investigaciones realizadas por Fonseca (2016) quien a partir de mediciones de penetración del ion cloruro superiores a los 300 días, logra identificar cambios dentro de la microestructura de las mezclas adicionadas con ceniza volante.

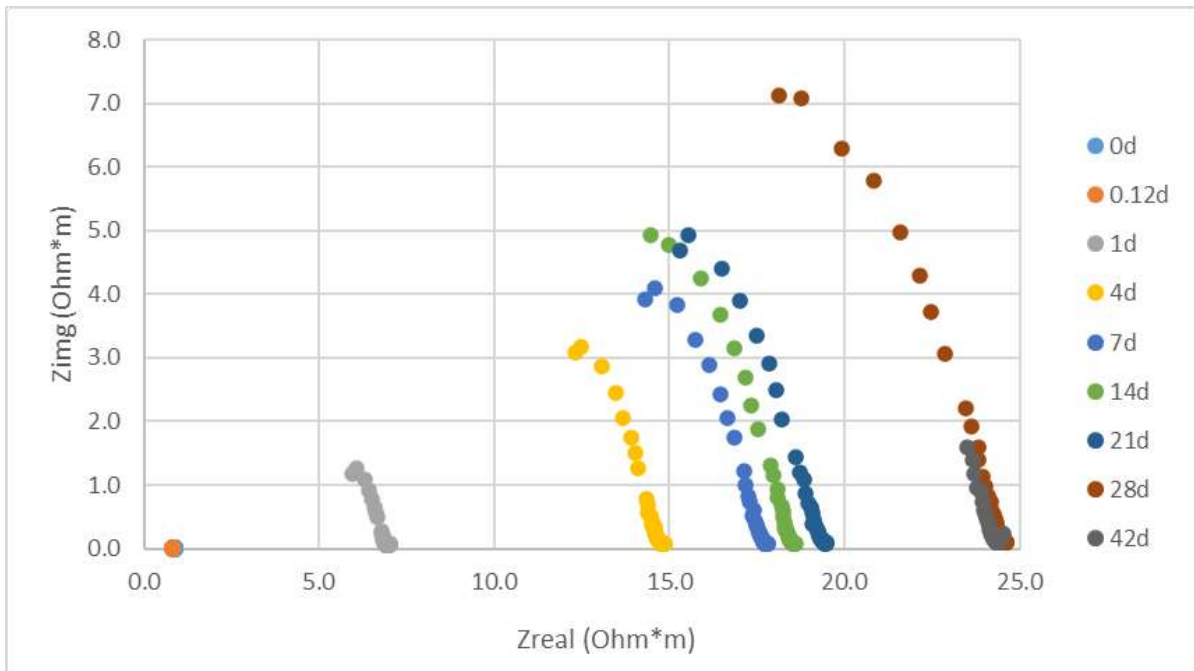
Pero, es de considerar que si tenemos en cuenta las mediciones de resistividad bulk presentadas en el numeral 4.3.2, donde se evidencio que durante las primeras 5 horas de mediciones fueron las muestras adicionadas con ceniza las que presentaron valores más altos de resistividad con respecto a la muestra M1, comportamiento que se atribuyó a las condiciones físicas (densidad la matriz) y químicas (iones disponibles en la solución). Vemos como esta condición se revierte por la muestra M1 al logran alcanzar su punto de saturación en un menor tiempo y por ende comenzar el proceso de precipitación de productos de la hidratación más rápido, lo cual implica un proceso de refinación de estructura interna de la muestra con mayor velocidad.

Comportamientos que nos llevan a confirmar que con mayores dosis de ceniza volante en las mezclas, se generara una mayor afectación en la velocidad del desarrollo de la estructura interna matriz. Esto, considerando que desde el inicio de la hidratación la cantidad de caminos de conducción continuos presentes en estas muestras es menor, los procesos para desarrollar los productos de hidratación se desarrollaran con menor velocidad. Sin embargo, estos procesos de transformación de la matriz se darán por largos periodos de tiempo con lo que se espera lograr superar el desarrollo interno de la matriz de la muestra realizada con 100% cemento.

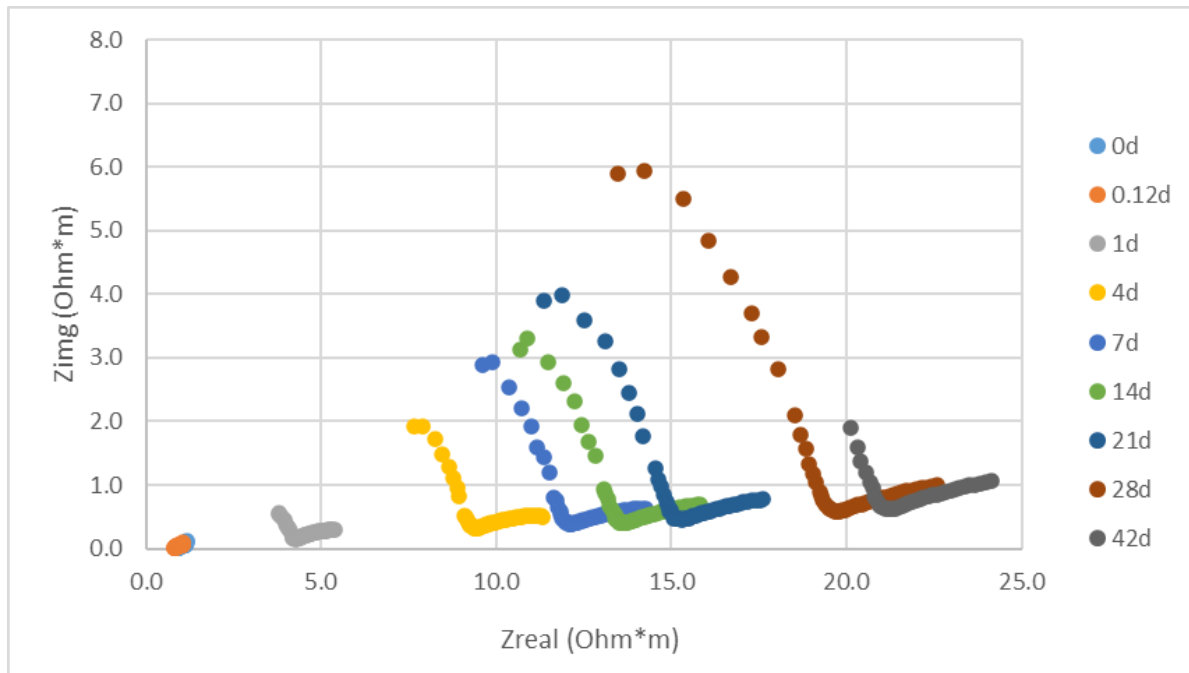
Resaltando, que esta condición la podemos evidenciar en la figura 4-40, donde se tiene en las edades de 28 y 42 días que las muestras adicionadas con ceniza no solo continúan presentando incrementos en los valores R_b , ya que la diferencia presentada en el valor de R_b a los 42 días es de $3.10 \text{ Ohm}\cdot\text{m}$ en el caso de M2 con respecto a M1 y de $4.35 \text{ Ohm}\cdot\text{m}$ en el caso de M3 con respecto a M1, siendo evidente una constante reducción en las brechas entre las lecturas de R_b .

Por último, se presenta el análisis de la evolución de la resistividad en el tiempo a partir de espectros de frecuencias de 100 Hz a 500000 Hz mediante el uso de diagramas de Nyquist (Fig 4-41 a-c y Fig 4-42 a-i), los cuales fueron obtenidos a partir de las mediciones realizadas con los electrodos de carbono. Esto debido a que como se menciona anteriormente, se evidenció una afectación considerable en las lecturas tomadas posterior a 3 días con los electrodos SPEs.

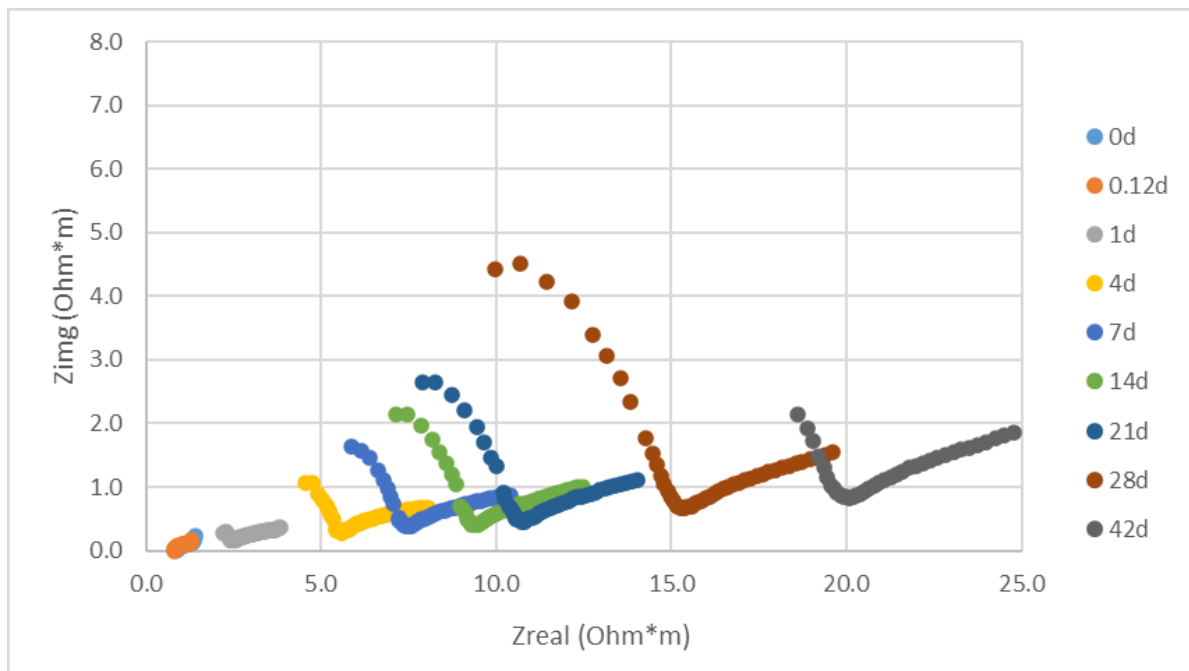
Figura. 4-41: Resumen por muestra de la evolución de la resistividad eléctrica en el tiempo con frecuencias variables (a) M1, (b) M2 y (c) M3.



(a) Muestra realizada con 100% cemento (M1)



(b) Muestra realizada con 85% cemento y 15% ceniza (M2)



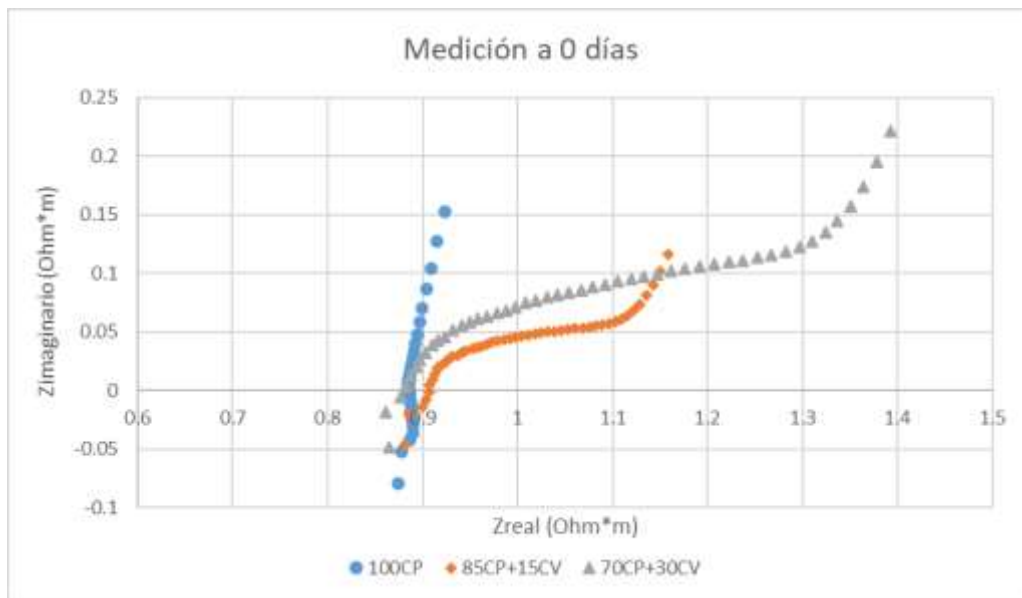
(c) Muestra realizada con 70% cemento y 30% ceniza (M3)

Inicialmente, analizadas las figuras 4-41 a, b y c, es importante evidenciar que adicional de la identificación de las mesetas mencionadas por McCarter (las cuales identificaremos más adelante), se observa como el efecto electrodo de la gráfica es más evidente en las muestras M2 y M3 (las cuales contienen ceniza), logrando tener la forma de los dos semicírculos con mayor facilidad en comparación con la muestra M1, en la cual no es evidente el crecimiento del semicírculo de alta frecuencia, el cual está ligado a la resistividad de la interfaz con el electrodo.

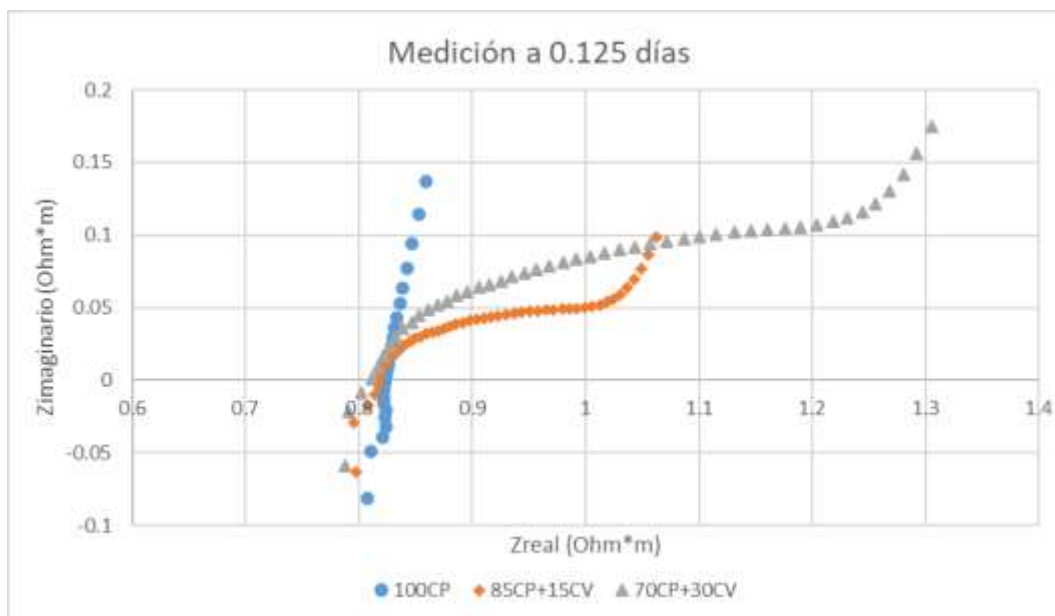
Sin embargo, es interesante ver como el diámetro del arco del semicírculo de alta frecuencia va en aumento gradual con la hidratación, lo que según algunos autores es debido a una disminución del contenido de electrolitos y la conectividad de los poros y un aumento en la formación de capas entre láminas de C-S-H. Llamando la atención la forma es que para las muestras 2 y 3 es hasta el día 21 en donde se comienzan a dar movimientos importantes en el aumento no solo de la resistencia (parte real) de las muestras, si no, que también se observa un ligero aumento de la reactancia (para imaginaria), pudiendo llegar a estimarse que es en este momento donde no solo el cemento aporta producto hidratación, si no que la ceniza inicia su proceso de precipitación de productos de hidratación; lo cual puede llegar a extenderse por varios meses.

A continuación, se presenta la comparación en el tiempo de los diagramas de Nyquist para cada una de las muestras en estudio.

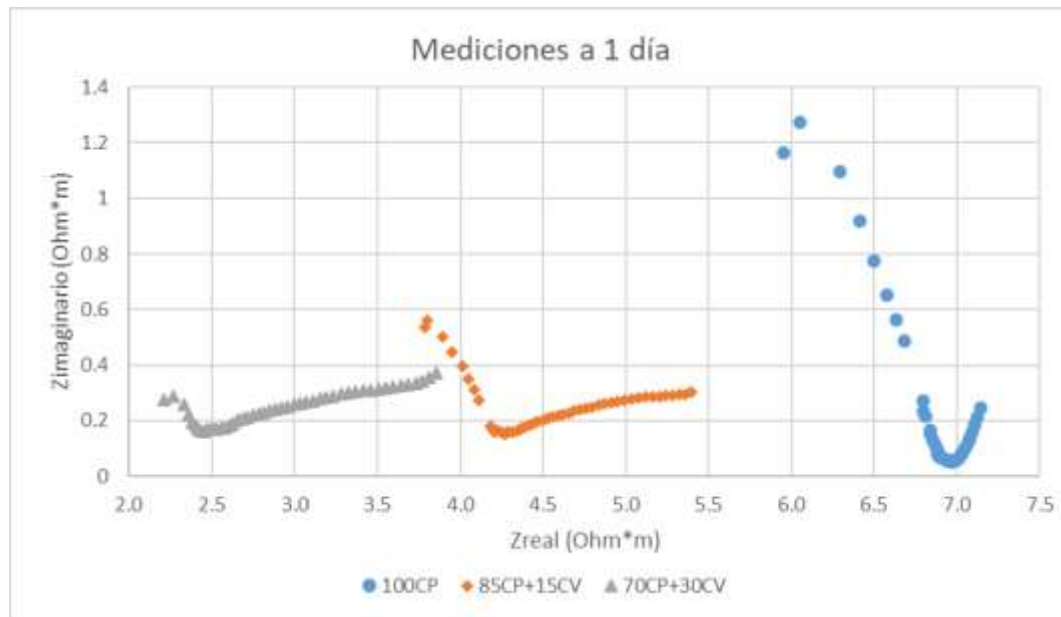
Figura. 4-42: Evolución de la resistividad eléctrica en el tiempo con frecuencias variables.



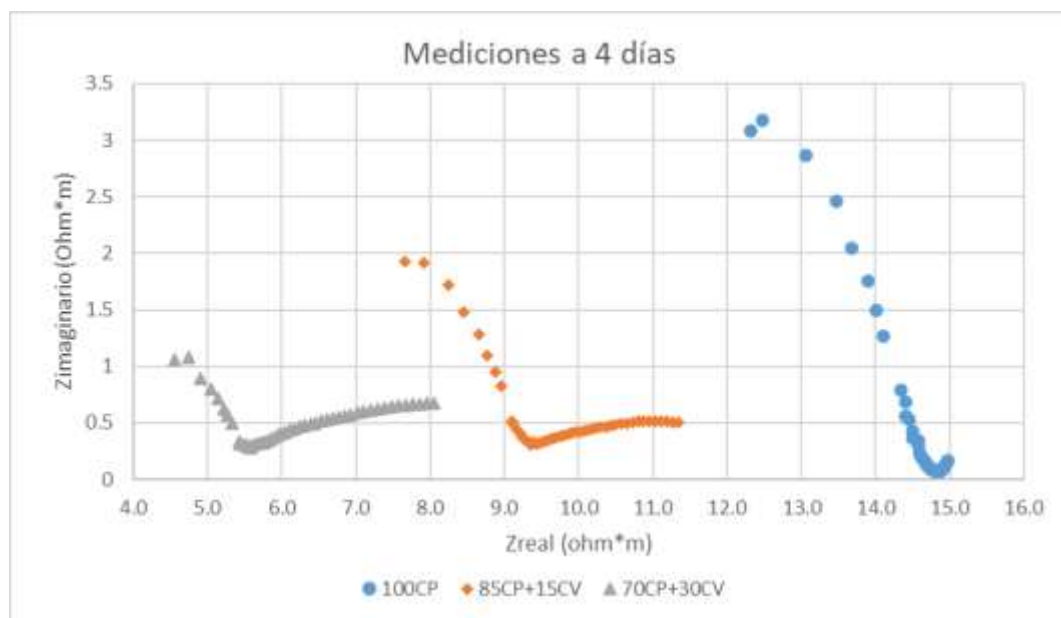
(a) Medición potenciostática para las muestras posterior a ser mezcladas.



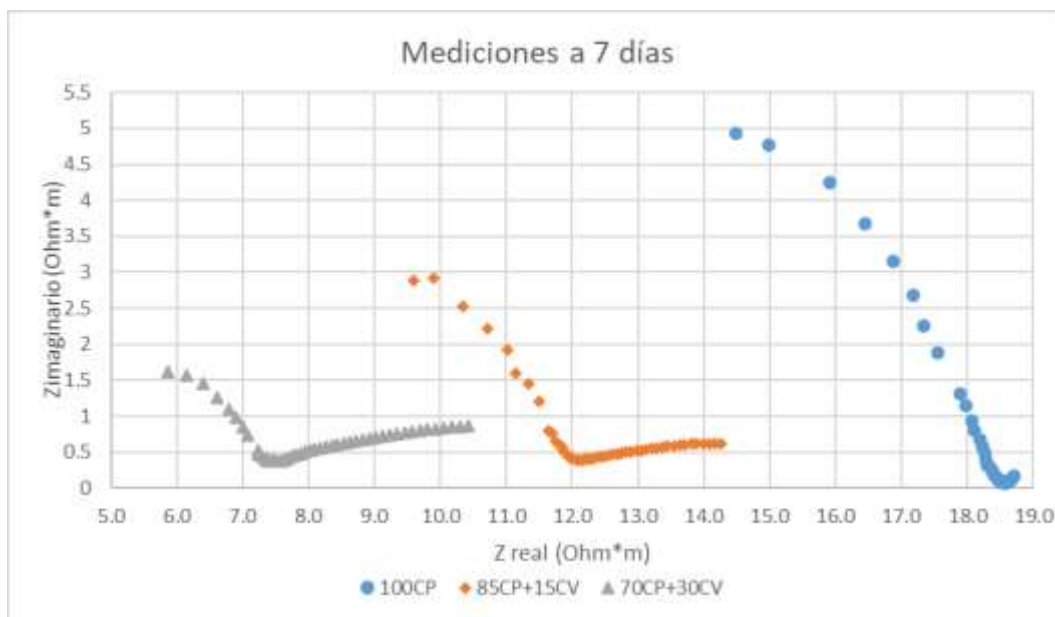
(b) Medición potenciostática para las muestras a 0.125 días (3 horas).



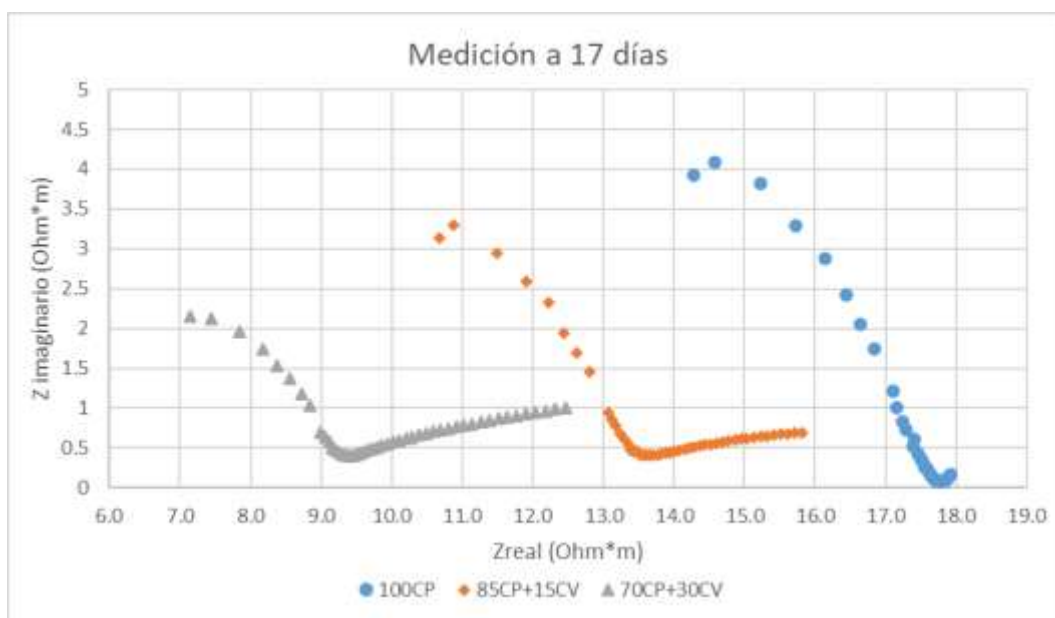
(c) Medición potenciostática para las muestras a 1 día.



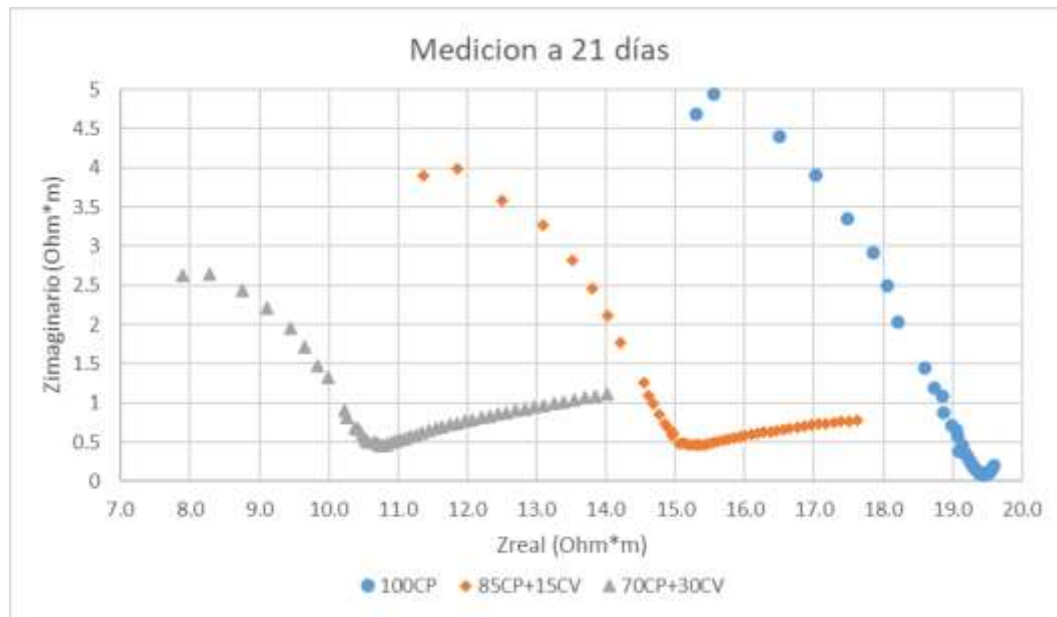
(d) Medición potenciostática para las muestras a 4 día.



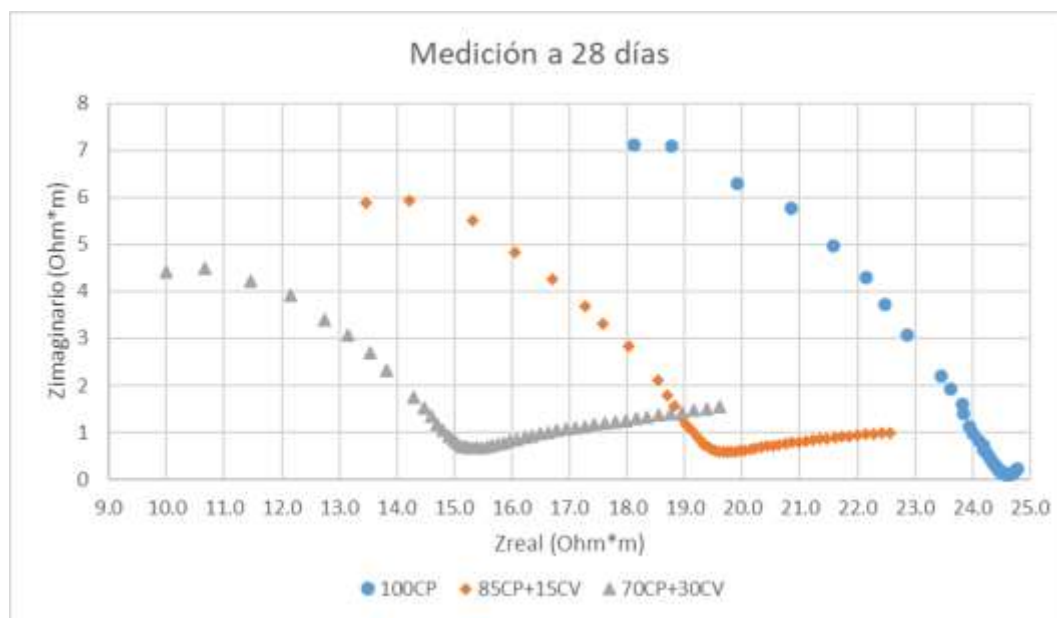
(e) Medición potenciostática para las muestras a 7 días.



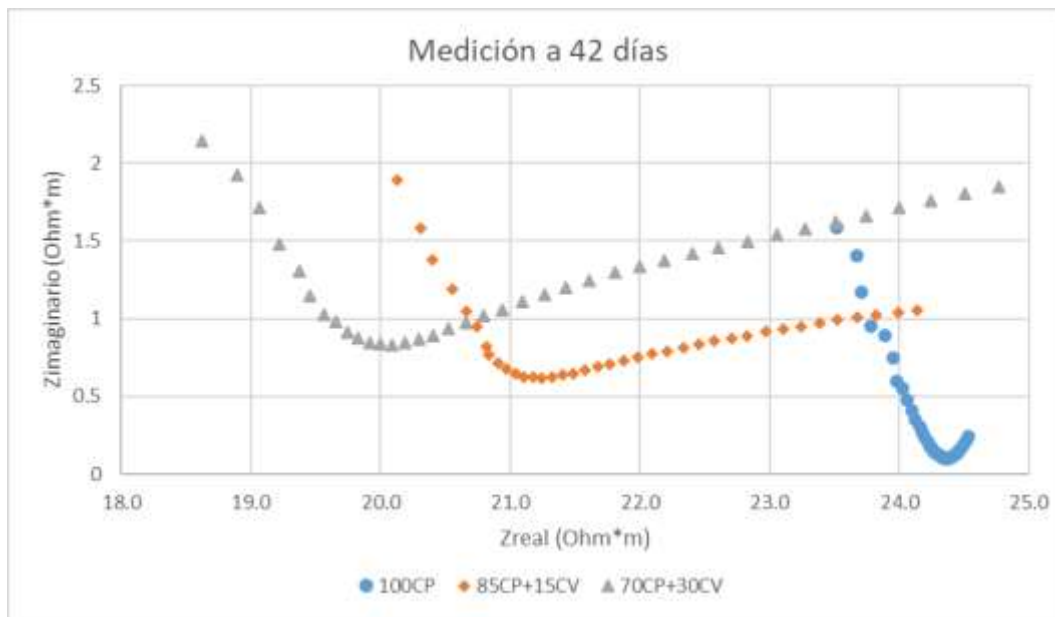
(f) Medición potenciostática para las muestras a 17 días.



(g) Medición potenciostática para las muestras a 21 días.



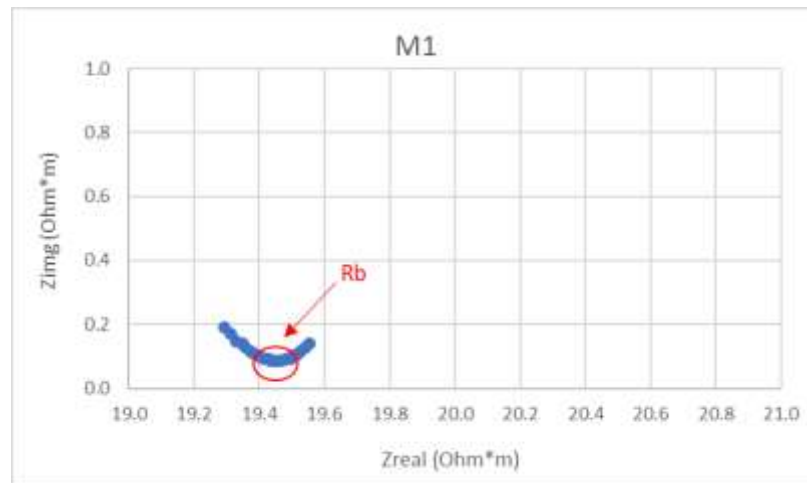
(h) Medición potenciostática para las muestras a 28 días.



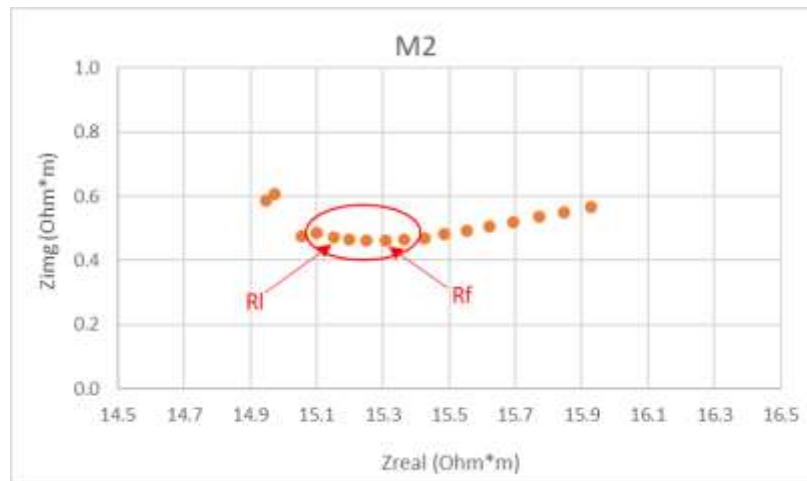
(i) Medición potenciostática para las muestras a 42 días.

Con base a lo mencionado anteriormente y de acuerdo con lo descrito por Hu et al. (2019), quien en sus investigaciones menciona que la mejora microestructural del concreto adicionado con ceniza puede no ocurrir a una edad temprana de hidratación debido a la baja reactividad puzolánica de las cenizas volantes; lo cual, como se describió anteriormente se ve reflejado en las mediciones realizadas pasado el día 21 de edad de la muestra, donde se tiene que la distancia de Rb entre cada una de las muestras comienza a disminuirse de forma considerable, lo cual estaría asociado principalmente a un inicio del mejoramiento de la microestructura de las muestras.

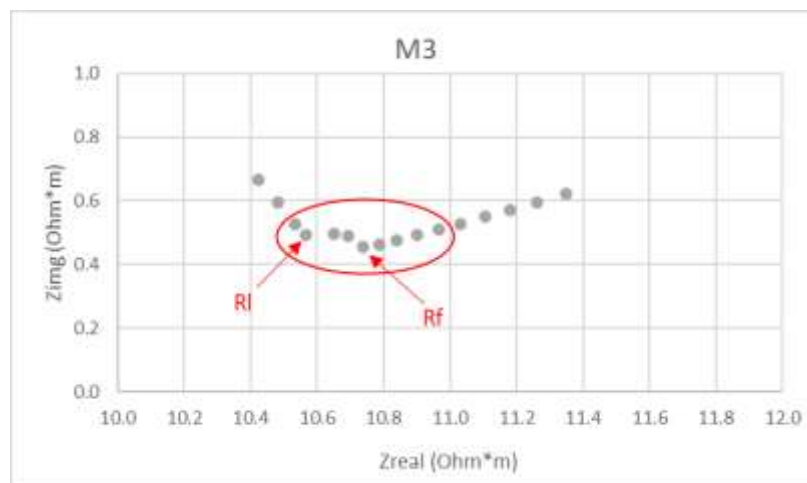
Finalmente, se analiza la formación de las mesetas intermedias generadas entre las zonas de afectación del material cementante y el electrodo; para lo cual se tomó como referencia las figuras 4-42 g (medición a 21 días), h (medición a 28 días) y i (medición a 42 días), las cuales se aumentan sobre la zona de la meseta para comprender mejor su desarrollo.

Figura. 4-43: Análisis del valle para las lecturas tomadas a 21 días.

(a) Muestra 100% CP (M1) evaluada a 21 días.

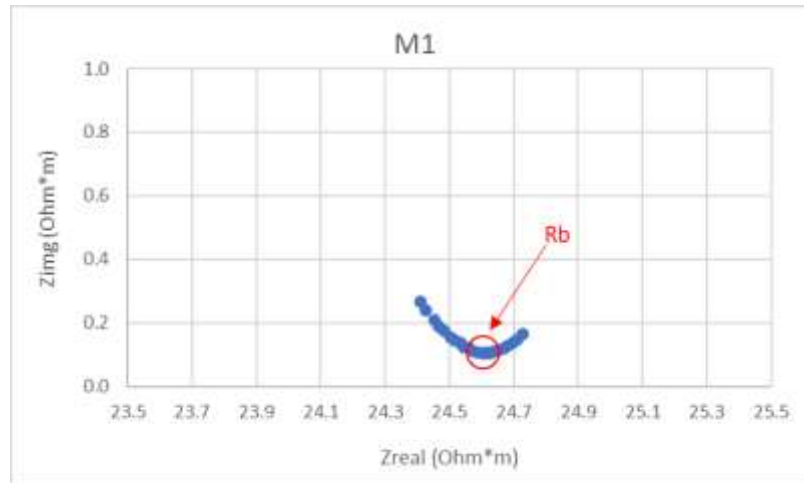


(b) Muestra 85%CP y 15%CV (M2) evaluada a 21 días.

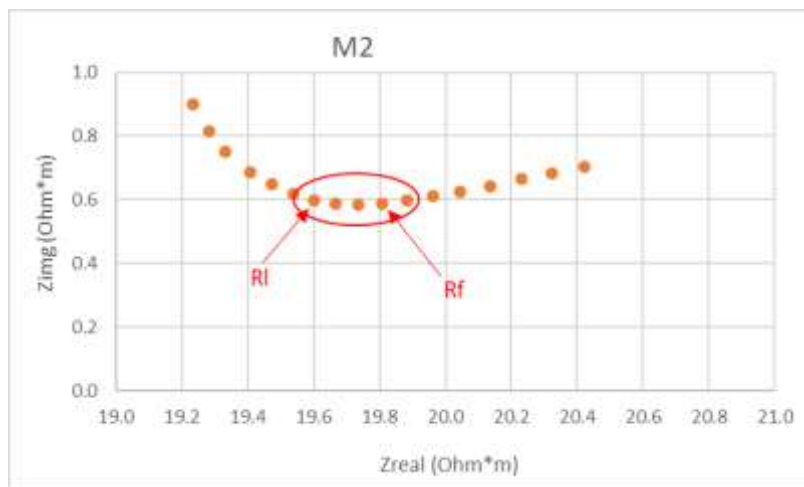


(c) Muestra 70%CP y 30%CV (M3) evaluada a 21 días.

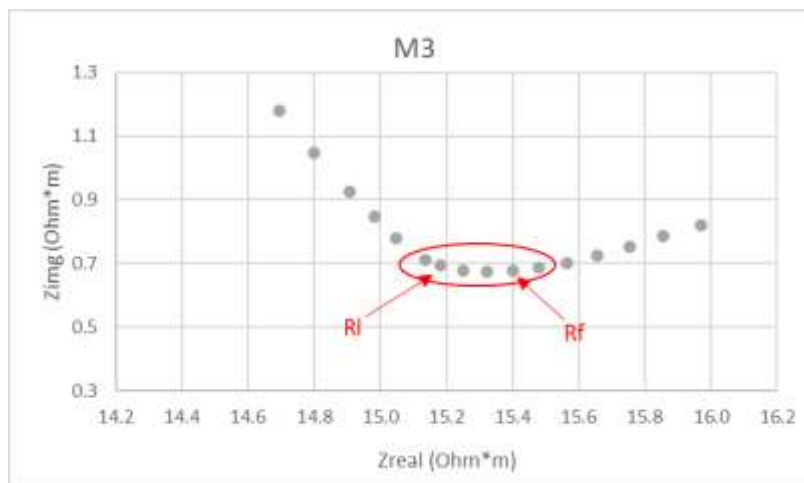
Figura. 4-44: Análisis del valle para las lecturas tomadas a 28 días.



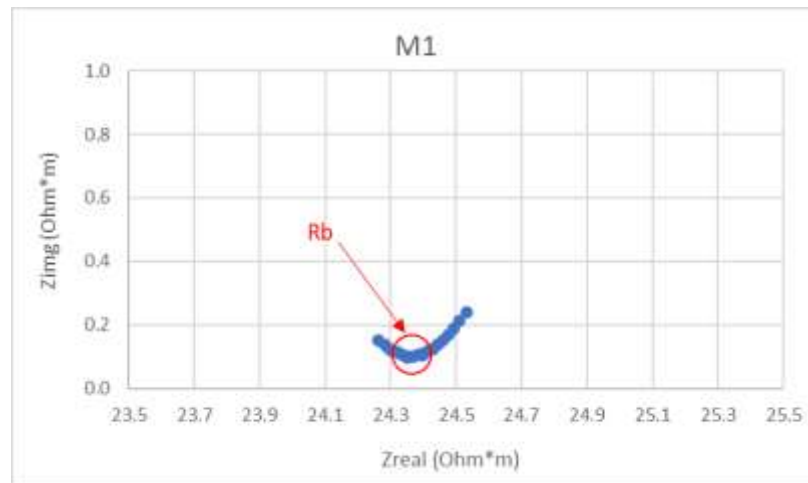
(a) Muestra 100% CP (M1) evaluada a 28 días.



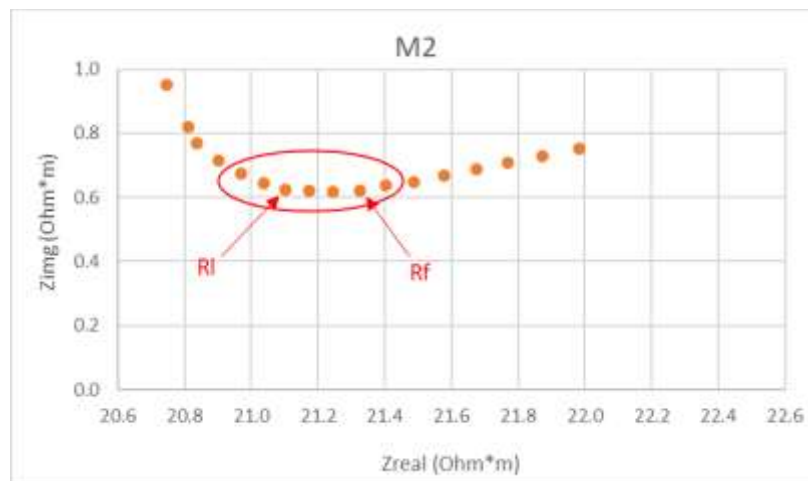
(b) Muestra 85%CP y 15%CV (M2) evaluada a 28 días.



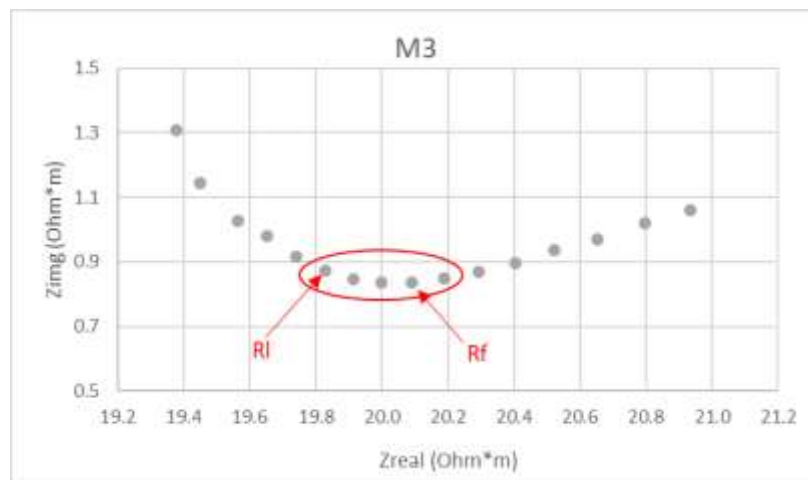
(c) Muestra 70%CP y 30%CV (M3) evaluada a 28 días.

Figura. 4-45: Análisis del valle para las lecturas tomadas a 42 días.

(a) Muestra 100% CP (M1) evaluada a 42 días.



(b) Muestra 85%CP y 15%CV (M2) evaluada a 42 días.



(c) Muestra 70%CP y 30%CV (M3) evaluada a 42 días.

En cuando a la característica del valle intermedio, se puede observar que las gráficas presentadas en las figuras 4-43 b y c, 4-44 b y c y 4-45 b y c presentan un valle con mayor tendencia a una forma de U. Lo cual algunos autores como Suryanto et al. (2016) han evidenciado que es una particularidad evidente cuando las mezclas presentan tiempos de hidratación más prolongados, siendo reportado como una característica típica de los sistemas cementantes que contienen adiciones de ceniza volante.

Sin embargo, la relevancia de esta característica está dictada por la proporción de carbono no quemado en las cenizas volantes que se cuantifica por la pérdida por ignición (LOI); siendo una característica que se vuelve más perceptible a medida que la cantidad LOI aumenta. Lo cual en el caso de la ceniza volante utilizada en este trabajo pudiese explicarse ya que esta tiene un LOI de 5.48% (valor que se encuentra dentro de los límites máximos permitidos por la NTC 3493) y podría explicar la forma de U en lugar de una región de meseta más distintiva (Suryanto et al., 2016).

5. Desarrollo y análisis de la metodología para la medición de la hidratación a partir de ensayos de resistividad eléctrica.

5.1 Metodología para el análisis de los tiempos de fraguado a partir de las mediciones de resistividad.

Para el desarrollo de una metodología que permita entender los procesos de hidratación del cemento a partir de las mediciones de la resistividad eléctrica, y una vez analizados los resultados presentados en el capítulo 4, se debe partir de que el cambio de la resistividad eléctrica con el tiempo se puede considerar como una huella del proceso de hidratación de los materiales a base de cemento. Y es que, a partir de monitorear la resistividad eléctrica del cemento fresco, se logran identificar diferentes puntos críticos y períodos en el proceso de hidratación (Li et al., 2003b), los cuales, una vez comparados con las mediciones tradicionales más comunes en el medio (metodologías de las agujas de Vicat y calor de hidratación) se puede identificar una buena correlación entre las mediciones (Yim et al., 2017).

Con base en lo descrito anteriormente, se inicio realizando una comparación entre los tiempos de fraguado definidos mediante las mediciones de agujas de Vicat y los valores de puntos críticos obtenidos a partir de los ensayos de la resistividad bulk de las muestras. En donde cómo se puede ver en la tabla 5-1 y las figuras 5-1, 5-2 y 5-3 la correlación para las 3 muestras en estudio es favorable, teniendo que a partir de ecuaciones de tendencia lineales se logran correlaciones que van del 0.93 al 0.99, lo cual nos indica que a partir de las mediciones de la resistividad eléctrica de una muestra y la definición de los puntos críticos P1 y P2, se pueden estimar fácilmente los tiempos de fraguado inicial, fraguado final y el punto de inicio de pérdida de manejabilidad de una muestra estudiada. Condición

que para el presente estudio se puede hacer mediante el uso de las ecuaciones lineales 5-1, 5-2 y 5-3 respectivamente, logrando valores aproximados muy cercanos a la realidad y estimando su valides sin importar el porcentaje de FA adicionado a la mezcla.

Tabla 5-1: Comparación de tiempos relevantes de tiempos de fraguado y puntos críticos según mediciones de resistividad.

	TIEMPOS RELEVANTES (HORAS)		
	M1 (100%CP)	M2 (85%CP+15%CV)	M3 (70%CP+30%CV)
Punto C1	0.66	0.90	1.01
Punto C2	3.85	5.45	5.85
Punto C3	10.05	10.36	10.46
F_i	1.91	2.58	3.23
F_f	3.67	4.33	4.87
P_m	1.50	2.00	2.17
$T_{m\acute{a}x}$	8.26	7.88	9.94

$$P_m = 1.9414 * P1 + 0.2266; (R^2 = 0.995) \quad \text{Ec. 5-1}$$

$$F_i = 3.6124 * P1 - 0.5213; (R^2 = 0.959) \quad \text{Ec. 5-2}$$

$$F_f = 0.5464 * P2 + 1.5305; (R^2 = 0.926) \quad \text{Ec. 5-3}$$

Figura. 5-1: Correlación punto crítico P1 Vs P_m .

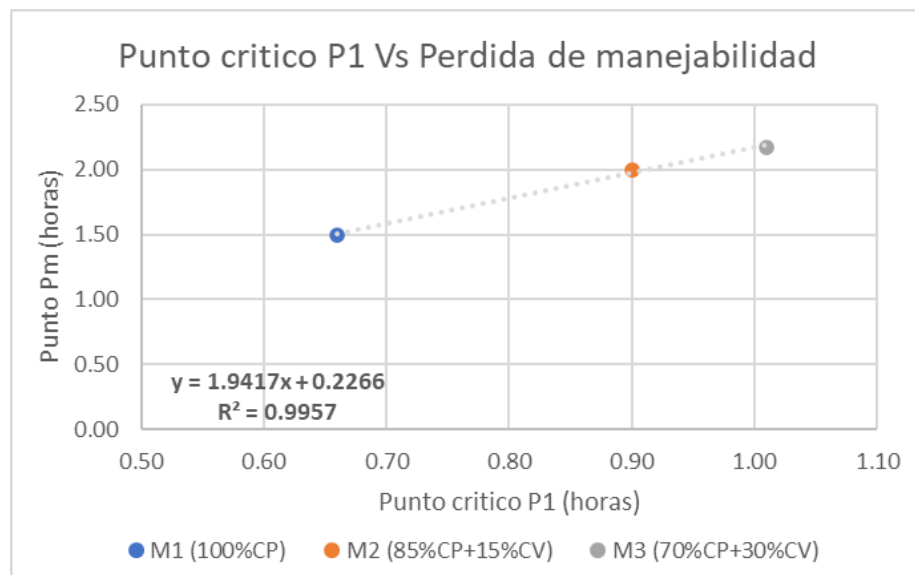
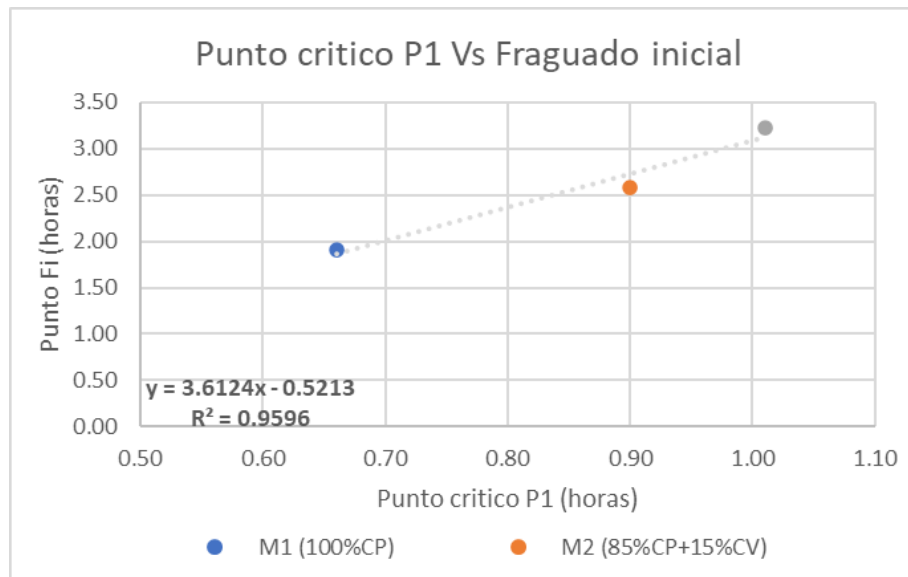
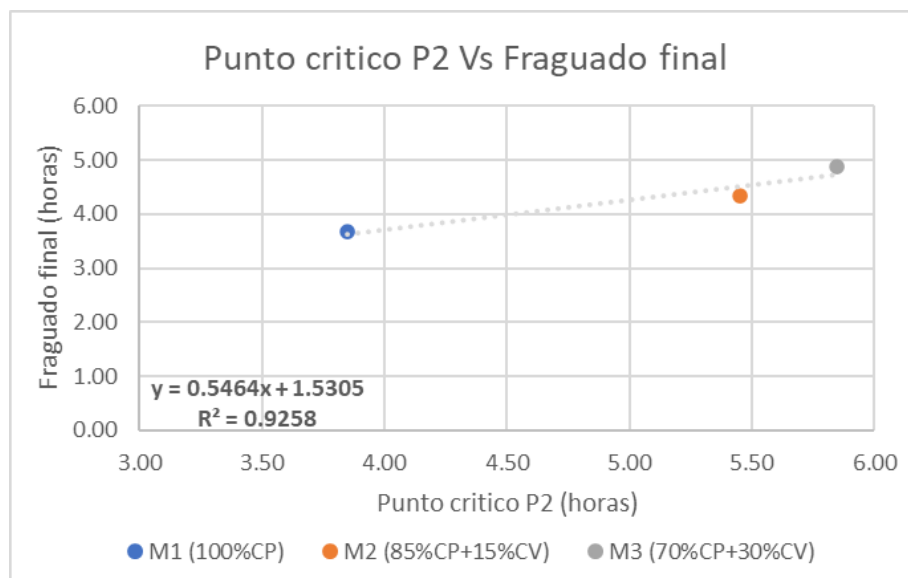


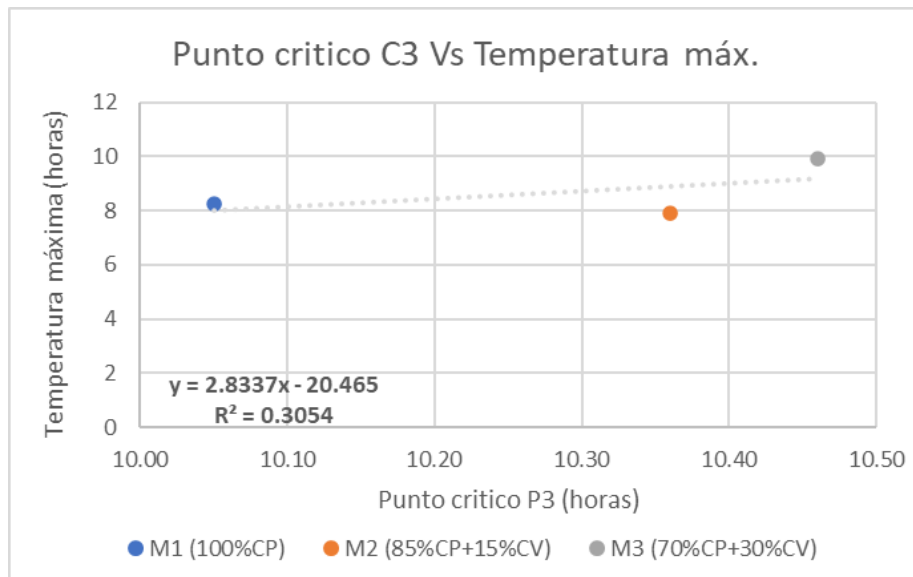
Figura. 5-2: Correlación punto crítico P1 Vs F_i .**Figura. 5-3:** Correlación punto crítico P2 Vs F_f .

Sin embargo, una vez analizado el punto crítico P1 y P2, se procede con el estudio del punto crítico P3; el cual según algunos autores hace referencia a el momento en el cual dentro de la muestra se estabilizan los procesos de precipitación de productos de la hidratación y la producción de iones OH^- . Razón por la cual se ve una relación entre la incidencia del punto crítico P3 y el tiempo donde se presenta el pico de temperatura máxima (figura. 5-4).

Obteniendo que, aunque dichos puntos se pueden representar con relación lineal (Ec. 5-4), la correlación de los datos para el presente caso es baja ($R^2=0.305$), lo cual como se puede en la figura 5-4, esta principalmente relacionado al tiempo para $T_{m\acute{a}x}$ obtenido para la muestra M2, en donde la calorimetría realizada en el equipo HidroCemUN se presentan un dato atípico en cuando a tiempo del pico de temperatura con respecto a los datos normales presentados en la literatura. Por lo que considerando lo descrito anteriormente, se puede llegar a proponer en función de la linealidad que se presenta en la ecuación 5-4, que la metodología con que esta es definida puede una estimación para el cálculo del tiempo de $T_{m\acute{a}x}$.

$$T_{m\acute{a}x} = 2.8337 * P3 - 20.465; (R^2 = 0.305) \quad \text{Ec. 5-4}$$

Figura. 5-4: Correlación punto crítico P3 Vs $T_{m\acute{a}x}$.



5.2 Metodología para el análisis del calor de hidratación y la resistividad eléctrica de la muestra

Adicionalmente, la impedancia ofrece un método prometedor para evaluar el grado de hidratación del cemento. Se ha logrado evidenciar que está medición en edades tempranas se puede considerar como un índice equivalente para la liberación total de calor, posición en la que como se puede observar en las figuras 5-5, 5-6 y 5-7, existe una relación directa

entre las dos variables, las cuales pueden ser explicadas a partir de un modelo bilineal conformado por dos regresiones, las cuales presenta correlaciones que van desde 0.89 a 0.99, dando una importante confiabilidad.

▪ **Muestra 1 (100%CP)**

$$Y1 = 0.0408 * X + 0.9738; R^2 = 0.977 \quad \text{Ec. 5-5}$$

$$Y2 = 0.2716 * X + 56.095; R^2 = 0.972 \quad \text{Ec. 5-6}$$

▪ **Muestra 2 (85%CP+15%CV)**

$$Y1 = 0.028 * X + 1.151; R^2 = 0.949 \quad \text{Ec. 5-7}$$

$$Y2 = 0.2963 * X + 56.985; R^2 = 0.886 \quad \text{Ec. 5-8}$$

▪ **Muestra 3 (70%CP+30%CV)**

$$Y1 = 0.0224 * X + 1.2895; R^2 = 0.996 \quad \text{Ec. 5-9}$$

$$Y2 = 0.1155 * X + 15.758; R^2 = 0.983 \quad \text{Ec. 5-10}$$

Figura. 5-5: Comparación calor liberado vs resistividad eléctrica de la muestra M1.



Figura. 5-6: Comparación calor liberado vs resistividad eléctrica de la muestra M2.

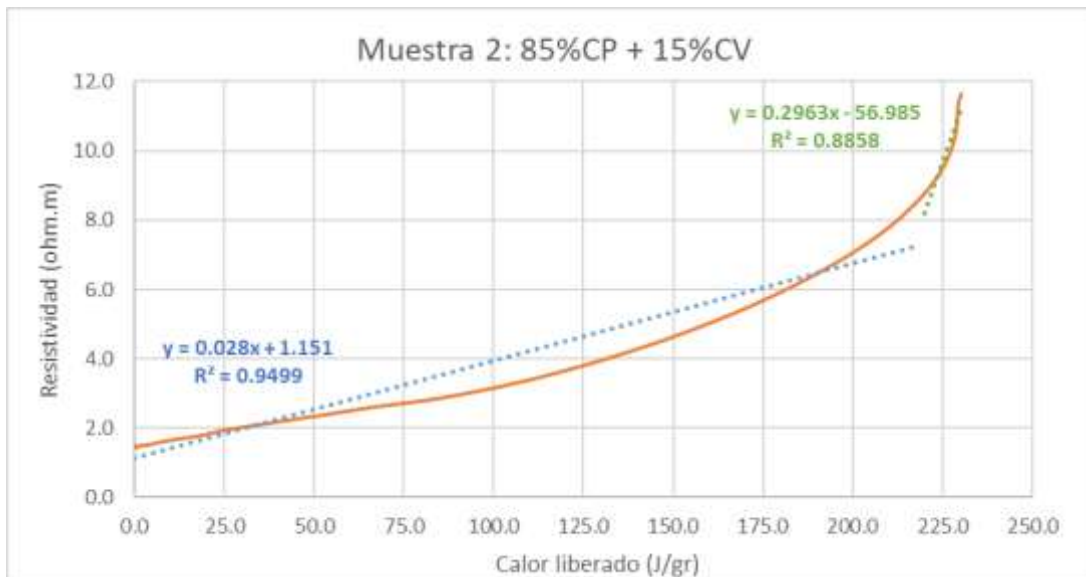
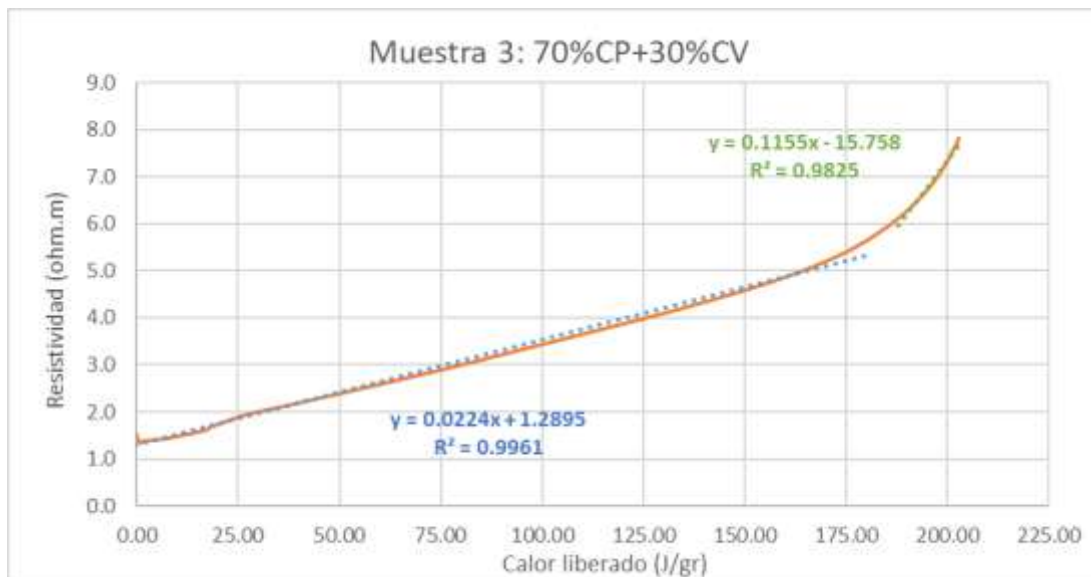


Figura. 5-7: Comparación calor liberado vs resistividad eléctrica de la muestra M3.



Una vez presentadas las figuras 5-5, 5-6 y 5-7, es importante tener presente que sobre la franja de 15 a 17 horas de fundida las muestras se identifica que estas entran en su etapa de estabilización en cuanto al calor de hidratación liberado, lo cual indica que a partir este momento, dicha variable se nos vuelve constante a diferencia que la resistividad eléctrica, la que como se puede evidenciar en el numeral 4.3.3 presenta aumentos constantes a

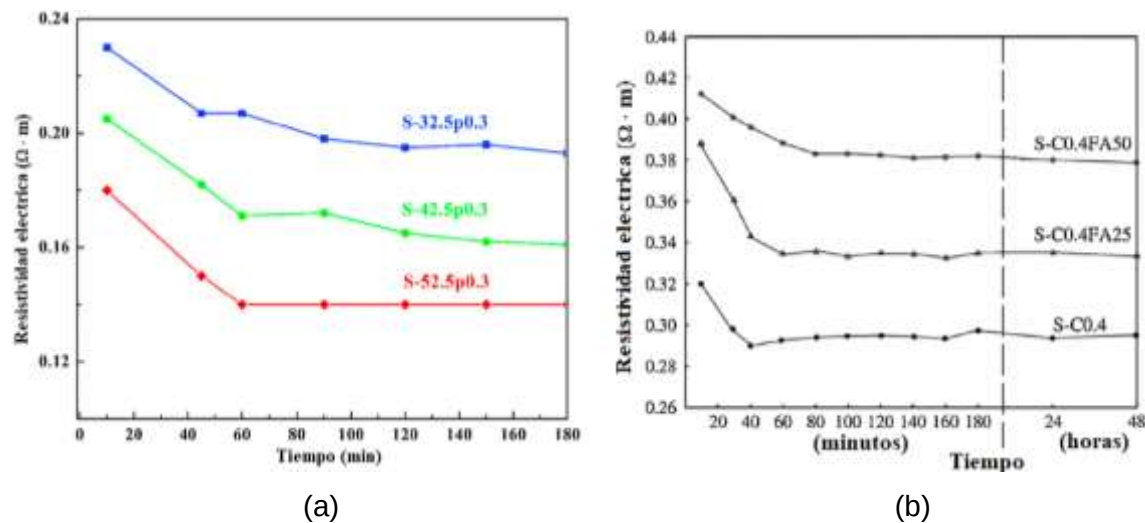
medida que la edad de especificación sigue en aumento, condición que se evidencia en la 3 graficas presentadas anteriormente, en donde ambas variables son directamente proporcionales en el desarrollo de las 3 tres primeras fases de su hidratación (disolución, aceleración y desaceleración).

5.3 Análisis de otras propiedades

Finalmente, se debe mencionar que la resistividad eléctrica se puede utilizar no solo para reflejar las reacciones de hidratación de una pasta de cemento, sino también para proporcionar una visión más precisa y completa de las características microestructurales de transporte y la forma en cómo estas características se va desarrollando con el avance del tiempo. Para lo cual se puede hacer uso de medición en edades tempranas de propiedades como el ángulo de fase y la capacitancia de la muestra con el fin de entender en mayor proporción la forma en que actúan las propiedades químicas (ligadas con la influencia de solución de poros) y físicas (asociadas con el desarrollo de la microestructura interna de la pasta) de las muestras para el desarrollo de la microestructura.

Por esta razón, en el presente análisis de la hipótesis plantea en el capítulo 4.3.2, donde se habla del punto crítico P1 o punto de saturación como el final de la fase de la hidratación donde la resistividad eléctrica depende netamente del comportamiento químico de la muestra. Condición que puede ser soportada en investigaciones como las realizadas por Tang et al. (2017) y Yousuf & Xiaosheng, (2020), quienes concluyeron que la resistividad eléctrica de la solución de poros tiende a presentar curvas decrecientes hasta una franja de tiempo comprendida entre los 50 a 80 minutos (figura 5-8 a y b), donde la resistividad toma valores constantes, franja de tiempo que presenta una correlación directa con los valores de punto crítico P1 definidos anteriormente. Representando lo descrito anteriormente que pasamos de valores que dependían principalmente de la química de la muestra (hasta el punto crítico P1), a una fase donde comienza a influir también el desarrollo físico de la microestructura de la pasta.

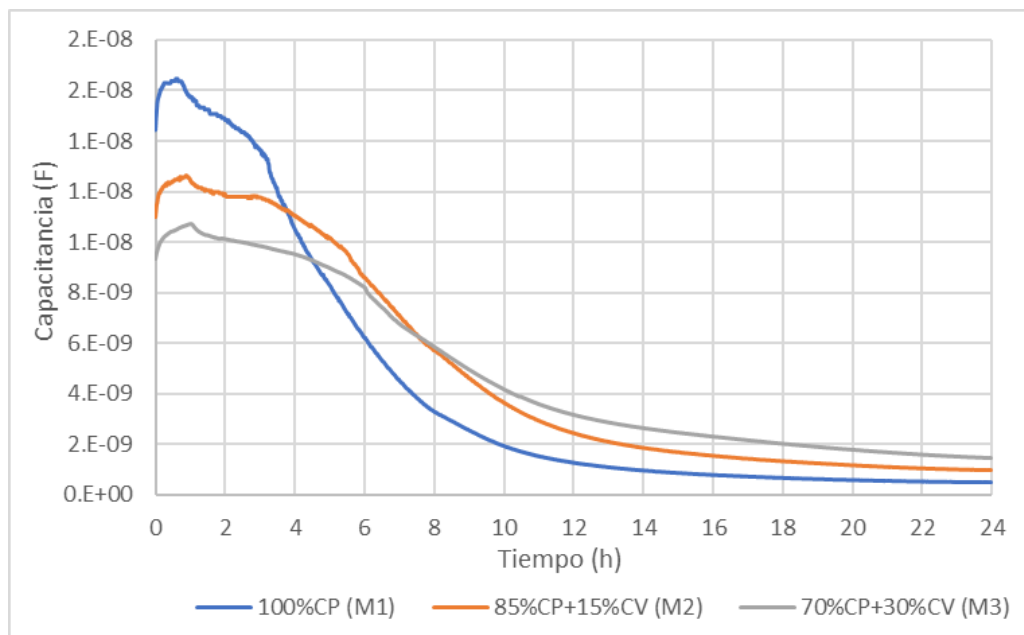
Figura. 5-8: Evaluación de la resistividad eléctrica en la solución de poros para: (a) muestras de 100% cemento y (b) muestras adicionadas con ceniza al 25% y 50%.



Nombre de la fuente: (a) (Yousuf & Xiaosheng, 2020) y (b) (Tang et al., 2017)

Comportamiento que, si comparamos con propiedad como la capacitancia de la mezcla evaluada, tiende a presentar un comportamiento similar al esperado cuando realizamos las mediciones de resistividad de la solución de poros. Sin embargo, es importante tener presente que la capacitancia es definida físicamente como la capacidad que puede presentar un cuerpo para mantener una carga eléctrica (Bosques, 2010); concepto que si evaluamos en paralelo puede estar directamente relacionado con el comportamiento resistivo de la solución de poros, ya que son los diferentes iones presentes dentro de esta solución los que tienen la capacidad de almacenamiento y transporte de la carga eléctrica a través de la muestra.

Por esta razón, cómo se observa en la figura 5-9 donde se presenta que al igual a lo descrito por Yousuf et al. (2020) durante las mediciones de solución de poros, se tiene que el comportamiento de la capacitancia de las muestras, presentan esa tendencia a la desaceleración una vez se alcanza el punto de saturación, para horas más tarde llegar a un punto de equilibrio constante dentro de la muestra.

Figura. 5-9: Medición de la capacitancia para las muestras en estudio.

Adicionalmente, es importante tener presente las tendencias reflejadas en la figura 5-9, donde se tiene una mayor capacitancia para la muestra realizada sin adición de ceniza volante, lo cual como lo hemos visto a lo largo del presente trabajo, es un comportamiento adecuado ya que es en esta donde se presentan la mayor cantidad de iones disueltos durante la etapa inicial. Condición que cambia considerablemente una vez la ceniza es adicionada a las muestras; ya que, sumado a la mayor densidad de la matriz, su proceso de liberaciones de iones se realiza de forma más lenta y durante un mayor lapso de tiempo. Situación que se ve representada en el último segmento de la gráfica donde es la muestra M3 la que presenta una mayor capacitancia con respecto a las demás muestras.

Sin embargo, es importante analizar esta propiedad cuando es comparada con otros parámetros como lo es el desarrollo de la resistividad bulk Figura.5-9, 5-10 y 5-11. Ya que aquí se puede plantear la hipótesis de la incidencia que 3 fases diferente de la hidratación, en donde primaran algunas propiedades por sobre otras de acuerdo con los procesos normales de hidratación mencionados en los capítulos anteriores.

Figura. 5-10: Comparación de resistividad eléctrica vs capacitancia para la muestra M1.

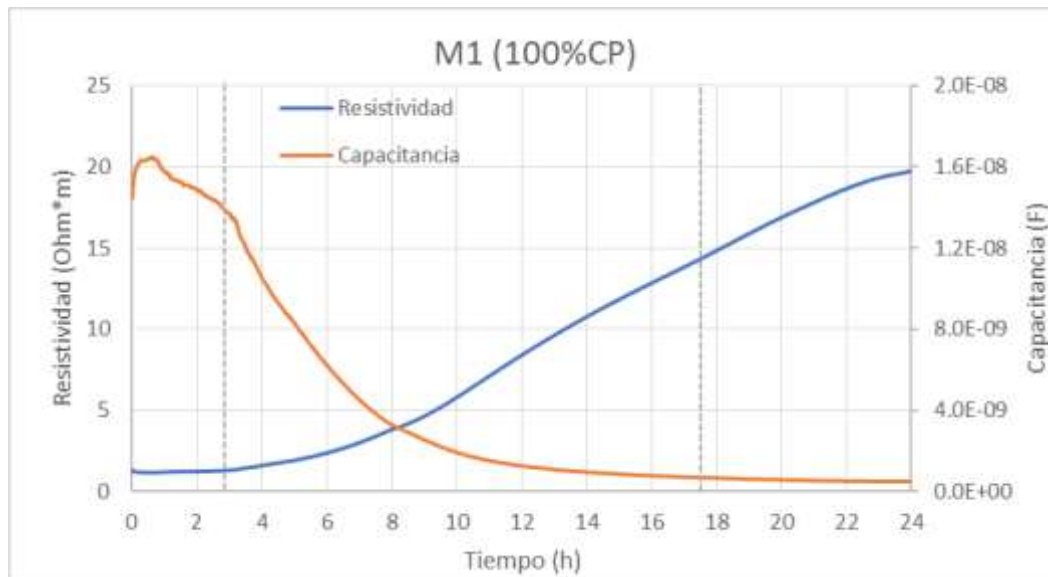


Figura. 5-11: Comparación de resistividad eléctrica vs capacitancia para la muestra M2.

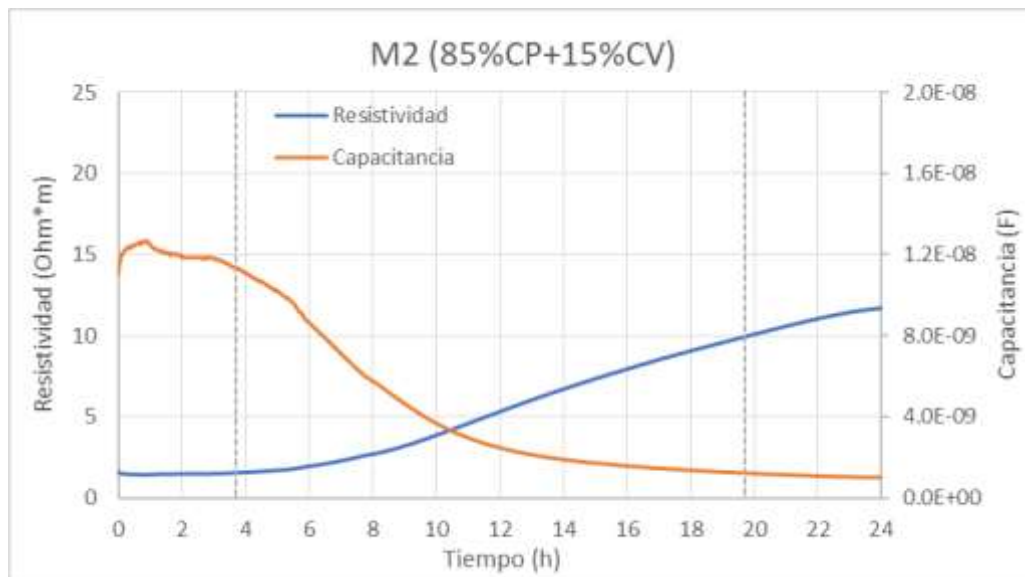
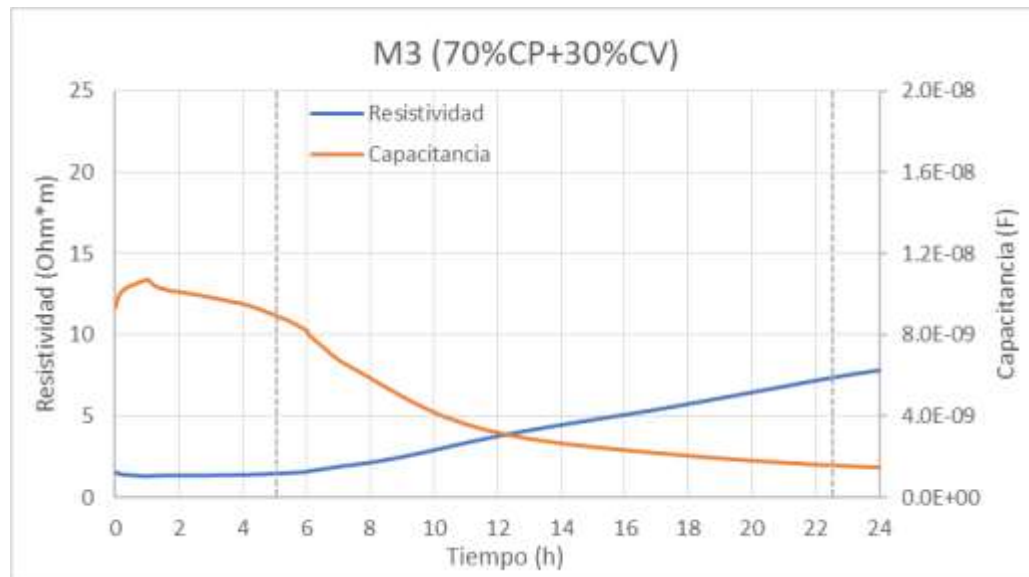


Figura. 5-12: Comparación de resistividad eléctrica vs capacitancia para la muestra M3.

Fases que de acuerdo con los comportamientos de hidratación de las pastas de cemento descritos a lo largo del presente se podrían describir como:

- Fase 1: hace referencia a ese periodo de disolución iónica donde la muestra se encuentra completamente saturada, esta fase pueda estar dominada por el comportamiento químico del cementante y la composición iónica de la solución de poros.
- Fase 2: durante este periodo se comienzan a presentar los primeros productos de hidratación, incidiendo en que se comienza a tener la formación de una microestructura al interior de la muestra; sin embargo, durante este periodo se logra un equilibrio entre la precipitación de productos de la hidratación y la liberación de iones a la solución. Por esta razón, se puede concluir que el comportamiento es tanto químico como físico.
- Fase 3: finalmente se define una tercera fase, en la cual se tiene un comportamiento asintótico de la capacitancia de la muestra, mientras que la resistividad presenta una tendencia al aumento, comportamiento que nos indican un dominio de la microestructura de la muestra sobre las propiedades químicas.

6. Conclusiones y recomendaciones

6.1 Conclusiones

- Partiendo del objetivo general definido para el presente trabajo de investigación, se puede concluir que la implementación de los electrodos serigráficos impresos en 3D, representan una buena alternativa para la investigación y estudio del proceso de hidratación de las pastas de cemento. Esto se puede definir, ya que se logra el desarrollo de una metodología que permite el estudio de la hidratación a partir de un modelo conceptual y la correlación de los puntos críticos identificados sobre las curvas de resistividad eléctrica y los tiempos de fraguado medidos por el método de las agujas de Vicat.

Adicionalmente, se tiene que a partir de estos modelos lineales basados en la estimación de los puntos críticos, se pueden encontrar los tiempos de importancia en el comportamiento de la muestra como lo es el fraguado inicial, fraguado final y momento de inicio de la pérdida de la manejabilidad.

- Por otra parte, una vez estudiados los datos del calor de hidratación acumulado de las muestras, se encuentra que estos presentan una correlación significativa con relación a los datos de resistividad. Resaltando, que este comportamiento se puede explicar a partir de un modelo de dos regresiones lineales que permiten inducir el calor liberado por la muestra durante su etapa inicial de la hidratación.
- Una vez estudiada la información obtenidos por diferentes autores con respecto a las mediciones de resistividad de la solución de poros extraída de una muestra, se puede identificar que dicha medición presenta una correlación directa con los valores de capacitancia estudiados. Esto nos permitió definir de forma sencilla las

fases de la hidratación que son predominadas por comportamientos químicos y/o físicos.

- Aunque se obtienen buenos resultado en el análisis de las propiedades evaluadas en edades tardías de las muestras; los cuales nos permitieron entender la forma en que se van desarrollando las microestructuras al interior de la pasta cuando esta es adicionada con ceniza volante. Se tiene que la referencia de electrodos SPES utilizados presentan algunas anomalías en su funcionamiento una vez se llega a las 72 horas de medición. Comportamiento que fue asociado a que la referencia de equipos utilizados esta principalmente diseñados para trabajos en medios acuosos, lo cual se evidencio por el avanzado deterioro presentado en los equipos al ser sometidos a un medio agresivo como el que se da durante la hidratación del cemento.
- Una vez analizados los datos de resistividad real obtenidos en las diferentes edades de estudio, se logra identificar la incidencia del uso de adiciones minerales como la ceniza volante en las diferentes propiedades como los tiempos de fraguado, el calor de hidratación liberado, el desarrollo de la resistividad eléctrica y la microestructura. Llamando particularmente la atención los periodos de resistividad constante que se desarrollaron en las muestras M2 y M3 y que dan indicios de las fases o momentos en que la ceniza produce sus “fases de retardo” en la mezcla, influyendo de esta forma en la liberación de energía dentro de la muestra e igualmente incidiendo directamente sobre los picos de temperatura.
- Con seguimiento de los lineamientos definidos por la por la normativa BS EN 196-9:210 y el desarrollo adecuado de un proceso de calibración, se logró la construcción de una cámara semi – adiabática la cual permite el estudio del calor de hidratación acumulado y evolución de la temperatura con buenas correlaciones en comparación a la cámara adiabática fabricada por la empresa Calorimetrix.

6.2 Recomendaciones

- Se recomienda la implementación y profundización en el uso de electrodos serigrafiados impresos en 3D realizados a base de materiales cerámicos; ya que estos permiten realizar mediciones durante mayores lapsos de tiempo gracias a que los materiales de su fabricación son más resistentes a medios alcalinos y de altas temperaturas.
- Es importante realizar la implementación de estas mediciones en campo con el uso de los electrodos impresos en 3D, ya que estos podrían presentar grandes beneficios al poder obtener datos en tiempo real del material cementante.
- Se recomienda realizar mediciones de repetitividad con el uso de los electros SPE, esto con el fin de validar su adecuado funcionamiento.
- Con el fin de validar el funcionamiento de los electros, se recomienda realizar mediciones en morteros y concretos.

A. Anexo: Ajuste y calibración del equipo de Calorimetría (HidroCemUn)

Como se menciona en el numeral 3.1.2, con el fin de dar continuidad al trabajo realizado en el año 2011 por la estudiante de pregrado Luisa Natalia Peña Leal, mediante el cual se construyó una cámara semi-adiabática para la medición de la evolución de la temperatura de hidratación de las mezclas de cemento hidráulico. El cual es un equipo que durante estos últimos años ha sido de gran ayuda para realizar proyectos de investigación de los estudiantes de pregrado y posgrado de la Universidad Nacional de Colombia. Razón por lo cual, surgió la necesidad de desarrollar algunas modificaciones al equipo, el cual con el paso del tiempo se fue deteriorando en algunas de sus partes como el armazón exterior u otros aspectos que pueden ser vistos como mejoras que facilitan el montaje y realización de ensayos.

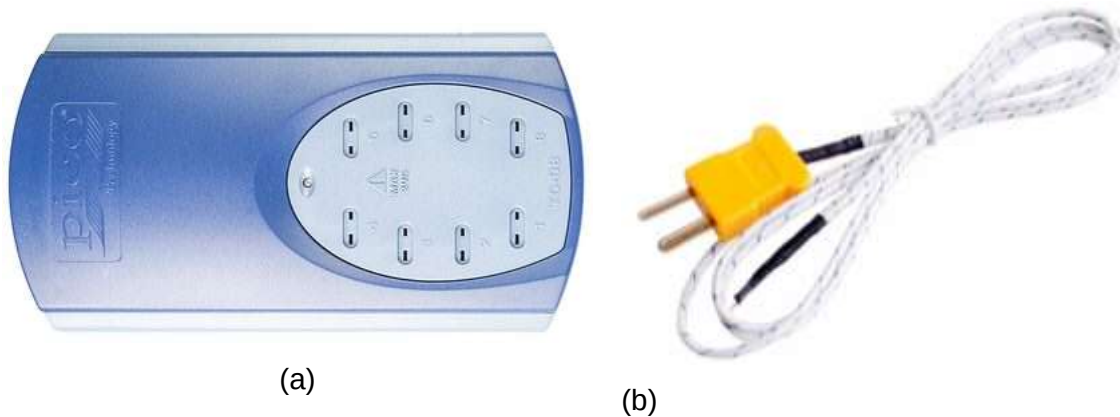
Esto implicó la adquisición de una nueva herramienta para el registro y control de los datos de temperatura, al igual que utilizar un mejor sistema aislante que disminuya las afectaciones de temperatura con el exterior. Dichas modificaciones implicaron realizar nuevamente la calibración del equipo. Por lo que a continuación, se describen las actividades que se realizaron para la adecuación y mejora de la cámara semi-adiabática.

1. Implementación de un nuevo sistema para la medición y registro de temperaturas.

Inicialmente, se realizó la compra de un datalogger de referencia PicLog 6 TC-08 (figura 6-1) producido por la compañía PicoTechnology; este equipo cuenta con 8 canales de lectura simultánea, los cuales funcionan con termocuplas tipo K. En comparación con el equipo usado anteriormente, el nuevo datalogger nos brinda mayor facilidad no solo a hora del montaje del ensayo ya que las termocuplas son de un solo uso, si no que el manejo de datos se realizará de forma más sencilla y rápida mediante el uso de la aplicación **PicoLog**

6 la cual es desarrollada por el proveedor del mismo y se puede descargar de forma gratuita en la página del proveedor.

Figura. 6-1: Sistema de registro de temperatura implementado en el equipo. (a) Datalogger TC-08. (b) Termocuplas tipo K.



Nombre de la fuente: Propia.

Figura. 6-2: Sistema de registro de temperatura utilizado anteriormente.



Nombre de la fuente: Peña, Luisa. 2011

2. Ajustes y adecuaciones de la cámara semi-adiabática

La cámara consta inicialmente de una nevera de poliestireno expandido con dimensiones externas de 0.34m * 0.55m * 0.35m e internas de 0.295m x 0.485m x 0.285m con tapa. En esta se ubican tres láminas de icopor con calibre de 0.075m cada una, las cuales fueron puestas una sobre la otra desde el fondo de la nevera. En las dos laminas superiores se perforaron 3 cámaras (figura 6.3 - a), dentro de las cuales se colocarán los vasos Dewar que contienen cada una de las muestras en estudio (Peña, 2011).

Por su parte, el uso de los vasos Dewar se da con el fin de garantizar un mejor aislamiento térmico de las muestras en estudio. Dichos vasos son metálicos y cuentan con unas dimensiones de 0.10m de alto y 0.05m de diámetro exterior, ver (figura 6.3 - b).

Figura. 6-3: Construcción de compartimientos interiores de la cámara semi-adiabática.



(a)



(b)

Nombre de la fuente: Peña, Luisa. 2011

Con el fin de mejorar el estado en que se encontraba la cámara, se construyó una caja dentro de la cual se instalara la cámara existente. Está fue hecha en láminas de madera de 0.05m de espesor y dimensiones internas de 0.40m x 0.60m x 0.40m, lo cual permitió aplicar una capa adicional de poliuretano expandido de 0.025m de espesor entre las láminas de madera y las paredes de la cámara inicial (figura 6.4). Con estos ajustes no solo se espera mejorar el estado físico de la cámara, ya que también se busca dar un mayor aislamiento termino a las muestras durante el tiempo de ejecución de los ensayos.

Figura. 6-4: (a) Vista interior de la cámara donde se identifican los 3 aislamientos utilizados. (b) Montaje de uso de la cámara semi-adiabática (HidroCemUn).



(a)



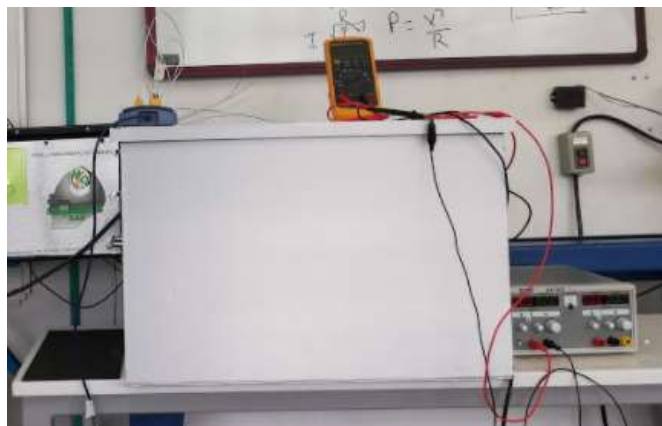
(b)

Nombre de la fuente: Propia.

3. Calibración del equipo

Una vez terminados los ajustes descritos en los numerales 1 y 2 del anexo en desarrollo, es necesario realizar nuevamente la calibración del equipo. Inicialmente, para el proceso de calibración se utilizó el procedimiento descrito por Luisa Peña en su tesis de pregrado, el cual consiste en realizar una comparación entre un valor de calor conocido (calor teórico) y el valor del calor medido por el equipo construido. Esto, nos permitió determinar una curva que relacionará la constante de calibración (K) con la temperatura pico alcanzada durante el ensayo, dándonos así, que dicha constante K será determinada en función de la temperatura pico alcanzada por cada muestra.

Figura. 6-5: Montaje de calibración del equipo HidroCemUn



Nombre de la fuente: Propia.

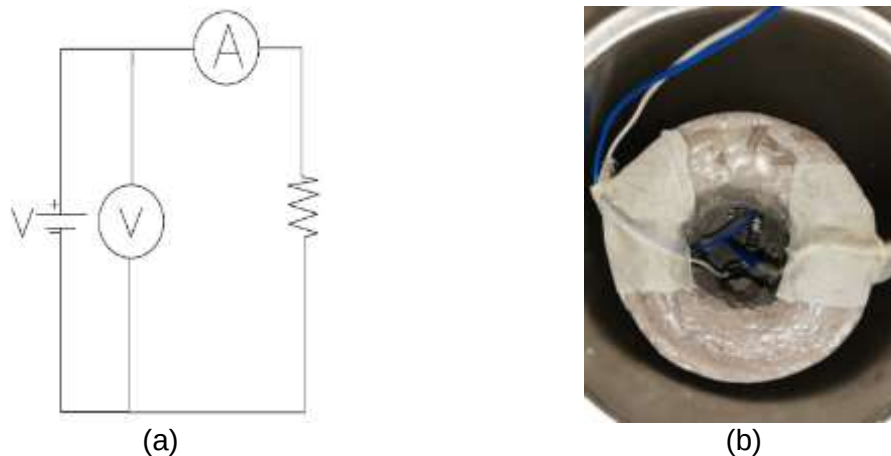
3.1 Desarrollo de la metodología.

Como se mencionó anteriormente, la metodología de calibración consiste en aplicar diferentes valores de calor conocido (calor teórico) en cada uno de los sub-compartimientos del equipo, para lo cual se tuvo como primicia el potencial eléctrico que realiza un circuito de calibración (figura 6-6), el cual consta de una fuente de poder, con la cual se aplicó un voltaje conocido y una resistencia conocida (5A y valores de resistividad que variaron de 82 a 150Ω) la cual se calienta una vez iniciado el flujo eléctrico generando así el calor indicado.

Se debe decir que las resistencias tienen un dipolo simétrico y lineal, lo que quiere decir que estos elementos cumplen con la ley de Ohm (Ec. 1).

$$V (v) = I(A) * R (\Omega) \quad (1)$$

Figura. 6-6: (a) Esquema del circuito eléctrico usado. (b) Resistencia dentro de la probeta de calibración.



Nombre de la fuente: Propia.

Considerando, que nuestro objetivo es conocer el potencial eléctrico del circuito, pero para conocer este valor, es muy importante que conozcamos primero el trabajo eléctrico que realiza el circuito (Ec. 2).

$$W_e(J) = I(A) * t(s) * \Delta V(v) \quad (2)$$

Una vez definido el trabajo eléctrico, el potencial eléctrico se define como:

$$P_e(W) = \frac{W_e(J)}{t(s)} = \frac{I(A) * t(s) * \Delta V(v)}{t(s)} = I(A) * \Delta V(v) \quad (3)$$

Reemplazando la ecuación (1) dentro de lo definido como potencial, tenemos:

$$P_e(W) = I^2(A) * R(\Omega) \quad (4)$$

3.2 Resultados obtenidos

El procedimiento de medición de estos valores en el equipo HidroCemUn, consistió en introducir una resistencia conocida junto con una termocupla (figura 6-6 - b) dentro de un espécimen de calibración (probeta de concreto endurecido con una perforación en el centro) el cual estaba dentro su respectivo vaso Dewar. Utilizando diferentes voltajes, resistencias y deltas de tiempo se obtuvieron los siguientes valores:

Tabla 6-1: Registros del calor teórico en la cámara N°1

Prueba N°	CAMARA N° 1				
	Corriente Voltios (V)	Intensidad Amperios (Amp)	Potencia Watts (W)	Δt (Total) s	Calor Teorico Joules (J)
1	15.00	0.100	1.50	2850	4275.00
2	20.00	0.130	2.60	2100	5460.00
3	15.00	0.105	1.58	1020	1606.50
4	19.99	0.130	2.60	1230	3196.40
5	10.00	0.118	1.18	1050	1239.00
6	15.00	0.150	2.25	780	1755.00
7	15.00	0.128	1.92	720	1382.40
8	20.00	0.165	3.30	780	2574.00
9	10.00	0.115	1.15	660	759.00
10	10.00	0.120	1.20	1140	1368.00
11	12.50	0.140	1.75	720	1260.00
12	17.50	0.195	3.41	720	2457.00
13	10.03	0.118	1.18	630	745.63
14	12.50	0.135	1.69	630	1063.13
15	10.00	0.100	1.00	570	570.00

Tabla 6-2: Registros del calor teórico en la cámara N°2

CAMARA N° 2					
Prueba N°	Corriente	Intensidad	Potencia	Δt (Total)	Calor Teorico
	Voltios (V)	Amperios (Amp)	Watts (W)	s	Joules (J)
1	10.00	0.165	1.65	1530	2524.50
2	10.11	0.066	0.67	1530	1020.91
3	15.09	0.098	1.48	1950	2883.70
4	20.00	0.110	2.20	810	1782.00
5	15.02	0.149	2.24	150	335.70
6	10.00	0.100	1.00	150	150.00
7	12.50	0.136	1.70	210	357.00
8	14.95	0.160	2.39	210	502.32
9	10.00	0.142	1.42	930	1320.60
10	12.50	0.170	2.13	630	1338.75
11	15.01	0.172	2.58	570	1471.58
12	17.51	0.200	3.50	540	1891.08
13	10.00	0.130	1.30	540	702.00
14	12.42	0.145	1.80	360	648.32
15	15.00	0.155	2.33	240	558.00

Tabla 6-3: Registros del calor teórico en la cámara N°3

CAMARA N° 3					
Prueba N°	Corriente	Intensidad	Potencia	Δt (Total)	Calor Teorico
	Voltios (V)	Amperios (Amp)	Watts (W)	s	Joules (J)
1	15.00	0.127	1.91	1350	2571.75
2	20.00	0.176	3.52	1300	4576.00
3	10.11	0.055	0.56	1650	917.48
4	15.10	0.085	1.28	1800	2310.30
5	15.00	0.1	1.50	750	1125.00
6	20.04	0.133	2.67	1890	5037.45
7	14.96	0.191	2.86	570	1628.70
8	12.50	0.170	2.13	450	956.25
9	10.00	0.115	1.15	750	862.50
10	10.03	0.130	1.30	1050	1369.10
11	12.49	0.160	2.00	810	1618.70
12	15.00	0.184	2.76	660	1821.60
13	17.50	0.191	3.34	420	1403.85
14	15.00	0.155	2.33	510	1185.75
15	10.00	0.090	0.90	810	729.00

*Para la definición de los valores establecidos en las tablas 6-1, 6-2 y 6-3, fue necesario realizar un análisis estadístico de la nube de valores que mejor se aproximó a media de la medición con una desviación estándar de aproximadamente el 5%. Esto considerando que para el presente ejercicio fue necesario realizar un aproximado de 30 mediciones donde fueron elegidas las que mejor se ajustaron al procedimiento anteriormente descrito.

Una vez definido el potencial eléctrico o calor teórico del sistema, se calculó el calor real medido por el equipo, este valor se define cómo el área bajo la curva de la gráfica del tiempo (s) vs el delta de temperatura (°C) que se registró durante el ensayo (es importante aclarar que este valor se tomara hasta el tiempo en que el circuito tuvo flujo eléctrico). Una vez calculado el calor medido por el equipo, la constante K se determinó como:

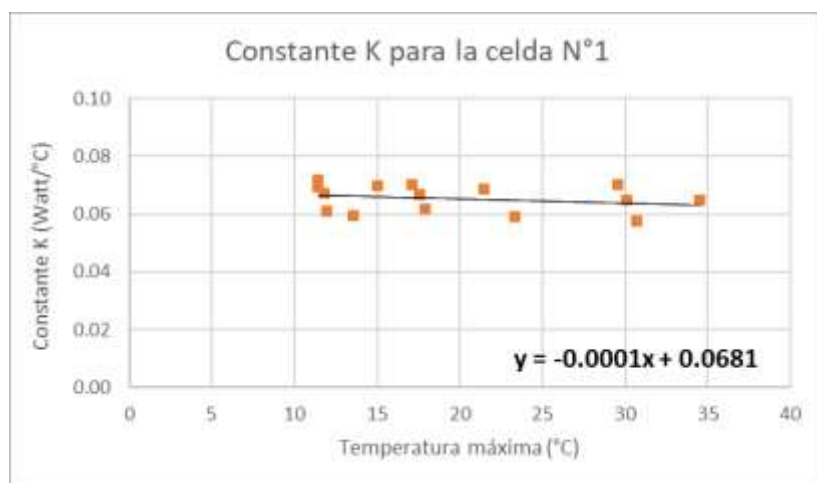
$$K = \frac{\text{Trabajo eléctrico}}{\text{Área bajo la curva}} \quad (5)$$

Debemos aclarar, que al realizar la nueva calibración del equipo y con el fin de seguir las recomendaciones hechas por la norma BS EN 196-9:210. *“Methods of testing cement. Heat of hydration. Semi-adiabatic method”*, dónde se aconseja utilizar la regresión lineal más sencilla en el momento de analizar las curvas de evolución de las constantes. Por esta razón, también se presentarán las gráficas de calibración anterior y así poder compararlas con las nuevas ecuaciones.

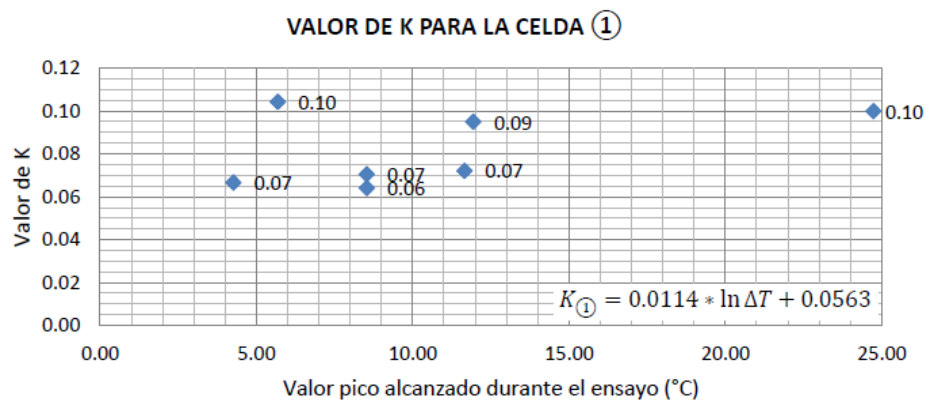
Finalmente, se obtuvieron los siguientes valores de K para cada una de las cámaras, con los que se espera mantener en contrarrestar los errores generados en el cálculo del calor medido a causa de los instrumentos usados.

Tabla 6-4: Cálculo curva de calibración en la cámara N°1.

CÁMARA DE MEDICIÓN N°1				
Prueba N°	Temperatura máx	Calor teórico	Área bajo la curva	Constante K
	(°C)	(J)	(°C*s)	Watt/°C
1	15.09	4275.00	61310.70	0.070
2	30.10	5460.00	84497.30	0.065
3	17.91	1606.50	26106.30	0.062
4	30.71	3196.40	55922.70	0.057
5	11.85	1239.00	18507.90	0.067
6	21.46	1755.00	25699.20	0.068
7	23.36	1382.40	23493.60	0.059
8	34.55	2574.00	39837.90	0.065
9	11.47	759.00	10992.90	0.069
10	13.59	1368.00	23053.50	0.059
11	17.15	1260.00	18009.30	0.070
12	29.62	2457.00	35076.60	0.070
13	11.43	745.63	10443.90	0.071
14	17.59	1063.13	15936.30	0.067
15	11.99	570.00	9398.10	0.061

Figura. 6-7: Curvas de calibración para cámara N°1. (a) Actual. (b) Desarrollada el 2011.

(a)



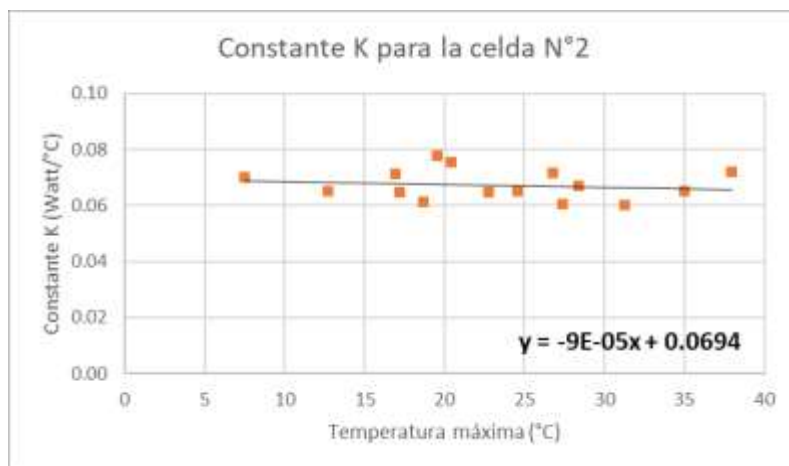
(b)

Nombre de la fuente: (b) Peña, Luisa. 2011

Tabla 6-5: Calculó curva de calibración en la cámara N°2.

CÁMARA DE MEDICIÓN N°2				
Prueba N°	Temperatur a máx	Calor teórico	Área bajo la curva	Constante K
	(°C)	(J)	(°C*s)	Watt/°C
1	27.40	2524.50	41763.82	0.060
2	7.54	1020.91	14643.27	0.070
3	16.94	2883.70	40501.09	0.071
4	20.42	1782.00	23702.18	0.075
5	38.00	335.70	4672.36	0.072
6	12.72	150.00	2309.45	0.065
7	19.56	357.00	4605.27	0.078
8	26.81	502.32	7018.64	0.072
9	18.74	1320.60	21669.27	0.061
10	24.58	1338.75	20580.00	0.065
11	31.28	1471.58	24566.73	0.060
12	35.04	1891.08	29162.18	0.065
13	17.26	702.00	10899.82	0.064
14	22.79	648.32	10069.91	0.064
15	28.40	558.00	8325.27	0.067

Figura. 6-8: Curvas de calibración para cámara N°2. (a) Actual. (b) Desarrollada el 2011.



(a)

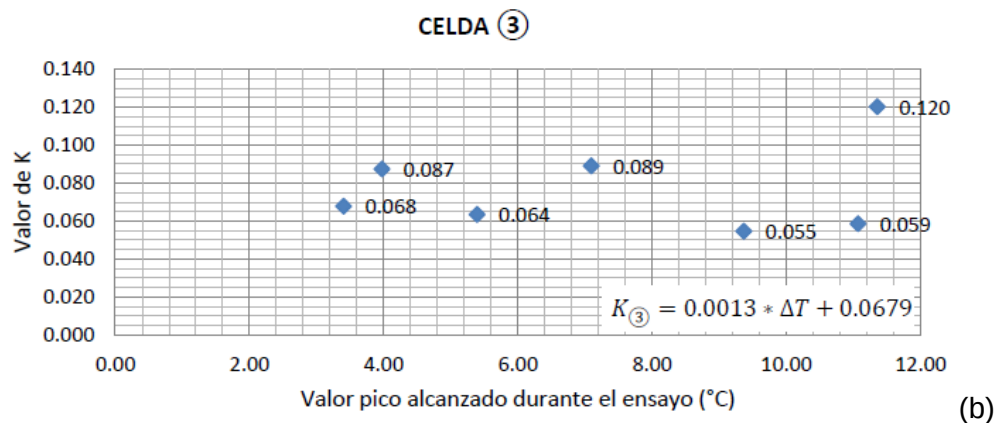
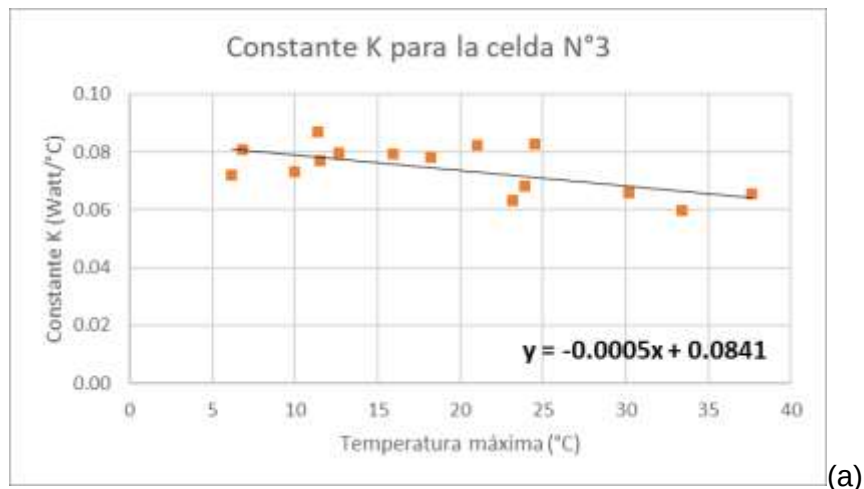


(b)

Nombre de la fuente: (b) Peña, Luisa. 2011

Tabla 6-6: Calculó curva de calibración en la cámara N°3.

CÁMARA DE MEDICIÓN N°3				
Prueba N°	Temperatur a máx	Calor teórico	Área bajo la curva	Constante K
	(°C)	(J)	(°C*s)	Watt/°C
1	23.22	2571.75	40830.63	0.063
2	37.68	4576.00	70230.37	0.065
3	6.16	917.48	12795.00	0.072
4	12.69	2310.30	29157.19	0.079
5	11.42	1125.00	12996.25	0.087
6	30.25	5037.45	76882.81	0.066
7	33.45	1628.70	27358.44	0.060
8	23.96	956.25	14111.88	0.068
9	9.98	862.50	11809.38	0.073
10	11.54	1369.10	17813.13	0.077
11	15.97	1618.70	20459.06	0.079
12	21.07	1821.60	22160.94	0.082
13	24.53	1403.85	17034.06	0.082
14	18.22	1185.75	15231.88	0.078
15	6.84	729.00	9046.25	0.081

Figura. 6-9: Curvas de calibración para cámara N°3. (a) Actual. (b) Desarrollada el 2011.

Nombre de la fuente: (b) Peña, Luisa. 2011

Como se puede observar, actualmente se utiliza un mayor número de datos para la calibración de las cámaras, al igual que un rango de temperaturas máximas más amplio, lo cual nos permitirá una mayor confiabilidad en el cálculo de la constante K a la hora de realizar diferentes ensayos.

3.3 Comparación de datos y ajuste de curvar de calibración.

Una vez terminada la calibración del equipo, se realizaron mediciones comparativas en 8 muestras de cemento Pórtland (tabla 6-7), las cuales fueron medidas bajo las mismas condiciones de ensayo en un calorímetro adiabático de referencia I-CAL 8000 fabricado por la compañía CALMETRIX, perteneciente a la empresa Toxement S.A., quienes nos facilitaron el uso de este para realizar algunos ensayos comparativos contra el equipo HidroCemUN, esto con el fin de validar los resultados obtenidos en el quipo desarrollado en la Universidad Nacional de Colombia.

Tabla 6-7: Muestras y relaciones A/C utilizadas para los ensayos de calibración

Muestra	Relación A/C
1	0.29
2	0.30
3	0.31
4	0.32
5	0.33
6	0.34
7	0.35
8	0.36

A continuación, se presentan los resultados de los dos equipos I-CAL 8000:

Figura. 6-10: Curvas de calor acumulado logradas en el equipo N°1 I-CAL 8000.

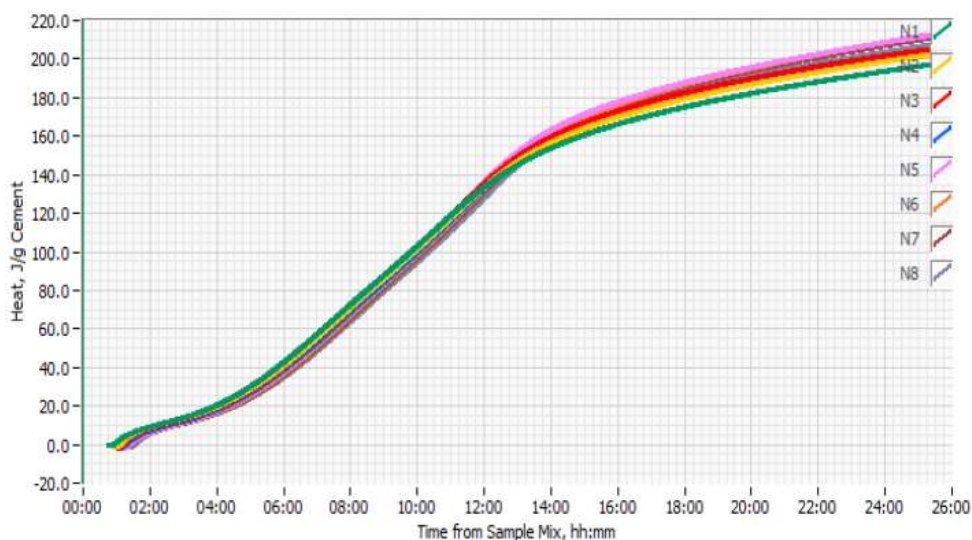
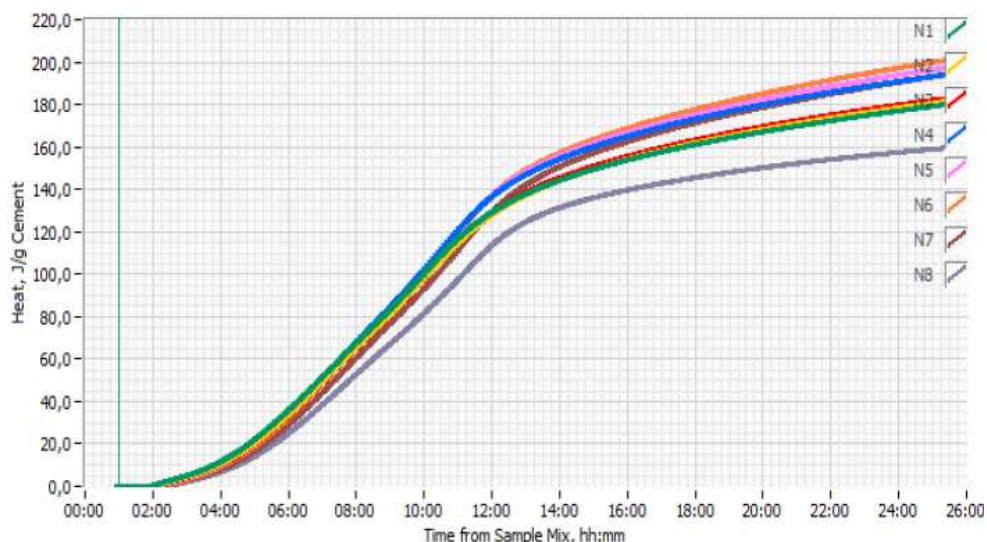
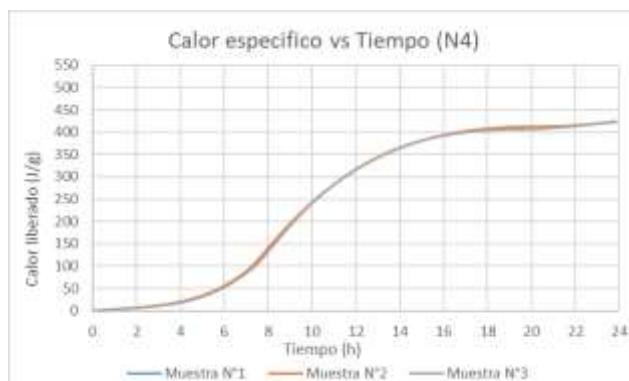


Figura. 6-11: Curvas de calor acumulado logradas en el equipo N°2 I-CAL 8000.



Valores que fueron comparados con los logrados por el equipo HidroCemUN. Es impórtate mencionar que, por disponibilidad de material al momento de realizar los ensayos, solo pudieron ser evaluadas las muestras n° 4, 5, 6, 7 y 8.

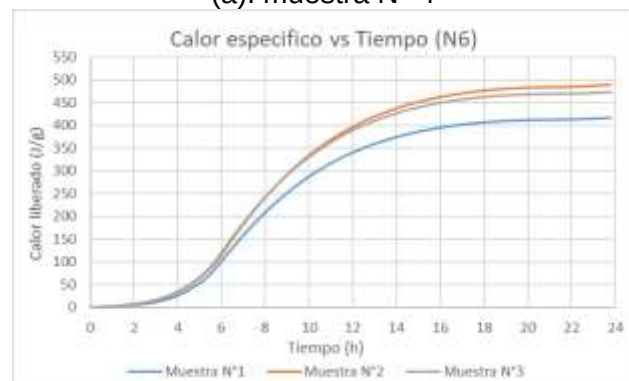
Figura. 6-12: Curvar de calor acumulado en el equipo HidroCemUN para las muestras N° 4, 5, 6, 7 y 8.



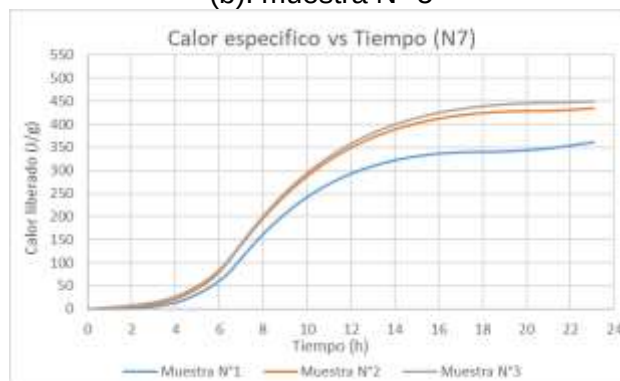
(a). muestra N° 4



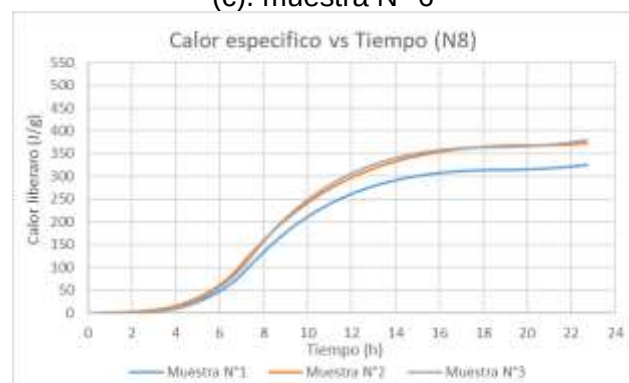
(b). muestra N° 5



(c). muestra N° 6



(d). muestra N° 7

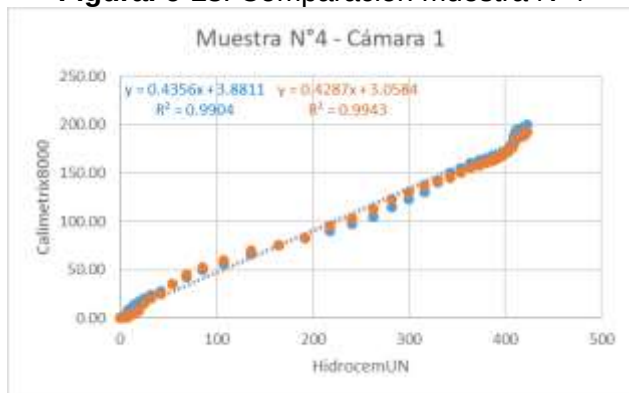


(e). muestra N° 8

Una vez terminado el ejercicio, se puede observar que a pesar de que tanto las curvas logradas por los equipos I-CAL 8000 como las logradas por HidroCemUN presentan una

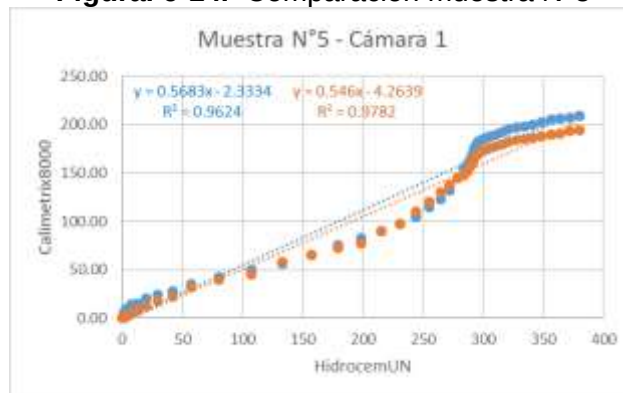
tendencia similar, donde las magnitudes del calor liberado alzadas por este último equipo pueden llegar a estar incrementadas en un factor de hasta 2 veces las alcanzadas por el I-CAL 8000. Razón por la cual se comparan los valores obtenidos por cada una de las cámaras para así analizar la correlación de resultados entre estos equipos y el factor de corrección entre ellos; obteniendo los siguientes resultados (figuras 6-13 a 6-17).

Figura. 6-13. Comparación muestra N°4

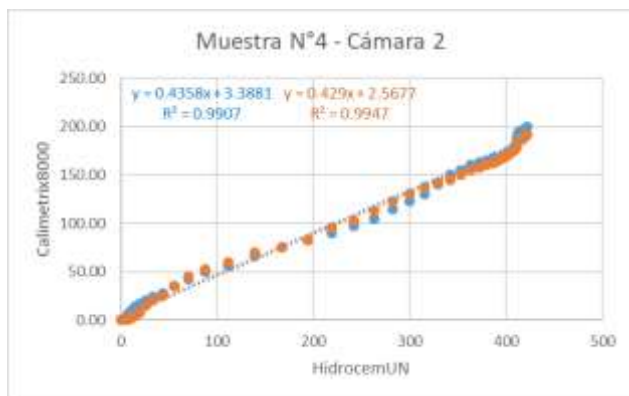


(a)

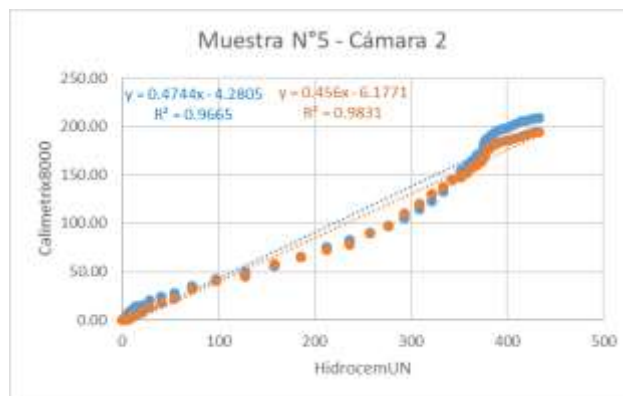
Figura. 6-14. Comparación muestra N°5



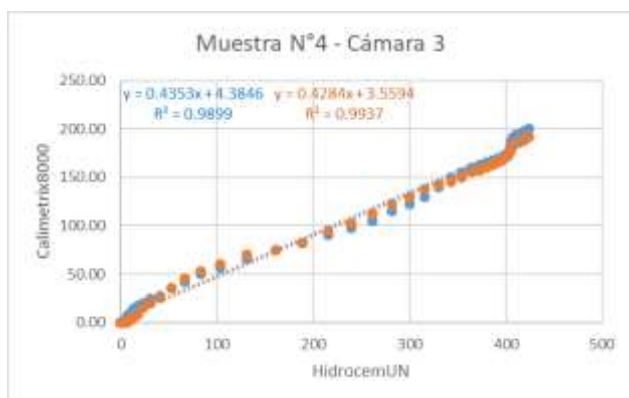
(a)



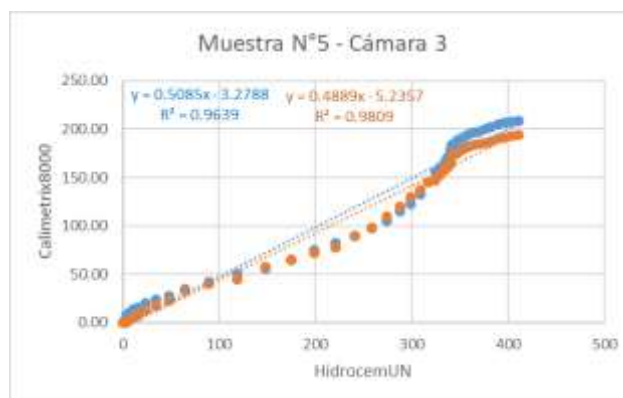
(b)



(b)

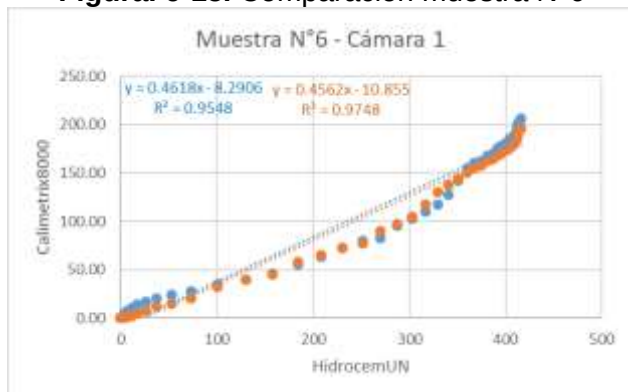


(c)



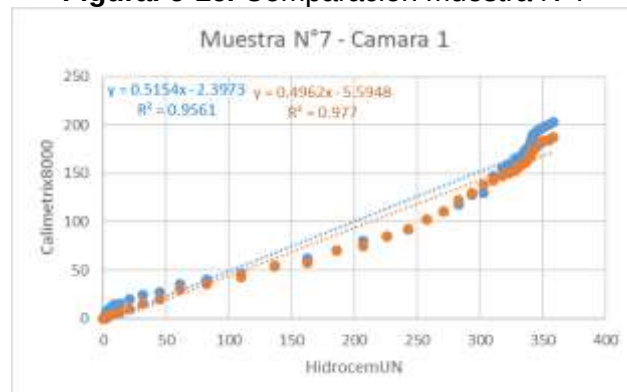
(c)

Figura. 6-15. Comparación muestra N°6

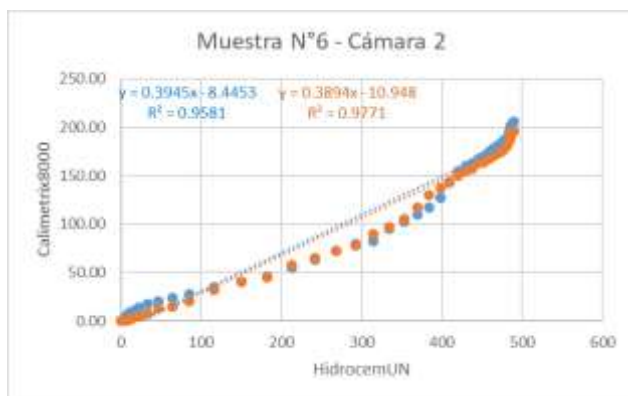


(a)

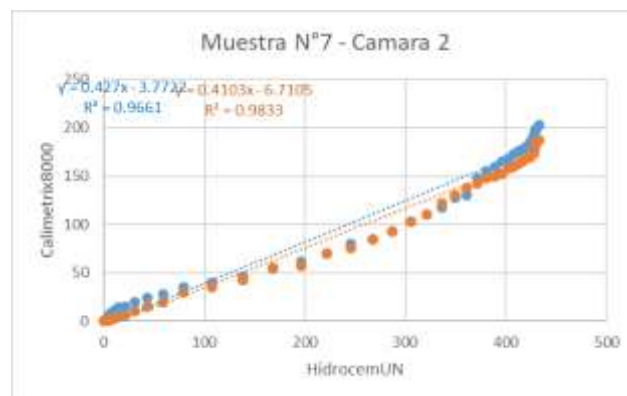
Figura. 6-16. Comparación muestra N°7



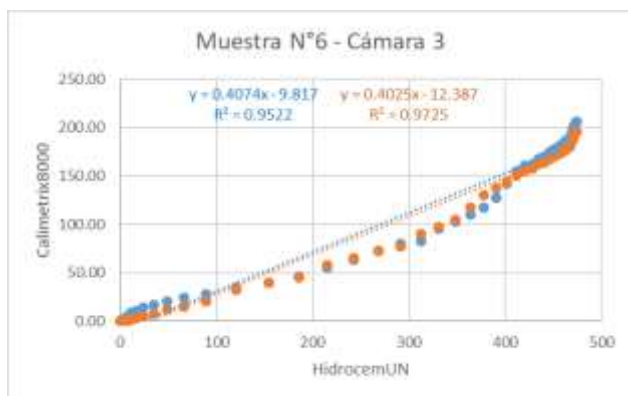
(a)



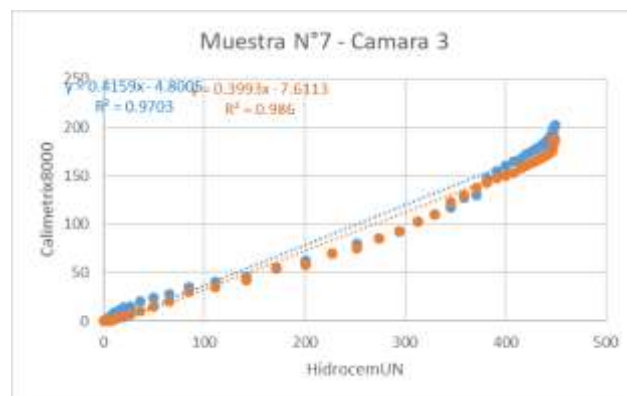
(b)



(b)

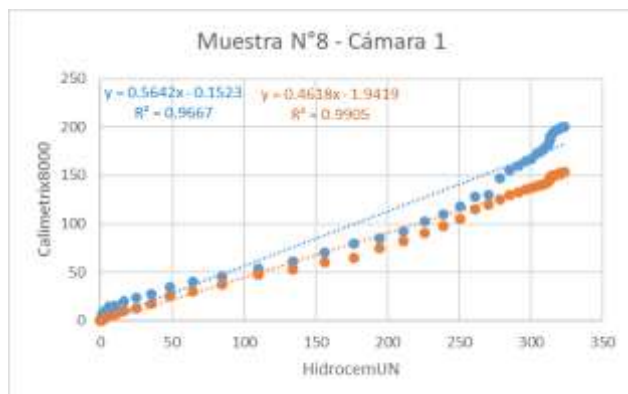


(c)

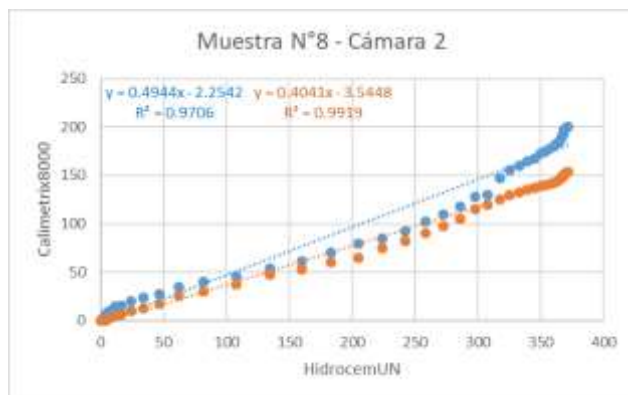


(c)

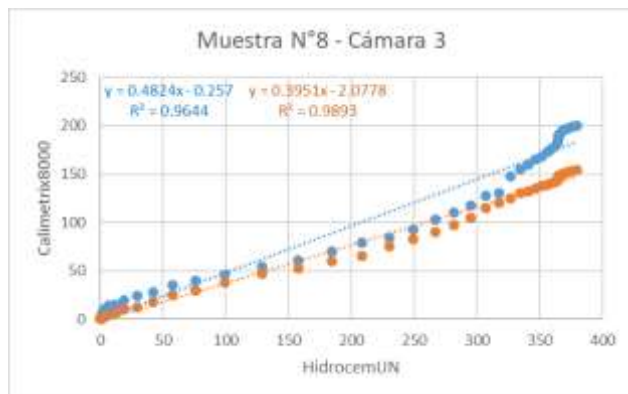
Figura. 6-17. Comparación muestra N°8



(a)



(b)



(c)

Este grupo de imágenes nos representan con la tendencia color azul la comparación entre los valores obtenidos en el calorímetro HidroCemUN (eje X) Vs los valores del Calorímetro I-CAL8000 N°1 (eje Y), por su parte con la tendencia de color anaranjado los valores obtenidos en calorímetro HidroCemUN (eje X) Vs los valores del Calorímetro I-CAL8000 N°2 (eje Y); donde a partir de dicho ejercicio se logra obtener para cada una de las cámaras una constante de corrección para los valores de calor liberado acumulado (ver tabla 6-1).

Tabla N° 6-1: Constantes de corrección para calorímetro HidroCemUN.

Prueba	Cámara 1	Cámara 2	Cámara 3
Prueba 1.1	0.5642	0.4944	0.4824
Prueba 2.1	0.5154	0.4270	0.4159
Prueba 3.1	0.4618	0.3945	0.4074
Prueba 4.1	0.5683	0.4744	0.5085
Prueba 5.1	0.4356	0.4358	0.4353
Prueba 1.2	0.4618	0.4041	0.3951
Prueba 2.2	0.4962	0.4103	0.3993
Prueba 3.2	0.4562	0.4560	0.4025
Prueba 4.2	0.5460	0.4290	0.4889
Prueba 5.2	0.4287	0.3894	0.4284
Constante K	0.4934	0.4315	0.4364

Una vez realizadas las pruebas con dichas constantes se procede a evaluar el porcentaje de error para cada una de las lecturas extraídas de las gráficas por equipos I-CAL 8000. Cómo se puede ver en la figura 6-18 el porcentaje de error entre los equipos tiende a variar en un promedio del 10%, donde durante las primeras 12 horas del ensayo este valor tiende a ser más variable. Es importante tener en cuenta que para la comparación de los datos logrado por el equipo I-CAL 8000, fue necesario realizar interpretación de las gráficas mediante programas como AutoCAD debido a que el software de dicho equipo no permite extraer las bases de datos; lo cual está fuertemente ligado con el generar que el valor de error pueda llegar a ser mayor del real debido a la interpretación que se dan a estos datos parte de un análisis netamente visual. Sin embargo, una vez estudiadas las gráficas finales de ambos equipos se tiene satisfacción en los resultados alcanzados.

Figura. 6-18: Variación del porcentaje de error para el equipo HidroCemUN vs los equipos I-CAL8000



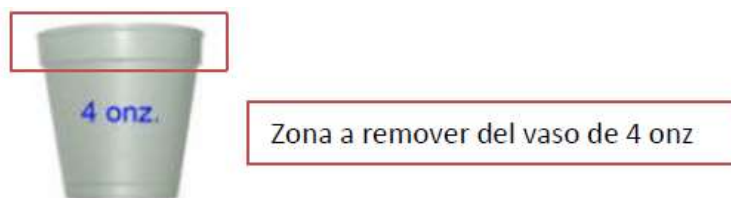
Nombre de la Fuente: Propia

4. Montaje de ensayo y análisis de resultados.

Las recomendaciones a tener en cuenta en el momento del ensayo son las siguientes:

- Utilizar vasos de icopor de 4oz, a los cuales deberá removérseles en anillo superior, tal como se indica en la figura 6-10.

Figura. 6-19: Vaso para realización de ensayos.



Nombre de la Fuente: Propia

- Encender el equipo de registro de datos de temperatura, preferiblemente de 10 a 15min antes de iniciar el ensayo, esto con el fin de estabilizar las lecturas (para la configuración de toma de datos, se recomienda la presentada en el tutorial anexo).

- Realizar el proceso de mezclado según la norma NTC 112. Teniendo en cuenta que las probetas serán de aproximadamente 4 onzas, por tanto, para cada probeta se estima una cantidad de muestra de aproximadamente 100gr.
- Depositar las muestras dentro de las respectivas cámaras, introduciendo las puntas de las termocuplas en el centro de la muestra y a una profundidad media de la misma.
- Cerrar los vasos Dewar poniendo dos parches de papel foamy (figura 6-11), para así cerrar el equipo con las dos tapas de seguridad. Por último, inicial el registro de datos.

Con el fin de facilitar el montaje, procesamiento y análisis de los datos obtenidos del ensayo, en el presente documento se encuentra anexo un tutorial y una hoja de cálculo programada para dichos procesos.



Macros
Calorimetrias.xlsm



Tutorial uso de hoja
de excel.pdf

B. Anexo: Ensayos de caracterización de los materiales

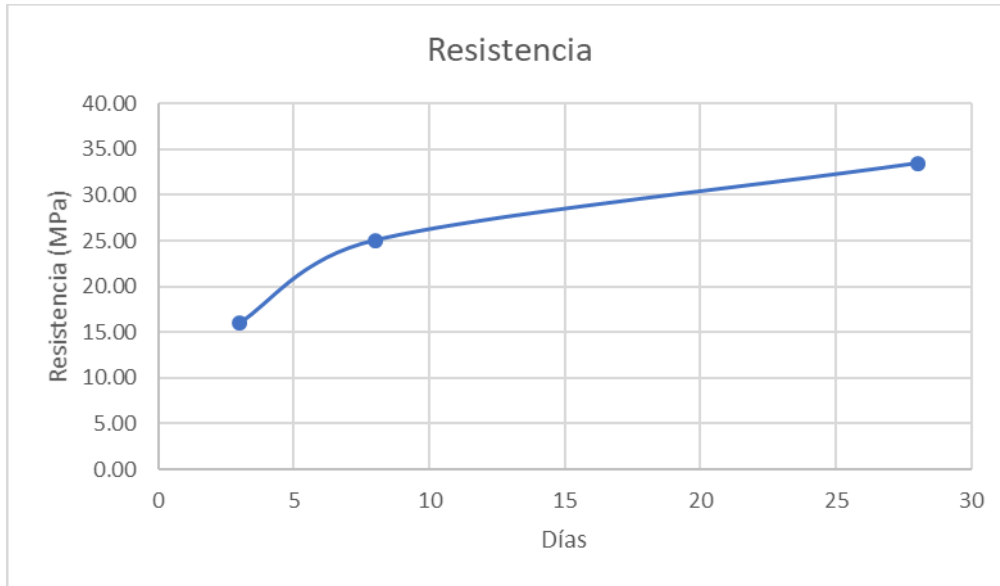
1. Caracterización del cemento

Para la caracterización del material cementante, se realizaron ensayos de resistencia a 3, 7 y 28 días, al igual que la verificación de porcentajes de inquemados en la muestra. Para esto se tomaron como referencia los requerimientos especificados en las tablas 1, 3 y 4 de la normativa ASTM C150 / C150M – 20 “Standard Specification for Portland Cement”. A continuación, se muestran los resultados obtenidos.

- Ensayos de resistencia a la compresión: para la verificación de la resistencia a compresión, se siguieron los lineamientos dados por las normas NTC 220 “Determinación de la resistencia de morteros de cemento hidráulico utilizando cubos de 50mm o 50.8mm de lado” y ASTM 109M - 2002 “*Standard test method for compressive strength of hydraulic cement mortars*”. En el ensayo, se utilizaron nueve (9) cubos de mortero de 50x50x50mm fabricados con el cemento hidráulico en estudio y una arena colombiana de características muy similares a la recomendada en dichas normas. Las muestras fueron ensayadas a 3, 8 y 28 días, obteniendo los siguientes resultados.

Tabla 6-8: Registro de control de resistencia a la compresión, muestra de cemento utilizada para calibración de la cámara semi-adiabática.

Día	Fuerza (kgf)	Esfuerzo (Mpa)	Esfuerzo (Mpa)
3	3555.46	14.22	16.08
	4485.24	17.94	
	1268.77	5.08	
8	6034.44	24.14	25.06
	6494.76	25.98	
	4327.53	17.31	
28	8611.48	34.45	33.42
	6787.49	27.15	
	8100.56	32.40	

Figura. 6-20: Evolución de la resistencia a la compresión.

- Prueba de inquinados: para la verificación del porcentaje de inquinados en una muestra de cemento hidráulico, se tomó como referencia la norma ASTM C114-18 “*Standard Test Methods for Chemical Analysis of Hydraulic Cement*”, la cual consiste en llevar una muestra del cemento a temperaturas de 950°C por aproximadamente 25min (realizando pesajes de la muestra a los 15, 20 y 25min), obteniendo los siguientes resultados:

Tabla 6-9: Registro de ensayo de inquinados en la muestra de concreto.

Muestra	Antes del ensayo			Desp. de 15 min. a 950°C		Desp. de 5 min. más a 950°C		Des. de 5 min. más a 950°C		% Inquemados
	Peso Tapa (g)	Peso Crisol (g)	Peso inicial muestra (g)	Peso tapa, crisol, muestra (g)	Peso muestra (g)	Peso tapa, crisol, muestra (g)	Peso muestra (g)	Peso tapa, crisol, muestra (g)	Peso muestra (g)	
1	2.627	11.444	1.000	15.035	0.965	15.037	0.967	15.036	0.966	3.39%
2	2.649	10.917	1.000	14.532	0.966	14.531	0.966	14.531	0.966	3.42%
3	4.127	11.076	1.000	16.166	0.963	16.168	0.965	16.166	0.963	3.72%

Obtenidos estos resultados, se consideran lo requerimientos de las tablas 1, 3 y 4 de la normativa ASTM C150 / C150M – 20.

Figura. 6-21: Requerimientos estándares de composición

Cement Type ^a	Applicable Test Method	I and IA	II and IIA	II(MH) and II(MH)A	III and IIIA	IV	V
Aluminum oxide (Al ₂ O ₃), max, %	C114	---	6.0	6.0	---	---	---
Ferric oxide (Fe ₂ O ₃), max, %	C114	---	6.0 ^b	6.0 ^{b,c}	---	6.5	---
Magnesium oxide (MgO), max, %	C114	6.0	6.0	6.0	6.0	6.0	6.0
Sulfur trioxide (SO ₃), ^d max, %	C114	---	---	---	---	---	---
When (C ₃ A) ^e is 8 % or less		3.0	3.0	3.0	3.5	2.3	2.3
When (C ₃ A) ^e is more than 8 %		3.5	^f	^f	4.5	^f	^f
Loss on ignition, max, %	C114	---	---	---	---	---	---
When limestone is not an ingredient		3.0	3.0	3.0	3.0	2.5	3.0
When limestone is an ingredient		3.5	3.5	3.5	3.5	3.5	3.5
Insoluble residue, max, %	C114	1.5	1.5	1.5	1.5	1.5	1.5
Equivalent alkalis (Na ₂ O + 0.658 K ₂ O), % †	C114	0	0	0	0	0	0
Tricalcium silicate (C ₃ S) ^g , max, %	See Annex A1	---	---	---	---	35 ^c	---
Dicalcium silicate (C ₂ S) ^g , min, %	See Annex A1	---	---	---	---	40 ^c	---
Tricalcium aluminate (C ₃ A) ^g , max, %	See Annex A1	---	8	8	15	7 ^c	5 ^h
Sum of C ₃ S + 4.75C ₃ A ^h , max, %	See Annex A1	---	---	100 ^{c,i}	---	---	---
Tetracalcium aluminoferrite plus twice the tricalcium aluminate (C ₄ AF + 2(C ₃ A)), or solid solution (C ₄ AF + C ₂ F), as applicable, max, %	See Annex A1	---	---	---	---	---	25 ^h

† Editorially corrected.

^a See Note 2.^b Does not apply when the sulfate resistance limit in Table 4 is specified.^c Does not apply when the cement complies with the heat of hydration limit in Table 4.^d It is permissible to exceed the values in the table for SO₃ content, provided it has been demonstrated by Test Method C1038/C1038M that the cement with the increased SO₃ will not develop expansion exceeding 0.020 % at 14 days. When the manufacturer supplies cement under this provision, supporting data shall be supplied to the purchaser. See Note 7.^e See Annex A1 for calculation.^f Not applicable.^g Report equivalent alkalis as part of the manufacturer's certification. See Note 5.^h See Note 6.

Nombre de la fuente: ASTM C150 / C150M – 20 “Standard Specification for Portland Cement”

Figura. 6-22: Requerimientos físicos estándares

Cement Type ^a	Applicable Test Method	I	IA	II	IIA	II(MH)	II(MH)A	III	IIIA	IV	V
Air content of mortar, ^b volume %:	C189										
max		12	22	12	22	12	22	12	22	12	12
min		---	16	---	16	---	16	---	16	---	---
Fineness, specific surface, m ² /kg											
Air permeability test	C204										
min		260	260	260	260	260	260	---	---	260	260
max		---	---	---	---	430 ^c	430 ^c	---	---	430	---
Autoclave expansion, max, %	C151/ C151M	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80
Strength, not less than the values shown for the ages indicated as follows:											
Compressive strength, MPa [psi]:	C109/ C109M										
1 day		---	---	---	---	---	---	12.0	10.0	---	---
								[1740]	[1450]		
3 days		12.0	10.0	10.0	8.0	10.0	8.0	24.0	19.0	---	8.0
		[1740]	[1450]	[1450]	[1160]	[1450]	[1160]	[3480]	[2760]		[1160]
7 days		19.0	16.0	17.0	14.0	17.0	14.0	---	---	7.0	15.0
		[2760]	[2320]	[2470]	[2030]	[2470]	[2030]			[1020]	[2180]
28 days		---	---	---	---	---	---	---	---	17.0	21.0
										[2470]	[3050]
Time of setting, Vicat test, ^d	C191										
Time of setting, minutes, not less than		45	45	45	45	45	45	45	45	45	45
Time of setting, minutes, not more than		375	375	375	375	375	375	375	375	375	375

^a See Note 2.^b Compliance with the requirements of this specification does not necessarily ensure that the desired air content will be obtained in concrete.^c Maximum fineness limits do not apply if the sum of C₃S + 4.75C₃A is less than or equal to 90, or the cement complies with the heat of hydration limit in Table 4.^d The strength at any specified test age shall be not less than that attained at any previous specified test age.^e The time of setting is that described as initial setting time in Test Method C191.

Nombre de la fuente: ASTM C150 / C150M – 20 “Standard Specification for Portland Cement”

Figura. 6-23: Requerimientos adicionales

Cement Type	Applicable Test Method	I and II	IA and IIA	II(MH)	II(MH)A	III	IIIA	IV	V
False set, final penetration, min, %	C451	50	50	50	50	50	50	50	50
Heat of hydration:									
Isothermal Conduction Calorimetry:									
3 days, max, kJ/kg [cal/g]	C1702	...	...	335 [80] [#]	335 [80] [#] †	...	...	200 [50] ^c	...
7 days, max, kJ/kg [cal/g]		...	...	...	...	...	...	225 [55] ^c	...
Strength, not less than the values shown:									
Compressive strength, MPa [psi]	C109/C109M								
28 days		28.0 [4060]	22.0 [3190]	28.0 [4060]	22.0 [3190]	...	...	...	...
Sulfate resistance, 74 days, max, % expansion	C822	...	...	...	...	...	...	...	0.040
Gilmore test:	C256								
Initial set, min, not less than		60	60	60	60	60	60	60	60
Final set, min, not more than		600	600	600	600	600	600	600	600

† Editorially corrected.

^a These optional requirements apply only when specifically requested. Verify availability before ordering. See Note 2.[#] The limit for the sum of $C_2S + 4.75C_2A$ in Table 1 shall not apply when the cement complies with this limit.^c The limits of C_2S , C_2S , C_2A , and Fe_2O_3 in Table 1 shall not apply when the cement complies with this limit.^d When the sulfate resistance is specified, it shall be instead of the limits of C_2A , $C_2AF + 2 C_2A$, and Fe_2O_3 listed in Table 1.^e Cement meeting the high sulfate resistance limit for Type V is deemed to meet the moderate sulfate resistance requirement of Type II and Type II(MH).

Nombre de la fuente: ASTM C150 / C150M – 20 “Standard Specification for Portland Cement”

Finalmente, para el material clasificando los resultados obtenidos de acuerdo con las tablas mostradas anteriormente, se puede clasificar el cemento como en el rango I – IA, lo que nos indica el alto contenido de pureza de esta muestra cementante.

2. Caracterización de la ceniza

Para la caracterización de la ceniza, se realizaron ensayos de resistencia a 7 y 28 días, al igual que la verificación de porcentajes de inquemados en la muestra. Para esto se tomaron como referencia los requerimientos especificados en las tablas 1 y 2 de la normativa ASTM C618 – 19 “Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete¹”. A continuación, se muestran los resultados obtenidos.

- Ensayos de resistencia a la compresión (caracterización física): para la verificación de la resistencia a compresión, se siguieron los lineamientos dados por las normas NTC 220 “Determinación de la resistencia de morteros de cemento hidráulico utilizando cubos de 50mm o 50.8mm de lado” y ASTM 109M - 2002 “Standard test method for compressive strength of hydraulic cement mortars”. En el ensayo, se utilizaron nueve (6) cubos de mortero de 50x50x50mm fabricados con el cemento hidráulico y adición de ceniza en estudio y una arena colombiana de características muy similares a la recomendada en dichas normas. Las muestras fueron ensayadas a 7 y 28 días, obteniendo los siguientes resultados.

Tabla 6-10: Resultados de resistencia a la compresión

MUESTRA PATRÓN				MUESTRA EN ESTUDIO			
Día	Fuerza (kgF)	Esfuerzo (Mpa)		Día	Fuerza (kgF)	Esfuerzo (Mpa)	
7	5072	19.1	19.3	7	3939	15.0	15.15
	5382	19.5			3976	15.3	
	5227	19.3			3958	15.2	
28	8405	31.9	30.87	28	8027	30.8	30.90
	8137	30.9			7901	31.0	
	8231	29.8			7964	30.9	

Día	% evolución
7	78.5%
28	100.1%

- Caracterización química: para la verificación del porcentaje de óxidos en la muestra de ceniza, se tomó como referencia la norma ASTM C618-19 “Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete1”, obteniendo los siguientes resultados:

Tabla 6-11: Caracterización de la ceniza.

ENSAYO	MÉTODO DE ANÁLISIS	UNIDADES	RESULTADOS
Contenido de óxido de silicio	XRF	%	55.14
Contenido de óxido de aluminio	XRF	%	25.33
Contenido de óxido de hierro	XRF	%	7.76
Contenido de óxido de calcio	XRF	%	1.43
Contenido de óxido de magnesio	XRF	%	0.78
Contenido de óxido de azufre	XRF	%	0.43
Contenido de óxido de sodio	XRF	%	0.18
Contenido de óxido de potasio	XRF	%	1.22
Pérdida al fuego	NTC 184	%	5.43
Finura en pasante en malla No. 325 (45µm)	Granulometría láser	%	76.53
Cóntenido de humedad	Gravimétrico	%	0.46
Densidad	NTC 221	g/mL	2.12

Obtenidos estos resultados, se consideran lo requerimientos de las tablas 1 y 2 de la normativa ASTM C618 - 19.

Tabla 6-12: Requerimientos químicos según ASTM C618.

	Class		
	N	F	C
Silicon dioxide (SiO_2) plus aluminum oxide (Al_2O_3) plus iron oxide (Fe_2O_3), min, %	70.0	70.0	50.0
Sulfur trioxide (SO_3), max, %	4.0	5.0	5.0
Moisture content, max, %	3.0	3.0	3.0
Loss on ignition, max, %	10.0	6.0 ^A	6.0

^AThe use of Class F pozzolan containing up to 12.0 % loss on ignition may be approved by the user if either acceptable performance records or laboratory test results are made available.

Nombre de la fuente: ASTM C618 – 19 “Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete”

Tabla 6-13: Requerimientos físicos según ASTM C618.

	Class		
	N	F	C
Fineness:			
Amount retained when wet-sieved on 45 μm (No. 325) sieve, max, %	34	34	34
Strength activity index: ^A			
With portland cement, at 7 days, min, percent of control	75 ^B	75 ^B	75 ^B
With portland cement, at 28 days, min, percent of control	75 ^B	75 ^B	75 ^B
Water requirement, max, percent of control	115	105	105
Soundness: ^C			
Autoclave expansion or contraction, max, %	0.8	0.8	0.8
Uniformity requirements:			
The density and fineness of individual samples shall not vary from the average established by the ten preceding tests, or by all preceding tests if the number is less than ten, by more than:			
Density, max variation from average, %	5	5	5
Percent retained on 45- μm (No. 325), max variation, percentage points from average	5	5	5

^A The strength activity index with portland cement is not to be considered a measure of the compressive strength of concrete containing the fly ash or natural pozzolan. The mass of fly ash or natural pozzolan specified for the test to determine the strength activity index with portland cement is not considered to be the proportion recommended for the concrete to be used in the work. The optimum amount of fly ash or natural pozzolan for any specific project is determined by the required properties of the concrete and other constituents of the concrete and is to be established by testing. Strength activity index with portland cement is a measure of reactivity with a given cement and is subject to variation depending on the source of both the fly ash or natural pozzolan and the cement.

^B Meeting the 7 day or 28 day strength activity index will indicate specification compliance.

^C If the fly ash or natural pozzolan will constitute more than 20 % by mass of the cementitious material in the project mixture, the test specimens for autoclave expansion shall contain that anticipated percentage. Excessive autoclave expansion is highly significant in cases where water to cementitious material ratios are low, for example, in block or shotcrete mixtures.

Nombre de la fuente: ASTM C618 – 19 “Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete”

Finalmente, para el material clasificando los resultados obtenidos de acuerdo con las tablas mostradas anteriormente, se puede clasificar la ceniza como material clase F.

Bibliografía

- Archie, G. E. (2003). The Electrical Resistivity Log as an Aid in Determining Some Reservoir Characteristics. *SPE Reprint Series*, 55, 9–16.
<https://doi.org/10.2118/942054-g>
- Bentz, D; Clifton, J; Ferraris, C; Garboczi, E. (1999). Transport properties and durability of concrete- Literature review. *Nacional Institute of Standards and Techonology*, May 2014, 709–721.
- Bosques, J. (2010). *Electricidad Básica autor FTS* (Vol. 1).
- Broomfield, J. P. (2007). *Corrosión of steel in croncrete* (Taylor & F).
- Chi, L., Wang, Z., Lu, S., Wang, H., Liu, K., & Liu, W. (2021). Early assessment of hydration and microstructure evolution of belite-calcium sulfoaluminate cement pastes by electrical impedance spectroscopy. *Electrochimica Acta*, 389.
<https://doi.org/10.1016/j.electacta.2021.138699>
- Chi, L., Wang, Z., Lu, S., Zhao, D., & Yao, Y. (2019). Development of mathematical models for predicting the compressive strength and hydration process using the EIS impedance of cementitious materials. *Construction and Building Materials*, 208, 659–668. <https://doi.org/10.1016/j.conbuildmat.2019.03.056>
- Christensen, B. (2001). Time of setting. *International Standards Worldwide STP 169D (ASTM)*, 68(C), 88–101. [https://doi.org/10.1016/S0080-8784\(01\)80194-0](https://doi.org/10.1016/S0080-8784(01)80194-0)
- Claisse, P. (2016). Civil engineering materials. *Elsevier, Oxford, UK*:
- Cruz, J. M., Fita, I. C., Soriano, L., Payá, J., & Borrachero, M. v. (2013). The use of electrical impedance spectroscopy for monitoring the hydration products of Portland cement mortars with high percentage of pozzolans. *Cement and Concrete Research*, 50, 51–61. <https://doi.org/10.1016/j.cemconres.2013.03.019>
- Dunn, R. C., Ross, R. A., & Davis, G. D. (2010). Corrosion monitoring of steel reinforced concrete structures using embedded instrumentation. *NACE - International Corrosion Conference Series*, July.

- Feliu, S., Andrade, C., González, J. A., & Alonso, C. (1996). A new method for in-situ measurement of electrical resistivity of reinforced concrete. *Materials and Structures/Materiaux et Constructions*, 29(6), 362–365.
<https://doi.org/10.1007/bf02486344>
- Fonseca Barrera, L. A. (2016). *Empleo de ceniza volante colombiana como material cementicio suplementario y sus efectos sobre la fijación de cloruros en concretos*. 319. <http://www.bdigital.unal.edu.co/53975/>
- Higuera Flórez, H. C. (2016). *Simulación multifísica y multifase del ensayo de migración del ión cloruro en el concreto (NT Build 492) teniendo en cuenta los fenómenos de adsorción e interacción iónica*. 339.
<http://bdigital.unal.edu.co/57316/1/1016014601.2017.pdf>
- Hu, X., Shi, C., Liu, X., Zhang, J., & de Schutter, G. (2019). A review on microstructural characterization of cement-based materials by AC impedance spectroscopy. *Cement and Concrete Composites*, 100, 1–14.
<https://doi.org/10.1016/j.cemconcomp.2019.03.018>
- Husain, A., Kupwade-Patil, K., Al-Aibani, A. F., & Abdulsalam, M. F. (2017). In situ electrochemical impedance characterization of cement paste with volcanic ash to examine early stage of hydration. *Construction and Building Materials*, 133, 107–117. <https://doi.org/10.1016/j.conbuildmat.2016.12.054>
- Kurdowski, W. (2014). Cement and concrete chemistry. En *Cement and Concrete Chemistry* (Springer, Vol. 9789400779). Springer. <https://doi.org/10.1007/978-94-007-7945-7>
- León, N., Eugenia, L., & Robles, R. (2012). *Evaluación de factores que afectan la aparición de etringita secundaria como simulación del envejecimiento de mezclas de concreto y su papel dentro de procesos de expansión y*. 1–8.
- Li, Z., Wei, X., & Li, W. (2003a). *Preliminary Interpretation of Portland Cement Hydration Process Using Resistivity Measurements*. 100, 253–257.
- Li, Z., Wei, X., & Li, W. (2003b). Preliminary interpretation of portland cement hydration process using resistivity measurements. *ACI Materials Journal*, 100(3), 253–257.
<https://doi.org/10.14359/12627>
- Liu, L., Yang, P., Zhang, B., Huan, C., Guo, L., Yang, Q., & Song, K. I. I. L. (2021). Study on hydration reaction and structure evolution of cemented paste backfill in early-age based on resistivity and hydration heat. *Construction and Building Materials*, 272.
<https://doi.org/10.1016/j.conbuildmat.2020.121827>
- Lizarazo, J. (2018). *CONCRETO HIDRAULICO APLICADO A LA CONSTRUCCIÓN CIVIL* (U. N. de COLOMBIA, Ed.; UNIVERSIDA).

- Lizarazo, J., Higuera, C., & Claisse, P. (2014). Measuring the effect of the ITZ on the transport related properties of mortar using electrochemical impedance. *Construction and Building Materials*, 52, 9–16. <https://doi.org/10.1016/j.conbuildmat.2013.10.077>
- Lizarazo, J., Ramírez, S., & Fonseca, L. (2016). *Influence of the Aggregate – Cement Ratio on the Electrical and Transport Properties of Cement Mortars*. 4901–4909. <https://doi.org/10.1007/s13369-016-2213-4>
- Lizarazo, J., Ramírez, S., & Fonseca, L. (2016). Influence of the Aggregate–Cement Ratio on the Electrical and Transport Properties of Cement Mortars. *Arabian Journal for Science and Engineering*, 41(12), 4901–4909. <https://doi.org/10.1007/s13369-016-2213-4>
- Lizarazo-marriaga, J. (2010). *CONCRETO HIDRAULICO APLICADO A LA CONSTRUCCIÓN CIVIL*.
- Long, G. C., Xie, Y. J., & Jiang, Z. W. (2005). Efficiency of fly ash in cementitious materials. *Advances in Cement Research*, 17(3), 113–119. <https://doi.org/10.1680/adcr.2005.17.3.113>
- Lura, P., Winnefeld, F., & Klemm, S. (2010). *Simultaneous measurements of heat of hydration and chemical shrinkage on hardening cement pastes*. c, 925–932. <https://doi.org/10.1007/s10973-009-0586-2>
- Mehta, P., & Monteiro, P. (2006). *Concrete: Microstructure, Properties and materials*. New York: The McGraw Hill Companies.
- Mendoza, J., Durán, R., & Genescá, J. (2010). Espectroscopía de impedancia electroquímica en corrosión. *Universidad Autonoma de Mexico*, 17(2), 255–258. <https://doi.org/10.3989/collectbot.1989.v17.143>
- Peña, L. N. (2011). Desarrollo de un sistema semi - adiabático para medir calor de hidratación de pastas de cemento y morteros. *Universidad Nacional de Colombia - Sede Bogotá*, 77.
- Ramezaniapour, A. A. (2013). *Cement replacement materials*. New York: Springer.
- Ramirez Arrienta, S. S. (2017). *Caracterización Hidráulica de Mezclas asfálticas Abiertas Mediante la Técnica de Espectroscopía de Impedancia Electroquímica (EIS)*. 163.
- Sadique, M., & Coakley, E. (2016). The influence of physico-chemical properties of fly ash and CKD on strength generation of high-volume fly ash concrete. *Advances in Cement Research*, 28(9), 595–605. <https://doi.org/10.1680/jadcr.15.00103>
- Sanchez de Guzmán, D. (2001). *Tecnología del concreto y del mortero*. (p. 334). BHANDAR EDITORES LTDA.

- Shen, P., Lu, L., He, Y., Wang, F., & Hu, S. (2016). Hydration monitoring and strength prediction of cement-based materials based on the dielectric properties. *Construction and Building Materials*, 126, 179–189. <https://doi.org/10.1016/j.conbuildmat.2016.09.030>
- Suryanto, B., McCarter, W. J., Starrs, G., & Chrisp, T. M. (2017). Characterization of Fly-ash using Electrochemical Impedance Spectroscopy. *Procedia Engineering*, 171, 705–714. <https://doi.org/10.1016/j.proeng.2017.01.414>
- Suryanto, B., McCarter, W. J., Starrs, G., & Ludford-Jones, G. v. (2016). Electrochemical immittance spectroscopy applied to a hybrid PVA/steel fiber engineered cementitious composite. *Materials and Design*, 105, 179–189. <https://doi.org/10.1016/j.matdes.2016.05.037>
- Swaddiwudhipong, S., Chen, D., & Zhang, M. H. (2002). Simulation of the exothermic hydration process of Portland cement. *Advances in Cement Research*, 14(2), 61–69. <https://doi.org/10.1680/adcr.2002.14.2.61>
- Tang, S. W., Cai, X. H., He, Z., Zhou, W., Shao, H. Y., Li, Z. J., Wu, T., & Chen, E. (2017). The review of early hydration of cement-based materials by electrical methods. En *Construction and Building Materials* (Vol. 146, pp. 15–29). Elsevier Ltd. <https://doi.org/10.1016/j.conbuildmat.2017.04.073>
- Tang, S. W., Li, Z. J., Shao, H. Y., & Chen, E. (2014). Characterization of early-age hydration process of cement pastes based on impedance measurement. *Construction and Building Materials*, 68, 491–500. <https://doi.org/10.1016/j.conbuildmat.2014.07.009>
- Wang, X., & Lee, H. (2012). *Heat of hydration models of cementitious materials*. 24(2).
- Wenner, F. (1916). A method of measuring earth resistivity. *Bulletin of the Bureau of Standards*, 12(4), 469. <https://doi.org/10.6028/bulletin.282>
- Whiting, D. A., & Nagi, M. A. (2003). Electrical Resistivity of Concrete - A Literature Review. *Portland Cement Association*, 57.
- Wolter, J. M., Schmeide, K., Huittinen, N., & Stumpf, T. (2019). Retention by calcium silicate hydrate (C-S-H) gel and secondary alteration phases in carbonate solutions with high ionic strength: A site-selective TRLFS study. *Scientific Reports*, 9(1), 1–32. <https://doi.org/10.1038/s41598-019-50402-x>
- Yim, H. J., Lee, H., & Kim, J. H. (2017). Evaluation of mortar setting time by using electrical resistivity measurements. *Construction and Building Materials*, 146, 679–686. <https://doi.org/10.1016/j.conbuildmat.2017.04.088>

- Yousuf, F., Wei, X., & Tao, J. (2017). *Evaluation of the influence of a superplasticizer on the hydration of varying composition cements by the electrical resistivity measurement method*. 144, 25–34.
<https://doi.org/10.1016/j.conbuildmat.2017.03.138>
- Yousuf, F., Wei, X., & Zhou, J. (2020). Monitoring the setting and hardening behaviour of cement paste by electrical resistivity measurement. *Construction and Building Materials*, 252. <https://doi.org/10.1016/j.conbuildmat.2020.118941>
- Yousuf, F., & Xiaosheng, W. (2020). Investigation of the early-age microstructural development of hydrating cement pastes through electrical resistivity measurements. *Case Studies in Construction Materials*, 13.
<https://doi.org/10.1016/j.cscm.2020.e00391>
- Zhu, Y., Zhang, H., Zhang, Z., & Yao, Y. (2017). Electrochemical impedance spectroscopy (EIS) of hydration process and drying shrinkage for cement paste with W/C of 0.25 affected by high range water reducer. *Construction and Building Materials*, 131, 536–541. <https://doi.org/10.1016/j.conbuildmat.2016.08.099>

INDUCED POLARIZATION, A STUDY OF ITS CAUSES*

DONALD J. MARSHALL† AND THEODORE R. MADDEN‡

ABSTRACT

The causes of induced electrical polarization include not only the polarization of metal-solution interfaces, but also effects associated with the coupling of different flows. Electro-osmotic, thermal electric, and ion diffusion effects are among such examples. A study of the physical properties of geologic materials indicates that only electrode interface and diffusion flow phenomena are important sources of induced polarization effects.

It was attempted to find characteristic differences between these two phenomena. Theoretical and experimental considerations show that the kinetic processes involved are quite similar in the two cases. This leads to difficulties in identifying the polarizing agent from electrical measurements, although the effects of well mineralized zones are easily recognized.

INTRODUCTION

During the past ten years an increasing interest has developed around the geophysical applications of electrical measurements that go under the popular name of induced polarization measurements.¹ The primary motivation has been the application of these measurements to the detection of sulphide mineralization. It is believed that the phenomenon taking place when electric current passes through a mineralized zone is essentially the same phenomenon that occurs at a polarized electrode (Bleil, 1953).

A polarized electrode is one that hinders the flow of electric current between the electrode and the solution in which it is immersed. Chemical or electrochemical barriers exist which the current carriers must overcome in order to allow current flow to pass across the electrode-liquid interface. Forcing a net current flow to take place across this interface entails an added voltage drop above and beyond that needed to overcome the ohmic losses in the solution and in the electrode. This additional voltage is called the overvoltage.

The picture that is used to explain the induced polarization effects in mineralized rocks involves the ionic current flow in the rock polarizing the metal particles within the rock. When the inducing current is turned off, the overvoltages that were set up fall off with time. Since it also takes a finite time to build up these overvoltages, one finds that the impedance of these zones decreases with increasing frequency, so that the measurements can also be made in the frequency domain. Qualitatively then, these effects behave somewhat as the ordinary

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† Nuclide Analysis Associates, State College, Pa.

‡ Dept. of Geology and Geophysics—M.I.T.

¹ The term induced polarization is perhaps a poor one, because it has so many different connotations. It would be more descriptive if we used the expression, induced electrical interfacial polarization, but we shall not attempt to force such a tongue-twisting term on our fellow geophysicists. In any case the measurements involve the transient or frequency dependent electrical properties of the ground.

dielectric property of the materials. These effects, however, occur at audio and sub-audio frequencies which are much too low for the ordinary displacement currents to be of any significance. Irrespective of the origin of the phenomenon, any time dependent or frequency dependent behavior of the electrical impedance of rock materials at these low frequencies is referred to as an induced polarization effect.

INDUCED POLARIZATION MEASUREMENTS AND PARAMETERS

Before going on to the more general study of the possible causes of these effects, it seems worthwhile at this point to review some of the measurement techniques and to correlate the widely different ways of describing the observed polarization effects that are in current use. Results of field measurements will also be presented to demonstrate the usefulness of these measurements in mining exploration.

Several different techniques are used to measure these effects, and several different parameters are used to describe the results of these measurements. When the measurements are made in the time domain, it is a common procedure to turn the current source on for a period, and then turn it off for a period before starting a new cycle with opposite polarity. The voltage remaining just at the beginning of the off-period, or at some fixed interval later, is often measured and compared with the on-period voltage. The polarization is evaluated in terms of mv/volt. Another practice consists of integrating the off-period voltage. The polarization effects are then evaluated in terms of mv-sec/volt.

When the measurements are made in the frequency domain it is the usual practice to compare the impedance at some alternating frequency with that of some very low frequency which is often referred to as the dc impedance. The effect is then evaluated as a certain percentage increase in the conductance at the ac frequency. Sometimes phase shift measurements are made. These phase shifts are usually very small, of the order of one degree or less.

The significance and the interrelation of these varied parameters can be better understood if the ideas used to describe the polarization of a mineralized zone are transcribed into an equivalent circuit (see Figure 1).

In Figure 1, R' stands for the impedance of the complete rock section. It is made up of unblocked ionic conduction paths, R , in parallel with other paths, Z , which are blocked by metallic or semi-conducting minerals. R_m stands for the resistance of the electronic conducting minerals, and R_e stands for the resistance of the electrolytes in the blocked pore paths. C is the electrode capacitance and R_{chem} represents the hindrance to the transfer of electrons between the metallic mineral and the solution.

Because time and frequency domain data are related to each other, when the phenomenon is linear, through the Fourier transform, we can expect to derive frequency information from the transient measurements or vice versa. There is not an exact one to one correspondence between a point in the frequency domain,

and a point in the time domain, but there is often an approximate one. The percentage frequency effect and the mv/volt parameters are found to be closely related (Madden and Marshall, 1958). If the percentage decrease in impedance is used instead of the percentage increase in conductivity, the relationship is given as percent effect at frequency $f_i \cong 0.1$ mv/volt value at $t_i = 1/2\pi f_i$.

The integrated decay voltage is a little different. It is not unlikely that $R_{\text{chem}} \gg R_m + R_e$, so that the dc impedance of $Z = Z(0)$ is approximately equal to R_{chem} . If we make this assumption, and if the integration is carried out long enough, we have that the mv-sec/volt value = 1,000 CR.

In all these parameters we have a large role played by R , the resistance of the unblocked paths. Small values of R can short circuit the polarizing effects,

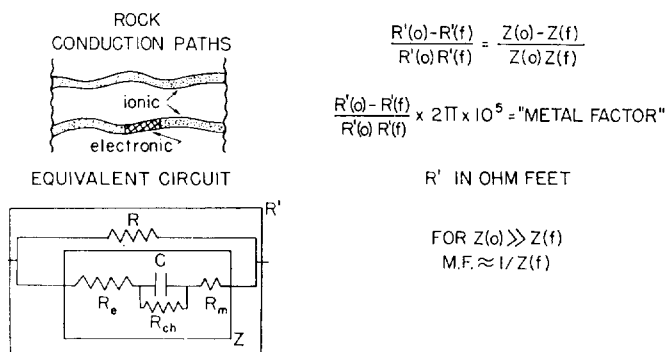


FIG. 1. Diagrammatic representation of section of conducting paths in rock, and the equivalent circuit.

resulting in very small parameter values, while a very tight rock may give large frequency effects without many metallic minerals being present. Since R is in parallel with Z it would seem advisable to look at the change in conductivity of R' , for this is just the change in conductivity of Z . When the measurements are made in the frequency domain, this is simply done, and is equal to the frequency effect divided by the dc resistivity. To give reasonably sized values, this parameter is defined as $2\pi[R'(0) - R'(f)] \times 10^5 / R'(0)R'(f)$ when using units of ohm-feet for the resistivity. It is called the "metal factor" because of its correlation with metallic mineral content. Using the assumptions previously mentioned concerning the parameters of Figure 1, the metal factor, m.f. $\cong 2\pi \times 10^5 / Z(f)$, and is therefore proportional to the ac conductivity of those paths that are blocked off by metallic minerals. The more metal a rock contains, the more paths that will be blocked, and thus the higher the metal factor. As the metallic content continues to increase, the individual paths will have higher ac conductivities, and this two-fold effect starts to skyrocket upwards the values of the metal factor. This is helpful in increasing the signal-to-noise ratio between

the effect of well mineralized zones, and the effect of only slightly mineralized zones. This parameter does favor the rocks having conductive electrolytes in the pores, and thus it is not quite so useful when dealing with sedimentary rocks.

Some typical metal factor values evaluated at 10 cps encountered in igneous areas are shown in Table I.

Similar parameters can be evaluated from the transient measurements. The only one that seems to be used is derived by dividing the mv-sec/volt value by the resistivity. This parameter is essentially proportional to C , the interfacial capacitance per unit cube, but is more often referred to as the effective zero frequency dielectric constant. The frequency measurements could also be evaluated in terms of an effective dielectric constant, but we specifically wish to avoid the

TABLE I
COMMON METAL FACTOR VALUES (10 cps)

Rock Type and Mineralization	Metal Factors
unmineralized granites	<1
unmineralized basic rocks	1-10
finely disseminated sulphides	10-100
disseminated sulphides (1-3%)	100-1,000
fracture filling sulphides (3-10%)	1,000-10,000
massive sulphides	>10,000

use of the term dielectric constant in describing these effects. First of all this so-called effective dielectric constant is not constant, but varies widely as a function of frequency. Secondly, the huge values obtained at low frequencies seem to imply a rather anomalous dielectric phenomenon, while actually the only really anomalous phenomenon is associated with the resistive properties. The interfacial capacitance C is quite large, being of the order of $30 \mu\text{f}/\text{cm}^2$, but is not anomalous since the interfacial layer is so thin. The anomalous feature of this layer is its high resistance, despite the fact that it is very thin. If it were not for this high resistance, the interfacial capacitance, which is no different from the capacitance of a similar thickness of ordinary fluid, would play no role.

In the remainder of this paper the polarizing effects will be referred to the frequency domain, and we shall speak of percentage frequency effects and metal factors.

FIELD RESULTS

The increased resolving power of induced polarization measurements over ordinary resistivity measurements in detecting sulphides is demonstrated by some field examples shown in Figures 2-7. The data were taken using 100 ft dipoles for both the sender and receiver. The data are plotted with the abscissa representing the separation between sender and receiver, and the ordinate repre-

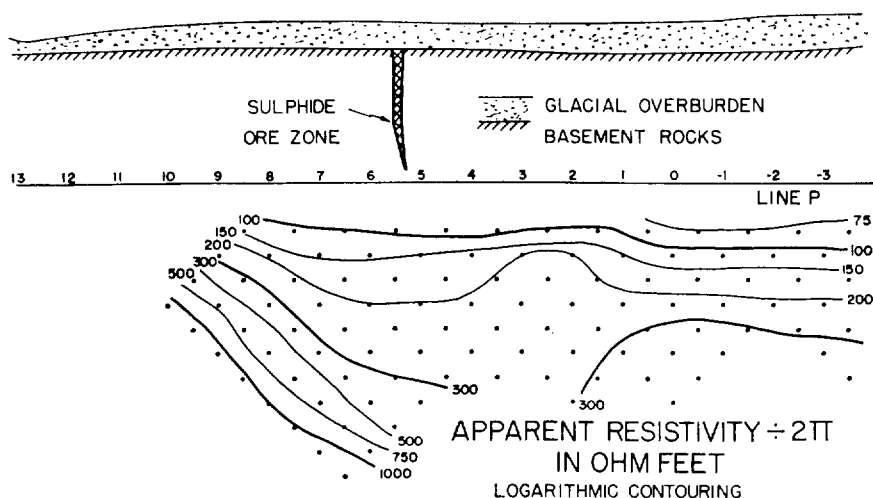


FIG. 2. Resistivity measurements using colinear dipoles.

senting the center position of the spread. In this way a sort of cross-section of the measured apparent electrical parameters are presented. Of course it is not an exact mapping, but it bears certain similarities (Hollof, 1957). The actual measured values have not been shown, merely being represented by dots, but the contours of these values are drawn in. In all these examples there is a clear cut advantage to the induced polarization measurements.

The frequency spectrum of these induced polarization effects are spread out smoothly enough so that the impedance, even though varying with frequency, remains almost purely resistive at any one frequency. This allows one to use

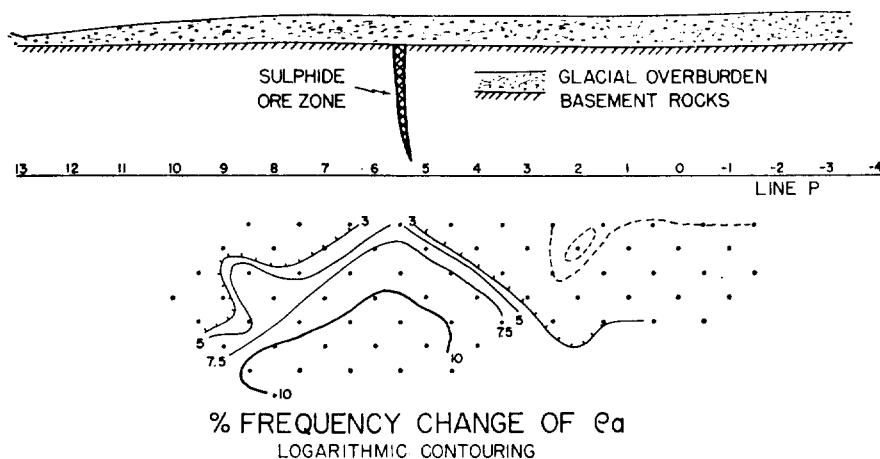


FIG. 3. Frequency variations of apparent resistivity at 60 cps.

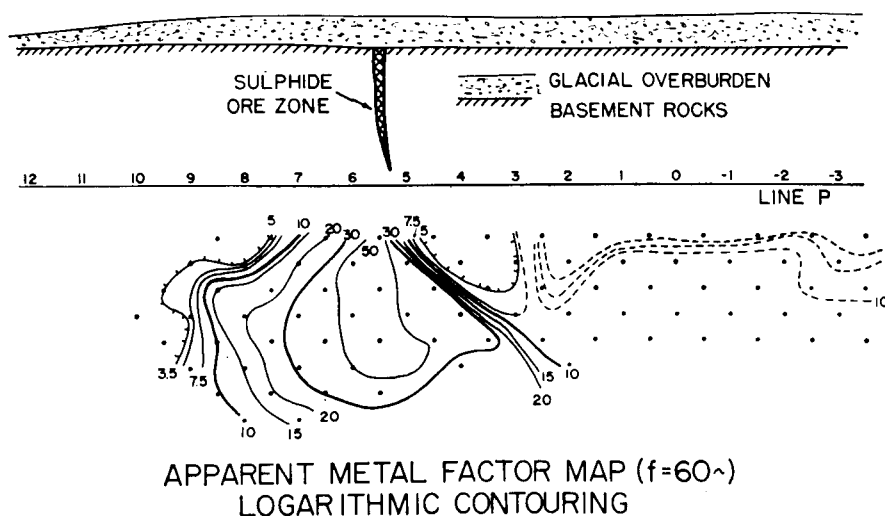


FIG. 4. Metal factor values computed from field measurements.

ordinary resistivity type curves to predict or interpret the apparent polarization effects of a given zone (Hallof, 1957).

COMPLICATIONS TO THE SIMPLE THEORY

The application of induced polarization measurements to the detection of sulphide mineralization is not as simple as it has been assumed in the previous discussion. One reason, of course, is that other semi-conducting minerals such as graphite, magnetite, and pyrolusite cause similar polarization effects. Polari-

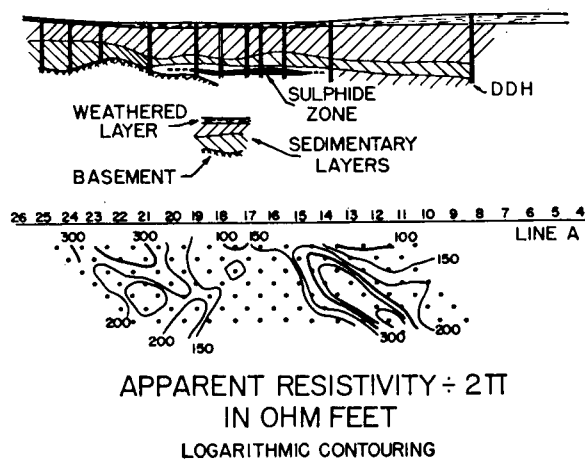


FIG. 5. Resistivity measurements using colinear dipoles.

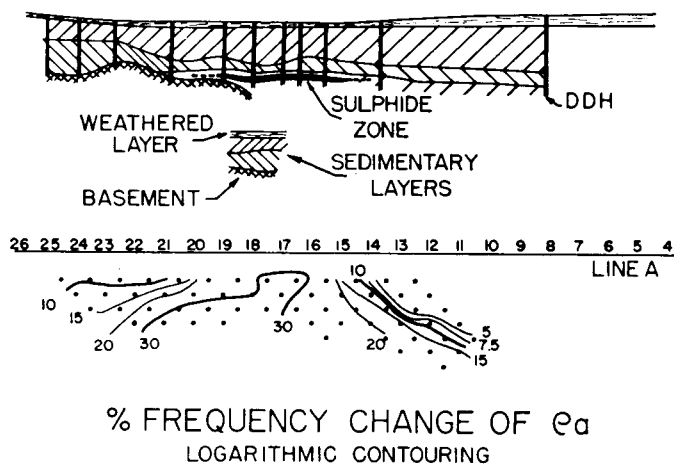


FIG. 6. Frequency variations of apparent resistivity at 60 cps.

zation can also occur, however, without the presence of any semi-conducting minerals. Schlumberger, in perhaps the first reference to induced polarization methods, stated that background polarization effects drowned out the effect of the sulphide minerals (Schlumberger, 1920). This view is somewhat exaggerated, but Vacquier in his work showed that clay minerals could lead to finite polarization effects (Vacquier et al., 1957). Other groups have also been observing these effects (Henkel and Van Nostrand, 1957; Brant, personal communication).

Because of the presence of these complicating factors, this present study was taken up. It was hoped that through an increased understanding of the causes of induced polarization in geologic materials, one could better interpret the electri-

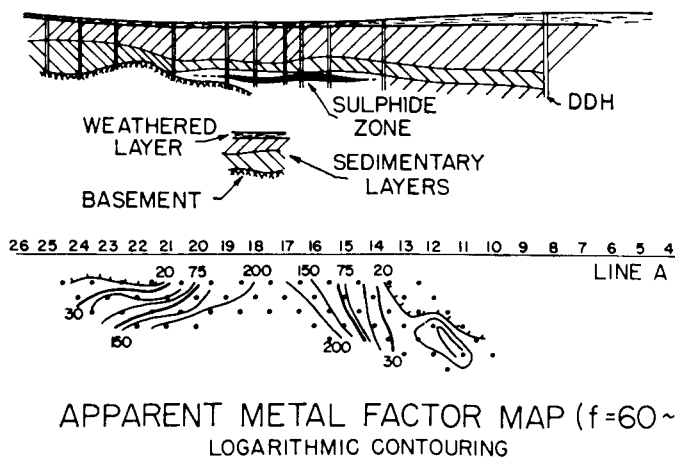


FIG. 7. Metal factor values computed from field measurements.

cal measurements. It seems appropriate then to attack this problem from a fresh point of view, starting with more general concepts rather than a specific model such as that given in Figure 1.

POLARIZATION AS ENERGY STORAGE

The observation that we have is that the polarizable materials are capable of maintaining, for a certain time, an electric current flow after an applied current field is turned off. Actually what is observed is a voltage gradient, and this does not guarantee that electric current is flowing, as will be apparent later, but some sort of flow will be taking place. This is a manifestation that energy is stored in the medium when current is passed through, and usually at least some of this energy is released by maintaining electric current flow after the driving field is turned off. The five ordinary forms of energy storage are, electric, magnetic, mechanical, thermal, and chemical. It is conceivable that any one or all of these are involved in the phenomenon. The picture presented in Figure 1 assumed that the energy was stored as electric energy in the capacitor C . This is the simplest and most direct form of setting up an electric polarization effect.

Another obvious form of energy storage is in the magnetic field. It is known that an electric current always sets up a magnetic field, and that when the current is turned off, the collapsing magnetic field returns its stored energy back into electrical energy. This effect can be very complicated when the geometry of the current flow is irregular, but for simple half-space geometries the solutions are well known (Sunde, 1949). When colinear spreads are used to make the field measurements, these electromagnetic effects behave qualitatively much like the polarization effects of mineralized zones. A two-fold increase of apparent conductivity takes place as the frequency is increased. The frequencies at which the effect is observed, however, are usually quite high, unless large separations in conductive areas are involved. For a separation of 2,000 ft in an area whose resistivity is 300 ohm-ft, no appreciable effect is observed until frequencies of 10 cps or higher are used. When the ground conductivities are not homogeneous, the electromagnetic effects can become quite different. This is especially true when horizontal variations of conductivity exist. Measurements have been made in the field where the apparent resistivity increased by more than three fold as the frequency was increased (Madden et al., 1957). These effects are again limited to the higher frequencies, however.

POLARIZATION THROUGH THE COUPLING OF FLOWS

The electromagnetic effects can be avoided, as has been mentioned, by using low enough frequencies, but anomalous polarization effects are still found when low frequencies are used. The possible effects of mechanical, thermal, and chemical energy storage must be investigated therefore. To store energy in these forms, a coupling must exist between the electric current flow and other flows such as heat and matter. The existence of such a coupling will then allow the stored

energy on its release to cause electric current flow again, and thus the phenomenon will be a form of induced electric polarization. A simple illustration of this idea is given in Figure 8. For instance, through electro-osmotic coupling, an electric current flow will cause a solvent flow through a capillary system. This flow may build up a hydrostatic pressure, either by building up a hydrostatic head against gravity, or perhaps through the blocking action of an impermeable bed. When the electric current is turned off, the hydrostatic pressure built up will drive the solvent back, and the solvent flow in turn will cause an electric current flow.

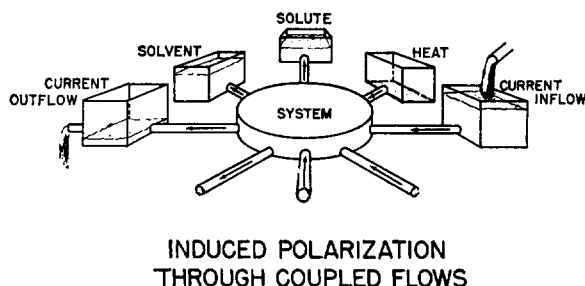


FIG. 8. Pictorial representation of the coupling of flows and the development of counter forces in a system.

This flow almost immediately builds up an electric potential which opposes any further current flow, and which is observed as a polarization voltage.

In a similar way any heat flow induced by the current flow can store energy thermally if temperature differences are set up. The solute flow can cause concentration differences to build up, and this represents a chemical energy.

The study of such couplings and their interrelations is the main area of investigation of steady-state thermodynamics. According to the principles of steady-state thermodynamics, the equations describing a general electrolyte system with coupling are²

$$\begin{aligned} \text{flow cations} &= J_p \\ \text{flow anions} &= J_n \\ \text{flow solvent} &= J_s \\ \text{flow heat} &= J_q \end{aligned} = \begin{bmatrix} L_{11} & L_{12} & L_{13} & L_{14} \\ L_{21} & L_{22} & L_{23} & L_{24} \\ L_{31} & L_{32} & L_{33} & L_{34} \\ L_{41} & L_{42} & L_{43} & L_{44} \end{bmatrix} \begin{bmatrix} -\nabla\mu_p - FZ_p\nabla\phi \\ -\nabla\mu_n - FZ_n\nabla\phi \\ -\nabla P \\ -\nabla T/T \end{bmatrix} \quad (1)$$

where

ϕ = electric potential; μ = chemical potential; P = pressure;
 T = temperature; $L_{ij} = L_{ji}$; F = Faraday's constant; and Z = ion valence.

² This is not the form of these equations usually found, but we follow Eckart here in an attempt to avoid the presence of arbitrary potential terms that appear in the formulations given in Denbigh and DeGroot (Eckart 1940, Denbigh 1951, DeGroot 1952).

The column on the left represents the flow vectors, while the column on the right of (1) represents the generalized forces. The L_{ij} matrix consists of the terms relating to the properties of the media such as the conductance and permeability.

These relationships assume a linear behavior of the medium, and this is always observed for small-flow densities. Onsager's relationships, $L_{ij} = L_{ji}$, is based on extensions of thermodynamics to situations involving slight deviations from equilibrium. It also appears to be verified experimentally for small-flow densities. This relationship is of great help in organizing the experimental results.

From these relationships the possible magnitudes of electrical polarization can be deduced in terms of the coupling coefficients. Many of these coefficients are well known from other studies. For instance, the solvent-electric current flow coupling known as electro-osmosis or streaming potential has been studied by geophysicists because of the effects on self-potential logging (Wylie, 1955). The soil mechanics engineers also have made such studies because of the possible applications in soil consolidation (Casagrande, 1952).

The thermo-electric properties are not as well known, although at present a great effort is being made to develop efficient thermo-electric materials because of their possible use in energy conversion. The measure of this efficiency is exactly the same parameter that evaluates its induced polarization capabilities. These studies are concentrating on semi-conducting materials. It is quite apparent, from data to be shown later, that geologic materials do not have the required properties.

The coupling effects between solute flow and electric currents are also very well known, being an important part of the subject matter of electro-chemistry. These effects are familiar to geophysicists using self-potential measurements, for they are represented by the familiar diffusion potential phenomena.

ELECTRO-OSMOTIC COUPLING

To investigate the possible influence of electro-osmotic coupling on induced electrical polarization we can rewrite equation (1), assuming that all the induced forces except ∇P are zero, and ignoring any dependence of μ on pressure. We also combine J_p and J_n to give us the current flow I .

$$I = F(Z_p J_p + Z_n J_n) = -F^2[L_{11}Z_p^2 + L_{22}Z_n^2 + (L_{12} + L_{21})Z_p Z_n]\nabla\phi \\ - F(L_{13}Z_p + L_{23}Z_n)\nabla P \quad (2)$$

$$J_s = -F(L_{31}Z_p + L_{32}Z_n)\nabla\phi - L_{33}\nabla P. \quad (3)$$

In the ultimate steady-state a hydrostatic pressure will build up to prevent any further electro-osmotic solvent flow. The pressure gradient needed to prevent the solvent flow can be calculated from (3) by setting $J_s = 0$. When this pressure gradient is introduced into (2) we obtain, using Onsager's relationship $L_{ij} = L_{ji}$,

$$I(\text{steady-state}) = - \left\{ F^2 [L_{11}Z_p^2 + L_{22}Z_n^2 + (L_{12} + L_{21})Z_pZ_n] - \frac{F^2(L_{13}Z_p + L_{23}Z_n)^2}{L_{33}} \right\} \nabla\phi. \quad (4)$$

The term multiplying $-\nabla\phi$ is the effective conductivity in the steady-state, σ_{dc} . If high frequencies are used, no pressure gradient can build up, and from (2) we have that the high frequency conductivity is

$$\sigma_{ac} = F^2 [L_{11}Z_p^2 + L_{22}Z_n^2 + (L_{12} + L_{21})Z_pZ_n]. \quad (5)$$

The maximum frequency effect is given by

$$\frac{\sigma_{ac}}{\sigma_{dc}} = \frac{1}{[1 - F^2(L_{13}Z_p + L_{23}Z_n)^2/\sigma_{ac}L_{33}]}. \quad (6)$$

The coefficients used are not the parameters in common usage. The streaming potential, which represents the potential induced by an applied pressure gradient, can be given in terms of these same coefficients, however.

$$\left(\frac{\nabla\phi}{\nabla P} \right)_{I=0} = \xi = - F(L_{13}Z_p + L_{23}Z_n)/\sigma_{ac}. \quad (7)$$

Thus it is possible to rewrite (6) as

$$\frac{\sigma_{ac}}{\sigma_{dc}} = 1 / \left[1 - \frac{\xi^2 \sigma_{ac}}{L_{33}} \right]. \quad (8)$$

The maximum polarization effect can, therefore, be determined in terms of the conductivity, the permeability, and the streaming potential, which are all easily measured parameters. In Table II are given some results of such measurements on geologic materials.

TABLE II
MEASURED ELECTRO-OSMOTIC COUPLING COEFFICIENTS
AND POSSIBLE POLARIZATION EFFECTS

Sample	Streaming Potential mv/atm.	Maximum % Frequency Effect	Investigator
quartz s.s.	8	.03	Kermabon
quartz s.s.	9	.0002	Kermabon
red s.s.	5.5	.02	Kermabon
shale	1	.05	Kermabon
limestone	2	.03	Kermabon
kaolinite (dispersed, Na)	30	2.0	Olsen
kaolinite (natural flocculated, Ca)	17	.02	Olsen
kaolinite (flocculated, .1 N NaCl)	0.7	.007	Olsen

The values listed in the table indicate that the electro-osmotic effects are relatively unimportant in causing induced electrical polarization.

THERMO-ELECTRIC COUPLING

A very similar analysis can be carried out to investigate the possible influence of thermo-electric effects. The thermo-electric coupling is more complicated because of the role played by the chemical potentials, which depend both on the temperature and on the concentration of the solute. In general, unlike the electro-osmotic case, to evaluate the polarization contribution it would be necessary to measure both cross-coupling effects; the thermally induced potential gradients and the electrically induced thermal gradients.

If we ignore pressure effects

$$\nabla\mu_i = \frac{\partial\mu_i}{\partial T} \nabla T + \sum_j \frac{\partial\mu_i}{\partial C_j} \nabla C_j.$$

The time scale for heat flow at ordinary temperatures is much shorter than the time scale for diffusion flow, so that for certain time scales we can assume the concentrations remain unchanged. This allows us to set

$$\nabla\mu_i \cong -S_i \nabla T = -TS_i \frac{\Delta T}{T} \quad (9)$$

S_i = partial molal entropy.

Our thermal coupling equations can then be written as

$$I = F(Z_p J_p + Z_n J_n) = -F^2[Z_p^2 L_{11} + Z_n^2 L_{22} + Z_p Z_n (L_{12} + L_{21})] \Delta\phi \\ - F[L_{14} Z_p + L_{24} Z_n - (Z_p L_{11} + Z_n L_{21}) TS_p - (Z_n L_{22} + Z_p L_{12}) TS_n] \frac{\nabla T}{T} \quad (10)$$

$$J_Q = -F[Z_p L_{14} + Z_n L_{24}] \nabla\phi - [L_{44} - L_{41} TS_p - L_{42} TS_n] \frac{\nabla T}{T}. \quad (11)$$

The quantity $-F[Z_p L_{14} + Z_n L_{24}] \nabla\phi$ appearing in (11) can be interpreted as the heat transported by the ions moving under the influence of the electrical field.

$$-F[Z_p L_{14} + Z_n L_{24}] \nabla\phi = Q_p J_p + Q_n J_n. \quad (12)$$

Q = heat of transport.

From the definition of transference number,

$$\tau^+ = F Z_p J_p / I; \quad \tau^- = 1 - \tau^+ = F Z_n J_n / I$$

we have therefore

$$FZ_p L_{14} = Q_p \tau^+ \sigma_{ac} / FZ_p \quad (13)$$

$$FZ_n L_{24} = Q_n \tau^- \sigma_{ac} / FZ_n. \quad (14)$$

We also have from (10)

$$F(Z_p L_{11} + Z_n L_{12}) = \tau^+ \sigma_{ac} / FZ_p \quad (15)$$

$$F(Z_n L_{22} + Z_p L_{21}) = \tau^- \sigma_{ac} / FZ_n. \quad (16)$$

A further simplification of notation is achieved by setting

$$[L_{44} - TS_p L_{41} - TS_n L_{42}] = TK; \quad K = \text{thermal conductivity}. \quad (17)$$

We can now rewrite (10) and (11), using Onsager's relation $L_{ij} = L_{ji}$, as

$$I = -\sigma_{ac} \nabla \phi - \left[(Q_p - TS_p) \frac{\tau^+}{FZ_p} + (Q_n - TS_n) \frac{\tau^-}{FZ_n} \right] \sigma_{ac} \frac{\nabla T}{T} \quad (18)$$

$$J_Q = - \left(Q_p \frac{\tau^+}{FZ_p} + Q_n \frac{\tau^-}{FZ_n} \right) \sigma_{ac} \nabla \phi - K \nabla T. \quad (19)$$

In the steady-state limit $J_Q = 0$, and solving for the temperature gradient set up we can obtain the steady-state conductivity.

$$\sigma_{dc} = \sigma_{ac} \left[1 - \frac{\left(\sum Q_i \frac{\tau_i}{FZ_i} \right) \left(\sum (Q_i - TS_i) \frac{\tau_i}{FZ_i} \right) \sigma_{ac}}{TK} \right]. \quad (20)$$

If the time scale is extended, the temperature gradients can also establish concentration gradients (Soret effect) which will modify the current flow because of the diffusion potentials set up. A greater practical difficulty in establishing the magnitudes of these effects is the fact that a measurement of the thermo-electric coefficient

$$\theta = \left(\frac{\nabla \phi}{\nabla T} \right)_{I=0} = - \sum \left(\frac{Q_i - TS_i}{T} \right) \frac{\tau_i}{FZ_i} \quad (21)$$

is not enough to establish the polarization effects of this coupling. In principle one would need to also measure the heat of transport. Such a calorimetric measurement is much more difficult to carry out than the electrical measurement involved in (21).

We can make estimates of the term $\sum Q_i (\tau_i / FZ_i)$ in (20) which are adequate for our purposes here. In most rocks the positive ion transference number is much larger than the negative ion transference number. Thus

$$\theta \cong \frac{Q_p - TS_p}{TFZ_p}. \quad (22)$$

The partial molal entropies for aqueous solutions of ions are known, so that if we can assume similar values for the ions in the pore fluids, (22) would give us an estimate of $\sum Q_i(\tau_i/FZ_i)$. Using KCl solutions typical values of θ ran around 0.3 mv/deg.C. These were measured in a cell where the thermo-electric voltages across the rock samples were bucked against the thermo-electric voltage across a water column (Uhri, 1958). The voltage between the measuring electrodes were then corrected for a semi-theoretical value of the water thermo-electric voltage (Eastman, 1928, and Wirtz, 1948). The partial entropy value for K^+ of 24 entropy units per mole, thus gave from (22) an estimate of Q_{K^+} of 3.6 Kcal/mole. The value of $\sum(Q_i - TS_i)\tau_i$ was about 2.3 Kcal/mole, so that we are within the correct order of magnitude to replace $\sum Q_i\tau_i$ by $\sum(Q_i - TS_i)\tau_i$. This allows us to rewrite (20) as

$$\frac{\sigma_{ac}}{\sigma_{dc}} \cong 1 / \left(1 - \frac{T\theta^2\sigma_{ac}}{K} \right). \quad (23)$$

In Table III are listed some of the thermo-electric data and the approximate maximum polarization effect possible from such coupling.

TABLE III
MEASURED THERMO-ELECTRIC COUPLING COEFFICIENTS
AND POSSIBLE POLARIZATION EFFECTS

Sample	Thermo-Electric Coef. mv/deg. C	Maximum % Frequency Effect
ss. with clay	.23	.000003
s.s.	.48	.00003
shale	.33	.00004
limestone	.27	.0000002

It is obvious that this coupling is of no importance in causing polarization effects. Actually the thermo-electric coefficients are quite high, but the ratio of electrical conductivity to thermal conductivity is very low, so that from (23) $\sigma_{ac} \cong \sigma_{dc}$. This is also the reason why semi-conductors are being used for thermo-electric power conversion; one must have a very good electrical conductor as well as strong thermo-electric effects.

DIFFUSION COUPLING

It is apparent by now that the cross terms appearing in the matrix describing the flow properties of the medium are not large enough in geologic materials to cause appreciable polarization effect. It is quite a different matter, however, when diffusion flow effects are investigated. This, of course, is because the ions themselves carry the electric current, so that the diffusion gradients and the electrical potential gradients have to appear together as primary driving forces for the ion motion.

If we drop all the off diagonal terms in (1) we are left with

$$J_p = -L_{11}\nabla\mu_p - L_{11}FZ_p\nabla\phi \quad (24)$$

$$J_n = -L_{22}\nabla\mu_n - L_{22}FZ_n\nabla\phi \quad (25)$$

as the flow equations for the anions and the cations. In place of the coefficients L_{ij} , we can introduce the more familiar electrochemical quantities of mobility, diffusion coefficients, and transference numbers. Thus we can rewrite the flow equations, assuming uni-valent ions, as

$$J_p = -D_p \frac{\partial p}{\partial x} + u_p p E \quad (26)$$

$$J_n = -D_n \frac{\partial n}{\partial x} - u_n n E \quad (27)$$

U = mobility

p = cation concentration

D = diffusion coef.

n = anion concentration

$t^+ = U_p/(U_p + U_n) = D_p/(D_p + D_n)$ = cation transference no.

$t^- = 1 - t^+$ = anion transference no.

If t^+ should vary along the current path, a divergence will result in the flow of the ions, causing concentration gradients to build up. This is illustrated in Figure 9 where zone II represents a strongly cation selective zone, and zone I represents a zone with no selectivity. Because of its selective transference properties zone II would be called a membrane zone. Since the current in zone II is largely carried by the cations, a surplus of these ions will occur at one end of zone II, and a deficiency at the other end. The anion flow will also be unbalanced, and the buildup of anions will equal that of the cations. The resulting concentration gradients will modify the ion flow until a balance is reached. In this steady-state

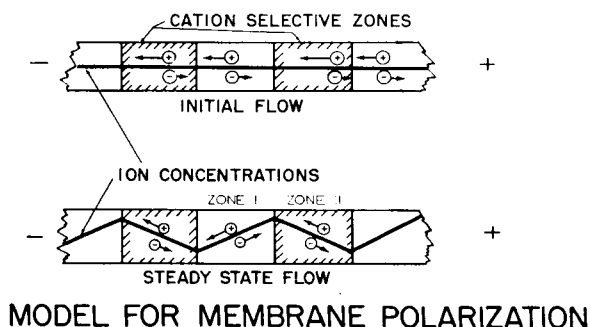


FIG. 9. Ion motion and concentration changes developed by current flow through a membrane system.

condition the net flow of cations and anions in zone II will equal that in zone I.

$$J_{pI} = J_{pII} \quad (28)$$

$$J_{nI} = J_{nII}. \quad (29)$$

The impedance of the zones will also be modified, because the concentration gradients developed will have diffusion potentials associated with them. These potentials arise because of a slight unbalance of charge, but since so little charge is needed to develop an electric field, one can say that the cation and anion concentrations are essentially equal.

$$p = n. \quad (30)$$

The electric field is assumed to be constant in each zone, so that the total potential difference across the pair of zones is given by

$$\Delta\phi = -E_I\Delta L_I - E_{II}\Delta L_{II}. \quad (31)$$

The concentrations, p and n , will be continuous across the boundaries, so that if the succeeding zones repeat the same pattern with the same geometry and electrical properties, we can put

$$\Delta p_I = -\Delta p_{II} \quad (32)$$

$$\Delta n_I = -\Delta n_{II}. \quad (33)$$

When it is also assumed that the concentration gradients are constant in each zone, (28), (29), (30), (32), and (33) can be considered as a system of algebraic equations for the unknown electric fields and concentration gradients. Solving these we obtain for the steady-state conductance

$$\sigma_{ss} = \frac{Fu_{pI}C\left(\frac{1}{\tau_{II}^-} + \frac{A}{B}\frac{1}{\tau_I^-}\right)S_II S_{II}}{\Delta L_I\left[S_I\left(1 + \frac{B}{A}\right) + S_{II}\left(1 + \frac{A}{B}\right)\right]} \quad (34)$$

c = net concentration

$$A = \Delta L_I/\Delta L_{II}$$

$$B = D_{pI}/D_{pII}$$

$$S_i = \tau_i^-/\tau_i^+.$$

At the high frequency limit, no concentration gradients develop, and the only unknowns are the potential gradients. Using the condition that the electric current is the same in the two zones as well as the condition on the total voltage given by (31) allows us to solve for the conductance

$$\sigma_{ac} = Fu_{pI}CA/\Delta L_I(A\tau_I^+ + B\tau_{II}^+). \quad (35)$$

TABLE IV
POLARIZATION OF MEMBRANE MODEL
percent frequency effect ($\tau_{I^+}=0.5$)

τ_{II^+}	$B =$ $A =$	1 .1	1 1	1 10	3 .1	3 1	3 10	D_P Ratio Length Ratio
.999		4.8	33	81	1.6	14	62	
.990		4.5	32	67	1.6	14	57	
.909		3.3	20	20	1.2	9.6	27	
.667		.7	2.8	1.2	1.9	1.8	2.5	

The maximum frequency effect that diffusion coupling can develop in our model is given as

$$\frac{\sigma_{ac}}{\sigma_{dc}} = \frac{(A + B) \left[\frac{A}{\tau_{I^-} \tau_{II^+}} + \frac{B}{\tau_{I^+} \tau_{II^-}} \right]}{\left[\frac{A}{\tau_{II^+}} + \frac{B}{\tau_{I^+}} \right] \left[\frac{A}{\tau_{I^-}} + \frac{B}{\tau_{II^-}} \right]}. \quad (36)$$

Typical values are given in Table IV. It will be observed that there is a finite limit to the frequency effect caused by this form of polarization.

For the time scales to be reasonable, the zones must be very small, and one would not be able to measure the electrical properties of each zone. The presence of membrane zones should make itself known when diffusion measurements are made, because the average transference value for the anions and cations will not be equal, and a diffusion potential will be set up. The information given in Table IV is therefore plotted in Figure 10 in terms of the effective transference number which would be determined from diffusion potential measurements. In Table V are shown the results of diffusion measurements on a few rock samples, and the possible polarizing effects that diffusion coupling could result in.

The values listed above show that these diffusion coupling effects are capable of causing considerable electrical polarization. There are two factors that should be pointed out here. First of all the possible polarization effects could increase without limit if we assumed that zones with positive ion blocking properties

TABLE V
MEASURED TRANSFERENCE VALUES AND POSSIBLE POLARIZATION EFFECTS

Sample	t^+ (Effective)	Maximum % Frequency Effect
tuff	.51	1
tuff	.72	40
tremolite limestone	.87	72
sandstone, medium grained	.48	1
sandstone, medium grained	.49	1
sandstone, fine grained	.89	78
dirty sandstone	.6	20

existed as well as those zones which can block negative ions. It is very unlikely that geologic materials will ever include such zones, but synthetic ion exchange resins can be found that do. The second factor that should be pointed out is the dependence of the polarization effect on the length ratio A . If a material has zones of very high t^+ values, but these zones are too numerous, so that most of the conduction path lies within such zones, little polarization will result. If has been

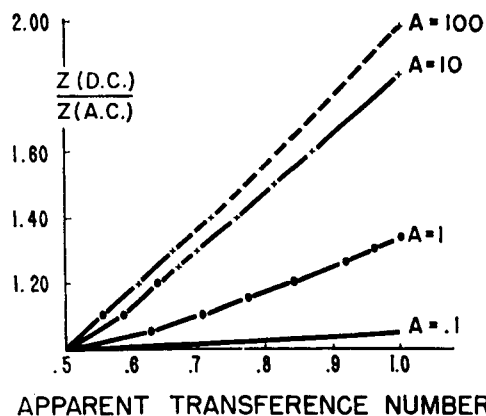


FIG. 10. Effect of the ratio of the zone lengths on the maximum polarization of a membrane system.

nected before that clays, despite their very striking t^+ values, often show extremely little polarization effect, especially if they have been mechanically manipulated. It is believed that this may be due to a tendency for the conduction paths to lie almost entirely within clay zones. Vacquier in his work reported that he could only obtain polarization effects from clays when he coated them on sand grains.

We have advanced no theory here as to why the pores of a rock or clay particles should show much membrane properties. This will involve a microscopic theory of the mineral solution interfaces. It is interesting to point out in this connection, however, that the presence of a surface charge on the minerals does not by itself cause any polarization.

A more detailed study showed that a fixed charge in a system can lead to a very slight polarization, but it is really necessary, if the system is to have a finite polarization effect, that the motion of certain ions be hindered in the system. Thus it appears that the electrostatic and Van der Waal forces that attract the positive ions to the surface of a polarizing clay system must also act to prevent negative ions from moving along the surface.

COMPARISON OF ELECTRODE AND MEMBRANE POLARIZATION

The model given for membrane polarization sets a finite limit to the polarization effect of membrane zones within a rock. The model given in Figure 1 for

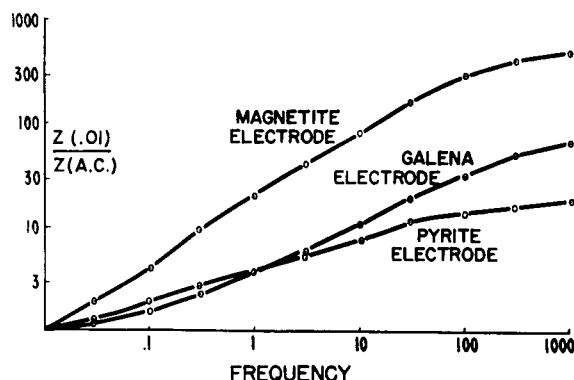


FIG. 11. Impedance of natural electrodes.

electrode polarization is not so limited however. In Figure 11 is shown some data on the interfacial impedance of a few natural electrodes. The frequency effects are much larger than 100 percent. We might expect, therefore, to find a difference between the magnitude of the observed polarization depending on the cause of the polarization. Because of the diluting, mentioned before, that is caused by parallel unpolarized conduction paths, it seems best to compare the metal factors rather than the percentage frequency effects. In Figure 12 there is shown in a somewhat generalized form, the results of laboratory measurements on a large number of rock samples, and field measurements in many areas. These results again indicate the pronounced polarization effects of well-mineralized rocks. However, in rocks with less mineralization, there begins to exist considerable ambiguity. This is especially true in sedimentary rocks of high conductivity. The geometry of the mineralization is also very critical. When the sulphides form connected veinlets

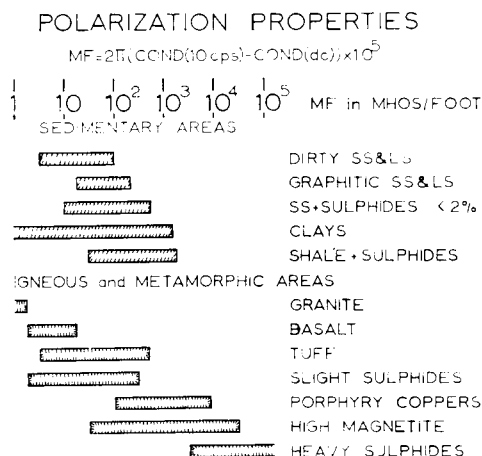


FIG. 12. Distribution of observed metal factor values.

their metal factor values are far superior to those of well-disseminated sulphides.

These results, although not eliminating the great usefulness of induced polarization measurements, do point out certain dangers in holding too simple a concept about the causes of the polarization. It also leads one to search for other criteria, besides a magnitude criterion, to help in separating out the various contributing factors.

FREQUENCY BEHAVIOR OF MEMBRANE POLARIZATION

The mechanism involved in membrane polarization appears to be quite different than that pictured in Figure 1 for electrode polarization. It might be hoped, therefore, that the details of the frequency spectrum for the two polarization effects would lead to some means of distinguishing between the different effects. For this reason the study of the membrane model was expanded.

From equations (26) and (27) we can obtain the equations of motion for the ion species.

$$\frac{\partial p}{\partial t} = D_p \frac{\partial^2 p}{\partial x^2} - U_p \frac{\partial}{\partial x} (pE) \quad (37)$$

$$\frac{\partial n}{\partial t} = SD_p \frac{\partial^2 n}{\partial x^2} + SU_p \frac{\partial}{\partial x} (nE). \quad (38)$$

Poisson's equation must also be satisfied

$$\frac{\partial E}{\partial x} = \frac{F}{\epsilon} (p - n). \quad (39)$$

These equations are non-linear but can be linearized for our case since we are interested in very small current densities. For small current densities we can consider the electric fields and the concentration changes to be small

$$\begin{aligned} P &= C + \Delta P \\ N &= C + \Delta N, \\ E &= \Delta E. \end{aligned} \quad (40)$$

If the product of these small terms is neglected and harmonic time dependence is assumed, (37), (38), and (39) become linear equations for Δp , Δn , and ΔE .

$$i\omega\Delta p = D_p \frac{d^2\Delta p}{dx^2} - U_p C \frac{d\Delta E}{dx} \quad (41)$$

$$i\omega\Delta n = SD_p \frac{d^2\Delta n}{dx^2} + SU_p C \frac{d\Delta E}{dx} \quad (42)$$

$$\frac{d\Delta E}{dx} = \frac{F}{\epsilon} (\Delta p - \Delta n). \quad (43)$$

These equations are the ones usually dealt with in treatments of space charge effects, and appear in discussions of solution-electrode interfaces as well as in this membrane model (MacDonald, 1953). The general solutions are simple exponentials with two characteristic lengths.

$$\begin{aligned}\Delta\phi &= L_p e^{\pm r_1 x} + M_p e^{\pm r_2 x} \\ \Delta n &= L_n e^{\pm r_1 x} + M_n e^{\pm r_2 x}.\end{aligned}\quad (44)$$

One of these characteristic lengths

$$\frac{1}{r_1} = 1/K \left(1 + \frac{j\omega(S+1)}{DSK^2} \right)^{1/2} \cong 1/K; \quad K^2 = \frac{2CF^2}{\epsilon RT} \gg 1 \quad (45)$$

is essentially frequency independent and represents a space charge set up at the boundaries of a zone. The other characteristic length

$$\frac{1}{r_2} = \left[\frac{2D_p S}{j\omega(S+1)} \right]^{1/2} \quad (46)$$

varies inversely with the square root of the frequency and represents a diffusion phenomenon.

The undetermined constants in the solution are solved for by imposing the boundary conditions. In the membrane model, because of the symmetry imposed by the regular spacing of the zones, there are six constants to be determined, three for each zone. The boundary conditions can be given as the continuity of concentration and of ion flow at the boundary, the continuity of E at the boundary, and the condition that $\int \Delta E dx = -\Delta\phi$, the applied potential. These conditions, when expressed in terms of the general solutions, give us six simultaneous algebraic equations for the undetermined constants. Their solution is straightforward, but laborious, since the system of equations is highly singular and many terms must be carried along. The impedance of the membrane zone can then be determined and is given by

$$\begin{aligned}Z &= \frac{\Delta L_I}{U_p CF} \left\{ t_1^+ + \frac{B}{A} t_{II}^+ + \frac{(S_{II} - S_I)^2}{\frac{X_I S_I}{(t_{II}^+)^2 t_1^+ \tanh X_I} + \frac{A}{B} \frac{X_{II} S_{II}}{t_{II}^+ (t_1^+)^2 \tanh X_{II}}} \right\} \\ X_i &= \left(\frac{j\omega}{2D_p t_i^-} \right)^{1/2} \frac{\Delta L_i}{2} \\ A &= \Delta L_I / \Delta L_{II} \\ B &= D_{pI} / D_{pII} \\ S_i &= t_i^- / t_i^+.\end{aligned}\quad (47)$$

Values of the membrane impedance as a function of frequency are shown in Figure 13, where it is assumed that zone I is a neutral zone. The value used for

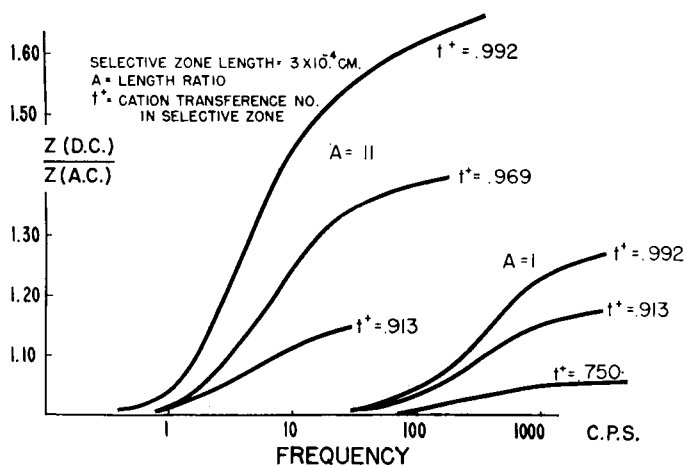


FIG. 13. Details of the impedance of a simple membrane system.

ΔL_2 of 0.3×10^{-3} cm. is typical of the size of clay particles. The frequency response for a given zone size is quite sharp. The frequencies at which the impedance is varying depend on the square of zone length, so that in an actual system with a distribution of zone lengths we can expect the frequency variations to be spread out over a wide range of frequencies. In Figure 14 are shown some impedance measurements on real membrane systems for comparison.

FREQUENCY BEHAVIOR OF ELECTRODE POLARIZATION

To compare electrode polarization to membrane polarization we must extend the simple model of Figure 1. In fact the data shown in Figure 11 are quite different from what we would expect from a simple RC circuit. Therefore, to better under-

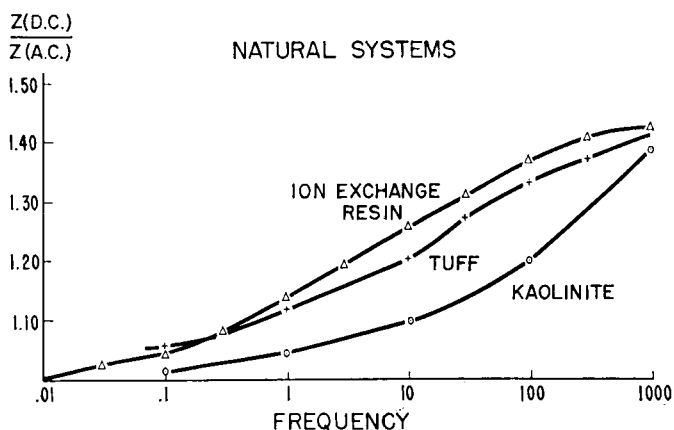


FIG. 14. Details of the impedance of a natural membrane system.

stand the nature of electrode impedances, detailed kinetic calculations were also carried out for the impedance of an electrode-solution interface. Equations (41) (42) and (43) can again be used to describe the ion motions, but the boundary conditions are very different. First of all there can be capacitive coupling between the solution and the electrode. Secondly any Faradaic current involving actual charge transfer must entail chemical reactions, and a whole chain of events may follow depending on the products of these reactions. These events are all reflected in the boundary conditions, and thus modify the electrode impedance (Madden and Marshall, 1959). When typical reaction rate values are used, however, the

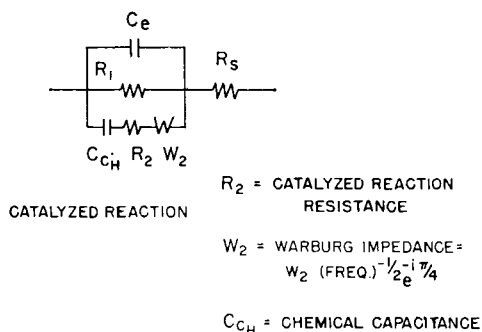
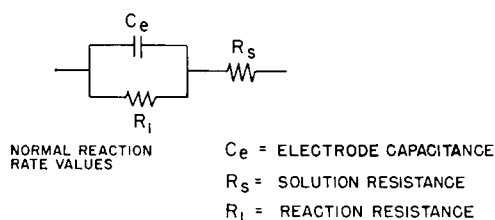


FIG. 15. Equivalent circuits for the impedance of polarized electrodes.

solutions obtained indicate that at audio and sub-audio frequencies the circuit of Figure 1 should well represent the electrode impedance. If a reaction occurring at the surface is well catalysed, a different behavior would be expected. There is little reaction resistance in such a case, but the depletion of the reacting species at the surface would require a diffusion of the ions to or away from the surface. This is responsible for an impedance which has the typical diffusion behavior and depends inversely on the square root of the frequency. It is called a Warburg impedance and is given the symbol W (Grahame 1952). The accumulation of the reaction product will in general tend to oppose any further reaction, and this causes an impedance just like that of a capacitance. We have given this capacitance the symbol C_{ch} . The equivalent circuits for an electrode impedance with normal reactions or with catalyzed reactions are shown in Figure 15.

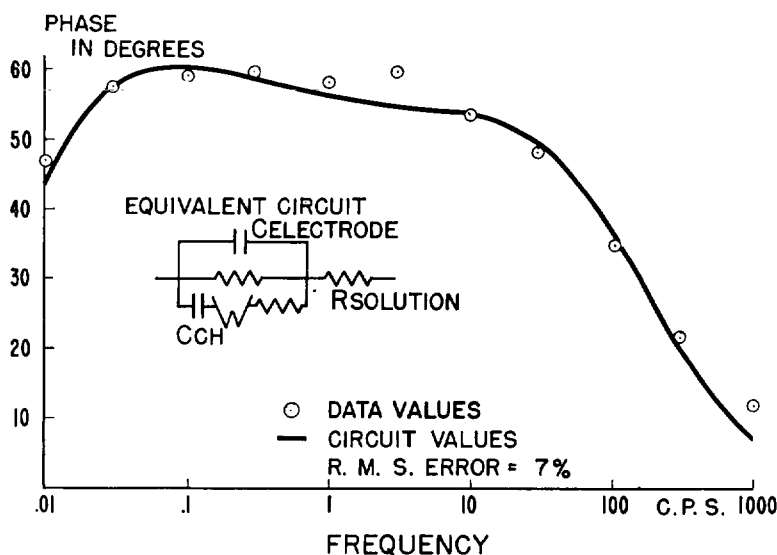


FIG. 16. Phase shift of the impedance of a magnetite electrode, and its comparison with the equivalent circuit.

Measurements were made on a wide variety of electrodes, and data were also obtained from the literature. Equivalent circuits such as those shown in Figure 15 were then adjusted to fit the data. In most cases the circuits could be made to fit the data within the experimental error. An example of such a fit is shown in Figure 16, where only the phase of the impedance is plotted. The magnitude of the impedance for this same electrode is shown in Figure 10, and on this logarithmic scale the errors of the equivalent circuit would not be discernible. The most striking result of these measurements was that in every case a well catalyzed reaction appeared to be taking place and contributed a large part of the current being passed. This means that the impedance was in a large part controlled by a diffusion phenomenon, just as in the membrane model. A summary of these electrode measurements is given in Table VI.

SUMMARY AND CONCLUSIONS

When using induced polarization measurements for the detection of metallic or semi-conducting minerals, one must be aware of other causes of the electrical polarization. An examination of the possible causes of these effects reduced the important factors to electromagnetic coupling effects and membrane polarization as well as the metallic mineral, i.e., electrode polarization that one is usually looking for. The electromagnetic effects can be avoided by using only information at very low frequencies. In typical mining exploration applications, one must work with frequencies of less than 10 cps.

TABLE VI

SUMMARY OF ELECTRODE IMPEDANCES

Values are given for 1 cm² of surface at room temp. Symbols are referred to Fig. 15.

Electrode	Reaction Resistance R_1	Electrode Capacitance C_e	Catalyzed Reaction Resistance R_2	Warburg Impedance W_2	Chemical Capacitance C_{ch}
ST steel	160,000	3.2	11	3,960	247
Ni	8,500	112	3	670	830
Cu	3,080	3.8	19	1,270	55,000
Graphite	49,500	80	26	2,700	170
Pyrite	3,900	15.	60	800	150,000
Galena*	3,800	8.5	6	760	43,000
Magnetite*	222,000	4.8	101	5,870	106

* Area not known accurately.

Resistances given in ohms.

Capacities given in μf . W_2 calculated at 1 cps.

The effects of membrane polarization are more difficult to separate out. The effects are more limited than the electrode polarization, but may still be significant. Detailed investigations of these two processes were made to look for fundamental differences in their electrical behavior. Unfortunately the investigations showed that the principal factor contributing to the electrode impedance is a diffusion phenomenon, and this is the same factor involved in membrane polarization. Both electrode and membrane impedances show a wide and gentle frequency variation making any discrimination on this basis very difficult. This result was borne out in the measurements on rock and clay samples. The phase shifts of the metal factor, which should represent approximately the phase shift of the blocked conduction paths, were computed from an analysis of transient measurements at one tenth, one and ten cps. for hundreds of samples. Although the results were often quite characteristic for a given rock or clay type, there seemed to be no clear separation between those samples with electrode polarization and those with membrane polarization. Some typical values are shown in Table VII.

This represents of course only a somewhat limited frequency range and a wider frequency analysis might prove to be more discriminating. It is difficult, however, to extend the measurements to higher frequencies in the field because of the electromagnetic coupling effects. The measurements at lower frequencies are in principle possible, but the natural earth currents make the noise problem an important practical factor.

The magnitude of the polarization effects still appears to be the best guide for the detection of electrode polarization in rocks. When the mineralized target is well below the surface, the observed effects are, of course, considerably reduced, and may be hardly different than a small background polarization. The authors are of the opinion that the greatest improvement in interpretation will come from

TABLE VII
METAL FACTOR PHASE IN DEGREES

Sample Type	Frequency cps.		
	10	1	0.1
Kaolinite	9	14	22
Mica	8	12	19
Ion exchange resins	13	29	51
Sediments from Colorado Plateau			
with sulphides	12	20	30
little or no sulphides	9	15	21
Sedimentary copper ore	12	18	19
Rhodesian copper ore	18	27	41
Field data, porphyry copper body		18	31
Lithic Tuff, little or no sulphides	12	22	34
Graphitic ss	13	16	22
Dirty sands from Dakota, ss sequence	11	17	30
Manganese ore	9	14	22
Highly magnetic altered, ultrabasic	42	41	47

an accurate handling of this geometric factor, rather than a subtle analysis of the frequency spectrum of the observed polarization.

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All of our early work on induced polarization was done with fellow students, Phil Hallof, now at McPhar Geophysics, Keeva Vozoff, now at the University of Alberta, and Norman Ness, and their contributions are firmly entrenched throughout this paper. We also wish to acknowledge a fruitful interchange of ideas and information with Dr. A. A. Brant, whose group at Newmont Exploration Co., developed and carried out perhaps the first wide scale use of induced polarization, and Dr. Ralph Holmer and George Rogers of the Bear Creek Mining Co., whose group also has considerable field experience in these measurements. We are indebted to Hal Olsen and Professor Martin of the Civil Engineering Department for their help in carrying out and interpreting measurements on clay systems.

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REFERENCES

- Bleil, D. F., 1953, Induced polarization, a method of geophysical prospecting: *Geophysics*, v. 18, p. 636-661.
- Brant, A. A., personal communication.
- Casagrande, L., 1952, Electro-osmotic stabilization of soils: *Jour. Boston Soc. Civil Eng.*, v. 39, p. 51.
- De Groot, S. R., 1951, *Thermodynamics of irreversible processes*: New York, Interscience Publishers.
- Denbigh, K. G., 1951, *The thermodynamics of the steady state*: London, Methuen & Co., Ltd.
- Eastman, 1928, Electromotive force of electrolyte thermocouples and thermocells and the entropy of transfer and absolute entropy of ions: *J. Amer. Chem. Soc.*, v. 50, p. 292.
- Eckart, 1940, Thermodynamics of irreversible processes: *Phys. Rev.*, v. 58, p. 267.
- Grahame, D. C., 1952, Mathematical theory of the faradaic admittance: *Jour. of Electrochem. Soc.*, v. 99, p. 370.
- Hallof, P., 1957, On the interpretation of resistivity and induced polarization results: Ph.D. thesis, M.I.T. Department of Geology and Geophysics.
- Henkel, J. H., and Van Nostrand, R. G., 1957, Experiments in induced polarization: *A.I.M.E. Trans.*, v. 9, p. 355-359.
- Kermabon, A. J., 1956, A study of some electro-kinetic properties of rocks: M.S. thesis, M.I.T. Department of Geology and Geophysics.
- MacDonald, J. R., 1953, Theory of A.C. space-charge polarization effects in photoconductors, semiconductors, and electrolytes: *Phys. Rev.*, v. 92, p. 4.
- Madden, Marshall, Fahlquist, and Neves, 1957, Background effects in the induced polarization method of geophysical exploration: A.E.C. report RME-3150.
- Madden, T. R., and Marshall, D. J., 1958, A laboratory investigation of induced polarization: A.E.C. report RME-3156.
- , 1959, Electrode and membrane polarization A.E.C. report RME-3157
- Olsen, H., 1958, Personal communication.
- Schlumberger, C., 1920, *Etude sur la prospection électrique du sous-sol*: Paris, Gauthier-Villars, Chapt. VIII.
- Sunde, E. D., 1949, *Earth conduction effects in transmission systems*: New York, D. Van Nostrand Co., Chapt. IV.
- Uhri, D., 1958, Personal communication.
- Vacquier, V., Holmes, C. R., Kintzinger, P. R., and Lavergne, M., 1957, Prospecting for ground water by induced electrical polarization: *Geophysics*, v. 22, p. 660-687.
- Wirtz, V. K., 1948, Platzwechselprozesse in Flüssigkeiten, *Zeit f. Naturforschung*, 3a, p. 672.
- Wyllie, M. R. J., 1955, The role of clay in well-log interpretation: State of California, Division of Mines, Bulletin 169, p. 282.

- 3) Wiener, O., Abh. Math. Phys. Kl. Sächs. Ges. Wiss. **32**, 509 (1912).
 4) Bruggeman, D. A. G., Ann. Physik **24**, 636 (1935).
 5) Pearce, C. A. R., Brit. J. Appl. Phys. **6**, 113 (1955).
 6) Rue, R. E. D. L. and C. W. Tobias, J. Electrochem. Soc. **106**, 827 (1959).

- 7) Fricke, H., Phys. Rev. **24**, 575 (1924).
 8) Sillars, R. W., J. Inst. Elec. Engrs. (London) **80**, 378 (1937); Kharadly, M. M. Z. and W. Jackson, Proc. Inst. Elec. Engrs. (London) **100**, (III), 199 (1953); Dryden, J. S. and R. J. Meakins, Proc. Phys. Soc. **70B**, 427 (1957).

From the Institute for Chemical Research, Kyoto University, Takatsuki, Osaka (Japan)

Theory of the Dielectric Dispersion due to the Interfacial Polarization and its Application to Emulsions

By Tetsuya Hanai

With 3 figures and 4 tables

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List of Symbols

ϵ , κ , ϵ^* , κ^* , γ : the dielectric constant, the electrical conductivity, the complex dielectric constant, the complex conductivity and the imaginary part of the complex dielectric constant¹⁾ of the dispersion system respectively.

Symbols with the subscript m or p such as ϵ_m , κ_m , ϵ_m^* , κ_m^* and γ_m or ϵ_p , κ_p , ϵ_p^* , κ_p^* and γ_p , refer to the continuous medium or to the disperse phase respectively.

ω : the angular frequency.

Among these quantities there hold the following relations:

$$\epsilon^* = \frac{q}{j\omega} \kappa^* = \epsilon + \frac{q}{j\omega} \kappa = \epsilon - j\gamma, \quad [1]$$

$$\epsilon_m^* = \frac{q}{j\omega} \kappa_m^* = \epsilon_m + \frac{q}{j\omega} \kappa_m = \epsilon_m - j\gamma_m \quad [2]$$

and

$$\epsilon_p^* = \frac{q}{j\omega} \kappa_p^* = \epsilon_p + \frac{q}{j\omega} \kappa_p = \epsilon_p - j\gamma_p, \quad [3]$$

where $j = \sqrt{-1}$, numerical factor $q = 112.94 \times 10^{11}$ and the conductivities are expressed in Ω/cm units.

Φ : the volume fraction of the disperse phase.

ϵ_h , ϵ_l : the limiting values of ϵ at higher and lower frequencies respectively.

κ_h , κ_l : the limiting values of κ at higher and lower frequencies respectively.

τ : the relaxation time of a dielectric dispersion.

1. Introduction

It is well known that a heterogeneous mixture composed of two phases which differ from each other in dielectric constant and electrical conductivity shows a dielectric dispersion due to the so-called interfacial polarization²⁾.

¹⁾ The imaginary part $\frac{q}{\omega} \kappa$ of the complex dielectric constant is usually denoted by ϵ'' . In this paper a symbol γ is used instead of ϵ'' in order to avoid confusion in complicated expressions.

²⁾ This phenomenon is different from the interfacial polarization which was pointed out by Fricke and Curtis (1).

Several studies have already been carried out experimentally and theoretically on this dielectric dispersion.

A possibility of the dielectric dispersion caused by a heterogeneous structure of dielectrics was first pointed out by Maxwell (2) by means of a stratified model. This stratified model is practically of no use, since the distinction between the dispersion medium and the disperse phase is left out of consideration in this model. Wagner (3) proposed a theory of the interfacial polarization for a sparsely distributed dispersion of spherical particles, and showed a solution of his equation only for a special case of lower concentrations and the non-conductive medium. The solution given by Wagner did not agree with experimental results especially at higher concentrations as shown later. Recently Fricke (4) has extended the Wagner theory to an ellipsoidal dispersion.

On the other hand, several investigators observed dielectric dispersions due to the interfacial polarization in emulsions.

In the earlier studies Sillars (5) found $\tan \delta$ due to the interfacial polarization for emulsions of water in wax, concentrations of the internal phase being less than 7% by volume. His results did not agree with the theoretical prediction given by Wagner. Hamon (6) measured the dielectric loss factors of mixtures of spherical particles of an organic semiconductor (copper phthalocyanine) in paraffin wax, and obtained reasonably good agreement with the Wagner theory, volume concentrations of the internal phase being less than 3%. Kharadly and Jackson (7) observed dielectric dispersions due to the interfacial polarization for emulsions of polystyrene and nitrobenzene in the mixture of paraffin wax and polythene at concentrations up to 46%. Their theoretical analysis, which is similar to the Wagner theory, failed to account for the experimental results of the emulsions at concentrations above 20%. Recently Dryden and Meakins (8) have reported the dielectric absorption of the Maxwell-Wagner type for the water-in-wool wax emulsions, concentrations being less than 38%. Their results are insufficient for the rigorous consideration of the interfacial polarization, since they gave no data on the real part of the dielectric constant.

In the previous investigations (9, 10), the dielectric constants and conductivities of nujol and carbon tetrachloride-in-water

emulsions were measured for high concentrations up to 85%, and no dielectric dispersion due to the interfacial polarization could be observed over a frequency range of 20 cps. to 3 mc. The observed dielectric constants and conductivities of these O/W emulsions did not agree with the *Wiener* equation, which is a special case of the *Wagner* theory, and were expressed better by the equation of the *Bruggeman* type (11) over the whole range of concentration.

Up to the present, dielectric dispersions due to the interfacial polarization have been observed only in the water-in-oil (W/O) type emulsions as mentioned above, and no example of the interfacial polarization has been found in the oil-in-water (O/W) type yet.

It should also be noted for characteristics of dispersion systems that O/W emulsions show different dielectric constants and conductivities from those of the W/O type with the same volume fraction of the oil. In practice, these characteristics are often utilized as a crude measure of the determination of emulsion type and the detection of phase inversion (12).

Although the interfacial polarization is of practical interest especially in concentrated emulsions, it appears to us that its theoretical treatment for a spherical dispersion has not been developed beyond the *Wagner* theory.

In the present paper, the author has intended to deal with the theories on the interfacial polarization of spherical dispersions. First the validity of the *Maxwell* theory is discussed briefly. Next the general solution of the *Wagner* theory is shown, and the characteristics of this solution are discussed for emulsions. Finally a new theory of the interfacial polarization is developed and is applied to the practical cases of emulsions.

2. The Maxwell Theory

Maxwell (2) explained the dielectric dispersion due to the interfacial polarization of a heterogeneous system by means of a stratified model.

He obtained the complex dielectric constant ϵ^* and the complex conductivity κ^* for a heterogeneous system in the stratified structure as follows:

$$\epsilon^* = \epsilon_m^* \frac{\epsilon_p^*}{\epsilon_p^* + \Phi(\epsilon_m^* - \epsilon_p^*)} \quad [4]$$

and

$$\kappa^* = \kappa_m^* \frac{\kappa_p^*}{\kappa_p^* + \Phi(\kappa_m^* - \kappa_p^*)} \quad [5]$$

Then we obtain

$$\epsilon^* = \epsilon_h + \frac{\epsilon_l - \epsilon_h}{1 + j\omega\tau} + \frac{q}{j\omega} \kappa_l \quad [6]$$

and

$$\kappa^* = \kappa_l + \frac{j\omega\tau(\kappa_h - \kappa_l)}{1 + j\omega\tau} + \frac{j\omega}{q} \epsilon_h, \quad [7]$$

where

$$\epsilon_h = \epsilon_m \frac{\epsilon_p}{\epsilon_p + \Phi(\epsilon_m - \epsilon_p)}, \quad [8]$$

$$\epsilon_l = \frac{\epsilon_m \kappa_p^2 + \Phi(\epsilon_p \kappa_m^2 - \epsilon_m \kappa_p^2)}{[\kappa_p + \Phi(\kappa_m - \kappa_p)]^2}, \quad [9]$$

$$\epsilon_l - \epsilon_p = \frac{(\epsilon_m \kappa_p - \epsilon_p \kappa_m)^2 \Phi(1 - \Phi)}{[\epsilon_p + \Phi(\epsilon_m - \epsilon_p)][\kappa_p + \Phi(\kappa_m - \kappa_p)]^2}, \quad [10]$$

$$\kappa_h = \frac{\kappa_m \epsilon_p^2 + \Phi(\kappa_p \epsilon_m^2 - \kappa_m \epsilon_p^2)}{[\epsilon_p + \Phi(\epsilon_m - \epsilon_p)]^2}, \quad [11]$$

$$\kappa_l = \kappa_m \frac{\kappa_p}{\kappa_p + \Phi(\kappa_m - \kappa_p)}, \quad [12]$$

$$\kappa_h - \kappa_l = \frac{(\epsilon_m \kappa_p - \epsilon_p \kappa_m)^2 \Phi(1 - \Phi)}{[\kappa_p + \Phi(\kappa_m - \kappa_p)][\epsilon_p + \Phi(\epsilon_m - \epsilon_p)]^2}, \quad [13]$$

$$\tau = \frac{\epsilon_p + \Phi(\epsilon_m - \epsilon_p)}{\kappa_p + \Phi(\kappa_m - \kappa_p)} \cdot \frac{1}{q} \quad [14]$$

and

$$(\kappa_h - \kappa_l)\tau q = \epsilon_l - \epsilon_h. \quad [15]$$

Eq. [10] shows that a dielectric dispersion may arise when $\epsilon_m \kappa_p \neq \epsilon_p \kappa_m$. As is easily known from the stratified model, the *Maxwell* theory is unable to explain considerable changes of dielectric constants and conductivities on phase inversion of emulsions, since no distinctions are made between the continuous medium and the disperse phase of emulsions.

This theory is also of no use to explain the experimental fact that dielectric dispersions due to the interfacial polarization could not be found in O/W emulsions, while such dispersions could be observed in W/O emulsions. Moreover ϵ_h and κ_l given by eqs. [8] and [12] show no agreement with experimental values.

3. The Wagner Theory

A) The general solution of the Wagner equation

Wagner (3) showed that, for a dispersion system where spherical particles are sparsely distributed throughout the dispersion medium, the complex dielectric constant and the complex conductivity of a heterogeneous mixture are given by the following expressions:

$$\epsilon^* = \epsilon_m^* \frac{2\epsilon_m^* + \epsilon_p^* - 2\Phi(\epsilon_m^* - \epsilon_p^*)}{2\epsilon_m^* + \epsilon_p^* + \Phi(\epsilon_m^* - \epsilon_p^*)} \quad [16]$$

and

$$\kappa^* = \kappa_m^* \frac{2\kappa_m^* + \kappa_p^* - 2\Phi(\kappa_m^* - \kappa_p^*)}{2\kappa_m^* + \kappa_p^* + \Phi(\kappa_m^* - \kappa_p^*)} \quad [17]$$

Eqs. [16] and [17] have hitherto been solved only when $\Phi \ll 1$ and $\kappa_m = 0$. General solutions of these equations are derived as shown below.

Inserting eqs. [1], [2] and [3] into eq. [16], we have

$$\varepsilon^* = \left(\varepsilon_m + \frac{q}{j\omega} \kappa_m \right) \frac{[2\varepsilon_m + \varepsilon_p - 2\Phi(\varepsilon_m - \varepsilon_p)] + \frac{q}{j\omega} [2\kappa_m + \kappa_p - 2\Phi(\kappa_m - \kappa_p)]}{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)] + \frac{q}{j\omega} [2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)]} \quad [18]$$

Of we put, for simplicity,

$$\left. \begin{aligned} A &= \varepsilon_m \\ B &= q \kappa_m \\ E &= 2\varepsilon_m + \varepsilon_p - 2\Phi(\varepsilon_m - \varepsilon_p) \\ F &= q[2\kappa_m + \kappa_p - 2\Phi(\kappa_m - \kappa_p)] \\ G &= 2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p) \\ H &= q[2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)], \end{aligned} \right\} \quad [19]$$

Eq. [18] is transformed to

$$\begin{aligned} \varepsilon^* &= \left(A + \frac{B}{j\omega} \right) \frac{E + \frac{F}{j\omega}}{G + \frac{H}{j\omega}}, \quad [20] \\ &= \frac{AF + BE + j\omega AE + \frac{1}{j\omega} BF}{H + j\omega G}, \\ &= \frac{AE}{G} + \frac{\frac{AF + BE}{H} - \frac{BFG}{H^2} - \frac{AE}{G}}{1 + j\omega \frac{G}{H}} + \frac{1}{j\omega} \cdot \frac{BF}{H}. \end{aligned}$$

Hence eq. [18] is rewritten in a simplified form:

$$\varepsilon^* = \varepsilon_h + \frac{\varepsilon_l - \varepsilon_h}{1 + j\omega \tau} + \frac{q}{j\omega} \kappa_l, \quad [22]$$

where

$$\varepsilon_h = \frac{AE}{G} = \varepsilon_m \frac{2\varepsilon_m + \varepsilon_p - 2\Phi(\varepsilon_m - \varepsilon_p)}{2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)}, \quad [23]$$

$$\varepsilon_l = \frac{AF + BE}{H} - \frac{BFG}{H^2} = \frac{\varepsilon_m(2\kappa_m + \kappa_p)^2 + [(9\varepsilon_p - 2\varepsilon_m)\kappa_m^2 - 8\varepsilon_m\kappa_m\kappa_p + \varepsilon_m\kappa_p^2]\Phi - 2\varepsilon_m(\kappa_m - \kappa_p)^2\Phi^2}{[2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)]^2}, \quad [24]$$

$$\varepsilon_l - \varepsilon_h = \frac{AF + BE}{H} - \frac{BFG}{H^2} - \frac{AE}{G} = \frac{9(\varepsilon_m\kappa_p - \varepsilon_p\kappa_m)^2\Phi(1-\Phi)}{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)][2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)]^2} \quad [25]$$

and

$$\tau = \frac{G}{H} = \frac{2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)}{2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)} \cdot \frac{1}{q}. \quad [26]$$

In a similar manner we obtain

$$\kappa^* = \kappa_l + \frac{j\omega \tau (\kappa_h - \kappa_l)}{1 + j\omega \tau} + \frac{j\omega}{q} \varepsilon_h, \quad [27]$$

in which

$$\kappa_l = \frac{BF}{qH} = \kappa_m \frac{2\kappa_m + \kappa_p - 2\Phi(\kappa_m - \kappa_p)}{2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)}, \quad [28]$$

$$\kappa_h = \frac{1}{q} \left[\frac{AF + BE}{G} - \frac{AEH}{G^2} \right] = \frac{\kappa_m(2\varepsilon_m + \varepsilon_p)^2 + [(9\varepsilon_p - 2\kappa_m)\varepsilon_m^2 - 8\kappa_m\varepsilon_m\varepsilon_p + \kappa_m\varepsilon_p^2]\Phi - 2\kappa_m(\varepsilon_m - \varepsilon_p)^2\Phi^2}{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)]^2}, \quad [29]$$

$$\kappa_h - \kappa_l = \frac{1}{q} \left[\frac{AF + BE}{G} - \frac{AEH}{G^2} - \frac{BF}{H} \right] = \frac{9(\kappa_m\varepsilon_p - \varepsilon_p\kappa_m)^2\Phi(1-\Phi)}{[2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)][2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)]^2} \quad [30]$$

and

$$(\kappa_h - \kappa_l) \tau q = \varepsilon_l - \varepsilon_h. \quad [31]$$

From eqs. [22] and [25] it is easily seen that a dielectric dispersion characterized by a single relaxation time arises when $\varepsilon_m \kappa_p \neq \varepsilon_p \kappa_m$. Therefore the Cole and Cole

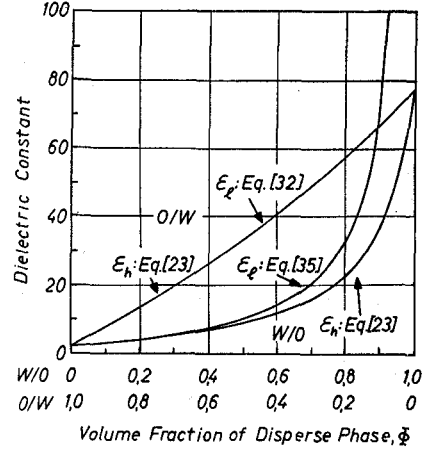


Fig. 1. Concentration dependence of dielectric constants for the O/W and W/O emulsions given by the Wagner theory

plot of eq. [22] in a complex plane is a semicircular-arc. Eq. [23] for the limiting dielectric constant ε_h at higher frequencies is the same as the Wiener equation (13) of the dielectric constant for a spherical

dispersion, and eq. [28] for κ_l is the same as the equation of the conductivity of dispersion systems used by Fricke (14).

B) Application to emulsions

We shall apply the results of the Wagner theory to the oil-water emulsions which are of practical interest. Since the conductivity of the water phase is, generally in emulsions, much larger than that of the oil phase,

eqs. [24], [28] and [29] can be simplified as follows:

i) Case $\kappa_m \gg \kappa_p$ (O/W type emulsion)

Here eqs. [24], [28] and [29] are reduced to the following respectively:

$$\varepsilon_l = \frac{2(1-\Phi)(2+\Phi)\varepsilon_m + 9\varepsilon_p\Phi}{(2+\Phi)^2}, \quad [32]$$

$$\frac{\kappa_l}{\kappa_m} = \frac{2(1-\Phi)}{2+\Phi} \quad [33]$$

and

$$\frac{\kappa_h}{\kappa_m} = \frac{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)][2\varepsilon_m + \varepsilon_p - 2\Phi(\varepsilon_m - \varepsilon_p)] - 9\varepsilon_m\varepsilon_p\Phi}{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)]^2}. \quad [34]$$

ii) Case $\kappa_p \gg \kappa_m$ (W/O type emulsion)

Here eqs. [24], [28] and [29] are reduced to the following respectively:

$$\varepsilon_l = \varepsilon_m \frac{1+2\Phi}{1-\Phi}, \quad [35]$$

$$\frac{\kappa_l}{\kappa_m} = \frac{1+2\Phi}{1-\Phi} \quad [36]$$

and

$$\frac{\kappa_h}{\kappa_p} = \left[\frac{3\varepsilon_m}{2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)} \right]^2 \Phi. \quad [37]$$

In both cases of O/W and W/O types, ε_h can be expressed by eq. [23].

In tables 1, 2 and fig. 1 are shown the theoretical values and curves obtained respectively by eqs. [23] and [32]–[37], where the dielectric constants of the oil and the water phases are taken to be 2.50 and 76.8 (values at 30°C) respectively.

It will be noticed from tables 1, 2 and fig. 1 that both differences between ε_l and ε_h and between κ_h and κ_l are very small over the whole range of concentration for the O/W type, while these differences are very large at higher concentrations of the W/O type. Hence it is expected from the general

solution of the *Wagner* equation that the dispersions of dielectric constants and conductivities are too small to be observed in O/W emulsions and are considerable in W/O emulsions. In fact, no dielectric dispersion due to the interfacial polarization could be found in O/W emulsions (9, 10), while several workers observed the dielectric dis-

persions of this type in W/O emulsions as mentioned in the introduction.

Moreover it is explained from tables 1, 2 and fig. 1 that the considerable changes of ε_h and κ_l may occur on phase inversion of emulsions.

However it has been recognized in the previous study (9) (10) that the observed dielectric constants and conductivities of O/W emulsions do not agree with the eqs. [23] and [33] especially at higher concentrations.

Therefore it is concluded that the *Wagner* theory is not valid quantitatively for the interfacial polarization of a spherical dispersion.

4. A New Theory of the Interfacial Polarization

A) Derivation of the basic equation

The *Wagner* theory does not hold for the concentrated dispersions, since this theory was derived for a dilute dispersion system.

Now we assume that the *Wagner* relation [16] holds for the infinitesimal process such that an infinitesimally small quantity of the disperse phase is added to the dispersion

Table 1

Dielectric constants and electrical conductivities of the O/W emulsion calculated from the *Wagner* theory

Volume fraction of disperse phase	Dielectric constant		Relative conductivity	
	ε_h	ε_l	$\frac{\kappa_h}{\kappa_m}$	$\frac{\kappa_l}{\kappa_m}$
	by Eq. [23]		by Eq. [34]	
Φ	Eq. [23]	Eq. [32]	Eq. [34]	Eq. [33]
0.0	76.80	76.80	1.0000	1.0000
0.1	66.33	66.34	0.8572	0.8571
0.2	56.77	56.78	0.7274	0.7273
0.3	48.01	48.02	0.6089	0.6087
0.4	39.95	39.96	0.5002	0.5000
0.5	32.51	32.52	0.4002	0.4000
0.6	25.62	25.63	0.3078	0.3077
0.7	19.22	19.23	0.2223	0.2222
0.8	13.26	13.27	0.1429	0.1429
0.85	10.44	10.44	0.1053	0.1053
0.9	7.702	7.704	0.06900	0.06897
0.95	5.058	5.060	0.03392	0.03390
1.0	2.500	2.500	0.00000	0.00000

Table 2

Dielectric constants and electrical conductivities of the W/O emulsion calculated from the *Wagner* theory

Volume fraction of disperse phase	Dielectric constant		Relative conductivity	
	ε_h	ε_l	$\frac{\kappa_h}{\kappa_p}$	$\frac{\kappa_l}{\kappa_m}$
	by Eq. [23]		by Eq. [37]	
Φ	Eq. [23]	Eq. [35]	Eq. [37]	Eq. [36]
0.0	2.500	2.500	0.000000	1.000
0.1	3.249	3.333	0.001017	1.333
0.2	4.165	4.375	0.002511	1.750
0.3	5.309	5.714	0.004765	2.286
0.4	6.780	7.500	0.008295	3.000
0.5	8.740	10.00	0.01411	4.000
0.6	11.48	13.75	0.02436	5.500
0.7	15.59	20.00	0.04437	8.000
0.8	22.44	32.50	0.09001	13.00
0.85	27.91	45.00	0.1375	18.00
0.9	36.09	70.00	0.2271	28.00
0.95	49.70	145.0	0.4249	58.00
1.0	76.80	∞	1.0000	∞

system. Namely, for such an infinitesimally additional process, ε_m^* , ε^* and Φ in eq. [16] should be replaced as follows:

$$\varepsilon_m^* \rightarrow \varepsilon^*, \quad \varepsilon^* \rightarrow \varepsilon^* + \Delta\varepsilon^*, \quad \Phi \rightarrow \frac{\Delta\Phi'}{1-\Phi'} \quad ^{3)}$$

where $\Delta\varepsilon^*$ denotes an increment of ε^* associated with an infinitesimal addition of the disperse phase, and Φ' means the volume fraction of the disperse phase in this additional process. Therefore we obtain, for the relation between $\Delta\varepsilon^*$ and $\Delta\Phi'$,

$$\frac{2\varepsilon^* + \varepsilon_p^*}{3\varepsilon^*(\varepsilon^* - \varepsilon_p^*)} \Delta\varepsilon^* = \frac{-\Delta\Phi'}{1-\Phi'}. \quad [38]$$

This mathematical procedure is similar to that employed by *Bruggeman* (11), who derived the equation of the dielectric constant for a concentrated dispersion composed of non-conductive materials.

By succession of these infinitesimally additional processes given by eq. [38], the system will reach the final concentration Φ and dielectric constant ε^* . Hence we obtain, as a basic equation of ε^* ,

$$\int_{\varepsilon_m^*}^{\varepsilon^*} \frac{2\varepsilon^* + \varepsilon_p^*}{3\varepsilon^*(\varepsilon^* - \varepsilon_p^*)} d\varepsilon^* = \int_0^\Phi \frac{-d\Phi'}{1-\Phi'} = \log(1-\Phi). \quad [39]$$

Let us consider in fig. 2 a complex plane of ε^* , its abscissa and ordinate denoting the real part ε and the imaginary part γ respectively. A complex integral of the left side of eq. [39] is integrated along a certain contour C from ε_m^* to ε^* in the complex plane shown in fig. 2.

A contour C is determined by the correlation between ε and γ at each stage in a series of successive additions. Hence we can not know a priori an analytical representation of the contour C .

³⁾ If the volume of the dispersion medium and the disperse phase are denoted by v_m and v_p respectively, the volume fraction Φ' of the disperse phase is given by

$$\Phi' = \frac{v_p}{v_m + v_p}.$$

When a small quantity Δv_p of the disperse phase is added to the dispersion system, a quantity Φ in eq. [16] is expressed by

$$\Phi = \frac{\Delta v_p}{(v_m + v_p) + \Delta v_p}.$$

Differentiating Φ' with respect to v_p , we obtain

$$\begin{aligned} \Delta\Phi' &= \frac{\Delta v_p (v_m + v_p) - v_p \Delta v_p}{(v_m + v_p)^2} \\ &= \frac{v_m}{v_m + v_p} \cdot \frac{\Delta v_p}{v_m + v_p} = (1 - \Phi') \Phi. \end{aligned}$$

Accordingly

$$\Phi = \frac{\Delta\Phi'}{1-\Phi'}.$$

B) Evaluation of the complex integral

In a complex plane shown in fig. 2, let us consider a rectangular domain KLMN bounded by four straight lines parallel to the abscissa or the ordinate respectively through the points ε_m^* or ε^* . Since, in view of physical meanings, both ε and γ are the monotonously increasing or decreasing functions of Φ , it is obvious that the contour C is in the rectangular domain KLMN.

According to *Cauchy's* theorem (15), if a function

$$f(\varepsilon^*) = \frac{2\varepsilon^* + \varepsilon_p^*}{3\varepsilon^*(\varepsilon^* - \varepsilon_p^*)} \quad [40]$$

is regular in a domain KLMN, the integral of $f(\varepsilon^*)$ along a contour C is equal to those along the contour $\vec{KLLM}$ or $\vec{KNNM}$. The function $f(\varepsilon^*)$ given by eq. [40] has two poles, $\varepsilon^* = 0$ and $\varepsilon^* = \varepsilon_p^*$. One pole $\varepsilon^* = 0$ is the origin

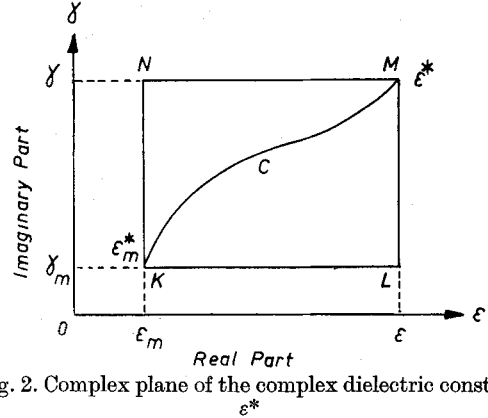


Fig. 2. Complex plane of the complex dielectric constant ε^* .

of the complex plane shown in fig. 2. Hence it is obvious that the pole $\varepsilon^* = 0$ is not in the domain KLMN. Whether another pole $\varepsilon^* = \varepsilon_p^*$ is in the domain KLMN or not, is still uncertain. If the integral of $f(\varepsilon^*)$ along the contour $\vec{KLLM}$ is equal to that along the contour $\vec{KNNM}$, then it is seen by *Morera's* theorem (15) that $f(\varepsilon^*)$ is a regular analytic function of ε^* in a domain KLMN, and a pole $\varepsilon^* = \varepsilon_p^*$ is not in this domain in consequence. The regularity of $f(\varepsilon^*)$ in the domain KLMN is proved as shown below.

The function $f(\varepsilon^*)$ is decomposed into its real part U and imaginary part V as follows.

$$f(\varepsilon^*) = -\frac{1}{3\varepsilon^*} + \frac{1}{\varepsilon^* - \varepsilon_p^*} \quad [41]$$

$$\begin{aligned} &= \left[-\frac{\varepsilon}{3(\varepsilon^2 + \gamma^2)} + \frac{\varepsilon - \varepsilon_p}{(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2} \right] \\ &\quad - j \left[\frac{\gamma}{3(\varepsilon^2 + \gamma^2)} - \frac{\gamma - \gamma_p}{(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2} \right] \\ &\equiv U - jV. \end{aligned}$$

[42]

Hence we have

$$U = -\frac{\varepsilon}{3(\varepsilon^2 + \gamma^2)} + \frac{\varepsilon - \varepsilon_p}{(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2} \quad [43]$$

and

$$V = \frac{\gamma}{3(\varepsilon^2 + \gamma^2)} - \frac{\gamma - \gamma_p}{(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2} \quad [44]$$

On the other hand, the integral of $f(\varepsilon^*)$ is decomposed into its real and imaginary parts as follows.

$$\int f(\varepsilon^*) d\varepsilon^* = \int U d\varepsilon - \int V d\gamma - j \left[\int U d\gamma + \int V d\varepsilon \right] \quad [45]$$

Each term of the integral [45] along the contour $\vec{KL}$ is given by

$$\int_{\varepsilon_m}^{\varepsilon} U d\varepsilon = \frac{1}{2} \log \frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma_m^2)^{1/3}} \quad [46]$$

$$\int_{\varepsilon_m}^{\varepsilon} V d\varepsilon = \frac{1}{3} \left[\tan^{-1} \frac{\varepsilon}{\gamma_m} - \tan^{-1} \frac{\varepsilon_m}{\gamma_m} \right] - \left[\tan^{-1} \frac{\varepsilon - \varepsilon_p}{\gamma_m - \gamma_p} - \tan^{-1} \frac{\varepsilon_m - \varepsilon_p}{\gamma_m - \gamma_p} \right] \quad [47]$$

and

$$\int V d\gamma = 0, \quad \int U d\gamma = 0. \quad [48]$$

Hence we obtain

$$\int_{\vec{KL}} f(\varepsilon^*) d\varepsilon^* = \frac{1}{2} \log \frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma_m^2)^{1/3}} + j \left[\frac{1}{3} \left(\tan^{-1} \frac{\varepsilon_m}{\gamma_m} - \tan^{-1} \frac{\varepsilon}{\gamma_m} \right) + \left(\tan^{-1} \frac{\varepsilon - \varepsilon_p}{\gamma_m - \gamma_p} - \tan^{-1} \frac{\varepsilon_m - \varepsilon_p}{\gamma_m - \gamma_p} \right) \right] \quad [49]$$

Similarly the integral [45] along the contour $\vec{LM}$ is

$$\int_{\vec{LM}} f(\varepsilon^*) d\varepsilon^* = \frac{1}{2} \log \frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2](\varepsilon^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma^2)^{1/3}} + j \left[\frac{1}{3} \left(\tan^{-1} \frac{\gamma}{\varepsilon} - \tan^{-1} \frac{\gamma_m}{\varepsilon} \right) + \left(\tan^{-1} \frac{\gamma_m - \gamma_p}{\varepsilon - \varepsilon_p} - \tan^{-1} \frac{\gamma - \gamma_p}{\varepsilon - \varepsilon_p} \right) \right] \quad [50]$$

Combining eq. [49] with eq. [50], we obtain the integral of $f(\varepsilon^*)$ along the contour $\vec{KLLM}$ as

$$\int_{\vec{KLLM}} f(\varepsilon^*) d\varepsilon^* = \frac{1}{2} \log \frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma^2)^{1/3}} - j \left[\frac{1}{3} \tan^{-1} \frac{\varepsilon \gamma_m - \varepsilon_m \gamma}{\varepsilon \varepsilon_m + \gamma \gamma_m} - \tan^{-1} \frac{(\varepsilon - \varepsilon_p)(\gamma_m - \gamma_p) - (\varepsilon_m - \varepsilon_p)(\gamma - \gamma_p)}{(\varepsilon - \varepsilon_p)(\varepsilon_m - \varepsilon_p) + (\gamma - \gamma_p)(\gamma_m - \gamma_p)} \right] \quad [51]$$

In a similar manner the integral of $f(\varepsilon^*)$ along the contour $\vec{KNNM}$ can be evaluated. As a consequence, it is seen that the integral of $f(\varepsilon^*)$ along the contour $\vec{KLLM}$ is equal to that along the contour $\vec{KNNM}$. Hence, in view of Morera's and Cauchy's theorems, it follows that

$$\int_{\vec{KLLM}} f(\varepsilon^*) d\varepsilon^* = \int_{\vec{KNNM}} f(\varepsilon^*) d\varepsilon^* = \int_{\varepsilon_m^*(C)} f(\varepsilon^*) d\varepsilon^* \quad [52]$$

Substitution of eq. [51] in eq. [39] and separation of real and imaginary parts give

$$\frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma^2)^{1/3}} = (1 - \Phi)^2 \quad [53]$$

and

$$\tan^{-1} \frac{\varepsilon \gamma_m - \varepsilon_m \gamma}{\varepsilon \varepsilon_m + \gamma \gamma_m} = 3 \tan^{-1} \frac{(\varepsilon - \varepsilon_p)(\gamma_m - \gamma_p) - (\varepsilon_m - \varepsilon_p)(\gamma - \gamma_p)}{(\varepsilon - \varepsilon_p)(\varepsilon_m - \varepsilon_p) + (\gamma - \gamma_p)(\gamma_m - \gamma_p)} \quad [54]$$

Two eqs. [53] and [54] are the general conditions to give ε and γ or κ .

Combining eq. [53] with eq. [54], we obtain the Bruggeman equation (11) extended to the complex numbers as

$$\frac{\varepsilon^* - \varepsilon_p^*}{\varepsilon_m^* - \varepsilon_p^*} \left(\frac{\varepsilon_m^*}{\varepsilon^*} \right)^{1/3} = 1 - \Phi \quad [55]$$

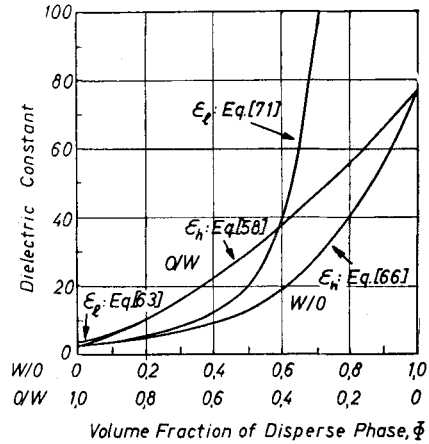


Fig. 3. Concentration dependence of dielectric constants for the O/W and W/O emulsions given by the new theory

C) Approximation for special cases

i) Case $\kappa_m \gg \kappa_p$ or $\gamma_m \gg \gamma_p$ (O/W type emulsion)

Here eqs. [53] and [54] lead to the following respectively:

$$\frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + \gamma_m^2](\varepsilon^2 + \gamma^2)^{1/3}} = (1 - \Phi)^2 \quad [56]$$

and

$$\tan^{-1} \frac{\varepsilon \gamma_m - \varepsilon_m \gamma}{\varepsilon \varepsilon_m + \gamma \gamma_m} = 3 \tan^{-1} \frac{(\varepsilon - \varepsilon_p)\gamma_m - (\gamma - \gamma_p)(\varepsilon_m - \varepsilon_p)}{(\varepsilon - \varepsilon_p)(\varepsilon_m - \varepsilon_p) + (\gamma - \gamma_p)\gamma_m} \quad [57]$$

a) At higher frequencies ($\omega \rightarrow \infty$)

By conditions $\varepsilon_m \gg \gamma_m$, $\varepsilon_p \gg \gamma_p$ and $\varepsilon \gg \gamma$, eq. [56] is reduced to

$$\frac{\varepsilon_h - \varepsilon_p}{\varepsilon_m - \varepsilon_p} \left(\frac{\varepsilon_m}{\varepsilon_h} \right)^{1/3} = 1 - \Phi \quad [58]$$

and eq. [57] is reduced to

$$\frac{\kappa_h}{\kappa_m} = \frac{\varepsilon_h (\varepsilon_h - \varepsilon_p) (2\varepsilon_m + \varepsilon_p)}{\varepsilon_m (\varepsilon_m - \varepsilon_p) (2\varepsilon_h + \varepsilon_p)}. \quad [59]$$

b) At lower frequencies ($\omega \rightarrow 0$)

By conditions $\gamma_m \gg \varepsilon_m$, $\gamma_p \gg \varepsilon_p$ and $\gamma \gg \varepsilon$, eqs. [56] and [57] are reduced to

$$\frac{\kappa_l - \kappa_p}{\kappa_m} \left(\frac{\kappa_m}{\kappa_l} \right)^{1/3} = 1 - \Phi \quad [60]$$

and

$$\varepsilon_l = \frac{\kappa_l [3\varepsilon_p \kappa_m + (2\varepsilon_m - 3\varepsilon_p) \kappa_l]}{\kappa_m (2\kappa_l + \varepsilon_p)}. \quad [61]$$

Especially if it is admissible to put $\kappa_l \gg \kappa_p$, eqs. [60] and [61] are reduced to

$$\frac{\kappa_l}{\kappa_m} = (1 - \Phi)^{3/2} \quad [62]$$

and

$$\frac{2\varepsilon_l - 3\varepsilon_p}{2\varepsilon_m - 3\varepsilon_p} = (1 - \Phi)^{3/2}. \quad [63]$$

ii) Case $\kappa_p \gg \kappa_m$ or $\gamma_p \gg \gamma_m$ (W/O type emulsion)

Here eqs. [53] and [54] lead to the following respectively:

$$\frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2] (\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + \gamma_p^2] (\varepsilon^2 + \gamma^2)^{1/3}} = (1 - \Phi)^2 \quad [64]$$

and

$$\begin{aligned} \tan^{-1} \frac{\varepsilon \gamma_m - \varepsilon_m \gamma}{\varepsilon \varepsilon_m + \gamma \gamma_m} \\ = 3 \tan^{-1} \frac{(\varepsilon_p - \varepsilon) \gamma_p - (\gamma_p - \gamma) (\varepsilon_p - \varepsilon_m)}{(\varepsilon_p - \varepsilon) (\varepsilon_p - \varepsilon_m) + (\gamma_p - \gamma) \gamma_p}. \end{aligned} \quad [65]$$

a) At higher frequencies ($\omega \rightarrow \infty$)

By conditions $\varepsilon_m \gg \gamma_m$, $\varepsilon_p \gg \gamma_p$ and $\varepsilon \gg \gamma$, eqs. [64] and [65] are reduced to

$$\frac{\varepsilon_p - \varepsilon_h}{\varepsilon_p - \varepsilon_m} \left(\frac{\varepsilon_m}{\varepsilon_h} \right)^{1/3} = 1 - \Phi \quad [66]$$

and

$$\frac{\kappa_h}{\kappa_p} = \frac{3\varepsilon_h (\varepsilon_h - \varepsilon_m)}{(\varepsilon_p + 2\varepsilon_h) (\varepsilon_p - \varepsilon_m)}. \quad [67]$$

b) At lower frequencies ($\omega \rightarrow 0$)

By conditions $\gamma_m \gg \varepsilon_m$, $\gamma_p \gg \varepsilon_p$ and $\gamma \gg \varepsilon$, eqs. [64] and [65] are reduced to

$$\frac{\kappa_p - \kappa_l}{\kappa_p} \left(\frac{\kappa_m}{\kappa_l} \right)^{1/3} = 1 - \Phi \quad [68]$$

and

$$\varepsilon_l = \varepsilon_m \frac{\kappa_l (\kappa_p - \kappa_l)}{\kappa_m (\kappa_p + 2\kappa_l)}. \quad [69]$$

Especially if it is admissible to put $\kappa_p \gg \kappa_l$, eqs. [68] and [69] are reduced to

$$\frac{\kappa_l}{\kappa_m} = \frac{1}{(1 - \Phi)^3} \quad [70]$$

and

$$\varepsilon_l = \varepsilon_m \frac{1}{(1 - \Phi)^3}. \quad [71]$$

Eqs. [58] and [66] are the same as the Bruggeman equation (11) of the dielectric constant for a spherical dispersion.

Table 3

Dielectric constants and electrical conductivities of the O/W emulsion calculated from the new theory

Volume fraction of disperse phase Φ	Dielectric constant		Relative conductivity	
	ε_h	ε_l	$\frac{\kappa_h}{\kappa_m}$	$\frac{\kappa_l}{\kappa_m}$
	Eq. [58]	Eq. [63]	Eq. [59]	Eq. [62]
0.0	76.80	76.80	1.0000	1.0000
0.1	66.11	66.12	0.8539	0.8538
0.2	56.00	56.02	0.7158	0.7155
0.3	46.50	46.54	0.5861	0.5857
0.4	37.65	37.70	0.4654	0.4648
0.5	29.51	29.58	0.3544	0.3536
0.6	22.13	22.23	0.2541	0.2530
0.7	15.60	15.75	0.1659	0.1643
0.8	10.04	10.28	0.09175	0.08944
0.85	7.671	7.994	0.06082	0.05809
0.9	5.605	6.060	0.03473	0.03162
0.95	3.873	4.567	0.01419	0.01118
1.0	2.500	2.500	0.00000	$\sim \kappa_p/\kappa_m$

Table 4

Dielectric constants and electrical conductivities of the W/O emulsion calculated from the new theory

Volume fraction of disperse phase Φ	Dielectric constant		Relative conductivity	
	ε_h	ε_l	$\frac{\kappa_h}{\kappa_p}$	$\frac{\kappa_l}{\kappa_m}$
	Eq. [66]	Eq. [71]	Eq. [67]	Eq. [70]
0.0	2.500	2.500	0.000000	1.000
0.1	3.317	3.429	0.001312	1.372
0.2	4.499	4.883	0.004231	1.953
0.3	6.242	7.289	0.01056	2.915
0.4	8.852	11.57	0.02402	4.630
0.5	12.79	20.00	0.05190	8.000
0.6	18.69	39.06	0.1070	15.63
0.7	27.33	92.59	0.2084	37.04
0.8	39.51	312.5	0.3789	125.0
0.85	47.14	740.7	0.4966	296.3
0.9	55.87	2500.	0.6386	1000.
0.95	65.75	20000.	0.8062	8000.
1.0	76.80	0.0	1.0000	$\sim \kappa_p/\kappa_m$

In tables 3, 4 and fig. 3 are shown the theoretical values and curves obtained respectively by eqs. [58]–[63] and [66]–[71], where the dielectric constants of the oil and the water phases are taken to be 2.50 and 76.8 (values at 30°C) respectively. As is easily known from the assumptions used, eqs. [62, 63, 70] and [71] do not hold in the vicinity of $\Phi = 1$. Since the function $f(\varepsilon^*)$ is not regular at $\varepsilon^* = \varepsilon_p^*$, the results of the present theory are not valid at $\Phi = 1$. Hence, in the lowest rows ($\Phi = 1$) of tables 3 and 4 are shown the limiting values given by eqs. [60, 61, 68] and [69] at higher concentrations ($\Phi \rightarrow 1$).

D) Discussion

It should be noticed from tables 3, 4 and fig. 3 that both differences between ε_l and

ϵ_h and between κ_h and κ_i are very small over the whole range of concentration for O/W emulsions, while these differences are very large at higher concentrations of W/O emulsions. Therefore it is concluded from the present theory that the dispersions of dielectric constants and conductivities are too small to be observed in O/W emulsions and are considerable in W/O emulsions. Strictly speaking, the differences between ϵ_i and ϵ_h are rather appreciable at very high concentrations above 90%, where it may be difficult to prepare actual emulsions.

In the previous dielectric measurements (9, 10) of O/W emulsions over a wide range of frequency, no dielectric dispersion due to the interfacial polarization could be observed, and the dielectric constants and the conductivities were expressed well by eqs. [58] and [62] respectively for concentrations up to 85%. Recently Rue and Tobias (16) have observed conductivities of suspensions of glass beads in ZnBr₂ solutions, concentrations of the disperse phase being less than 40%. It was found from their results that the dependence of the conductivity of random suspensions on volume fraction is expressed better by eq. [62] rather than by eq. [33].

Author's results observed on the dielectric dispersions due to the interfacial polarization in W/O emulsions will be published in the later paper, where the validity of eqs. [66, 71, 70] and [67] would be examined.

The small differences between ϵ_i and ϵ_h for the O/W emulsion are attributable to the approximate equality between eqs. [63] and [58] for smaller values of ϵ_p . Therefore it may be suggested that, if the dielectric constant of the oil phase is much larger, the dielectric dispersion due to the interfacial polarization may be observed even in emulsions of the O/W type. Such an example of the dielectric dispersion in emulsions of the O/W type will be reported later.

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Summary

The interfacial polarization of spherical dispersions is dealt with theoretically, and is discussed for the actual cases of emulsions.

First it is shown that the *Maxwell* theory is not valid qualitatively for the dielectric properties of emulsions. Next the general solution of the *Wagner* theory is derived. This solution shows that a dielectric dispersion due to the interfacial polarization is characterized by a single relaxation time. It is found that the *Wagner* theory is not valid quantitatively for the actual emulsions.

Finally a new theory of the interfacial polarization is developed on the assumption that the *Wagner* equation holds for an infinitesimally increasing process in concentration of the dispersion system. Several dielectric properties of emulsions are interpreted quantitatively by this new theory. Dielectric dispersions due to the interfacial polarization have so far been observed only in the W/O emulsions, and no example of the interfacial polarization has been found in usual emulsions of the O/W type yet. According to the new theory it is suggested that, if the dielectric constant of the oil phase is much larger, the dielectric dispersion due to the interfacial polarization may be observed even in emulsions of the O/W type.

Zusammenfassung

Die Grenzflächenpolarisation kugelförmiger Dispersionen wird theoretisch behandelt und an Fällen von Emulsionen diskutiert.

Zuerst wird gezeigt, daß die *Maxwellsche* Theorie für die dielektrischen Eigenschaften von Emulsionen nicht einmal qualitativ gilt. Als nächstes wird eine allgemeinere Lösung der *Wagner-Theorie* abgeleitet. Diese zeigt, daß eine dielektrische Dispersion einer Grenzflächenpolarisation, durch eine einzelne Relaxationszeit charakterisiert, zuzuschreiben ist. Auch die *Wagnersche* Theorie gilt für wirkliche Emulsionen nicht quantitativ.

Schließlich wird eine neue Theorie der Grenzflächenpolarisation auf der Basis entwickelt, daß die *Wagnersche* Gleichung für ein infinitesimales Anwachsen der Konzentrationen von dispersen Systemen gilt. Zahlreiche dielektrische Eigenschaften von Emulsionen sind mit dieser neuen Theorie quantitativ interpretierbar.

Dielektrische Dispersion durch Grenzflächenpolarisation wurde bisher nur in W/O-Emulsionen beobachtet, und kein Beispiel von Grenzflächenpolarisation ließ sich in gebräuchlichen Emulsionen des O/W-Typs bis heute auffinden. Nach der neuen Theorie wird vermutet, daß die dielektrische Dispersion auf Grund von Grenzflächenpolarisation auch in Emulsionen des O/W-Typs beobachtet werden könnte, falls die Dielektrizitätskonstante der Ölphase viel größer wäre.

References

- 1) Fricke, H. and H. J. Curtis, *J. Phys. Chem.* **41**, 729 (1937).
- 2) Maxwell, J. C., *Electricity and Magnetism* Vol. I, p. 452 (London 1892); von Hippel, A. R., *Dielectrics and Waves*, p. 228 (New York 1954).
- 3) Wagner, K. W., *Arch. Electrotechn.* **2**, 371 (1914).
- 4) Fricke, H., *J. Phys. Chem.* **57**, 934 (1953).
- 5) Sillars, R. W., *J. Inst. Elec. Engrs. (London)* **80**, 378 (1937).
- 6) Hamon, B. V., *Aust. J. Phys.* **6**, 304 (1953).
- 7) Kharadly, M. M. Z. and W. Jackson, *Proc. Inst. Elec. Engrs. (London)* **100** (III), 199 (1953).

- 8) Dryden, J. S. and R. J. Meakins, Proc. Phys. Soc. **70B**, 427 (1957).
 9) Hamai, T., N. Koizumi and R. Gotoh, Kolloid-Z. **167**, 41 (1959).
 10) Hamai, T., N. Koizumi, T. Sugano and R. Gotoh, Kolloid-Z. **171**, 20 (1960).
 11) Bruggeman, D. A. G., Ann. Physik **24**, 636 (1935).
 12) Becher, P., Emulsions, p. 327 (New York 1957).
 13) Wiener, O., Abh. Math. Phys. kgl. Sächs. Ges. Wiss. **32**, 509 (1912).
 14) Fricke, H., Phys. Rev. **24**, 575 (1924).
 15) See, for example, Titchmarsh, E. C., The Theory of Functions Chap. II (London 1932).
 16) Rue, R. E. D. L. and C. W. Tobias, J. Electrochem. Soc. **106**, 827 (1959).

From the Chemical Laboratories, University of Allahabad, Allahabad (India)

The Influence of Ultrasonic Waves on the p_H , Conductivity and Gelling Time of Some Succinate Sols

By Y. D. Upadhyaya and S. P. Mushran

With 6 tables

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The influence of ultrasonic radiations on various colloidal systems has been investigated by several workers (1). In recent publications from this laboratory Prakash and associates (2) have made a detailed study of the influence of the ultrasonic waves on the stability of a number of sols.

In this paper our results with some gel forming succinate sols are reported. The sols investigated are (A) Zirconium Succinate, (B) Ferric Succinate and (C) Titanium Succinate.

Preparation of the Sols

(A) Zirconium succinate sol was prepared by adding sodium succinate, taken short of precipitation, to a slight excess of zirconium nitrate solution. 150 mls. of 0.357 M zirconium nitrate solution and 90 mls. of 0.25 M sodium succinate with a total volume of 990.00 mls. gave a clear sol, which was purified by dialysis for a period of 96 hrs. The sol set to transparent gels with suitable concentration of potassium chloride.

(B) Ferric succinate sol was prepared by method described by Bose and Mushran (3). Freshly precipitated and thoroughly washed ferric succinate was dissolved in 300 mls. of M/2 succinic acid by shaking in a Microid shaker. The total volume was kept 575 mls. A wine red coloured transparent sol was obtained. The sol set to transparent gels with potassium-chloride or potassium sulphate solutions of suitable concentration.

(C) Titanium succinate was prepared by adding sodium succinate, taken short of precipitation, to a slight excess of titanium tetra-chloride solution. 160 mls. of $TiCl_4$ containing 11.834 gms. of titanium per litre and 80 mls. of 0.25 M sodium succinate, with a total volume of 320 mls. gave a clear sol, which was purified by dialysis for a period of 18 hours. The sol gave transparent gels with suitable concentration of potassium chloride.

Ultrasonic Apparatus

The source of ultrasonic power was a Mullard high frequency generator with a barium titanate transducer working at 225 watts, the transducer furnishing a frequency of 1 MC/s. The transducer was immersed in a circulation bath maintained at a constant temperature. The sols were kept in jena glass vessel suitably clamped and immersed in the circulation bath. The distance between the vessel and the transducer was carefully adjusted in a manner that a good fountain was obtained.

Electrometric Measurements

The conductivity measurements were made using a L and N drum type bridge, length 1000 cms, an audio-oscillator with a tuning fork giving a constant frequency of 1000 c/sec, a tunable head phone and an immersion cell of the type GM 4221.

p_H measurements were carried out using a L and N p_H meter with a glass electrode. The instrument was properly standardised by the determination of its asymmetric potential using a standard solution of potassium hydrogen phthalate.

All electrometric measurements were made at a bath temperature of 35° C.

Gelling Time

Gelling time was determined by the method described by Bose and Mushran (4) using drops of barium sulphate suspension. The results on the influence of ultrasonic waves on the p_H and conductivity are given in tables 1-3, and in tables 4-6 are recorded the influences on gelling time.

Table 1

Showing the effect on p_H and conductivity of Zirconium Succinate Sol

Exposures	0	1	2	3	4 hrs.
p_H	3.10	3.08	3.06	3.04	3.00
Sp. Cond. $\times 10^4$	4.363	4.526	4.751	5.010	5.328 mhos.

Table 2

Showing the effect on p_H and conductivity of Ferric Succinate Sol

Exposures	0	1	2	3	4 hrs.
p_H	2.65	2.62	2.60	2.59	2.58
Sp. Cond. $\times 10^4$	2.224	2.279	2.325	2.366	2.398 mhos.

Table 3

Showing the effect on p_H and conductivity of Titanium Succinate Sol

Exposures	0	1	2	3	4 hrs.
p_H	2.55	2.50	2.45	2.41	2.38
Sp. Cond. $\times 10^3$	4.929	5.032	5.093	5.132	5.159 mhos.

A THEORY OF THE LOW-FREQUENCY DIELECTRIC DISPERSION OF COLLOIDAL PARTICLES IN ELECTROLYTE SOLUTION^{1,2}

BY GERHARD SCHWARZ

Electromedical Division, Moore School of Electrical Engineering, University of Pennsylvania, Philadelphia 4, Pa.

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It is proposed that the low-frequency dielectric dispersion observed with colloidal particles suspended in electrolytes is due to the polarization of the counterion atmosphere around the particles by an external electric field. A simplified theory of this effect is developed for particles of spherical shape. The displacement of counterions in the double layer is shown to be equivalent to the existence of a surface capacitance displaying a diffusion-controlled relaxation. This effect can be expressed by an additional "apparent" dielectric constant of the suspended particles, exceeding their actual dielectric constant at low frequencies by many orders of magnitude. The theory is applied to the special case of particles suspended in a fairly well conducting electrolyte. Such colloidal systems display a particularly well pronounced low frequency dispersion effect. The results of previous experimental investigations with suspensions of uniformly sized polystyrene spheres are used in order to test the theory. Assuming reasonable values for the density and the mobility of the counterions in the double layer, the theory agrees very well with the static dielectric increment and the characteristic frequency of the dispersion as experimentally observed. A distribution of activation energies of the counterion motion on the particle surface is introduced to account for the relaxation spectrum of the dispersion curves.

Introduction

Suspensions of colloidal particles in aqueous electrolyte solutions generally display unusually high dielectric constants at low frequencies.³ Thus, for sphere-like particles about 1μ in diameter, the dielectric constant may rise well above 10^4 below 1 kc.^4 However, the experimental difficulties are considerable. Due to the appreciable electrolytic conductivity and the low frequencies used, it was possible only recently to achieve sufficient accuracy in the dielectric measurements and in the control of electrode polarization to take complete curves of the low frequency dielectric dispersion. For example, while investigating the dielectric behavior of biological material down to about 10 c.p.s. , Schwan and co-workers found a well pronounced dispersion in the low frequency range.⁵ This so-called α -dispersion perhaps is the same effect which occurs in the case of non-biological colloids. In fact, similar dispersion curves for fat particles and polystyrene spheres suspended in electrolyte solution have been measured by Schwan, *et al.*^{3,5,6}

It has been suggested that displacements of electrical charges within the interface between the particle and the electrolyte may be responsible for the α -dispersion (*e.g.*, Murphy and Lowry⁷ and Schwan⁵). Colloidal particles are electrically charged by fixed or adsorbed ions and are surrounded by counterions, forming an electric double layer or ionic atmosphere. A shift of ions within this layer by an external electrical field would result in a dielectric dispersion. Yet merely introducing an ohmic surface conductance around the particle into the Maxwell-Wagner theory of

heterogeneous dielectrics,⁸ as done by Miles and Robertson⁹ and more recently by O'Konski,¹⁰ cannot account for the experimental facts.³ As has been demonstrated by Fricke and Curtis⁴ and Schwan, *et al.*³, only a frequency-dependent capacitive part of the surface admittance can explain the measured high dielectric constants. Unfortunately there has been no detailed justification of such a term so far.

Apparently it is possible to explain the effect only by a consideration of the microphysical structure of the double layer and the behavior of an individual charge carrier. Kirkwood and Shumaker¹¹ have considered this effect in the special case of the proton fluctuation around protein molecules by employing a statistical method. They demonstrated that the dielectric properties of protein solutions at radiofrequencies may be explained in this way just as well as by orientation of permanent dipoles, the latter being the commonly adopted interpretation.¹² In the general case of non-polar, spherical, and dielectrically-isotropic particles, an orientation mechanism is out of the question.¹³ But a deformation of the ionic atmosphere due to an applied field can be anticipated and must be reflected in a corresponding dielectric relaxation. Whether this model explains the above-quoted dielectric phenomenon will be the subject of this paper.

Counterions on the surface of the highly-charged colloidal particle are strongly bound by electrostatic attraction. In order to escape from the surface into the free solution, they have to overcome a high potential barrier. Along the surface, however, they can be moved much more easily. Thus

(1) Presented before the Division of Physical Chemistry of the American Chemical Society, 140th National Meeting, Chicago, Illinois, Sept., 1961.

(2) This investigation was supported by National Institute of Health Grant H 1253 (C9).

(3) H. P. Schwan, G. Schwarz, J. Maczuz, and H. Pauly, *J. Phys. Chem.*, **66**, 2626 (1962).

(4) H. Fricke and H. J. Curtis, *ibid.*, **41**, 729 (1937). Further reference of older papers will be found here.

(5) H. P. Schwan, *Advan. Biol. Med. Phys.*, **5**, 147 (1957).

(6) H. P. Schwan and J. Maczuz, *Proc. Natl. Biophys. Conf.*, 1st, Columbus, Ohio, 348 (1959).

(7) E. J. Murphy and H. H. Lowry, *J. Phys. Chem.*, **34**, 598 (1930).

(8) K. W. Wagner, *Arch. Elektrotech.*, **3**, 83 (1914).

(9) J. B. Miles, Jr., and H. P. Robertson, *Phys. Rev.*, **40**, 583 (1932).

(10) C. T. O'Konski, *J. Phys. Chem.*, **64**, 605 (1960).

(11) J. G. Kirkwood and J. B. Shumaker, *Proc. Natl. Acad. Sci. U. S.*, **38**, 855 (1952).

(12) J. L. Oncley, in Cohn and Edsall, "Proteins, Amino Acids and Peptides," Reinhold Publ. Corp., New York, N. Y., 1943.

(13) Moreover, an orientation mechanism would not account for the dielectric dispersion in question. The characteristic frequency of such an effect is inversely proportional to the third power of the diameter, according to Debye's well known equation of the relaxation time of oriented, spherical dipoles. This is in contrast to the experimental evidence discussed by Schwan, *et al.*³ Furthermore, the quantitative value of such a characteristic frequency is found to be far below those of interest here.

they will be moved tangentially by an external field, polarizing the ion atmosphere and inducing an electric dipole moment of the particle. This concept has proved to be successful for rod-like particles in recent investigations on the orientation field effect of polyelectrolytes.¹⁴⁻¹⁷ In place of a rigorous and difficult theory of the polarization of the counterion atmosphere, a fairly simplified model of the colloidal system is used in the calculations. Despite this rather crude model, the theoretical results yield a surprisingly good qualitative and quantitative representation of the experimental facts previously reported.³

Theory

Effect of an Electric Field on a Sphere with a Counterion Layer.—The colloidal particle will be introduced as a sphere of radius R and complex conductivity (in the presence of an alternating field of angular frequency ω)

$$K_i = \kappa_i + i\omega\epsilon_r\epsilon_i \quad (1)$$

suspended in an electrolyte solution of complex conductivity

$$K_a = \kappa_a + i\omega\epsilon_r\epsilon_a \quad (2)$$

where κ_i , κ_a and ϵ_i , ϵ_a are the conductivities and the (relative) dielectric constants of the particle and the electrolyte, respectively. $\epsilon_r = 8.854 \times 10^{-12}$ f./m. is the absolute dielectric constant of free space. The sphere is electrically charged by fixed or adsorbed ions and is surrounded by counterions of electric charge e_0 and mechanical mobility u (i.e., velocity per unit force), which can be moved only along the surface but not perpendicular to it. The electric double layer is assumed to be infinitely thin (in comparison with the radius of the particle) with a total net charge equal to zero. When the colloidal particle is not affected by external forces, the counterions are randomly distributed, so that the surface density σ (ions per unit area) is a constant σ_0 . An applied external electric field E will result in a shift of the counterions along the surface, thus disturbing the random distribution. The counterion surface density may generally be denoted

$$\sigma = \sigma_0 + \bar{\sigma} \quad (3)$$

where $\bar{\sigma}$ represents the counterion surface density in excess of the constant value σ_0 .

It is suitable now to introduce spherical coordinates r and θ with $r = 0$ at the center of the sphere and $\theta = 0$ in the direction of E . If ψ_s is the electric potential on the surface of the sphere, the corresponding electric field

$$E_s = -\frac{1}{R} \times \frac{\partial \psi_s}{\partial \theta} \quad (4)$$

causes a surface flux of counterions

$$j_E = -\frac{e_0\sigma\mu}{R} \times \frac{\partial \psi_s}{\partial \theta} \quad (5)$$

In addition to the flux induced by the impressed electric field there is a counteracting diffusion flux which tends to re-establish the random distribution

$$j_D = -\frac{ukT}{R} \frac{\partial \sigma}{\partial \theta} \quad (6)$$

Since the counterions can move only tangentially to the surface, any change of the counterion density must be due to those fluxes. Consequently the equation of continuity can be written as

$$\frac{\partial \sigma}{\partial t} - \text{div} (j_E + j_D) = 0 \quad (7)$$

This is a partial differential equation for σ , namely

$$\frac{\partial \sigma}{\partial t} = \frac{ukT}{R^2} \times \frac{1}{\sin \theta} \times \frac{\partial}{\partial \theta} \left\{ \sin \theta \frac{\partial \sigma}{\partial \theta} + \frac{e_0}{kT} \frac{\partial \psi_s}{\partial \theta} \sigma \sin \theta \right\} \quad (8)$$

To compare the possible energy gain of a counterion responding to the external field with the energy due to thermal movement, the quantity

$$\gamma = \frac{e_0RE}{kT} \quad (9)$$

may be introduced. For all cases discussed here

$$\gamma \ll 1 \quad (10)$$

It stands to reason that the electric field along the surface is of the same order of magnitude as the external field or even smaller, therefore the factor of $\sigma \sin \theta$ in eq. 8 is also small compared with unity

$$\left| \frac{e_0}{kT} \times \frac{\partial \psi_s}{\partial \theta} \right| = \frac{e_0R}{kT} \cdot |E_s| \lesssim \gamma \ll 1 \quad (11)$$

This indicates that the counterion density will deviate only little from the random distribution, that is

$$\sigma - \sigma_0 = \bar{\sigma} \ll \sigma_0 \quad (12)$$

Because of relations 11 and 12, σ_0 can be substituted for σ in the second term on the right-hand side of eq. 8. Further simplification is possible if the external field is assumed to be alternating with the angular frequency ω , so that

$$E = E_0 e^{i\omega t} \quad (13)$$

Under these circumstances one may write

$$\frac{\partial \sigma}{\partial t} = \frac{\partial \bar{\sigma}}{\partial t} = i\omega \bar{\sigma} \quad (14)$$

Using this result the equation for σ (8) takes the form

$$i\omega \bar{\sigma} = \frac{ukT}{R^2} \frac{1}{\sin \theta} \frac{d}{d\theta} \left\{ \sin \theta \frac{d\bar{\sigma}}{d\theta} + \frac{e_0\sigma_0}{kT} \sin \theta \frac{d\psi_s}{d\theta} \right\} \quad (15)$$

(14) G. Schwarz, *Z. Physik*, **145**, 563 (1956).

(15) M. Eigen and G. Schwarz, *J. Colloid Sci.*, **12**, 181 (1957).

(16) C. T. O'Konski and A. J. Haltner, *J. Am. Chem. Soc.*, **79**, 5634 (1957).

(17) G. Schwarz, *Z. physik. Chem. (Frankfurt)*, **19**, 286 (1959).

The potential ψ_s must be derived from the over-all distribution of the electric potential. The potential in the electrolyte ψ_a as well as the potential inside the sphere ψ_i has to satisfy the Laplace equation

$$\Delta\psi_a = 0 \quad (r > R) \quad (16)$$

$$\Delta\psi_i = 0 \quad (r < R) \quad (17)$$

The boundary conditions are

$$1. \quad \lim_{r \rightarrow R} \psi_a = \lim_{r \rightarrow R} \psi_i = \psi_s \quad (18)$$

$$2. \quad \psi_a \rightarrow -Er \cos \theta \text{ for } r \rightarrow \infty \quad (19)$$

$$3. \quad \psi_i \text{ must remain finite for } r \rightarrow 0 \quad (20)$$

$$4. \quad K_a \frac{\partial \psi_a}{\partial r} - K_i \frac{\partial \psi_i}{\partial r} = -i\omega\epsilon_0\bar{\sigma} \text{ for } r \rightarrow R \quad (21)$$

The last condition results from the fact that the temporal change of the total surface density of charge depends completely upon the net influx of electrical charges perpendicular and parallel to the surface. The parallel flux is provided by the counterion motion, while the conductivities of the electrolyte and the particle are responsible for the flux in the perpendicular direction.

Now the three equations 15, 16, and 17 can be solved by standard methods. Using a representation of $\bar{\sigma}$ and ψ_s by means of Legendre polynomials $P_n(\cos \theta)$

$$\bar{\sigma} = \sum \alpha_n P_n(\cos \theta) \quad (22)$$

$$\psi_s = \sum \beta_n P_n(\cos \theta) \quad (23)$$

eq. 15 yields the following relation between the coefficients α_n and β_n of these series

$$\alpha_n = - \frac{1}{1 + i\omega \frac{R^2}{n(n+1)ukT}} \times \frac{\epsilon_0\sigma_0}{kT} \times \beta_n \quad (24)$$

As is well known the solutions of eq. 16 and 17 can be represented as

$$\psi_a = \sum \{A_{1n}r^n + B_{1n}r^{-n-1}\} P_n(\cos \theta) \quad (25)$$

and

$$\psi_i = \sum \{A_{2n}r^n + B_{2n}r^{-n-1}\} P_n(\cos \theta) \quad (26)$$

Taking into account the above-mentioned boundary conditions as well as eq. 22, 23, and 24 we find that all but the terms with $n = 1$ vanish. Under these circumstances eq. 24 is equivalent to

$$\bar{\sigma} = - \frac{1}{1 + i\omega\tau} \times \frac{\epsilon_0\sigma_0}{kT} \times \psi_s \quad (27)$$

where

$$\tau = \frac{R^2}{2ukT} \quad (28)$$

Thus we obtain a simple relaxation of the change in the counterion surface density. Obviously the relaxation time τ is determined by the rate of diffusion of the counterions along the particle. With

respect to the actual electric potential ψ_s on the particle surface, the relaxation of the counterion layer is always diffusion-controlled. The same is not necessarily true with regard to the applied field, since there is in general a frequency-dependent phase shift between E_s and E .

According to eq. 5 and 6 the surface current density is

$$i_s = e_0(j_E + j_D) = - \frac{e_0^2 u \sigma}{R} \frac{\partial \psi_s}{\partial \theta} - \frac{e_0 u k T}{R} \times \frac{\partial \sigma}{\partial \theta} \quad (29)$$

In the first term σ_0 may be substituted for σ and in the second term $\bar{\sigma}$ for σ . Taking advantage of eq. 27

$$i_s = - \frac{i\omega\tau}{1 + i\omega\tau} \times \lambda_0 \frac{\partial \psi_s}{R \partial \theta} = \frac{i\omega\tau}{1 + i\omega\tau} \lambda_0 E_s \quad (30)$$

The quantity

$$\lambda_0 = e_0^2 \sigma_0 u \quad (31)$$

has been called surface conductivity by O'Konski,¹⁰ who used the relation $i_s = \lambda_0 E_s$. As can be seen from eq. 30, this relation is indicated only at very high frequencies or in cases where the diffusion can be neglected. In general the diffusion effect results in a phase shift between current and field on the surface. Thus λ_0 can be replaced by a complex surface conductivity

$$\lambda = \frac{i\omega\tau}{1 + i\omega\tau} \lambda_0 \quad (32)$$

It appears to be more suitable, however, to represent the current-field correlation by a surface dielectric constant ϵ_s^* instead of a surface conductivity, so that

$$\epsilon_s^* = \frac{1}{1 + i\omega\tau} \times \frac{\lambda_0\tau}{\epsilon_r} \quad (33)$$

The electric potentials ψ_a and ψ_i , as evaluated from the above-mentioned system of equations, can now be written

$$\psi_a = -Er \cos \theta + \frac{\bar{K}_i - K_a}{\bar{K}_i + 2K_a} R^3 E \frac{\cos \theta}{r^2} \quad (r \geq R) \quad (34)$$

$$\psi_i = - \frac{3K_a}{2K_a + \bar{K}_i} Er \cos \theta \quad (r \leq R) \quad (35)$$

where

$$\bar{K}_i = K_i + i\omega\epsilon_r \frac{2\epsilon_s^*}{R} \quad (36)$$

ψ_i becomes ψ_s by merely substituting R for r . Evidently the dielectric behavior of the sphere with the counterion layer is identical with that of a sphere without such a layer but an apparent dielectric constant

$$\epsilon_i = \epsilon_i + \frac{2\epsilon_a^*}{R} \quad (37)$$

The additional dielectric constant

$$\Delta\epsilon_i = \frac{2\epsilon_a^*}{R} = \frac{1}{1 + i\omega\tau} \frac{e_0^2 R \sigma_0}{\epsilon_r kT} \quad (38)$$

may rise far above the actual dielectric constant ϵ_i .

The Dielectric Behavior of a Suspension of Spheres with a Counterion Layer.—If p is the volume concentration of spheres with a complex conductivity $\bar{K}_i$ suspended in material of complex conductivity K_a , the complex conductivity K of the system can be evaluated from the relation^{3,18}

$$\frac{K - K_a}{K + 2K_a} = p \frac{\bar{K}_i - K_a}{\bar{K}_i + 2K_a} \quad (39)$$

This equation was essentially derived by Maxwell¹⁹ for small values of p but has been found useful even for a fairly high volume concentration. The theoretical determination of K by means of eq. 39 can be eased if we assume a fairly well conducting electrolyte with

$$\kappa_a \gg \frac{\lambda_0}{R} \quad (40)$$

Since ϵ_i and κ_i are much smaller than ϵ_a and κ_a , respectively, this relation is equivalent to

$$|\bar{K}_i| \ll |K_a| \quad (41)$$

Under these circumstances K follows from eq. 39 as

$$K = \frac{1-p}{1+\frac{p}{2}} K_a + \frac{9}{4} \frac{p}{\left(1+\frac{p}{2}\right)^2} \bar{K}_i \quad (42)$$

This expression deviates from the complex conductivity K_∞ due to the simple Maxwell-Wagner effect only by the term $\bar{K}_i$ in place of K_i . Thus

$$K - K_\infty = \frac{9}{4} \frac{p}{\left(1+\frac{p}{2}\right)^2} (\bar{K}_i - K_i) \quad (43)$$

is the additional complex conductivity resulting from the polarization of the counterion layer. It is suitable, however, to transform this relation into the corresponding equation for complex dielectric constants

$$\epsilon^* = \epsilon' - i\epsilon'' \quad (44)$$

where ϵ' is the real dielectric constant ϵ , and ϵ'' is the dielectric loss $\kappa/\epsilon_r\omega$, κ being the ohmic conductivity. Taking into account eq. 36 and 38, we obtain for the complex dielectric increment

$$\epsilon^* - \epsilon_\infty^* = \Delta\epsilon^* = \frac{\Delta\epsilon_0}{1 + i\omega\tau} \quad (45)$$

with the static dielectric increment

$$\Delta\epsilon_0 = \frac{9}{4} \frac{p}{\left(1+\frac{p}{2}\right)^2} \frac{e_0^2 R \sigma_0}{\epsilon_r kT} \quad (46)$$

$\Delta\epsilon^*$ displays a simple Debye-type dispersion and the characteristic frequency

$$f = \frac{1}{2\pi\tau} = \frac{ukT}{\pi R^2} \quad (47)$$

The counterion surface mobility u should be smaller than the mobility u_0 in free solution. In general the counterion motion along the surface is subject to an additional activation energy α , so that

$$u = u_0 e^{-\alpha/kT} \quad (48)$$

This activation energy α is of electrostatic origin. Since the fixed charges on the surface of the sphere are discretely distributed, the electric potential is changing periodically parallel to the surface. A counterion within the double layer must overcome potential barriers in order to follow an applied electric field. The height of the potential wall multiplied by the charge of a counterion yields the additional activation energy α . Generally the minimum distance δ between an individual counterion and its counter charge on the surface can be assumed to be at least several times smaller than the distance between the surface charges. In this case

$$\alpha \approx \frac{e_0^2}{\epsilon_a \delta} \quad (49)$$

For a δ of a few Angstrom units this equation yields an activation energy α of the order of magnitude of 1 kcal./mole.

Activation Energy Distribution and the Relaxation Spectrum.—On account of eq. 48, we can write the relaxation time τ (eq. 28 and 47) as

$$\tau = \tau_0 e^{\alpha/kT} \quad (50)$$

with

$$\tau_0 = \frac{R^2}{2u_0 kT} \quad (51)$$

If the suspended particles are not of uniform size, a relaxation spectrum according to the particle size distribution will occur. The resulting dispersion curves are flatter than those corresponding to a single relaxation time. However, as shown by experimental evidence, a relaxation spectrum may exist even in the case of particles of uniform size where all counterions are of the same kind. This discrepancy between our model theory and the experimental data simply may be due to the mathematical simplification of the problem. A more rigorous theoretical treatment of the diffusion-controlled relaxation of the counterion atmosphere possibly may result in a relaxation spectrum. Such an idea is suggested in view of the relaxation

(18) K. S. Cole, *J. Gen. Physiol.*, **12**, 29 (1928).

(19) J. C. Maxwell, "A Treatise on Electricity and Magnetism (1873)," Oxford Univ. Press, article 314.

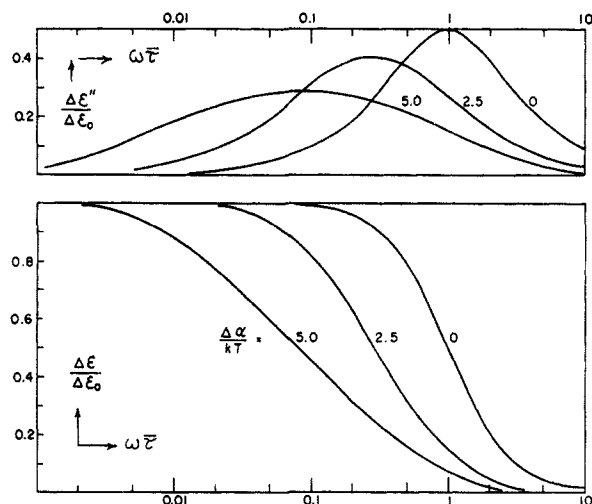


Fig. 1.—Effect of a statistical distribution of activation energies over a range $\Delta\alpha$ on the dispersion curves ($\bar{\tau}$ = relaxation time corresponding to the minimum activation energy $\bar{\alpha}$).

spectrum found for the ionic cloud of simple ions.^{20,21} But the mechanism which is responsible for the relaxation spectrum will not be discussed here. Only a formal extension of the theory outlined above will be given in order to describe the experimental dispersion curves. This extension can only be done by a superposition of independent relaxation steps due to counterions with different activation energies, as is obvious from eq. 50 and 51. We define a distribution function $\rho(\alpha)$, so that there are in the mean

$$d\sigma_0 = \sigma_0 \rho(\alpha) d\alpha \quad (52)$$

counterions per unit area which are subject to an activation energy between α and $\alpha + d\alpha$. Then the relation

$$\int_0^\infty \rho(\alpha) d\alpha = 1 \quad (53)$$

must hold. Now the total complex dielectric increment is derived simply from eq. 45 by integrating over the distribution of activation energies

$$\Delta\epsilon^* = \Delta\epsilon_0 \int_0^\infty \frac{\rho(\alpha) d\alpha}{1 + i\omega\tau_0 e^{\alpha/kT}} \quad (54)$$

Since the static dielectric increment $\Delta\epsilon_0$ is not affected by the activation energy, it is unchanged (eq. 46). Naturally the function must be known in order to evaluate eq. 54 rigorously. However, the shape of the distribution curve is not very crucial with regard to the result of the integration. For example, it has been shown by Schwan⁵ that several distribution functions which are symmetrical when plotted vs. $\ln \tau$ (which is proportional to α in our case) yield almost identical dispersion curves. Hence for practical purposes we use a function $\rho(\alpha)$ which is a constant in an activation energy range from $\bar{\alpha}$ up to a maximum $\bar{\alpha} + \Delta\alpha$, but is zero otherwise. Because of eq. 53 the constant has to be $1/\Delta\alpha$. Then if

(20) P. Debye and H. Falkenhagen, *Z. Elektrochem.*, **34**, 562 (1928).

(21) M. Eigen, *Discussions Faraday Soc.*, **24**, 24 (1957).

$$\xi = e^{-(\alpha - \bar{\alpha})/kT}, \quad \xi_0 = e^{-\Delta\alpha/kT} \quad (55)$$

and

$$\bar{\tau} = \tau_0 e^{\bar{\alpha}/kT} \quad (56)$$

eq. 54 gives

$$\Delta\epsilon^* = \frac{\Delta\epsilon_0}{\ln \xi_0^{-1}} \int_{\xi_0}^1 \frac{d\xi}{\xi + i\omega\bar{\tau}} = \frac{\Delta\epsilon_0}{\ln \xi_0^{-1}} \ln \frac{1 + i\omega\bar{\tau}}{\xi_0 + i\omega\bar{\tau}} \quad (57)$$

Separating the real and the imaginary part of this expression, we obtain for the dielectric increment

$$\Delta\epsilon' = \frac{\Delta\epsilon_0}{2 \ln \xi_0^{-1}} \ln \frac{1 + \omega^2 \bar{\tau}^2}{\xi_0^2 + \omega^2 \bar{\tau}^2} \quad (58)$$

and for the increment of the dielectric loss

$$\Delta\epsilon'' = \frac{\Delta\epsilon_0}{\ln \xi_0^{-1}} \tan^{-1} \frac{(1 - \xi_0)\omega\bar{\tau}}{\xi_0 + \omega^2 \bar{\tau}^2} \quad (59)$$

Dispersion curves of this kind are shown in Fig. 1 for several values of $\Delta\alpha/kT$. With increasing width $\Delta\alpha$ of the activation energy range, they flatten out and differ more and more from the Debye-type curves ($\Delta\alpha = 0$). Simultaneously there is a decrease of the characteristic frequency f defined as the frequency for which $\Delta\epsilon' = 1/2 \Delta\epsilon_0$. This behavior also is shown by the relation

$$f = \frac{1}{2\pi\bar{\tau}} \sqrt{\xi_0} = \frac{1}{2\pi\bar{\tau}} e^{-\Delta\alpha/2kT} \quad (60)$$

as evaluated from eq. 58.

The maximum of the dielectric loss $\Delta\epsilon''$ always is found at the characteristic frequency, but its absolute magnitude declines for increasing values of $\Delta\alpha$ while the shape of the curve becomes broader. The quantitative relation

$$\frac{2\Delta\epsilon''}{\Delta\epsilon_0} = \frac{2 \tan^{-1} \frac{1 - \xi_0}{\sqrt{\xi_0}}}{\ln \xi_0^{-1}} \quad (61)$$

as derived from eq. 58 and 59 is presented in Fig. 2 as a function of $\ln \xi_0^{-1} = \Delta\alpha/kT$. This curve can be used to determine $\Delta\alpha$ from measurements of $\Delta\epsilon'$ and $\Delta\epsilon''$.

The quantity $\Delta\alpha$, which characterizes the width of the relaxation spectrum, can be determined even if only one of the dispersion curves (either that for $\Delta\epsilon'$ or that for $\Delta\epsilon''$) has been measured. Since the degree of flattening depends on $\Delta\alpha$, one has to find out which value of ξ_0 in eq. 58 or 59, respectively, best fits the measured curve. For this purpose it is suitable to express $\bar{\tau}$, according to eq. 60, by a mean relaxation time

$$\tau_m = \frac{1}{2\pi f} = \frac{\bar{\tau}}{\sqrt{\xi_0}} \quad (62)$$

because f and therefore τ_m can be obtained experimentally from either curve in a very simple way. Equations 58 and 59 then are written

$$\Delta\epsilon = \frac{\Delta\epsilon_0}{2 \ln \xi_0^{-1}} \ln \frac{\xi_0^{-1} + \omega^2 \tau_m^2}{\xi_0 + \omega^2 \tau_m^2} \quad (63)$$

$$\Delta\epsilon'' = \frac{\Delta\epsilon_0}{\ln \xi_0^{-1}} \tan^{-1} \frac{1 - \xi_0}{\sqrt{\xi_0}} \frac{\omega \tau_m}{1 + \omega^2 \tau_m^2} \quad (64)$$

Discussion

The theory presented in this paper has been developed on the basis of a model which is to a certain extent rather crude. Nevertheless the experimental data can be interpreted surprisingly well in the light of this theory. On the assumption of reasonable values for the basic parameters, *i.e.*, the counterion surface density σ_0 and the activation energy α , the theoretical results agree more than satisfactorily with the measurements.

Sufficient experimental data for a qualitative and quantitative test of the theoretical results can be found in the article of Schwan, *et al.*³ The suspensions of polystyrene spheres used by these authors are most suitable for our purpose because the colloidal particles are non-polar spheres of extremely uniform and well known size. We shall discuss in particular the measurements carried out with polystyrene spheres in KCl solution. The diameters were either 1.17 or 0.557 μ . Experimental results for various concentrations of the polystyrene and the electrolyte, respectively, are shown in Fig. 8, 9, 10, and 11 of the mentioned article.

In order to apply the theory to a colloidal system, the electrolyte has to be fairly well conducting in accordance with relation 40. In fact, this inequality is generally satisfied in the case of the suspensions which are to be discussed. The quantity λ_0 can be determined from the conductivity of the system at higher frequencies (where $\lambda \rightarrow \lambda_0$ according to eq. 32). With regard to the previous investigation,³ λ_0 has to be interpreted as the high frequency limit of the frequency-dependent part of the surface conductance (see footnote 23). It always has been found in the order of magnitude of 10^{-9} mho.^{3,4} Moreover we can estimate λ_0 by means of eq. 31, using a result of Sieglaff and Mazur.²² From electrophoretic measurements these authors determined the density of charge units on the surface of polystyrene spheres in a comparable system to be 1.9×10^{13} cm.⁻². Thus we may reasonably assume a σ_0 in the order of magnitude of 10^{13} cm.⁻². With potassium ions as the counterions, the calculation likewise yields a λ_0 around 10^{-9} mho. Hence for the particle sizes of interest, λ_0/R has to be expected to be in the order of magnitude of 10^{-5} mho/cm. This value is indeed well below the conductivity of the electrolyte solution in virtually all the suspensions in consideration.

Now we may compare the experimental results with the theoretical equations. Below about $p = 15\%$ the measured values of the static dielectric increment $\Delta\epsilon_0$ for the 1.17 μ sample, as shown in Fig. 10 of ref. 3, are in good agreement with eq. 46 if the counterion surface density is set as $\sigma_0 = 2.0 \times 10^{13}$ cm.⁻² for $\kappa_a = 0.7$ mmho/cm. and $\sigma_0 = 1.5 \times 10^{13}$ cm.⁻² for $\kappa_a = 1.4$ mmhos/cm. In view of the above remarks on λ_0 and σ_0 , these

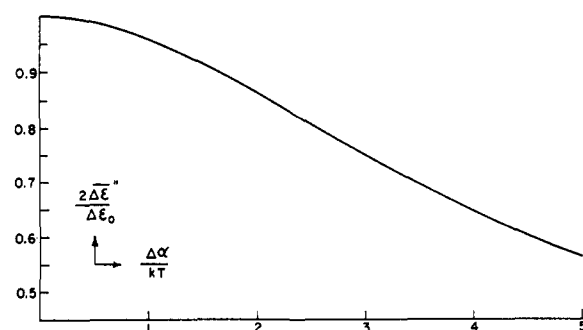


Fig. 2.—Effect of the activation energy distribution on the ratio of the maximum increment of the dielectric loss and the low-frequency limit of the dielectric increment.

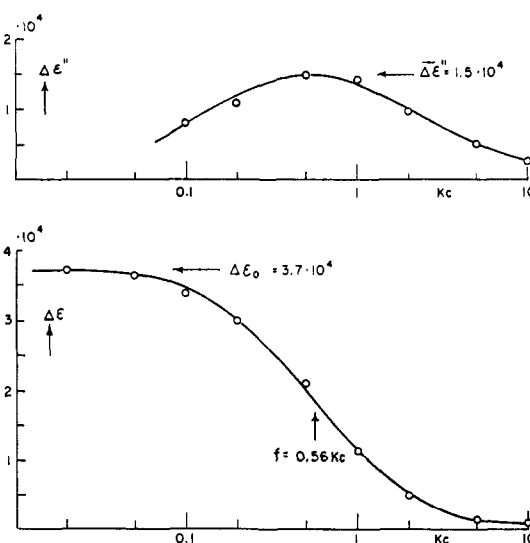


Fig. 3.—Low-frequency dielectric dispersion of $\Delta\epsilon$ and $\Delta\epsilon''$ as measured with a special sample: $2R = 1.17 \mu$, $p = 22\%$, $K_a = 0.81$ mmho/cm. (KCl). The curves are computed theoretically with $\alpha = 0$, $\Delta\alpha = 2.6 kT$.

σ_0 values obviously are very reasonable. At higher volume concentrations the measured dielectric increments exceed those computed from eq. 46 with the same σ_0 as before. However, since the basic eq. 39 has been proven only for small values of p , a disagreement between experiment and theory in this range of p is not surprising. Moreover, the quantity σ_0 may become a function of p here; thus the deviations between measurement and theory could be due also to an increase of σ_0 with increasing volume concentration. Such an interaction effect is not unlikely in view of the fact that other parameters of the suspension evidently affect the surface density of the counterions. This is shown with regard to the electrolyte concentration (conductivity) by the fact that it was necessary to use different values of σ_0 for different electrolyte conductivities. There are also indications of a slight dependence of σ_0 on the particle diameter, since according to Fig. 11 of ref. 3 the initial slope $\Delta\epsilon_0/p$ is not an exact linear function of R as it should be for a σ_0 independent of the diameter. Strictly speaking the counterion density σ_0 which appears in the theory represents only the ions bound to the particle surface, that is the "undissociated" counterions in contrast to the "dissociated" ones in free solution.²³ The degree

of "dissociation" and thereby σ_0 may well depend on the parameters κ_a , p , and R . Fortunately the previous assumption of a zero net charge of the double layer is not crucial here, so it may be dropped in compliance with this picture. An adequate extension of the basic theory will be discussed in a future paper.

An especially good agreement between our theory and the experiments is encountered in the case of the characteristic frequency f . Even the simple eq. 47 yields a proportionality to the inverse square of the radius of the particles, exactly as found experimentally (Fig. 9 of ref. 3). Furthermore it yields the quantitative results of the measurements by employing a counterion mobility $u = 1.35 \times 10^8$ c.g.s. units. Since the u_0 -value for a potassium ion in water at 25° is 5×10^8 c.g.s. units, an activation energy $\alpha = 1.3 kT = 0.8$ kcal./mole has to be assumed. This agrees very well with the estimation in the discussion of eq. 49. The successful interpretation of the characteristic frequency is not affected if an activation energy distribution is introduced in order to account for the shape of the dispersion curves. Then we have to use eq. 60 for the characteristic frequency f . The activation energy range $\Delta\alpha$ can be obtained from the measured ratio $2\Delta\epsilon''/\Delta\epsilon_0 \approx 0.8$. According to Fig. 2 that results in $\Delta\alpha = 2.6 kT$. With an $\bar{\alpha} = 0$, that is $\bar{\tau} = \tau_0$, eq. 60 results in the same quantitative relation as eq. 47. Dispersion curves computed on the assumption of such an activation energy distribution fit the experimental data very well as demonstrated by a set of measured curves in Fig. 3. According to eq. 49 the upper limit of the activation energy $\Delta\alpha = 2.6 kT \approx 1.5$ kcal./mole corresponds to a minimum distance between opposite charges of $\delta = 3.0 \times 10^{-8}$ cm., obviously a very reasonable value.

In view of the very encouraging results obtained

(23) According to Cole,²⁴ Fricke and Curtis,⁴ and Schwan, *et al.*,³ the surface conductance of a colloidal particle is composed of two parts; one is frequency independent while the other one depends upon the frequency. The latter one apparently is due to the counterions which are "undissociated." The frequency independent part of the surface conductance causes virtually no dielectric dispersion in the cases of interest here³ and is not discussed in this paper.

(24) K. S. Cole, *Cold Spring Harbor Symp. Quant. Biol.*, **1**, 1 (1933).

in comparing the experimental data with the theory, we may well conclude that the basic model of the colloidal particle used in this paper is essentially true. That means a diffusion-controlled polarization of a counterion layer around the colloidal particles is responsible for the low frequency dielectric dispersion under consideration. Particularly striking evidence for this conclusion is furnished by the following facts: (1) The characteristic frequency can be excellently expressed by the diffusion time of a counterion around the particle. (2) A very reasonable value of the counterion density on the surface of the particle results in a quantitative explanation of the increment of the dielectric dispersion.

Outlook

The theory as presented in this paper has been applied only to a special type of colloidal system. However, there can be no doubt that the basic effect of the polarization of the counterion layer is of significance for all kinds of systems including those in the biological field, where ion layers on interfaces are involved. Obviously there are no essential difficulties in extending the theory to the more general case of suspensions of spherical particles with no restrictions concerning the electrical properties of the spheres and the surrounding medium. Furthermore, the theoretical considerations are suitable to be adapted to more complicated colloidal particles which may be shaped like ellipsoids or carrying a non-conductive shell. Particles of the latter kind are of special biological interest, since they represent the living cell with its cell wall. Investigations dealing with all these generalizations for the fundamental theory, as well as with the question of how the density σ_0 and the mobility u of the counterions in the double layer depend on the structure of the system, are in progress. They are to be discussed in future articles.

Acknowledgment.—The author wishes to thank Dr. H. P. Schwan for many interesting and valuable discussions concerning the subject of this paper.

Electrical conductance in a porous medium

A. E. Bussian*

ABSTRACT

The Hanai-Bruggeman equation is applied to a model of a conductive rock matrix immersed in water. The result is a general two-component equation which relates the electrical properties of a formation to the electrical properties of the rock matrix and the water at any frequency. The low-frequency limit is compared with previous theoretical and experimental work on sand formations. From a comparison of the three-resistor model of Wyllie and Southwick, the Waxman-Smits model, the dual water model of Clavier et al (1977) and the proposed model, it is found that the only representation which is viable, conceptually sound, and in agreement with experiment, is the last one.

INTRODUCTION

The interpretation of electrical conductivity measurements in porous material has applications in many areas of science and technology. In particular, it is of considerable importance to the oil industry for distinguishing hydrocarbons from water in formation pores. Although this paper has applications in other areas, it is primarily concerned with the interpretation of electrical well logging measurements.

The basic relationship for interpreting the resistivity log is Archie's law which was formulated in 1942. However, since its application is limited to "clean" sands, i.e., formation matrices which are poor conductors, it has been modified and expanded to "explain" resistivity measurements over a wide range of rock formations. In 1954, Wyllie and Southwick suggested that an aggregate of conductive particles saturated with a conducting electrolyte could be modeled with a three resistor network. In 1957, de Witte discovered an empirical relationship which is a special case of the Wyllie and Southwick model but is more general than Archie's law. In 1968, Waxman and Smits proposed the first extensive model of shaly sand formations. They proposed that clay particles contribute exchange cations to the electrolyte thereby increasing the conductivity of the formation. In 1977 Clavier et al suggested a model based on the assumption that the exchange cations contribute to the conductivity of a clay water which is spatially separated from the bulk water. Consequently, it is called the dual water model. In all this work Archie's empirical law is assumed.

Recently, however, Sen (1980) showed how Archie's law follows from the work of Bruggeman (1935) and Hanai (1960, 1961) assuming that nonconducting rock particles are dispersed in a continuous phase of saline water. In the present work this approach is reinterpreted and expanded to include a conducting lattice-like matrix.

This paper is divided into three major parts. The first is a critical review of the basic assumptions which form the foundation of the earlier models. In the second part a model, which is based on the work of Bruggeman and Hanai, is formulated. The model is of a general nature which relates the electrical properties of any heterogeneous two-component mixture to the properties of the individual components at any frequency. However, only the low-frequency limit is considered in detail. In the final section, this low-frequency case is applied to shaly sand formations and compared with previous work.

PREVIOUS THEORETICAL MODELS OF SHALY SAND FORMATIONS

The three-resistor model

One of the first attempts to understand the electrical conductivity of a water-saturated shaly sand formation is due to Wyllie and Southwick (1954). Their model is based on the assumption that the conductivity of an aggregate of conductive particles saturated with a conducting electrolyte can be represented by three resistive elements in parallel (Figure 1c). They are (1) an element consisting of particles in electrical contact with each other, (2) an element consisting of the electrolyte, and (3) an element consisting of particles and electrolyte in series with each other. Mathematically one can then write the conductance of a formation as

$$\sigma_0 = \frac{\sigma_r \sigma_w}{W \sigma_r + Z \sigma_w} + X \sigma_r + Y \sigma_w, \quad (1)$$

where σ_0 = conductivity of formation, σ_r = conductivity of rock matrix, σ_w = conductivity of electrolyte, and W , X , Y , Z = geometrical factors.

Sauer et al (1955) performed an experiment with cation exchange resin spheres immersed in four different electrolytic solutions. Such a system is electrically similar to shaly sand formations. They analyzed their results in terms of equation (1). They found a good fit to their data and therefore concluded that the curvature in Figure 2 could be explained by the geome-

try of the system. That is, the curvature at small σ_w can be explained by the first term of equation (1) which describes in series electrical current flow through the solution and the rock matrix. Although this model gave good agreement with experiment, it was not developed further. The difficulty lay in not being able to define the geometrical factors in equation (1) more precisely. As we shall see, the importance of the model proposed later lies in its success in overcoming this deficiency.

The Waxman and Smits (W-S) model

Waxman and Smits (1968) proposed a model which has become the most fully developed and widely accepted approach

to the understanding of the electrical conductivity of shaly sand formations. It is based on three assumptions.

(1) The most basic assumption which serves as the groundwork for the model is that the electrical conductance of shaly sand formations can be represented by two resistive elements in parallel. One element represents the resistance associated with the free electrolyte in the pore volume of the rock. The other element represents the resistance associated with the clay whose conductance arises from exchange cations believed to originate from lattice imperfections. Equation (1), therefore, reduces to

$$\sigma_0 = X\sigma_r + Y\sigma_w \quad (2)$$

(2) The second assumption of Waxman and Smits is that the current attributed to the clay exchange ions and the current attributed to the electrolytic ions follow the same path through the formation. It follows that $X = Y$ in equation (2). This means that the two conduction mechanisms must use the same water, i.e., a distinction is not made between, say, bound clay water and pore water. Here, then, one has a single water model where the clay exchange cations and the ions of the electrolyte are uniformly distributed and share the same pore space. This is schematically shown in Figure 1a.

The second assumption is superfluous since it follows from the first assumption. If there are no conduction paths which go through both the rock matrix and the water, then the path through the water must be the same as the path associated with the clay cations, i.e., $X = Y$. Conversely if $X \neq Y$, then the two paths must be spatially separated and a conduction path must also exist which goes through both the rock matrix and the water, i.e., the first term in equation (1) must be taken into account. In any case, if $X = Y$ a conceptual difficulty arises when our current understanding of the electrochemistry of clay minerals, such as smectite, is considered (Van Olphen, 1963). The principal building elements of these minerals are two-

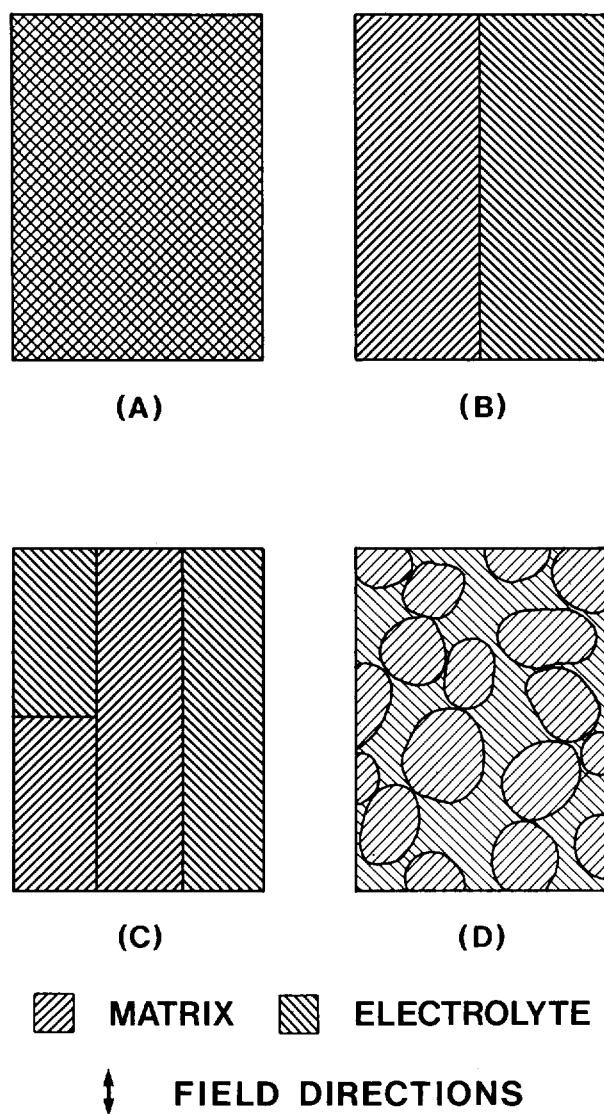


FIG. 1. Geometrical schematics of models discussed in text. The electric field direction is shown by the arrow. (a) Waxman-Smits model assumes matrix and electrolytic conduction paths are identical. (b) Dual water model assumes matrix and electrolytic conduction paths are spatially separated. (c) Three-resistor model is the same as (b) but includes a matrix-electrolyte in series element. (d) Proposed model is a more realistic version of three-resistor model.

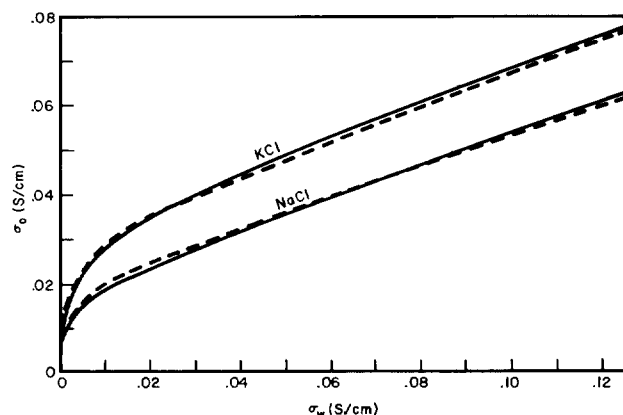


FIG. 2. Conductivity of resin spheres saturated with KCl and NaCl. The dashed curves are fits of equation (1) using the geometrical factors determined by Sauer et al. (1955). Since these fits are in good agreement with experiment, the data points are not shown. The solid curves are equation (8) for $\phi = 0.43$ and $m = 1.3$, with $\sigma_r = 0.047$ S/cm for KCl and $\sigma_r = 0.026$ S/cm for NaCl.

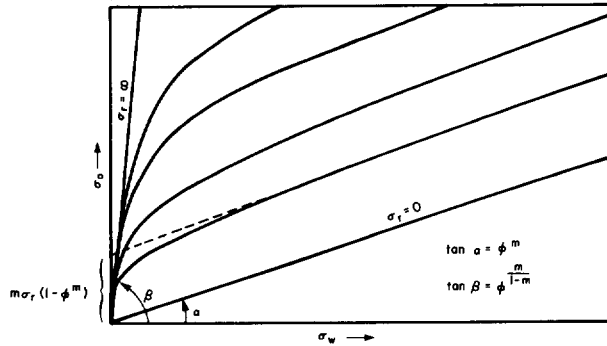


FIG. 3. σ_0 versus σ_w . Family of curves generated from equation (8) with $\phi = 0.2$, $m = 2$, and σ_r from 0 to ∞ .

dimensional (2-D) arrays of silicon-oxygen tetrahedra and 2-D arrays of aluminum- or magnesium-oxygen-hydroxyl octahedra. The combination of an octahedral sheet and one or two tetrahedral sheets is called a unit layer. Most clay minerals consist of a large number of such unit layers which are stacked parallel to each other. This results in an immense surface area. Associated with this multilayered surface is a negative charge which arises from the substitution of aluminum ions for silicon ions within the framework of the clay mineral. To balance this negative charge, exchange cations are distributed near the clay surface and, therefore, cannot be distributed uniformly throughout the electrolyte as Waxman and Smits assumed.

(3) The last assumption of the model is that σ_r is dependent upon σ_w . Now, equation (2) must be written as

$$\sigma_0 = (\sigma_r + \sigma_w)\phi^m, \quad (3)$$

where ϕ = porosity and m = a geometrical factor which characterizes the formation. This result follows if equation (2) is to be consistent with assumption (2) and Archie's empirical law (Archie, 1942), and if, as will be discussed, X and Y are to depend upon geometry only. The model is developed further by assuming that σ_r arises from cation exchange, which when treated as a volume phenomenon gives

$$\sigma_r = \Lambda_+ Q_v, \quad (4)$$

where Λ_+ = equivalent cation conductance in $S\text{-cm}^2/\text{g-equiv}$, and Q_v = concentration of cations in g-equiv/l . It was found that if equations (3) and (4) are to agree with the previously discussed empirical work of Sauer et al, then

$$\Lambda_+ \propto 1 - \beta \exp(-\sigma_w/\gamma),$$

where β and γ are constants. In other words, this functional dependence was introduced to obtain the curvature at small σ_w in Figure 2 and has no theoretical justification. In the three-resistor model and in the proposed model, this curvature follows in a more natural way from the geometry of the system.

In their work, Waxman and Smits assumed that

$$X = Y = 1/F^*,$$

where

$$\lim_{\sigma_r \rightarrow 0} F^* = \phi^{-m}.$$

From their analysis of shaly sands, they found

$$F^* = \phi^{-m}.$$

In general, however, their model allows for F^* to be a function of σ_r . This contradicts the basic assumption that X and Y are geometrical factors. Hence, F^* must be equal to ϕ^{-m} .

The dual water model

An alternative to the Waxman and Smits model was proposed by Clavier et al (1977) in an attempt to take into account indications that two kinds of water are present in shaly sand formations. Their model is based on two major assumptions. (1) The starting point of the dual water model is identical to assumption (1) of the W-S model. (2) The second assumption is that a shaly sand formation behaves as if two kinds of water are present. The first kind is called clay water and is closely associated with clay particles. As in the W-S model, its conductive nature is attributed to clay exchange cations. The second kind of water is called far water, which is the bulk of the water found in the pore space.

The dual water model differs from the W-S model in the manner in which the clay exchange cations are distributed in the formation. In keeping with current concepts of clay electrochemistry, the exchange cations are assumed to be concentrated near the surface of the unit layers which are the building blocks of clay particles. The electrolytic ions, on the other hand, are distributed throughout the pore space. The two conductive mechanisms are, therefore, spatially separated. This is schematically shown in Figure 1b. This separation into two kinds of water which are spatially separated introduces a conceptual difficulty when one visualizes current flow through the formation. Assumption (1) does not allow for current paths which pass through both kinds of water, i.e., in series resistance paths. For a complete description of current flow, such paths must be considered. In other words, the dual water model assumes that $X \neq Y$. It follows, as was previously discussed, that the first term of equation (1) must, therefore, be taken into account. As noted earlier in the three-resistor model, it is the in-series resistance paths which give rise to the curvature in Figure 2. Thus, not only is a conceptual difficulty introduced if one does not consider such paths, but such paths can explain the curvature of Figure 2 in a straightforward way from the geometry of the system alone.

There is another conceptual problem with the dual water model. X and Y are defined to be geometrical factors which should depend upon the geometry of the formation only. However, the dual water model is formulated in such a way that X and Y are also functions of Q_v and σ_w .

PROPOSED MODEL

The basis of the present approach to the problem of electrical conductance in shaly sands lies with the work of Bruggeman (1935) and Hanai (1960, 1961) who derived an equation which relates the electrical properties of a heterogeneous mixture to the properties of the individual components. If one defines the complex conductivity as

$$\sigma^* = \sigma + i\omega\epsilon_0\kappa, \quad (5)$$

the Hanai-Bruggeman equation is given by

$$\frac{\sigma_0^* - \sigma_r^*}{\sigma_w^* - \sigma_r^*} \left(\frac{\sigma_w^*}{\sigma_0^*} \right)^d = 1 - \phi_r, \quad (6)$$

where

- σ_0^* = complex conductivity of the mixture,
- σ_r^* = complex conductivity of dispersed particles,
- σ_w^* = complex conductivity of continuous medium,
- ϕ_r = fractional volume of dispersed particles,
- d = depolarization factor of dispersed particles,
- ω = angular frequency,
- σ = conductivity,
- ϵ_0 = permittivity of free space,
- κ = relative dielectric constant, and
- $i = \sqrt{-1}$.

The original work of Bruggeman and Hanai considered spheres dispersed in a continuous medium, in which case $d = 1/3$. More generally, d depends upon the shape of the particles and can have values between 0 and 1 (Stratton, 1941; Von Hippel, 1954; Sen et al, 1981; Sen, 1981; Mendelson and Cohen, 1982). The value of 0 corresponds to infinitely long cylindrical rods (needles) whose axes are parallel to the electrical field. The value of 1 corresponds to infinitely short cylindrical rods (plates) whose axes are parallel to the electrical field.

It is important to note that a particular value of d does not specify a unique geometrical configuration. For example, the right combination of infinitely long rods parallel to the electric field ($d = 0$) and perpendicular to the electric field ($d = 1/2$) will give a value identical to that of dispersed spheres ($d = 1/3$). It is

apparent, then, that a lattice-like structure can give rise to d values in the same range as dispersed particles. This means that the definitions of σ_r^* and ϕ_r in equation (6) should be broadened to include an immersed network of interconnected particles, as well as dispersed individual particles. This should be kept in mind later when a geometrical factor, defined as $m = 1/(1 - d)$, is empirically determined for core samples saturated with saline water. m is often called the cementation factor.

Equation (6) can be written as

$$\sigma_0^* = \sigma_w^* \phi^m \left(\frac{1 - \sigma_r^*/\sigma_w^*}{1 - \sigma_r^*/\sigma_0^*} \right)^m, \quad (7)$$

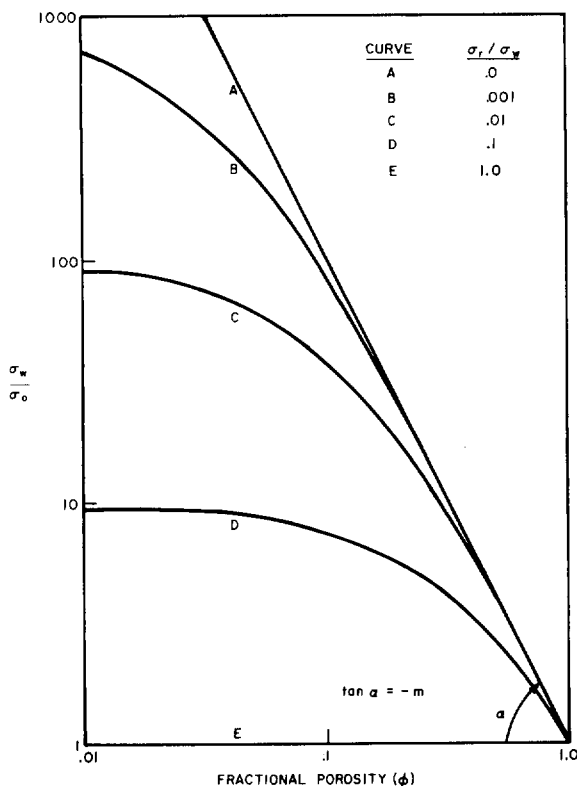
where $\phi = 1 - \phi_r$. Equation (7) is the basic and most general equation of the proposed model. The full implications of this equation are beyond the scope of this paper. In the remainder of this paper discussion will be limited to the low-frequency case with application to shaly sand formations.

At low frequencies equation (7) reduces to

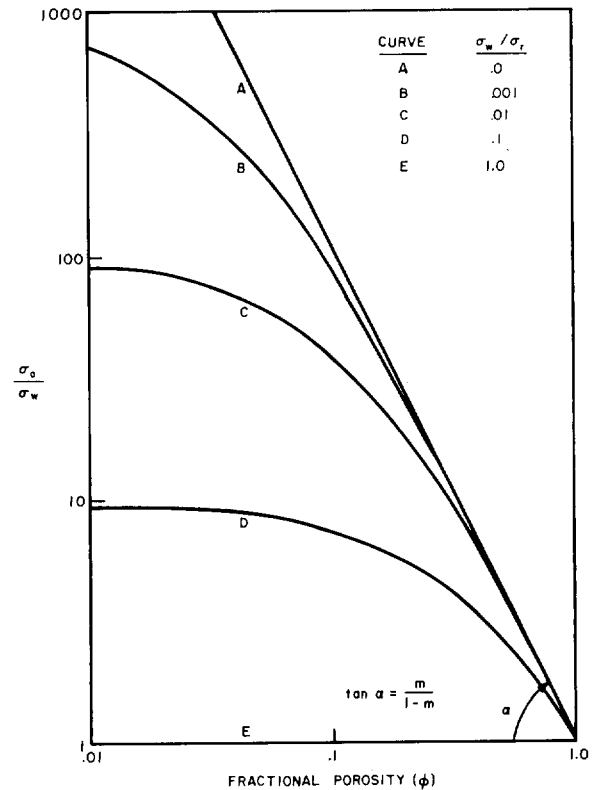
$$\sigma_0 = \sigma_w \phi^m \left(\frac{1 - \sigma_r/\sigma_w}{1 - \sigma_r/\sigma_0} \right)^m. \quad (8a)$$

Equation (8a) can also be written as

$$\sigma_0 = \sigma_w \phi^{m/(1-m)} \left(\frac{1 - \sigma_w/\sigma_r}{1 - \sigma_0/\sigma_r} \right)^{m/(1-m)}. \quad (8b)$$



(a)



(b)

FIG. 4. (a) σ_w/σ_0 versus ϕ . Family of curves generated from equation (13a) for the values of σ_r/σ_w specified and $m = 2$. (b) σ_0/σ_w versus ϕ . Family of curves generated from equation (13b) for the values of σ_w/σ_r specified and $m = 2$.

A σ_0 versus σ_w plot of equation (8) for different σ_r values is shown in Figure 3. The family of curves results from values of σ_r between 0 and ∞ . The limiting cases shown are

$$\lim_{\sigma_r \rightarrow 0} \frac{\sigma_0}{\sigma_w} = \phi^m \quad (9a)$$

and

$$\lim_{\sigma_r \rightarrow \infty} \frac{\sigma_0}{\sigma_w} = \phi^{m/(1-m)} \quad (9b)$$

For the purpose of studying shaly sand formations, it is useful to consider in some detail the case where the water conductivity is greater than the matrix conductivity, i.e., $\sigma_w > \sigma_0 > \sigma_r$. This is the case commonly found in formations as is shown later. Using the binomial expansion, equation (8a) can be written

$$\sigma_0 = \sigma_w \phi^m \frac{1 - m \frac{\sigma_r}{\sigma_w} + \frac{m(m-1)}{2!} \left(\frac{\sigma_r}{\sigma_w}\right)^2 - \dots}{1 - m \frac{\sigma_r}{\sigma_0} + \frac{m(m-1)}{2!} \left(\frac{\sigma_r}{\sigma_0}\right)^2 - \dots} \quad (10)$$

Retaining first-order terms only gives

$$\sigma_0 \approx m\sigma_r(1 - \phi^m) + \phi^m\sigma_w \quad (11)$$

An approximate criterion for the validity of equation (11) can be obtained from equation (10) and is given by

$$m \frac{\sigma_r}{\sigma_0} \gtrsim \frac{10m(m-1)}{2!} \left(\frac{\sigma_r}{\sigma_0}\right)^2, \quad (12)$$

or

$$\sigma_0 \gtrsim 5(m-1)\sigma_r.$$

In some experiments relevant to this paper, data are plotted on a σ_w/σ_0 versus ϕ graph. In order to compare these experiments with the proposed model, one can write equation (8a) in the form

$$\phi = \frac{\left(\frac{\sigma_w}{\sigma_0}\right)^{-1/m} - \left(\frac{\sigma_r}{\sigma_w}\right)\left(\frac{\sigma_w}{\sigma_0}\right)^{(m-1)/m}}{1 - \sigma_r/\sigma_w} \quad (13a)$$

A σ_w/σ_0 versus ϕ plot of equation (13a) for different values of σ_r/σ_w is shown in Figure 4a for the case $m = 2$. The family of curves results from values of σ_r/σ_w between 0 and 1. The importance of this plot will become evident later.

Of academic interest is a symmetry property of equation (13a) which can also be written as

$$\phi = \frac{\left(\frac{\sigma_0}{\sigma_w}\right)^{(1-m)/m} - \left(\frac{\sigma_w}{\sigma_r}\right)\left(\frac{\sigma_0}{\sigma_w}\right)^{1/m}}{1 - \sigma_w/\sigma_r} \quad (13b)$$

A σ_0/σ_w versus ϕ plot of equation (13b) for different values of σ_w/σ_r is shown in Figure 4b for the case $m = 2$. The family of curves results from values of σ_w/σ_r between 0 and 1. The apparent symmetry of Figures 4a and 4b is only true for the case $m = 2$, i.e., for $m = 2$ equation (8) is symmetric under the inverse operations

$$\frac{\sigma_w}{\sigma_0} \rightarrow \frac{\sigma_0}{\sigma_w}$$

and

$$\frac{\sigma_r}{\sigma_w} \rightarrow \frac{\sigma_w}{\sigma_r}.$$

This symmetry is broken for any other value of m . However, symmetry is retained for any value of m if the operation $m \rightarrow m/(m-1)$ is carried out along with the inverse operations.

SHALY SAND FORMATIONS

Earlier the Hanai-Bruggeman equation was reinterpreted and expanded to any two component mixture where one component can be considered to be immersed in the other. This includes the case of a conductive, lattice-like structure immersed in a fluid. Therefore, it is a more geometrically realistic representation of a water-saturated rock formation than any of the models discussed above (Figure 1d).

In equation (8) it is implicitly assumed that σ_r and σ_w are independent variables. If this equation is applied to a shaly sand formation, the implication is that two distinct kinds of water are present. σ_r is then the conductivity resulting, at least in part, from exchange cations in water bound between the unit layers of the clay particles. σ_w is the conductivity arising from the electrolytic ions of the pore water. That is, equation (8) with σ_r and σ_w independent presumes assumption (2) of the dual water model. However, in contrast to the dual water model, the present formulation properly takes into account the spatial separation of these two kinds of water by allowing for current paths through both. The three-resistor model, which did take these series electrical paths into account, was not sufficiently formulated to be of practical value.

In this section experimental results for shaly sand formations are compared with the predictions of equation (8). In spite of the conceptual difficulties discussed earlier, the predictions of the W-S and the dual water models will also be compared with this empirical work.

de Witte (1957) concluded from empirical considerations that the conductive behavior of water-bearing shaly sands could adequately be represented by

$$\sigma_0 = A + B\sigma_w,$$

where A and B are formation parameters independent of σ_w . From equation (11) de Witte's parameters are given by

$$A = m\sigma_r(1 - \phi)^m$$

and

$$B = \phi^m.$$

Not only does this result demonstrate agreement of the proposed model with experiment, but it also enhances the usefulness of de Witte's work. In particular, de Witte has argued on empirical grounds that the water saturation of oil-bearing sands can be determined from the relationship

$$B\sigma_w S_w^2 + A S_w = \sigma_t, \quad (14)$$

where S_w = water saturation and σ_t = measured conductivity of a formation. Since the three-component mixture problem has not yet been solved theoretically, one must still resort to an empirical approach for determining S_w . However, it must be emphasized that although equation (14) was empirically derived, the expressions for A and B are theoretically justified.

The work of de Witte demonstrates that the model agrees with experiment when the brine conductivity is high compared to the rock conductivity, i.e., when equation (12) is satisfied. In

the Sauer et al experiment, which was discussed earlier, a wider range of brine conductivities was used and is a more rigorous test of the model. The conductivity of the resin spheres is believed to be the result of exchangeable ions within the water-filled gel (Spiegler and Coryell, 1951) and in this respect is analogous to clay particles.

The three-resistor model, although it is in agreement with the resin sphere data and suggests that geometry is responsible for the curvature in Figure 2, is not sufficiently developed for a more penetrating interpretation of experimental results. For example, it is unable satisfactorily to explain the dependence on electrolyte type. The proposed model, on the other hand, provides a ready explanation. In Figure 2 approximate fits to equation (8) are compared with fits to equation (1) of the three-resistor model for KCl and NaCl data. It is important to note that in the equation (8) fits the two electrolytes have the same geometrical factors (ϕ and m) but different values for σ_r . The larger σ_r value for the KCl electrolyte can be attributed to the larger ion conductance of K^+ as compared with that of Na^+ . This follows from the procedure used in performing the experiment. Since the resin spheres were thoroughly flushed with the electrolyte of interest before the conductivity measurements were made, K^+ was the exchange ion for the KCl elec-

trolyte and Na^+ was the exchange ion for the NaCl electrolyte.

It should be mentioned that the W-S and dual water models can also explain the shape of the curves in Figure 2, although the latter model appears to breakdown at low values of σ_w . However, a careful comparison of the models in terms of the shape of the curves in Figure 2 is not made here. As will now be shown, a consideration of the σ_w/σ_0 versus ϕ is a more decisive comparison.

Archie (1942) originally proposed the relationship

$$\frac{\sigma_w}{\sigma_0} = a\phi^{-m}, \quad (15)$$

with $a = 1$. Although equation (15) with $a = 1$ is found to hold for "clean" sands, many core measurements require $a \neq 1$ to obtain satisfactory fits (Carothers, 1968; Porter and Carothers, 1970; Gomez-Rivera, 1976). Note that when $a \neq 1$, an inconsistency occurs in the limit

$$\lim_{\phi \rightarrow 1} \sigma_0 = \sigma_w/a.$$

Nevertheless, it has become the accepted practice to analyze core samples by fitting conductivity measurements to equation (15) where a and m are determined. On a log-log plot of σ_w/σ_0

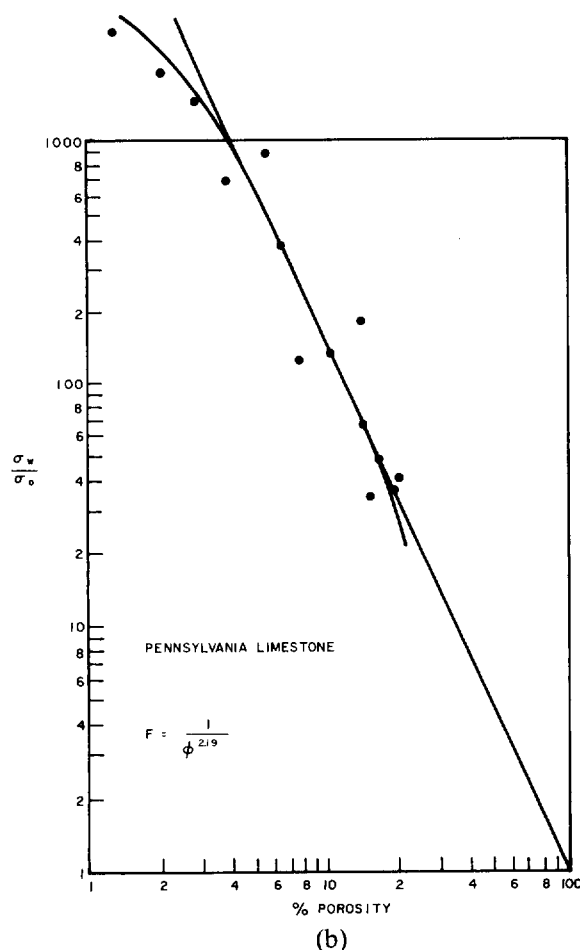
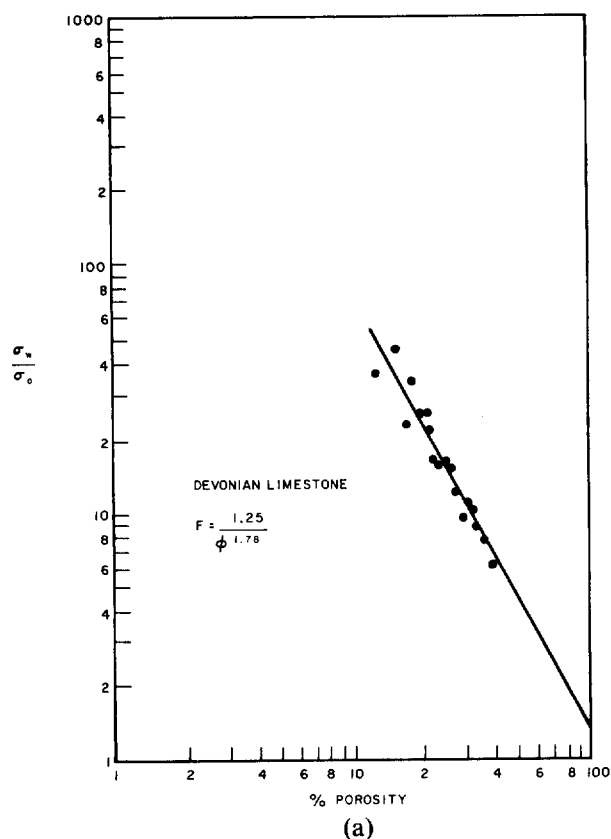


FIG. 5. σ_w/σ_0 versus ϕ . Data and curves are from Carothers (1968).

Table 1. m and σ_r values of cores.
(The first three columns are from Waxman and Smits, 1968)

Sample no.	ϕ (%)	Q_v (equiv/liter)	σ_r (mS/cm)	m	Sample no.	ϕ (%)	Q_v (equiv/liter)	σ_r (mS/cm)	m
25	20.3	.076	.05	1.74	12	11.6	.281	.07	2.00
31	21.8	.081	.09	1.90	13	18.6	.100	.07	1.92
34	21.9	.060	.14	1.87	14	17.4	.206	.13	1.97
38	23.2	.018	.04	1.57	15	16.3	.185	.19	1.97
16	11.8	.008	.01	1.86	1	5.4	.433	.16	2.42
21	13.9	.247	.14	1.75	2	26.1	.191	.25	2.06
25	15.7	.225	.14	1.85	3	25.9	.135	.18	1.98
39	21.2	.033	.12	1.67	4	23.9	.453	.56	1.98
48	22.4	.088	.21	1.79	5	24.8	.200	.31	1.92
5	9.5	.578	.10	1.78	6	25.7	.175	.18	1.97
44	16.5	.232	.08	1.94	1	15.0	2.494	.92	2.50
47	17.5	.215	.07	1.90	2	11.9	2.057	.93	1.71
51	18.6	.073	.03	1.80	3	8.7	1.578	1.09	3.54
52	18.7	.239	.09	1.80	4	25.5	1.436	1.06	2.55
19	20.3	.293	.08	2.19	5	25.3	1.440	1.23	2.63
28	20.1	.327	.11	1.99	6	12.2	1.669	1.10	12.71
36	23.1	.075	.06	1.92	7	27.2	1.332	1.13	2.53
4	16.2	.347	.08	2.12	8	29.0	.689	.70	2.39
13	23.1	.225	.13	2.02	9	26.7	1.074	.85	2.37
17	23.9	.249	.15	2.04	10	26.2	1.228	.91	2.44
18	24.1	.171	.08	1.91	11	28.4	.857	.82	2.29
23	24.5	.195	.11	1.96	12	27.7	1.085	.88	2.40
24	24.5	.146	.12	1.84	13	25.3	1.530	1.07	2.59
27	24.8	.188	.17	1.86	14	23.2	1.541	1.18	2.42
16	23.9	.699	.61	1.99	15	29.8	.776	.85	2.34
22	25.6	.887	.99	2.23	16	12.8	1.587	.91	2.11
27	26.8	.767	.73	2.19	17	27.5	.898	.78	2.34
33	27.4	.798	.71	2.10	18	31.4	.639	.58	2.33
38	28.2	.750	.72	2.19	19	29.0	.922	.81	2.35
42	28.4	.617	.63	2.11	20	32.2	.860	.72	2.61
43	28.6	.645	.68	2.18	21	30.2	.887	.80	2.54
45	29.3	.605	.66	2.15	22	29.0	1.066	.88	2.55
47	30.8	.502	.67	2.17	23	27.2	1.316	1.11	2.51
64	26.5	.148	.09	1.84	1	23.9	.017	.04	1.75
21	17.5	.045	.003	1.95	2	21.2	.052	.06	1.71
1	21.7	.093	.09	1.88	3	23.1	.052	.07	1.65
2	16.6	.302	.16	2.00	4	8.0	.26	.03	1.53
3	24.4	.083	.13	1.85	5	15.4	.20	.13	1.44
4	20.2	.036	.08	1.76	6	21.5	.095	.08	1.90
5	24.2	.051	.09	1.80	7	17.1	.053	.07	1.86
6	21.2	.104	.13	1.82	8	17.1	.053	.06	1.83
7	20.5	.097	.14	1.81	9	19.9	.085	.10	1.79
8	20.8	.069	.17	1.83	10	12.5	.253	.11	7.59
9	20.4	.076	.09	1.79	11	12.5	.253	.01	2.47
10	24.5	.049	.06	1.80	12	11.0	.28	.13	1.71
11	14.5	.454	.20	2.03	13	11.0	.28	.15	1.72
1	20.2	.085	.13	1.86	14	11.0	.28	.21	1.57
2	21.1	.102	.15	1.87	15	9.2	.41	.11	2.24
3	25.8	.050	.08	1.79	16	10.3	.67	.25	1.81
4	19.3	.112	.08	1.95	17	14.0	.33	.36	1.94
5	20.7	.062	.13	1.86	18	25.9	.59	.67	2.16
6	20.0	.067	.12	1.88	19	25.9	.59	.49	2.09
7	18.9	.065	.18	1.86	20	25.9	.59	.54	1.90
8	18.2	.123	.16	1.96	21	23.8	.29	.33	1.87
9	16.1	.158	.09	1.92	22	22.5	.72	.60	2.44
10	14.9	.298	.14	1.89	23	24.2	1.04	.95	2.35
11	15.9	.254	.17	1.97	24	21.6	.81	.66	2.54
					25	18.7	1.27	.79	2.48
					26	22.9	1.47	1.19	2.67
					27	20.9	1.48	1.12	2.53

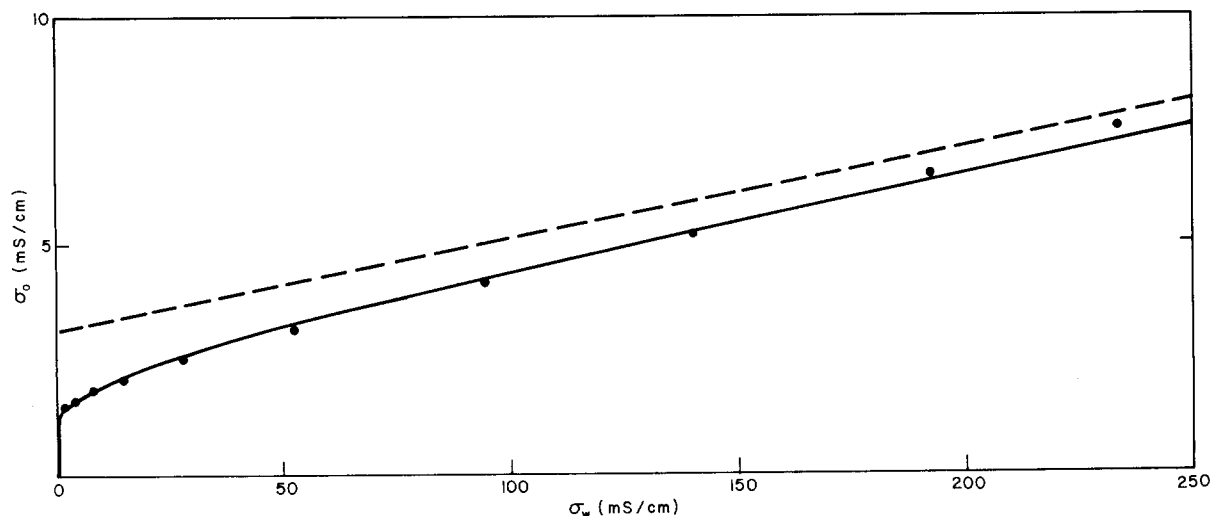


FIG. 6. σ_0 versus σ_w . Data points are from core no. 26 of Table 7 in Waxman and Smits paper. The solid line is equation (8) with $m = 2.67$ and $\sigma_r = 1.19$ mS/cm. The dashed line is the asymptotic limit, i.e., equation (11).

versus ϕ , equation (15) gives a straight line. Two examples taken from Carothers (1968) are shown in Figure 5. These measurements are typical of core data in that $a \geq 1$. Note that for low porosities there appears to be a deviation of the data from equation (15), as is emphasized by the author's curve in Figure 5b. In those cases where core measurements give $a < 1$, a fit with $a = 1$ is within experimental error. For example, a careful analysis of the work of Winsauer et al (1952), reveals that their measurements give a best fit with $a = 0.62$ and $m = 2.15$. Equation (15) with these values is known as the "Humble formula." However, an equally good fit is found with $a = 1$, i.e., the laboratory measurements do not justify $a < 1$.

These effects can be understood in terms of Figure 4a. Since formation porosity values are usually in the 10–40 percent range, the curvature of the family of curves with $\sigma_r \neq 0$ will not be evident in most cases, an example of which is shown in Figure 5a. On the other hand, data which follow these curves but are fit to equation (15) will give values of $a \geq 1$.

In marked contrast, the W-S and dual water models predict $a \geq 1$. This will be shown for the W-S model, although a similar argument can be given for the dual water model. Equation (3) can be written as

$$\frac{\sigma_w}{\sigma_0} = \left(\frac{\sigma_w}{\sigma_r + \sigma_w} \right) \phi^{-m}.$$

A comparison of this equation with equation (15) gives

$$a = \frac{\sigma_w}{\sigma_r + \sigma_w} \leq 1.$$

It has become common practice to classify formations according to the a and m values of equation (15). As seen above, a is an ad hoc parameter with no theoretical basis. Equation (8) suggests a more meaningful classification scheme based on a specification of m and σ_r . Such a program was carried out using conductivity measurements on 117 core samples which were published by Waxman and Smits (1968). A number of conduc-

tivity measurements were made on each core sample, where each measurement was made with the sample being saturated with water of a different salinity. From the published measurements of each core sample, a least-squares fitting program (Bevington, 1969) was applied to equation (8) to determine the values of m and σ_r . These results, as well as independently determined experimental values of ϕ and Q_v , taken from the Waxman and Smits paper, are listed in Table 1.

It is important to note that the exact equation [equation (8)] was used to determine the formation parameters. Since the more conductive matrix cores in Table 1 do not satisfy equation (12), the asymptotic limit given in equation (11) is not justified. An example is shown in Figure 6 with data from a core with a relatively high matrix conductivity. As the curves indicate, the asymptotic limit has not been reached. Furthermore, if equation (11) is used to fit to the data points which appear to lie on a straight line, one obtains values for m and σ_r which are less than their true values.

In Table 2 the average values of σ_r and m are listed for those cases (Tables 1 and 3 of Waxman and Smits, 1968) where core samples were qualitatively classified by the authors. It is clear from this analysis that these categories are only crudely descriptive of the formation. A determination of m and σ_r , on the other hand, provides a more precise method of classification.

The σ_r and m values from Table 1 are cross-plotted in Figure 7. The suggested correlation is probably related to the clay content which can affect both the conductivity of the matrix and its geometrical structure. The correlation is more apparent in Table 3 where the average values of m and σ_r have been determined from core samples of known smectite content. This is an example which illustrates how the formation constitution is reflected in the values of m and σ_r .

Since at least part of the shaly sand matrix conductivity is believed to be due to exchange cations one might expect a correlation between Q_v and σ_r . In Figure 8 a cross plot of these variables for core samples listed in Table 7 of the W-S paper is

Table 2. Average m and σ_r values of cores.

Formation type ¹	Age	Average m^2	Average σ_r (mS/cm)	Average R_r (Ω -m)	Number of samples	Table ⁴
Clean Sandstone Weeks Island, LA	Miocene	$1.80 \pm .08$	$.08 \pm .04$	125	4	1
Clean Sandstone Weeks Island, LA	Miocene	$1.85 \pm .08$	$.12 \pm .04$	83	11	3
Clean Sandstone Mithcell Creek, TX Quitman, TX	Cretaceous	$1.78 \pm .07$	$.12 \pm .06$	83	5	1
Shaly Sandstone Sheridan, TX	Eocene	$1.90 \pm .14$	$.08 \pm .02$	125	6	1
Shaly Sandstone Sheridan, TX	Eocene	$1.91 \pm .06$	$.13 \pm .04$	77	15	3
Shaly Sandstone Seeligson, TX	Oligocene	$1.96 \pm .08$	$.11 \pm .03$	91	9	1
Shaly Sandstone Seeligson, TX	Oligocene	$2.06 \pm .17$	$.27 \pm .14$	37	6	3
Very Shaly Sandstone Big Foot, TX	Cretaceous	$2.15 \pm .07$	$.71 \pm .11$	14	9	1
Very Shaly Sandstone Big Foot, TX	Cretaceous	$2.46 \pm .31$	$.92 \pm .16$	11	22 ³	3
Limestone Chapel Hill, TX	Cretaceous	1.84	.09	111	1	1
Limestone Magnolia, AR	Jurassic	1.95	.003	3333	1	1

¹As quoted by the authors
²Error quoted is one standard deviation
³One sample not included: $m = 12.71$, $\sigma_r = 1.10$ mS/cm
⁴Table number in Waxman and Smits paper (1968).

shown. If matrix conduction is depicted as a volume cation exchange phenomenon, one can use equation (4), but where Λ_+ is now independent of σ_w . For the data shown in Figure 8, one obtains $\Lambda_+ \sim 1$ S-cm²/g-equiv. For comparison $\Lambda_+ = 50$ S-cm²/g-equiv for Na⁺ cations in an electrolyte at infinite dilution, i.e., the limiting equivalent conductance as the concentration goes to zero. One might attribute this reduced exchange cation mobility to the close proximity of charges associated with the clay structure. A knowledge of Λ_+ values, therefore, may be valuable in furthering an understanding of clay structure. Finally, the broad scatter of the data suggests that the matrix conductivity may depend upon additional factors which can only be determined through a more complete understanding of the electro-chemistry of shaly sand formations.

Table 3. Effect of smectite (montmorillonite) on formation parameters.

Smectite (%)	Average σ_r (mS/cm)	Average m	No. of samples
0	$.12 \pm .09$	$1.81 \pm .25$	16
20	$.57 \pm .08$	$2.05 \pm .11$	3
80	$.64 \pm .22$	$2.30 \pm .26$	4
100	$1.03 \pm .17$	$2.56 \pm .08$	3

SUMMARY AND CONCLUSIONS

A model of a rock formation is proposed which is based on the work of Bruggeman and Hanai. By broadening the interpretation of their equation to include a conducting lattice-like matrix immersed in a fluid, a realistic mathematical representation of a rock formation is obtained. In its most general form, both the dielectric constant and the conductivity are taken into account. In the low-frequency limit where displacement current is unimportant, a comparison with the Waxman-Smits and dual water models shows that only the proposed model is free of conceptual difficulties and in agreement with experimental results. This comparison is summarized in Table 4. Column 1 recapitulates the discussion on the conceptual problems of the three models. The number of variables in the equation corresponding to each model is listed in column 2. The fact that the proposed model has the fewest number of variables is not only aesthetic, but it also means that it can be more readily used. As summarized in column 3, only the proposed model agrees with those experiments where the dependence on porosity is studied. Finally, as column 4 indicates, only the proposed model has the potential for expansion to include dielectric constant phenomena which must be incorporated in any fully developed model.

Table 4. Comparison of models.

Model	(1) Conceptual Problems	(2) No. of Variables	(3) Archie's a	(4) Dielectric Applicability
Waxman-Smiths	Yes	8	≤ 1	No
Dual water	Yes	9	≤ 1	No
Proposed	No	5	≥ 1	Yes
Required	No	Minimum	≥ 1	Desirable

The independence of σ_r and σ_w has important implications for understanding shaly sand formations. When taken together with knowledge of clay structure, it supports one of the basic tenets of the dual water model—the presence of two kinds of water. However, in the proposed model the assumption of two resistive elements in parallel is replaced with a more realistic geometrical picture which properly accounts for conducting paths which pass through both kinds of water in series.

The proposed model is applicable to any formation where a distinct conductive property can be ascribed to the matrix. It is fortuitous that shaly sand formations are included in this general, yet simple, representation.

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REFERENCES

- Archie, G. E., 1942, The electrical resistivity log as an aid in determining some reservoir characteristics: *Petr. Tech.* v. 1, p. 55–62.
 Bevington, P. R., 1969, *Data reduction and error analysis for the physical sciences*: New York, McGraw-Hill, Inc.
 Bruggeman, D. A. G., 1935, Berechnung verschiedener physikalischer konstanten von heterogenen Substanzen: *Ann. Physik*, v. 24, p. 636–664.
 Carothers, J. E., 1968, A statistical study of the formation factor relation: *The Log Analyst*, v. 9, p. 13–20.
 Clavier, C., Coates, G., and Dumanoir, 1977, The theoretical and experimental bases for the "Dual water" model for the interpretation of shaly sands: 52nd Annual Fall Technical Conference and

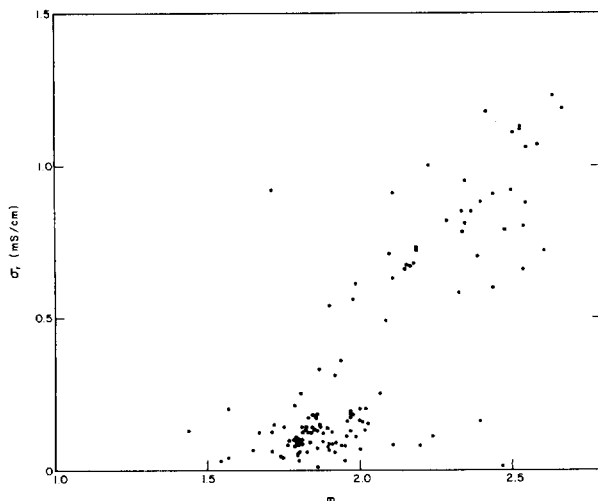


FIG. 7. Matrix conductivity versus cementation factor. Data plotted are from Table 1.

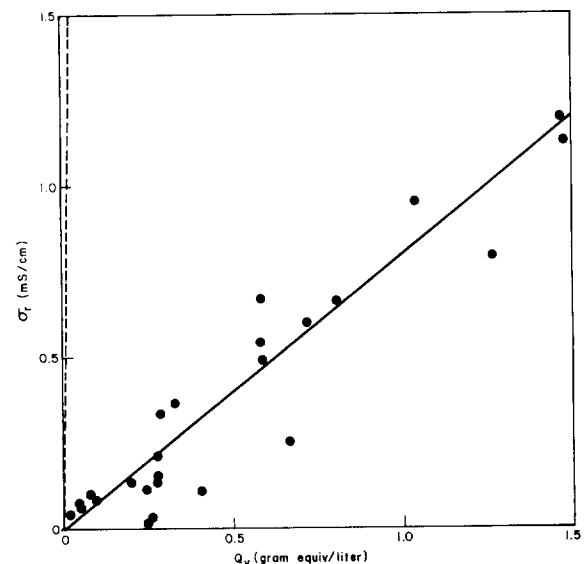


FIG. 8. σ_r versus Q_v . Data plotted are the last group of Table 1. The solid line is equation (4) with $\Lambda_+ = 0.8 \text{ S-cm}^2/\text{g-equiv.}$ The dashed line is equation (4) with $\Lambda_+ = 50 \text{ S-cm}^2/\text{g-equiv.}$

- Exhibition of the SPE of AIME, Denver, October 9–12.
- de Witte, A. J., 1957, Saturation and porosity from electric logs in shaly sands: *Oil and Gas J.*, v. 55, p. 89–93.
- Gomez-Rivera, O., 1976, A practical method for determining cementation exponents and some other parameters as an aid in well log analysis: *The Log Analyst*, v. 16, p. 8–24.
- Hanai, T., 1960a, Theory of the dielectric dispersion due to the interfacial polarization and its application to emulsions: *Kolloid-Zeitschrift*, v. 171, p. 23–31.
- 1960b, a remark on the “Theory of the dielectric dispersion due to the interfacial polarization:” *Kolloid-Zeitschrift*, v. 175, p. 61–62.
- 1961, Dielectric theory on the interfacial polarization for two-phase mixtures: *Bull. Inst. Chem. Res., Kyoto Univ.*, v. 39, p. 341–367.
- Mendelson, K. S., and Cohen, M. H., 1982, The effect of grain anisotropy on the electrical properties of sedimentary rocks: *Geophysics*, v. 47, p. 257–263.
- Porter, C. R. and Carothers, J. E., 1970, Formation factor-porosity relation derived from well log data. SPWLA 11th Annual Logging Symp. Los Angeles, May 3–6.
- Sauer, M. C., Jr., Southwick, P. F., Spiegler, K. S., and Wyllie, M. R. J., 1955, Electrical conductance of porous plugs: *Ind. Eng. Chem.*, v. 47, p. 2187–2193.
- Sen, P. N., 1980, The dielectric and conductivity response of sedimentary rocks: 55th Annual Fall Technical Conference and Exhibition of the SPE of AIME, Dallas, September 21–24.
- 1981, Relation of certain geometrical features to the dielectric anomaly of rocks: *Geophysics*, v. 46, p. 1714–1720.
- Sen, P. N., Scala, C., and Cohen, M. H., 1981, Self-similar model for sedimentary rocks with application to the dielectric constant of fused glass beads: *Geophysics*, v. 46, p. 781–795.
- Spiegler, K. S. and Coryell, C. D., 1951, Electromigration in a cation exchange resin: *Science*, v. 113, p. 546–547.
- Stratton, J. A., 1941, *Electromagnetic theory*: New York, McGraw-Hill, Inc.
- Van Olphen, H., 1963, *An introduction to clay colloid chemistry*: New York, Interscience Publ.
- Von Hippel, A. R., 1954, *Dielectrics and waves*: Cambridge, The MIT Press.
- Waxman, M. H., and Smits, L. J. M., 1968, Electrical conductivities in oil-bearing shaly sand: *Soc. Petr. Eng. J.*, v. 8, p. 107–122.
- Winsauer, W. O., Shearin, H. M., Jr., Masson, P. H., and Williams, M., 1952, Resistivity of brine saturated sands in relation to pore geometry: *AAPG Bull.*, v. 36, p. 253–277.
- Wyllie, M. R. J., and Southwick, P. F., 1954, An experimental investigation of the S. P. and resistivity phenomena in dirty sands: *J. Petr. Tech.*, v. 6, p. 44–57.

THE RELAXATION OF THE DOUBLE LAYER AROUND COLLOIDAL PARTICLES AND THE LOW-FREQUENCY DIELECTRIC DISPERSION

PART I. THEORETICAL CONSIDERATIONS

J. LYKLEMA *

Laboratory for Physical and Colloid Chemistry, Agricultural University, De Dreijen 6, 6703 BC Wageningen (The Netherlands)

S.S. DUKHIN and V.N. SHILOV

Institute of Colloid and Water Chemistry, Academy of Sciences of the Ukrainian Soviet Republic, Vernadskogo Prospekt 42, Kiev-252180 (U.S.S.R.)

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ABSTRACT

This paper discusses the relaxation of electrical double layers around colloid particles and its role in low-frequency dielectric dispersion $\Delta\epsilon(\omega)$. As this dispersion is very sensitive to the distribution of ions over the various double-layer parts, relaxation measurements can provide information on the double-layer composition. A picture is elaborated in which the Schwarz theory is corrected for the contribution of polarization in the diffuse part of the double layer; the potential drop over this part under non-equilibrium conditions is computed from the polarization of the Stern layer. There are two contributions to $\Delta\epsilon(\omega)$, one stemming from bound counterions and one from concentration polarization in the diffuse part of the double layer. The conditions under which these two mechanisms are operative are discussed.

(I) INTRODUCTION

Over the past decades various models for the relaxation of the double layer around colloidal particles have been developed. Of these Overbeek's contribution in connection with his improved theory of electrophoresis [1] deserves special attention. Since the particle and the countercharge bear opposite charges, movement in opposite directions is effected by an applied outer electric field. Details of the hydrodynamics depend among other things on the position of the slipping plane, or rather on the mechanism of the slipping process. Owing to the two opposite movements, the original symmetry of the double layer is perturbed; conduction and diffusion processes try to restore the equilibrium distribution, but this restoration

* To whom correspondence should be addressed.

requires a certain time, the *relaxation time*, so that the ions in the outer double-layer part lag behind the particle. This leads to an additional electric field, directed opposite to the outer one, which reduces the electrophoretic mobility [2].

The most important characteristics of such a relaxation process are (1) the spatial deformation of the double layer, (2) the ensuing counteracting additional field (which we will designate as a “polarization field”) and (3) the important role of diffusion in the region beyond the double layer, the rate of which is coupled to the time of relaxation.

Double-layer relaxation has also been considered in connection with the theory of conductivity of suspensions [3,4], because current flow through double layers leads to their deformation. Recently, the important role of double-layer relaxation in the collision between charged particles (and hence its influence on colloid stability) has also been recognised [5,6]. It is mandatory to take double-layer relaxation into account if coagulation proceeds under non-equilibrium conditions [7], for instance under the influence of an applied electric field [8] or in a concentration gradient [9].

The relation between double-layer relaxation and these phenomena and the connection between this relaxation and the dielectric properties of suspensions [10–12] are related phenomena that should be treated from the same view point. For instance, it may be noted that the Dukhin–Shilov theory for the polarization of double layers [13,14] and its application to the dielectric properties of suspensions [10–12] and Stigter’s theory for the electric conductivity of suspensions [4], are both developed on the basis of Overbeek’s representation [1]. On the other hand, the Schwarz–Schurr theory [15,16] ignores the relaxation of the diffuse part of the double layer and is based on a quite different mechanism. The substantial discrepancies between the polarization mechanisms of Dukhin and Shilov [9–14] and that of Schwarz and Schurr [15,16] are due to different kinetic models of the double layer, and their respective qualities depend on the quality of the model assumptions.

It is the purpose of the present study to further our insight in this matter, combining new theoretical and experimental developments. In this first paper we discuss the present state of the theory, thereby proposing an improvement. We intend to follow this article with an experimental one (Part II) on model colloids (namely homodisperse polystyrene latices in various electrolytes), and one in which the modified theory is applied to the measurements (Part III).

(II) GENERAL CONSIDERATIONS. THE MODELS OF O’KONSKI AND SCHWARZ

In dilute suspensions, the contribution $\Delta\epsilon^*$ of dispersed particles to the complex dielectric permittivity ϵ^* in an alternating field of cyclic frequency ω can be shown to obey [11,17]

$$\Delta\epsilon^*/\epsilon_a^* = 4\pi n\mu^*/E \quad (1)$$

where n is the number of particles per unit volume, ϵ_a^* the complex dielectric permittivity of the dispersion medium and μ^* the effective induced dipole moment of the particle, i.e. the dipole moment divided by $4\pi\epsilon_0\epsilon_a$. It also occurs in the

expression for the potential ϕ_a far from the particles:

$$\phi_a = (-Er + \mu^*/r^2) \cos \theta \quad (2)$$

Furthermore,

$$E = E_0 e^{i\omega t} \quad (3)$$

is the field strength of the applied external uniform ac field, t the time and r and θ the spherical coordinates. The particle is assumed to be spherical, and θ is referred to the direction of the external field ($\theta = 0$).

Because of eqn. (1), those particle properties that are related to μ^* are also relevant for the dielectric permittivity of suspensions. Two of them deserve special attention. First, as a rule, the particle conductivity k_i and the dielectric permittivity ϵ_i of the particles differ from the corresponding quantities in the continuous medium, k_a and ϵ_a^* . The results of this are described in the Maxwell–Wagner theory [18]. According to this, the dispersion of $\Delta\epsilon^*$ is determined by the finite time required for the formation of a polarization charge.

The second important factor is the surface (excess) conductivity. In O’Konski’s theory this has been accounted for by introducing an integral surface conductivity [17]. The theory allows for free exchange of charges between the surface conductance layer and the bulk. Charge conservation along non-conducting particles at any angle θ with the direction of E is expressed through

$$\text{div}_s \kappa^\sigma E_s(\theta) + k_a E_n(\theta) = -\partial\sigma(\theta)/\partial t \quad (4)$$

where E_s and E_n are the tangential and normal component of the field strength near the particle surface respectively, κ^σ is the surface conductivity, and σ the surface charge density.

For a suspension of spheres, the O’Konski theory leads to expressions for $\Delta\epsilon(0)$ and the relaxation time that are formally equivalent to those of Maxwell–Wagner, provided in the latter k_i is replaced by $2\kappa^\sigma/a$, where a is the particle radius. The O’Konski model successfully explains $\Delta\epsilon^*(\omega)$ near the characteristic frequency of Maxwell–Wagner, but offers no satisfactory interpretation of the high values for the low-frequency dispersion of the dielectric permittivity that often occurs in suspensions [11,12].

An attempt to account for the low-frequency polarization was put forward by Schwarz [15]. This approach was not based on a specified model of the double layer. Schwarz attributed the double-layer polarization entirely to counterions bound to the spherical particle in a thin layer. Exchange between the bulk and the layer of surface conductivity was assumed to be absent. Schwarz’s charge conservation condition contains no normal flows and is written

$$\text{div}_s I_b(\theta) = -\partial\sigma_{bE}(\theta)/\partial t \quad (5)$$

* In order to avoid confusion with the reciprocal Debye length κ^{-1} , we shall use the symbol k for conductivities.

where I_b and σ_{bE} are the current and surface charge density (in the presence of the field E) respectively, in the layer of bound counterions. The charge σ_{bE} in eqn. (5) need not be identical to σ in eqn. (4). Because of the absence of exchange, I_b must contain a diffusion contribution. In dc fields ($\omega \rightarrow 0$), $I_b \rightarrow 0$; the surface current in the layer of bound ions vanishes because diffusion and conduction compensate for each other.

According to Schwarz, the expression for $I_b(\theta)$ contains a conduction (migration) and a diffusion part and can be written

$$I_b(\theta) = \frac{D_b}{a} \left[\frac{\partial \Delta \sigma_{bE}}{\partial \theta} + \frac{F \sigma_b}{RT} \frac{\partial \phi_{bE}}{\partial \theta} \right] \quad (6)$$

where $\Delta \sigma_{bE}$ is the deviation of the bound charge σ_{bE} at given θ and t in the presence of a field E from the same σ_b in the absence of a field, i.e.

$$\Delta \sigma_{bE}(\theta, t) = \sigma_{bE}(\theta, t) - \sigma_b \quad (7)$$

where D_b is the surface diffusion coefficient for bound ions and ϕ_{bE} is the electric potential in the layer of bound ions

$$\Delta \phi_{bE}(\theta, t) = \phi_{bE}(\theta, t) - \phi_b \quad (8)$$

In Schwarz's picture the time required to establish the distribution of bound charges in the stationary state is $a^2/2D_b$. The corresponding critical frequency of dielectric dispersion is

$$\omega_{b,cr} = 2D_b/a^2 \quad (9)$$

As stated above, the Schwarz model applies only in the absence of normal flow. As these flows are suppressed at high frequencies, the picture is applicable if

$$\omega \gg \tau_{bn}^{-1} \quad (10)$$

where τ_{bn} is the time required to establish equilibrium between the layer of bound ions and the bulk. Because of eqn. (9) the applicability range of the Schwarz theory can also be written

$$(1/\tau_{bn})(a^2/2D_b) \ll 1 \quad (11)$$

Condition (11) is not likely to be met in practice. First, eqn. (11) requires large values of D_b , but if the bound ions are so tightly associated with the surface that they do not exchange, this is unlikely. Second, the conditions do not concur with the kinetics of a diffuse double-layer part either. The relaxation time of a diffuse layer τ_d is on the order of

$$\tau_d \sim (2\kappa^2 D_d)^{-1} \quad (12)$$

where κ is the reciprocal Debye length and D_d the diffusion coefficient of ions in this layer (which may be lower than the corresponding value D in bulk). Schwarz developed his theory for thin layers, implying

$$\kappa a \gg 1 \quad (13)$$

so that, if the bound ions were distributed as in a diffuse double layer,

$$(1/\tau_d)(a^2/2D_d) \sim (\kappa a)^2 \gg 1 \quad (14)$$

which is in conflict with eqn. (11).

In addition to these objections, it may be repeated that Schwarz's double-layer theory is far too simple in that he only considers bound counterions. In line with the present view that the solution side of a double layer generally consists of two parts, a diffuse and a non-diffuse one, we shall generalize below the Schwarz theory by explicitly accounting for three categories of charges (Fig. 1):

- (1) fixed surface charges (e.g. covalently bound ionic groups), charge density σ_f ;
- (2) a layer of bound counterions (e.g. as ion pairs), charge density σ_b ;
- (3) a diffuse layer, charge density σ_d .

It would have been even more rigorous to subdivide the layer of bound ions into an inner and an outer Helmholtz layer, but we shall not pursue this refinement, partly because it would considerably complicate the mathematics and partly because the present picture already appears adequate to explain recent experimental results (Part III of this series).

The term "fixed charge" is interpreted as a charge with a negligible mobility along the surface. This is an important condition. With respect to counterions with surface diffusion coefficient $D_s < D$, for frequencies

$$\omega \gg 2D_s a^{-2} \quad (15)$$

they may be considered as fixed, whereas for

$$\omega \lesssim 2D_s a^{-2} \quad (16)$$

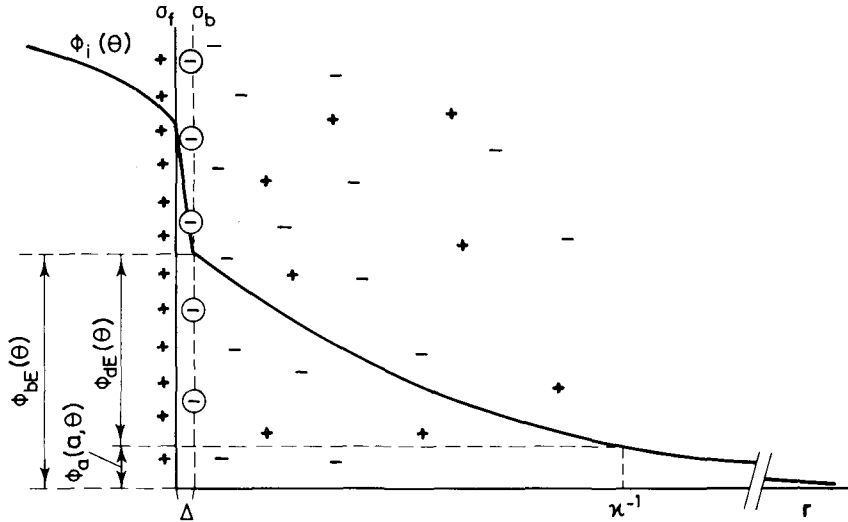


Fig. 1. The model for a polarized double layer used in the present study. Explanation in text. (At large distances ϕ equals Er because of eqn. 2.)

they are mobile, and if they do not exchange against the bulk they would be equivalent to the “bound ions” in Schwarz’s model. We recall that a Schwarz-type polarization layer *can* be realized if condition (11) applies.

In passing, we note that in this and following equations strictly $(a + \Delta)$ should be used instead of a , but as the difference is trivial we shall ignore it for simplicity.

If the ions in the inner Helmholtz plane are also fixed, the double-layer picture is again simple, with all ions outside the outer Helmholtz plane constituting the countercharge. However, if these ions are not considered as fixed, i.e. if eqn. (15) is not satisfied, the Schwarz model requires generalization. In that case the polarization of two layers of ions must be considered, with charge densities σ_b and σ_d .

(III) GENERALIZATION OF THE SCHWARZ MODEL

The application of an outer electric field leads to a shift in the distribution of bound charges, upon which the spherical symmetry is lost (Fig. 2a).

The diagram of Fig. 2a is the result of the tendency of all counterions to move in the direction opposite to that of the field E . It is typical for the bound ions that they

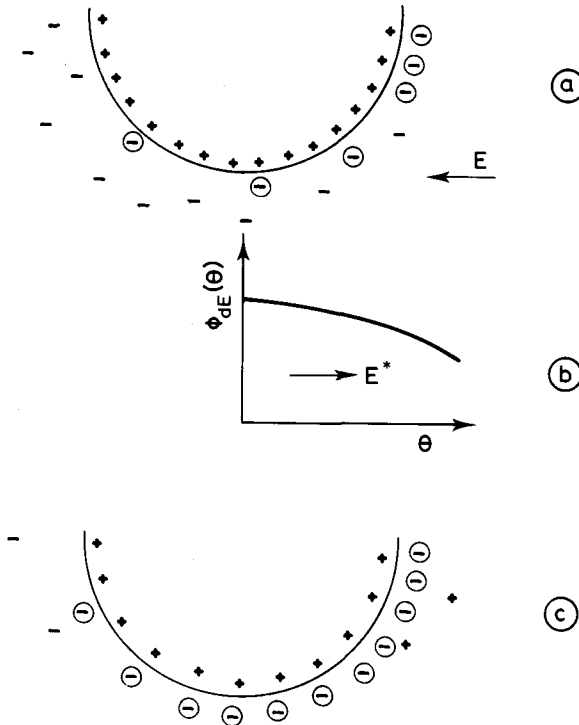


Fig. 2. (a) Illustration of polarization mechanism in a double layer; (b) illustration of the ensuing variation of $\phi_{dE}(\theta)$ with θ . The polarization field E^* is directed to the right; (c) polarization mechanism if $\sigma_d = 0$.

enrich the positive side (to the right in Fig. 2a), whereas diffuse ions can exchange with the bulk. To maintain (almost) electroneutrality at each θ , they enrich the negative side (to the left in Fig. 2a).

We assume that the applied field is uniform and harmonic (eqn. 3) and sufficiently weak, so that all quantities related to the field are linear with the field and vary as a function of time according to the same law as the field. Let us first consider the bound counterions, i.e. the only ions considered by Schwarz.

For σ_{bE} we have eqn. (7) with

$$\Delta\sigma_{be}(\theta, t) = \Delta\sigma_{bE}(\theta)e^{i\omega t} \quad (17)$$

Substituting eqn. (6) in eqn. (5), writing σ_b according to eqn. (7), $\Delta\sigma_{bE}$ following eqn. (17), and elaborating the second term on the rhs of eqn. (6) by using eqn. (8) and

$$\Delta\phi_{bE}(\theta, t) = \Delta\phi_{bE}(\theta)e^{i\omega t} \quad (18)$$

the following expression is obtained:

$$i\omega\Delta\sigma_{bE}(\theta) = \frac{D_b}{a^2} \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left\{ \sin\theta \left[\frac{\partial\Delta\sigma_{bE}(\theta)}{\partial\theta} + \frac{F\sigma_b}{RT} \frac{\partial\phi_{bE}}{\partial\theta} \right] \right\} \quad (19)$$

which is identical to eqn. (15) in ref. 15. From this equation Schwarz obtained the following general form for the relationship between $\Delta\sigma_{bE}(\theta)$ and $\Delta\phi_{bE}(\theta)$:

$$\Delta\sigma_{bE}(\theta) = - \frac{F\sigma_b/RT}{1 + i\omega\tau_{bs}} \Delta\phi_{bE}(\theta) \quad (20)$$

with

$$\tau_{bs} = a^2/2D_b \quad (21)$$

The next step is to find the functions $\Delta\phi_{bE}(\theta)$ and $\Delta\sigma_{bE}(\theta, t)$, i.e. the oscillations of the bound and the diffuse ions with the outer field. The space beyond the boundaries of the double layer is electroneutral, so that according to Laplace

$$\nabla^2\phi_a(r, \theta) = 0 \quad (r > a + \kappa^{-1}) \quad (22)$$

which leads to eqn. (2).

As Schwarz only considers bound counterions, he identifies $\phi_{bE}(\theta)$ with $\phi_a(r, \theta)$ and substitutes the latter for the former in eqn. (19) and, hence, in eqn. (20). Let us discuss the result of this oversimplification, restricting ourselves to the most interesting low-frequency range, determined by $\omega\tau_{bs} \lesssim 1$. As in the Schwarz picture $\kappa a \gg 1$ (eqn. 13), τ_d in eqn. (12) is much smaller than τ_{bs} , so that $\omega\tau_d \ll \omega\tau_{bs} \lesssim 1$, implying that under these conditions redistribution of diffuse ions (if there is also a diffuse double-layer part) is sufficiently fast almost to retain equilibrium between the diffuse and the bound part of the double layer at any moment during polarization. Consequently, the surface and bound charges are continually almost completely screened by the diffuse part and hence the polarization of the diffuse layer cannot be ignored. In Appendix I the situation is rigorously analysed for small perturbations; there it will be proved that such perturbations may be neglected.

We conclude from the foregoing that the layer of bound charges is in fact continually screened by the diffuse charge, so that a more rigorous analysis is needed in which both the diffuse and the bound counterions are considered.

To achieve this, we write the surface charge distribution in the polarized diffuse part of the double layer, σ_{dE} , as the sum of the original symmetrical distribution σ_d , and a small excess $\Delta\sigma_{dE}(\theta, t)$ in a way similar to that for the bound ions eqns. (7) and (17)

$$\sigma_{dE}(\theta, t) = \sigma_d + \Delta\sigma_{dE}(\theta, t) \quad (23)$$

$$\Delta\sigma_{dE}(\theta, t) = \Delta\sigma_{dE}(\theta)e^{i\omega t} \quad (24)$$

The double layer as a whole is always approximately electroneutral at any θ

$$\sigma_f + \sigma_b(\theta) + \sigma_d(\theta) = 0 \quad (25)$$

$$\sigma_f + \sigma_{bE}(\theta, t) + \sigma_{dE}(\theta, t) = 0 \quad (26)$$

Using eqns. (7), (23) and (25), it follows from eqn. (26) that

$$\Delta\sigma_{dE}(\theta) = -\Delta\sigma_{bE}(\theta) \quad (27)$$

The Laplace equation in the form (22) does not hold for parts within the diffuse double layer, but it does apply for the potential distribution along the outer periphery of this layer, say, for potentials $\leq \phi_a(a + \kappa^{-1}, \theta)$. We shall identify the potential drop over the diffuse layer ϕ_{dE} as the drop between $r = a$, where $\phi = \phi_{bE}(\theta, t) = \phi_{bE}(a, \theta, t)$ and $r = a + \kappa^{-1}$ (Fig. 1b); hence

$$\phi_{bE}(\theta, t) = \phi_a(a + \kappa^{-1}, \theta, t) + \phi_{dE}(\theta, t) \quad (28)$$

in which by analogy to eqn. (8):

$$\phi_{dE}(\theta, t) = \phi_d + \Delta\phi_{dE}(\theta, t) \quad (29)$$

Strictly, the potentials should be referred to $\phi_a = 0$ attained at very high values of r (in principle, $r \rightarrow \infty$). However, in non-equilibrium double layers the difficulty arises that ϕ also keeps changing in the region of electroneutrality, so that the unambiguous establishment of a reference for ϕ becomes difficult. Therefore, we accept for these purposes $r = a + \kappa^{-1}$ as the outer double-layer boundary, considering the quasi-equilibrium of the part $r > a + \kappa^{-1}$.

Transformation of potential perturbations into charge perturbations proceeds by introducing differential capacitances. With eqn. (27)

$$\Delta\phi_{dE}(\theta) = -\Delta\sigma_{dE}(\theta)/C_d = \Delta\sigma_{bE}(\theta)/C_d \quad (30)$$

where

$$C_d = \epsilon_a \epsilon_0 \kappa \cosh(F\phi_d/2RT) \quad (31)$$

for a symmetrical electrolyte. Here, ϵ_a is the relative dielectric constant of the bulk solution. Strictly, for C_d we require only the capacitance of the part between $r = a$ and $r = a + \kappa^{-1}$, but substitution of the complete capacitance introduces only negligible error, since $\kappa a \gg 1$.

As illustrated by Fig. 2a, the quantities $|\sigma_{dE}(\theta)|$ and $|\Delta\sigma_{dE}(\theta)|$ decrease with increasing θ . Also, $\phi_{dE}(\theta)$ and $\Delta\phi_{dE}(\theta)$ decrease with θ . The polarization field E^* is directed towards decreasing ϕ_{dE} (Fig. 2b), i.e. opposite to the outer field E . This polarization field reduces the amplitude of the shifts of the bound counterions and hence the magnitudes of μ^* and $\Delta\epsilon^*$. The polarization of the diffuse part reduces the relaxation time of the bound ions. As Schwarz ignores the diffuse part, he overestimates the relaxation time and $\Delta\epsilon^*$. In passing, it may be added that this also applies for the case that $\sigma_d = 0$, because it is the oscillating part of the diffuse part ($\Delta\sigma_{dE}(\theta, t)$) that counts, and because of eqn. (27) this must persist if there is no net overall diffuse charge. Figure 2c illustrates this.

Differentiating eqn. (28) with respect to θ , using eqns. (30) and (27), gives

$$\frac{\partial\phi_{bE}}{\partial\theta} = \frac{\partial\phi_a}{\partial\theta} + \frac{\partial\phi_{dE}}{\partial\theta} = \frac{\partial\phi_a}{\partial\theta} - \frac{1}{C_d} \frac{\partial\Delta\sigma_{dE}}{\partial\theta} = \frac{\partial\phi_a}{\partial\theta} + \frac{1}{C_d} \frac{\partial\Delta\sigma_{bE}}{\partial\theta} \quad (32)$$

illustrating how the polarization of bound ions is influenced by the tangential oscillations of the diffuse atmosphere ($\partial\phi_{dE}/\partial\theta$). For very high diffuse capacitances the last term vanishes; in that case the angle dependence of ϕ_b is fully determined by that in ϕ_a .

Substitution of eqn. (32) in eqn. (19) leads to

$$i\omega\Delta\sigma_{bE}(\theta) = \frac{FD_bM}{a^2 \sin \theta} \frac{\partial}{\partial\theta} \left\{ \sin \theta \left[\frac{\partial\Delta\sigma_{bE}}{\partial\theta} + \frac{F\sigma_b}{RT} \frac{1}{M} \frac{\partial\phi_a}{\partial\theta} \right] \right\} \quad (33)$$

with

$$\begin{aligned} M &= 1 + \frac{F\sigma_b}{RTC_d} = 1 + \frac{\kappa\sigma_b}{2Fc \cosh(F\phi_d/2RT)} \\ &= \frac{2}{\cosh(F\phi_d/2RT)} \left[\frac{\sigma_f\kappa}{4Fc} - \sinh\left(\frac{F\phi_d}{2RT}\right) \right] \end{aligned} \quad (34)$$

where c is the electrolyte concentration entering the equation through $\kappa^2 = 2F^2c/\epsilon_a\epsilon_0RT$. We used eqns. (25) and (31) in the derivation. The factor M corrects the Schwarz equation for the contribution of the diffuse part of the double layer.

Equation (33) is an improvement of eqn. (19). The former follows from the latter by substituting D_bM for D_b and $\phi_a(\theta)/M$ for $\phi_{bE}(\theta)$. By a similar argument, one can derive

$$\Delta\sigma_{bE}(\theta) = - \frac{F\sigma_b/RT}{1 + i\omega\tau_{bs}} \frac{\phi_a(\theta)}{M} \quad (35)$$

and

$$\tau_{bs} = a^2/2D_bM \quad (36)$$

It is easily verified that these two expressions follow from eqns. (20) and (21) respectively, after application of the same substitutions as above.

In order to find the dipole moment we use a procedure equivalent to that applied

by Schwarz. Consider first the boundary condition for the conservation of free charge σ_{dE} in the absence of tangential transport in the diffuse layer, a situation concurring with the condition

$$R_{el} = k_d^\sigma / k_a a \ll 1 \quad (37)$$

which reads

$$k_a \frac{\partial \phi_a}{\partial r} \Big|_{r=a+\kappa^{-1}} = \frac{\partial \Delta \sigma_{dE}}{\partial t} = i\omega \Delta \sigma_{dE} \quad (38)$$

Let us designate the sum of the charge densities in all layers at a given θ as $\sigma_t(\theta)$. Under non-equilibrium conditions $\sigma_t(\theta) \neq 0$, implying that $\Delta \sigma_{dE}(\theta)$ and $\Delta \sigma_{bE}(\theta)$ vary in a different fashion tangentially to the surface, so that at any specific θ the double layer is unbalanced, although, of course, taken over the entire particle, electroneutrality prevails*. The non-equilibrium total charge gives rise to an electric field beyond its boundaries, or, for that matter, to a potential distribution. The two excess surface charges are connected through Gauss's theorem, applied to the double-layer part subtending θ , and can be expressed in terms of σ_t as follows (see also the note following eqn. A2 in Appendix A):

$$-\epsilon_0 \epsilon_i \frac{\partial \phi_i}{\partial r} \Big|_{r=a} + \epsilon_0 \epsilon_a \frac{\partial \phi_a}{\partial r} \Big|_{r=a+\kappa^{-1}} = -[\Delta \sigma_{bE}(\theta) + \Delta \sigma_{dE}(\theta)] = -\Delta \sigma_t(\theta) \quad (39)$$

Multiplication of eqn. (39) by $i\omega$ and adding the outcome to eqn. (38) yields

$$-K_i \frac{\partial \phi_i}{\partial r} \Big|_{r=a} + K_a \frac{\partial \phi_a}{\partial r} \Big|_{r=a+\kappa^{-1}} = -i\omega \Delta \sigma_{bE} \quad (40)$$

with $K_a = k_a + i\omega \epsilon_0 \epsilon_a$ and $K_i = i\omega \epsilon_0 \epsilon_i$. In this way we arrive at the boundary condition that was also used by Schwarz (eqn. 21 in ref. 15). Equation (40) does not contain σ_{dE} explicitly, but from the procedure to derive it it is clear that $\Delta \sigma_{dE}$ has been accounted for, i.e. that the screening of the bound charge density changes by the diffuse double-layer part is taken into account.

That Schwarz, although not using a specific double-layer representation, arrived at the same boundary condition (40) is due to the fact that he, because he ignored $\Delta \sigma_{dE}$, had to introduce into eqns. (38) and (39) the charge σ_{MW} , generated by Maxwell-Wagner polarization. This charge σ_{MW} is eliminated in a later stage because of the conditions corresponding to eqns. (39) and (40).

The equation for μ^* is obtained by substitution of $\phi_a(r, \theta)$ according to eqn. (2) and $\Delta \sigma_{bE}(\theta)$ according to eqn. (35) in eqn. (40). Solving the equation leads to

$$\text{Im } \mu^* = \frac{3}{2} \frac{k_b^\sigma}{k_a a} \frac{\omega \tau_{bs}}{1 + \omega^2 \tau_{bs}^2 (1 + k_b^\sigma / k_a a)^2} a^3 E \quad (41)$$

* In passing, it may be as well to point out that $\sigma_t(\theta) \ll \sigma_b$ (see also Appendix A). Hence, σ_t could be neglected in eqn. (26) but not in eqn. (39), because in the latter expression it governs the potential distribution sought.

where $k_b^0 = F^2 D_b |\sigma_b| / RT$. On the basis of eqns. (1) and (41) we obtain the following corrected Schwarz equation:

$$\Delta\epsilon(0)/\epsilon_a = \frac{9}{2} p \kappa a \frac{\sigma_f \kappa / 4Fc - \sinh(F\phi_d / 2RT)}{1 + \frac{2}{\cosh(F\phi_d / 2RT)} \left[\frac{\sigma_f \kappa}{4Fc} - \sinh\left(\frac{F\phi_d}{2RT}\right) \right]} \quad (42)$$

where p is the volume fraction of the dispersed material. This equation is also obtainable from eqn. (45) in ref. 15, replacing σ_f by $(\sigma_f - \sigma_d)/M$.

The contribution to $\Delta\epsilon(0)$ of the polarization of the bound ions at fixed σ_f depends strongly on the distribution of counterions over the Stern and the diffuse layer. It passes through a maximum as a function of σ_d because:

- (1) at high σ_d (i.e. $\sigma_d \rightarrow \sigma_f$) σ_b has to go down;
- (2) at low σ_d (i.e. $\sigma_b \rightarrow \sigma_f$) I_b goes down because of the opposing electric field, determined by the angular dependency $\Delta\phi_d(\theta)$; this latter quantity rises at low σ_d because of the low capacitance of the diffuse part.

It follows that the relative contribution to $\Delta\epsilon(0)$ of bound ions is very sensitive to the nature and concentration of the indifferent electrolyte and is a maximum around $\sigma_b \sim \sigma_d = \sigma_f/2$.

(IV) LOW-FREQUENCY DIELECTRIC DISPERSION, DETERMINED BY CONCENTRATION POLARIZATION OF A DIFFUSE ELECTRICAL DOUBLE LAYER

In section (III) the creation of μ^* due to bound ions was considered. In that case there were only normal flows in the diffuse part of the double layer, no tangential ones (eqn. 37). Now we consider the opposite case where Stern ions are absent ($\sigma_b = 0$) and polarization is only due to the diffuse part. It is typical for this new situation that electro-osmotic flow around the particle is to occur. Consequently, under these conditions we must introduce the notion of a slipping plane, delimiting the hydrodynamically movable and stationary parts of the liquid.

We shall assume that all ions are mobile, including those between the surface and the slipping plane, so that they contribute to the surface conductance, to μ^* and $\Delta\epsilon^*$. By way of approximation we shall not consider a detailed structure of this layer. Clearly it is bound by two equipotential surfaces. The outer surface is the slipping plane, characterized by the potential ζ . For the inner layers we choose the plane delimiting the fixed surface charge (including fixed counterions, if any). This is the Stern plane, characterized by ϕ_d . In other words, we now allow for the possibility that there is a layer in which the ions are mobile, even if the liquid is not. The assumption implies a possible difference between ϕ_d and ζ .

For frequencies $\omega \ll \tau_d^{-1}$ (eqn. 12) the diffuse layer is quasi-stationary, i.e. it can

* For a double layer at rest it is known that it expels electrolyte due to negative adsorption of *co-ions* (the Donnan effect). Under non-equilibrium conditions this distribution of neutral electrolyte around the particle becomes asymmetrical and it is due to the polarization of the *counterions*.

be described by the same equations that Overbeek used for the electrophoresis in a dc field [1]. It follows from Overbeek's theory that in an applied field the concentration of the neutral electrolyte beyond the limits of the double layer is changed *. In a dc field this change depends on the coordinate (as $\cos \theta/r^2$). The occurrence of a concentration drop in the outer field is very important for the low-frequency impedance of suspensions. According to refs. 10–12, 14 for $\omega \ll \tau_d^{-1}$ and $\kappa a \gg 1$ the diffuse double layer is not only quasi-stationary but is also in local equilibrium with the neutral electrolyte close to this part. This consideration permits an analytical approach to the relaxation in an ac field on the same basis as Overbeek [1], without making restrictions on the value of the potential drop over the diffuse layer [14].

According to refs. 10–12, the finite time to establish the concentration drop beyond the double-layer limits is the determining factor for the $\mu^*(\omega)$ dependence. This time is close to the time required to forward a diffusion front over a distance of the order of the particle radius ($\tau_c \sim a^2/2D$). In this case the induced dipole moment takes the following shape [10]:

$$\mu^* = -\frac{3}{2} E a^3 \frac{a_1 + a_1 W + i(a_1 W + a_2 W^2)}{A_1 + A_1 W + i(A_1 W + A_2 W^2)} \quad (43)$$

where

$$W = a\sqrt{\omega/2D} \quad (44)$$

$$a_1 = a^+(1 + \alpha_c^-) - \alpha^-(1 + \alpha_c^+) \quad (45)$$

$$A_1 = (z^- - \alpha^-)(1 + \alpha_c^+) + (z^+ + \alpha^+)(1 + \alpha_c^-) \quad (46)$$

$$a_2 = \alpha^+(1 + \alpha_1^-) - \alpha^-(1 + \alpha_1^+) \quad (47)$$

$$A_2 = (1 + \alpha_1^+)(z^- - \alpha^-) + (1 + \alpha_1^-)(z^+ + \alpha^+) \quad (48)$$

Here, z^+ and z^- are the valencies. Furthermore,

$$\alpha^\pm = \left[\frac{3}{2} m I_2^\pm \mp z^\pm I_1^\pm \right] / \kappa a \quad (49)$$

$$\alpha_c^\pm = \left[\frac{3}{2} m I_{2c}^\pm - I_1^\pm \right] / \kappa a \quad (50)$$

$$\alpha_1^\pm = z^\pm \left[I_1^\pm + I_3^\pm \right] / \kappa a \quad (51)$$

For symmetrical electrolytes * ($z^+ = z^- = z$)

$$I_1^\pm = -2 \left[\exp \left(\mp \frac{zF\phi_d}{RT} - 1 \right) \right] \quad (52)$$

$$I_2^\pm = \frac{2F\xi}{RT} \mp \frac{4}{z} \left[1 - \exp \left(\mp \frac{zF\xi}{2RT} \right) \right] \quad (53)$$

$$I_{2c}^\pm = \frac{2F\xi}{zRT} - \frac{4}{z^2} \left[1 - \exp \left(\mp \frac{zF\xi}{2RT} \right) \right] - 16 \ln \cosh \left(\frac{zF\xi}{4RT} \right) \quad (54)$$

* These and the following expressions apply for the simplifying condition $D^+ = D^- = D$. The more general case $D^+ \neq D^-$ can be found in ref. 12, p. 80ff. We intend to revert to this matter in part III.

$$I_3^\pm = - \frac{3 \cosh(zF\phi_d/2RT) - \exp(\mp zF\phi_d/RT) - 2}{\cosh(zF\phi_d/2RT)} \quad (55)$$

$$m = 2\epsilon_0\epsilon_a (RT/F)^2 / 3\eta D \quad (56)$$

We note that eqn. (52) is a conduction term, hence it contains ϕ_d ; on the other hand, eqns. (53) and (54) are electro-osmotic terms and therefore contain ζ .

The expression for the dielectric increment now reads

$$\frac{\Delta\epsilon(\omega)}{\epsilon_a} = \frac{9}{4} p (\kappa a)^2 \frac{(A_1 a_2 - A_2 a_1)(1 + W + W^2)}{(A_1 + A_1 W)^2 + (A_1 W + A_2 W^2)^2} \quad (57)$$

$$\frac{\Delta\epsilon(0)}{\epsilon_a} = \frac{9}{4} p (\kappa a)^2 \frac{A_1 a_2 - A_2 a_1}{A_1^2} \quad (58)$$

For sufficiently negatively charged particles, $I_1^+ \gg I_1^-$, $I_2^+ \gg I_2^-$, $I_{2c}^+ \gg I_{2c}^-$, and then eqn. (58) can be elaborated to

$$\frac{\Delta\epsilon(0)}{\epsilon_a} = \frac{9p(1+z)}{8} \left\{ \frac{\exp[zF|\phi_d|/2RT] + 3m/z^2 \exp[zF|\zeta|/2RT]}{1 + \frac{2 \exp[zF|\phi_d|/2RT]}{\kappa a} + \frac{6m \exp[zF|\zeta|/2RT]}{\kappa a z^2}} \right\}^2 \quad (59)$$

Thus, consideration of the change in the electrolyte concentration outside the double layer in an applied ac field enables us to formulate an expression for the low-frequency dielectric dispersion without having to introduce the notion of bound ions. However, this notion will be required if the Stern ions behave as if bound.

It follows from the considerations of sections (III) and (IV) that ion specificity in $\Delta\epsilon(0)$ can have two causes, depending on the question whether or not bound ions are present. If they are present, different counterions will bind to a differing extent and hence lead to different values of the ratio σ_d/σ_b . On the other hand, if they are absent, differences between D^+ and D^- (i.e. differences between m^+ and m^-) have to be invoked. In Part III of this series we intend to return to this matter.

(V) RESULTS AND COMPARISON OF POLARIZATION MECHANISMS

Some typical examples of dispersion curves are collected in Fig. 3a–d. These diagrams illustrate the points previously made, viz. that the low-frequency dispersion tends to be mainly due to polarization of the diffuse part of the double layer and that the contribution of bound counterions is a maximum around $\sigma_b \sim \sigma_f/2$. At given κa , the polarization of the purely diffuse double layer increases much more strongly with σ_f than polarization due to bound ions. At given σ_f there is a general increase with κa .

In Fig. 4 the boundary is given delimiting the areas where the contribution to the polarization of the diffuse double layer and the Stern layer respectively, prevail. To that end, $\Delta\epsilon(0)$ has been computed on the basis of eqns. (42) and (58) for the

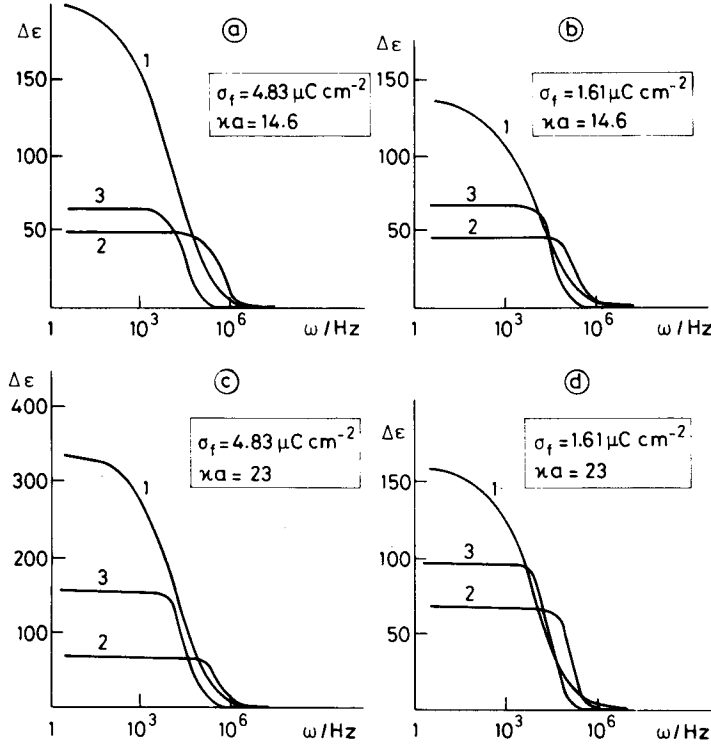


Fig. 3. Dielectric dispersion curves at different values of κa and the fixed charge σ_f . Volume fraction 0.02, $a = 0.22 \mu$, $D^+ = D^- = 2 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$. Curve 1 polarization of diffuse atmosphere only (eqn. 57); curve 2: polarization due to bound counterions (improved Schwarz picture, eqn. (42); $\sigma_d = 0$; $\sigma_b = \sigma_f$; curve 3: as curve 2; $\sigma_d = \sigma_b = \sigma_f/2$. (Note the different ordinate axis scale in Fig. 3c.)

condition

$$\sigma_b = \sigma_d = \sigma_f/2 \quad (60)$$

at various σ_f and κa , taking $z = 1$. Curve 1 of this diagram is the relationship between σ_f and κa that, according to eqns. (42) and (58), leads to the same $\Delta\epsilon(0)$. Because of eqn. (60) this can also be written in terms of a relation between a certain critical value ϕ_d^{cr} and κa . Approximately

$$\sinh(zF\phi_d^{\text{cr}}/2RT) \sim 0.03 \kappa a \quad (61)$$

In the area left of this curve, the influence of the polarization of the diffuse layer prevails, whereas to the right of it an appreciable influence of the bound counterions must be expected (provided eqn. 13 is valid). Hence, the influence of bound counterions plays an important role only at moderate values of $\phi_d < \phi_d^{\text{cr}}$. Then $\Delta\epsilon(0)$ is much lower than its maximally attainable value attained at the same κa if $\phi_d \rightarrow \infty$ (this maximum is then determined by concentration polarization). To estimate the

maximum value of $\Delta\epsilon(0)$ at fixed σ_f and given κa , as determined by the bound counterions, computations on the basis of eqn. (42) have been carried out for the combination of σ_f and κa values satisfying curve 1. Results are given in curve 2.

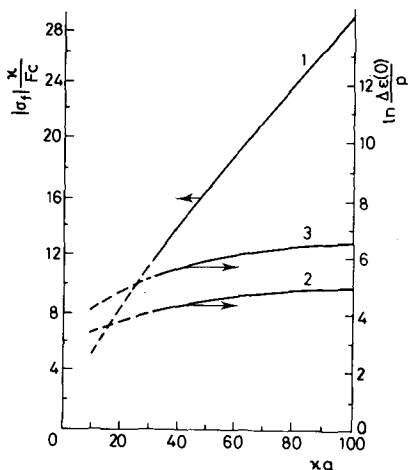


Fig. 4. Curve 1 represents pairs of values σ_f , κa for which $\epsilon(0)$ calculated according to eqns. (58) and (42) is the same, assuming $\sigma_d = \sigma_b = \sigma_f/2$. In the (σ_f, a) range left of the curve, eqn. (58) gives a higher value for $\Delta\epsilon(0)$ than eqn. (42); to the right of the curve it is the other way round. Curve 2 gives the pertaining $\Delta\epsilon(0)$ per unit volume fraction p according to eqn. (42), and curve 3 is the maximally attainable value of $\Delta\epsilon(0)$ according to eqn. (58). Below $\kappa a \sim 30$, the theory is only approximately valid.

Curve 3 characterizes the maximum value if $\Delta\epsilon(0)$ is only determined by concentration polarization according to eqn. (59) with $\exp(F\phi_d/2RT) \gg \kappa a$.

The extent of counterion binding can be described by a Stern-type equation [19] that we shall use in the simplified form

$$\frac{\sigma_b}{\sigma_f} = - \frac{\beta x \exp(-F\phi_d/RT)}{1 + \beta x \exp(-F\phi_d/RT)} \quad (62)$$

where β is the binding constant ($= \exp(-F\Phi_i/RT)$ if Φ_i is the specific adsorption potential), $x = 0.018 c$, with c in mol l^{-1} . An equal distribution (eqn. 60) is obtained if $\beta x \exp(-F\phi_d/2RT) = 1$ or, considering eqn. (61), if $\beta \sim x^{-1} \sim 50/c$. Complications due to electrode polarization limit measurements of the low-frequency dielectric dispersion to concentrations above $10^{-2} \text{ mol l}^{-1}$. Hence, it follows that a strong influence of bound counterions (relative to that of polarization of the diffuse part of the double layer) is possibly only at high values of β , of the order of 10^4 . As in most systems (AgI, Hg, lattices) $F\Phi_i$ is only a few kT , such high β -values are not usually attained except in the case of strongly specifically adsorbing counterions.

From all of this it follows that under most experimental situations $\Delta\epsilon(0)$ is determined by the concentration polarization mechanism.

(VI) COMMENTS ON OTHER TREATMENTS

Schurr [16] has proposed some modifications of the Schwarz model, but the modified picture is still far from perfect. He distinguishes two types of surface conductivity, one frequency-dependent and due to bound charges, and one frequency-independent contribution. This second contribution resembles the O'Konski treatment [17] and Henry's approach to electrophoresis [20]. However, the diffusion current of ions in the diffuse layer is ignored, as it is in the Schwarz theory, and this implies that the tangential electric field at its inner and outer boundary are identified. This means that Schurr's model is internally inconsistent, because it does take into account electric conduction in the diffuse layer, but not the ensuing concentration drop beyond the boundaries of the double layer.

The simultaneous accounting for the polarization of the layer of bound ions and that of the diffuse part is a rather elaborate and as yet unsolved problem. However, in the special case of low k_b^g and low k_d^g (both $\ll 2k_a a$), there is no mutual influence of the tangential flows in the bound and diffuse layer part, allowing the relaxations of these layers to be treated independently. In that case, $\Delta\epsilon^*(\omega)$ is simply a superposition of the contributions of these two layers.

If condition $k_d^g \ll 2k_a a$ is not fulfilled, the radial current in eqn. (40) is not only determined by the tangential flow of the bound counterions but also by that of the diffuse-layer ions. Schurr took this into account by the introduction in eqn. (40) of the extra term

$$\frac{k_d^g}{a^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi_{bE}}{\partial \theta} \right) \quad (63)$$

Solving eqn. (40) after this addition leads to his formula for $\Delta\epsilon(0)$. However, as follows from estimation (61), the condition $k_d^g \ll 2k_a a$ is very well obeyed for values of κa and σ_f , for which the effect of bound counterions on $\Delta\epsilon(0)$ is not small in comparison with the contribution of concentration polarization of the diffuse double-layer part. This condition breaks down only for substantially higher values of σ_d and ϕ_d when concentration polarization prevails, so that the addition of eqn. (63) is ineffective.

Ballario et al. [21] applied the Schwarz-Schurr theory to their data in solutions of very low c , but such a procedure is defective because it demands the use of Bikerman's surface conductivity equation in eqn. (63) which is allowed only if eqn. (13) is satisfied and this, in turn, is not the case at very low c . In a later paper [22] dealing with not so extremely low c , in an attempt to account for discrepancies with the Schwarz-Schurr theory, a phenomenologically adjustable parameter was introduced to account for the distribution of relaxation times. No physical meaning could be assigned to this parameter.

The theory for the polarization of the diffuse double layer by Dukhin and Shilov recently found a sequel in publications by O'Brien, White and DeLacey [23,24]. The physical picture of the two theories is identical, both regarding electrophoresis and dielectric phenomena, although O'Brien used only numerical methods. Both use the

law of the conservation of ions of every type, Poisson's law, and hydrodynamic equations. However, a difference emerges in that Dukhin and Shilov in their dielectric theory do not consider electrophoretic oscillations of the particles in ac fields. In addition, the O'Brien–White theory for dielectric dispersion is not restricted to large κa . This difference, however, does not lead to differences in the outcome, because at high κa , which is the applicability range of the Dukhin–Shilov theory, the contribution to μ^* by the double-layer deformation due to electrophoretic oscillations is much smaller than the contribution following from the deformation due to the applied field E (Appendix B). This has already been demonstrated in a theory dealing with the contribution of electrophoretic oscillations to the conductivity of suspensions [25].

Systematic comparison of the two theories is thwarted by the fact that the O'Brien–White theory has a numerical character; these authors present curves for the real and imaginary part of the dielectric permittivity only for one single large κa value, viz. $\kappa a = 50$ (Figs. 2 and 7 in ref. 24). In any case, the curves computed by our eqn. (57) for $\kappa a = 50$, $\phi_d = 50$ mV, $z = 1$, are very close. Three parameters, namely τ , $\epsilon(0)$ and $\epsilon''_{\max}$ can act as characteristic quantities. For $\kappa a = 50$, these parameters are identical in the two theories. It is important that for fixed a and D for $\kappa a = 50$ and 10, τ is the same in the two approaches and independent of ϕ_d . Also, the frequency at which ϵ'' attains its maximum value does not depend on κa , nor does it on ϕ_d . Inspection of Figs. 1, 2, 7, 8 or O'Brien and White [23] leads to the conclusion that for $\kappa a = 10$ or 50 and various values of ϕ_d the relation

$$\epsilon''_{\max} = 0.28 \epsilon(0) \quad (64)$$

is valid. This equation also follows from the Dukhin–Shilov theory as was demonstrated by Springer [26].

For lower values of κa , but still $\gg 1$, the discrepancies between the two theories grow with respect to $\Delta\epsilon(0)$ but not with the other characteristics.

Summarizing, the work by O'Brien, White and DeLacey [23,24], if made more accessible, would provide a picture over a wide κa -range provided relaxation is determined by the diffuse layer only. However, as the role of tangential flow in the dense layer or hydrodynamically stagnant part of the diffuse double layer is underestimated, this approach is as yet less suitable to exploit dielectric and related studies on colloidal systems to investigate the double-layer structure.

(VII) CONCLUSION

In this paper it was demonstrated that $\Delta\epsilon(\omega)$ in the Schwarz and Schurr theory is overestimated, because these authors neglect the fact that in order to maintain near-electroneutrality of the double layer as a whole at any section between θ and $\theta + \Delta\theta$, the diffuse part of the double layer becomes polarized in a direction opposite to that of the applied field. The associated polarization field acts as a brake on the flow of bound ions and, hence, reduces their contribution to the polarization. Incorporation of the diffuse double-layer contribution into the theory not only leads

to lower values of $\Delta\epsilon(0)$ but also renders the relaxation time dependent on double-layer parameters.

The influence of bound (i.e. Stern layer) ions, if any, on $\Delta\epsilon(\omega)$ is felt only in a restricted area of ϕ_d and κa ; only below a certain critical value of ϕ_d do bound ions contribute; above this value polarization is governed solely by the diffuse part. The conclusions on the dominating role of the diffuse-layer polarization was also independently obtained by DeLacey and White.

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APPENDIX A

From eqn. (39) it follows that

$$\Delta\sigma_{dE}(\theta) = -\Delta\sigma_{bE}(\theta) - \epsilon_0 \left(\epsilon_a \frac{\partial\phi_a}{\partial r} - \epsilon_i \frac{\partial\phi_i}{\partial r} \right) \quad (A1)$$

The second term in the rhs can be estimated by writing the condition for potential continuity upon transfer through the layer of bound ions (henceforth we shall assume that there are no free charge carriers inside the particle, i.e. $K_i = 0$).

$$\phi_i|_{r=a} = \phi_a|_{r=a} + \Delta\phi_{dE}(\theta) \quad (A2)$$

In this expression the potential drop $\phi_i(\theta) - \Delta\phi_{bE}(\theta)$ contribution of the inner part has been neglected because there is no screening of the charge due to bound ions by the particle side of the double layer. In other words, for the purpose of the description of the non-equilibrium situation, variations in ϕ_{bE} can be ignored since $|(\phi_i - \phi_{bE})/\phi_i| \sim \Delta/a \ll 1$.

Furthermore, we can write

$$\phi_i = \frac{\phi(a) + \Delta\phi_{dE}(\theta)}{a} \cdot r \cos \theta \quad (A3)$$

$$\phi_a = -Er \cos \theta + \mu^* \cos \theta / r^2 \quad (A4)$$

From eqn. (A2) it follows that

$$\frac{\partial\phi_i}{\partial r}|_{r=a} = \frac{\phi_a + \Delta\phi_{dE}(\theta)}{a} \quad (A5)$$

and from eqn. (A4),

$$\frac{\partial\phi_a}{\partial r}|_{r=a} = -\frac{2}{a}\phi_a(a) - 3E \cos \theta \quad (A6)$$

Substituting eqns. (A5) and (A6) in eqn. (A1), using eqn. (31), we obtain

$$\Delta\phi_{dE}(\theta) = \frac{\Delta\sigma_{bE}(\theta)}{C} - \frac{1}{\kappa a \cosh(F\phi_d/2RT)} \left\{ \frac{\epsilon_i \phi_a(a)}{\epsilon_a} + 2\phi_a(a) - 3Ea + \frac{\epsilon_i}{\epsilon_a} \Delta\phi_{dE}(\theta) \right\} \quad (A7)$$

The last term on the rhs of this expression may be neglected against the lhs because $\kappa a \gg 1$ and $\epsilon_i < \epsilon_a$ for aqueous systems. For the same reasons it is also allowed to neglect the first term in brackets against the second.

Under condition $2\kappa_b^0 \ll \kappa a$ and very low frequency $\omega \ll a^2/2D_b$, the current in the layer of bound ions is virtually zero (as it is in a constant field), so that for a non-conducting particle

$$\phi_a = -Er \cos \theta - Ea^3 \cos \theta / 2r^2$$

so that

$$\phi_a(a) - 3Ea/2 \quad (A8)$$

At sufficiently high frequencies, if $(2D_b\kappa^2)^{-1} \gg \omega \gg a^2/2D_b$, the diffusional counterflow is irrelevant so that the contribution of the conductivity of bound ions becomes more prominent. In that case

$$\phi_a = -Er \cos \theta + \frac{2k_b^0/a - k_a}{2k_b^0/a + 2k_a} \cdot \frac{a^3}{r^2} E \cos \theta$$

$$\phi_a(a) = -\frac{3Ea}{2 + 2k_b^0/k_a a} \quad (A9)$$

The variation of ϕ_a with ω is given by a Debye-type curve, therefore at $\omega \simeq \tau^{-1}$ one may write

$$\phi_a(a)|_{\omega=\tau^{-1}} \simeq \phi_a(a)|_{\omega=0} - \frac{1}{2} \langle \phi_a(a)|_{\omega=0} - \phi_a(a)|_{\omega=\infty} \rangle \quad (A10)$$

Substitution of eqns. (A8) and (A9) in eqn. (A10) leads to

$$\phi_a(a)|_{\omega=\tau^{-1}} \simeq -\frac{3Ea}{2} \left\{ 1 - \frac{k_b^0/2k_a a}{1 + k_b^0/k_a a} \right\} \quad (A11)$$

Using (A11) it is easy to see that

$$|2\phi_a(a) + 3Ea|_{\omega=\tau^{-1}} \leq |\phi_a(a)|_{\omega=\tau^{-1}} \quad (A12)$$

for all k_b^0 , provided

$$0 < \omega \leq 2D_b/a^2 \quad (A13)$$

Hence, upon substitution of eqn. (A7) in eqn. (28), it is again permissible to neglect $[2\phi_a(a) + 3Ea]/\kappa a \cosh(F\phi_d/2RT)$ against $\phi_a(a)$, so that

$$\Delta\phi_{dE}(\theta) \simeq \Delta\sigma_{bE}(\theta)/C_d \quad (A14)$$

implying that for the screening of the bound charges it is not necessary to account for the (small) deviations from the equilibrium situation.

APPENDIX B

That electrophoretic oscillations do not significantly contribute to $\Delta\epsilon(\omega)$ can also be understood on the basis of the following argument.

Motion of the particle brings about a double-layer deformation, creating an additional dipole moment μ_v^* , proportional to the electrophoretic mobility u , which in turn can be estimated from Smoluchowski's corresponding expression for the sedimentation potential [24]

$$\mu_v^* = \frac{3\epsilon\epsilon_0\zeta a}{2k_a} u = \frac{3}{2} \frac{a(\epsilon\epsilon_0\zeta)^2}{2\eta k_a} E^* \quad (B1)$$

The value of μ_v^* can be estimated by considering its influence on the electric conductivity k_{susp} of suspensions, which is related to $Re\mu_v^*$. In ref. 24 it was proven that if eqn. (13) is valid, μ_v^* is smaller than μ^* , the contribution to μ^* from the direct action of E on the double layer. Next, it was also ascertained that electrophoretic oscillations do not influence k_{susp} . Dielectric phenomena being connected with Im^* , the same reasoning can be applied to assess the influence of μ_v^* . Because of eqn. (2),

$$Im E^*(\theta) \simeq Im(\mu^*/a^3) \quad (B2)$$

Substitution of $Im(E^*)$ in eqn. (B1) instead of E^* allows an estimation of the ratio

$$\frac{Im \mu_v^*}{Im \mu^*} \sim \frac{3a(\epsilon\epsilon_0\zeta)^2}{2\eta k_a a^3} \quad (B3)$$

which, by using $k_a = \epsilon\epsilon_0 D\kappa^2$, $m = \epsilon\epsilon_0(\kappa T)^2/\eta De^2$, can be rewritten

$$\frac{Im \mu_v^*}{Im \mu^*} = \frac{3m}{2(\kappa a)^2} \left(\frac{e\zeta}{kT} \right)^2 \quad (B4)$$

As $m \sim 0.2$, it follows that for $\kappa a \gg 1$ electrophoretic oscillations can be neglected, also for dielectric phenomena, since they are a second-order feature.

REFERENCES

- 1 J.Th.G. Overbeek, *Kolloid Beih.*, 54 (1943) 287.
- 2 J.Th.G. Overbeek, *Adv. Colloid Sci.*, 3 (1950) 97.
- 3 S.S. Dukhin, *Croat. Chem. Acta*, 48 (1976) 574.
- 4 D. Stigter, *J. Phys. Chem.*, 82 (1978) 1424.
- 5 J.Th.G. Overbeek, *J. Colloid Interface Sci.*, 58 (1977) 408.
- 6 J. Lyklema, *Pure Appl. Chem.*, 52 (1980) 1221; J. Lyklema and H.P. van Leeuwen, *Adv. Colloid Interface Sci.*, 16 (1982) 127.
- 7 S.S. Dukhin, *Croat. Chem. Acta*, 52 (1980) 167.
- 8 V.N. Shilov and V.R. Estrela-Lopis, in *Ustoichivost tonkikh plenok*, Nauka, Moscow, 1979, p. 35.
- 9 S.S. Dukhin in B.V. Deryagin (Ed.), *Issledovaniya b oblasti poverkhnostnykh sil*, Academy of Sciences, Moscow, 1961 p. 738 (trans. by Consultants Bureau, N.Y., as *Research in Surface Forces*, Vol. I, 1963, p. 27).

- 10 V.N. Shilov and S.S. Dukhin, *Kolloid Zh.*, 32 (1970) 293.
- 11 S.S. Dukhin in E. Matijević (Ed.), *Surface and Colloid Science Vol. 3*, Wiley-Interscience, New York, p. 293.
- 12 S.S. Dukhin and V.N. Shilov, *Dielectric Phenomena and the Double Layer in Disperse Systems and Polyelectrolytes*, Wiley-Interscience, New York, 1974.
- 13 S.S. Dukhin in B.V. Deryagin (Ed.), *Issledovaniya b oblasti poverkhnostnykh sil*, Academy of Sciences, Moscow, 1967, p. 335 (trans. by Consultants Bureau, NY, as *Research in Surface Forces*, Vol. III, 1971, p. 287).
- 14 V.N. Shilov and S.S. Dukhin, *Kolloid Zh.*, 32 (1970) 117.
- 15 G. Schwarz, *J. Phys. Chem.*, 66 (1962) 2636.
- 16 J.M. Schurr, *J. Phys. Chem.*, 68 (1964) 2407.
- 17 C.T. O'Konski, *J. Phys. Chem.*, 64 (1960) 605.
- 18 J.C. Maxwell, *A Treatise on Electricity and Magnetism*, Clarendon Press, Oxford, 1881.
- 19 O. Stern, *Z. Elektrochem.*, 30 (1924) 508.
- 20 D.C. Henry, *Trans. Faraday Soc.*, 44 (1948) 955.
- 21 C. Ballario, A. Bonincontro and C. Cametti, *J. Colloid Interface Sci.*, 54 (1976) 415.
- 22 C. Ballario, A. Bonincontro and C. Cametti, *J. Colloid Interface Sci.*, 72 (1979) 304.
- 23 R.W. O'Brien and L.R. White, *J. Chem. Soc. Faraday Trans. II*, 74 (1978) 1607.
- 24 E.H.B. DeLacey and L.R. White, *J. Chem. Soc. Faraday Trans. II*, 77 (1981) 2007.
- 25 S.S. Dukhin and B.V. Deryagin in E. Matijević (Ed.), *Surface and Colloid Science*, Vol. 7, Wiley, 1974, Ch. 2, especially eqn. (226) ff.
- 26 M.M. Springer, Thesis, Agricultural University, Wageningen, The Netherlands, 1979.

Induced polarization of shaly sands

H. J. Vinegar* and M. H. Waxman*

ABSTRACT

Equations have been developed that relate induced polarization (IP) in shaly sands to measurable petrophysical parameters. The induced-polarization process has been modeled in terms of two mechanisms: clay counterion displacement and membrane blockage. The resulting equations can be used to determine shaliness, brine conductivity, and oil saturation from in-phase and out-of-phase conductivities. Laboratory measurements have confirmed the IP dependence on these variables, as well as on temperature.

INTRODUCTION

Electrical resistivity logs are the oil industry's basic tool for the in-situ determination of hydrocarbon saturation. For clean sands, Archie's empirical equations (1942) have provided the basis for the fluid saturation calculations. In shaly sands, however, exchange counterions associated with clay minerals increase rock conductivity over that of clean sand, and the Archie relations are no longer valid. For shaly sands containing dispersed clay, the Waxman-Smits equation has proven successful in accounting for the additional clay conductance with measurable petrophysical parameters (Waxman and Smits, 1968; Waxman and Thomas, 1974).

In order to use the Waxman-Smits equations, the shaliness parameter Q_v , defined as the cation exchange capacity of the sandstone per unit pore volume, must be obtained. However, Q_v cannot be determined directly from present wireline logs, although various indirect techniques have been proposed using core-derived empirical relations in conjunction with log responses (Lavers et al., 1975; Juhasz, 1979, 1981). If core samples or drill cuttings are available, Q_v can be measured in the laboratory by (1) a *destructive method* using standard cation exchange capacity, grain density, and porosity determinations (Waxman and Smits, 1968) and (2) a *nondestructive method* using membrane potential measurements (Thomas, 1976).

The Waxman-Smits equations apply only to in-phase conductivity, that is, conduction which occurs in phase with the reference current. Electrical impedance is, however, a complex quantity with a reactive as well as a resistive component. Out-of-phase or quadrature conduction at low frequencies is commonly referred to as induced polarization (IP).

Recently, attention has been focused on induced polarization in shaly sands as a means of determining cation exchange capacity (Snyder et al., 1977; Sumner, 1976; Sill, 1964; Hoyer and Rumble, 1976). Although these workers have observed that IP is proportional to shaliness, what has been lacking is a model which relates IP response in shaly sands to basic petrophysical quantities, such as electrolyte concentration and oil saturation.

We present equations for the out-of-phase conductivity of shaly sands, consistent with the Waxman-Smits equations and using the same petrophysical variables. Unlike in-phase conductivity, the out-of-phase response is a weak function of salinity, predictable from clay counterion and electrolyte ion mobilities. The IP equations can be solved simultaneously with the Waxman-Smits equations for one of the variables which is often unknown, for example, shaliness or electrolyte conductance. Then hydrocarbon saturation can be determined independent of shaliness or the salinity of the formation waters.

The organization of this paper is as follows. The first section reviews parameters used to describe induced polarization. The model for out-of-phase (quadrature) conductivity is presented in the second section. Predictions from the model are compared with laboratory measurements of shaly sand quadrature conductivities as functions of shaliness and sodium chloride electrolyte concentration (the third section), and oil saturation (the final section). Apparatus and experimental procedures are described in Appendices A and B.

INDUCED-POLARIZATION PARAMETERS

For a linear, homogeneous, isotropic and time-invariant medium, the general constitutive equations are

$$\mathbf{J} = C^*(\omega)\mathbf{E} = [C_I(\omega) + iC_Q(\omega)]\mathbf{E},$$

and

$$\mathbf{D} = K^*(\omega)\epsilon_0 \mathbf{E} = [K_I(\omega) + iK_Q(\omega)]\epsilon_0 \mathbf{E},$$

where $\mathbf{J}$ is the current density, $\mathbf{D}$ the displacement field, $\mathbf{E}$ the electric field, C^* the complex conductivity, K^* the complex dielectric constant, and ϵ_0 the permittivity of free space (8.85×10^{-12} F/m). $\{C_I, C_Q\}$ and $\{K_I, K_Q\}$ are the in-phase and quadrature components of C^* and K^* , respectively. For linear, causal systems, $\{C_I, C_Q\}$ and $\{K_I, K_Q\}$ each satisfy the Kramers-Kronig relations, and are in general frequency-dependent.

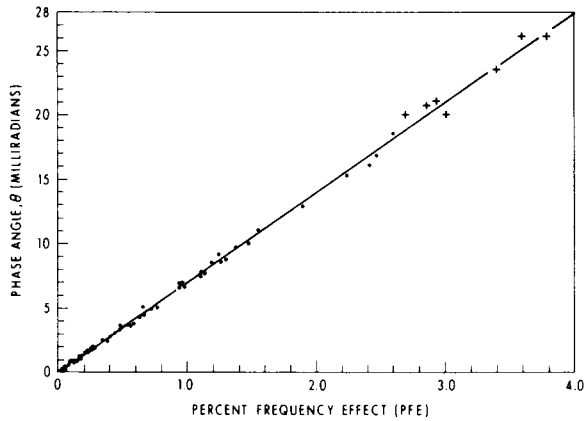


FIG. 1. Relationship between phase angle and percent frequency effect (PFE).

The phase angle θ is defined as the angle between in-phase and quadrature conductivities:

$$\theta(\omega) = \tan^{-1} \frac{C_Q(\omega)}{C_I(\omega)} \cong \frac{C_Q}{C_I} \text{ for small } \theta. \quad (2)$$

When the constitutive relations (1) are incorporated into the Maxwell equation for the curl of the magnetic field $\mathbf{H}$, i.e.,

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (3)$$

and assuming harmonic excitation of the electric field, $\mathbf{E} = \mathbf{E}_0 e^{i\omega t}$, then

$$\nabla \times \mathbf{H} = [C_{\text{eff}} + i\omega\epsilon_0 K_{\text{eff}}]\mathbf{E}, \quad (4)$$

where C_{eff} and K_{eff} are the "effective" conductivity and dielectric constant:

$$C_{\text{eff}}(\omega) = C_I(\omega) - \omega\epsilon_0 K_Q(\omega), \quad (5)$$

$$K_{\text{eff}}(\omega) = \frac{C_Q(\omega)}{\omega\epsilon_0} + K_I(\omega).$$

As indicated by Fuller and Ward (1970), $C_{\text{eff}}(\omega)$ and $K_{\text{eff}}(\omega)$ are the electrical quantities actually measured on rocks.

In the following discussion, the general equations (5) are specialized to the case of brine-saturated rocks with connected pore spaces, measured at low frequencies. Under these conditions, ionic conduction is expected to be the dominant conduction mechanism and dielectric loss is therefore neglected. Following Fuller and Ward, we assume

$$C_I \gg \omega\epsilon_0 K_Q \text{ and } C_{\text{eff}} \cong C_I. \quad (6)$$

Also, if C_Q is nearly independent of frequency, then at low frequencies in shaly sands

$$\frac{C_Q}{\omega\epsilon_0} \gg K_I \text{ and } K_{\text{eff}} \cong \frac{C_Q}{\omega\epsilon_0}. \quad (7)$$

This is the basis for observations by Hoyer and Rumble (1976)

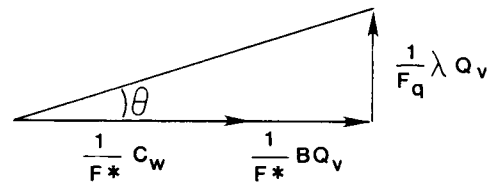
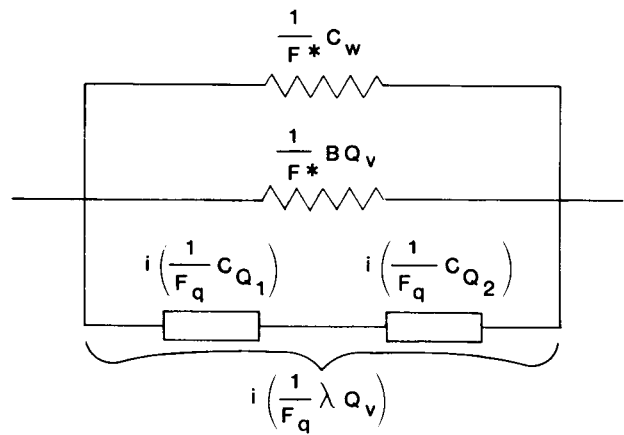


FIG. 2. Complex conductance model for shaly sands.

that K_{eff} increases as $1/\omega$ in shaly sands for frequencies below 10^5 Hz.

The complex resistivity ρ^* is defined as

$$\rho^*(\omega) = \rho_I(\omega) + i\rho_Q(\omega) = 1/C^*(\omega).$$

Frequency effect (FE) is the fractional rate of change of $\rho_I(\omega)$ between two low frequencies, ω and $A\omega$

$$\text{FE} \equiv \frac{\rho_I(\omega) - \rho_I(A\omega)}{\rho_I(\omega)}. \quad (8)$$

For phase angles which are slowly varying functions of frequency, FE is proportional to phase shift. In particular, Shuey and Johnson (1973) and Van Voorhis et al. (1973) have shown that the relationship between FE and θ at the geometric mean frequency $\sqrt{A\omega}$ is

$$\text{FE} = \left(1 - A \frac{2}{\pi} \theta\right) \cong -\frac{2}{\pi} \theta \log_e A, \quad \theta \ll 1. \quad (9)$$

When a frequency difference of one decade is chosen, $A = 10$; the percent frequency effect (PFE) then is

$$\text{PFE} \equiv 100 \times \text{FE} = 0.1466\theta \text{ (mrad)}. \quad (10)$$

Equation (10) may be used to check complex resistivity measurements for self-consistency. In Figure 1 phase shift and PFE are plotted for our twenty sandstone samples at five different salinities, as reported in the third section. The PFE was computed between 10 and 100 Hz, and the phase angle θ was measured near the geometric mean frequency (30 Hz). The crosses in Figure 1 are field data from Van Voorhis et al. (1973); their PFE was taken between 0.1 and 1.0 Hz. A linear regres-

sion of PFE on θ using our sandstone data gives

$$\text{PFE} = [0.144 \pm 0.003]\theta \quad (\text{mrad}),$$

in agreement with equation (10).

Because FE is typically a few percent or less in shaly sands, the frequency variation of $C_I(\omega)$ will not be shown explicitly in the following discussion. Nevertheless, $C_I(\omega)$ satisfies the Kramers-Kronig relations and has the frequency dependence shown in equation (10).

THE MODEL

The complex conductance model can be represented by the equivalent circuit shown in Figure 2. We describe first the in-phase and then the quadrature elements.

In-phase conductivity model

The Waxman-Smits model (1968) consists of two conductive elements in parallel, one consisting of the conductance of free electrolyte contained in the pores and the other the conductance contributed by the exchange cations of the clay surfaces (Figure 2). Waxman and Smits proposed the following equation for the in-phase conductivity of brine-saturated shaly sands:

$$C_I = \frac{1}{F^*} (C_w + BQ_v), \quad (11)$$

where

C_I = specific in-phase conductivity of water-bearing shaly sand (mho-cm^{-1}),¹

F^* = formation resistivity factor defined by Waxman-Smits,

C_w = specific conductivity of the equilibrating electrolyte solution (mho-cm^{-1}),

Q_v = cation exchange capacity of the rock/unit pore volume (meq-ml^{-1}),

and

B = equivalent ionic conductance of clay exchange cations as function of C_w ($\text{mho-cm}^2\text{-meq}^{-1}$).

From their data, Waxman and Smits obtained an empirical expression for B :

$$B = 0.0383[1 - 0.83 \exp(-50C_w)] \quad (\text{mho-cm}^2\text{-meq}^{-1}). \quad (12)$$

The constants in equation (12) are for shaly sand in equilibrium with NaCl solution at 25°C. Equation (12) shows that the equivalent ionic conductance of the Na^+ exchange cation increases with increasing electrolyte conductivity up to a maximum value of $B_{\max} = 0.0383 \text{ mho-cm}^2\text{-meq}^{-1}$. $B_{\max}$, as well as the other parameters in equation (12), depends upon cation type.

For oil-bearing shaly sands, Waxman and Smits proposed the equation

$$C_I = \frac{S_w^{n^*}}{F^*} (C_w + BQ_v/S_w), \quad (13)$$

¹In this paper, C_I and C'_I denote in-phase conductivities for water- and oil-bearing sands. Notations were C_0 and C_i respectively in previous papers (Waxman and Smits, 1968; Waxman and Thomas, 1974).

where

C'_I = specific in-phase conductance of oil-bearing sand (mho-cm^{-1}),

S_w = fractional water saturation,

and

n^* = saturation exponent.

Equation (13) is developed from equation (11) with the additional assumptions that the formation factor for oil-bearing sands is $F^*/S_w^{n^*}$ and the volume concentration of clay exchange cations is $Q'_v = Q_v/S_w$. The validity of equation (13) was demonstrated by Waxman and Thomas (1974).

Quadrature conductivity model

The mechanism proposed here for the quadrature conductivity is similar to that of Madden and Marshall (1959), where the rock pore is modeled as an alternating series of clay-rich and clay-free zones. The clay-rich zones have high concentrations of fixed negative charge sites which enhance cation relative to anion transport. In the steady-state condition after constant current is applied, the inequality of cation and anion flows causes local concentration excesses and deficiencies to form at clay sites, alternating throughout the pore length. When the applied current is terminated, the diffusion voltages resulting from these concentration gradients decay with time as the ions redistribute themselves to their equilibrium positions.

The general solution for the equations of motion of each ion species subject to diffusion and electric forces shows two contributions to the local concentration changes (Madden and Marshall, 1959; Buchheim and Irmer, 1979). The first contribution is a space charge developed at the boundaries of a clay-rich zone, which, at low frequencies, has a characteristic length equal to the Debye-Hückel distance. The second contribution is from ionic diffusion to the clay-rich zones, which act as cation-selective membranes for electrolyte transport.

In this spirit, Figure 2 models the induced concentration gradients as arising from two mechanisms: (1) clay counterion displacement, and (2) electrolyte blockage by the clay-site membranes. Both mechanisms are assumed to be proportional to Q_v , i.e.,

$$\Delta c_1 = A_1 Q_v \quad \text{and} \quad \Delta c_2 = A_2 Q_v, \quad (14)$$

where $\Delta c_{1,2}$ are the concentration excesses, and $A_{1,2}$ are the proportionality constants for each mechanism. We first consider the specific quadrature conductivity in a straight section of pore; the tortuosity and geometry factors are added later through the quadrature formation factor. Multiplying Δc_1 by the equivalent conductance of the clay counterions B gives C_{Q_1} , the specific quadrature conductance of the displaced clay exchange cations as

$$C_{Q_1} = A_1 B Q_v. \quad (15)$$

Similarly, multiplying Δc_2 by the equivalent conductance of electrolyte solution Λ gives C_{Q_2} , the specific quadrature conductance of the blocked electrolyte, as

$$C_{Q_2} = A_2 \Lambda Q_v. \quad (16)$$

In order to account for membrane efficiency on C_{Q_2} , equation (16) is multiplied by the factor $(2t_+ - 1)$, where t_+ is the cation transport number through the clay-rich section, i.e.,

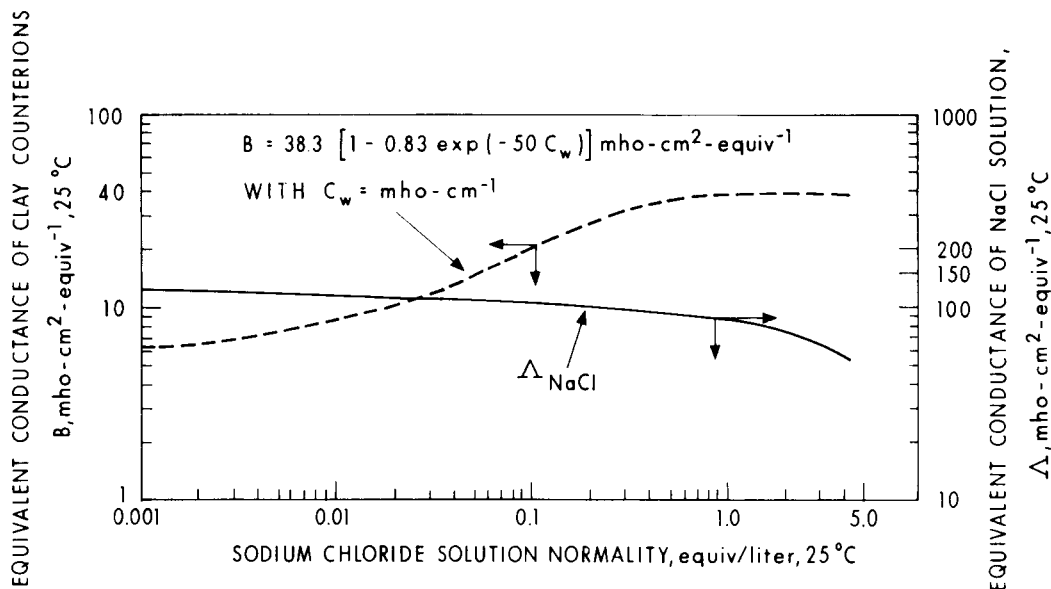


FIG. 3. Equivalent conductances of NaCl solution and sodium exchange cations versus NaCl concentration at 25°C.

$$C_{Q_2} = A_2(2t_+ - 1)\Lambda Q_v. \quad (17)$$

If no transport of anions occurs through the clay-rich zones (perfect membrane behavior), $t_+ = 1$ and equation (17) reduces to equation (16).

In the complex quadrature model in Figure 2, the quadrature conductances C_{Q_1} and C_{Q_2} are modeled in series. Thus

$$\frac{1}{(C_Q)_{\text{series}}} = \frac{1}{C_{Q_1}} + \frac{1}{C_{Q_2}}, \quad (18)$$

or

$$(C_Q)_{\text{series}} = \frac{A_1 A_2 (2t_+ - 1) \Lambda B}{[A_1 B + A_2 (2t_+ - 1) \Lambda]} Q_v. \quad (19)$$

An equivalent quadrature conductance λ is defined by

$$\lambda = \frac{(A_1 B)[A_2 (2t_+ - 1) \Lambda]}{[A_1 B + A_2 (2t_+ - 1) \Lambda]}, \quad (20)$$

or for t_+ equal to unity,

$$\lambda = \frac{A_1 B A_2 \Lambda}{A_1 B + A_2 \Lambda}, \quad (21)$$

so that

$$(C_Q)_{\text{series}} = \lambda Q_v. \quad (22)$$

The behavior of Λ_{NaCl} and B (sodium exchange cation) as a function of sodium chloride solution concentration is shown in Figure 3. Because Λ_{NaCl} and B are weak functions of salt concentration, λ will also have a weak dependence on salinity.

The effects of pore cross-sectional area and tortuosity (effective path length) are now considered. For in-phase conductivity, a reduction in pore cross-sectional area or an increase in tortuosity reduces the mean current flow through the rock. Both effects are usually included in the formation resistivity

factor, which is F for the Archie clean-sand case and F^* in the Waxman-Smiths notation.

Empirical studies by Archie (1942) indicated that the formation resistivity factor may be represented by $F = \phi^{-m}$ where ϕ is the fractional porosity and the cementation factor m is approximately equal to 2 for consolidated rocks. For shaly sands, the analogous Waxman-Smiths notations are F^* and m^* . Since the quadrature current travels along the same tortuous path as the current attributed to the ions in the pore water, C_Q should depend upon the same formation resistivity factor F^* as does the Waxman-Smiths equation. That is, if F_q is the formation factor for quadrature conductivity, then we have

$$F_q = F^*. \quad (23a)$$

However, quadrature conductivity may have additional dependence upon pore size because constriction of pores increases both the membrane blockage effect and displacement of clay counterions. In the next section we show that, although equation (23a) represents a good empirical fit to our data for typical reservoir rocks, a better fit is obtained with

$$F_q = F^* \phi. \quad (23b)$$

This form for the quadrature formation factor is used throughout this paper.

The general equation for the quadrature conductivity of water-saturated shaly sands then becomes

$$C_Q = \frac{1}{F_q} \lambda Q_v. \quad (24)$$

When the sand is oil-bearing, the formation factor F'_q (primes indicate presence of oil) will increase according to Archie's second empirical equation (1942) and we obtain

$$F'_q = F_q S_w^{-n^*}. \quad (25)$$

Waxman and Thomas (1974) showed that in oil-bearing sands

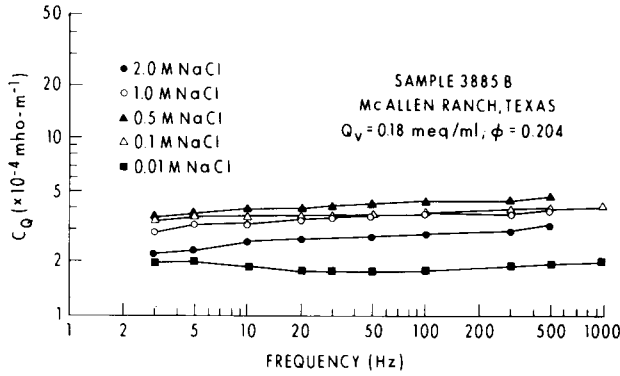


FIG. 4. Frequency independence of quadrature conductivity, sample 3885B.

the shaliness parameter Q'_v is increased by

$$Q'_v = Q_v / S_w \quad (26)$$

Substitution of equations (25) and (26) into equation (24) yields the equation for quadrature conductivity of oil-bearing shaly sands as

$$C'_Q = \frac{1}{F_q} \lambda Q_v S_w^{n^*-1} \quad (27)$$

When equations (2), (11), and (24) are combined, the phase angle in brine-saturated sands is

$$\theta = \frac{C_Q}{C_I} = \frac{\lambda Q_v}{\phi(C_w + BQ_v)} \quad (28)$$

Figure 2 shows the vectorial relations between C_I and C_Q . Similarly, when equations (2), (13), and (27) are combined, the

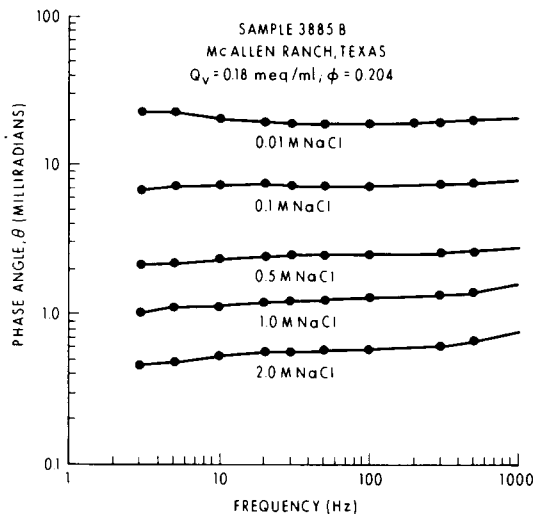


FIG. 5. Frequency independence of phase angle, sample 3885B.

phase angle θ' in oil-bearing shaly sands is

$$\theta' = \frac{C'_Q}{C'_I} = \frac{\lambda Q_v}{\phi S_w \left[C_w + \frac{BQ_v}{S_w} \right]} \quad (29)$$

The ratio of phase angles is then

$$\frac{\theta}{\theta'} = \frac{S_w [C_w + (BQ_v/S_w)]}{(C_w + BQ_v)} \quad (30)$$

Solving equation (30) for S_w leads to

$$S_w = \frac{\theta}{\theta'} \left[1 + \frac{BQ_v}{C_w} \right] - \frac{BQ_v}{C_w} \quad (31)$$

At high salinities and/or in clean sands, i.e., $BQ_v \ll C_w$, equation (31) reduces to

$$S_w = \frac{\theta}{\theta'} \quad (32)$$

Thus, measurements of the phase angles θ and θ' permit the determination of oil/water saturations without knowledge of the Waxman-Smiths saturation exponent n^* , if it is assumed that only S_w varies in traversing from brine-saturated to oil-bearing zones in the same shaly sand formation.

In summary, the model predicts that quadrature conductivity is (1) directly proportional to the shaliness parameter Q_v , (2) weakly dependent on salinity, and (3) reduced in oil-bearing sands by the factor $S_w^{n^*-1}$.

Frequency dependence and limits of applicability

The distance an ion is displaced by the electric field determines the time required for the ion to return to its equilibrium position. Therefore, the spectrum of relaxation distances determines the frequency dependence of the quadrature conductance. The broad distribution of relaxation lengths in naturally occurring sands results in quadrature conductance essentially independent of frequency over a wide range of frequencies.

For all our samples, the quadrature conductivities and phase angles were approximately independent of frequency in the range 3–1000 Hz, as shown in Figures 4 and 5 for a typical sand. Since C_Q is frequency-independent in this range, the effective dielectric constant K_{eff} increases as $1/\omega$ when frequency is lowered [equation (7)]. This was also reported by Hoyer and Rumble (1976), as well as Poley et al. (1978).

Frequency-independent IP behavior cannot be valid over all frequencies, however. The frequency limits f_{min} and f_{max} of the frequency-independent IP spectrum can be estimated from the diffusion constant D of sodium and chloride ions in aqueous solution (Jost, 1952) as

$$D \sim \frac{r^2}{t} = 1.5 \times 10^{-5} \text{ cm}^2/\text{s}, \quad (33)$$

where r is displacement length (cm) and t is relaxation time (s).

For frequencies below f_{min} , the ions have ample time to diffuse across the longest clay-free zones and larger concentration gradients do not develop. A scale for longest relaxation length is set by the distance between adjacent but not connected clay aggregates. If the clay occurs as a uniform grain coating, then this distance should be the largest intergrain

distance, approximately 100 microns. This yields $t \sim 6.7$ s and $f_{\min} = 1/(2\pi t) \sim 0.024$ Hz. For higher clay concentrations or smaller pore sizes, $f_{\min}$ may be higher than this estimate. As discussed later, this partly explains IP behavior in shales.

A general mathematical argument shows that $C_Q(\omega)$ must approach zero at low enough frequency (Fuller and Ward, 1970). Conductance is a physical response function, so the imaginary part of its Fourier transform, $C_Q(\omega)$, is an odd function of ω . $C_Q(\omega)$ must therefore go to zero at $\omega = 0$. Moreover, if the effective dielectric constant in equation (5) is to remain

finite as $\omega \rightarrow 0$, $C_Q(\omega)$ must approach zero at least as fast as $\omega^{1.0}$ at the low-frequency end, as expected for capacity-limited behavior.

Diffusion-limiting occurs at the high-frequency end of the IP spectrum. At frequencies above $f_{\max}$, the ions cannot diffuse fast enough to develop concentration gradients over the shortest relaxation distances. If the microporous regions between clay aggregates are ~ 100 Å, then $f_{\max} \sim 2.4 \times 10^6$ Hz. The shortest possible spatial variation between charged and uncharged zones is the average distance between charge sites on a clay

Table 1. Petrophysical properties of the samples.

Sample	Number	Porosity ϕ (Percent)	Brine Permeability* k (md)	Q_v meq/ml	X-Ray Weight Percent Clay	Clay Type
1. Benton Field, IL	3477	20.1	833	.034	2	Kaolinite with Chlorite
2. North Slope, AK	3336A	20.5	71.7	.042	10	Kaolinite
3. Benton Field, IL	3478	18.1	78.2	.055	6	Kaolinite with Chlorite
4. Berea Outcrop	101	21.0	?	.055	10	Kaolinite with Illite
5. Berea Outcrop	102	20.5	?	.064	10	Kaolinite with Illite
6. Crown Zell, Tuscaloosa	CZ10	23.3	116	.084	~ 5	Chlorite
7. Little Creek Pike County, MI	3833A	26.8	118	.10	22.4	Chlorite
8. Bay Marchand, Blk 2 Offshore, LA	3126B	24.9	468	.12	~ 5	Illite, Montmorillonite, Chlorite, Kaolinite
9. Camp Shelby Outpost, MI	3847A	12.6	.66	.14	~ 10	Illite, Chlorite with Kaolinite
10. Helen Gohlke, Dewitt County, TX	3283A	18.6	40	.16	?	Kaolinite, Chlorite, Illite
11. McAllen Ranch, TX	3885B	20.4	4.95	.18	~ 25	Illite with Chlorite
12. Gulf of Alaska	3972E	18.7	407	.185	5	Chlorite, Montmorillonite with Illite
13. Provident City, TX	3258A	12.8	0.61	.30	~ 18	Illite
14. McAllen Ranch, TX	3891A	19.2	0.26	.385	~ 32	Chlorite with Illite
15. Big Foot, Frio County, TX	3308A	27.0	5.0	.409	~ 35	Montmorillonite, Chlorite with Kaolinite, Illite
16. Big Foot, Frio County, TX	3323F	28.8	3.1	.59	?	Same as 15
17. Big Foot, Frio County, TX	3324A	22.0	1.48	.65	?	Same as 15
18. Big Foot, Frio County, TX	3323E	28.1	1.26	.67	?	Same as 15
19. Big Foot, Frio County, TX	3324B	20.4	1.14	.84	?	Same as 15
20. Big Foot, Frio County, TX	3306F	17.1	.70	.95	?	Same as 15
21. Big Foot, Frio County, TX	3323C	29.6	3.0	.55* .61§	35	Montmorillonite, Chlorite with Kaolinite, Illite

*100 grams NaCl per liter.

#Analytical determination.

§Multiple Salinity Method - Conductivity Plot.

platelet, about 10 Å. For montmorillonite this distance is 8.25 Å and it does not vary by more than a factor of 1.5 between clay mineral species (Van Olphen, 1977). Then $t \sim 6.7 \times 10^{-10}$ s and $f_{\max} \sim 2.4 \times 10^8$ Hz. Clay effects will therefore be a significant component of the effective dielectric constant even at radio frequencies.

Equation (24) indicates that C_Q should continue to increase as the Q_v parameter of the rock increases. Experiments have shown, however, that clay-rich shales do not exhibit induced polarization at low frequencies (Madden and Marshall, 1959; Dakhnov et al., 1952; Dakhnov, 1959). Therefore, the IP model becomes invalid somewhere in the transition from shaly sands to shales. Where this transition zone begins is not known, but it is clear from data presented in the next section that the model applies for typical reservoir sands for $Q_v \leq 1.0$ meq/ml and $\phi \geq 0.12$.

The model does not apply in the case of shales for at least two reasons.

(1) The abundance of clay fixed-charge sites amounts to a clay conduction path which is spatially continuous. Since there are few clay-free pores, concentration gradients do not develop in the presence of the applied electric field.

(2) The relaxation time, which is determined by the length of the clay-free zones, is so short that measurable polarization should exist only at higher frequencies. We have observed a steady increase of polarization in samples of Pierre shale as the frequency was increased to 10 kHz.

IP MEASUREMENTS OF SHALINESS AND ELECTROLYTE CONCENTRATION

This section describes laboratory measurements of IP in brine-saturated shaly sands.

Equations (20) and (21) define the equivalent quadrature conductance λ in terms of the equivalent conductance of sodium chloride solution Λ , the equivalent conductance of clay exchange cations B , and A_1 and A_2 , which are parameters for each sandstone. The term $A_1 B$ arose from displacement of the counterions associated with the clay minerals by the applied current and the term $A_2 \Lambda$ from electrolyte blockage in the clay-rich zones. A_1 and A_2 are determined by measuring quadrature conductivity in typical shaly sands. In particular, we have obtained an average value of λ which applies to all sands (denoted as $\bar{\lambda}$ with parameters $\bar{A}_1$ and $\bar{A}_2$) and which is a good approximation in equations (20) and (21).

Experimental procedures

Complex conductivity measurements were made on twenty sandstone plugs (obtained from eleven formations) at five different salinities: 2.0, 1.0, 0.5, 0.1, and 0.01 M NaCl. Petrophysical and geologic descriptions of the sandstones are in Table 1. The samples varied widely in cation exchange capacity and Q_v ($0.03 \leq Q_v \leq 0.95$ meq/ml), type of clay, and clay distribution in

Table 2.1. In-phase and quadrature conductivities (30 Hz, 25°C) with supporting petrophysical data for shaly sand samples.

Sample	Salinity	C_1 (mho-m ⁻¹)	C_Q (10 ⁻⁴ mho-m ⁻¹)	θ (Milli-radians)	Q_v Waxman-Smiths meq/ml	Q_v Analysis	F^*	$\phi(\%)$ Porosity	$C_Q F^* \phi$ (10 ⁻⁴ mho-m ⁻¹)
3477	2 M	.9624	2.06	.21	.034	.026*	15.43	20.1	6.39
	1 M	.5630	1.50	.27					4.65
	0.5 M	.3144	1.10	.35					3.41
	0.1 M	.0741	.669	.90					2.07
	0.01 M	.0109	.418	3.84					1.30
3336A	2 M	.6281	1.73	.28	.042	.042*	24.10	20.5	8.55
	1 M	.3579	1.17	.33					5.78
	0.5 M	.1985	.930	.47					4.59
	0.1 M	.0526	.806	1.53					3.98
	0.01 M	.00741	.651	8.78					3.22
3478	2 M	.7472	1.71	.23	.055	.052*	20.08	18.1	6.21
	1 M	.4407	1.41	.32					5.12
	0.5 M	.2382	1.26	.53					4.58
	0.1 M	.0640	1.09	1.70					3.96
	0.01 M	.00832	.752	9.03					2.73
101	2 M	.8602	2.14	.25	.068	.055	17.51	21.0	7.87
	1 M	.5119	-	-					-
	0.5 M	.2826	-	-					-
	0.25 M	.1580	-	-					-
	0.1 M	.0713	1.88	2.64					6.91
	0.05 M	.0403	1.76	4.36					6.47
	0.01 M	.0113	1.14	10.1					4.19
102	1 M	.4909	1.29	.26	.070	.064	17.95	20.5	4.75
	0.5 M	.2789	1.57	.56					5.78
	0.25 M	.1530	1.82	1.19					6.70
	0.1 M	.0695	1.85	2.66					6.81
	0.05 M	.0384	1.69	4.39					6.22
	0.01 M	.00845	.833	9.86					3.07

*From adjacent plugs.

the rock. The samples contained only trace amounts of conductive metallic minerals.

Plugs were 1 inch in diameter and between $1\frac{1}{2}$ and 2 inches in length. Each sample was flushed with a 3:1 azeotrope of chloroform/methanol to remove all hydrocarbons, then with methanol to remove all soluble salts.

Standard core analyses of these samples yielded values for bulk volume, grain volume, porosity, grain density, and permeability to air. Values of Q_v were determined by the destructive analytical technique described by Waxman and Smits (1968). Cation exchange capacities were determined by the procedure described by Mortland and Mellor (1954).

Since shaly sands often disintegrate in low-salinity brines, the conductance measurements were started in 2.0 M NaCl and salinity was reduced in sequence. The plugs were saturated under vacuum with 2.0 M NaCl solution for 24 hours. Very shaly sandstones were equilibrated with this solution for several months before conductivities were recorded. Electrolyte concentration was changed by slowly flushing each plug with 2 liters (about 400 pore volumes) of the next, less-concentrated NaCl solution in the series. A conductivity cell was used to check to 1 percent accuracy the salinities of the input and effluent brines. Conductivities for the very shaly Big Foot sandstones were not obtained at 0.01 M NaCl due to sample disaggregation.

After measurements were completed at 0.01 M NaCl, several samples were reequilibrated at higher salinities; typically values of C_I and C_Q were repeatable to 2 percent or better.

In Appendices A and B, experimental design considerations, sample IP cells, electrodes, the frequency-domain bridge circuit, and tests of IP measurement accuracy are described. The accuracy of the measurements is ± 1.0 percent after correction for temperature and cell constants. Measurements were made from 3 to 1000 Hz.

Measurements were carried out at ambient laboratory temperatures, typically $25 \pm 1^\circ\text{C}$. Small temperature corrections to conductivities were made to obtain C_I and C_Q at 25°C , using Arps' formula (1953). Some conductivity measurements were made at temperatures up to 100°C with the entire IP sample cell immersed in a temperature-controlled oil bath.

Presentation of data

In-phase and quadrature conductivities at 25°C and 30 Hz for the twenty brine-saturated sandstones are given in Table 2. Plots of C_I versus the equilibrating brine conductivity C_w were made for each plug; a typical example is in Figure 6. Values of F^* and $(Q_v)_{\text{WS}}$ in Table 2 were computed from the slopes and intercepts, respectively, of the straight-line portions of the C_I versus C_w plots using the Waxman-Smits (WS) multiple-salinity method (1968). A value of B_{max} equal to $38.3 \text{ mho-cm}^2\text{-meq}^{-1}$ was used in computing $(Q_v)_{\text{WS}}$. Also included in Table 2 are the Q_v values determined by the analytical procedures described previously, obtained either by destructive measurement on the particular sample or, where indicated, by measurement on adjacent plugs.

Table 2.2. In-phase and quadrature conductivities (30 Hz, 25°C) with supporting petrophysical data for shaly sand samples.

Sample	Salinity	C_I (mho-m ⁻¹)	C_Q (10 ⁻⁴ mho-m ⁻¹)	θ (Milli-radians)	Q_v Waxman-Smits meq/ml	Q_v Analysis	F^*	$\phi(\%)$ Porosity	$C_Q F^* \phi$ (10 ⁻⁴ mho-m ⁻¹)
C210	2 M	.8110	2.79	.34	.084	.07*	18.83	23.3	12.24
	1 M	.4723	2.13	.45					9.35
	0.5 M	.2689	1.73	.64					7.59
	0.1 M	.0698	1.33	1.90					5.84
	0.01 M	.0138	.935	6.78					4.10
3833A	2 M	1.0106	3.57	.35	.10	.110*	15.38	22.4	12.30
	1 M	.6240	2.23	.36					7.68
	0.5 M	.3246	1.95	.60					6.72
	0.1 M	.0932	1.54	1.65					5.31
	0.01 M	.0144	.997	6.92					3.43
3I26B	2 M	1.1999	5.13	.43	.14	.120	12.82	24.9	16.38
	1 M	.7193	4.76	.66					15.19
	0.5 M	.4094	4.36	1.07					13.92
	0.1 M	.1045	3.89	3.72					12.42
	0.01 M	.0195	2.14	11.0					6.83
3847A	1 M	.1704	1.42	.83	.136	.144	53.48	12.6	9.57
	0.5 M	.0944	1.73	1.81					11.66
	0.25 M	.0545	1.80	3.32					12.13
	0.1 M	.0269	1.77	6.58					11.93
	0.05 M	.0159	1.60	10.05					10.78
3283A	0.01 M	.00525	1.06	20.2					7.14
	2 M	.6906	3.20	.46	.159	.15*	22.57	18.6	13.43
	1 M	.4466	3.43	.77					14.40
	0.5 M	.2376	3.74	1.57					15.70
	0.1 M	.0675	3.45	5.11					14.48
	0.01 M	.0103	1.74	16.9					7.30

*From adjacent plugs.

Table 2.3. In-phase and quadrature conductivities (30 Hz, 25°C) with supporting petrophysical data for shaly sand samples.

Sample	Salinity	C_I (mho-m ⁻¹)	C_Q (10 ⁻⁴ mho-m ⁻¹)	θ (Milli-radians)	$\frac{Q_V}{\text{Waxman-Smits}}$ meq/ml	$\frac{Q_V}{\text{Analysis}}$	F*	ϕ (%) Porosity	$C_Q F^* \phi$ (10 ⁻⁴ mho-m ⁻¹)
3885B	2 M	.4767	2.76	.58	.157	.180	32.36	20.4	18.22
	1 M	.2838	3.59	1.27					23.70
	0.5 M	.1675	4.16	2.48					27.46
	0.1 M	.0492	3.68	7.48					24.29
	0.01 M	.00889	1.76	19.8					11.62
3972E	2 M	.6597	3.56	.54	.11	.185	23.31	18.7	15.52
	1 M	.3866	3.52	.91					15.34
	0.5 M	.2169	3.38	1.56					14.73
	0.1 M	.0564	2.47	4.38					10.77
	0.01 M	.00983	1.28	13.0					5.58
3258A	2 M	.2790	2.47	.89	.30	.372*	58.82	12.8	18.60
	1 M	.1685	3.69	2.19					27.78
	0.5 M	.1016	4.17	4.10					31.40
	0.1 M	.0364	3.50	9.61					26.35
	0.01 M	.0103	1.58	15.3					11.90
3891A	2 M	.3668	2.74	.75	.40	.385	43.67	19.2	22.97
	1 M	.2354	4.36	1.85					36.56
	0.5 M	.1442	5.33	3.70					44.69
	0.1 M	.0547	4.64	8.48					38.90
	0.01 M	.0240	2.19	9.12					18.36
3308A	2 M	1.0768	8.15	.76	.384	.409	15.24	27.0	33.54
	1 M	.6587	10.8	1.64					44.44
	0.5 M	.4002	12.4	3.10					51.02
	0.1 M	.1446	11.2	7.75					46.09
	0.01 M	.0443	4.90	11.1					20.16
3323F	2 M	1.0969	7.52	.69	.59	.69*	15.67	28.8	33.94
	1 M	.6920	12.0	1.74					54.16
	0.5 M	.4404	14.6	3.32					65.89
	0.1 M	.1789	12.4	6.91					55.96

*From adjacent plugs.

Table 2.4. In-phase and quadrature conductivities (30 Hz, 25°C) with supporting petrophysical data for shaly sand samples.

Sample	Salinity	C_I (mho-m ⁻¹)	C_Q (10 ⁻⁴ mho-m ⁻¹)	θ (Milli-radians)	$\frac{Q_V}{\text{Waxman-Smits}}$ meq/ml	$\frac{Q_V}{\text{Analysis}}$	F*	ϕ (%) Porosity	$C_Q F^* \phi$ (10 ⁻⁴ mho-m ⁻¹)
3324A	2 M	.5647	6.25	1.11	.80	.646	32.26	22.0	44.36
	1 M	.3808	9.53	2.50					67.64
	0.5 M	.2547	11.5	4.53					81.62
	0.1 M	.1268	9.87	7.79					70.05
3323E	2 M	1.0534	9.69	.92	.67	.69*	16.56	28.1	45.09
	1 M	.6728	13.5	2.00					62.82
	0.5 M	.4376	15.5	3.53					72.13
	0.1 M	.1807	12.1	6.69					56.31
3324B	2 M	.5699	7.34	1.29	.94	.842	32.47	20.4	48.62
	1 M	.3776	10.6	2.81					70.21
	0.5 M	.2560	12.8	5.00					84.79
	0.1 M	.1247	10.7	8.60					70.88
3306F	2 M	.3898	6.86	1.76	.95	.95	47.17	17.1	55.33
	1 M	.2681	10.9	4.07					87.92
	0.5 M	.1781	13.0	7.30					104.86

*From adjacent plugs.

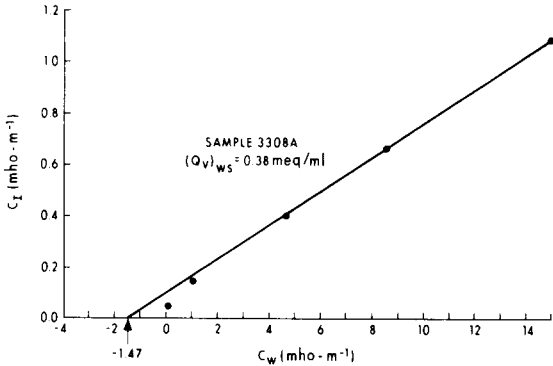


FIG. 6. In-phase sand conductivity versus NaCl conductivity, sample 3308A.

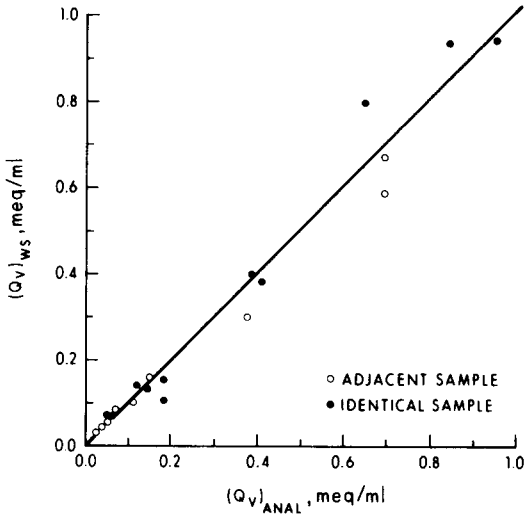


FIG. 7. Regression of Q_v from Waxman-Smiths equation on Q_v from exchange capacity measurements.

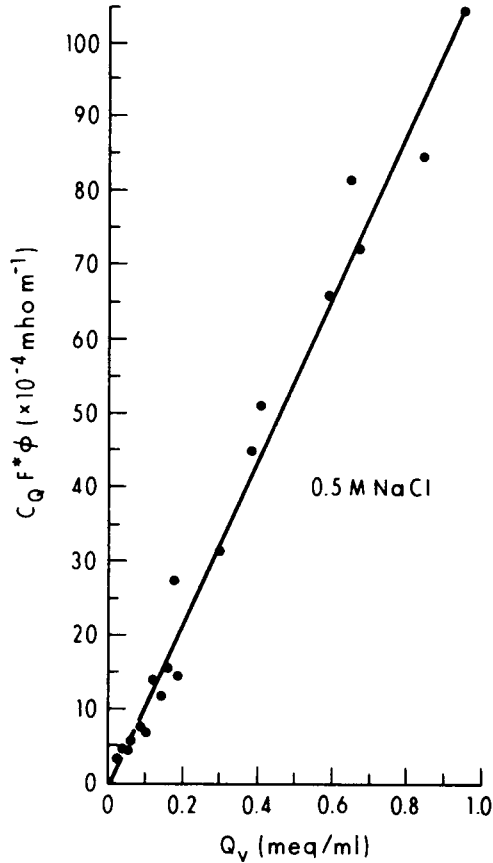


FIG. 8. Regression of $C_Q F^* \phi$ on Q_v at 0.5 M NaCl.

Figure 7 plots $(Q_v)_{ws}$ versus $(Q_v)_{anal}$. There appears to be no systematic difference between analytical determinations of Q_v on the same or from adjacent samples. A linear regression of the form

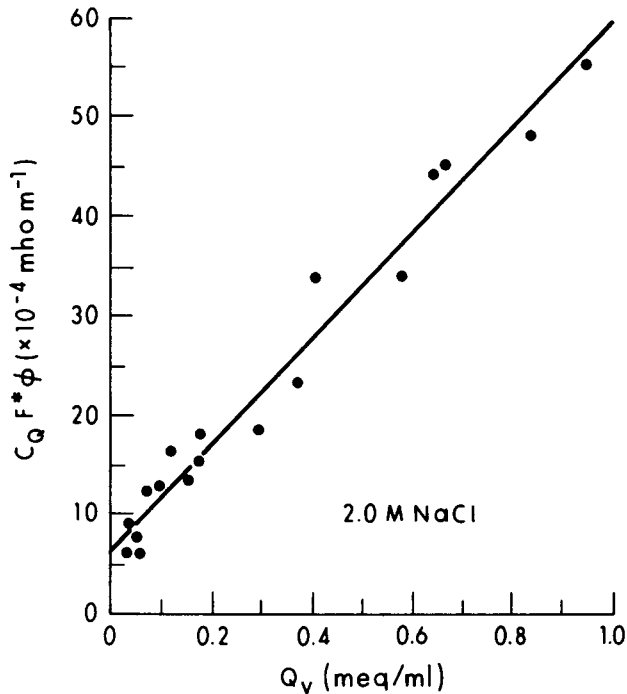
$$(Q_v)_{ws} = a(Q_v)_{anal} + b$$

yields values of $a = 1.031 \pm 0.068$ and $b = -0.009 \pm 0.021$ meq/ml with a correlation coefficient of 0.985.

To test the validity of equation (24), $C_Q F_q = C_Q F^* \phi$ versus Q_v was plotted for each rock at each of five salinities. Figures 8

Table 3. Regression analysis of the quadrature conductivity data. $C_Q F^* \phi = \bar{\lambda} Q_v + b$.

Salinity	Number of Samples	Correlation Coefficient, r	$\bar{\lambda}$ (10^{-4} mho-m $^{-1}$ -meq $^{-1}$ -ml)	b (10^{-4} mho-m $^{-1}$)
2.0 M	18	.983	53.03 ± 4.28	6.20 ± 1.40
1.0 M	19	.990	91.17 ± 5.33	1.25 ± 1.70
0.5 M	19	.991	111.79 ± 6.35	$-.72 \pm 2.04$
0.1 M	19	.980	92.79 ± 7.85	$.18 \pm 2.40$
0.01 M	15	.964	45.51 ± 6.13	$.37 \pm 0.93$

FIG. 9. Regression of $C_Q F^* \phi$ on Q_v at 2.0 M NaCl.

and 9 are examples of these plots. For these plots (Q_v)_{anal} was used for the particular sample when available; otherwise (Q_v)_{ws} was substituted. A linear regression, $C_Q F^* \phi = aQ_v + b$, was performed on the data sets for each of the five salt concentrations. The slopes and intercepts of the regression lines are given in Table 3, as well as the correlation coefficients and the 95 percent confidence belts for the regression coefficients.

If the slope of each regression line is interpreted as $\bar{\lambda}$, then it is clear that equations (23b) and (24) accurately describe the data at all salinities except 2.0 M NaCl. The reasons for the positive intercept with 2.0 M NaCl will be discussed in the next subsection.

In order to test the consistency of Q_v values obtained from Waxman-Smiths conductivity data and those obtained from quadrature conductivity, we performed an orthonormal regression of BQ_v from

$$BQ_v = F^*C_l - C_w \quad (34)$$

against $\bar{\lambda}Q_v$, from

$$\bar{\lambda}Q_v = C_Q F^* \phi. \quad (35)$$

Results are tabulated in Table 4; the correlation coefficient is 0.96 or greater for each salinity except 0.01 M NaCl.

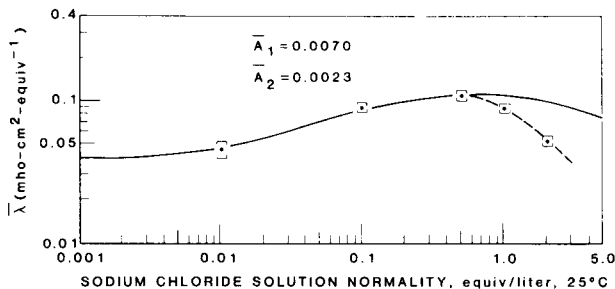
Table 4 also includes values of $\bar{\lambda}/B_{\text{calc}}$ for comparison, where the $\bar{\lambda}$ values were obtained from the slopes given in Table 3 and the B_{calc} values were calculated from equation (12).

Discussion of data

A plot of $\bar{\lambda}$ versus sodium chloride solution normality (taken from Table 3) is given in Figure 10. It is evident that equations (23b) and (24) are a good description of the behavior of quadrature conductivity over a wide range of salinity (0.01 M–1.0 M NaCl). As expected, $\bar{\lambda}$ is only a weak function of salinity.

In order to determine values of the parameters $\bar{A}_1$ and $\bar{A}_2$ in equation (21), the best fit for $\bar{\lambda}$ was obtained using only salinities ≤ 0.5 M NaCl. The solid curve in Figure 10 is obtained with parameters $\bar{A}_1 = 0.0070$ and $\bar{A}_2 = 0.0023$. From these values of $\bar{A}_1$ and $\bar{A}_2$, it appears that the contribution to quadrature conductivity from displacement of clay exchange cations is of the same order of magnitude as that due to displacement of electrolyte solute by membrane blockage.

Although equation (21) can be well fit in the low-salinity range, it deviates from experimental results at concentrations greater than 1.0 M NaCl. The reason is that equation (21)

FIG. 10. Equivalent quadrature conductance $\bar{\lambda}$ versus NaCl solution normality at 25°C.Table 4. Orthonormal regression, $\bar{\lambda}Q_v = a(BQ_v) + b$.

Salinity	Correlation Coefficient, r	$a(x 10^{-4})^*$	$b(x 10^{-4} \text{ mho-m}^{-1})$	$\bar{\lambda}/B_{\text{calc}}^\dagger$
2.0 M	.969	12.887 ± 1.942	7.381 ± 2.401	13.9 ± 1.1
1.0 M	.956	20.632 ± 3.554	1.022 ± 4.965	24.1 ± 1.4
0.5 M	.983	25.929 ± 2.797	1.440 ± 3.549	31.7 ± 1.8
0.1 M	.970	31.754 ± 4.858	-0.135 ± 3.012	47.2 ± 4.0
0.01 M	.817	61.146 ± 29.450	-1.072 ± 2.945	54.6 ± 7.4

* $a = \Delta(\bar{\lambda}Q_v)/\Delta(BQ_v)$

$^\dagger \bar{\lambda}$ from slopes, Table 3.

B_{calc} from equation (12).

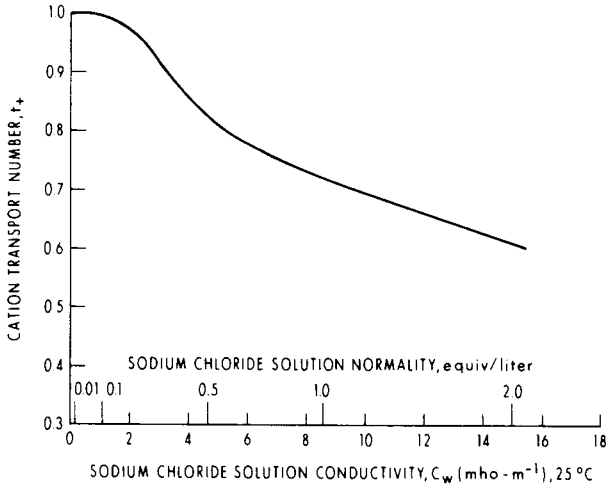


FIG. 11. Variation of cation transport number t_+ with electrolyte conductivity.

assumes a cation transport number t_+ equal to unity through the clay-rich zone, i.e., no anion leak occurs through this zone. In fact, the diffuse portion of the electrical double layer responsible for restrictive anion transport is increasingly compressed at higher salinities (Grahame, 1947); hence the magnitude of the membrane blockage effect and of $\bar{A}_2$ is reduced. At high-salinity levels, the anion leak through pores in clay-rich zones must increase with increasing salinity because the ability of these zones to restrict chloride ion transport is progressively swamped out by higher interstitial chloride ion concentrations. The more general equation (20) must then be used to allow for cation transport numbers less than unity. One approach is to set t_+ equal to unity at low salinities (0.01–0.1 M NaCl) and to determine empirically its dependence at higher salinities. This results in values of the 25°C parameters $\bar{A}_1 = 0.00604$ and $\bar{A}_2 = 0.00397$ and the dependence of t_+ on salinity shown in Figure 11. The decrease of t_+ at high salinities is qualitatively consistent with other laboratory measurements of chloride and sodium ion transport in cation exchange membranes, shaly sands, and shales (e.g., Smits, 1969). The preceding treatment of quadrature conductivity data yields average values for $\bar{\lambda}$ and the $\bar{A}_1$, $\bar{A}_2$, and t_+ parameters.

Alternatively, these data can be treated—albeit less conveniently—as dichotomous, depending upon the relative importance of the clay-exchange conductance and membrane-blockage polarization mechanisms.

The quadrature conductivity data in Table 2 indicate that, for “clean” sandstones with $Q_v \leq 0.1$ meq/ml, C_Q ’s and hence individual λ s continue to increase with increasing NaCl concentration from 0.5 to 2.0 M. By contrast, for the shalier samples with $Q_v \geq 0.15$ meq/ml, C_Q values increase to maxima at concentrations of ~ 0.4 –0.5 M NaCl, but then decrease as concentrations increase to 2.0 M NaCl.

These results may be interpreted as follows. For the low Q_v samples, the membrane blockage contribution to the polarization is negligible and the clay counterion conductance term dominates, i.e., $\lambda \approx A_1 B$. For concentrations greater than 0.4 M NaCl, the clay counterion equivalent conductivity B is constant. However, with increased clay contents and higher Q_v ’s, the membrane blockage effect becomes significant and λ is represented by the formalism of equation (20).

This interpretation is supported by the regression analysis cited in Table 3. At 2.0 M NaCl (Figure 9), the linear regression $(C_Q F^* \phi) = a Q_v + b$ results in a nonzero and positive b intercept. To a large extent, this is caused by reduction in membrane blockage effect in the shalier samples at high salinities.

In addition, part of the positive intercept of $C_Q F^* \phi$ versus Q_v at 2.0 M NaCl (Figure 9) probably is due to residual sample polarization of about 0.1 mrad from sources other than clays. This residual polarization is absent in samples of silicon carbide and clay-free limestone and dolomite. This allows us to exclude both inductive coupling and polarization of electrodes and sample holders as sources of residual polarization (refer to Appendices A and B). These background effects are due to specific material in the sandstones themselves, a response from either quartz sand or from trace amounts of conductive metallic minerals. An extrapolation of laboratory measurements by Collett (1959) yields an estimated phase angle of 0.15 mrad for artificial sand samples containing 0.1 percent by volume pyrite.

Metallic minerals in rocks, particularly metallic sulfides such as pyrite, chalcopyrite, and galena, generate a large IP response caused by oxidation-reduction reactions at the metal surfaces activated by applied current (Wait, 1959).

Since the presence of pyrite is a source of noise for IP applications in shaly sands, we surveyed the frequency and amounts of pyrite in typical sandstone reservoirs. Of 50 sandstones described in Shell data, pyrite was reported in only eight samples and, of those, only three contained more than 0.5

Table 5. Regression analysis. $C_Q F^* = \bar{\lambda}_a Q_v + b$.

Salinity	Correlation Coefficient, r	$\bar{\lambda}_a (x 10^{-4} \text{ mho-m}^{-1}\text{-meq}^{-1}\text{-ml})$	$b (x 10^{-4} \text{ mho-m}^{-1})$
2.0 M	.951	254.98 ± 36.38	28.54 ± 11.86
1.0 M	.952	436.97 ± 58.77	7.12 ± 18.62
0.5 M	.954	536.19 ± 70.37	-2.17 ± 22.30
0.1 M	.941	390.59 ± 59.15	11.96 ± 15.98
0.01 M	.917	225.12 ± 48.25	3.81 ± 7.34

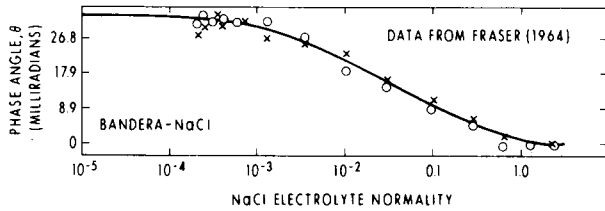


FIG. 12. IP model fit to data of Fraser (1964) for two Bandera sandstone samples.

percent by bulk volume. However, the presence of redox reactions at metal/water interfaces can be differentiated from clay ionic mechanisms by examination of harmonic content (Olhoeft and Scott, 1980; Anderson, 1981; Klein and Sill, 1982). Pyrite concentrations greater than a few percent can also be detected using the Litholog[®] (photoelectric bulk density tool) because of the high photoelectric cross-section of FeS₂ (Clavier et al., 1976).

Equation (24) represents a good fit to the data (for sands with porosity range of about 12–24 percent) when $F_q = F^* \phi$ [equation (23b)] is used. However, it is still possible that the formation factor for quadrature conductivity F_q is the same as for in-phase conductivity F^* , i.e.,

$$C_Q = \frac{1}{F^*} \lambda_a Q_v. \quad (36)$$

The results of the linear regression $C_Q F^* = a Q_v + b$ are given in Table 5. Although there is more scatter in the $C_Q F^*$ versus Q_v plots than when $F_q = F^* \phi$ is used, nevertheless this represents a good fit for a petrophysical data set. The salinity dependence of λ_a is nearly identical to that of $\tilde{\lambda}$; in fact, $\lambda_a = 4.71 \tilde{\lambda}$.

Low electrolyte concentrations

In the limit of low electrolyte concentration ($C_w \ll B Q_v$), $C_w \rightarrow 0$ and θ and θ' in equations (28) and (29) approach

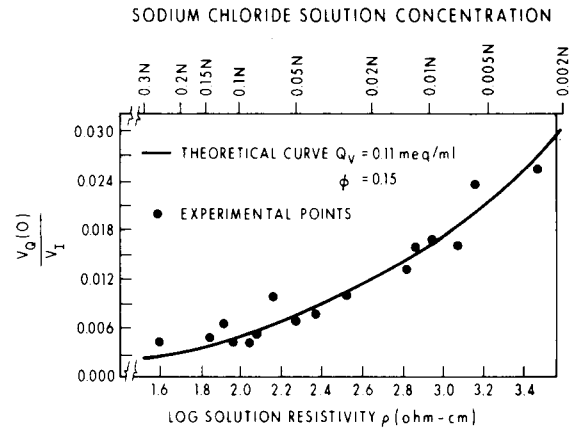


FIG. 13. IP model fit to data of Henkel and Collins (1961).

limiting values

$$\theta, \theta' \rightarrow \frac{(\tilde{\lambda})_{C_w=0}}{\phi B_{C_w=0}}, \quad (37)$$

where $(\tilde{\lambda})_{C_w=0}$ and $B_{C_w=0}$ are the values of $\tilde{\lambda}$ and B for shaly sand in equilibrium with distilled water. The phase angle in equation (37) is independent of both Q_v and S_w in this limit. From Figures 3 and 10 the values of $B_{C_w=0}$ and $(\tilde{\lambda})_{C_w=0}$ are 6.51 and 0.0395 mho-cm²-equiv⁻¹, respectively. When these values are substituted, equation (37) predicts that in all shaly sands the phase angle will approach the universal value in fresh water of

$$\theta_{\max} = \theta'_{\max} = \frac{0.0061}{\phi}. \quad (38)$$

This is about 30 mrad for a rock of 20 percent porosity. (This analysis assumes that $F_q = F^* \phi$.)

Although the lowest salinity used in our study was 0.01 N NaCl, Fraser (1964) reported IP measurements in sandstone cores where salinity was varied from 1.0 N to 10⁻⁴ N NaCl. Figure 12 reproduces his data for two samples of Bandera sandstone. The solid-line curve is our fit of equation (28) to his data using $Q_v = 0.35$ meq/ml and $\phi = 0.19$, reasonable values for Bandera sandstone. All of Fraser's samples (Berea, Boise, and Bandera sandstones) reached maximum polarizations of 27–30 mrad of phase at 10⁻⁴ N NaCl. This agrees with the prediction of equation (38) in the limit of distilled water as the pore electrolyte.

Henkel and Collins (1961) also published IP data extending to very low electrolyte concentrations. Figure 13 shows their data for the chargeability of a sandstone core as a function of salinity; the solid-line curve is a fit of equation (28) where $Q_v = 0.11$ meq/ml and $\phi = 0.15$ were assumed.

Temperature dependence

The temperature dependence of in-phase and quadrature conductivities of two sandstones was measured between -5° and 95°C (Table 6). These samples, Berea V107 and Big Foot 3306F, represent the extremes of a "clean" and a very shaly

Table 6. Variation of conductance with temperature.

T (°C)	C _I (mho-m ⁻¹)	C _Q (10 ⁻⁴ mho-m ⁻¹)	C _Q (T)/C _Q (25°C)
BEREA V107; 0.01 N NaCl, 25°C Q _v = 0.06 meq/ml; φ = 0.169			
25	.0207	1.83	1.00
50	.0320	3.02	1.65
75	.0424	4.26	2.33
100	.0531	5.60	3.06
BIG FOOT 3306F; 0.46 N NaCl (from C _I versus C _w data), 25°C Q _v = 0.95 meq/ml; φ = 0.171			
25	.170	15.0	1.00
50	.304	26.1	1.74
75	.469	39.0	2.59
100	.585	47.8	3.18

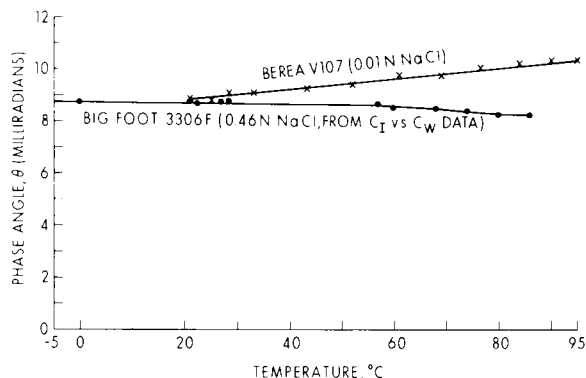


FIG. 14. Variation of phase angle with temperature.

sandstone. The Berea sand was equilibrated with 0.01 N NaCl at 25°C; the Big Foot sand was equilibrated with 0.46 N NaCl at 25°C, as estimated from C_I versus C_w data. For these samples, as shown in Figure 21, the phase angle is practically independent of temperature. The temperature independence of the phase angle in shaly sands has also been reported by Kern et al. (1977) and by Keevil and Ward (1962).

In fact, a small increase in the phase angle was observed for the Berea sample, from 8.84 to 10.55 mrad, as temperature increased from 25° to 100°C. Conversely, over the same temperature interval, the phase angle for the shaly Big Foot sample was reduced from 8.82 to 8.17 mrad.

These trends are explained in terms of the two polarization mechanisms: clay exchange-cation displacement and membrane blockage of electrolyte anions. As described previously, clay cation displacement is the dominant polarization effect for "clean" sandstones. If $C_w \gg BQ_v$ for the low- Q_v rock, the phase angle for a given sample is essentially proportional to the ratio B/Λ [equations (15) and (28)]. Conversely, for an extremely shaly, high- Q_v rock, the membrane blockage effect dominates with $BQ_v \geq C_w$. The phase angle for this rock will approach proportionally to Λ/B [equations (16) and (28)]. Waxman and Thomas (1974) observed that the equivalent conductance of the clay exchange cations B increases more rapidly with increasing temperature than the equivalent conductivity of the electrolyte Λ . Therefore, with increasing temperature, some increase or decrease in phase angle is expected for the limiting cases of clean or shaly sands, respectively.

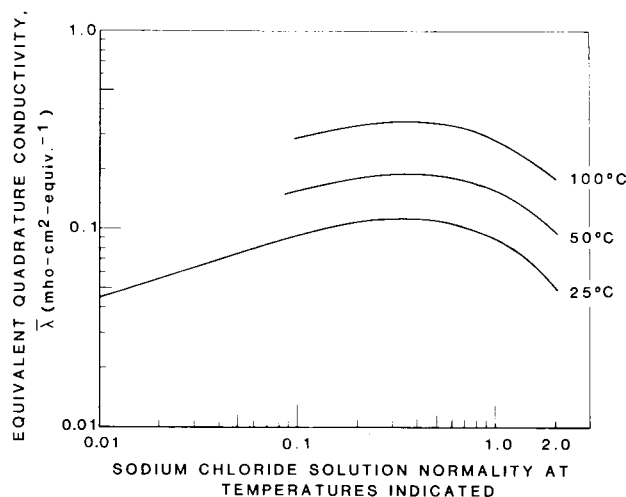
Values of $\bar{\lambda}$ as a function of sodium chloride concentration at 50° and 100°C are given in Figure 15. The calculation, based on the limited data of Table 6, is described in Appendix C.

IP MEASUREMENTS OF OIL SATURATION

For oil-bearing shaly sands ($S_w < 1$), equation (27) can be written as

$$C_Q = \frac{S_w^{n^*}}{F^* \phi} \bar{\lambda} \frac{Q_v}{S_w} = \frac{S_w^p}{F^* \phi} \bar{\lambda} Q_v, \quad (39)$$

where n^* is the saturation exponent in the Waxman-Smiths equation (13) and $p = n^* - 1$ is the quadrature saturation exponent. Note that, because $n^* \approx 2$ in most instances, the model

FIG. 15. Variation of $\bar{\lambda}$ with temperature.

predicts that quadrature conductivities in oil-bearing sands are reduced by the factor S_w from their values at 100 percent water-saturated conditions.

The dependence of C_Q on oil saturation as described by equation (39) is supported by two previous studies of IP in shaly sands. Sill (1964) measured the effects of oil saturation on IP in three cores having $Q_v \leq 0.7$ meq/ml. He concluded that the increase in percent frequency effect was proportional to the increase in "effective b -factor," i.e., the modified b -factor of Hill and Milburn 1956 in partially oil-saturated sands, obtained by replacing b with b/S_w . Since the Hill-Milburn b -factor in water-saturated sands is approximately equal to $-Q_v/7$, this is a verification of the assumption described by equation (26) and incorporated in equations (29) and (39).

More recently, Hoyer and Rumble (1976) and Kern et al. (1977) reported that the low-frequency effective dielectric constant K_{eff} varied with water saturation as

$$K_{eff} = K_{100} S_w^p, \quad (40)$$

where K_{100} is the effective dielectric constant at 100 percent water saturation and $p = 1.00 \pm 0.20$. Data for three cores were presented where the range of water saturation was $0.3 \leq S_w \leq 1.0$. Since the low-frequency dielectric constant is related to C_Q by $K_{eff} = C_Q/\omega\epsilon_0$ [equation (7)], these data also confirm the quadrature saturation dependence in equation (39), based on the assumption that $n^* \approx 2$.

While the results of both Hoyer and Sill are consistent with equation (39), neither experiment was conducted at capillary equilibrium oil saturation. In addition, in-phase conductivity data were not reported, so the saturation exponents n^* are not available to verify the relation

$$p = n^* - 1. \quad (41)$$

The present study was undertaken to examine the dependence of C_Q on oil saturation at capillary equilibrium conditions. Plugs from four formations with a wide range of shaliness were used to measure in-phase and quadrature conductivities, so that the saturation exponents n^* and p could be determined independently.

Table 7. Petrophysical properties of samples.

Sample No.	Well	Age	Formation	Depth Ft.	Porosity	Grain Density, g/cc	Q_v meq/ml
V104	-	-	Berea Outcrop	-	0.188	2.659	0.06
V106	-	-	Berea Outcrop	-	0.188	2.659	0.066
V110	-	-	Berea Outcrop	-	0.189	2.662	0.06
V112	-	-	Berea Outcrop	-	0.188	2.659	0.06
1883	Shell Plow Realty, Texas	Lower Eocene	Lower Wilcox	~10,825	0.133	2.646	0.120
3371	Middle Ground Shoal, Alaska	Oligocene	Kenai	~8,259	0.142	2.645	0.321
3372C	Middle Ground Shoal, Alaska	Oligocene	Kenai	~8,259	0.127	2.637	0.349
3374	Middle Ground Shoal, Alaska	Oligocene	Kenai	~8,259	0.146	2.649	0.236
3307C	Unit No. 169, Frio County, Texas	Cretaceous	Big Foot	~3,580	0.118	2.698	1.25

Experimental procedures and data presentation

Petrophysical and geologic descriptions of the various sandstones are given in Table 7.

The measurement cells and experimental procedures related to the establishment of various oil saturations in the sand plugs are described in Appendices A and B.

A sample was assumed to be in capillary equilibrium at a given oil saturation when the electrical resistivity and the oil volume introduced into the plug were constant within experi-

mental error over a 24-hour period. The measurement of each saturation curve required 4–8 weeks. Only limited data ($S_w \cong 0.85$ to 0.90) could be obtained for two of the more shaly samples (3374, 3307C) due to the high oil pressures required to attain lower water saturations and the restriction of the oil displacement pressure (<20 psi) of the capillary diaphragms. After other measurements were completed, Q_v determinations were made on crushed samples as described previously. Oil saturations were determined from burette readings and from petrophysical determinations of pore volumes; the estimated error is $\Delta S_w/S_w = \pm 2$ percent.

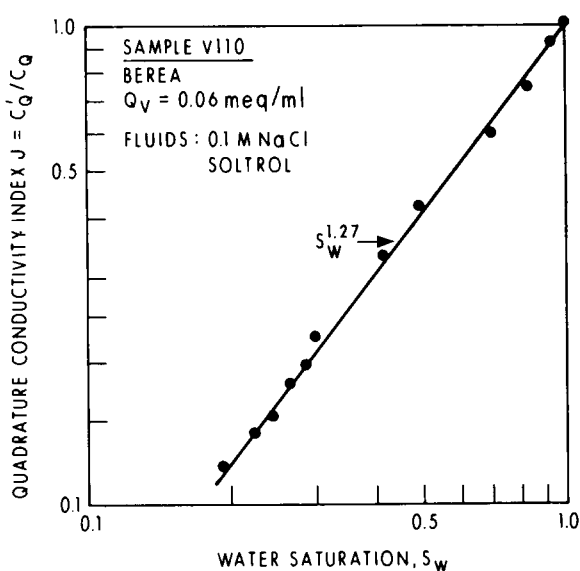


FIG. 16. Quadrature conductivity index versus water saturation, sample V110.

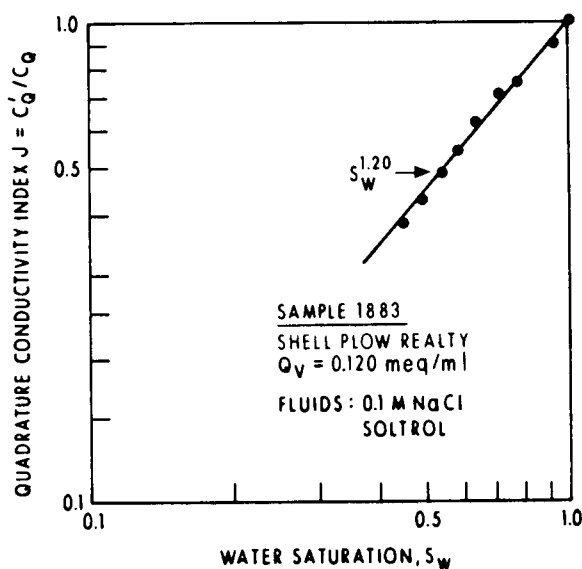


FIG. 17. Quadrature conductivity index versus water saturation, sample 1883.

Table 8 includes the brine salinity and type of oil used for each plug. Most of the samples were saturated with 0.1 N NaCl solution, except Berea V112 (0.01 N NaCl) and Big Foot 3307C (1.0 N NaCl). Soltrol was used as the nonwetting phase except in Berea samples V104 and V106, where El Dorado and Ventura crude oils were substituted.

Discussion of data

By analogy with the resistivity index I ($I = R_i/R_o = C_i/C'_i$), we define the quadrature conductivity index J as the ratio of C'_Q for $S_w < 1$ to C_Q for $S_w = 1$. Complex conductivity data taken at 30 Hz are reported in Table 9 for fully brine-saturated ($S_w = 1$) sands as well as the values of I and J as a function of S_w . Figures 16 and 17 are typical examples of J versus S_w plotted on log-log coordinates. The solid lines are the least-squares regressions of the function $J = aS_w^p$ on S_w .

Table 8 lists the number of data points for each sample, the coefficients a and quadrature saturation exponents p , correlation coefficients r , and standard errors of estimate for a and p . For the sample set, $p = 1.11 \pm 0.17$.

The consistency between $p \equiv n^* - 1$ predicted by equation (39) and the in-phase saturation exponent n^* in the Waxman-Smiths equation was checked by computing n^* from the in-phase conductivity data. The known values of C_w and Q_v were used to solve the Waxman-Smiths equation for I_{calc} at each equilibrium saturation. The value of n^* was determined by minimizing the sum of the squares of the deviations of I_{calc} from I_{meas} .

Table 10 compares n^* computed in this way with $(p + 1)$ obtained from the quadrature conductivity data. For the sample set, the mean value of $[n^* - (p + 1)] = -0.047$ with a standard deviation of ± 0.204 .

These results indicate that the saturation dependence $S_w^{(n^*-1)} = S_w^p$ predicted for C'_Q in equation (39) is in good agreement with the Waxman-Smiths saturation exponent n^* . The mean value of p for our data set agrees with that reported by Hoyer and coworkers under nonequilibrium conditions. Within experimental error, the quadrature saturation exponents show no dependence on salinity or type of oil.

CONCLUSIONS

A model and equations have been developed to account for the effects of shaliness, brine conductivity, and oil saturation on the reactive part of the electrical conductivity of shaly sands. These equations use the same petrophysical variables as the Waxman-Smiths in-phase conductivity equation.

Several aspects of the theory have been verified experimentally. For water-bearing sands, the quadrature conductivity C_Q is directly proportional to the shaliness parameter Q_v and only weakly dependent upon electrolyte concentration. Empirical values of $\bar{\lambda}$, the equivalent quadrature conductivity, were determined as a function of sodium chloride concentration.

Both quadrature conductivities and phase angles are practically independent of frequency. Phase angles are approximately independent of temperature.

Table 8. Quadrature saturation exponents and associated error analysis.

Sample	Salinity	Oil Type	No. of Data Points	Correlation Coefficient, r	p	Standard Error of Estimate, S_p	a	Standard Error of Estimate, S_a
V104 Berea	0.1 N	Eldorado Crude	6	0.99	1.180	0.0237	0.994	0.016
V106 Berea	0.1 N	Ventura Crude	8	0.99	1.284	0.0439	0.980	0.013
V110 Berea	0.1 N	Soltrol	14	0.99	1.274	0.0242	0.981	0.025
V112 Berea	0.01 N	Soltrol	7	0.99	0.777	0.012	1.006	0.012
1883 Shell Plow Realty	0.1 N	Soltrol	9	0.98	1.202	0.023	1.017	0.012
3371 Middle Ground Shoal	0.1 N	Soltrol	7	0.99	1.170	0.028	0.993	0.011
3372C Middle Ground Shoal	0.1 N	Soltrol	6	0.99	1.159	0.021	1.002	0.006
3374 Middle Ground Shoal	0.1 N	Soltrol	2	1.00	0.93	-	1.00	-
3307C Big Foot	1.0 N	Soltrol	2	1.00	1.03	-	1.00	-

Table 9.1. In-phase and quadrature conductivities for brine-saturated sands; in-phase resistivity and quadrature conductivity indices, water saturations for oil-bearing sands.

Sample Number	S_w	I	J (at 30 Hz)
V104	1.000	1.000	1.000
$C_I = 0.0601 \text{ mho-m}^{-1}$	0.630	2.895	0.565
$C_Q = 0.000118 \text{ mho-m}^{-1}$	0.601	3.167	0.543
	0.489	4.295	0.440
	0.436	5.313	0.373
	0.412	5.865	0.347
V106	1.000	1.000	1.000
$C_I = 0.0561 \text{ mho-m}^{-1}$	0.958	1.126	0.945
$C_Q = 0.000142 \text{ mho-m}^{-1}$	0.917	1.356	0.872
	0.804	1.871	0.718
	0.792	1.940	0.710
	0.662	2.697	0.574
	0.636	2.768	0.557
	0.614	2.897	0.537

Table 9.3. In-phase and quadrature conductivities for brine-saturated sands; in-phase resistivity and quadrature conductivity indices, water saturations for oil-bearing sands.

Sample Number	S_w	I	J (at 30 Hz)
1883	1.000	1.000	1.000
$C_I = 0.0282 \text{ mho-m}^{-1}$	0.913	1.160	0.903
$C_Q = 0.000118 \text{ mho-m}^{-1}$	0.778	1.490	0.754
	0.721	1.680	0.699
	0.644	1.980	0.614
	0.586	2.320	0.543
	0.542	2.650	0.484
	0.489	3.030	0.431
	0.451	3.490	0.381
3371	1.000	1.000	1.000
$C_I = 0.0467 \text{ mho-m}^{-1}$	0.848	1.468	0.804
$C_Q = 0.000363 \text{ mho-m}^{-1}$	0.755	1.755	0.722
	0.697	2.047	0.656
	0.632	2.522	0.571
	0.592	2.738	0.544
	0.570	2.994	0.513
3372C	1.000	1.000	1.000
$C_I = 0.0407 \text{ mho-m}^{-1}$	0.977	1.105	0.976
$C_Q = 0.000335 \text{ mho-m}^{-1}$	0.826	1.502	0.797
	0.759	1.707	0.737
	0.679	2.079	0.643
	0.639	2.331	0.591

Table 9.2. In-phase and quadrature conductivities for brine-saturated sands; in-phase resistivity and quadrature conductivity indices, water saturations for oil-bearing sands.

Sample Number	S_w	I	J (at 30 Hz)
V110	1.000	1.000	1.000
$C_I = 0.0656 \text{ mho-m}^{-1}$	0.945	1.196	0.911
$C_Q = 0.000154 \text{ mho-m}^{-1}$	0.836	1.741	0.738
	0.701	2.374	0.592
	0.494	3.932	0.414
	0.420	5.279	0.327
	0.364	6.658	0.295
	0.333	7.987	0.259
	0.302	9.612	0.224
	0.287	11.259	0.196
	0.266	12.838	0.177
	0.243	15.714	0.154
	0.224	17.835	0.140
	0.196	21.391	0.120
V112	1.000	1.000	1.000
$C_I = 0.0105 \text{ mho-m}^{-1}$	0.878	1.187	0.906
$C_Q = 0.000103 \text{ mho-m}^{-1}$	0.472	2.458	0.570
	0.426	2.755	0.525
	0.381	3.105	0.474
	0.345	3.562	0.442
	0.300	4.066	0.387

Table 9.4. In-phase and quadrature conductivities for brine-saturated sands; in-phase resistivity and quadrature conductivity indices, water saturations for oil-bearing sands.

Sample Number	S_w	I	J (at 30 Hz)
3374	1.000	1.000	1.000
$C_I = 0.0565 \text{ mho-m}^{-1}$	0.895	1.193	0.902
$C_Q = 0.000489 \text{ mho-m}^{-1}$			
3307C	1.000	1.000	1.000
$C_I = 0.208 \text{ mho-m}^{-1}$	0.852	1.293	0.848
$C_Q = 0.000790 \text{ mho-m}^{-1}$			

Table 10. Comparison of $(p + 1)$ derived from IP measurements with n^* from in-phase conductivity measurements.

Sample	R_w , ohm-m	$B^\dagger$	Q_v , Equiv/Liter	$R_w B Q_v$	n^*	$(p + 1)$
V104	0.937	2.033	0.06	0.114	2.17	2.18
V106	0.937	2.033	0.066	0.126	2.43	2.28
V110	0.937	2.033	0.06	0.114	2.11	2.27
V112	8.44	0.942	0.06	0.477	1.63	1.78
1883	0.937	2.033	0.120	0.229	1.81	2.20
3371	0.937	2.033	0.321	0.611	2.40	2.17
3372C	0.937	2.033	0.349	0.665	2.35	2.16
3374	0.937	2.033	0.236	0.450	1.79	1.93
3307C	0.094	3.82	1.25	0.458	1.89	2.03
Average					2.06	2.11

† Units of B; liter equiv⁻¹ohm⁻¹m⁻¹

For oil-bearing sands, the IP model indicates that quadrature conductivities should be reduced from their values at $S_w = 1$ by the factor $S_w^{n^*-1}$ or S_w^p , where n^* is the in-phase saturation exponent and p is the quadrature saturation exponent. The relation $p = n^* - 1$ was confirmed by independent experimental determinations of n^* and p at capillary equilibrium.

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REFERENCES

- Anderson, R., 1981, Nonlinear induced polarization spectra: M. S. thesis, Univ. of Utah, Dec.
- Archie, G. E., 1942, The electrical resistivity log as an aid in determining some reservoir characteristics: *Trans. Am. Inst. Min., Metall. & Petr. Eng.*, **146**, 54–67.
- Arps, J. J., 1953, The effect of temperature on the density and electrical resistivity of sodium chloride solutions: *Petr. Trans. Am. Inst. Min., Metall. & Petr. Eng.*, **198**, 327–330.
- Buchheim, W., and Irmer, G., 1979, Theory of induced galvanic polarization in solids with electrolyte-filled pores: *Gerlands Beitr. Geophys.*, **88**, 53–72.
- Chambers, J. F., 1958, The conductance of concentrated solutions of potassium iodide at 25°C and of potassium and sodium chlorides at 50°C: *J. Phys. Chem.*, **62**, 1136–1137.
- Clavier, C., Heim, A., and Scala, C., 1976, Effect of pyrite on resistivity and other logging measurements: Paper HH, Soc. Prof. Well Log Analysts 17th Annual Logging Symp., June 9–12, Denver, CO.
- Collett, L. S., 1959, Laboratory investigation of overvoltage, in J. R. Wait, Ed., *Overvoltage research and geophysical applications*: Pergamon Press.
- Dakhnov, V. N., 1959, The application of geophysical methods; electrical well logging: Moscow Petroleum Institute; trans. by G. V. Keller (1962), Colorado School of Mines Quart., 57.
- Dakhnov, V. N., Latishova, M. G., and Ryapolov, V. A., 1952, Investigation of wells by the induced polarization method (electrolytic well logging): *Sb. Promislovaya Geofizika, Vnetoneft*; trans. by G. V. Keller: *The Log Analyst*, November–December 1967, 46–82.
- Fraser, D. C., 1964, Electrical properties of clay-containing sandstone, B. Electrical polarization: Report MT-64-4, Inst. of Eng. Res., Univ. of California, Berkeley.
- Fuller, B. D., and Ward, S. H., 1970, Linear system description of the electrical parameters of rocks: *Inst. Elect. and Electron. Eng. Trans. Geosci. Electr.*, **GE-8**, Jan., 7–18.
- Grahame, D. C., 1947, The electrical double layer and the theory of electrocapillarity: *Chem. Rev.*, **41**, 3–27.
- Henkel, J. H., and Collins, T. C., 1961, Induced polarization in electrolyte saturated earth plugs: *Geophysics*, **26**, 205–210.
- Hill, H. J., and Milburn, J. D., 1956, Effect of clay and water salinity on electrochemical behavior of reservoir rocks: *Trans. Am. Inst. Min., Metall., & Petr. Eng.*, **207**, 65–72.
- Hoyer, W. A., and Rumble, R. C., 1976, Dielectric constant of rocks as a petrophysical parameter: Soc. Prof. Well Log Analysts 17th Annual Logging Symp., June 9–12, Denver, CO, paper O.
- Jost, W., 1952, *Diffusion*: Academic Press Inc., 444.
- Juhasz, I., 1979, The central role of Q_v and formation water salinity in the evaluation of shaly formations: 20th Annual Soc. Prof. Well Log Analysts Logging Symposium, Tulsa, OK, June 3–6, Trans. I, paper AA.
- , 1981, Normalized Q_v —the key to shaly sand evaluation using the Waxman-Smiths equation in the absence of core data: 22nd Annual Soc. Prof. Well Log Analysts Logging Symp., June 23–26, Mexico City, Trans. I, paper Z.
- Keevil, N. B., and Ward, S. H., 1962, Electrolyte activity—its effect on induced polarization: *Geophysics*, **27**, 677–690.
- Kern, J. W., Hoyer, W. A., and Spann, M. M., 1977, High temperature electrical conductivity of shaly sands: 18th Annual Soc. Prof. Well Log Analysts Logging Symp., June 5–8, Houston, TX, paper U.
- Klein, J. D., and Sill, W. R., 1982, Electrical properties of artificial clay-bearing sandstone: *Geophysics*, **47**, 1593–1605.
- Lavers, B. A., Smits, L. J. M., and Van Baaren, C., 1975, Some fundamental problems of formation evaluation in the North Sea: *Log Analyst*, **16**, 3–13, May–June.
- Madden, T. R., and Marshall, D. J., 1959, Induced polarization, a study of its causes: *Geophysics*, **24**, 790–816.
- Mortland, M. M., and Mellor, J. L., 1954, Conductometric titration of soils for cation exchange capacity: *Proc., Soil. Sci. Soc. Am.*, **18**, 363–364.
- Noyes, A. A., 1908, The conductivity and ionization of salts, acids and bases in aqueous solutions at high temperatures: *J. Amer. Chem. Soc.*, **30**, 335–353.
- Olhoeft, G. R., and Scott, J. H., 1980, Nonlinear complex resistivity logging, 21st Annual Soc. Prof. Well Log Analysts Logging Symp., July 8–11, Lafayette, LA, Trans. I, paper T.
- Poley, J. Ph., Nooteboom, J. J., and de Waal, P. J., 1978, Use of VHF dielectric measurements for borehole formation analysis: Soc. Prof. Well Log Analysts, *Log Analyst*, May–June, 8–30.
- Shuey, R. T., and Johnson, M., 1973, On the phenomenology of electrical relaxation in rocks: *Geophysics*, **38**, 37–48.
- Sill, W. R., 1964, Induced polarization in clay-bearing sandstones and the effects of oil saturation: *Res. Rept. Inst. of Geophys. and Planet. Sci., Univ. of California, San Diego*, August.
- Smits, L. J. M., 1969, Propriétés électrochimiques de sables argileux et de schistes argileux: *Revue de l'Institut Français du Pétrole*, XXIV, January, 91–120.
- Snyder, D. D., Merkel, R. H., and Williams, J. T., 1977, Complex formation resistivity—the forgotten half of the resistivity log: Soc. Prof. Well Log Analysts 18th Annual Logging Symp., June 5–8, Houston, TX, paper Z.
- Sumner, J. S., 1976, Principles of induced polarization for geophysical exploration: Elsevier Scientific Publ. Co.
- Thomas, E. C., 1976, Determination of Q_v from membrane potential measurements on shaly sands: *Trans., Am. Inst. Min., Metall., & Petr. Eng.*, **261**, Part I, 1087–1096.
- Van Olphen, H., 1977, *An introduction to clay colloid chemistry*, 2nd ed.: John Wiley and Sons.
- Van Voorhis, G. D., Nelson, P. H., and Drake, T. L., 1973, Complex resistivity spectra of porphyry copper mineralization: *Geophysics*, **38**, 49.
- Wait, J. R., Ed., 1959, *Overvoltage research and geophysical applications*: Pergamon Press.
- Washburn, W., Ed., 1929, *International critical tables*, v. VI: McGraw-Hill Book Company Inc.
- Waxman, M. H., and Smits, L. J. M., 1968, Electrical conductivities in oil-bearing shaly sands: *Trans. Am. Inst. Min., Metall. & Petr. Eng.*, **243**, Part II, 107–122.
- Waxman, M. H., and Thomas, E. C., 1974, Electrical conductivities in shaly sands: I. Relation between hydrocarbon saturation and resistivity index; II. The temperature coefficient of electrical conductivity: *Trans. Am. Inst. Min., Metall., & Petr. Eng.*, **257**, Part I, 213–225.

APPENDIX A EXPERIMENTAL IP SYSTEM

This section describes the measurement system, sample holders, electrodes, and flow apparatus.

Measurement system

The frequency-domain measurements reported here were

made with a four-electrode bridge in order to eliminate electrode polarization problems inherent in two-electrode systems at low frequencies. As shown in Figure A-1, a sinusoidal current is applied to a series circuit consisting of the sample and a variable RC network. The voltage across the sample and the RC network is equal when the impedance of the RC network is

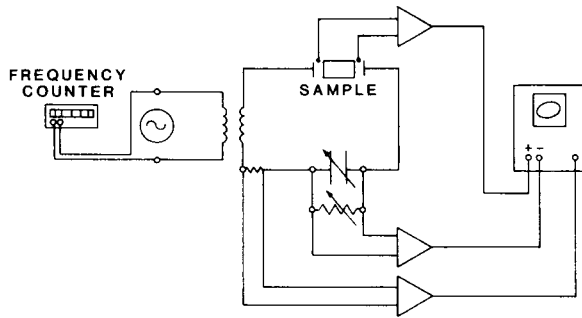


FIG. A-1. Bridge circuit for measuring complex resistivity in frequency domain.

adjusted to equal the sample impedance. The Lissajous pattern on the oscilloscope is used for phase comparison and null detection: at resistive null the oscilloscope trace is an ellipse with horizontal major axis; at both resistive and capacitive balance the ellipse becomes a straight line. Bridge accuracy is 0.01 percent from 1 Hz to 1 KHz.

FET input instrumentation amplifiers are used to isolate the current and potential circuits. The very high input impedance ($10^{13} \Omega$) prevents current flow through the potential circuit, which would polarize the voltage electrodes.

Transformer coupling of the oscillator to the bridge circuit guarantees that no dc bias is applied to the sample and pro-

vides balanced drive for shielding the sample and RC boxes. This minimizes stray coupling at the higher frequencies.

Conductivity measurements were made in the frequency range of 3 Hz to 1 KHz. Inductive coupling limits the upper frequency range to about 1 KHz, without correction.

Sample holders

The basic sample holder used in these experiments is shown in Figure A-2. The sample holder is constructed entirely of Plexiglas® to prevent metallic polarization. Electrical contact with the sample is made through the electrolyte solution in the endpieces. The solution is maintained at the same concentration as the solution saturating the core. While inside the sample holder, the rock can be flushed with brine up to 500 psi internal pressure.

The voltage electrodes are reversible Ag-AgCl, made by chloridizing annealed silver rods using standard electroplating techniques.

The samples are sealed first with a thick epoxy butter to prevent imbibition, then cemented into roughened Plexiglas cylinders with a less-viscous epoxy. The ends of the samples are faced flat with a carbide-tipped cutting tool; no cutting fluid is used.

For the experiments at capillary equilibrium oil saturation, the oil was introduced through a hole drilled into the side of the Plexiglas sleeve encapsulating the sample. Capillary diaphragms through which the water phase was displaced were

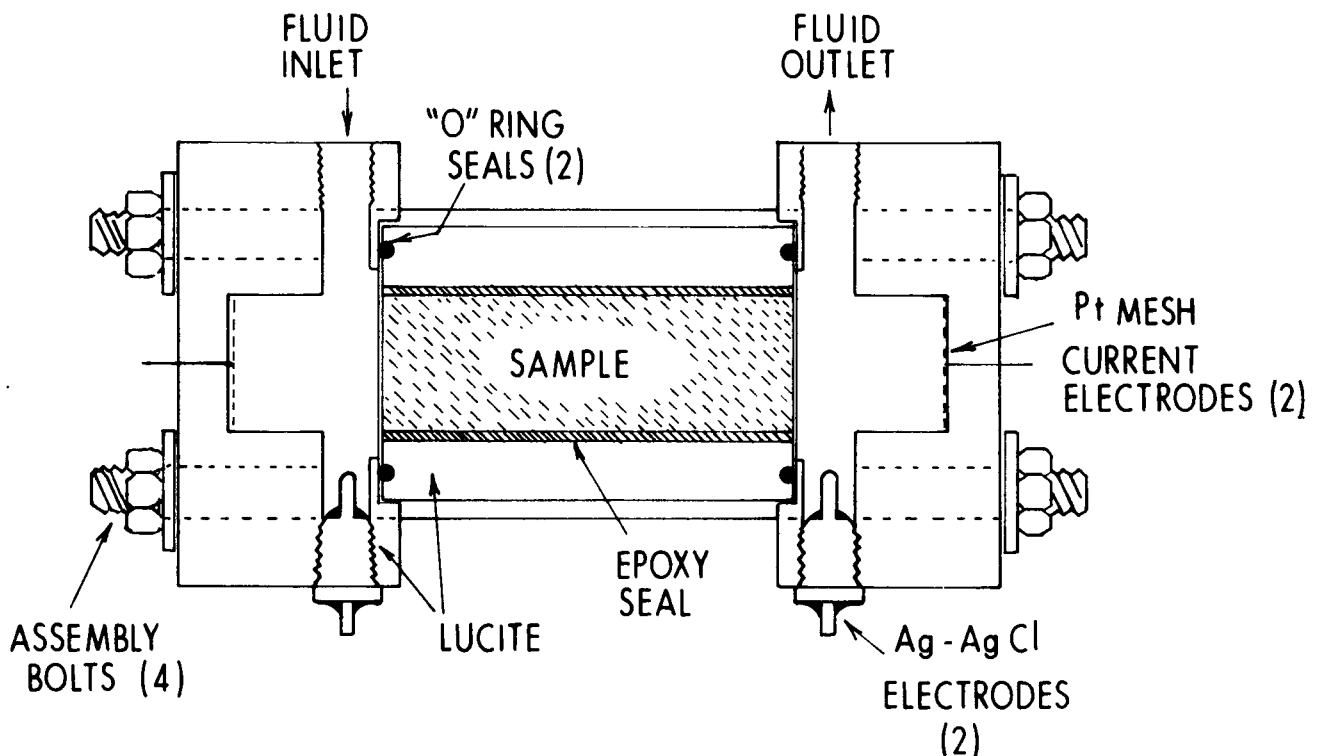


FIG. A-2. Core sample holder for IP measurements.

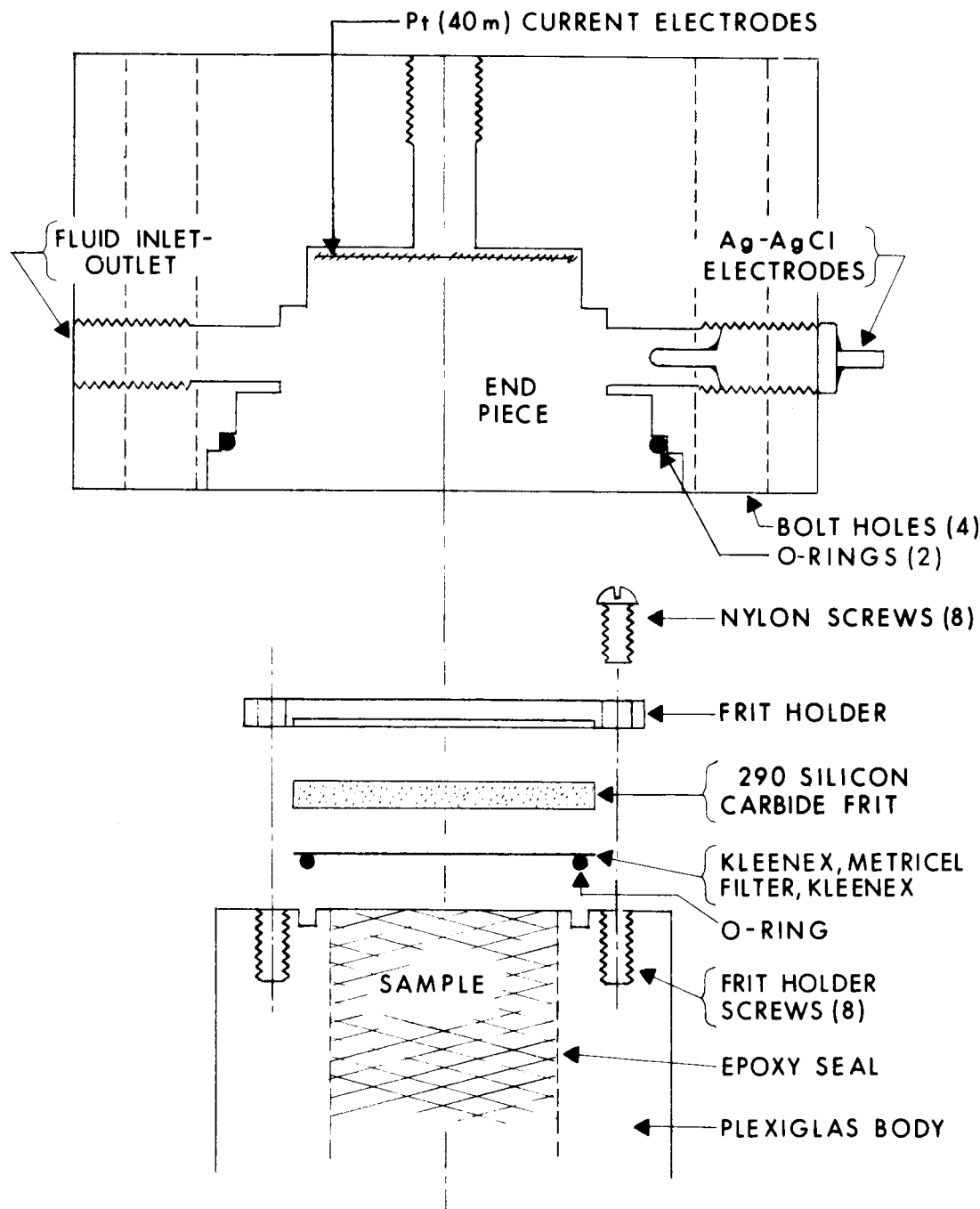


FIG. A-3. Sample and holder, exploded view.

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mounted on each end of the core (see Figure A-3). The capillary diaphragms are brine-saturated Alpha Metrical® filters (Alpha-8, 0.2 micron pore size) with an oil displacement pressure of about 15–20 psi when wet. These thin diaphragms introduce negligible polarization; when wet, the Metrical filters swell and improve core contact. Kleenex® discs are added to ensure capillary continuity. The filters are squeezed against O-rings and sample by a 3/32-inch thick silicon carbide frit backed by a Lucite® ring. Polishing the rock faces and seating the filter

firmly are necessary to prevent an oil layer from developing across the face of the sample.

Corrections are made to the raw data for the silicon carbide frits and capillary diaphragms. As determined in separate experiments, these frits have a series resistance but no polarizability.

The entire sample holder was immersed in a temperature-regulated oil bath in order to take measurements at different temperatures.

APPENDIX B ACCURACY AND VALIDITY CHECKS

Spurious polarization

It is desirable to test a porous but electrically inert sample to verify that the sample holder and electrodes do not generate spurious polarization. Cores of porous silicon carbide, and limestone and dolomite cores free of clay, have been found suitable for this purpose. They exhibit no detectable polarization over the measured frequency range.

As another check, the IP of a uniform sample of Berea sandstone was measured. The sample was then sliced in half and each half remeasured. The phase measurement was unchanged, verifying that a bulk property of the core is measured and sample edge effects are not significant.

Reciprocity and nonpolarizing electrodes

Erroneous results are obtained if there is polarizable material in the current path between the voltage electrodes or if the voltage electrodes are themselves polarizable. This was studied in a series of ten experiments with interchangeable platinum mesh (polarizable) and Ag-AgCl electrodes (nonpolarizable). The results are summarized in Figure B-1 where the current and voltage connections are designated I and V , respectively.

Note that because the rock-plus-electrodes material behaves as a linear system, the function of current and voltage electrodes can be reversed without changing the measured values. This consequence of reciprocity for linear systems is shown by comparing the left and right sides of Figure B-1.

Linearity

The oscilloscope display of bridge balance gives visual indication of deviations from linearity; nonlinear behavior appears as deviation from a simple ellipse or straight line. Current densities were maintained below 10^{-7} A/cm² to ensure linear behavior.

The electrical behavior of silicon carbide plugs and clay-free limestone and dolomite was observed to be linear from 1 Hz to 10 KHz. Small nonlinearities (~ 0.1 – 0.5 percent THD) were observed in shaly sands under certain conditions. The amount of nonlinearity increased with shaliness, suggesting that the nonlinear mechanism is associated with interactions at the clay surfaces. The nonlinearity also increased with decreasing fre-

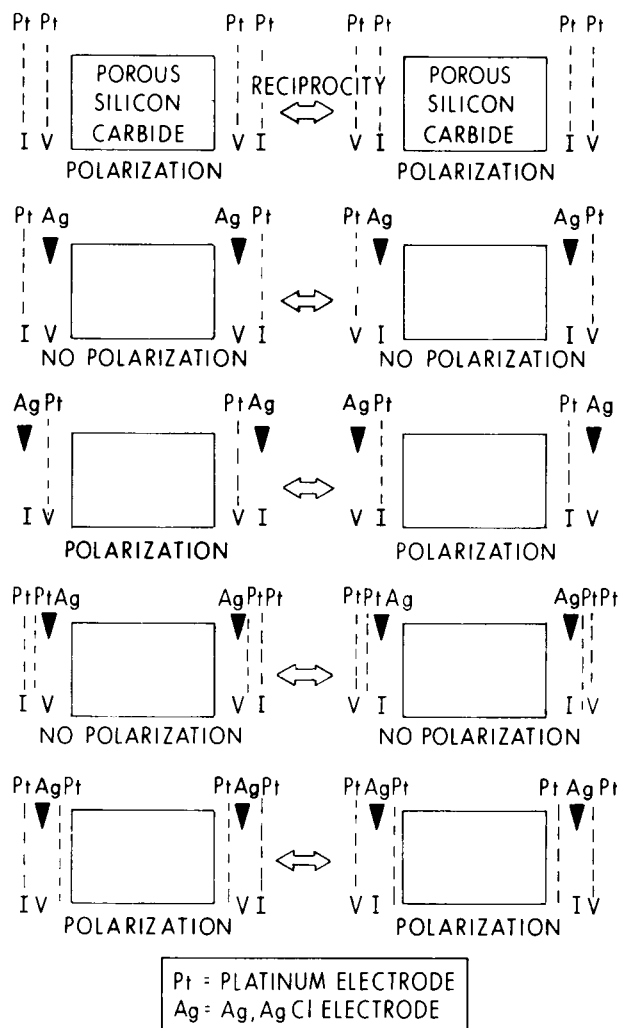


FIG. B-1. Summary of experiments on electrode polarization.

quency below 10 Hz, increasing current density, and increasing oil saturation.

APPENDIX C CALCULATION OF TEMPERATURE DEPENDENCE OF $\bar{\lambda}$

Average values of the ratio $(C_Q)_T/(C_Q)_{25}$ and hence $(\bar{\lambda})_T/(\bar{\lambda})_{25}$ are equal to 1.70 and 3.12 at 50° and 100°C, respectively (Table 6). From these ratios and the $(\bar{\lambda})_{25}$ values at 0.1 and 0.5 N NaCl, the 50° and 100°C $\bar{\lambda}$ values for 0.1 and 0.5 N NaCl were obtained. These were used in equation (20) to solve for $\bar{A}_1$ and $\bar{A}_2$ at both 50° and 100°C. For the two temperatures, values of $\bar{A}_1$ and $\bar{A}_2$ are identical: $\bar{A}_1 = 0.0032$ and $\bar{A}_2 = 0.0058$. These were used in equation (20) to calculate $\bar{\lambda}$ at 50° and 100°C at higher salt concentrations.

Other assumptions and required data were obtained as follows. The cation transport numbers t_+ (Figure 11) were assumed to be temperature independent. Slightly revised values of the clay counterion equivalent conductance B at 50° and 100°C were obtained from Waxman-Thomas data (1974). $B_{\max}(50^\circ\text{C})$ and $B_{\max}(100^\circ\text{C})$ are 94 ± 12 and 166 ± 11 mho-cm²-eq⁻¹, respectively. The equivalent conductivities of sodium chloride solutions as a function of salt concentration were taken from the literature (Chambers, 1958; Noyes, 1908; Washburn, 1929).

A generalized Maxwell-Wagner theory for membrane polarization in shaly sands

Olivar A. L. de Lima* and Mukul M. Sharma‡

ABSTRACT

The effects of charged clay platelets on the frequency dependent electrical properties of shaly materials are analyzed using simplified models for the membrane polarization around charged spheres immersed in electrolytic solutions, under a thin double layer approximation. The polarization is defined through two possible mechanisms: (1) a surface conductivity related with a modified Stern double layer model (S-model) according to Schurr-Schwarz theory; (2) a coupled electro-diffusional mechanism occurring in a Guoy-Chapman double layer using Fixman's approach (D-model). By comparing the electric potential in such microscopic models with the external potentials derived for the equivalent homogeneous sphere using a Maxwell-Wagner approach, we obtain the total current conductivity functions for these two models. The theory, therefore, provides explicit expressions relating the total conductivity functions to the model parameters.

The behavior of the S-model is described by a complex conductivity exhibiting a simple Debye characteristic. In the D-model, both the conductivity and the dielectric permittivity are given as complex properties, showing similar but much wider dispersion than that of a Debye substance. Our representation of the grains and their associated ionic double layers by an equivalent sphere with effective properties allows us to extend our results to simulate rocks containing clays. This is accomplished using generalized mixture equations written in terms of the total conductivity functions. The frequency behavior of both models are compared and their fit to experimental data on clay-water systems and shaly materials suggest that the D-model is more appropriate for representing the dielectric behavior of clay bearing rocks. The theory can be adapted to estimate the clay parameters of a shaly sandstone using electromagnetic borehole measurements.

INTRODUCTION

Systematic experimental measurements of the electrical properties of shaly materials have shown that both their conductivity (σ) and permittivity (ϵ) are dispersive. Although individually for water and most silicate minerals these properties are almost independent of frequency (ω), the composite properties vary appreciably with ω (Olhoeft, 1985). In the frequency range from 1 to 10^6 Hz, the effective conductivity is a slowly increasing function of ω . The effective dielectric constant $k(k = \epsilon/\epsilon_0$, where ϵ_0 is the permittivity of vacuum), on the other hand, is much more strongly dependent on ω , starting with values above 10^3 at a few Hertz and decreasing to an asymptotic value of less than 80 at high frequencies (Fuller and Ward, 1970; Sen et al., 1984; Vinegar and Waxman, 1984). Furthermore, σ and k are

also dependent upon the salinity of the electrolyte (Olhoeft, 1985; Johnson and Sen, 1988; Sen et al., 1988).

These reported characteristics have been interpreted as being caused by geometric or textural heterogeneities of the rock system (Sen, 1981; Sen et al., 1984; Schwartz et al., 1989; Knight and Endres, 1990), or were related to electrical and electrochemical processes developed at the interfaces between clay minerals and electrolytic solutions (Marshall and Madden, 1959; Dias, 1972; Sen et al., 1984). Interactions between charged clays and aqueous electrolytes give rise to an ionic double layer around these particles. Polarization of such a layer by an applied electric field, has been pointed out as the main mechanism for the anomalous behavior observed in rocks, soils, and other biphasic systems of charged particles (Dukhin and Shilov, 1974; Fixman, 1980; Chew and Sen, 1982; Grosse and Foster, 1987).

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*PPPG/UFBA—Instituto de Geociências, Rua Caetano Moura, 123—Federação, 40210, Salvador, Bahia, Brazil.

‡Department of Petroleum Engineering, The University of Texas at Austin, Austin, TX 78712.

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Several previous models have attempted to explain the frequency dependent electric behavior of rocks on the basis of a Maxwell-Wagner polarization mechanism (Wobschall, 1977; Lysne, 1983; Feng and Sen, 1985). The rock grains were assumed to have fixed (frequency independent) dielectric properties as was the saturating fluid. Large inconsistencies between such models and experimental data are observed. These models are not adequate in simulating shaly sand measurements mainly because they do not properly account for the electromigration and diffusion of ions which occur at the particle-fluid interface (Lyklema et al., 1983).

In a previous paper (Lima and Sharma, 1990), we developed a practical procedure to describe the low-frequency conductivity of a shaly sandstone by using simple granular models where the clays occur either as continuous coatings over the sand grains or as isolated conducting particles. The essence of that method was a procedure to express the surface conductivity of clay particles by an equivalent volume conductivity. This procedure gave excellent agreement with laboratory and well log resistivity data.

The objective of the present work is to use a similar approach and apply it in the frequency domain. We choose to represent the clay effects by total current conductivity functions as proposed by Dias (1972). These functions are derivable from solutions for the electric potentials around charged grains and their counterion atmosphere (Schurr, 1964; Fixman, 1980), in conjunction with a generalized Maxwell-Wagner treatment of the suspended sphere problem. In the frequency range considered here (1 Hz to 10 MHz), the diffusion coupling mechanism is the most important effect in the electrical dispersion of shaly rocks. Other coupling processes, such as the electro-osmotic and electrokinetic effects that may have significant contributions if the sphere were moving, are not considered here.

To simulate shales and shaly sands, we construct composite materials using generalized effective medium theories (Hanai, 1968; Sen et al., 1981; Feng and Sen, 1985). For each particular model, the total current conductivity functions are decomposed into the corresponding complex physical properties and the effective experimental parameters that describe the behavior of the binary system. Through this procedure it is possible to represent the response of a wet charged sphere aggregate in terms of equivalent homogeneous complex electrical properties. In this way, we present an integrated approach to describe the dispersive behavior of both the conductivity and permittivity of shaly rocks.

The analytical results can be used to describe the phenomenon of frequency dependent charge polarization in clay rich water-saturated rocks and to explain the observed frequency and salinity dependence of the electrical behavior of these materials. They also serve as a fundamental basis for explaining and designing simple equivalent circuit analogs for rocks. Furthermore, the low-frequency limits of these functions support our previous results for dc currents (Lima and Sharma, 1990).

Electromagnetic logging tools now in use, operate within a frequency range where the effects of distributed clays in rocks are most prominent. By having an appropriate physical model for such rocks, one can extract from well log measurements useful information regarding the clay type, its proportion, and distribution through a porous reservoir.

BASIC THEORY

The macroscopic electrical behavior of conducting dielectrics subjected to harmonic sinusoidal fields is described by Maxwell's equations and certain constitutive relationships. For a time dependence $\exp(-i\omega t)$, the constitutive dielectric equation, Ohm's law and Maxwell's equation for the total current density ($\mathbf{J}_T$) in a medium can be combined to obtain,

$$\mathbf{J}_T = (\sigma^* - i\omega\epsilon^*)\mathbf{E}, \quad (1)$$

where σ^* and ϵ^* are, in general, complex scalars dependent upon frequency (Fuller and Ward, 1970; Dias, 1972). $\mathbf{J}_T$ represents the sum of two current densities: one due to the drift of free charges under the action of the applied electric field and of local physico-chemical potentials; the other is related to the oscillations in the distribution of bound charges. In the frequency domain, $\mathbf{E}$ is the applied electric field.

Writing $\sigma^* = \sigma' + i\sigma''$ and $\epsilon^* = \epsilon' + i\epsilon''$, in terms of their real and imaginary parts, equation (1) is rewritten as

$$\mathbf{J}_T = (\sigma_{ef} - i\omega\epsilon_{ef})\mathbf{E} = \bar{\sigma}^*\mathbf{E}, \quad (2)$$

where $\sigma_{ef} = \sigma' + \omega\epsilon''$ and $\epsilon_{ef} = \epsilon' - \sigma''/\omega$ are real functions of frequency that constitute the experimentally measured rock parameters, and $\bar{\sigma}^*$ represents the total current conductivity function proposed by Dias (1972). Thus, the real effective conductivity of a substance includes contributions from free charge conduction as well as some

LIST OF SYMBOLS

a	= Particle radius
D	= Ionic diffusion coefficient
$\mathbf{E}$	= Electric field
E_o	= Electric field intensity
e_o	= Electronic charge = 1.6022×10^{-19} C
$\mathbf{J}$	= Current density
k_B	= Boltzmann constant = 1.38×10^{-23} J/K
T	= Absolute temperature
p	= Volume clay fraction
Ψ	= Electric potential
ϵ^*	= Dielectric permittivity
σ^*	= Electric conductivity
ϵ_o	= Permittivity of free space = 8.854×10^{-12} F/m
$\bar{\sigma}^*$	= Total current conductivity
μ	= Ionic mobility
λ	= Surface conductivity
ρ	= Charge density
τ	= Relaxation time
β	= Surface counterion density
ϕ	= Porosity
κ	= Debye screening distance

Single and double primes denote real and imaginary parts of complex functions. Subscripts w , c , denote pore water and clay, respectively. Additional subscripts ef , 0 , and ∞ denote the effective bulk, low-frequency and high-frequency properties, respectively.

effects due to polarization mechanisms. Conversely, the effective permittivity has contributions from both bound and mobile charge carriers.

Using equation (2) and applying the classical Maxwell-Wagner treatment, it can be shown that the external potential distribution (Ψ_w) around a spherical particle of arbitrary total current conductivity $\bar{\sigma}_c^*$ suspended in a medium with total current conductivity $\bar{\sigma}_w^*$ and subject to an alternating electric field, can be written as

$$\Psi_w = \left[-r + \frac{(\bar{\sigma}_c^* - \bar{\sigma}_w^*)a^3}{(\bar{\sigma}_c^* + 2\bar{\sigma}_w^*)r^2} \right] E_o \cos \theta, \quad (3)$$

where E_o is the electric field intensity, a the particle radius, and r and θ the spherical coordinates. It is important to note that in the present case, $\bar{\sigma}_c^*$ is the total current conductivity function of the solid grain and its associated electrical double layer. It incorporates within it, double layer polarization effects. Equation (3) is employed in the following subsections to compute the total current conductivity function of a suspended charged sphere by solving for the ionic fluxes in the double layer. Two limiting conditions are considered: (a) bound layer polarization; and (b) diffuse layer polarization.

Bound layer polarization

A simplified theory for the dielectric dispersion of a system of charged particles based on a surface conductivity mechanism has been considered by O'Konski (1960), Schwarz (1962), and Schurr (1964).

O'Konski (1960) introduced an integral surface conductivity operating at the interface between two different media to represent its excess electrical conduction. His boundary value problem for spheres within electrolytes was solved by allowing a free exchange of charges between the surface layer and the electrolyte, but neglecting any ionic diffusion along this layer. His constant conductivity is satisfactory at high frequencies, but it is not adequate in representing the dielectric behavior of charged systems below the characteristic frequency of the Maxwell-Wagner effect (Schurr, 1964; Lyklema et al., 1983).

Schwarz's approach (Schwarz, 1962) incorporates a strong capacitive effect by just considering the diffusive fluxes of counterions in a spherical ionic atmosphere. In fact, his double layer was treated as a thin layer of tightly bound counterions so as to prevent their exchange with ions in the bulk electrolyte. These bound ions are, however, free to move tangentially along the surface of the sphere. Schwarz's theory agrees satisfactorily with experimental effective permittivity data on polystyrene spheres even for frequencies below 10 Hz (Chew and Sen, 1982). However, his neglect of continuous charge transport across the double layer leads to an inconsistent dc result of zero equivalent conductivity for a charged insulating particle having a counterion layer.

In Schurr's theory (Schurr, 1964) the ionic atmosphere around a charged sphere immersed in an aqueous electrolyte is divided into two parts, one represented as a thin layer of fixed counterions with density ρ_b , and another as a diffuse layer with ionic density ρ_d . The composite double layer is thin in comparison to the particle radius and has zero total net charge. The following conditions were additionally introduced: (1) the fixed or Stern layer charges are only free to

move tangentially; the electromigration currents are relaxed by opposing currents due to ionic concentration gradients; (2) the diffuse layer charges exchange readily with the bulk electrolyte; thus, there is no polarization in the diffuse atmosphere; (3) no ions are exchanged between the fixed and the diffuse layers even in the presence of an alternating field. Under these assumptions, Schurr's solution for the external potential of an insulating sphere is equivalent to that of a homogeneous sphere with a total current conductivity given by

$$\bar{\sigma}_c^* = \left(\lambda + \frac{\lambda_o \omega^2 \tau^2}{1 + \omega^2 \tau^2} \right) \frac{2}{a} - i \omega \left[\epsilon'_c + \frac{2\lambda_o \tau}{a(1 + \omega^2 \tau^2)} \right], \quad (4)$$

where, $\lambda = e_o \rho_d \mu_l$ and $\lambda_o = e_o \rho_b \mu_l$, are surface conductivities and $\tau = a^2/2D_1$, is the relaxation time of the charged particle or, equivalently, the time required to establish a stationary ionic distribution in its Stern layer under the action of a static field. Parameters μ_l , and D_1 , are the mobility and diffusion coefficient of the counterion in the charged layer, e_o the charge of the counterion and ϵ'_c the real dielectric permittivity of the particle. Equation (4) reduces to the O'Konski result if $\rho_b = 0$, and to Schwarz's equation if $\rho_d = 0$.

According to equation (2), we may write from equation (4) that

$$\sigma_{c,ef} = \frac{2\lambda}{a} + \frac{2\lambda_o \omega^2 \tau^2}{a(1 + \omega^2 \tau^2)}; \quad (5a)$$

$$\epsilon_{c,ef} = \epsilon'_c + \frac{2\lambda_o \tau}{a(1 + \omega^2 \tau^2)}. \quad (5b)$$

Separating the real and imaginary components and using Kronig-Kramers' relation by which if $\Re(\epsilon^*) = \epsilon'_c = \text{constant}$, then $\Im(\epsilon^*) = \epsilon''_c = 0$, we obtain after some algebra,

$$\sigma_c^* = \frac{2(\lambda + \lambda_o)}{a} - \frac{2\lambda_o}{a(1 - i\omega\tau)}. \quad (6)$$

Thus, the dielectric response of a charged sphere and its associated Stern layer ionic atmosphere is uniquely described by a complex conductivity that exhibits a simple Debye relaxation with characteristic time τ and limits $\sigma_{c,0} = 2\lambda/a$ and $\sigma_{c,\infty} = 2(\lambda + \lambda_o)/a$. Thus, an Argand diagram of equation (6) is represented as a semicircle centered on the real axis, with intercepts at $\sigma_{c,0}$ and $\sigma_{c,\infty}$ on this axis.

The S-model polarization is a satisfactory approximation for cases where most counterions are tightly bound to the solid particles. However, neglecting the effect of ionic diffusion in the external double layer, produces a stronger enhancement in σ_{ef} and ϵ_{ef} than that experimentally observed in clay suspensions (Arulanandan, 1969; Lockhart, 1980). To take into account this screening effect we have also considered a purely diffuse model for the ionic polarization of clays.

Diffuse layer polarization

To account for the motion of ions both tangentially and radially, Fixman (1980) developed an approximate theory based on the polarization of a diffuse Guoy-Chapman model. The thickness of the double layer was again assumed to be much smaller than the particle size. Fixman's formulation is based on a "far field" approximation for both the electro-

static and concentration potentials outside the diffuse layer and a pair of boundary conditions specified at its external limit. These boundary conditions were derived by integrating, over the thin diffuse layer, the difference between the exact ionic density distribution and the one resulting from the far field approximation. Next, he also required the tangential components of his diffusion or velocity potential function (defined as the sum of the electrostatic and the concentration potentials), to be continuous across the double layer. This results in the dielectric relaxation around a charged sphere being controlled by the diffusive fluxes of counterions within the double layer, but which extend in the bulk electrolyte beyond the limits of this layer. A similar observation has also been made by Dukhin and Shilov (1974) and Chew and Sen (1982) working on this problem using the same general theoretical approach.

With these approximations, Fixman's solution for the potential outside an insulating-charged sphere immersed in a symmetric monovalent electrolyte, can be written as

$$\Psi_w = \left[-r - \frac{[1 - \delta_1(1 + Y)]a^3}{[2 + \delta_1(1 - 2Y)]r^2} \right] E_o \cos \theta, \quad (7)$$

with

$$Y = -\frac{1 + (1 - i)\sqrt{\omega\tau}}{2 + 2(1 - i)\sqrt{\omega\tau} - 2i\omega\tau}, \quad (8)$$

where $\delta_1 = \beta_1/aC_1$, β_1 being the surface counterion density necessary to balance the immobile charges on the particle and C_1 the cationic concentration in the electrolyte far from the sphere. In equation (8), τ has the same definition as in equation (4), representing the characteristic time required to build up a stationary charge configuration surrounding the sphere.

Equating equations (3) and (7), and using equation (2), we can show that

$$\bar{\sigma}_c^* = \frac{\delta_1 \bar{\sigma}_w^*}{1 - \delta_1 Y}, \quad (9)$$

or in terms of the effective properties

$$\sigma_{c,ef} = \frac{\delta_1(1 + \delta_1 A)\sigma_w - \omega\delta_1^2 B\epsilon_w}{(1 + \delta_1 A)^2 + \delta_1^2 B^2}, \quad (10a)$$

$$\epsilon_{c,ef} = \frac{\delta_1(1 + \delta_1 A)\epsilon_w + \delta_1^2 B\sigma_w/\omega}{(1 + \delta_1 A)^2 + \delta_1^2 B^2}, \quad (10b)$$

where $A = (1 - y^3)/[2(1 - y^4)]$, $B = y^2(1 - y)/[2(1 - y^4)]$ and $y = \sqrt{\omega\tau}$. Rearranging the components of equation (10a,b) using equation (2), we can write

$$\sigma_c^* = \frac{\delta_1 \sigma_w}{1 - \delta_1 Y}; \quad (11a)$$

$$\epsilon_c^* = \frac{\delta_1 \epsilon_w}{1 - \delta_1 Y}, \quad (11b)$$

where σ_w and ϵ_w are the real parts of the conductivity and the permittivity of the aqueous electrolyte. Thus, in the case of a Guoy-Chapman ionic atmosphere, both the conductivity

and the dielectric permittivity of the system are represented as complex functions of frequency. From the behavior of Y for small and large ω it can easily be shown that equation (11a) can be written as

$$\sigma_c^* = \sigma_{c,\infty} + \frac{\sigma_{c,0} - \sigma_{c,\infty}}{1 - 2i\omega\tau/(2 + \delta_1)[1 + (1 - i)\sqrt{\omega\tau}]}, \quad (12)$$

with the limits

$$\sigma_{c,0} = \frac{2\delta_1 \sigma_w}{2 + \delta_1}; \quad (13a)$$

$$\sigma_{c,\infty} = \delta_1 \sigma_w. \quad (13b)$$

Equation (12) has a slightly different dispersion pattern as compared with that of a Debye substance. It admits a Cole-Cole representation with the form (Cole and Cole, 1941)

$$\sigma_c^* = \delta_{c,\infty} + \frac{\sigma_{c,0} - \sigma_{c,\infty}}{1 + f(\omega, \tau)i^{-(1-\alpha)}} \quad (14)$$

where $f(\omega, \tau) = 2\omega\tau/(2 + \delta_1)[(1 + \sqrt{\omega\tau})^2 + \omega\tau]$. It is easy to show that for high $\omega\tau$ values the complex (σ' , σ'') plane plot or Argand diagram of equation (14) tends to a skewed circular arc with its center located below the real axis. The angle between a radius through $\sigma_{c,0}$ and the real axis is equal to 45 degrees ($\alpha = 1/2$). For small $\omega\tau$, however, the plot resembles that of an ideal Debye substance.

Equation (13a) results in a wet-particle dc-conductivity that is a nonlinear function of cationic concentration C_1 and hence of σ_w . From Einstein's relation, $\sigma_w = \mu_e C_1 e_o$, (where μ_e is the cationic mobility and e_o the electronic charge) we may verify that $\sigma_{c,0}$ increases with σ_w up to a constant value given by $\sigma_{c,0} = \beta_1 \mu_e e_o/a$. This behavior is consistent with the observed curvature (at dilute electrolyte concentrations) of experimental plots of the bulk conductivity of shaly sands as a function of the conductivity of the electrolyte (Johnson and Sen, 1988). Plots of σ_o versus σ_w generated using this model clearly show the nonlinear behavior discussed above (Lima and Sharma, 1991). Expressions similar to equations (12) and (13a, b) can also be written for the complex permittivity by simply replacing σ by ϵ in these equations. The corresponding low- and high-frequency limits of ϵ_{ef} are given by

$$\epsilon_{c,0} = \frac{2\delta_1}{2 + \delta_1} \left(\epsilon_w + \frac{\delta_1 \tau \sigma_w}{2 + \delta_1} \right); \quad (15a)$$

$$\epsilon_{c,\infty} = \delta_1 \epsilon_w, \quad (15b)$$

respectively. Thus, the observed electrolyte concentration dependence of the electrical properties of shaly sandstones can be accounted for with this model.

The boundary conditions introduced in the above models are accurate for values of $\delta_1 < a/\kappa$, where κ is the Debye screening distance ($\kappa = [\epsilon_w k_B T / (2C_1 e_o^2)]^{1/2}$, where k_B is the Boltzmann constant and T the absolute temperature). According to Sen (1987), such an approximation gives satisfactory results for δ_1 values below 16. We have restricted our calculations to this high limit.

Extension to shaly materials

From the total current conductivity functions of a single particle we can derive the electrical properties of composite materials using generalized forms of Wagner and Bruggeman-Hanai theories (Hanai, 1968). Thus, for a dilute-disperse suspension of spherical particles we have from Wagner's theory

$$\bar{\sigma}_o^* = \bar{\sigma}_w^* \left[\frac{2\bar{\sigma}_w^* + \bar{\sigma}_c^* - 2p(\bar{\sigma}_w^* - \bar{\sigma}_c^*)}{2\bar{\sigma}_w^* + \bar{\sigma}_c^* + p(\bar{\sigma}_w^* - \bar{\sigma}_c^*)} \right], \quad (16)$$

where $\bar{\sigma}_o^* = \sigma_{o,ef} - i\omega\epsilon_{o,ef}$, $\bar{\sigma}_c^* = \sigma_{c,ef} - i\omega\epsilon_{c,ef}$, and $\bar{\sigma}_w^* = \sigma_w - i\omega\epsilon_w$ are the total current conductivities of the suspension, individual particle and electrolyte, respectively. The volume fraction of solids in the mixture is p . It is important to note that our incorporation of the membrane polarization phenomena in the total current conductivity function ($\bar{\sigma}_c^*$) allows us to use the above equation directly. This simplifies matters considerably. Equation (16) was derived under the assumption of a weak interaction between the spheres. So, it holds for small values of p . In practice, however, it has been found that equation (16) can be consistently applied even for fairly large p . After substitution of the above definitions and considering the limits of the effective properties at small and large ω , equation (16) is rewritten as

$$\bar{\sigma}_o^* = \sigma_{o,0} - \frac{i\omega\tau_m[\sigma_{o,\infty} - \sigma_{o,0}]}{1 - i\omega\tau_m} - i\omega\epsilon_{o,\infty}, \quad (17)$$

where

$$\sigma_{o,0} = \sigma_w \left[\frac{2\sigma_w + \sigma_{c,0} - 2p(\sigma_w - \sigma_{c,0})}{2\sigma_w + \sigma_{c,0} + p(\sigma_w - \sigma_{c,0})} \right]; \quad (18a)$$

$$\sigma_{o,\infty} = \epsilon_{o,\infty} \frac{\sigma_w}{\epsilon_w} - \frac{9p[\sigma_w\epsilon_{c,\infty} - \epsilon_w\sigma_{c,\infty}]}{[2\epsilon_w + \epsilon_{c,\infty} + p(\epsilon_w - \epsilon_{c,\infty})]^2}; \quad (18b)$$

$$\epsilon_{o,\infty} = \epsilon_w \left[\frac{2\epsilon + \epsilon_{c,\infty} - 2p(\epsilon_w - \epsilon_{c,\infty})}{2\epsilon_w + \epsilon_{c,\infty} + p(\epsilon_w - \epsilon_{c,\infty})} \right]; \quad (18c)$$

and

$$\tau_m = \frac{2\epsilon_w + \epsilon_{c,ef} + p(\epsilon_w - \epsilon_{c,ef})}{2\sigma_w + \sigma_{c,ef} + p(\sigma_w - \sigma_{c,ef})}. \quad (18d)$$

Complex plane plots of the real versus imaginary part of equation (17) produce circular arcs with skewed spurs. Equation (18d) shows that the dielectric behavior of a charged suspension of spheres is characterized by a continuous distribution of relaxation times, since both $\sigma_{c,ef}$ and $\epsilon_{c,ef}$ are smooth functions of frequency. Although equations (16) and (17) are not appropriate to model precisely the dielectric response of shales (since they are only valid for dilute suspensions of solids), they are explicit expressions and are therefore useful to study the effects of the various governing factors on the behavior of clay-water mixtures. They are also helpful in designing representative circuit analogs for such materials.

For more concentrated suspensions the Bruggeman-Hanai equation (Hanai, 1968; Sen et al., 1981) is more appropriate,

$$\frac{\bar{\sigma}_o^* - \bar{\sigma}_c^*}{\bar{\sigma}_w^* - \bar{\sigma}_c^*} \left(\frac{\bar{\sigma}_w^*}{\bar{\sigma}_o^*} \right)^{1/3} = \phi, \quad (19)$$

where ϕ is the volume fraction of the electrolyte or the porosity of the mixture. Although some doubts have been raised against the theoretical validity of Bruggeman's approach (Wobschall, 1977), equation (19) has been validated experimentally for concentrated mixtures of various kinds of materials (Hanai, 1968; Sen et al., 1981; Feng and Sen, 1985).

Equation (19) has the following frequency limits. When $\omega \rightarrow 0$

$$\frac{\sigma_{o,0} - \sigma_{c,0}}{\sigma_w - \sigma_{c,0}} \left(\frac{\sigma_w}{\sigma_{o,0}} \right)^{1/3} = \phi, \quad (20a)$$

and

$$\begin{aligned} & \epsilon_{o,0} \left[\frac{3}{\sigma_{o,0} - \sigma_{c,0}} - \frac{1}{\sigma_{o,0}} \right] \\ &= 3 \left[\frac{\epsilon_w - \epsilon_{c,0}}{\sigma_w - \sigma_{c,0}} + \frac{\epsilon_{c,0}}{\sigma_{o,0} - \sigma_{c,0}} \right] - \frac{\epsilon_w}{\sigma_w}. \end{aligned} \quad (20b)$$

When $\omega \rightarrow \infty$

$$\frac{\epsilon_{o,\infty} - \epsilon_{c,\infty}}{\epsilon_w - \epsilon_{c,\infty}} \left(\frac{\epsilon_w}{\epsilon_{o,\infty}} \right)^{1/3} = \phi, \quad (21a)$$

$$\begin{aligned} & \sigma_{o,\infty} \left[\frac{3}{\epsilon_{o,\infty}} - \epsilon_{o,\infty} - \frac{1}{\epsilon_{o,\infty}} \right] \\ &= 3 \left[\frac{\sigma_w - \sigma_{c,\infty}}{\epsilon_w - \epsilon_{c,\infty}} + \frac{\sigma_{c,\infty}}{\epsilon_{o,\infty} - \epsilon_{c,\infty}} \right] - \frac{\sigma_w}{\epsilon_w}. \end{aligned} \quad (21b)$$

In general, equation (19) cannot be explicitly written in a simple closed form, in terms of the high- and low-frequency limits of the effective electrical parameters of the mixture. Therefore, it was solved numerically for specific combinations of the component properties, using the Laguerre method of finding the roots of polynomials with complex coefficients. One of such complex roots, yields the effective properties of the composite material. We used a Fortran subroutine implemented by Press et al. (1986). The ambiguity in identifying the correct solutions was removed by considering the functional behavior of the solutions at the high and low frequency limits.

Equations (16) and (19) can also be used to describe electrolytic suspensions of spherical sand grains coated by an aggregate of wet clays. For this, $\bar{\sigma}_c^*$ has to be replaced by an effective total current conductivity ($\bar{\sigma}_{CS}^*$) for the uniformly coated sphere. To compute the effective total current conductivity of a clay-coated sphere we compare the external electric potential around the layered sphere immersed in a homogeneous electrolyte of total current conductivity $\bar{\sigma}_w^*$ and under the influence of a uniform electric field, with that of a single equivalent homogeneous sphere under the same conditions (Lima and Sharma, 1990). Thus, we get

$$\bar{\sigma}_{CS}^* = \bar{\sigma}_c^* \left[\frac{2p\bar{\sigma}_c^* + (3 - 2p)\bar{\sigma}_a^*}{(3 - p)\bar{\sigma}_c^* + p\bar{\sigma}_a^*} \right], \quad (22)$$

where $\bar{\sigma}_c^*$ and $\bar{\sigma}_a^*$ are the total current conductivities of the coating clay aggregate and spherical nucleus, respectively. In equation (22) $p = 1 - (a/b)^3$, is the volume fraction of the clay shell, with a and b as the radius of the core and of the composite sphere. Thus, by assuming the rock grains as spheres of arbitrary sizes coated with clays, we can compose a self-similar mixture whose electrical properties are described by equation (19). Such an approach has been formally validated by Sen et al. (1981) in a model for the electrical response of clean wet sandstones.

Equivalent expressions for a dispersed model where the clays occur as discrete particles in a sand matrix have to rely on the more complicated three-component effective medium formulation (Feng and Sen, 1985; Lima and Sharma, 1990) and will not be discussed here.

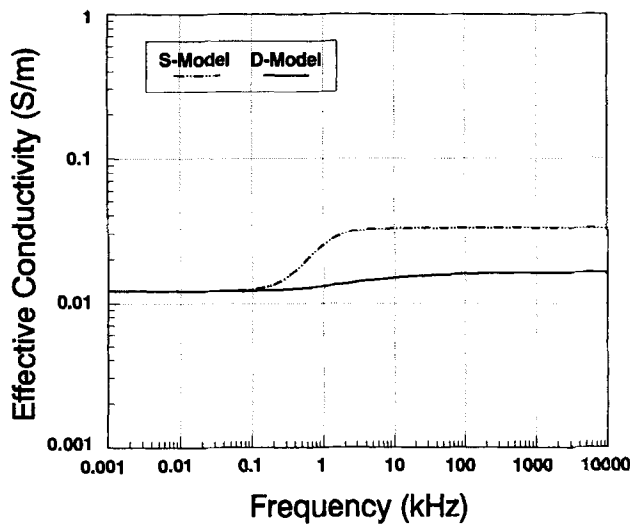
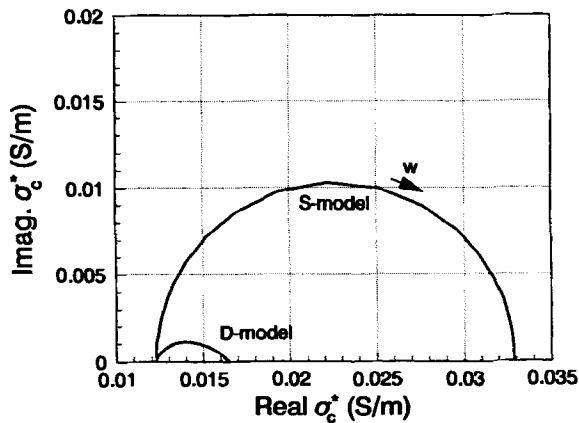


FIG. 1. The dependence of the effective conductivity on frequency for a single-charged wet sphere. Parameters: $a = 9.0 \times 10^{-7}$ m; $\beta_a = 1.2 \times 10^{18}$ /m²; $\sigma_w = 0.024$ S/m; $\epsilon_w = 6.9 \times 10^{-10}$ F/m; $\epsilon_c = 3.1 \times 10^{-11}$ F/m; $D_1 = 2.0 \times 10^{-9}$ m²/s.



a)

NUMERICAL AND EXPERIMENTAL RESULTS

In Figures 1 and 2 we compare theoretical results obtained with equations (5a, b) and (10a, b) for the effective properties of a single charged sphere, using data typical of kaolinite clays as summarized in Figure 1. We have assumed a ρ_b value for the Stern layer model (S-model) such that in the low-frequency range, both models show the same $\sigma_{c,ef}$. This particular situation required about 63 percent of the compensating counterions to be stored in the fixed part of the double layer. The figures show that the diffuse model (D-model) has a lower amplitude of variation in its effective parameters, as compared with the S-model, and also has a broader dispersion spectrum, with a characteristic frequency in the kilocycle range.

The Argand diagrams of Figure 3a, b show the complex conductivities of a single sphere according to equations (6a) and (12). The S-model plots as a semicircle centered on the

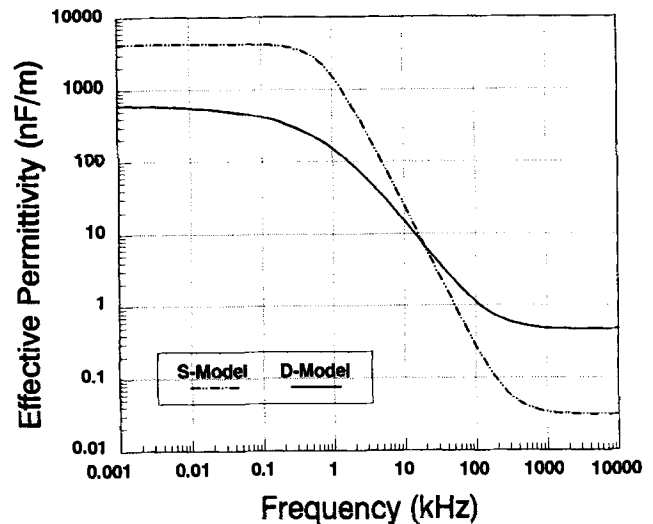
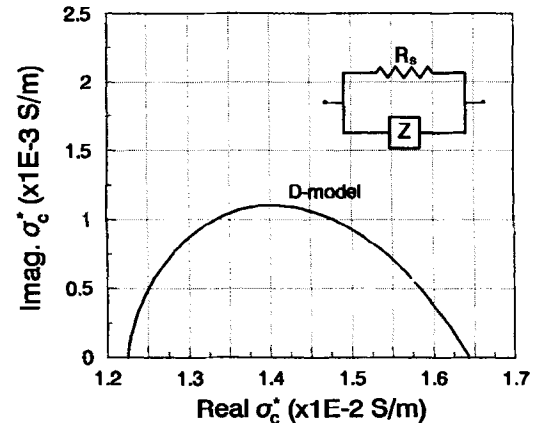


FIG. 2. Dependence of effective permittivity on frequency for a single-charged wet sphere. Parameters as in Figure 1.



b)

FIG. 3. Argand diagrams for the complex conductivity of a single-charged wet sphere. Parameters as in Figure 1.

real axis. The D-model diagram, on the other hand, is depicted as an asymmetric curve with its high-frequency portion being represented by a depressed circular arc, as discussed before. The enhancement depicted on the S-model curve is due to an increase in the imaginary component of its electrical conductivity, as a result of neglecting the ionic diffusion in the external part of the double layer (Fixman, 1980; Lyklema et al., 1983). The equivalent circuit inserted in Figure 3b simulates the complex conductivity of the D-model. The resistor R_s , proportional to $\sigma_{e,\infty}$, corresponds to a steady ionic transport through the double layer that is active even at high frequencies. The complex impedance Z , given by the second term on the right side of equation (12), constitutes a parallel ionic path that is more effective at low frequencies. It provides both an in-phase current due to the applied electric field and an out-of-phase current due to charge polarization and ionic concentration gradients. Figure 4 contains equivalent representation of the same basic data in terms of the total current conductivity functions given by equations (4) and (9). The vertical straight segments in these diagrams allows us to introduce in the circuit analog, the additional capacitance (proportional to $\epsilon_{e,\infty}$) connected in parallel with R_s as depicted in Figure 7.

Figures 5 and 6 show the modeling of clay-water mixtures treated as dilute-disperse suspensions in accordance with equation (16). A pattern of amplitude variations with frequency similar to that for a single sphere is observed. The S-model curves show the existence of two relaxation times: one due to diffusion in the fixed counterion layer and another resulting from the permittivity contrast between the media (the conventional Maxwell-Wagner effect). The D-model, however, as for the isolated particle case, has a wide spectrum simulating a continuous distribution of relaxation times between these two extremes. These figures also contain, for comparison purposes, experimental data for a sodium kaolinite suspension with 36.5 percent and 39 percent of clay by weight, as reported by Arulanandan (1969) and by Lockhart (1980). Although the individual clays are far from being spherical, we find good agreement with the D-model simulation using a clay parameter $\beta_1/a = 1.333 \times 10^{24} \text{ m}^3$. A water conductivity $\sigma_w = 1.2 \times 10^{-3} \text{ S/m}$ was used in both our simulations and their experiments. This

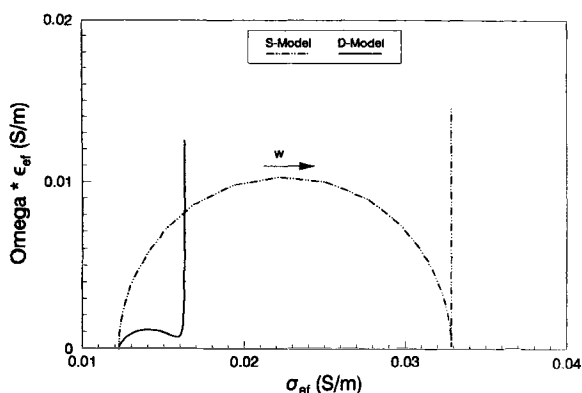


FIG. 4. Complex plane plots of the total current conductivity functions for the single charged sphere. Parameters as in Figure 1.

gives a characteristic relaxation time $\tau \approx 2.0 \times 10^{-4} \text{ s}$. Numerical results for the same input data were also obtained using the Bruggeman-Hanai equation (19). The results were for all practical purposes indistinguishable from those plotted in Figures 5 and 6 using equation 16.

The data from the total current conductivity functions (equation 17) of these suspensions were also plotted as Argand diagrams in the complex plane (Figure 7). The curves are represented as depressed circular arcs with spurs at the high-frequency limit, similar to the ones experimentally observed in nonideal admittance plots of dielectric solids (Jonscher, 1975). A maximum in the imaginary part of equation (17), located within the frequency range of each circular arc, occurs at $\omega \approx 1/2\pi\tau'_m$, with τ'_m being a mean characteristic time for the relaxation of that suspension. In

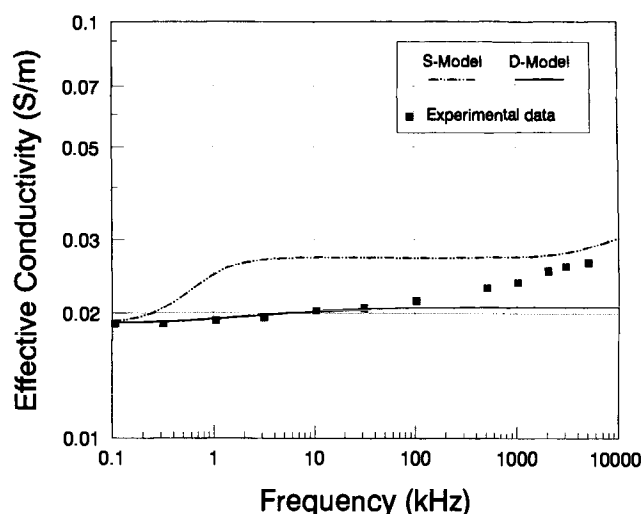


FIG. 5. Conductivity dispersion of a dilute suspension of charged spheres. Comparison with experimental data for a kaolinite gel with 39 percent of clays according to Arulanandan (1969). $\beta_1/a = 1.333 \times 10^{24} \text{ m}^3$, $p = 0.39$.

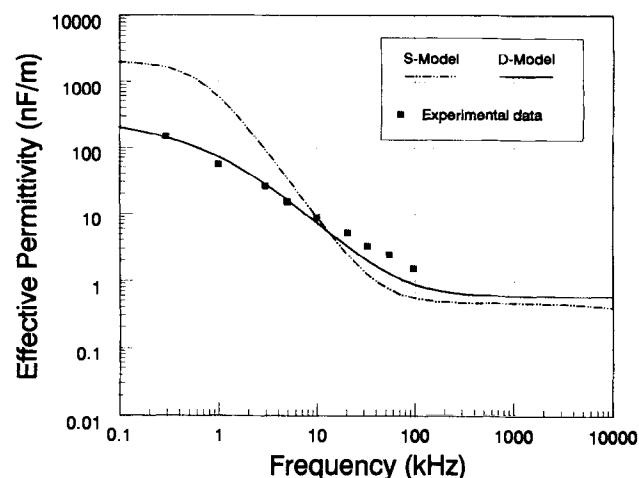


FIG. 6. Permittivity dispersion of a dilute suspension of charged spheres. Comparison with experimental data of a kaolinite gel with 36.7 percent of clay according to Lockhart (1980). $\beta_1/a = 1.333 \times 10^{24} \text{ m}^3$, $p = 0.39$.

the high-frequency limits, the capacitive effect represented by $\omega\epsilon_{o,\infty}$ gives rise to the ascending spurs in the diagrams. These features are duplicated by the circuit analog depicted in Figure 7b. This circuit is quite similar to the one used by Dias (1972) to fit experimental data on the membrane polarization of rocks. It also corresponds, in essence, to the equivalent circuit used by Knight (1984) for modeling the dielectric response of sandstones. The present analysis of course provides us with an explicit physical and mathematical representation for each component in the equivalent circuit in terms of basic model parameters.

The Bruggeman-Hanai equation (19) was also used to model typical shale data as displayed in Figure 8. The data was reported by Scott et al. (1967) for the Mancos shale, an Early Cretaceous dark gray carbonaceous shale from eastern Utah. The experimental values were obtained for a distilled water content of 3.8 percent by volume. We assume this shale to be a compacted clay-water mixture having the following component properties: water conductivity $\sigma_w = 2 \times 10^{-2}$ S/m and clay parameter $\beta_1/a = 6.0 \times 10^{23}/\text{m}^3$ (which is a typical value for an illite clay). For these input data, we predict a frequency dependence of the effective conductivity and permittivity that is remarkably close to the measured curves, using a D-model.

Theoretical results using equations (19) and (22) for shales and sandstones with a clay coating structure are shown in Figures 9 to 11. Figure 9 illustrates the effects of changing the clay parameter β_1/a on the electrical dispersion of idealized shales. It is evident that as β_1/a increases the characteristic time for double layer relaxation (τ) decreases and the frequency at which large changes in dielectric constant occur correspondingly increases.

The predicted variation in the effective properties of a shaly sandstone with 20 percent of porosity, as a function of frequency and clay proportion in the solid matrix, is shown in Figure 10. These overall trends are in qualitative agree-

ment with many experimental measurements described in the literature (Mehran and Arulanandan, 1977; De Loor, 1983; Kenyon, 1984).

In Figure 11 we show the frequency behavior of the dielectric parameters of a shaly sandstone also depending on the saturating electrolyte conductivity. These simulations reveal that the magnitude of the bulk effective conductivity increases with the electrolyte concentration in conformity with many experimental observations. The effective permittivity, however, may behave as a decreasing function of σ_w . Although an adequate set of dielectric permittivity data was not available to us, some data reported by Vinegar and Waxman (1984) for a shaly sand equivalent quadrature conductivity reflect this same behavior. However, some sandstone measurements in the frequency range from 1 to 10^4 MHz (Kenyon, 1983) show that, in fact, $\epsilon_{o,ef}$ increases with σ_w . This behavior can also be reproduced by the present approach, assuming the water salinity inside the shale coating is almost constant and independent of the free pore electrolyte (Lima and Sharma, 1991).

Table 1 provides a potential method for the application of the theory to wireline measurements. Obtaining the six reservoir parameters listed from the six wireline measurements will require inverting the equations in this paper. Since the equations are coupled and nonlinear this will be a nontrivial task. This would be a fruitful avenue for future research in applying the theory presented.

CONCLUSIONS

Two simplified approaches have been considered to simulate the frequency-dependent electrical properties of materials containing charged particles such as clays in soils and rocks. The clays were assumed to consist of an insulating-charged nucleus surrounded by a thin layer of electrolyte containing a double layer counterion atmosphere. For the

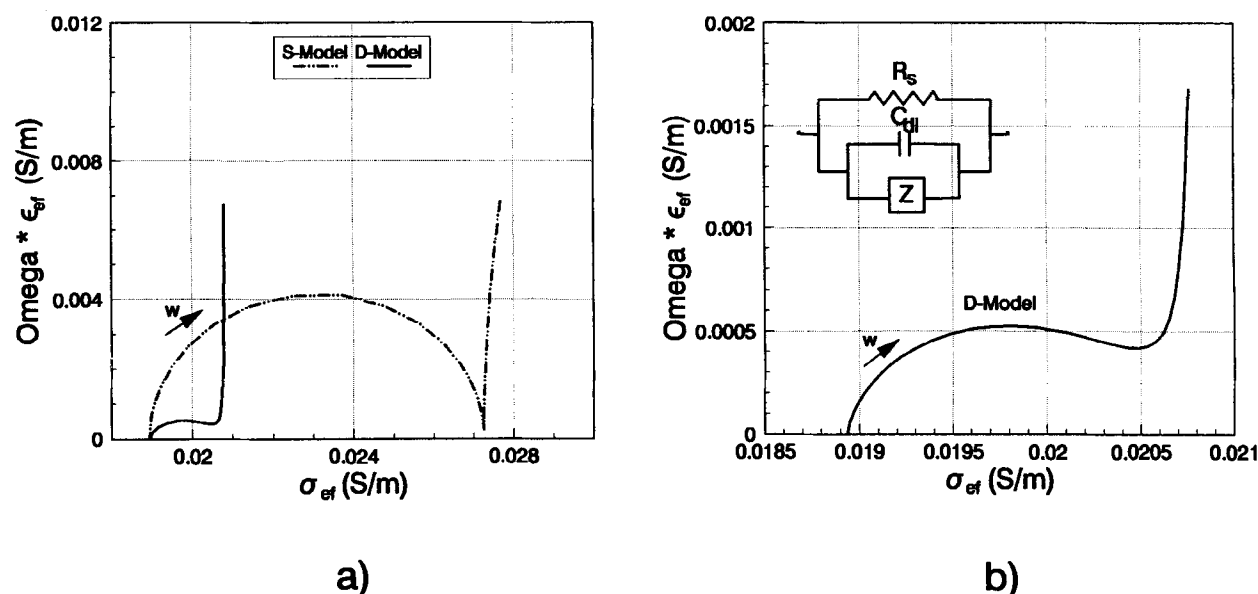


FIG. 7. Argand diagram for the total current conductivity functions of dilute suspensions of charged spheres. Parameters as in Figures 6 and 7.

Stern layer (S-model), we use the results from Schurr (1964), whereas for the diffuse layer D-model we use Fixman's (1980) model as a starting point. The effective conductivity and permittivity functions of the grains and their associated double layers were derived. These effective properties allow us to use generalized mixture theories to obtain the behavior of shaly sandstones.

A macroscopic representation in terms of a total (volume) current conductivity for a charged particle was shown to be a sound approach to describe the dielectric dispersion of shaly sands. From the total current conductivity of wet particles derived for both models we found that the electrical properties of heterogeneous mixtures can be characterized as complex functions of frequency by applying general mixture theories.

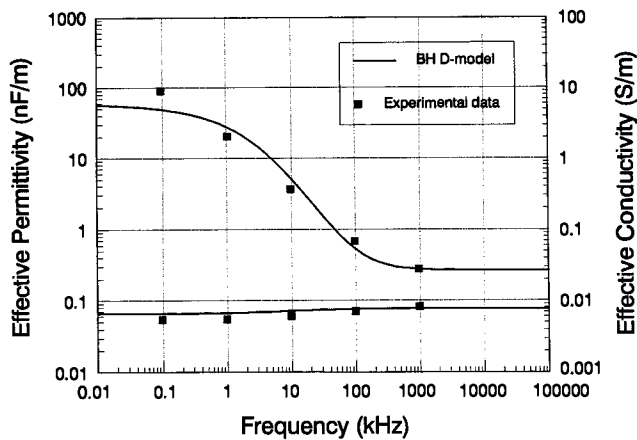


FIG. 8. Modeling the effective conductivity and permittivity of the Mancos shale. Experimental data according to Scott et al., (1967). Parameters: $\beta_1/a = 6.0 \times 10^{23} \text{ /m}^3$; $\sigma_w = 0.02 \text{ S/m}$; $\epsilon_w = 6.9 \times 10^{-10} \text{ F/m}$; $D_1 = 2.0 \times 10^{-9} \text{ m}^2/\text{s}$; $\phi = 4 \text{ percent}$.

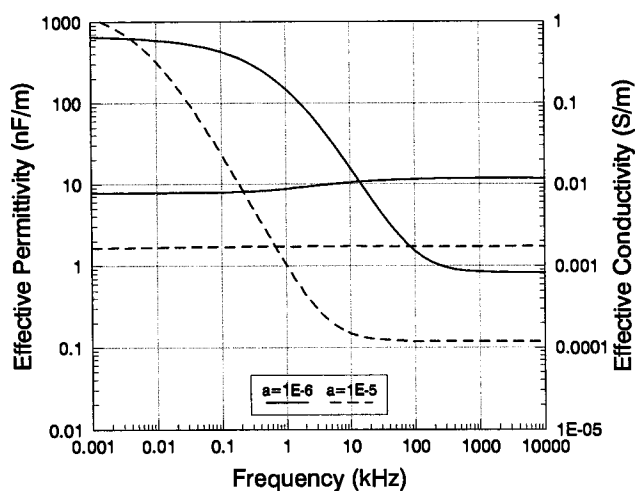


FIG. 9. Effective conductivity and permittivity of an idealized shale of porosity $\phi = 10 \text{ percent}$ for two different clay volume fractions p . Further parameters: $\sigma_w = 0.01 \text{ S/m}$; $\epsilon_w = 6.9 \times 10^{-10} \text{ F/m}$; $D_1 = 2.0 \times 10^{-9} \text{ m}^2/\text{s}$.

On the basis of this approach, we have expressed the effective electrical parameters of a shaly sand in terms of the geometrical and physical properties of its components. These properties include the charge density and grain size of the clays, and the conductivity and permittivity of the electrolyte. The D-model with its wider dispersion appears to be more appropriate for describing the electrical response of clay-water suspensions, shales and shaly sandstones in the frequency range from 1 to 10^7 Hz . The zero frequency limit of their conductivity functions are nonlinear functions of the electrolyte salinity at low concentrations, in conformity with experimental evidence.

Although the assumed spherical geometry does not represent the real shape of mineral grains in sedimentary rocks, the results obtained with this representation describe fairly

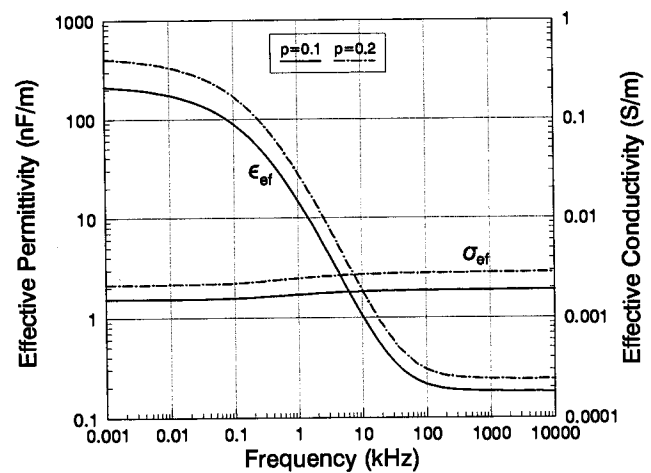


FIG. 10. Effective properties of idealized shaly sandstones of porosity $\phi = 20 \text{ percent}$, for two different clay volume fractions p . $\beta_1/a = 1.0 \times 10^{24} \text{ /m}^3$. Other parameters as in Figure 9.

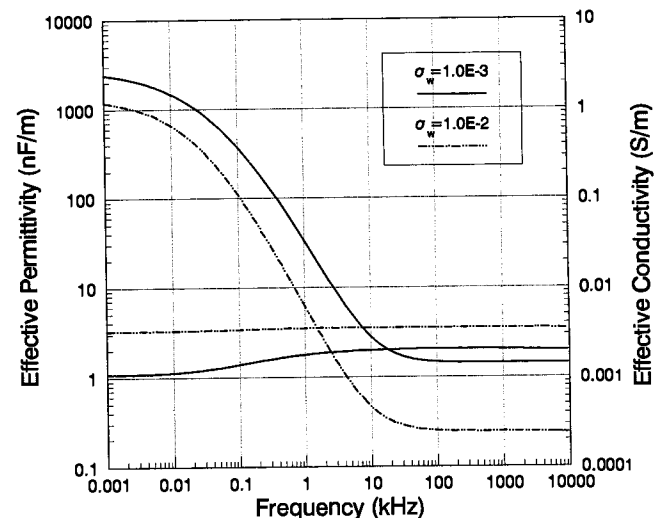


FIG. 11. Effective properties of a shaly sandstone of porosity $\phi = 20 \text{ percent}$ and $p = 10 \text{ percent}$ for two different electrolyte conductivities σ_w . Parameters as in Figure 10.

Table 1. Potential application of the theory.

6 Measurements	6 Unknowns
Low Frequency (< 100 Hz) σ_o Flushed zone σ_o Virgin formation High Frequency (> 1MHz) σ_o, ϵ_o Flushed zone σ_o, ϵ_o Virgin formation	Φ (Porosity) S_{xo} (Water saturation) S_w (Water saturation) p (Clay fraction) B/a (Clay parameter) α_w (Water resistivity)

well the essential features of the dispersion spectra of the electrical properties of shaly materials and other charged systems. The results are also useful in simulating the electrical behavior of rocks in terms of ideal circuit elements such as resistors and capacitors combined in parallel with a frequency-dependent complex element that is similar to a diffusion controlled Warburg impedance. Finally, on the basis of the equations presented here, it is possible to solve for shaly sandstone parameters using the multifrequency electromagnetic data measured in boreholes with the available electric logging tools. Additional experimental data is needed to validate the model. In addition, the model equations need to be inverted so that reservoir parameters can be estimated from measurements of conductivity and dielectric constant at various frequencies.

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REFERENCES

- Arulanandan, K., 1969, Hydraulic and electrical flows in clays: *Clays and Clay Min.*, **17**, 63–76.
- Chew, W. C., and Sen, P. N., 1982, Dielectric enhancement due to electrochemical double layer: Thin double layer approximation: *J. Chem. Phys.*, **77**, 4683–4693.
- Cole, K. S., and Cole, R. H., 1941, Dispersion and absorption in dielectrics. I. Alternating current characteristics: *J. Chem. Phys.*, **9**, 341–351.
- De Loor, G. R., 1983, The dielectric properties of wet materials: *IEEE Trans. on Geosci. and Remote Sensing*, **GE-21**, 364–369.
- Dias, C. A., 1972, Analytical model for a polarizable medium at radio and lower frequencies: *J. Geophys. Res.*, **77**, 4945–4956.
- Dukhin, S. S., and Shilov, V. N., 1974, Dielectric phenomena and the double layer in disperse systems and polyelectrolytes: John Wiley & Sons, Inc.
- Feng, S., and Sen, P. N., 1985, Geometrical model of conductive and dielectric properties of partially saturated rocks: *J. Appl. Phys.*, **58**, 3226–3243.
- Fixman, M., 1980, Charged macromolecules in external fields. I. The sphere: *J. Chem. Phys.*, **72** (9), 5177–5186.

- Fuller, B. D., and Ward, S. H., 1970, Linear system description of the electrical parameters of rocks: *IEEE Trans. Geosci. Electr.*, **GE-8**, 7–15.
- Grosse, C., and Foster, K. R., 1987, Permittivity of a suspension of charged spherical particles in electrolyte solutions: *J. Phys. Chem.*, **91**, 3073–3076.
- Hanai, T., 1968, Electrical properties of emulsions; in, P. Sherman, Ed., *Emulsion science*: Academic Press Inc.
- Johnson, D. L., and Sen, P. N., 1988, Dependence of the conductivity of a porous medium on electrolyte conductivity: *Phys. Rev. B*, **37**, 3052–30.
- Jonscher, A. K., 1975, The interpretation of non-ideal dielectric admittance and impedance diagrams: *Phys. Stat. Sol. (A)*, **32**, 665–676.
- Kenyon, W. E., 1983, Dielectric dispersion in laboratory measurements on cores: Schlumberger Wireline Interpretation Symp.
- , 1984, Texture effects on megahertz dielectric properties of calcite rock samples: *J. Appl. Phys.*, **55**, 3153–3159.
- Knight, R. J., 1984, The dielectric constant of sandstones, 5 Hz to 13 MHz: Ph.D. Dissertation, Stanford University, 119.
- Knight, R. J., and Endres, A., 1990, A new concept in modeling the dielectric response of sandstones: *Geophysics*, **55**, 586–594.
- Lima, O. A. L., and Sharma, M. M., 1990, A grain conductivity approach to shaly sandstones: *Geophysics*, **55**, 1347–1356.
- , 1991, Water conductivity and saturation effects on the dielectric response of shaly sands: *Soc. Prof. Well Log Analysts 32th Ann. Logging Symp.*, Paper G.
- Lyklema, J., Dukhin, S. S., and Shilov, V. N., 1983, The relaxation of the double layer around colloidal particles and the low-frequency dielectric dispersion. Part I. Theoretical considerations: *J. Electroanal. Chem.*, **143**, 1–21.
- Lysne, P. C., 1983, A model for the high-frequency electrical response of wet rocks: *Geophysics*, **48**, 775–786.
- Lockhart, N. C., 1980, Electrical properties and the surface characteristics and structure of clays: *J. Colloid Interface Sci.*, **74**, 520–529.
- Marshall, D. J., and Madden, T. R., 1959, Induced polarization, a study of its causes: *Geophysics*, **24**, 790–816.
- Mehran, M., and Arulanandan, K., 1977, Low frequency conductivity dispersion in clay-water-electrolyte systems: *Clays and Clay Min.*, **25**, 39–48.
- O'Konski, C. T., 1960, Electric properties of macromolecules. V. Theory of ionic polarization in polyelectrolytes: *J. Phys. Chem.*, **64**, 605–619.
- Olhoeft, G. R., 1985, Low-frequency electrical properties: *Geophysics*, **50**, 2492–2503.
- Press, W. H., Flannery, B. P., Teukolsky, S. A., and Vetterling, W. T., 1986, Numerical recipes. The art of scientific computing: Cambridge Univ. Press.
- Schurr, J. M., 1964, On the theory of the dielectric dispersion of spherical colloidal particles in electrolyte solutions: *J. Phys. Chem.*, **68**, 2407–2413.
- Schwarz, G., 1962, A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solutions: *J. Phys. Chem.*, **66**, 2636–2642.
- Schwartz, L. M., Sen, P. N., and Johnson, D. L., 1989, Influence of rough surfaces on electrolytic conduction: *Phys. Rev. B*, **40**, 2450–54.
- Scott, J. H., Carroll, R. D., and Cunningham, D. R., 1967, Dielectric constant and electrical conductivity measurements of moist rock: A new laboratory method: *J. Geophys. Res.*, **72**, 5101–5115.
- Sen, P. N., 1981, Relation of certain geometrical features to the dielectric anomaly of rocks: *Geophysics*, **46**, 1714–1720.
- , 1987, Electrolytic conduction past arrays of charged insulating spheres: *J. Chem. Phys.*, **87**, 4100–4107.
- Sen, P. N., Chew, W. C., and Wilkinson, D., 1984, Dielectric enhancement due to geometrical and electrochemical effects: in, Johnson, D. L., and Sen, P. N., Eds., *Physics and chemistry of porous media*: Proc. Am. Inst. Phys., 107.
- Sen, P. N., Goode, P. A., and Sibbit, A., 1988, Electrical conduction in clay-bearing sandstones at low and high salinities: *J. Appl. Phys.*, **63**, 4832–4840.
- Sen, P. N., Scala, C., and Cohen, M. H., 1981, A self-similar model for sedimentary rocks with application to the dielectric constant of fused glass beads: *Geophysics*, **45**, 781–795.
- Vinegar, H. J., and Waxman, M. H., 1984, Induced polarization of shaly sands: *Geophysics*, **49**, 1267–1287.
- Wobschall, D., 1977, A theory of the complex dielectric permittivity of soil containing water: The semidisperse model: *IEEE Trans. Geosci. Electr.*, **GE-15**, 49–58.

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H. Cheng and S. Torquato

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References

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Effective conductivity of periodic arrays of spheres with interfacial resistance

BY H. CHENG AND S. TORQUATO

Department of Civil Engineering and Operations Research and Princeton Materials Institute, Princeton University, Princeton, NJ 08544, USA

We consider the problem of analytically determining the effective thermal conductivity of a composite material consisting of periodic arrays of spheres with interfacial resistance. We applied Rayleigh's method which has been used extensively for such calculations in the perfect interface case, i.e. no jump in the temperature across the interface. Results are presented for simple, body-centred and face-centred cubic arrays, and each for a wide range of volume fractions. Our calculations were based on very accurate lattice sums in all three lattice cases and temperature fields were resolved to very high multipole moments, including all azimuthal terms. In the case of zero interfacial strength, our formulation recovers all previously reported benchmark results.

1. Introduction

There exists a vast literature on predicting the effective conductivity of two-phase composite materials (Rayleigh 1892; Batchelor 1974; Willis 1981; Hashin 1983; Torquato 1991). Virtually all of these studies adopt the assumption of a perfect-interface between the constituents, implying that the potential field and the normal component of the flux are continuous across the two-phase interface. In a variety of applications, the perfect interface assumption cannot be made. For example, thermal resistance at the interface (due to roughness or acoustic mismatch between the phases) can dramatically decrease the effective conductivity of the composite (Chiew & Glandt 1983; Benveniste 1987; Every *et al.* 1992; Lipton & Vernescu 1996). This interfacial resistance is measured by a jump in the temperature across the interface. Recently, Torquato & Rintoul (1995) used classical minimum energy principles to find very sharp, rigorous bounds on the effective properties of a class of composites with imperfect interfaces that incorporates crucial microstructural information about the interface.

In this paper, we have generalized an analytical tool—Rayleigh's method—to find the effective conductivity of periodic arrays of spherical inclusions with imperfect interfaces. Despite the fact that it was devised more than 100 years ago, Rayleigh's method has received a great deal of attention in recent years due to the dramatic increase in computational power. It has been applied by a variety of investigators (McPhedran & McKenzie 1978; McKenzie *et al.* 1978; Sangani & Acrivos 1983; Hinson & Felderhof 1992; Greengard & Moura 1994) to study the perfect-interface case.

We extend Rayleigh's method in the present work to take into account the effect of the interface in terms of a dimensionless parameter R which characterizes the inter-

facial resistance. Our approach is essentially an extension of the work of McKenzie *et al.* (1978), but we take into account all of the azimuthal terms, and we choose a different unit cell in the body-centred and face-centred cubic lattice cases which results in different lattice sums and different linear systems to be solved numerically. Another feature of our calculation is that we have used extremely accurate lattice sums (Berman & Greengard 1995), and the solution is resolved until it exhibits a spectral decaying rate in non-singular cases. Hence our results achieved spectral accuracy in describing both the field and the effective conductivity.

The remainder of the paper is organized as follows: in §2, we describe in detail the boundary-value problem for the temperature field and the solution techniques. In §3, an exact formula for effective conductivity has been derived to give results to the order of $O(f^9)$, where f is the volume fraction of the spheres. In §4, we present all of our numerical results and discuss them. In §5, we show that an approximation expression for the effective conductivity that incorporates a three-point microstructural parameter (see Torquato & Rintoul 1996) predicts the effective conductivity remarkably well.

2. Formulation and solution technique

(a) Formulation of the problem

We consider a composite material consisting of periodic arrays of equal-sized spherical inclusions of radius a and conductivity σ_2 embedded in a uniform matrix of conductivity σ_1 , in which there exists a thermal resistance at the interface between the two phases. The strength of this resistance is characterized by a dimensionless parameter R which can be defined through a limiting process by first examining a three-phase composite in which each inclusion possesses a concentric coating of thickness δ and conductivity σ_c and then letting $\delta \rightarrow 0$ and $\sigma_c \rightarrow 0$ (see Torquato & Rintoul 1996), i.e.

$$R = \frac{\tilde{R}\sigma_2}{a}, \quad \text{with} \quad \tilde{R} = \lim_{\substack{\delta \rightarrow 0 \\ \sigma_c \rightarrow 0}} \frac{\delta}{\sigma_c}. \quad (2.1)$$

Unlike the perfect interface case, this is equivalent to prescribing a temperature jump across the interface, namely,

$$J_n = \frac{1}{R}[T_- - T_+]. \quad (2.2)$$

where T_+ and T_- are the temperatures as one approaches the inclusion surface from the matrix and particle side respectively, and

$$J_n = -\sigma_1 \frac{\partial T_+}{\partial n} = -\sigma_2 \frac{\partial T_-}{\partial n} \quad (2.3)$$

is the normal component of the heat flux at the interface which is continuous across the interface.

Without loss of generality, let us apply a uniform field with strength E_0 along the negative z -axis to the composite material. The harmonic temperature field is governed by the following boundary value problem inside the cubic unit periodic cell Ω with sides of length 1:

$$\nabla \cdot (\sigma \nabla T) = 0 \quad \text{in} \quad \Omega, \quad (2.4)$$

$$\sigma(\mathbf{x}) = \begin{cases} \sigma_2 & \text{if } \mathbf{x} \in V, \\ \sigma_1 & \text{if } \mathbf{x} \in \Omega \setminus V, \end{cases}$$

$$\frac{\sigma_2}{Ra}(T_+ - T_-) = \sigma_2 \frac{\partial T_-}{\partial n} = \sigma_1 \frac{\partial T_+}{\partial n} \quad \text{on } \partial V, \quad (2.5)$$

with periodic boundary conditions given by

$$\begin{aligned} T(x+1, y, z) &= T(x, y, z), \\ T(x, y+1, z) &= T(x, y, z), \\ T(x, y, z+1) &= T(x, y, z) - E_0. \end{aligned} \quad (2.6)$$

Here V is the volume occupied by the spherical inclusion, ∂V is the sphere surface, and $\partial/\partial n$ is the outward normal derivative on ∂V .

In the case of a perfect interface ($R = 0$), a formal solution to the above partial differential equation can be obtained analytically using the spherical-harmonics-expansion procedure devised by Lord Rayleigh (1891) in his pioneering work. For our case ($R > 0$), we will show in the next subsection that the similar procedure can be carried out to obtain an analytical solution with its coefficients satisfying an infinite-dimensional linear system of equations.

After solving equations (2.4)–(2.6), our remaining task is to find the effective conductivity for the composite material. In the perfect interface case, Lord Rayleigh and many others showed that a simple application of Green's theorem could accomplish this, yielding a formula for the effective conductivity in terms of the dipole moment of the solution. For the imperfect interface case, this becomes much more complicated because the definition of proper average quantities in the composite is different from that in a composite with perfect interfaces, which will now includes a surface integral involving the temperature jump across the interface, and it is impossible to rely on Green's theorem again in our case. But we will show in §2e that the effective conductivity can be derived again from the temperature field by using the field representations directly and surprisingly we arrived at the same formula as in the perfect interface case.

(b) Solution technique

To solve equations (2.4)–(2.6), we represent the temperature field in terms of spherical harmonics (Hobson 1955; Jackson 1975; Wallace 1985; Greengard 1988). The potential inside the sphere is given by

$$T_2(r, \theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n C_{nm} r^n Y_n^m(\theta, \phi), \quad (2.7)$$

and the potential outside the sphere by

$$T_1(r, \theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n \left(A_{nm} r^n + \frac{B_{nm}}{r^{n+1}} \right) Y_n^m(\theta, \phi), \quad (2.8)$$

where functions of the form $Y_n^m(\theta, \phi)r^n$ are the regular solid harmonics of degree n , while those of form $Y_n^m(\theta, \phi)/r^{n+1}$ are called irregular solid harmonics of degree $-n-1$. Here (r, θ, ϕ) is the spherical coordinate of point $\mathbf{x}$. Following the notation of Greengard (1988), the Y_n^m are defined in terms of the associated Legendre functions

by

$$Y_n^m(\theta, \phi) = \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} \cdot P_n^{|m|}(\cos \theta) e^{im\phi}. \quad (2.9)$$

With these representations, equation (2.4) is automatically satisfied. Imposition of the boundary condition (2.5) on the sphere surface and the use of orthogonal properties of spherical harmonics yields

$$\begin{aligned} \sigma_2 C_{nm} n a^{n-1} &= \sigma_1 \left(A_{nm} n a^{n-1} - (n+1) \frac{B_{nm}}{a^{n+2}} \right) \\ &= \frac{\sigma_2}{Ra} \left[(A_{nm} - C_{nm}) a^n + \frac{B_{nm}}{a^{n+1}} \right], \end{aligned}$$

from which we obtain

$$A_{nm} = \frac{B_{nm}}{a^{2n+1}} \cdot \frac{\alpha + (n+1)/n + (n+1)R}{1 - \alpha + nR}, \quad (2.10)$$

$$C_{nm} = \frac{B_{nm}}{a^{2n+1}} \cdot \frac{2n+1}{n(1 - \alpha + nR)}, \quad (2.11)$$

where $\alpha = \sigma_2/\sigma_1$ is the conductivity ratio. However, the relations above are not enough to uniquely determine the coefficients A_{nm} , B_{nm} and C_{nm} . These coefficients are determined by requiring that our solution representations satisfy a consistency condition, which also enforces the periodic boundary conditions (2.6):

$$\sum_{n=0}^{\infty} \sum_{m=-n}^n A_{nm} r^n Y_n^m(\theta, \phi) = E_0 z + \sum_{k \neq 0} \sum_{n=0}^{\infty} \sum_{m=-n}^n \frac{B_{nm}}{\rho_k^{n+1}} Y_n^m(\theta_k, \phi_k), \quad (2.12)$$

where the index k runs over all spheres except the one at the origin, and $(\rho_k, \theta_k, \phi_k)$ is the spherical coordinate of the point (r, θ, ϕ) when measured in the local coordinate system centred at the k^{th} sphere. The last term can be expanded in terms of coordinates (r, θ, ϕ) by using the addition theorem (Greengard 1988) for spherical harmonics:

$$\frac{Y_n^m(\theta_k, \phi_k)}{\rho_k^{n+1}} = \sum_{n=0}^{\infty} \sum_{m=-n}^n G(n, m; i, j; k, 0) r^n Y_n^m(\theta, \phi), \quad (2.13)$$

where

$$G(n, m; i, j; k, 0) = (-1)^n J_j^m \cdot \frac{H_i^j H_n^m}{H_{n+i}^{j-m}} \cdot \frac{Y_{n+i}^{m-j}(\alpha_{k0}, \beta_{k0})}{\rho_{k0}^{n+i+1}},$$

and $(\rho_{k0}, \alpha_{k0}, \beta_{k0})$ is the spherical coordinate of the k th sphere centre in the current spherical coordinate system, and $H_n^m, J_m^{m'}$ are defined by

$$H_n^m = \frac{(-1)^n}{\sqrt{(n-m)! \cdot (n+m)!}}, \quad (2.14)$$

$$J_m^{m'} = \begin{cases} (-1)^{\min(|m'|, |m|)}, & \text{if } m \cdot m' > 0; \\ 1, & \text{otherwise.} \end{cases} \quad (2.15)$$

Substituting (2.13) and (2.10) to (2.12), using the orthogonal properties of the spherical harmonics once again, we arrive at (after some rearrangements) the following

infinite linear system of equations for B_{nm} :

$$\frac{B_{nm}}{a^{2n+1}} \frac{\alpha + (n+1)/n + (n+1)R}{1 - \alpha + nR} = E_0 \delta_{n1} \delta_{m0} + \sum_{i=0}^{\infty} \sum_{j=-i}^i (-1)^i J_m^j \mathcal{L}_{n+i}^{j-m} \Upsilon_{n+m}^{i-j} \Upsilon_{n-m}^{i+j} B_{ij} \quad (2.16)$$

for $n = 0, 1, \dots, \infty$, $m = -n, \dots, n$. Here δ_{ij} is the standard Kronecker delta,

$$\mathcal{L}_n^m = \sum_{p \in \Lambda_3} Y_n^m(\theta_p, \phi_p) / r_p^{n+1}, \quad n \geq 1, \quad m = -n, \dots, 0, \dots, n \quad (2.17)$$

are three-dimensional lattice sums, where Λ_3 denotes the set

$$\{(k_1, k_2, k_3) \mid k_i \in \mathbf{Z}, (k_1, k_2, k_3) \neq (0, 0, 0)\}.$$

And Υ_k^l represents

$$\Upsilon_k^l = \sqrt{\binom{k+l}{k}} \quad (2.18)$$

for any integer k, l . The linear system (2.16) is valid in all three cubic lattice cases that we are going to deal with. The difference among them lies at the fact that (2.17) will have different values for each lattice case.

From (2.16), we can see that the effect of the interface is to change its left-hand side. When $R = 0$, (2.16) reduces to the linear systems of references McPhedran & McKenzie (1978) and McKenzie *et al.* (1978).

(c) Numerical solutions

In practice, the infinite system (2.16) has to be made finite by ignoring all multipole moments beyond a given order. In addition, the full symmetry of cubic arrays makes it possible to greatly reduce the number of unknowns of (2.16). First we observe that, for our choice of the unit cell described in the next subsection, the temperature is an odd function of θ , and hence only those spherical harmonics with odd degree in n are needed. Moreover, it must be a symmetric function of ϕ about $\phi = \pi/4$, thus only those spherical harmonics whose order m is an exact multiple of 4 can contribute. After truncation and some simplification, (2.16) becomes

$$\begin{aligned} & \frac{B_{2n-1,m}}{a^{4n-1}} \frac{\alpha + 2n/(2n-1) + 2nR}{1 - \alpha + (2n-1)R} \\ & + \sum_{i=1}^K \sum_{j=0}^{2i-1} B_{2i-1,j} \frac{\epsilon_j}{2} [\mathcal{L}_{2(n+i-1)}^{j-m} \Upsilon_{2n-1+m}^{2i-1-j} \Upsilon_{2n-1-m}^{2i-1+j} + \mathcal{L}_{2(n+i-1)}^{j+m} \Upsilon_{2n-1-m}^{2i-1-j} \Upsilon_{2n-1+m}^{2i-1+j}] \\ & = E_0 \delta_{n1} \delta_{m0}, \end{aligned} \quad (2.19)$$

for $n = 1, 2, \dots, K$, $m = 0, 4, \dots, m \leq 2n-1$. Here m, j run over only numbers which are exact multiples of 4, and ϵ_j is the Neumann symbol,

$$\epsilon_j = \begin{cases} 1, & \text{if } j = 0; \\ 2, & \text{otherwise.} \end{cases}$$

Notice also that the linear system (2.19) is symmetric. Relation (2.19) is the final linear system which we solve numerically for simple, body-centred and face-centred cubic lattice of spherical inclusions.

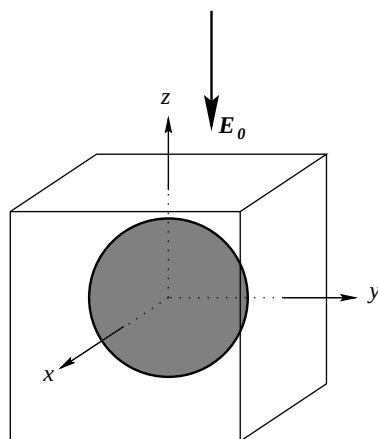


Figure 1. The simple cubic lattice structure.

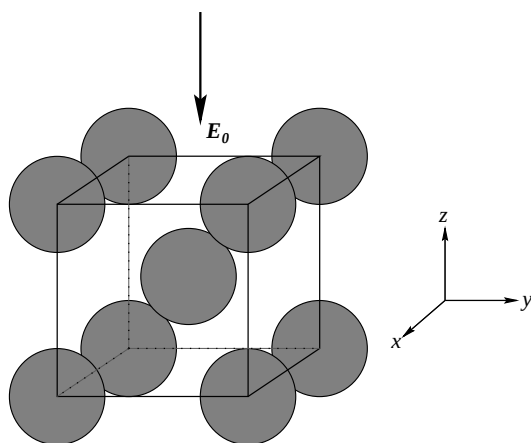


Figure 2. The body-centred cubic lattice structure.

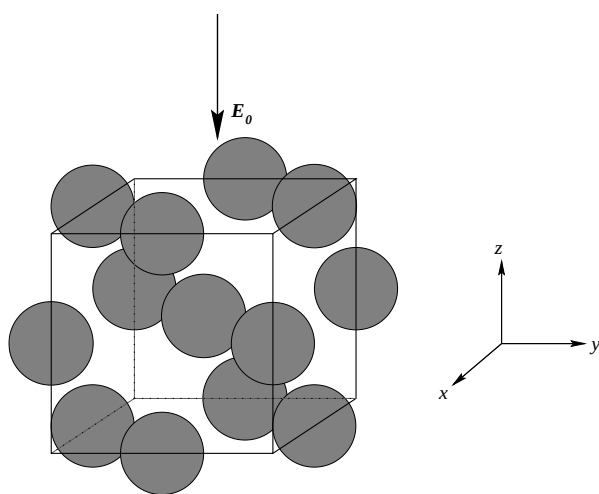


Figure 3. The face-centred cubic lattice structure.

(d) Choice of unit cell and lattice sums

Before we present our numerical results, we need to clarify our choices of the unit computational cell and the corresponding lattice sums. Certainly, one can not change material properties by choosing different computational cells, but the choice can affect the computing process.

Figure 1, 2 and 3 show the unit cells and the location of the spherical inclusions used in our calculations. They are all conventional, but differ from the approach taken in McPhedran & McKenzie (1978) and McKenzie *et al.* (1978) for body and face-centred cubic lattice cases. The lattice sums for the simple cubic lattice have been computed very accurately by Berman & Greengard (1995) using the renormalization method. We have generalized this method in order to generate the lattice sums for body and face-centred cubic lattice cases. The non-zero lattice sums for $n \leq 20$ and for all three lattice cases are listed in the table in the Appendix.

(e) Effective conductivity

The question we want to address now is that of obtaining the effective conductivity from the solution of (2.16). Extra attention is needed here for our composite materials whose constituent interfaces exhibit contact resistance, since the definition of the average intensity, $\langle \mathbf{E} \rangle$, and the average flux, $\langle \mathbf{J} \rangle$, are now given by

$$\langle \mathbf{E} \rangle = -\frac{1}{V} \int_V \nabla T \, dV + \frac{1}{V} \int_{\partial S} (T_1 - T_2) \mathbf{n} \, dS, \quad (2.20)$$

$$\langle \mathbf{J} \rangle = \frac{1}{V} \int_V \mathbf{J} \, dV. \quad (2.21)$$

Notice that the average potential gradient, $\langle \mathbf{E} \rangle$, in (2.20) contains an additional integral involving the temperature jump across the interface ∂S . Of course, if the temperature field is continuous across the interface (perfect interface case), then (2.20) reduces to the well-known classical form.

The effective thermal conductivity tensor of the material is now defined via the relation,

$$\langle \mathbf{J} \rangle = \boldsymbol{\sigma}_{\text{eff}} \langle \mathbf{E} \rangle, \quad (2.22)$$

where $\boldsymbol{\sigma}_{\text{eff}}$ is generally a tensor, but for our isotropic material, it is characterized by a single scalar parameter σ_{eff} through $\boldsymbol{\sigma}_{\text{eff}} = \sigma_{\text{eff}} \mathbf{I}$, where $\mathbf{I}$ is the 3×3 unit matrix. We now show that the quantity σ_{eff} can be determined from the solution of (2.19) by the same formula as in McPhedran & McKenzie (1978) and McKenzie *et al.* (1978), even though the definition of $\langle \mathbf{E} \rangle$ has been changed to include the imperfect interface. We will use a different approach by applying the representation of T directly.

Consider the simple cubic lattice case for which $V = 1$, thus

$$\begin{aligned} \langle \mathbf{J} \rangle &= \int_{V_1} \mathbf{J} \, dV + \int_{V_2} \mathbf{J} \, dV \\ &= \sigma_1 \int_{V_1} \nabla T_1 \, dV + \sigma_2 \int_{V_2} \nabla T_2 \, dV \\ &= \sigma_1 \left[\langle \mathbf{E} \rangle - (1 - \alpha) \int_{V_2} \nabla T_2 \, dV - \int_{\partial S} (T_1 - T_2) \mathbf{n} \, dS \right], \end{aligned}$$

using the fact that $\nabla^2 T_2 = 0$ and the boundary condition (2.5), the above equation

Table 1. Numerical constants $c_1, \dots, c_6$

	simple cubic	body-centred cubic	face-centred cubic
c_1	1.30472	0.12930	0.07529
c_2	0.07232	0.25696	0.24195
c_3	-0.52895	-0.09881	0.05583
c_4	0.15256	0.01313	0.02311
c_5^*	-0.30667	0.05338	-0.05244
c_6	0.01045	0.00562	9.16×10^{-7}

Table 2. Dimensionless effective thermal conductivity $\sigma_{\text{eff}}/\sigma_1$ of simple cubic lattice of spherical inclusions with imperfect interfaces versus the volume fraction f (Here $f_{\text{max}} = \pi/6$. The critical value $R_c = 9$ for $\alpha = 10$ and $R_c = 9999$ for $\alpha = 10000$.)

f	conductivity ratio $\alpha = 10$		conductivity ratio $\alpha = 10000$	
	$R = 5$	$R = 30$	$R = 5000$	$R = 20000$
0.00	1.00000	1.00000	1.00000	1.00000
0.05	1.02752	0.95688	1.03796	0.97030
0.10	1.05555	0.91497	1.07690	0.94117
0.15	1.08410	0.87420	1.11683	0.91259
0.20	1.11318	0.83447	1.15780	0.88453
0.25	1.14279	0.79567	1.19982	0.85694
0.30	1.17292	0.75766	1.24289	0.82975
0.35	1.20356	0.72027	1.28702	0.80290
0.40	1.23470	0.68329	1.33218	0.77628
0.425	1.25045	0.66487	1.35514	0.76303
0.45	1.26630	0.64643	1.37833	0.74978
0.475	1.28225	0.62790	1.40174	0.73651
0.50	1.29829	0.60913	1.42535	0.72316
0.51	1.30473	0.60151	1.43485	0.71778
0.515	1.30795	0.59767	1.43961	0.71507
0.52	1.31118	0.59379	1.44437	0.71235
0.523	1.31311	0.59145	1.44723	0.71072

can be rewritten as

$$\langle \mathbf{J} \rangle = \sigma_1 \left[\langle \mathbf{E} \rangle - (1 - \alpha) \int_{\partial S} \mathbf{x} \frac{\partial T_2}{\partial n} dS - Ra \int_{\partial S} \mathbf{n} \frac{\partial T_2}{\partial n} dS \right],$$

where $\mathbf{x}$ is the vector (x, y, z) . Now substitute (2.7) into the above equation and note that $\langle \mathbf{E} \rangle = (0, 0, E_0)$, after some computation, one obtains for the simple cubic lattice case,

$$\langle \mathbf{J} \rangle = \sigma_1 (1 - 4\pi B_{1,0}/E_0) \langle \mathbf{E} \rangle, \quad (2.23)$$

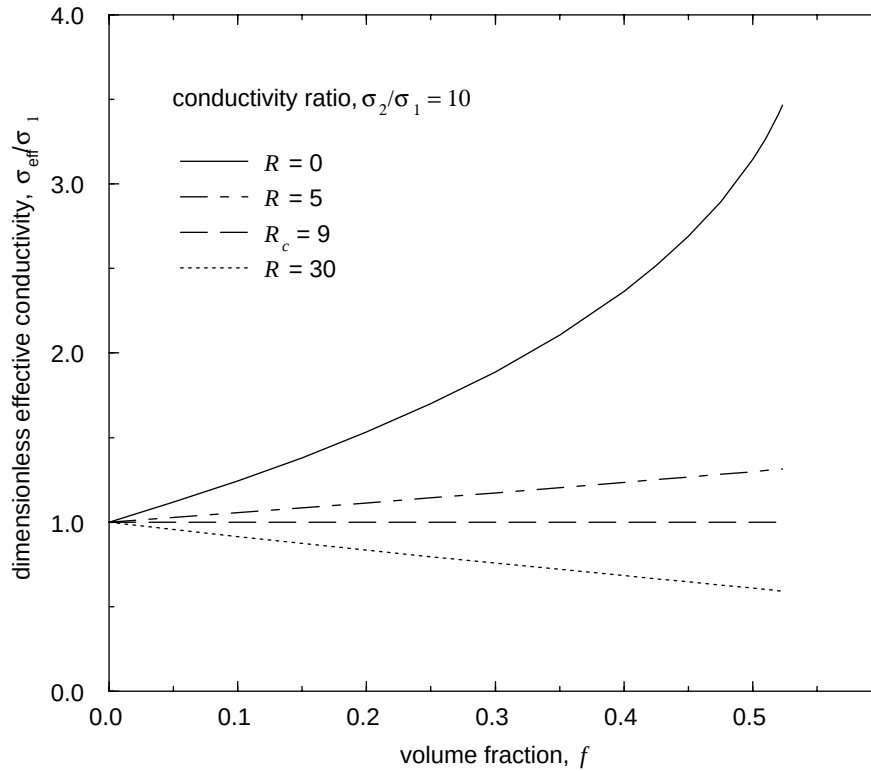


Figure 4. Dimensionless effective conductivity $\sigma_{\text{eff}}/\sigma_1$ versus volume fraction f for a simple cubic lattice with sphere to matrix conductivity ratio $\alpha = \sigma_2/\sigma_1 = 10$. The solid line is the previously known result for the perfect-interface case ($R = 0$). The dash-dotted line is for a moderate interfacial resistance ($R = 5$), the long-dashed line is for the critical interfacial resistance value ($R_c = 9$), and the dotted line for a large interfacial resistance ($R = 30$).

from which one can easily obtain the formula to compute σ_{eff}

$$\sigma_{\text{eff}} = \sigma_1(1 - 4\pi B_{1,0}/E_0). \quad (2.24)$$

Similarly, one can show that for the body-centred cubic lattice,

$$\sigma_{\text{eff}} = \sigma_1(1 - 8\pi B_{1,0}/E_0), \quad (2.25)$$

and for the face-centred cubic lattice,

$$\sigma_{\text{eff}} = \sigma_1(1 - 16\pi B_{1,0}/E_0). \quad (2.26)$$

Despite the modified definition (2.20) for the average intensity, relations (2.24)–(2.26) have exactly the same forms as in the perfect interface case, but here the coefficient $B_{1,0}$ incorporates the effect of the imperfect interface.

3. The exact solution for σ_{eff} to $O(f^9)$

In this section, we shall solve the exact solution for σ_{eff} from (2.19) to all multipoles of order 2^9 , including all the azimuthal terms. We obtained

$$\frac{\sigma_{\text{eff}}}{\sigma_1} = 1 - \frac{3f}{D}, \quad (3.1)$$

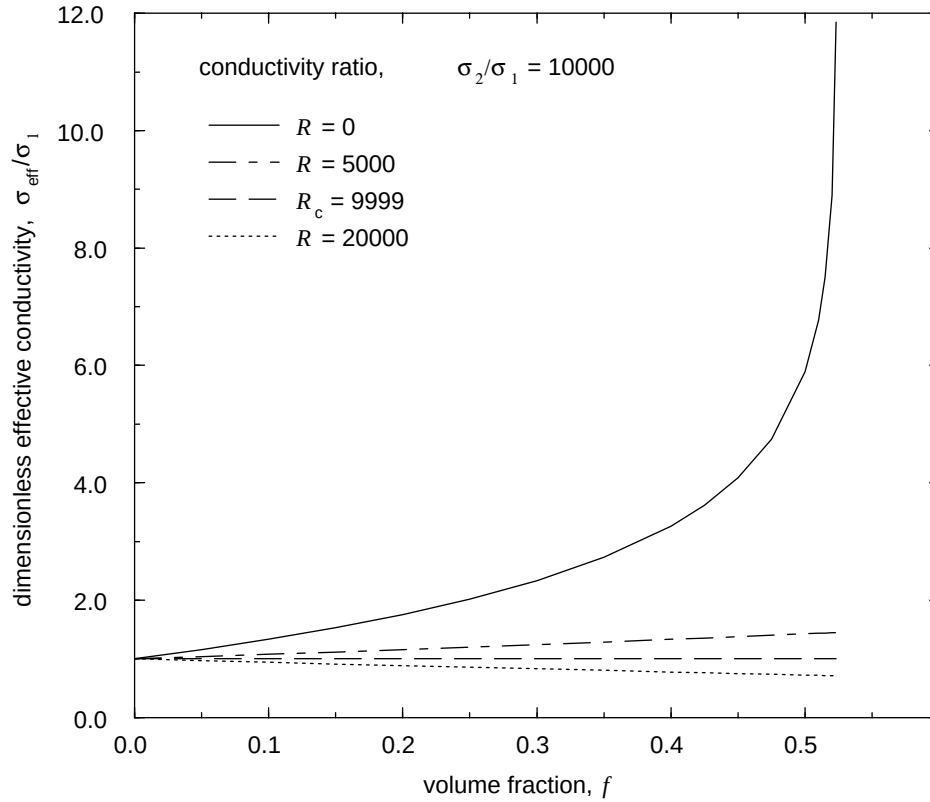


Figure 5. Dimensionless effective conductivity $\sigma_{\text{eff}}/\sigma_1$ versus volume fraction f for a simple cubic lattice with sphere to matrix conductivity ratio $\alpha = \sigma_2/\sigma_1 = 10000$. The solid line is the previously known result for the perfect-interface case ($R = 0$). The dash-dotted line is for a moderate interfacial resistance ($R = 5000$), the long-dashed line is for the critical interfacial resistance value ($R_c = 9999$), and the dotted line for a strong interfacial resistance ($R = 20000$).

where

$$D = -\beta_1^{-1} + f + c_1\beta_3f^{10/3} + c_2\beta_5f^{14/3} + c_3\beta_3^2f^{17/3} \\ + c_4\beta_7f^6 + c_5\beta_3\beta_5f^7 + c_6\beta_9f^{22/3} + O(f^{25/3}),$$

in which, β_i is defined by

$$\beta_i = \frac{\alpha - 1 - iR}{\alpha + (i+1)/i + (i+1)R}$$

and numerical constants $c_1, \dots, c_6$ for three cubic lattice arrays are computed and listed in table 1.

Formula (3.1) enables one to estimate the effective conductivity to the order of $O(f^9)$ for composite materials with imperfect interfaces, with the interfacial resistance incorporated in the constants β_i . Our calculated results of constants $c_1, \dots, c_6$ agreed extremely well with the results reported in Sangani & Acrivos (1982) for the perfect interface case, except a sign error occurred at c_5 in Sangani & Acrivos (1982). The results in McPhedran & McKenzie (1978) and McKenzie *et al.* (1978) were in-

Table 3. *Dimensionless effective thermal conductivity $\sigma_{\text{eff}}/\sigma_1$ of body-centred cubic lattice of spherical inclusions with imperfect interfaces versus the volume fraction f*
 (Here $f_{\text{max}} = \sqrt{3}\pi/8$. The critical value $R_c = 9$ for $\alpha = 10$ and $R_c = 9999$ for $\alpha = 10000$.)

f	conductivity ratio $\alpha = 10$		conductivity ratio $\alpha = 10000$	
	$R = 5$	$R = 30$	$R = 5000$	$R = 20000$
0.00	1.00000	1.00000	1.00000	1.00000
0.05	1.02752	0.95688	1.03796	0.97030
0.10	1.05556	0.91498	1.07690	0.94117
0.15	1.08411	0.87424	1.11684	0.91261
0.20	1.11320	0.83462	1.15784	0.88460
0.25	1.14284	0.79604	1.19991	0.85710
0.30	1.17304	0.75845	1.24310	0.83010
0.35	1.20380	0.72176	1.28742	0.80356
0.40	1.23513	0.68589	1.33291	0.77745
0.425	1.25100	0.66822	1.35608	0.76454
0.45	1.26701	0.65072	1.37955	0.75171
0.475	1.28315	0.63337	1.40330	0.73897
0.50	1.29943	0.61615	1.42732	0.72630
0.525	1.31583	0.59903	1.45163	0.71368
0.55	1.33236	0.58200	1.47621	0.70113
0.575	1.34901	0.56501	1.50104	0.68860
0.60	1.36576	0.54803	1.52611	0.67609
0.625	1.38261	0.53099	1.55140	0.66356
0.65	1.39954	0.51378	1.57688	0.65098
0.66	1.40633	0.50682	1.58712	0.64591
0.67	1.41312	0.49978	1.59737	0.64082
0.68	1.41993	0.49263	1.60764	0.63568

complete due to the fact that all the azimuthal terms were neglected, as pointed out by Sangani & Acrivos (1982).

4. Results and discussion

The general method that we have used here is well-established and has been proven to give accurate results in the perfect interface case (McPhedran & McKenzie 1978; McKenzie *et al.* 1978). We have checked our algorithm with the results reported there and found excellent agreement in general. We will not repeat the perfect interface results here but instead will focus on the imperfect interface case. We implemented the method in double precision in FORTRAN with the finite linear system (2.19) being solved directly by LAPACK subroutine DSYSV, which is an indefinite symmetric linear system solver, a well suited tool for our purpose.

The results for simple cubic lattice arrays are presented in table 2 and figures 4 and 5. Columns 2 and 3 in table 2 list the dimensionless effective thermal conductivity $\sigma_{\text{eff}}/\sigma_1$ with conductivity ratio $\alpha = \sigma_2/\sigma_1 = 10$ for two different values of the

Table 4. *Dimensionless effective thermal conductivity $\sigma_{\text{eff}}/\sigma_1$ of face-centred cubic lattice of spherical inclusions with imperfect interfaces versus the volume fraction f*
 (Here $f_{\text{max}} = \sqrt{3}\pi/8$. The critical value $R_c = 9$ for $\alpha = 10$ and $R_c = 9999$ for $\alpha = 10000$.)

f	conductivity ratio $\alpha = 10$		conductivity ratio $\alpha = 10000$	
	$R = 5$	$R = 30$	$R = 5000$	$R = 20000$
0.00	1.00000	1.00000	1.00000	1.00000
0.05	1.02752	0.95688	1.03796	0.97030
0.10	1.05556	0.91498	1.07690	0.94117
0.15	1.08411	0.87425	1.11684	0.91261
0.20	1.11320	0.83462	1.15784	0.88460
0.25	1.14285	0.79606	1.19991	0.85711
0.30	1.17305	0.75849	1.24311	0.83012
0.35	1.20382	0.72183	1.28744	0.80359
0.40	1.23515	0.68600	1.33295	0.77750
0.425	1.25103	0.66837	1.35614	0.76461
0.45	1.26705	0.65091	1.37963	0.75180
0.475	1.28321	0.63360	1.40340	0.73908
0.50	1.29950	0.61642	1.42746	0.72643
0.525	1.31593	0.59936	1.45181	0.71385
0.55	1.33247	0.58240	1.47643	0.70132
0.575	1.34915	0.56550	1.50131	0.68884
0.60	1.36594	0.54864	1.52646	0.67639
0.625	1.38284	0.53179	1.55184	0.66395
0.65	1.39983	0.51489	1.57744	0.65150
0.675	1.41691	0.49789	1.60324	0.63902
0.70	1.43406	0.48071	1.62922	0.62647
0.71	1.44093	0.47375	1.63965	0.62142
0.72	1.44781	0.46674	1.65009	0.61634
0.73	1.45469	0.45964	1.66066	0.61123
0.74	1.46158	0.45244	1.67101	0.60608

dimensionless resistance R and a wide range of volume fractions. Figure 4 shows corresponding plots of these results. Columns 4 and 5 of table 2 and figure 5 are for arrays with an virtually infinite contrast ratio, i.e. $\alpha = 10000$. The figures also include results for the perfect interface case (solid lines).

It can be seen that the interfacial resistance has a dramatic effect on the effective property. It has been shown that there exists a critical value $R_c = \alpha - 1$ of the dimensionless parameter R such that when $R = R_c$ the effective conductivity of the composite material equals exactly the matrix conductivity (Torquato & Rintoul 1995), i.e. the inclusions are effectively ‘hidden’. More significantly, when $R > R_c$, the more conducting spherical inclusions actually reduce the conductivity of the composite material below that of the matrix. This effect can be seen from the figure where the long dashed lines represent the results when R equals the critical value R_c . The critical value R_c is directly related to the notion of a critical radius first

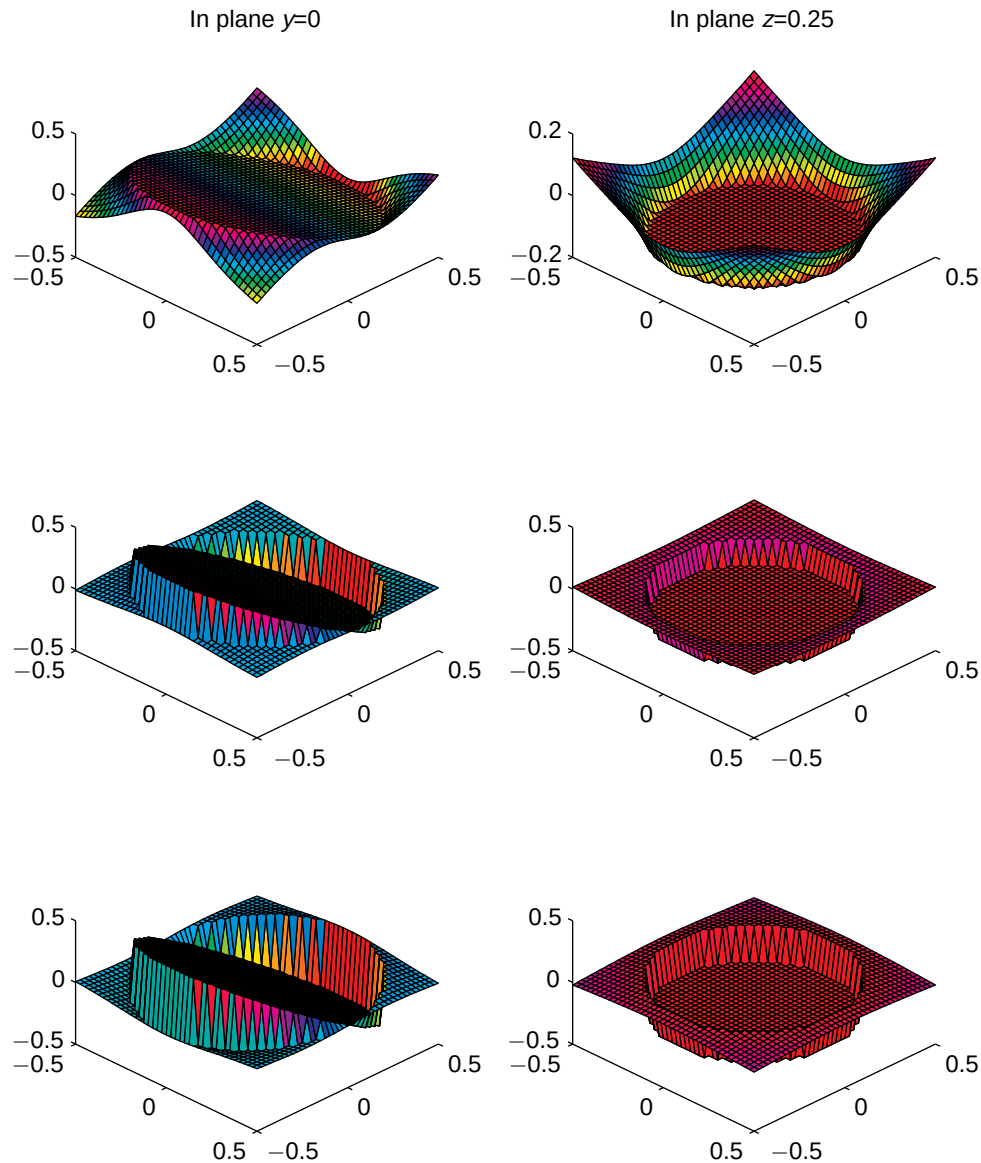


Figure 6. The surface plot of potential in plane $y = 0$ (first column) and plane $z = 0.25$ (second column). The material is composed of a simple cubic lattice of spheres with conductivity ratio $\alpha = 10$ and volume fraction $f = 0.5$. The first row is for $R = 0$ (no interfacial resistance), the second is for $R = 5$ and the third for $R = 30$. The jump of the potential at the interface is shown clearly in the latter two cases.

observed by others (Chiew & Glandt 1983; Every *et al.* 1991; Lipton & Vernescu 1996).

Tables 3 and 4 give corresponding results for body-centred and face-centred cubic arrays, respectively. Qualitatively, these results are similar to those for the simple cubic array illustrated in figures 4 and 5 and hence are not depicted graphically.

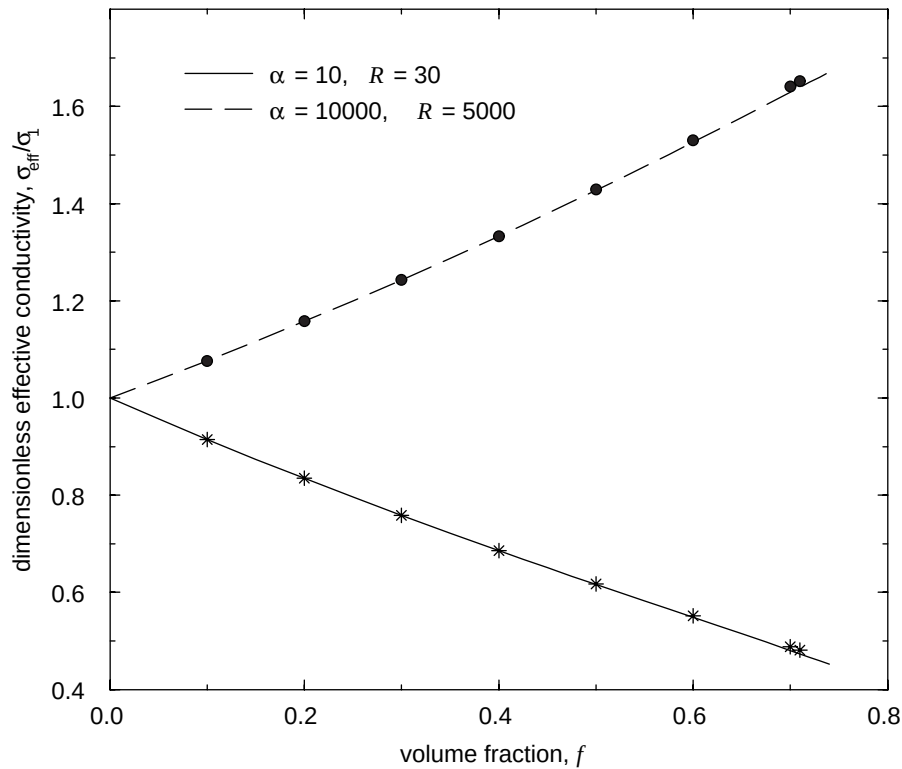


Figure 7. Dimensionless effective conductivity $\sigma_{\text{eff}}/\sigma_1$ versus volume fraction f for face-centred cubic lattices of spheres. The solid line is the exact numerical result from §4 for conductivity ratio $\alpha = 10$ and interfacial resistance $R = 30$, while the dashed line is for $\alpha = 10000$ and $R = 5000$. The stars and filled circles represent the respective predicted values of formula (5.1).

Finally, we present in figure 6 surface plots of the temperature fields in the plane $y = 0$ and $z = 0.25$ for the simple cubic lattice case for the conductivity ratio $\alpha = 10$ and volume fraction $f = 0.5$. The first, second and third rows of Figure 6 corresponds to $R = 0$, $R = 5$ and $R = 30$ respectively. The results for body-centred and face-centred cubic arrays are similar and hence are not plotted.

5. An accurate approximation formula

The asymptotic formula (3.1) we presented in §3 can be used to estimate the effective conductivity, but it yields accurate results only for moderate values of the volume fraction. For example, in the simple cubic lattice case, we examined and found that (3.1) provides useful results only when $f < 0.3$ for the imperfect interface case. In this section, we will present a simpler but much more powerful formula (5.1), based not just on the volume fraction alone, but also on a three-point microstructural parameter ζ_2 . In the perfect interface case, such a formula has been shown (Torquato 1985) to be able to predict the effective conductivity remarkably well for both periodic and random arrays of inclusions and for a wide range of volume fractions and conductivity ratios. We will show here that an appropriately generalized

Table 5. The non-zero lattice sums $\mathcal{L}_n^m$ for $n \leq 20$

n	m	simple cubic	body-centred cubic	face-centred cubic
2	0	$2\pi/3$	$4\pi/3$	$8\pi/3$
4	0	0.310822E+01	-0.310646E+01	-0.752569E+01
4	4	0.185752E+01	-0.185647E+01	-0.449746E+01
6	0	0.573329E+00	0.544656E+01	-0.266349E+02
6	4	-0.107260E+01	-0.101896E+02	0.498293E+02
8	0	0.325929E+01	0.764839E+01	0.811865E+02
8	4	0.122566E+01	0.287618E+01	0.305302E+02
8	8	0.186744E+01	0.438222E+01	0.465166E+02
10	0	0.100922E+01	-0.939666E+01	-0.152435E+01
10	4	-0.101696E+01	0.946866E+01	0.153603E+01
10	8	-0.121042E+01	0.112700E+02	0.182824E+01
12	0	0.289125E+01	0.486715E+01	-0.938077E+02
12	4	0.891680E+00	-0.692780E+01	-0.281749E+03
12	8	0.106997E+01	0.139589E+02	0.329947E+03
12	12	0.158878E+01	-0.244497E+00	-0.139104E+03
14	0	0.115364E+01	0.152759E+02	-0.291581E+03
14	4	-0.848340E+00	-0.112333E+02	0.214418E+03
14	8	-0.910342E+00	-0.120543E+02	0.230089E+03
14	12	-0.110537E+01	-0.146368E+02	0.279381E+03
16	0	0.279236E+01	-0.120685E+02	0.909996E+03
16	4	0.800719E+00	0.197893E+02	0.406632E+03
16	8	0.841132E+00	-0.156788E+02	0.198649E+03
16	12	0.947940E+00	-0.277459E+02	0.160735E+03
16	16	0.150058E+01	0.284767E+01	0.547506E+03
18	0	0.125802E+01	-0.847180E+01	-0.391631E+02
18	4	-0.749851E+00	0.146809E+01	0.492731E+03
18	8	-0.784928E+00	0.206436E+02	-0.198831E+04
18	12	-0.845512E+00	-0.117252E+02	0.230920E+04
18	16	-0.104366E+01	0.123007E+02	-0.658506E+03
20	0	0.270422E+01	0.346397E+02	-0.121337E+04
20	4	0.709961E+00	-0.309409E+02	-0.350522E+04
20	8	0.734552E+00	0.162299E+02	0.213314E+03
20	12	0.783313E+00	0.358415E+02	0.170274E+04
20	16	0.891188E+00	0.433054E+02	0.213845E+04
20	20	0.141533E+01	0.408147E+00	-0.204563E+04

version of the formula for $R > 0$ (Torquato & Rintoul 1996) can also provide very good estimate of the effective conductivity in the imperfect interface case.

Torquato & Rintoul (1996) derived the following approximate formula for the effective conductivity σ_{eff} of a suspension (random or periodic) with the dimensionless

resistance R ,

$$\frac{\sigma_{\text{eff}}}{\sigma_1} = \frac{1 + 2f\beta_1 - 2(1-f)\zeta_2\beta_1^2}{1 - f\beta_1 - 2(1-f)\zeta_2\beta_1^2}, \quad (5.1)$$

where the parameter β_1 is given in §3, and ζ_2 is the aforementioned three-point microstructural parameter. In the perfect interface case, β_1 reduces to the parameter β_{21} contained in the approximate formula of Torquato (1985). For the three cubic lattices of spheres, ζ_2 has been tabulated as a function of the volume fraction f by McPhedran & Milton (1981).

Figure 7 compares the exact numerical results for $\sigma_{\text{eff}}/\sigma_1$ from §4 to corresponding results predicted by formula (5.1) for the face-centred cubic lattice case. It is seen that formula (5.1) provides an excellent estimate of $\sigma_{\text{eff}}/\sigma_1$ for different values of conductivity ratio, interfacial resistance and a wide range of volume fractions. The predictions of (5.1) for the simple and body-centred cubic lattice cases are equally good and hence are not shown graphically. In the perfect-interface case (Torquato 1985), (5.1) was shown to give good estimates for random configurations as well. Torquato & Rintoul (1996) have shown that (5.1) yields good estimates of the effective conductivity of random dispersions with imperfect interfaces.

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Appendix A. The lattice sums

We list in table 5 the non-zero lattice sums for $n \leq 20$ and for all three cubic lattice arrays. For details of the renormalization method that we used to calculate these sums and more complete results for the simple cubic lattice sums, we refer to the original paper by Berman & Greengard (1995).

References

- Batchelor, G. K. 1974 Transport properties of two-phase materials with random structure. *A. Rev. Fluid Mech.* **6**, 227–255.
- Benveniste, Y. 1987 Effective thermal conductivity of composites with a thermal contact resistance between the constituents: non-dilute case. *J. Appl. Phys.* **61**, 2840–2843.
- Berman, C. L. & Greengard, L. 1994 A renormalization method for the evaluation of lattice sums. *J. Math. Phys.* **35**, 6036–6048.
- Bonnecaze, R. T. & Brady, J. F. 1990 A method for determine the effective conductivity of dispersions of particles. *Proc. R. Soc. Lond. A* **430**, 285–313.
- Chiew, Y. C. & Glandt, E. D. 1983 The effect of structure on the conductivity of a dispersion. *J. Coll. Interface Sci.* **94**, 90–104.
- Every, A. G., Tzou, Y., Hasselman, D. P. H. & Raj, R. 1992 The effect of particle size on the thermal conductivity of ZnS/diamond composites. *Acta Metall. Mater.* **40**, 123–129.
- Greengard, L. 1988 *The rapid evaluation of potential fields in particle systems*. Cambridge, MA: MIT Press.
- Greengard, L. & Moura, M. 1994 On the numerical evaluation of the electrostatic fields in composite materials. *Acta Numerica*, pp. 379–410. Cambridge University Press.
- Hashin, Z. 1983 Analysis of composite materials. *J. Appl. Mech.* **50**, 481–504.
- Hinsen, K. & Felderhof, B. U. 1992 Dielectric constant of a suspension of uniform spheres. *Phys. Rev. B* **46**, 12955–12963.
- Hobson, E. W. 1955 *The theory of spherical and ellipsoidal harmonics*. New York: Chelsea.
- Proc. R. Soc. Lond. A* (1997)

- Jackson, J. D. 1975 *Classical electrodynamics*. New York: Wiley.
- Lipton, R. & Vernescu, B. 1996 Composites with imperfect interfaces. *Proc. R. Soc. Lond. A* **452**, 329–358.
- McPhedran, R. C. & Milton, G. W. 1981 Bounds and exact theories for the transport properties of inhomogeneous media. *Appl. Phys. A* **26**, 207–220.
- McPhedran, R. C. & McKenzie, D. R. 1978 The conductivity of lattices of spheres. I. The simple cubic lattice. *Proc. R. Soc. Lond. A* **359**, 45–63.
- McKenzie, D., McPhedran, R. & Derrick, G. 1978 The conductivity of lattices of spheres. II. The body centred and face centred cubic lattices. *Proc. R. Soc. Lond. A* **362**, 211–232.
- McPhedran, R. C., McKenzie, D. R. & Phan-Thien, N. 1983 Transport properties of two-phase composite materials. In *Advances in the mechanics and the flow of granular materials*, pp. 413–482. Clausthal: Trans. Tech. Publications.
- Rayleigh, Lord 1892 On the influence of obstacles arranged in rectangular order upon the properties of a medium. *Phil. Mag.* **34**, 481–502.
- Sangani, A. S. & Acrivos, A. 1982 The effective conductivity of a periodic array of spheres. *Proc. R. Soc. Lond. A* **386**, 263–275.
- Torquato, S. 1985 Effective electrical conductivity of two-phase disordered composite media. *J. Appl. Phys.* **58**, 3790–2797.
- Torquato, S. 1991 Random heterogeneous media: microstructure and improved bounds on effective properties. *Appl. Mech. Rev.* **44**, 37–76.
- Torquato, S. & Lado, F. 1988 Bounds on the conductivity of a random array of cylinders. *Proc. R. Soc. Lond. A* **438**, 591–604.
- Torquato, S. & Rintoul, M. 1995 Effects of the interface on the properties of composite media. *Phys. Rev. Lett.* **52**, 2635–2638.
- Torquato, S. & Rintoul, M. 1996 Conductivities of suspensions with imperfect interfaces. (In preparation.)
- Van Bladel, J. 1964 *Electromagnetic fields*. New York: McGraw-Hill.
- Wallace P. R. 1985 *Mathematical analysis of physical problems*. New York: Dover.
- Willis, J. R. 1981 Variational and related methods for the overall properties of composite materials. In *Advances in applied mechanics* (ed. C.-S. Yih), vol. 21, pp. 2–78. New York: Academic Press.

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Nature of surface electrical conductivity in natural sands, sandstones, and clays

A. Revil

Laboratoire de Mesures en Forage, Aix-en-Provence, France

P. W. J. Glover

University of Aberdeen, Dept. of Geology and Petroleum Geology, Aberdeen, Scotland

Abstract. The electrical conductivity of rocks results from conduction through the bulk solution occupying the pores and from surface conduction occurring at the fluid/grain interface. The nature of the surface electrical conductivity of shaly and clean sands and sandstones is examined. Surface conduction is characterized by the specific surface conductance which is the sum of three contributions: (i) Conduction within the electrical diffuse layer, which makes a negligible contribution to the total specific surface conductance. (ii) Conduction in the Stern layer, which is shown to vary significantly with the salinity of the pore fluid at low salinities (10^{-6} to 10^{-3} mol l^{-1}), but becomes independent of salinity at higher salinities. (iii) A mechanism operating directly on the mineral surface, independent of salinity, and perhaps associated with proton transfer. At salinities higher than 10^{-3} mol l^{-1} and at $25^{\circ}C$, the specific surface conductance of quartz and clays is equal to 8.9×10^{-9} S and 2.5×10^{-9} S respectively. Equations describing the influence of surface conductivity and microstructure upon the macroscopic electrical conductivity of sands, sandstones, and shales are also developed.

Introduction

Surface electrical conductivity in rocks is an important parameter commonly used in characterizing hydrocarbon reservoirs [Waxman and Smits, 1968]. It is also closely related to electrokinetic parameters [Pride, 1994] that are beginning to be used to monitor earthquake and volcanic processes. The macroscopic conductivity of a saturated porous medium with insulating grains results from two mechanisms: (i) bulk conduction, which corresponds to electromigration of the ions in the interconnected pore space, and (ii) surface conduction, which corresponds to electrical conduction in the vicinity of the fluid/grain interface [e.g., Pride, 1994; Revil and Glover, 1997]. The question concerning the importance of surface electrical conductivity relative to the electrical conductivity of the electrolyte in the pore space was studied by Waxman and Smits [1968], Watillon and De Backer [1970], and Bussian [1983]. The conclusion reached by these authors was that surface conductivity is an important contribution to macroscopic rock conductivity at low salinity or/and high temperature. However, the nature of surface conductivity remains poorly understood [e.g., Zukoski and Saville, 1986].

Our purpose is to establish the basis of a theoretical framework able to predict surface electrical conductivity in porous media formed with silica and aluminosilicates.

The Specific Surface Conductance of Quartz

The surface of grains which compose the solid matrix of sands, sandstones or clays is typically charged when it is in contact with an electrolyte. The counterions required to balance the mineral surface charge form an "electrical diffuse layer" (EDL). A theoretical model of electrical conduction in the EDL was reported by Revil and Glover [1997]. However, surface conduction in the vicinity of the mineral grains can have other contributions (Pride, personal communication). We write the specific surface conductance Σ_s in the vicinity of a mineral surface as the sum of three contributions:

$$\Sigma_s \equiv \int_0^{\infty} (\sigma(\chi) - \sigma_f) d\chi = \Sigma_s^{EDL} + \Sigma_s^{Stern} + \Sigma_s^{Prot}, \quad (1)$$

where Σ_s^{EDL} , Σ_s^{Stern} , and Σ_s^{Prot} are the contributions to the surface conductivity from ionic conduction in the EDL and in the Stern layer, and associated with proton transfer, respectively. The variable χ is the local distance from the mineral surface considered locally flat, $\sigma(\chi)$ is the local conductivity at the distance χ from the surface, and σ_f is conductivity of the bulk electrolyte. At pH > 3, in presence of a 1:1 salt (e.g., NaCl) plus an acid or a base (e.g., HCl or NaOH), Σ_s^{EDL} is given by [Revil and Glover, 1997, their Eq. (55)],

$$\Sigma_s^{EDL} = 2\chi_d e \left\{ \left(B_{(+)} C_f + B_{H^+} 10^{-pH} \right) \left[\exp \left(-\frac{e\phi_d}{2k_b T} \right) - 1 \right] + \left(B_{(-)} C_f + B_{OH^-} 10^{pH - pK_f} \right) \left[\exp \left(\frac{e\phi_d}{2k_b T} \right) - 1 \right] \right\}. \quad (2)$$

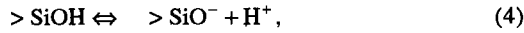
Here C_f and pH are respectively the salinity and the pH of the free electrolyte, e is the elementary charge (1.6×10^{-19} C), k_b is Boltzmann's constant (1.381×10^{-23} J K^{-1}), T is the temperature in Kelvin, $pK_f = -\log_{10} K_f \approx 13.8$ at $25^{\circ}C$ (where K_f is the dissociation constant of water), ϕ_d is the Stern plane potential, i.e., the electrical potential of the inner part of the EDL, and B_i are the equivalent ionic mobilities for ion "i" [Revil and Glover, 1997] $B_i = \beta_i + 2\epsilon_f k_b T / (\eta_f e Z_i)$ where β_i is the ionic mobility of ion "i" which is related to the self-diffusion coefficient D_i by the Nernst-Einstein relationship: $D_i = \beta_i (k_b T) / (e Z_i)$, Z_i is the valence of the ionic species "i", and η_f is the dynamic viscosity of water. The subscripts (+) and (-) refer to the electrolyte cation and anion, respectively. The parameter χ_d is the Debye screening length:

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$$\chi_d = \left(\frac{1}{2} \frac{\epsilon_f k_b T}{10^3 N e^2 (C_f + 10^{-\text{pH}})} \right)^{1/2}, \quad (3)$$

where N is Avogadro's number ($6.02 \times 10^{23} \text{ mol}^{-1}$), ϵ_f is the dielectric constant of water ($\epsilon_f = 80 \epsilon_0$, $\epsilon_0 \cong 8.84 \times 10^{-12} \text{ F}$). The chemical reactions needed to determine the Stern potential in the pH range 5–8 are given by Revil and Glover [1997] as,



where $>$ refers to the mineral lattice, M^+ is the electrolyte cation (e.g., Na^+ or K^+). We note $K_{(-)}$ and K_M the equilibrium constants of Eqs. (4) and (5) [$K_{(-)} = 10^{-7.5}$ at 25°C , *Hiemstra and Van Riemsdijk*, 1990]. In the pH range 5–8, Eqs. (104) and (105) in Revil and Glover [1997] lead to:

$$\varphi_d = \frac{2k_b T}{3e} \ln \Theta, \quad (6)$$

where Θ is defined by,

$$\Theta \equiv \left[8 \times 10^3 N \epsilon_f k_b T (C_f + C) \right]^{1/2} \left(\frac{10^{-\text{pH}} + C_f K_M}{2e \Gamma_s^0 K_{(-)}} \right), \quad (7)$$

where $C \equiv \text{Max}[10^{-\text{pH}}, 10^{\text{pH}-\text{pK}_f}]$, $\text{Max}[a; b]$ is the maximum value of a and b , and Γ_s^0 is the total surface site density. Combining Eqs. (2), (3), and (6), we have,

$$\Sigma_S^{\text{EDL}} = \left(\frac{2 \times 10^3 N \epsilon_f k_b T}{C_f + 10^{-\text{pH}}} \right)^{1/2} \left[(B_{(+)} C_f + B_{\text{H}^+} 10^{-\text{pH}}) (\Theta^{-1/3} - 1) + (B_{(-)} C_f + B_{\text{OH}^-} 10^{\text{pH}-\text{pK}_f}) (\Theta^{1/3} - 1) \right] \quad (8)$$

The Stern layer is formed by ions adsorbed on the mineral surface. These ions can move along the surface when the porous medium is submitted to an electrical field [*O'Konski*, 1960]. The Stern contribution to Σ_S can be written:

$$\Sigma_S^{\text{Stern}} = e \beta_S \Omega_M^0 \Gamma_S^0, \quad (9)$$

where β_S is the equivalent mobility of the adsorbed ions in the Stern layer and Ω_M^0 is the relative surface density of adsorbed ions, given by Eq. (84) of Revil and Glover [1997] as,

$$\Omega_M^0 = \frac{K_M C_f}{10^{-\text{pH}} + K_{(-)} + K_M C_f}. \quad (10)$$

The third contribution, Σ_S^{Prot} , is thought to be directly related to a surface conduction phenomenon such as migration of protons or electrons as suggested by *O'Konski* [1960] (also *S. Pride*, personal communication). This contribution is probably confined in a thin region on or near the surface, and is assumed independent of the salinity but dependent of the total site density of the mineral surface.

We compare the previous equations to the ζ -potential and specific surface conductance data made by *Watillon and de Backer* [1969] on a silica capillary (Fig. 1). Their data results from streaming potential and electrical conductivity measurements ($T \approx 20\text{--}25^\circ\text{C}$, electrolyte: KNO_3). We take for Σ_S^{Prot} the value of the specific surface conductance given by

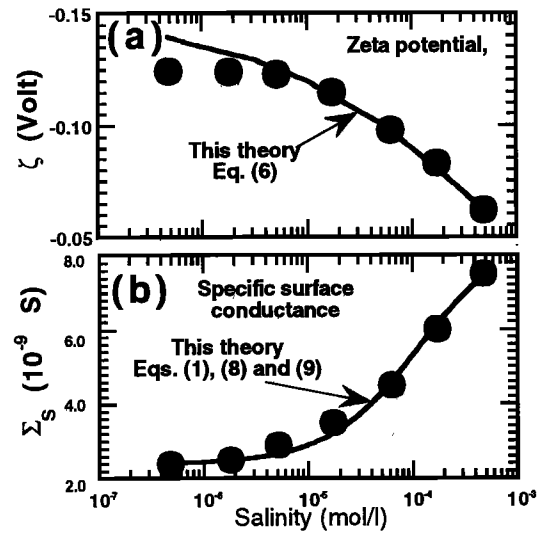


Figure 1. (a) Zeta-potential and (b) specific surface conductance as a function of salinity. The dots represent the experimental data from *Watillon and de Backer* [1970]. The zeta-potential and the specific surface conductance results from streaming potential and electrical conductivity measurements respectively. The solid lines represent the result of the model presented in this paper.

Watillon and de Backer [1969] in pure water: $\Sigma_S^{\text{Prot}} = 2.4 \times 10^{-9} \text{ S}$. We consider $\zeta \approx \varphi_d$ [*Pride*, 1994]. We use the data shown in Fig. 1 to calculate pH, Γ_s^0 , K_M and β_S by doing non-linear fitting of Eq. (1) [in which we have incorporated Eqs. (8) and (9)], and Eqs. (6) to the experimental data. The algorithm used is a standard Newton-Raphson method to solve non-linear equations. We have tested the uniqueness of the solution using a Monte-Carlo procedure in order to assess the sensitivity of the result to the initial guess for the parameters inverted. The initial values for a non-linear inversion are sampled in the possible range of each of the four parameters. The solution is unique, and the theory fits the experimental data very well over the range of salinities for which experimental observations are available ($\sim 10^{-7}\text{--}10^{-3} \text{ mol l}^{-1}$) except for the ζ -potential at the lowest values of salinity. We found from this procedure $\text{pH} = 6.64 \pm 0.04$, $\Gamma_s^0 = 10.0 \pm 0.6 \text{ sites nm}^{-2}$, $K_M = 10^{-(2.77 \pm 0.07)}$ and $\beta_S = (0.40 \pm 0.02) \times 10^{-8} \text{ m}^2 \text{s}^{-1} \text{V}^{-1}$. This result is convincing because the pH value from the inversion compared very well with the measured value reported by *Watillon and de Backer* [1969] in their experiment [6.8 ± 0.5]. The surface site density is close to the values reported by *Hiemstra and Van Riemsdijk* [1990] from geometrical considerations ($\Gamma_s^0 = 9.6 \text{ sites nm}^{-2}$ for the 001 cleavage plane, and $\Gamma_s^0 = 6.0 \text{ sites nm}^{-2}$ for the 101 cleavage plane of quartz crystals). The value of K_M agrees very well with values in the range $10^{-2.5}\text{--}10^{-3.2}$ reported by *Scales et al.* [1990]. The value of β_S is in good agreement with the value of β_S derived in the next section for clay minerals. At salinities such that $C_f \gg \text{Max}(10^{-\text{pH}}/K_M; K_{(-)}/K_M)$ (i.e., larger than $10^{-3} \text{ mol l}^{-1}$), Σ_S reaches a fairly constant value given by $\sim 8.9 \times 10^{-9} \text{ S}$. We found that the EDL contribution to the overall surface conductivity Σ_S^{EDL} is a minor component compared to the other two components (Σ_S^{Stern} and Σ_S^{Prot}), accounting for less than few percent for salinities varying between $10^{-7}\text{--}1 \text{ mol l}^{-1}$.

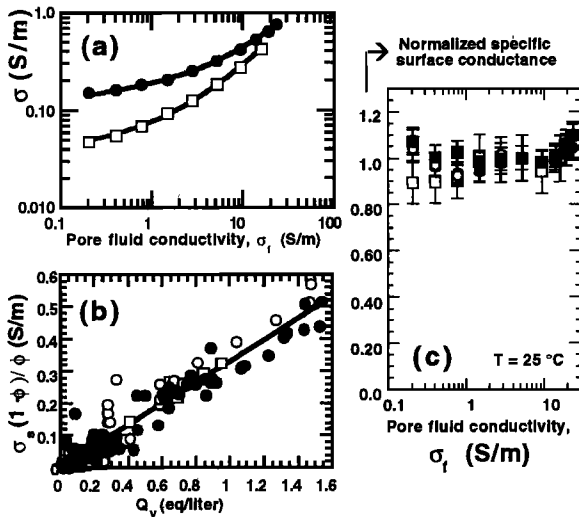


Figure 2. Electrical conductivity of shaly sands. (a) Macroscopic electrical conductivity, σ , as a function of pore fluid electrical conductivity, σ_f [data from Waxman and Smits, 1968]. Eq. (12) is used to determine the electrical formation factor, F , and the surface conductivity, σ_s ($\square$ #WS-17: $F = 46.48 \pm 0.42$, $\sigma_s = (426 \pm 11) \times 10^{-4} \text{ S m}^{-1}$; $\bullet$ #WS-26: $F = 46.95 \pm 0.63$, $\sigma_s = (1521 \pm 29) \times 10^{-4} \text{ S m}^{-1}$). (b) Surface conductivity, σ_s , scaled by $(1-\phi)/\phi$ as a function of the excess charge density per unit pore volume, Q_v , in eq l^{-1} (1 $\text{eq} = 96320 \text{ C}$). Q_v is estimated from CEC values which are determined by cation exchange experiments. Data from $\square$ Vinegar and Waxman [1984], $\bullet$ Waxman and Smits [1968], $\circ$ Hill and Milburn [1956]. The line represents a linear regression forced through the origin ($r = 0.959$). The slope of this regression gives the surface mobility of the counterions: $\beta_s = 0.51 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. (c) Normalized specific surface conductance $\Sigma_s(\sigma_f)/\Sigma_s(5.249 \text{ S/m})$ as a function of the pore fluid conductivity, σ_f . The specific surface conductance is calculated using the method described in the text from data for 4 sands samples with high clay content [#17, #23, #24 and #26, Table 7 from Waxman and Smits, 1968]. The clay fractions are: $\square$ WS#17 (K:100%, I: 0%, S: 0%), $\circ$ WS#23 (K:8%, I: 8%, S: 84%), $\bullet$ WS#24 (K:12%, I: 8%, S: 84%), $\blacksquare$ WS#26 (K:0%, I: 0%, S: 100%), (K, kaolinite; I, illite; and S, smectite).

Specific Surface Conductance of Clay Minerals

We have taken a “microscopic approach” in the previous section of this paper for quartz because it is relatively easy to have measurements of electrical conductivity (and ζ -potential) in a capillary with smoothed walls of silica. Such data are available in the literature [e.g., Watillon and De Backer, 1970]. In this case, the microgeometry is known and can be removed from the macroscopic measurements. For clays, the problem is far more complex due to the geometry of the clay particles themselves. We consider here a macroscopic approach in which the microgeometry is explicitly incorporated in the equation for the macroscopic electrical conductivity. The validity of this approach is tested using a large database of experimental results. We combine the work of Bussian [1983] based on the Hanai-Bruggeman equation with the work of Revil and Glover [1997] which takes into account the different behavior of anions and cations. We note

$t_{(+)}^f$ and $t_{(-)}^f = 1 - t_{(+)}^f$, the fraction of electrical current carried in the free electrolyte by the cations and the anions respectively. These parameters are called the “Hittorf numbers” of the cations and anions in the free electrolyte [e.g., Revil and Glover, 1997] and can be calculated by $t_{(+)}^f = \beta_{(+)}^f / (\beta_{(+)}^f + \beta_{(-)}^f)$, where $\beta_{(\pm)}^f$ (in $\text{m}^2 \text{ s}^{-1} \text{ V}^{-1}$) are the mobilities of the ions in the bulk electrolyte [for NaCl, $t_{(+)}^f \approx 0.38$]. Similarly, we note $t_{(\pm)}^s$ the fraction of current carried by anions and cations for surface conduction. The macroscopic electrical conductivity, σ , of a saturated sand or sandstone is given by $\sigma = \sigma_{(+)} + \sigma_{(-)}$ where each ionic contribution $\sigma_{(\pm)}$ is given by the Hanai-Bruggeman equation [Bussian, 1983]:

$$\sigma_{(\pm)} = \frac{t_{(\pm)}^f \sigma_f}{F} \left(1 - \frac{t_{(\pm)}^s \sigma_s}{t_{(\pm)}^f \sigma_f} \right)^m \left(1 - \frac{t_{(\pm)}^s \sigma_s}{\sigma_{(\pm)}} \right)^{-m}, \quad (11)$$

where σ_s denotes surface conductivity, F is the electrical formation factor related to the porosity ϕ by Archie’s law $F = \phi^{-m}$, m is the cementation exponent [e.g., Waxman and Smits, 1968]. For a granular porous medium, σ_s is related to the specific surface conductance by $\sigma_s = (2 \Sigma_s / R)$ where R is the radius of the grains [O’Konski, 1960]. Taking $m \approx 2$ in Eq. (11) [which is a good approximation for the cementation exponent, e.g., Waxman and Smits, 1968], and $t_{(+)}^f = 1$ (and therefore $t_{(-)}^f = 0$), i.e., the counterions are essentially cations [Revil and Glover, 1997], Eq. (11) can be rewritten as a second degree equation in $\sigma_{(\pm)}$ which can be easily solved to give σ :

$$\sigma = \frac{\sigma_f}{F} \left[1 - t_{(+)}^f + F \xi + \frac{1}{2} (t_{(+)}^f - \xi) \left(1 - \frac{\xi}{t_{(+)}^f} + \sqrt{\left(1 - \frac{\xi}{t_{(+)}^f} \right)^2 + \frac{4F\xi}{t_{(+)}^f}} \right) \right] \quad (12)$$

where the dimensionless parameter ξ is defined by [Kan and Sen, 1987],

$$\xi \equiv \frac{\sigma_s}{\sigma_f} \approx \frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \frac{\beta_s Q_v}{\sigma_f}. \quad (13)$$

Here β_s is the surface mobility of the counterions, and Q_v is the excess surface charge per unit pore volume, which is related to the “Cation Exchange Capacity” (CEC) [Waxman and Smits, 1968] by $Q_v \equiv \rho_m [(1-\phi)/\phi] \text{CEC}$, where ρ_m is the grain density (in kg m^{-3}). The CEC (in C kg^{-1} of matrix weight) indicates the maximum number of surface exchangeable cations per unit mass of shaly rock. It strongly depends on clay mineral type. Patchett [1975] gives: $\text{CEC}(\text{kaolinite}) = 0.04 \pm 0.02 \text{ meq g}^{-1}$, $\text{CEC}(\text{illite}) = 0.22 \pm 0.02 \text{ meq g}^{-1}$ and $\text{CEC}(\text{smectite}) = 1.5 \pm 0.5 \text{ meq g}^{-1}$ (1 $\text{meq g}^{-1} = (e N) \text{ C kg}^{-1} = 96320 \text{ C kg}^{-1}$). For a mixture of clay, we can take an arithmetic average of the CEC weighted by the corresponding mass fraction of each clay mineral. Eq. (12) is a new equation relating the macroscopic electrical conductivity to the microstructural parameters, F and ϕ , and surface properties, $\beta_s Q_v$. Eq. (12) is compared with experimental data in Fig. 2a. Clearly the theoretical predictions accurately match the experimental data at all electrolyte salinities. It should be noted that this is achieved with a single constant value for the formation factor and the surface conductivity.

We test Eq. (12) against experimental data at 25°C from databases in Hill and Milburn [1956], Waxman and Smits [1968], and Vinegar and Waxman [1984]. We use all the 129 samples from these databases with $0 \leq Q_v \leq 1.60 \text{ eq l}^{-1}$ (1 $\text{eq l}^{-1} = 96.32 \times 10^6 \text{ C m}^{-3}$). We calculate F and σ_s by fitting Eq. (12)

non-linearly to the experimental rock conductivity as a function of fluid conductivity data for each sample. Surface conductivity is commonly plotted directly as a function of Q_V [e.g., Waxman and Smits, 1968; Bussian, 1983]. If this is done for the surface conductivity derived above, the data show a considerable degree a scatter. By contrast, our model [Eqs. (12) and (13)] implies a plot of $\sigma_S(1-\phi)/\phi$ as a function of Q_V as shown in Fig. 2b. We use linear regression of the data and Eq. (13) to derive $\beta_S(25^\circ\text{C}) = 5.14 \times 10^{-9} \text{ m}^2\text{s}^{-1}\text{V}^{-1}$. The ionic mobility of Na^+ at 25°C in water is $\beta_{(+)}'(25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2\text{s}^{-1}\text{V}^{-1}$ [Waxman and Smits, 1968]. The ratio of the surface mobility to that in the bulk electrolyte for Na^+ is therefore 0.098. The relationship between the specific surface conductance Σ_S and the surface mobility β_S is $\Sigma_S = e\beta_S\Gamma_S^0$ [Patchett, 1975]. Using $\Gamma_S^0 = 3$ charges per nm^2 [Patchett, 1975], the specific surface conductance of clays is $\Sigma_S(25^\circ\text{C}) = 2.47 \times 10^{-9} \text{ S}$.

We examine the dependence of the surface conduction on salinity. The database of Waxman and Smits [1968, Table 7] is well suited for that purpose since it covers a large range of Q_V and of pore fluid conductivities (~ 0.2 to 20 S m^{-1}). First, we calculate F and σ_S by fitting Eq. (12) non-linearly to the experimental conductivity data as a function of fluid conductivity in the range 2.8–16 S m^{-1} for each sample. Second, keeping the formation factor constant, we recalculate the surface conductivity σ_S for the complete salinity range using Eq. (12). Consequently, we can reconstruct the dependence of the surface conductivity on the pore fluid conductivity in the range 0.2 to 20 S m^{-1} (Fig. 2c). This figure shows that the surface conductivity is approximately constant within the salinity range investigated (~ 0.02 – 3 mol l^{-1}).

The independence of the specific surface conductance with salinity for salinities $>10^{-2} \text{ mol l}^{-1}$ has physical implications for the conduction mechanism. Indeed, independence on salinity is only possible if the conduction takes place behind the Stern plane as suggested by Zukovski and Saville [1986]. In the dynamic Stern layer model of Zukovski and Saville [1986], the ratio between the mobility in the Stern layer and the mobility in the free electrolyte for K^+ is 0.10 independent of salinity. Our results provide the same ratio for the mobility of Na^+ , and demonstrate the independence of surface mobility with the salinity (Fig. 4). These results strongly suggest a mechanism for surface electrical conduction in clay-rich sediments similar to that postulated by Zukovski and Saville [1986]. Application of Eq. (8) to clay minerals shows that the contribution of the EDL to Σ_S can be safely ignored. The consistency of the results found with the macroscopic approach that we used and the striking ability of Eq. (12) to describe electrical conductivity data (Fig. 2) makes us confident in the ability of Eq. (12) to account correctly for the microstructure in the electrical conductivity problem applied to shaly sands.

Conclusion

The conclusions of this paper are the following. (1) The contribution of the electrical diffuse layer to the surface conductivity can be neglected for silica and aluminosilicates.

(2) the main contribution to surface conductivity is probably due to counterions located in the Stern layer with surface mobilities approximately ten times smaller than in the bulk electrolyte. (3) A third contribution is also required for quartz, which is salinity independent, and possibly related to proton transfer directly on the mineral surface. (4) A new electrical conductivity equation has been developed which can be used to quantify the influence of surface conductivity at the pore-grains interface and the microstructure upon the macroscopic electrical conductivity of natural sands and sandstones.

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References

- Bussian, A.E., Electrical conductance in a porous medium, *Geophysics*, **48**, 1258–1268, 1983.
- Hiemstra, T., and Van Riemsdijk, W.H., Multisite proton adsorption modelling at the solid/solution interface of (hydr)oxides: a new approach, *J. Coll. Interf. Sci.*, **133**, 91–117, 1989.
- Hiemstra, T., and Van Riemsdijk, W.H., Multiple activated complex dissolution of metal (hydr)oxide: a thermodynamic approach applied to quartz, *J. Coll. Interf. Sci.*, **136**, 132–150, 1990.
- Hill, H.J., and Milburn, J.D., Effect of clay and water salinity on electrochemical behavior of reservoir rocks, *Trans., AIME*, **207**, 65–72, 1956.
- Kan, R., and Sen, P.N., Electrolytic conduction in periodic arrays of insulators with charges, *J. Chem. Phys.*, **86**, 5748–5756, 1987.
- O'Konski, C.T., Electric properties of macromolecules. V. Theory of ionic polarization in polyelectrolytes, *J. Chem. Phys.*, **64**, 605–619, 1960.
- Patchett, J.G., An investigation of shale conductivity, *SPWLA, 16th Annual Logging Symposium*, Paper U, 40 pp., 1975.
- Pride, S., Governing equations for the coupled electromagnetics and acoustics of porous media, *Phys. Rev. B*, **50**, 15678–15696, 1994.
- Revil, A., and Glover, P.W.J., Theory of ionic surface electrical conduction in porous media, *Phys. Rev. B*, **55**, 1757–1773, 1997.
- Scales, P.J., Grieser, F., and Healy, T.W., Electrokinetics of the muscovite mica-aqueous solution interface, *Langmuir*, **6**, 582–289, 1990.
- Vinegar, H.J., and Waxman, M.H., Induced polarization of shaly sands, *Geophysics*, **49**, 1267–1287, 1984.
- Watillon, A., and de Backer, R., Potentiel d'écoulement, courant d'écoulement et conductance de surface à l'interface eau-verre, *J. Electroanal. Chem.*, **25**, 181–196, 1970.
- Waxman, M.H. and Smits, L.J.M., Electrical conductivities in oil-bearing shaly sands, *Soc. Pet. Eng. J.*, **8**, 107–122, 1968.
- Zukoski, C.F., and Saville, D.A., The interpretation of electrokinetic measurements using a dynamic model of the Stern layer. I. The dynamic model., *J. Coll. Interf. Sci.*, **114**, 32–44, 1986.

A. Revil, LMF, ODP-NEB, BP 72, 13545 Aix-en-Provence, Cedex 4, France (e-mail: revil@lmf-air.gulliver.fr). P.W.J.

Glover, Dept. of Geology and Petroleum Geology, University of Aberdeen, King's College, Aberdeen, AB24 3UE, Scotland, UK (e-mail: p.glover@abdn.ac.uk).

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Electrical spectroscopy of porous rocks: a review—I. Theoretical models

T. L. Chelidze¹ and Y. Gueguen²

¹*Institute of Geophysics, Georgian Ac. Sci. 1, Alexidze str. 380093, Tbilisi, Georgia and Geosciences Rennes, Université de Rennes 1, 35042, Rennes, France*

²*Ecole Normale Supérieure, Laboratoire de Géologie, Dep. TAO, 24, rue Lhomond, Paris, France. E-mail: gueguen@geologie.ens.fr*

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SUMMARY

The complex dielectric permittivity ϵ^* of porous water-bearing rocks in the frequency range from a few to hundreds of megahertz reveals several intensive relaxation effects and a non-trivial dependence on the water content. At high frequencies, $f > 10$ MHz, both the real part of the complex dielectric permittivity ϵ' and the conductivity σ of water-bearing rocks are correctly predicted by the Maxwell–Wagner–Bruggeman–Hanai (MWBH) theory of composite dielectrics. This theory takes into account only the bulk properties of components, their partial volumes and the configuration of particles. The theory ignores two important factors: the surface contribution to polarization and the effect of clustering of components. At frequencies $f < 10$ MHz there are certain frequency domains which exhibit relaxation processes not predicted by MWBH theory. The characteristic times of these processes range from 10^{-6} to 10 s. These relaxation effects are related to different surface polarization processes which are, in order of increasing water content, (i) orientational polarization of bound water, (ii) polarization of liquid films or pockets, producing a polarization catastrophe effect, (iii) polarization of rough fractal surfaces, (iv) polarization of the ‘closed’ electrical double layer (EDL), when the displacement of the excess surface charges is limited by the external boundary of the EDL, and (v) polarization of the ‘open’ double layer, implying free exchange of excess ions with the bulk electrolyte and generation of transient diffusional potentials, which lag behind the applied field. Some theoretical models predict large effective values of relative dielectric constants in the range 10^5 – 10^6 at low frequencies. Knowledge of the characteristic signatures of these physical mechanisms is important for the correct interpretation of experimental data.

Analysis of existing theories of polarization of heterogeneous media shows that electrical spectroscopy can be useful for the interpretation of frequency spectra of complex dielectric permittivity or conductivity of water-bearing rocks and porous materials in general, and for the determination of water content, its thermodynamic state, the connectivity of water-bearing channels and their correlation lengths and the surface to volume ratio and surface charge in particular, in addition to the traditional formation factor, which is obtained from ohmic conductivity measurements.

Key words: dielectric permittivity, electrical conductivity, electromagnetism, fractals, inhomogeneous media, polarization.

1 INTRODUCTION

Electrical properties of rocks are of great interest for both exploration of the deep Earth and exploration of the shallow crust. Due to the intensive development of magnetotelluric and alternating current methods, the electrical imaging of geological structures has become decisive in solving some

prospecting problems. The monitoring of electrical properties can be very useful for the control of pollution (Roberts & Lin 1997), oil reservoir monitoring, hydrothermal energy and earthquake prediction studies (Mogi 1985) and so on. Correct interpretation of field data calls for a clear understanding of possible sources of variations of electrical properties. This is a very delicate problem as electrical properties of porous media

such as conductivity, dielectric permittivity (DP) and electrochemical potentials are extremely sensitive to physical conditions (temperature, pressure), the chemical composition of solid and liquid phases, and microstructural factors (porosity, its distribution and its connectivity).

The main purpose of this paper is to discuss the principal physical models and to consider critically possible physical mechanisms of dielectric losses and polarization under various conditions in order to review and explain the principal experimental results obtained by methods of low-frequency alternating current spectroscopy of rocks (Chelidze *et al.* 1999, hereafter referred to as paper II). We restrict ourselves to the range of relatively weak applied fields, thus excluding non-linear effects, which are covered by the studies of Olhoeft (1981, 1985). The frequency domain of electrical spectroscopy considered in this paper covers the interval from 1 mHz to 100 MHz, thus we do not consider here the domain of ultra-high frequencies, which is considered by Hoekstra & Doyle (1971) and Shen *et al.* (1985).

Typical rock in a vacuum is a good insulator with a conductivity of the order of $10^{-10} \text{ S m}^{-1}$ or less, but brine-saturated samples can have conductivities of the order of 1 S m^{-1} . This makes the measurement of rock electric properties a difficult technical problem. In addition, the possible superposition of various types of conduction and polarization, both in volume and on the solid–liquid interface, complicates dielectric spectra and calls for model experiments, where the influence of a single factor can be traced reliably.

The principles of solid-state physics, the physics of disordered media and the physics and chemistry of surfaces are now intensively applied to the interpretation of electrical spectra of rocks. The first study in this area was that of Smith-Rose (1934), who was surprised to discover a strong dependence of the real part of complex dielectric permittivity ε^* on frequency, and very high values of DP at low frequencies for samples of wet rocks and soil, ε of the samples being larger than that of the components. He ascribed this effect to the granular structure of samples. This effect of large low-frequency polarization has recently been a subject of discussion: in general the question has been posed whether it is possible for a mixture of components to have greater DP than that of any of its components. Some authors explain high dielectric constant values of the order of thousands and more by electrode polarization. Electrode impedances can greatly distort the measured electrical parameters of rocks at low frequencies, and special methods of separation and exclusion of the electrode contribution to measured electrical characteristics have been developed (see Paper II). As a result, very high values of DP for water bearing rocks have been confirmed (Chelidze 1963). It is worth noting that the electrical behaviour of dry rocks and bulk water is quite trivial and follows well-known theories of solid dielectrics and polar liquids. At the same time, water-bearing rocks demonstrate unusual dielectric spectra, enormous static polarization and anomalously large relaxation times. All of these non-trivial effects result, as shown below from physical and chemical processes on the solid–liquid interface, and can be considered as surface contributions to the complex electrical conductivity of a porous system.

Contributions to ac electrical studies of rocks are presented in the papers of Keller & Licastro (1959), Howell & Licastro (1961), Parkhomenko (1967), Chelidze (1971), Chelidze *et al.* (1977, 1979), Olhoeft (1981, 1985), Keller (1982), Knight (1983,

1991), Lockner & Byerlee (1985), Sen (1989), Ruffet *et al.* (1991a,b), Knight & Dvorkin (1992), Ruffet (1993), Garrouch & Sharma (1994), Glover *et al.* (1994a,b, 1996), Roberts & Tyburczy (1994), etc.

2 GENERAL DEFINITIONS

Application of an electric field to a rock causes both irreversible migration and reversible displacement of electric charges. The first process is characterized by the specific conductivity σ , and the latter by the dielectric permittivity ε .

The polarization vector $\mathbf{P}$ of a material is defined as the induced electric dipole moment per unit volume. The electrical displacement vector $\mathbf{D}_e$ is related to the electric field $\mathbf{E}$ and to $\mathbf{P}$ by

$$\mathbf{D}_e = \varepsilon \mathbf{E} + \mathbf{P}, \quad (1)$$

where ε is the dielectric permittivity of the material. The quantities $\mathbf{E}$, $\mathbf{D}_e$, σ and ε (or, alternatively, ε') are expressed in SI units as follows: V m^{-1} , C m^{-2} , S m^{-1} and F m^{-1} , respectively (see Appendix A for symbols and abbreviations).

The total current $\mathbf{J}_t$ resulting from the applied voltage $V = V_0 \exp(i\omega t)$ is the sum of a conduction current $\mathbf{j}_c$ and a displacement current $\mathbf{j}_d$:

$$\mathbf{J}_t = \mathbf{J}_c + \mathbf{J}_d, \quad (2)$$

$$\mathbf{J}_c = \sigma \mathbf{E}, \quad (3)$$

$$\mathbf{J}_d = \varepsilon_d^* \partial \mathbf{E} / \partial t = i\omega \varepsilon_d^* \mathbf{E}, \quad (4)$$

$$\mathbf{J}_t = (\sigma + i\omega \varepsilon_d^*) \mathbf{E} = \sigma^* \mathbf{E}, \quad (5)$$

where ω is the angular frequency and $\varepsilon_d^* = (\varepsilon' - i\varepsilon'')$ is the complex dielectric permittivity *excluding* ohmic losses. The term ε_d'' is caused by energy dissipation during polarization, for example by the rotation of dipole molecules in a viscous solvent. Eq. (5) can be rewritten as

$$\mathbf{J}_t = i\omega \varepsilon^* \mathbf{E} \quad \text{or} \quad \mathbf{J}_t = \sigma^* \mathbf{E}, \quad (6)$$

where

$$\varepsilon^* = \varepsilon_d^* + \sigma / i\omega = \varepsilon' - i(\varepsilon_d'' + \sigma / \omega) = \varepsilon' - i\varepsilon'', \quad (7)$$

$$\sigma^* = \sigma' + i\sigma'' = i\omega \varepsilon^* = i\omega(\varepsilon' - i\varepsilon''). \quad (8)$$

The ratio of the imaginary part of ε^* to the real part $\varepsilon''/\varepsilon'$ (or the ratio σ'/σ'') is defined as the loss angle tangent, $\tan \delta$:

$$\tan \delta = \varepsilon''/\varepsilon' = \sigma'/\sigma''. \quad (9)$$

The loss angle tangent due only to relaxation processes (see Fig. 1a) is

$$\tan \delta_d = \varepsilon_d''/\varepsilon'. \quad (10)$$

In the following text ε will stand for ε' in order to simplify expressions.

The electrical spectrum of a given material is defined as the plot of ε' and ε'' (or $\tan \delta$) versus frequency $f = \omega/2\pi$. In physical terms, it is the Fourier transformation of a transient process, namely of the induced polarization decay function. Polarization processes are quite often characterized by an exponential decay with a relaxation time τ , and maximal energy dissipation is observed at the critical frequency $f_c = 1/2\pi\tau$ (Debye 1929; Frölich 1990). In this case, which is called Debye relaxation, the complex dielectric permittivity is

$$\varepsilon^* = \varepsilon - i\varepsilon'' = \varepsilon_\infty + \frac{(\varepsilon_s - \varepsilon_\infty)}{(1 + i\omega\tau)} - i\frac{\sigma}{\omega}, \quad (11)$$

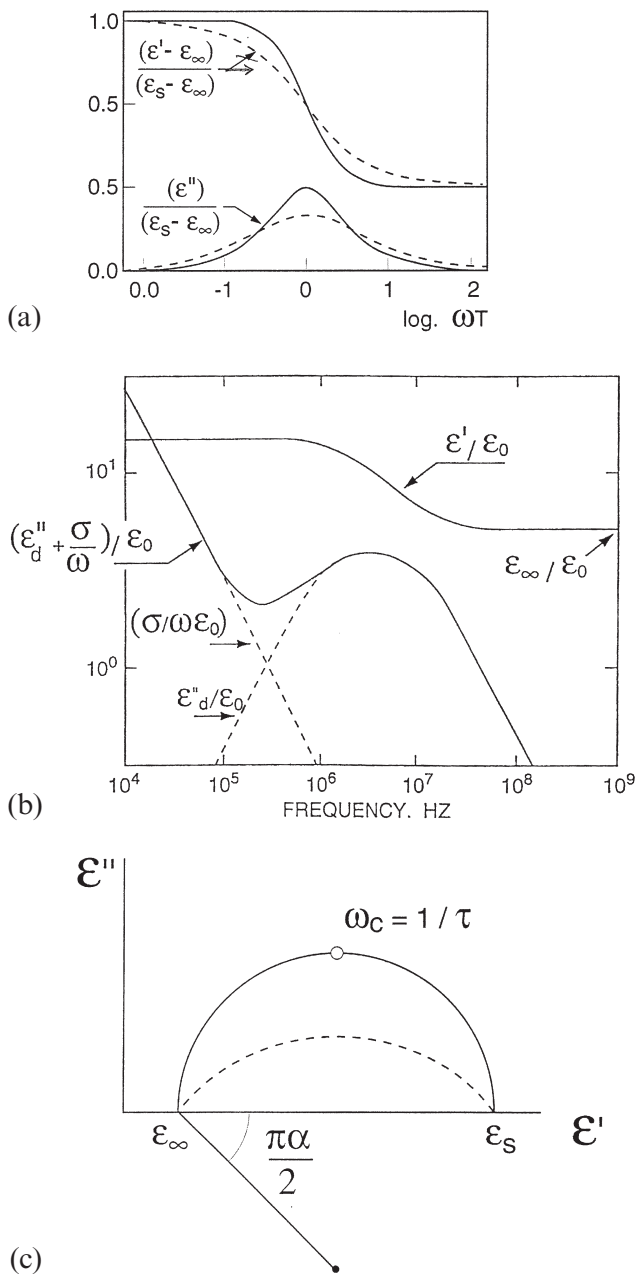


Figure 1. (a) The real and imaginary parts of relative dielectric permittivity (dielectric constant) for a rock with $\sigma = 0$. Solid curves: Debye model; dashed curves: Cole–Cole model (Gueguen & Palciauskas, 1994). (b) General behaviour of the real (ϵ'/ϵ_0) and imaginary ($(\epsilon'' + \sigma/\omega)/\epsilon_0$) parts of the complex dielectric constant ϵ^*/ϵ_0 for a rock with ohmic conductivity σ . Higher saturation and conductivity of brine increase ϵ^*/ϵ_0 by a term $\sigma/\omega\epsilon_0$ (Gueguen & Palciauskas, 1994). (c) Cole–Cole plot of ϵ'' as a function of ϵ' for a rock with $\sigma = 0$. Solid line: a single relaxation time; dashed line: Cole–Cole distribution of relaxation times with $\alpha = 0.5$.

where ϵ_∞ and ϵ_s correspond to $f \gg f_c$ and $f \ll f_c$ respectively (Figs 1a and b).

In real dielectrics, polarization is the sum effect of many relaxation processes with different relaxation times, and instead of the Debye equation, an empirical expression, proposed by Cole & Cole (1941), is usually used (Fig. 1a):

$$\epsilon^* = \epsilon_{cc} - i\sigma/\omega, \quad (12)$$

with

$$\epsilon_{cc} = \epsilon_\infty + \frac{(\epsilon_s - \epsilon_\infty)}{1 + (i\omega\tau)^{1-\alpha}}, \quad (13)$$

where α is the Cole–Cole distribution function exponent. The plot of ϵ'' versus ϵ' can be more complicated (skewed arc), and this can also be taken into account by additional exponents in the Cole–Cole equation (Chelidze *et al.* 1977).

For a very broad distribution of relaxation times there is no dominant characteristic time constant, with all frequencies contributing to the polarization process. The scale-invariant relaxation manifests itself by an inverse power-law frequency spectrum of ϵ' and frequency-independent $\tan \delta$ (Hanai 1968; Clerc *et al.* 1990).

The type of relaxation time distribution functions can be easily determined from plots of ϵ'' versus ϵ' or σ' versus σ'' for a broad range of frequencies, which are known as Cole–Cole or Argand plots (Fig. 1c). The Cole–Cole plot for a single relaxation time is a semi-circle between $\epsilon' = \epsilon_\infty$ and $\epsilon' = \epsilon_s$, centred on the ϵ' -axis. For the symmetrical distribution of relaxation times around a dominant time constant, it is the section of a circle with its centre below the ϵ' -axis; the value $\pi\alpha/2$ is the angle between the ϵ' -axis and a straight line from the point $\epsilon' = \epsilon_\infty$ to the centre of the circle. Larger values of α represent broader relaxation time distribution functions. α can thus in principle characterize the degree of heterogeneity of the material.

Any type of frequency behaviour of ϵ' and ϵ'' can be readily mimicked by a properly arranged combination of resistances R and capacitances C , which is called an equivalent circuit. In particular, water-bearing rock can be considered as a network of bonds with complex resistances R^* associated with porous space, saturated partially or fully by the fluid; mineral grains are good dielectrics and thus the main conductive paths follow pores and cracks. A well-designed equivalent circuit can give some indication of the general model of conduction and polarization (Ruffet *et al.* 1991; Ruffet 1993; Roberts & Tyburczy 1994). Nevertheless, it should be borne in mind that different equivalent circuits can demonstrate practically identical frequency behaviour (Von Hippel 1954). This means that the equivalent circuit approach does not give a unique solution, and only physical experiments under various conditions can reveal the true physical mechanisms of conduction and polarization.

It is commonly assumed that ohmic conduction is an irreversible process and that dielectric polarization, on the other hand, can be caused only by the reversible displacement of charges. The situation is somewhat different in thermodynamically open systems, where various flows are generated by the application of a physical field (thermodynamic force). According to the thermodynamics of irreversible processes (De Groot & Mazur 1967), the interaction of flows produces some reversible contribution to the process such as electrical polarization. For example, interference of irreversible processes of migration and diffusion can produce electrical polarization in rocks, which in principle can explain such unusual effects as the simultaneous increase of both the dielectric constant and conductivity. Such phenomena are impossible to explain from the static approach as, according to this approach, the higher the ohmic conductivity of a system is, the less the chances for a local accumulation of charges are, i.e. for polarization. The detailed physical mechanism and kinetics of such processes are considered in Section 3.2.2.

3 THEORIES OF ELECTRIC PROPERTIES OF HETEROGENEOUS SYSTEMS

3.1 Bulk effects (Maxwell–Wagner, Bruggeman, Hanai)

3.1.1 First-order models: layers and dilute suspensions

Theories of effective electrical and other transport properties of composite media are founded on Maxwell's approach (Maxwell 1891). In fact, Maxwell derived the expression for the dielectric constant and conductivity of a two-layer dielectric, approximating it by imperfect capacitors (RC circuits in series). It was later extended to a dilute suspension of spheres by Wagner (1924) and to a suspension of ellipsoidal particles by Fricke (1924) and Sillars (1937).

The effective medium approach (EMA) can be formulated in the following way. Consider a mixture of two ideal dielectrics with dielectric permittivities ε_1 and ε_2 and partial volumes $\Phi_1 = V_1/V$ and $\Phi_2 = V_2/V$, where V is the total volume of the mixture and V_1 and V_2 are those of the first and the second components. The mean electric field in the mixture $\langle \mathbf{E} \rangle$ can be calculated as

$$\langle \mathbf{E} \rangle = (1/V) \int_v \mathbf{E} dv, \quad (14)$$

and those in the components as

$$\langle \mathbf{E}_1 \rangle = (1/V_1) \int_{v_1} \mathbf{E}_1 dv, \quad (15)$$

$$\langle \mathbf{E}_2 \rangle = (1/V_2) \int_{v_2} \mathbf{E}_2 dv. \quad (16)$$

Thus,

$$\langle \mathbf{E} \rangle = \Phi_1 \langle \mathbf{E}_1 \rangle + \Phi_2 \langle \mathbf{E}_2 \rangle. \quad (17)$$

Similarly, for the mean electrical displacement vector $\langle \mathbf{D}_e \rangle$,

$$\langle \mathbf{D}_e \rangle = \Phi_1 \langle \mathbf{D}_1 \rangle + \Phi_2 \langle \mathbf{D}_2 \rangle. \quad (18)$$

The effective dielectric permittivity of mixture ε is

$$\varepsilon = \langle \mathbf{D}_e \rangle / \langle \mathbf{E} \rangle, \quad (19)$$

and those of the components are

$$\varepsilon_1 = \langle \mathbf{D}_1 \rangle / \langle \mathbf{E}_1 \rangle, \quad (20)$$

$$\varepsilon_2 = \langle \mathbf{D}_2 \rangle / \langle \mathbf{E}_2 \rangle. \quad (21)$$

Introducing the field ratios f of the components as

$$f_1 = \langle \mathbf{E}_1 \rangle / \langle \mathbf{E} \rangle, \quad (22)$$

$$f_2 = \langle \mathbf{E}_2 \rangle / \langle \mathbf{E} \rangle, \quad (23)$$

it follows from the equations for displacement vector (18), dielectric permittivity (19–21) and field ratio (22, 23) that

$$\varepsilon = \varepsilon_1 + (\varepsilon_2 - \varepsilon_1) \Phi_2 f_2. \quad (24)$$

In the following sections we will assume that the mixture is a matrix with dielectric permittivity ε_1 , containing inclusions with dielectric permittivity ε_2 . We also assume that the size of the inclusions is much less than that of the sample so that the sample can be considered as macroscopically homogeneous. This is not true, however, in systems with a fractal structure, where the homogeneous regime is attained only when the sample size L is much greater than the correlation length L_c ;

that is, greater than the largest cluster of heterogeneities (Chelidze *et al.* 1998).

The main difficulty is in the calculation of the field ratio f_2 . Considering a dilute suspension of spheres, Wagner (1924) derived the field ratio as

$$f_2 = \frac{3\varepsilon_1}{2\varepsilon_1 + \varepsilon_2}. \quad (25)$$

Substituting eq. (25) into eq. (24), we obtain the simplest formula for a suspension of spheres:

$$\varepsilon = \varepsilon_1 + 3\varepsilon_1 \frac{\varepsilon_2 - \varepsilon_1}{2\varepsilon_1 + \varepsilon_2} \Phi_2. \quad (26)$$

The increment of dielectric permittivity of a matrix due to the addition of inclusions is,

$$\Delta\varepsilon = \varepsilon - \varepsilon_1 = 3\varepsilon_1 \frac{\varepsilon_2 - \varepsilon_1}{2\varepsilon_1 + \varepsilon_2} \Phi_2. \quad (27)$$

Similarly, f_2 can be calculated exactly for ellipsoidal inclusions (Reinolds & Hough 1957):

$$f_2 = \sum_{i=1}^3 \frac{\cos^2 \Theta_i}{1 + A_i[(\varepsilon_2/\varepsilon_1) - 1]}, \quad (28)$$

where A_i are three depolarization factors (Dukhin & Shilov 1974), Θ_i are three angles between the ellipsoidal axis and the field $\mathbf{E}$, and ε_2 and ε_1 are the dielectric permittivities of the inclusions and the matrix, respectively. For spheres, $A_i = 1/3$, $\cos^2 \Theta_i = 1/3$ and from eq. (28) we recover eq. (25).

The main difference between the numerous effective medium theories (Reinolds & Hough 1957; Hanai 1968; Chelidze *et al.* 1977) is the assumption made about the value of the dielectric constant of the embedding substance, ε_e , which differs from the dielectric constant of the matrix due to the interaction of the electrical fields of inclusions. The simplest assumption is $\varepsilon_e = \varepsilon_1$, as above (eq. 28); that is, the dielectric permittivity of the equivalent embedding medium is that of the first component (matrix). This approximation is valid only for very dilute mixtures and relatively small contrasts between ε_1 and ε_2 when the electrical fields of components do not overlap significantly (Gueguen *et al.* 1997). Other approximations are examined below.

3.1.2 Self-consistent and differential self-consistent models

Some authors (see Reinolds & Hough 1957; Chelidze *et al.* 1977) substitute in eq. (28) instead of $\varepsilon_e = \varepsilon_1$ the value of the effective dielectric permittivity of the mixture itself, that is $\varepsilon_e = \varepsilon$; this approach results in a so-called self-consistent model. However, it seems that the best way to take into account the interaction of the inclusions is a differential self-consistent approach, which has been developed by Bruggeman (1935). In this approach the final concentration of inclusions is approached step by step by addition of infinitesimal portions of inclusions $\delta\Phi$. The increment $\delta\varepsilon$ of dielectric permittivity of the mixture ε due to the n th addition of such a small portion of inclusions is then given correctly by (27):

$$\delta\varepsilon_n = \varepsilon - \varepsilon_n = 3\varepsilon_n \frac{\varepsilon_2 - \varepsilon_n}{2\varepsilon_n + \varepsilon_2} \delta\Phi_n, \quad (29)$$

which can be considered as a differential equation linking $\delta\varepsilon_n$ and $\delta\Phi_n$. After integration from $\varepsilon = \varepsilon_1$ and $\Phi_2 = 0$ to $\varepsilon_n = \varepsilon$ and

$\Phi_n = \Phi_2$, we obtain Bruggeman's formula:

$$\frac{\varepsilon - \varepsilon_2}{\varepsilon_1 - \varepsilon_2} \left(\frac{\varepsilon_1}{\varepsilon} \right)^{1/3} = 1 - \Phi_2. \quad (30)$$

Hanai (1968) developed a comprehensive theory of the complex dielectric constant of a concentrated suspension of spheres, where each component is characterized by a complex dielectric constant ε_i^* , and derived the complex analogue of Bruggeman's equation,

$$\frac{\varepsilon^* - \varepsilon_2^*}{\varepsilon_1^* - \varepsilon_2^*} \left(\frac{\varepsilon_1^*}{\varepsilon^*} \right)^{1/3} = 1 - \Phi_2. \quad (30a)$$

This expression can be considered as the final result of the development of the effective media approach to the analysis of suspension-type composite dielectrics. We will refer to it as Maxwell–Wagner–Bruggeman–Hanai (MWBH) theory. From the complex equation (30a), the simple expressions for some practically important limiting cases, namely for low or high frequency, high conductivity of inclusions or high conductivity of matrix, can be obtained. For example, after cumbersome transformations of eq. (30a) in the low-frequency (LF) limit at $\sigma_2 \gg \sigma_1$, the real part of the complex dielectric permittivity of the mixture can be obtained as (Hanai 1968; Dukhin & Shilov 1974)

$$\varepsilon = \frac{\varepsilon_1}{(1 - \Phi_2)^3}, \quad (31)$$

from which we can conclude that conducting particles embedded in the dielectric cause a large increment of the dielectric permittivity as $\Phi_2 \rightarrow 1$. Thus, although water inclusions in a dielectric (say oil) at high frequencies (HF) have a standard value of $\varepsilon = 80$, at low enough frequencies the inclusions succeed in screening the external electric field due to the displacement of charges over a distance of the order of the Debye length, which means that in the LF limit the water inclusions in the dielectric have infinite dielectric permittivity (compare Figs 2a and b). Note that in the above formulae the ε^* and consequently σ^* of a suspension of spherical particles do not depend on the size of inclusions. Theories for ellipsoidal particles predict a strong dependence of ε^* on the aspect ratio, but again ε^* does not depend on the particle size. This means that the surface contribution to polarization and conductivity is ignored in MWBH theory. This is also evident from the derivation of eqs (14)–(31), where only bulk properties and partial volumes of components are taken into consideration.

Experiments confirm the MWBH theory of composite dielectrics in a broad range of Φ_2 for many materials (Fig. 2). Nevertheless, there are some factors that are not taken into account in this theory. For example, the spatial distribution of inclusions is assumed to be homogeneous; this is unrealistic in many cases. Aggregation of inclusions affects the properties of a mixture significantly. This effect is considered by Dukhin & Shilov (1974). The second principal problem with MWBH theory is that in real systems the interface between different phases is not as inert as it is assumed to be in eqs (14)–(31).

3.2 Surface effects

It is clear from a thermodynamical viewpoint (Chelidze 1971) that the surface contribution to electrical processes can be quite significant, and even dominating in heterogeneous media

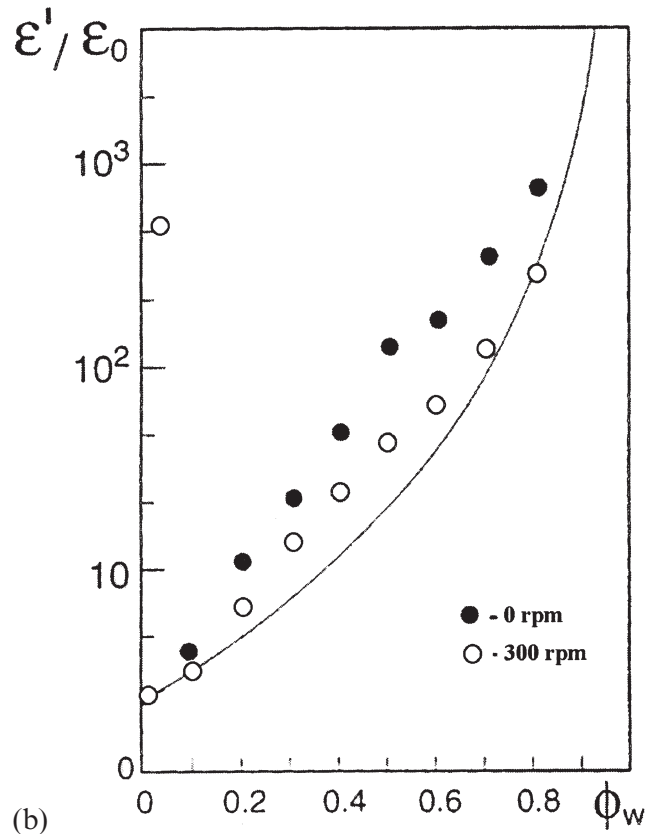
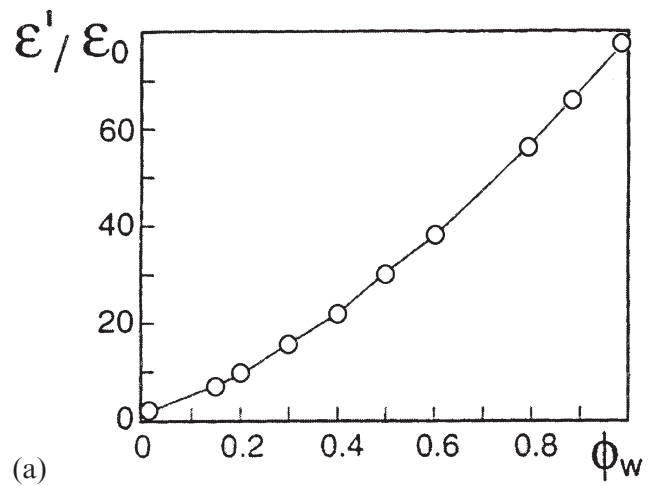


Figure 2. (a) The HF dielectric constant of water in oil emulsions as a function of the dispersed phase (water) content Φ_w at 30 °C (MHz range). The theoretical curve corresponds exactly to the formula of Bruggeman (solid line). Open dots are experimental data (Hanai 1968). (b) The dielectric constant of the same system at 100 Hz at different stirring velocities in rpm: solid dots—at rest; open dots—at 300 rpm. The solid line is the theoretical prediction by the formula of Hanai (1968). The difference in experimental plots for various rpm values comes from the effect of aggregation: at high stirring velocities the emulsion is less aggregated and hence closer to the assumptions of the effective medium theory.

with highly developed interfaces between components. The magnitude of this contribution depends on the following factors: (1) the surface-to-volume ratio of the components; (ii) the surface and volume physical and chemical properties of the

components; (iii) the mode of coupling between surface and volume processes; (iv) the characteristics of the electrical double layer at the solid–liquid interface; and (v) the geometrical, chemical or electrical heterogeneity of real interfaces.

As a rule, many processes develop on surfaces in response to the application of an electric field. At very low water contents almost all water molecules are in an adsorbed state. The properties of adsorbed water differ significantly from those of the bulk liquid: the relaxation time is longer and the energy of activation is higher in the adsorbed state. This means that the properties of adsorbed molecules are intermediate between bulk water and ice (Chelidze *et al.* 1977; Hoekstra & Doyle 1971). The quantitative analysis of dielectric spectra of the first adsorbed layers of water can be performed in the framework of the theory of absolute rates of reactions. The theory allows the estimation of thermodynamic properties of adsorbed water from the experimental data on temperature dependence of relaxation times (Chelidze 1977). The physical mechanism of the polarization of adsorbed layers can be either the orientational polarization of dipole molecules of water or the transfer of charges (protons) along hydroxilic groups or chains, so-called surface charge transfer complexes (Chelidze *et al.* 1977; Pride 1994).

The contribution of surface processes to the ohmic conductivity of porous systems is considered by Dukhin (1975), Pride (1994), Glover *et al.* (1994a,b) and Revil & Glover (1997). In the last three papers a new fundamental theory of ionic surface conductivity is presented which is based on first principles of the physics and chemistry of surfaces.

Surface migration and diffusion also evoke polarization mechanisms, which are often much stronger than the MWBH polarization. The contribution of surface processes to conductivity and polarization can be experimentally estimated by calculating deviations of electrical spectra from MWBH theory.

3.2.1 Geometrical surface effects

Le Mehaute & Crepy (1983), Wong (1987), Ruffet *et al.* (1991a,b), Ruffet (1993) and Glover *et al.* (1994a) considered the geometrical effects resulting from the existence of fractal surfaces. They assumed that the internal surfaces of rocks are rough and examined how the charge layer is built up at the interface. When an electric field is applied along the *Y*-axis normal to the surface, ions have to migrate in this direction. As minerals are mainly dielectrics, ions do not penetrate into them and have to migrate along the rough surface in the *X*-direction in order to progress further in the *Y*-direction.

Assuming that the electrode surface is a self-affine fractal, the time *t* needed to cover the horizontal diffusion distance *r* is $t \sim r^2$. During that time the average displacement *w* along the *Y*-axis is

$$w = r^\beta, \quad (32)$$

where for a self-affine surface $\beta = d - D_f$, D_f being the fractal dimension of the surface and *d* that of the embedding space (i.e. $d = 3$ in our case). At time *t* the polarization is proportional to *w*:

$$P \sim w = r^\beta = r^{d-D_f} = t^{(d-D_f)/2}. \quad (33)$$

The dielectric permittivity has the same frequency dependence as polarization, so introducing the angular frequency ω from

$t \sim \omega^{-1}$ and $d = 3$ the frequency response of ϵ^* is

$$\epsilon^* \sim \omega^{-(3-D_f)/2}. \quad (34)$$

The model predicts a power-law frequency dependence for ϵ with an exponent $(D_f - 3)/2$, and a similar behaviour for σ^* with an exponent $(D_f - 1)/2$. Wong (1987) presented results for the Portland sandstone which agree quite well with these predictions in the megahertz range. In this case the high-frequency behaviour is $\epsilon^* \sim \omega^{-x}$, with $x = 0.29$, so $D_f = 2.42$.

The earlier and more formal model of Le Mehaute & Crepy (1983) gives for a 2-D space an exponent $D_f = 2/(1 - \alpha)$, where α is the Cole–Cole parameter; accordingly, the value of D_f tends to infinity as $\alpha \rightarrow 1$, which makes this approach unrealistic at large α .

For these models, the frequency dependence exponent reflects the distribution of relaxation times, which in turn is connected with ion fluxes passing obstacles of various sizes on the rough surface. This exponent can then be related to the Cole–Cole exponent α . This, in principle, gives a recipe for calculating surface roughness (exponent D_f) from the experimentally determined Cole–Cole parameter α . It should also be noted that according to some numerical studies (Cao *et al.* 1994) surface roughness results in fractal behaviour only at very high frequencies, in the gigahertz range.

Glover *et al.* (1994a) give D_f of saturated sandstones, calculated from experimental HF values of ϵ^* , as a function of fluid salinity and ionic type. These experiments show that purely geometrical fractal surface models of polarization do not fit the experimental data, as it has been found that variations of the electrolyte concentration and cation type change the apparent fractal surface dimension: it decreases with decreasing concentration and increasing ionic radius of the dominant cations, which seems strange because the fractal dimension of the solid surface should be independent of the properties of the electrolyte. The most natural explanation is that the electrical method of assessment of surface roughness is not a direct one; it actually gives apparent ‘electrical roughness’ (Glover *et al.* 1994a). The further from the surface of solid, the less the ‘roughness’ of the equipotential lines of a heterogeneous surface electric field is (Bunde & Havlin 1992). The less the electrolyte concentration is and the larger the ionic radius of the dominant cations in solution is, the further from the grain surface (and hence along a less tortuous route) the transport of dominant charge carriers occurs. This can qualitatively explain the phenomenon of variability of apparent fractal surface dimensions as a result of the multifractal nature (see e.g. Korvin 1992) of electrically measured surface roughness.

3.2.2 Smooth surfaces with homogeneous charges

(a) *Surface polarization: conducting films*: Surface conductivity is due to the high concentration and different mobilities of charge carriers in the electrical double layer (EDL). O’Konsky (1960) calculated the polarization field of spherical particles of radius *a* and bulk conductivity σ_2 . He introduced the frequency-independent surface conductivity σ_s , and assumed that the double-layer ions, in response to the applied electrical field, can move normal to the interface over a distance which is of the order of the Debye length. He concluded that the surface conductivity σ_s causes the same effect on the distribution of the polarization field as an increase in the particle’s

bulk conductivity σ_2 to a value σ_{2e} :

$$\sigma_{2e} = (\sigma_2 + 2\sigma_s/a). \quad (35)$$

Hence, eq. (30) adequately describes the surface conductivity effect if the bulk conductivity σ_2 is replaced by the effective conductivity σ_{2e} . The relaxation time for establishing the polarization is (O'Konsky 1960)

$$\tau = \frac{a(\varepsilon_2 + 2\varepsilon_1)}{8\pi\sigma_s}, \quad (36)$$

where ε_1 and ε_2 are the DP of the electrolyte and the particle respectively.

Schwarz (1962) allowed for the frequency-dependent surface conductivity of bound ions within the double layer. In his model, the application of an electric field resulted in only tangential migration of charges; displacement normal to the interface was forbidden. In this case, the equilibrium is reached through diffusion of ions along the interface. The relaxation time for attaining equilibrium is then obtained as a typical diffusion time constant (Wert & Thomson 1964; Dukhin & Shilov 1974):

$$\tau_s = \frac{a^2}{2D_s}, \quad (37)$$

where D_s is the surface diffusion coefficient of bound ions.

A general solution to the problem of polarization of a non-conducting particle of radius a and dielectric permittivity ε_2 coated with a film of surface conductivity σ_s and surface diffusion coefficient D_s and embedded in a medium with complex dielectric permittivity ε^* in an ac field leads to the following expression for the oscillating dipole moment d_e of the particle (Dukhin *et al.* 1969; Chelidze *et al.* 1977):

$$d_e = \frac{a^3}{\varepsilon_2 + 2\varepsilon_1} \left[\varepsilon_2 + \varepsilon_1 + 3\varepsilon_1 \frac{1 - (\tau/\tau_s)}{1 + i\omega\tau} \right] E, \quad (38)$$

where

$$\tau = \frac{\tau_s}{1 + \frac{4\pi a \sigma_s}{D_s(2\varepsilon_1 + \varepsilon_2)}}. \quad (39)$$

For small particles the second term in the denominator of eq. (39) is small, and the relaxation is controlled by diffusion along the surface $\tau = \tau_s$. For large particles the second term dominates and we obtain O'Konsky's relaxation time, eq. (36).

At low frequencies ($\omega\tau \ll 1$) and for high σ_s , a dielectric particle coated with a conducting film and embedded in a non-conducting medium is polarized as a conducting inclusion. This results in $\varepsilon \rightarrow \infty$ when $\Phi_2 \rightarrow 1$, according to eq. (31).

This explains why the addition of small quantities of water to a porous rock causes a strong dielectric increment at low frequencies, much larger than that expected from MWBH theory or from the orientational polarization of adsorbed molecules (Chelidze *et al.* 1977).

(b) *Surface polarization: insulating films*: The case of insulating films covering conducting particles is the reverse of the previous case. It corresponds to a system such as oxidized metal inclusions dispersed in a conductive medium. Such a situation could be observed with metallic ores.

The corresponding theoretical model has been developed by Pauly & Schwan (1959). They modified the two-component

effective medium theory by taking into account the existence of a third component, an insulating envelope of thickness h with dielectric permittivity ε_3 and conductivity σ_3 coating spherical particle of radius a with electrical parameters ε_2 and σ_2 . The envelope is assumed to be thin, $h \ll a$. The coated particles are embedded in a medium with characteristic parameters ε_1 and σ_1 . The equation for a complex DP is

$$\varepsilon^* = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_i}{1 + i\omega\tau_p} + \frac{\varepsilon_i - \varepsilon_\infty}{1 + i\omega\tau_q} - \frac{i\sigma_0}{\omega}, \quad (40)$$

where $\varepsilon_s, \varepsilon_i, \varepsilon_\infty$ are functions of components ε and σ , σ_0 is the dc conductivity of the system, τ_q is the relaxation time for a suspension of non-coated particles and τ_p is that for a suspension of coated particles, $\tau_q < \tau_p$ (Pauly & Schwan 1959). Since there are two relaxation times there are accordingly two 'static' values of dielectric permittivity—one at very low frequencies (ε_s) and the second in the intermediate-frequency range (ε_i).

Generally, $\sigma_3 \ll \sigma_1$, $\sigma_3 \ll \sigma_2$ and $h \ll a$. After the introduction of coating conductance $G_s = \sigma_3/4\pi h$ and its capacitance $C_s = \varepsilon_3/4\pi h$, eq. (40) simplifies greatly in the LF limit, where in most cases the condition $G_s \ll \sigma_2/(a+h)$ is fulfilled and we obtain

$$\sigma = \sigma_1 \frac{2(1 - \Phi)}{2 + \Phi},$$

$$\varepsilon_s - \varepsilon_i = 4\pi \frac{9\Phi a C_s}{(2 + \Phi)^2}, \quad (41)$$

$$\tau_p = (a+h)C_s \left(\frac{1}{\sigma_2} + \frac{1}{\sigma_1} \frac{1 - \Phi}{2 + \Phi} \right),$$

where Φ is the partial volume of coated particles and $\Phi \cong \Phi_2$, because the envelopes are negligibly thin.

Such systems undergo strong frequency dispersion of ε and σ , which is explained by the shunting of insulating films at sufficiently high frequencies. Shunting incorporates the bulk conducting volume of particles into the global conduction process, causing an increment of conductivity and a drastic decrease of the dielectric constant to MWBH values at high frequencies.

(c) *Open electrical double layer*: The 'gigantic low frequency polarization' theory (GLFP) considers transient processes in the bulk of the pore electrolyte caused by polarization of the electrical double layer (EDL) at the solid-liquid interface. The potential ψ within the EDL is shown in Fig. 3, where the total potential drop ψ_0 between the solid and the bulk of the electrolyte includes that in the diffuse, mobile part of the EDL, which is usually called the ζ potential, and that in the dense, immobile part of the EDL, which in turn includes the Stern potential ψ_s and the potential drop $(\psi_0 - \psi_s)$, due to chemically bound non-solvated ions which cannot be exchanged with the electrolyte. It is expected that all ions which are at a potential field $\psi < \psi_s$ can be exchanged with the bulk of the electrolyte.

Thus, unlike the Schwarz approach, the GLFP theory allows for a free exchange of EDL excess (transient) charges with the electrolyte, which is accomplished through the displacement of ions normal to the interface. This makes the polarization process much more effective. Application of an electric field to a charged particle embedded in an electrolyte leads first to a distortion of the initially symmetrical EDL due to migration

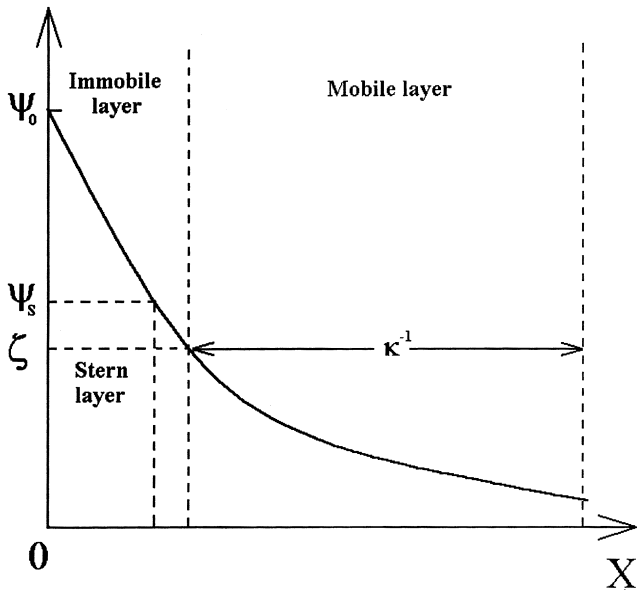


Figure 3. Surface potential ψ versus distance from the solid surface x . ψ_0 is the total potential drop between the surface of the solid and the bulk of electrolyte, ψ_s and ζ are respectively potential drops in the Stern layer and in the mobile part of the EDL (modified from Adamson 1982).

and diffusion of charges in it. In the following treatment only sufficiently large particles will be considered, satisfying the condition $a \gg \kappa^{-1}$, where a is the particle radius and κ^{-1} is the Debye screening length or thickness of the diffuse double layer. As the charge carriers in EDL are transported in a different ratio to those in the bulk electrolyte, and as the surface fluxes of anions and cations are not limited to flow parallel to the interface as in previous models, the asymmetry of anion and cation fluxes and the electroneutrality condition for a pore electrolyte result in the appearance of concentration gradients near the particle, outside the EDL, in addition to the electric polarization field of the deformed EDL. We can analyse this behaviour through a sequence of 4 steps (Fig. 4). The relaxation time for establishing a steady-state polarization of EDL of a negatively charged particle (Fig. 4a) is small: several microseconds for brine conductivities of about 10^{-4} S m^{-1} , and this corresponds to the first step. Due to the free exchange between the open EDL and the bulk electrolyte, the non-equilibrium concentration of charges, developed in EDL, results in the appearance of local transient clouds with the excess of positive charges on one side of the particle and with their deficit on the other side within the bulk electrolyte: this is the second step (Fig. 4b). The third step (Fig. 4c) involves the time τ_{dl} needed to attain the electrical equilibrium state between charged diffusion clouds and the adjoining volume of macroscopically neutral bulk electrolyte. This time is also small because the charges diffuse only over a distance κ^{-1} , so (Dukhin & Shilov 1974)

$$\tau_{dl} = \frac{1}{\kappa^2 D_{ef}}, \quad (42)$$

where

$$D_{ef} = \frac{D^+ D^- (z^+ + z^-)}{z^+ D^+ + z^- D^-}, \quad (43)$$

z^- and z^+ are valences and D^- and D^+ are diffusion coefficients of anions and cations. Hence at frequencies $\omega \ll 2\pi/\tau_{dl}$, the bulk electrolyte near the EDL regains its electroneutrality, and only asymmetry in concentration remains from charged clouds outside the EDL. However, steady-state polarization due to the induced concentration gradients (the last step, Fig. 4d) is established over a much longer interval of time, as it is controlled by diffusion over distances of the order of the linear dimension of the particle. The relaxation time τ_a in this case is the time needed for a diffusion front to propagate over a distance a (Dukhin & Shilov 1974):

$$\tau_a = \frac{a^2}{2D}, \quad (44)$$

where D is the bulk diffusion coefficient of the electrolyte. Because of this, the polarization field of the particles lags behind the applied electric field at low frequencies.

The key point is that in a conducting medium the polarization field induces conduction currents which lag behind the applied field. These out-of-phase currents appear as displacement currents. Such large displacement currents are seen experimentally as high values of the dielectric constant. Thus, on the one hand a large electric charge of a particle produces a significant asymmetry of ion fluxes and diffusion gradients, and on the other hand a large electrolyte conductivity produces large out-of-phase currents. Only together can these effects result in a large low-frequency dielectric constant, which can be considered as an effective parameter of a system. It can be shown (Dukhin & Shilov 1974; Sen & Chew 1984) that high values of DP are obtained only when the surface charge is sufficiently large and $\kappa a \gg 1$. If $\kappa a \ll 1$, the dielectric constant of the mixture cannot exceed the MWBH values of ϵ (Trukhan 1962).

A simplified version of the GLFP theory can be written as follows. From Maxwell–Wagner theory, we use eq. (27), which we rewrite in terms of the dipole moment of a spherical particle d_e^* and the number of particles per unit volume N . The partial volume of inclusions, Φ_2 , for spherical particles of radius a is $4\pi a^3 N/3$, thus eq. (27) can be written in complex form as

$$\Delta \epsilon^* = 4\pi \epsilon_1^* N d_e^* / E, \quad (45)$$

where d_e^* is the complex dipole moment of a polarized sphere,

$$d_e^* = \frac{\epsilon_2^* - \epsilon_1^*}{2\epsilon_1^* + \epsilon_2^*} a^3 E. \quad (46)$$

The real part of the complex dielectric permittivity from eq. (45) is

$$\text{Re } \Delta \epsilon^* = 4\pi N [\text{Re}(\epsilon_1^*) \text{Re}(d_e^*/E) + \text{Im}(\epsilon_1^*) \text{Im}(d_e^*/E)]. \quad (47)$$

Here, $\text{Im}(\epsilon_1^*)$ is the imaginary part of the electrolyte complex dielectric permittivity. It increases greatly when frequency decreases, and varies as σ/ω . Hence, if $\text{Im}(d_e^*/E)$ is sufficiently large, the product $\text{Im}(\epsilon_1^*) \text{Im}(d_e^*/E)$ can be very large at low frequencies, so the value $\text{Re } \Delta \epsilon^*$ as a whole becomes large.

It seems that the above model provides a universal explanation of the amplification of polarization of any origin in a moderately conducting media.

A comprehensive analysis of the processes in EDL involves the solution of the equations of convective diffusion, Poisson and Navier–Stokes equations with the assumption that κa is large. This corresponds to the case of a moderate electric field which means that the Boltzmann's distribution holds in EDL.

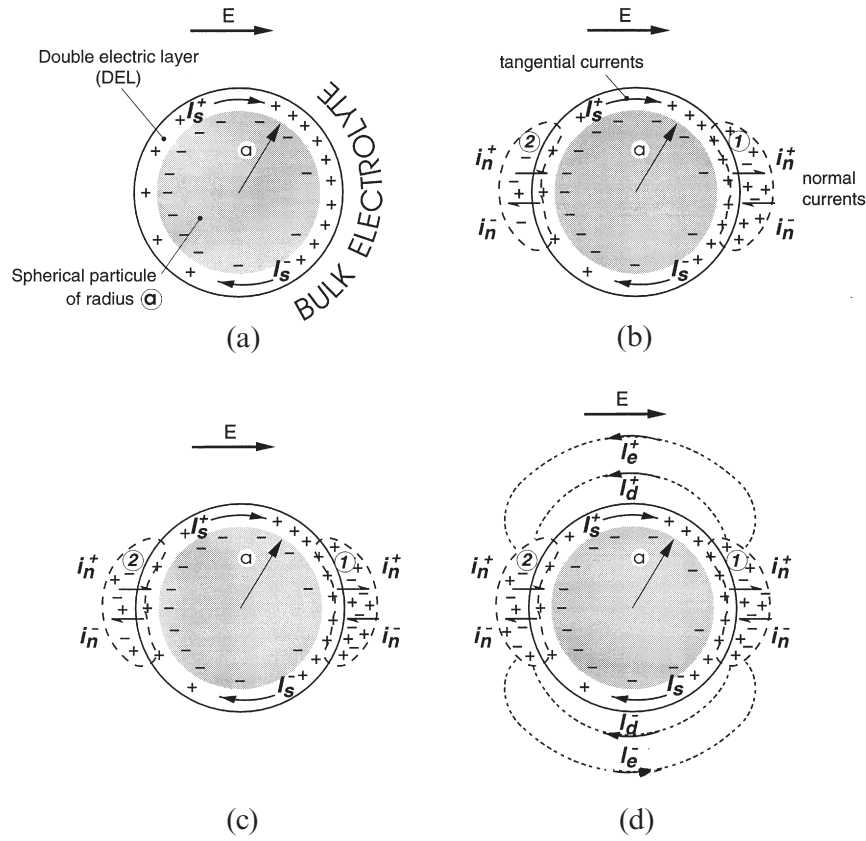


Figure 4. Model of evolution of EDL polarization for a negatively charged spherical particle of radius a under an electric field E . The particle is immersed in the electrolyte and EDL is present on its surface. (a) Tangential fluxes of positive (I_s^+) and negative (I_s^-) ions in the double layer result in an asymmetric charge distribution within the EDL. (b) Normal fluxes of anions and cations (i_n^+ , i_n^-) between the EDL and the bulk electrolyte result in high- and low-concentration areas or diffuse charged clouds on both sides of a particle in the bulk electrolyte. Electroneutrality is not achieved at this stage. (c) Electroneutrality is achieved within a short time interval because the time constant $\tau_{dl} = 1/\kappa^2 D_e$ in both diffuse clouds. (d) Concentration differences and electrical potential differences between areas (1) and (2) generates both diffusion (i_d^+ , i_d^-) and migration (i_e^+ , i_e^-) fluxes of anions and cations which lag behind the applied field E . The time constant of this process is large, $\tau = a^2/D$.

The dielectric increment in the static limit $\Delta\epsilon_s$ for highly (negatively) charged particles in an electrolyte with $z^+ = z^- = z$ is (Dukhin & Shilov 1974; Chelidze *et al.* 1977)

$$\Delta\epsilon_s = \frac{9\Phi_2\epsilon_1 \left[3m \exp\left(\frac{z|\zeta'|}{2}\right) + \exp\left(\frac{z|\psi'_s|}{2}\right) \right]^2}{4z \left[1 + 3m \left(\frac{1}{z} + \frac{1}{z^2} \right) \frac{\exp\left(\frac{z|\zeta'|}{2}\right)}{\kappa a} + (z+1) \frac{\exp\left(\frac{z|\psi'_s|}{2}\right)}{\kappa a} \right]^2}, \quad (48)$$

where Φ_2 is the partial volume of inclusions, $\kappa a \gg 1$:

$$m = \frac{\epsilon_1 \left(\frac{kT}{e} \right)^2}{6\pi\eta D^+}, \quad (49)$$

$$\zeta' = ze\zeta/kT, \quad (50)$$

$$\psi'_s = ze\psi_s/kT. \quad (51)$$

Here, k is the Boltzmann constant, a is the particle size, T is the absolute temperature, D^+ is the diffusion coefficient of cations, η is the dynamic viscosity of the fluid, ζ is the potential

of the shear plane, ψ_s is the Stern potential, e is the electron charge and z is the valence of anions. The value of m is about 0.1 for aqueous suspensions.

The maximum value of $\Delta\epsilon_s$, attained at $\kappa a \gg 1$, is

$$\Delta\epsilon_s^{\max} = \frac{9\Phi_2\epsilon_1}{4z} \left(3m \exp\left(\frac{z|\zeta'|}{2}\right) + \exp\left(\frac{z|\psi'_s|}{2}\right) \right)^2. \quad (52)$$

Thus, if the particle is large enough, the static DP does not depend on their size.

In order to check this theory quantitatively, complex dielectric constant measurements and surface charge determinations by electrokinetic methods were carried out simultaneously on a suspension of negatively charged spherical non-conducting particles with relatively smooth homogeneous surfaces (latex spheres). The frequency range is 300 Hz–100 kHz. The experimental results are presented in Table 1.

It can be seen from Table 1 that the ζ potential values, corrected for the retardation of the ionic atmosphere of particles (ζ'_{cor}), obtained from electrophoretal measurements are close the values of Stern potential ψ'_s , calculated by GLFP theory from (48). This means that ion fluxes from the immobile part of EDL are insignificant. This agrees with the results obtained by Revil & Glover (1997). The average particle radii, calculated from the critical frequency using (44), are also in

Table 1. Comparison of dielectric, electrophoretic and microscopic measurements on the model system: suspension of spherical particles in an electrolyte (Chelidze *et al.* 1977).

Substance	Φ_2	τ_a , s	a_d , μm	a_m , μm	κa_d	ζ'_m	ζ'_{cor}	ψ'_s	$\Delta\epsilon_m$	$\epsilon_\infty/\epsilon_0$
1	0.02	8×10^{-6}	0.13		15	2.30	3.50	3.50	68	76
1	0.04	8×10^{-6}	0.13		15				136	75
1	0.08	8×10^{-6}	0.13		15				272	70
1	0.18	8×10^{-6}	0.13		15				612	57
2	0.05	10^{-3}	0.45		60	2.60	3.00	3.20	700	76
3	0.10	5×10^{-6}	0.10		20	2.90	4.00	3.80	780	66
3	0.025	5×10^{-6}	0.10	0.09	20				120	75
3	0.013	6×10^{-4}	1.10	1.60	95	3.00			9750	76

Substances are: 1—chlorophren latex, 2—aged chlorophren latex, 3—polyvinylchlorid latex; Φ_2 is the volume fraction of inclusions; τ_a is the relaxation time from dielectric spectra in s; a_m is the particle radius value determined from microscopic measurements in μm ; a_d is the particle radius predicted by eq. (44) in μm ; κa_d is the product of the inverse Debye screening length and particle radius; ζ'_m is the reduced value of the measured ζ potential (see eq. 50); ζ'_{cor} is a value of ζ'_m corrected for the additional retardation of particles at electrophoresis resulting from the relaxation of their ionic atmosphere (Dukhin & Shilov 1974); ψ'_s is the reduced Stern potential or total potential drop in both mobile and immobile parts of EDL; $\Delta\epsilon_m$ is the experimental (relative) dielectric increment, i.e. the difference between LF and HF values of the relative dielectric constant; ϵ_∞ is the relative dielectric constant attained at high frequencies. The ohmic conductivity of all samples was of the order of 10^{-2} S m^{-1} .

agreement with independent electron microscopy measurements. Thus both relaxation times and the magnitude of the polarization $\Delta\epsilon_m/\Phi_2$ conform GLFP theory for this simple model system.

3.2.3 Surfaces with non-homogeneous charges

In the Dukhin–Shilov theory it is implicitly assumed that the surfaces of particles are homogeneously charged. This condition is evidently fulfilled in latex suspensions where surface charges are distributed uniformly. The surfaces of minerals, however, are electrically heterogeneous. This can result from surface roughness, for example, because the asperities can be considered as small charged particles glued to the surface. Moreover, many studies have shown that even very finely polished surfaces of minerals are electrically heterogeneous. Mineral surfaces are actually mosaics of negatively and positively charged areas. The heterogeneity of the surface depends strongly on the mineral type and the pH of the solution (Revil & Glover 1997).

Chelidze *et al.* (1977) proposed a model of polarization of electrically heterogeneous surfaces of solids in which clusters of charges are considered as analogues of charged particles in the GLFP theory. The relaxation time in this model is controlled by the mean size of charged surface clusters a_{cl} and not by particle size a :

$$\tau_{\text{cl}} = a_{\text{cl}}^2 / 2D, \quad (53)$$

and the LF dielectric increment can be approximately calculated from eq. (48) using a_{cl} instead of a and Φ_{cl} instead of Φ_2 . Here Φ_{cl} is the concentration of surface clusters in the unit volume $\Phi_{\text{cl}} \sim (S_{\text{cl}}/S)\Phi_2$, where S is the surface area of the particle and S_{cl} is the mean surface area of the cluster.

A similar model was advanced earlier by Marshall & Madden (1959). These authors described a two-mineral mixture where mineral surfaces are a mosaic of areas which are either covered by clay (zone 1) or free of it (zone 2). In zone 1 the EDL contains a large number of cations (or anions); zone 2 is depleted in them. Application of an electric field generates a concentration gradient around zones. The relaxation time τ

needed to attain equilibrium is $\tau \propto L_2$, where L is an average distance between clay islands.

The clay coating model (De Lima & Sharma 1992; Garrouch & Sharma 1994) is also similar to GLFP theory; here the clay envelope plays the role of a polarized EDL. The relaxation time derived in this case is identical to expression (44).

3.3 Partial saturation effects: percolation models

The MWBH theory assumes that the embedding media can be filled by inclusions up to high values of Φ_2 homogeneously, without clustering of particles. However, experiments show that the random addition of conductive inclusions to an insulating matrix leads to the formation of clusters of conductors, and eventually to a transition from insulating to conducting behaviour at some critical concentration of the conducting phase Φ_c . This critical threshold is reached when a continuous conducting path is formed in the system. Gueguen *et al.* (1997) showed that the critical concentration, predicted by effective medium theory, results from an attempt to apply the theory beyond its limits of validity, and does not correspond to real data.

Conductor–insulator transitions are correctly described by percolation theory (Broadbent & Hammersley 1957; Shante & Kirkpatrick 1971; Stauffer & Aharony 1992; Sahimi 1994; Gouyet 1992). The critical concentration is defined as a percolation threshold and the system-spanning cluster of conductive inclusions as an infinite cluster (IC). This approach is obviously of great interest for rocks, which are mixtures of conductors and insulators.

Shante & Kirkpatrick (1971) developed a self-consistent model in order to calculate the conductivity of a network of bonds with random conductances. Percolation theory predicts the behaviour of ϵ and σ as a power function of the quantity $\Delta\Phi = (\Phi - \Phi_c)$ in the critical region of conductor concentration Φ , namely, near the percolation threshold.

Geometrically, bond percolation can be defined on a regular lattice of bonds, which are either occupied with probability p , or empty with probability $(1 - p)$. A set of connected occupied bonds forms a cluster. Clusters are of finite size at $p < p_c$ and

form an infinite cluster at $p = p_c$. As percolation is similar to critical transitions the 'distance' Δp from the critical point p_c is the most important parameter in percolation theory:

$$\Delta p = p - p_c. \quad (54)$$

The percolation probability $P(p)$ defines the probability that a given bond belongs to the IC. $P(p) = 0$ at $p < p_c$, but at $p > p_c$, percolation probability P increases as

$$P(p) \sim \Delta p^\beta, \quad (55)$$

where β is the critical exponent of percolation probability.

The second important parameter is the correlation length L_c , which can be loosely defined as the maximal size of finite clusters of occupied bonds for $p < p_c$. A similar definition is true for the correlation length of empty bonds above the percolation threshold ($p > p_c$). The correlation length diverges as $p \rightarrow p_c$:

$$L_c(p) \sim \Delta p^{-\nu}, \quad (56)$$

where ν is the critical exponent of correlation length.

The occupied bonds can be identified as conductors of conductance b and the empty ones as insulators of conductance a . Thus geometrical percolation maps on to the problem of the electrical conductivity of composites. The electrical properties of such systems are completely defined by the ratio $b/a = h_c$ and the increment of p , i.e. by the parameter Δp . The critical behaviour of conductivity near the percolation threshold allows the introduction of the following scaling expressions for the effective conductivity of the network, σ (Efros & Shklovskii, 1976; Sahimi 1994):

$$\sigma = a|\Delta p|^t f_\pm(x), \quad (57)$$

where

$$x = h_c|\Delta p|^{-t-s} \quad (58)$$

and t and s are the critical exponents of conductivity for the transitions metal–insulator and conductor–superconductor respectively, and f_+ and f_- are homogeneous functions, corresponding to $p > p_c$ and $p < p_c$ respectively.

The most interesting contribution of the percolation approach to the theory of composites may be the prediction of scale dependence of σ near the percolation threshold. In contrast to effective medium theories, which imply the existence in a heterogeneous system of a representative elementary volume (REV), percolation theory predicts that at $p = p_c$ the effective conductivity depends on the sample size in a regular manner (Havlin & Bunde 1991):

$$G \sim L^{-t}, \quad (59)$$

where L is the size of the sample and t is the conductivity exponent. The point is that near the percolation transition the system is fractal and contains non-conducting clusters (holes). The size of the holes increases as the system approaches the threshold from above p_c , causing a regular decrease of σ . On the other hand, in finite conducting systems at $p > p_c$ the fragmentation of the sample will cause a transition from a conducting to an insulating state when the scale of partition becomes less than the correlation length of non-conducting clusters L_{cn} ; in most cases, for a small part of the system the conductivity path present in the large sample will be destroyed. This means that the hypothesis of REV, which is the basic assumption of the effective medium theories, is not valid for

systems near percolation transitions. Scale dependence of specific physical properties, obtained from the first principles of modern theory of disordered media, is a striking result, which may even affect the interpretation of geophysical data (Jones 1995; Chelidze 1987).

Eq. (57) can be generalized for complex conductances a^* and b^* using either lattice gas or equivalent circuit models (Efros & Shklovskii 1976; Havlin & Bunde 1991; Sahimi 1994). Here we follow the latter approach. If p is the fraction of bonds which are purely resistive, where R is the resistance of the bond and $(1 - p)$ is the fraction of purely capacitive bonds, where the capacitance of the bond is C , then, according to Sahimi (1994),

$$a = 1/R, \quad (60)$$

$$b = i\omega C. \quad (61)$$

The conductance ratio $h_c = b/a$ is given by

$$h_c = i\omega/\omega_0, \quad (62)$$

where $\omega_0 = 1/RC$.

In the static limit $\omega = 0$, the capacitors become insulators and the model reduces to a conductor–insulator mixture; in this case the σ^* is calculated by eq. (57), where

$$a = 1/R, \quad (63)$$

$$x = i(\omega/\omega_0)|\Delta p|^{-t-s}. \quad (64)$$

The frequency-dependent complex dielectric permittivity is obtained as follows:

$$\varepsilon^* = \sigma^*/i\omega. \quad (65)$$

This equation predicts unusual behaviour of the static DP as $\omega \rightarrow 0$. Using a Taylor expansion of $f(x)$ around $x = 0$ (Clerc *et al.* 1990; Sahimi 1994), we obtain from (65) and the complex analogue of eq. (57)

$$\varepsilon \sim A|\Delta p|^{-s}, \quad (66)$$

where A is the amplitude factor. This equation predicts that ε diverges when $p \rightarrow p_c$. This effect is called polarization catastrophe. Eq. (66) is similar to eq. (31), but there is a difference in the critical concentration value and the exponent, which is explained by a different approach to the structure of the composite: the former uses a fractal approximation and the latter a continuous media approximation.

The third striking result is for loss angle. As $p \rightarrow p_c$, the effective conductivity becomes

$$\sigma^* \approx \frac{f(0)}{R} \left(\frac{i\omega}{\omega_0} \right)^{t+t+s}. \quad (67)$$

This means that the frequency dependences of both σ' and σ'' are identical. Then, as $p \rightarrow p_c$, the ratio $\sigma'/\sigma'' = \varepsilon''/\varepsilon' = \tan \delta$ is independent of frequency (Fig. 5a); that is, the loss angle in the critical state is universal ($\delta_c = \text{const}$):

$$\delta_c = \frac{\pi}{2} \frac{s}{t+s}, \quad (68)$$

and for 3-D systems it should be equal to 0.45. The Cole–Cole plot is asymmetric and at $p = p_c$ degrades to a straight line with a slope of 45°. It may be difficult to reveal these peculiarities experimentally because, as p deviates from p_c , the unusual behaviour of δ is observable only in a limited frequency range.

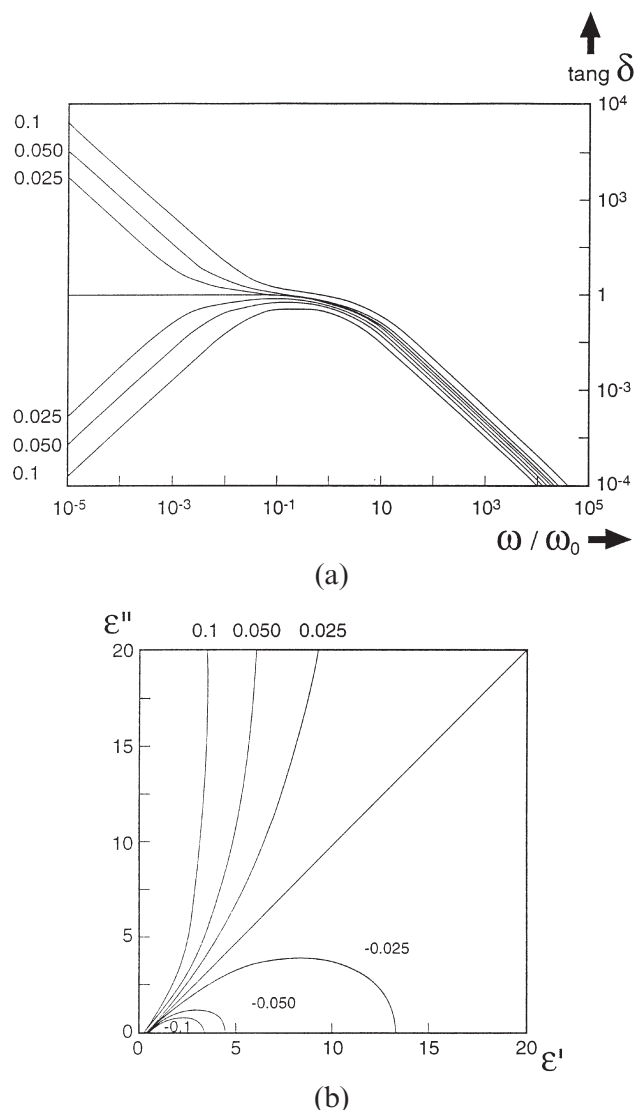


Figure 5. (a) A log-log plot of $\tan \delta$ versus reduced frequency ω/ω_0 according to the effective network approximation (ENA) for the 3-D RC model on a cubic lattice. Values of distance from the percolation threshold Δp are indicated on the curves. (b) Cole-Cole plot of the complex dielectric constant according to ENA for the same model; values of Δp are indicated (Clerc *et al.* 1990).

The percolation model can be useful in interpreting interesting hysteresis phenomena in electrical and other transport properties of partially saturated porous rocks (Knight & Nur 1987; Knight 1991) because it takes into consideration not only the content of water but also the details of fluid distribution in the porous space, which can be different at imbibition and drainage for the same water content. It seems that the site-bond percolation model is an appropriate tool for the explanation of the hysteresis effect (Paper II).

We show in Paper II how the above theories may be applied to real rocks with various water contents.

4 CONCLUSIONS

Electrical spectra of porous rocks in the 1 mHz–10 MHz range reflect numerous polarization processes resulting mainly

from rock heterogeneity. The first is the Maxwell–Wagner polarization due to differences in the electric bulk properties of rock components. Maxwell–Wagner theory allows the prediction of electrical spectra of mixtures from bulk partial volumes, the properties of their components and their microstructure. However, Maxwell–Wagner theory, even when extended further by Bruggeman for concentrated systems and by Hanai for the complex dielectric constant or complex conductivity of components, proves unable to explain the anomalously large polarization in water-bearing rocks at low frequencies. The problem is that the MWBH theory does not take into account important points, such as surface conductivity and polarization, as well as clustering of components in disordered mixtures. Nevertheless, the MWBH theory is a starting point for any theoretical interpretation of experimental electrical spectra.

The simplest surface effect is due to surface migration and diffusion of charges under an applied field limited by a thin surface layer; this can be a conductive film or a closed electrical double layer. Film polarization is especially strong at moderate partial saturations, where it can produce strong LF polarization.

Another three-component system, a conducting matrix containing conductive grains with a thin insulating envelope, also shows a strong LF polarization. An example of such a composite is a system of oxidized metal particles embedded in brine-saturated rock. In this case, polarization is due to the large capacitance of thin dielectric shells, which dies out at high frequency when shells are shunted by displacement currents.

Very strong dielectric effects are caused by the polarization of ‘open’ electric double layers on the solid–liquid interface at large saturations. Whereas polarization of bound charges in ‘closed’ double layers produces relatively weak polarization, akin to that of films, diffusion potentials generated by the free exchange of ions between ‘open’ deformed EDL and the adjoining electrolyte produce out-of-phase currents, and hence large values of effective dielectric constant. This model is known as GLFP; it can also be applied to heterogeneously charged surfaces.

The anomalous dielectric properties of partially saturated rocks can alternatively be interpreted using percolation theory. This theory predicts that, when the conductive fraction increases, clustering of conductive inclusions develops, and the thickness of insulating gaps between conductive clusters decreases, causing a large increment in the capacitance of the sample. The dielectric catastrophe is expected near the percolation threshold where the dielectric constant diverges. Further increases in the conductive component cause the shunting of insulating capacitive gaps and the decrease of the dielectric constant. The difference amongst the equations which describe the effects of dielectric catastrophe in MWBH and percolation theories illustrates the difference between continuous and fractal approaches to disordered media. The fractal approach also allows an understanding of the scale dependence of specific electric properties of heterogeneous materials; this is not possible in MWBH theory.

The multitude of possible relaxation processes and available theories call for model experiments on simple systems under controlled conditions using independent methods in addition to electrical spectroscopy. Experimental electrical spectra of real geological objects and geophysical applications are considered in the Paper II.

As the electrical spectra are actually Fourier transforms of the time variation of polarization decay, all the theories quoted can be applied to the interpretation of both ac and induced-polarization data.

REFERENCES

- Adamson, A.W., 1982. *Physical Chemistry of Surfaces*, John Wiley & Sons, New York.
- Broadbent, S.R. & Hammersley, I.M., 1957. Percolation processes, *Proc. Cambridge Phil. Soc.*, **53**, 629–642.
- Bruggeman, D., 1935. Berechnung Verschiedeneren Physikalischer Konstanten von Heterogenen Substanzen, *Ann. Phys.*, **24**, 636–679.
- Bunde, A. & Havlin, S., eds, 1992. *Fractals and Disordered Systems*, Springer-Verlag, Berlin.
- Cao, Q.-Z., Wong, P.-Z. & Schwartz, L.M., 1994. Numerical studies of the impedance of blocking electrodes with fractal surfaces, *Phys. Rev. B*, **50**, 5771–5774.
- Chelidze, T.L., 1963. On the frequency dependence of dielectric properties of rocks, *Proc. Inst. Geophys. Academy of Sciences of Georgian SSR*, **21**, 161–173 (in Russian).
- Chelidze, T.L., 1971. Electrical Properties of Rocks, in *Rocks in Physical Fields*, pp. 53–109, ed. Chanturishvili, L., Metsniereba Publ. House, Tbilisi (in Russian).
- Chelidze, T., 1987. *Percolation Theory in Mechanics of Geomaterials*, Nauka, Moscow (in Russian).
- Chelidze, T.L., Derevjanko, A.I. & Kurilenko, O.D., 1977. *Electrical Spectroscopy of Heterogeneous Systems*, Naukova Dumka, Kiev (in Russian).
- Chelidze, T.L., Chelishvili, M., Togonidze, D. & Khatiasvili, N., 1979. *Electrical and Magnetic Properties of Rocks at Elevated Temperatures*, Metsniereba Publ. House, Tbilisi (in Russian).
- Chelidze, T., Gueguen, Y. & Le Ravalec, M., 1998. From classic to fractal mechanics of disordered media: self-consistency versus self-similarity, in *PROBAMAT—21st Century: Probabilities and Materials*, pp. 197–231, ed. Frantzisonis, G.N., Kluwer, the Netherlands.
- Chelidze, T.L., Gueguen, Y. & Ruffet, C., 1999. Electrical spectroscopy of porous rocks: a review—II. Experimental results and interpretation, *Geophys. J. Int.*, **137**, 16–34 (this issue).
- Clerc, J.P., Giraud, G., Laugier, J.M. & Luck, J.M., 1990. The electrical conductivity of binary disordered percolation clusters, fractals and related models, *Adv. Phys.*, **39**, 193–306.
- Cole, K.S. & Cole, R.H., 1941. Dispersion and adsorption in dielectrics (I), *J. Chem. Phys.*, **9**, 341–351.
- Debye, P., 1929. *Polar Molecules*, Dover, New York.
- De Groot, S. & Mazur, P., 1967. *Thermodynamics of Irreversible Processes*, Mir, Moscow (Russian edition).
- De Lima, O.A.I. & Sharma, M.M., 1992. A generalized Maxwell-Wagner theory for membrane polarization in shaly sands, *Geophysics*, **57**, 431–440.
- Dukhin, S.S., 1975. *Electrical Conductivity and Electrokinetic Properties of Disperse Systems*, Naukova Dumka, Kiev (in Russian).
- Dukhin, S.S. & Shilov, V.N., 1974. *Dielectric Phenomena and the Double Layer in Disperse Systems and Polyelectrolytes*, John Wiley & Sons, New York.
- Dukhin, S.S., Sorokina, T.V. & Chelidze, T.L., 1969. On the theory of low-frequency polarization of coarse-grained disperse system, *Kolloid Zhurnal (Moscow)*, **31**, 823–830 (in Russian).
- Efros, A.L. & Shklovskii, B.I., 1976. Critical behaviour of conductivity and dielectric constant near the metal-non-metal transition threshold, *Phys. Status Sol.*, **76**, 475–489.
- Fricke, H., 1924. A mathematical treatment of the electrical conductivity and capacity of disperse systems, *Phys. Rev.*, **24**, 575–587.
- Frolich, H., 1990. *Theory of Dielectrics*, Oxford University Press, Oxford.
- Garrouch, A.A. & Sharma, M.M., 1994. The influence of clay content, salinity, stress and wettability on the dielectric properties of brine-saturated rocks: 10 Hz to 10 MHz, *Geophysics*, **59**, 909–917.
- Glover, P., Gomez, J., Heredish, P., Boon, S., Sammonds, P. & Murrel, S., 1996. Modelling of stress-strain behaviour of saturated rocks using complex electrical conduction measurements, *Surv. Geophys.*, **17**, 307–330.
- Glover, P.W.J., Meredith, P.G., Sammonds, P.R. & Murrel, S.A.F., 1994a. Ionic surface electrical conductivity in sandstone, *J. geophys. Res.*, **99** (B11), 21 635–21 650.
- Glover, P.W.J., Meredith, P.G., Sammonds, P.R. & Murrel, S.A.F., 1994b. Measurements of complex electrical conductivity and fluid permeabilities in porous rocks at raised confining pressures, in *Rock Mechanics in Petroleum Engineering*, pp. 29–36, Proc. EUROROCK94, Delft, Balkema, Amsterdam.
- Gouyet, J.-F., 1992. *Physique et Structures Fractals*, Masson, Paris.
- Gueguen, Y. & Palciauskas, V., 1994. *Introduction to the Physics of Rocks*, Princeton University Press, Princeton.
- Gueguen, Y., Chelidze, T. & Le Ravalec, M., 1997. Microstructures, percolation thresholds and rock physical properties, **279**, 23–35.
- Hanai, T., 1968. Electrical properties of emulsions, in *Emulsion Science*, pp. 354–478, ed. Sherman, P., Academic Press, New York.
- Havlin, S. & Bunde, A., 1991. Percolation in *Fractals in Disordered Systems*, pp. 97–149, eds Bunde, A. & Havlin, S., Springer-Verlag, Berlin.
- Hoekstra, P. & Doyle, W.T., 1971. Dielectric relaxation of surface adsorbed water, *J. Colloid Interface Sci.*, **36**, 513–521.
- Howell, B.F. & Licastro, P.H., 1961. Dielectric behaviour of rocks and minerals, *Am. Mineral.*, **64**, 269–288.
- Jones, J., 1995. Scale-dependent resistivity measurements of Oracle granite, *Geophys. Res. Lett.*, **22**, 1453–1456.
- Keller, G.V., 1982. Electrical properties of rocks and minerals, in *Handbook of Physical Properties of Rocks*, Vol. 1, pp. 000–000, ed. Carmichael, R. S., CRC Press, Boca Raton.
- Keller, G.V. & Licastro, P.H., 1959. Dielectric constant and electrical resistivity of natural state cores, *USGS Bull.*, **1052-H**, 257–285.
- Knight, R., 1983. The use of complex plane plots in studying the electrical response of rocks, *J. Geomag. Geoelectr.*, **35**, 767–776.
- Knight, R., 1991. Hysteresis in the electrical resistivity of partially saturated sandstones, *Geophysics*, **56**, 2139–2147.
- Knight, R. & Dvorkin, J., 1992. Seismic and electrical properties of sandstones at low saturations, *J. geophys. Res.*, **97**, (B12), 17 425–17 432.
- Knight, R.J. & Nur, A.N., 1987. The dielectric constant of sandstones, 60 kHz to 4 MHz, *Geophysics*, **52**, 644–654.
- Korvin, G., 1992. *Fractal Models in the Earth Sciences*, Elsevier, Amsterdam.
- Le Mehaute, A. & Crepy, G., 1983. Introduction to transfer and motion in fractal media; the geometry of kinetics, *Solid State Ionics*, **9&10**, 17–30.
- Lockner, D. & Byerlee, J., 1985. Complex resistivity measurements of confined rocks, *J. geophys. Res.*, **90**, 7837–7847.
- Marshall, D.J. & Madden, T.R., 1959. Induced polarization, a study of its causes, *Geophysics*, **24**, 790–813.
- Maxwell, J.C., 1891. *A Treatise on Electricity and Magnetism*, Dover Publ. Inc., New York.
- Mogi, K., 1985. *Earthquake Prediction*, Academic Press, New York.
- O'Konsky, C.T., 1960. Electrical properties of macromolecules. V. Theory of ionic polarization of polyelectrolytes, *J. Phys. Chem.*, **64**, 605–619.
- Olhoeft, G.R., 1981. Electrical properties of rocks, in *Physical Properties of Rocks and Minerals*, pp. 257–330, eds Touloukian, Y.S., Judd, W.R. & Roy, R.F., McGraw Hill, New York.
- Olhoeft, G.R., 1985. Low frequency electrical properties, *Geophysics*, **50**, 2492–2503.
- Parkhomenko, E.I., 1967. *Electrical Properties of Rocks*, Plenum, New York.

- Pauly, H. & Schwan, H.S., 1959. Über die impedanz einer suspension von kugelförmigen teilchen mit einer schale, *Zeitschr. Naturforsch.*, **14b**, 125–131.
- Pride, S., 1994. Governing equations for the coupled electro-magnetics and acoustics of porous media, *Phys. Rev. B*, **50**, 15 678–15 696.
- Reinolds, J.A. & Hough, J.M., 1957. Formulae for dielectric constant of mixtures, *Proc. Phys. Soc. B*, **70**, 769–775.
- Revil, A. & Glover, P., 1997. Theory of ionic electrical conduction in porous media, *Phys. Rev. B*, **55**, 1757–1773.
- Roberts, J.J. & Lin, W., 1997. Electrical properties of partially saturated Topopah Spring tuff: water distribution as a function of saturation, *Water Res. Research*, **33**, 577–587.
- Roberts, J.J. & Tyburczy, J.A., 1994. Frequency dependent electrical properties of minerals and partial melts, *Surv. Geophys.*, **15**, 239–262.
- Ruffet, C., 1993. La conductivite electrique complexe dans quelques roches cristales, *PhD thesis*, University of Strasbourg.
- Ruffet, C., Gueguen, Y. & Darot, M., 1991a. Rock conductivity and fractal nature of porosity, *Terra Nova*, **3**, 265–275.
- Ruffet, C., Gueguen, Y. & Darot, M., 1991b. Complex measurements and fractal nature of porosity, *Geophysics*, **56**, 758–768.
- Sahimi, M., 1994. *Applications of Percolation Theory*, Taylor & Francis, London.
- Schwarz, G., 1962. A theory of low frequency dielectric dispersion of colloidal particles, *J. Phys. Chem.*, **66**, 2636–2642.
- Sen, P.N., 1989. Unified models of conductivity and membrane potential of porous media, *Phys. Rev. B*, **39**, 9508–9517.
- Sen, P.N. & Chew, W.C., 1984. Dielectric enhancement due to geometrical and electrochemical effects, *Proc. Am. Inst. Phys. Conf.*, **107**, 52–65.
- Shante, V.K.J. & Kirkpatrick, S., 1971. An introduction to percolation theory, *Adv. Phys.*, **20**, 325–357.
- Shen, L.C., Savre, W.C., Price, J.M. & Athavale, K., 1985. Dielectric properties of reservoir rocks at ultra-high frequencies, *Geophysics*, **50**, 692–704.
- Sillars, R.W., 1937. The properties of dielectric, containing semi-conducting particles of various shapes, *J. Inst. Electr. Engineers (London)*, **80**, 379–394.
- Smith-Rose, R.L., 1934. Electrical measurements on soil with alternating currents, *J. Inst. Electr. Engineers (London)*, **72**, 452.
- Stauffer, D. & Aharony, A., 1992. *Introduction to Percolation Theory*, Taylor & Francis, London.
- Trukhan, E.M., 1962. Dispersion of dielectric constant of disperse systems, *Solid State Physics (Moscow)*, **4**, 3496–3511 (in Russian).
- Von Hippel, A.R., 1954. *Dielectrics and Waves*, John Wiley & Sons, New York.
- Wagner, K.W., 1924. Erklärung der Dielectricischen Nachwirkungsvorgänge auf grund Maxwell'scher vorstellungen, *Archiv Electrotechnik*, **2**, 371–387.
- Wert, A. & Thomson, R., 1964. *Physics of Solids*, McGraw-Hill, New York.
- Wong, P.Z., 1987. Fractal surfaces in porous media, in *Physics and Chemistry of Porous Media*, Vol. 2, eds Bahavar, J.P., Koplik, J. & Winkler, K.W., *Am. Inst. Phys.*, **154**, 304–318.

APPENDIX A: SYMBOLS AND ABBREVIATIONS

- a —radius of particle
 a_{cl} —characteristic size of charged surface cluster
 C —capacitance
 d —Euclidian dimension
 d_e —dipole moment of particle
 D —bulk diffusion coefficient of electrolyte

- D_e —electrical induction
 D_f —fractal dimension
 D_s —surface diffusion coefficient
 E —electrical field intensity
 f —frequency (Hz)
 f_c —critical frequency
 f_1, f_2 —field ratio in composite dielectric
 h —conductivity ratio in composite dielectric
 J —electrical flux
 k —Boltzmann constant
 L —sample size
 L_c —correlation length in percolation theory
 p —probability of site (bond) occupation in percolation theory
 p_c —probability p at percolation threshold
 P —polarization
 P —percolation probability
 R —resistance
 s —critical index of conductor–superconductor transition in percolation theory
 t —conductivity critical index in percolation theory
 T —temperature, °C
 z —valency
 Z —impedance
- α —Cole–Cole exponent
 β —critical exponent of percolation probability
 δ —loss angle
 ϵ —dielectric permittivity
 ϵ_0 —dielectric permittivity of vacuum
 ϵ_d^* —complex dielectric permittivity excluding ohmic conductivity
 ϵ_r —relative dielectric permittivity (dielectric constant); $\epsilon_r = \epsilon/\epsilon_0$
 ϵ^* —complex dielectric permittivity
 ϵ_s —static dielectric permittivity
 ϵ_∞ —high-frequency (unrelaxed) dielectric permittivity
 ϵ' —real part of complex dielectric permittivity
 ϵ'' —imaginary part of complex dielectric permittivity, including σ
 ϵ_d'' —imaginary part of complex dielectric permittivity, excluding σ
 ϵ_1 —dielectric permittivity of embedding substance in composite
 ϵ_2 —dielectric permittivity of inclusions
 ϵ_3 —dielectric permittivity of shell
 η —dynamic viscosity
 κ^{-1} —Debye screening length
 ν —critical index of correlation length
 ρ —specific electrical resistivity
 σ —specific conductivity
 σ_0 —dc conductivity
 σ^* —complex conductivity
 σ' —real part of complex conductivity
 σ'' —imaginary part of complex conductivity
 σ_1 —conductivity of matrix
 σ_2 —conductivity of inclusions
 σ_3 —conductivity of shell ($S\ m^{-1}$)
 σ_s —specific surface conductivity (S)
 σ_e —brine conductivity
 τ —relaxation time
 τ_a —relaxation time of particle of radius a
 τ_{cl} —relaxation time of charged surface cluster

τ_s —relaxation time for surface diffusion of particle of radius a
 Φ_2 —partial volume of inclusions
 Φ_c —critical partial volume of inclusions
 ω —angular frequency
 ω_c —critical angular frequency
 ψ —surface potential at distance x from surface
 ψ_s —Stern potential

$\mathcal{I}m$ —imaginary part
 $\mathcal{R}e$ —real part
 EDL—electrical double layer
 DP—dielectric permittivity
 DC—dielectric constant
 MWBH—Maxwell–Wagner–Bruggeman–Hanai
 GLFP—gigantic low-frequency polarization